A Wilson Loop after the First Renormalisation Step

22 October 2026

Contents

1 The Insertion at a Fixed Cutoff **2**
1.1 Normalisation . . . 5
1.2 The Main Result . . . 5

2 Extraction from the Effective Interaction **18**

3 The Sourced Effective Interaction at the First Step **19**
3.1 Branch Data, the Outer Operation and the Conditional Integral . . . 23
3.2 The Constrained Hessian at a Regular Background and Its Stencils . . . 39
3.3 The Source Phase and the Local Primitives . . . 43
3.4 The Boundary Law, the Exterior Interpolation and the Collar . . . 49
3.5 A Fixed Walk Expansion of the Centred Kernel . . . 58
3.6 The Selected Integrals and the Polymer Representation . . . 61
3.7 Probe Kernels and the Centred Expectation . . . 64
3.8 Joint Extraction of the Scalar Reference . . . 67

4 The Flat Regular-Region Calculation **71**
4.1 The Covariance and the Normalised Insertion . . . 71

5 The Straight-Edge Coefficient from Gaussian Marginals **74**
5.1 Connected Terms of the Normalised Integral . . . 75
5.2 The Integrated Density Estimate . . . 78
5.3 Extent Weights and the Square Geometry . . . 80
5.4 The Perimeter Coefficient . . . 83

6 The Corner Coefficient in the Same First Step **85**
6.1 A Local Corner Subtraction . . . 85
6.2 Identification with the Square Constant . . . 87

7 Identification in a Regular Integration Region **89**
7.1 The Wilson Source before the Flat Evaluation . . . 89
7.2 Regional Integrals and Their Connected Densities . . . 90
7.3 Periodic Regions and Bałaban's Kernel . . . 93
7.4 The Regional Square Scalar . . . 98

8 The Insertion Retained after One Step **102**
8.1 The Conditional Integral . . . 103
8.2 The Closed Flux Insertions . . . 108
8.3 The Numerator Carried to the Next Transformation . . . 108

9 The Exponential Moment of the Leading Error **110**
9.1 Scales . . . 111
9.2 The Refined Split . . . 111
9.3 The Joint Witness Bound . . . 111
9.4 The Exponential Moment of the Boundary Sum . . . 112
9.5 The Moment of the Leading Error . . . 113

10 Two Disjoint Squares 114
10.1 Setting . . . 114
10.2 Groups for Two Squares . . . 115
10.3 Reference Scalars and the Joint Extraction . . . 118
10.4 The Defects of a Family with Cross Groups . . . 119
10.5 Pins for the Groups of Two Squares . . . 120
10.6 The Cross Terms . . . 123
10.7 Proof of Theorem 1.13 . . . 125

11 Towards the Next Step 127

A The Flat Covariance 130

B Background Estimates at the First Step 136
B.1 Cubes, Localisation Domains and Tree Lengths . . . 136
B.2 The Regular-Background Bounds . . . 138
B.3 Uniform Charts and Primitive Bounds . . . 143
B.4 Proof of the Gaussian Payment . . . 152
B.5 The Mayer Constants and the Activity Bounds . . . 153
B.6 Proof of the Vacuum-Cluster Bounds . . . 156
B.7 The Reference Data . . . 157
B.8 The Thresholds . . . 158

C The Local Interaction at the Identity 162

D The Exterior Response and Its History Average 163
D.1 Objects and the Conditional Law . . . 164
D.2 Local Derivative Bounds . . . 165
D.3 An Extension within the Proved Chart . . . 166
D.4 Covariance Decay at the Hard Boundary . . . 167
D.5 The Response . . . 169
D.6 The Physical Substitution . . . 170
D.7 The History Average by One-Point Witnesses . . . 171
D.8 The Minimum Comparison from Local Data . . . 181

E The Signed Integrated Wilson Defects 189
E.1 Operand and Retained Normalisation . . . 189
E.2 Histories near the Bad Region . . . 189
E.3 The Interior Operand . . . 190
E.4 The Joint Signed Selected-Integral Estimate . . . 191
E.5 The Closed Group Subtraction . . . 194
E.6 The Joint Kernel Payment . . . 200
E.7 Quantitative Group Estimates . . . 203
E.8 The Coefficient Extension and the Countable-Gas Limit . . . 209
E.9 Analytic Inputs for the Coefficient Extension . . . 211
E.10 Identification of the Centred Coefficient and Its Local Extension . . . 216
E.11 Quantitative Assembly . . . 219

F The Regional Scalar of the Covered Branch 221
F.1 Conventions . . . 222
F.2 Identification with the Regional Square Integral . . . 222
F.3 The Scalar of the Branch Identity . . . 223

1 The Insertion at a Fixed Cutoff

This paper follows a Wilson loop through the first renormalisation-group step of Bałaban's construction of four-dimensional lattice Yang–Mills theory [9, 8]. That step is the block-spin transformation T of (1.3): the bond variables of a periodic fine lattice of spacing ε are averaged over blocks of side L, and the bare measure is mapped to the induced measure of the

bond variables of the coarse lattice of spacing $L\varepsilon$. The loop is inserted in the bare measure and carried through T; the result (Section 1.2) writes its bare expectation as a perimeter factor times a corner factor times an average, over the coarse ensemble, of the loop at the background field reconstructed from the coarse field, plus two error terms that it bounds.

The gauge group G is a compact connected simple Lie group, g_0 is the bare coupling, and $\chi\colon G \to \mathbb{C}$ is a normalised irreducible character, so that $\chi(\mathbf{1}) = 1$. Let $\varepsilon > 0$ be the fine lattice spacing and $L > 1$ the blocking factor. The fine and coarse bond fields are U and V, respectively.

C is an oriented square contour on the fine lattice and $\bar{C}$ an oriented square on the lattice of spacing $L\varepsilon$, each side of $\bar{C}$ being represented by the corresponding straight fine path in C; the two contours describe the same square in the underlying torus.

For a field U and an ordered contour $C = (b_1, \dots, b_n)$, holonomy and the Wilson loop are

$$\mathrm{Hol}_U C \coloneqq U(b_1)\cdots U(b_n), \qquad W_\chi(C;U) \coloneqq \chi(\mathrm{Hol}_U C)\,. \tag{1.1}$$

The bond field satisfies $U(b^{-1}) = U(b)^{-1}$ when a contour traverses a bond backwards. The character makes W_χ gauge invariant.

Let E and E_1 be the positive fine and coarse bonds of the four-dimensional periodic lattices, and let $\mathcal{P}_T$ be the unoriented fine plaquettes. The action uses a fixed unitary realisation $G \subset U(N)$ and $\mathrm{tr}_N = N^{-1}\,\mathrm{Tr}$. With ∂p denoting the oriented boundary of p, the bare action and density are

$$S(U) \coloneqq \sum_{p\in\mathcal{P}_T} (1 - \mathrm{Re}\,\mathrm{tr}_N\, U(\partial p)), \qquad \rho_0(U) \coloneqq \exp(-g_0^{-2}S(U) - E_\varepsilon)\,. \tag{1.2}$$

Here E_ε is a field-independent vacuum scalar. Product normalised Haar measures on G^E and G^{E_1} are denoted by $\mathrm{d}U$ and $\mathrm{d}V$.

The full first-step operator, before the small-field/large-field decomposition, is [9, (0.1)]:

$$(T\rho)(V) = \int \left(\prod_{c\in E_1} \delta_G(M(U,c)V(c)^{-1}) \right) \rho(U)\,\mathrm{d}U\,. \tag{1.3}$$

Here δ_G is the group Dirac measure normalised against Haar measure, $\int \delta_G(WV^{-1})\,\mathrm{d}V = 1$ for every $W \in G$, and M is the averaging operation, defined as follows. For a positive coarse bond c let $P_c(U)$ be the transport along the straight fine path from c_- to c_+, let $\beta_c(U)$ be the logarithmic average of [8, (0.4)], displayed in (4.1) below, and let $\mathcal{O}_c$ be the set of fine configurations U at which every logarithm in $\beta_c(U)$ has its argument in the chart domain (the chart is the inverse of $A \mapsto e^{iA}$ on the Hilbert ball $h(A) < \varrho_G$, fixed in the conventions paragraph of Section 3). The averaging operation is

$$M(U,c) \coloneqq \begin{cases} e^{i\beta_c(U)}P_c(U), & U \in \mathcal{O}_c, \\ P_c(U), & U \notin \mathcal{O}_c. \end{cases} \tag{1.4}$$

[9, (0.1)] integrates over every fine configuration, so it needs an averaging operation on all of G^E; Bałaban's own formula is [8, (0.4)], defined where its logarithms are, and Bałaban states in [9, p. 243] that any averaging operation satisfying general properties analogous to [8, (0.5)–(0.9)] may be taken.

The properties of (1.4) used in this paper are those of Lemma 1.1; (ii) is the analogue of the gauge covariance [8, (0.6)]. On the branches the averaging operation is evaluated on

$\mathcal{O}_c$ only, where it is [8, (0.4)] ((B1), Section 3.1); its value off $\mathcal{O}_c$ enters only through the constraint on the rows of Ω_1^c.

Lemma 1.1 (The Averaging Operation)**.** *(i) $\mathcal{O}_c$ is open, and $M(\bullet, c)\colon G^E \to G$ is Borel, analytic on $\mathcal{O}_c$, where it is [8, (0.4)], and a function of the bonds with both endpoints in the two blocks of c alone.*

(ii) For every fine gauge transformation $g\colon T \to G$, $U^g(b) = g(b_-)U(b)g(b_+)^{-1}$, one has $U^g \in \mathcal{O}_c$ if and only if $U \in \mathcal{O}_c$, and $M(U^g, c) = g(c_-)M(U, c)g(c_+)^{-1}$; in particular $M(U^g) = M(U)$ when g is the identity at the block centres.

Proof. (i) The chart domain is open, being the image of the open ball under $A \mapsto e^{iA}$, which is injective with invertible differential there (the conventions paragraph of Section 3). Each loop $U(\gamma)U([x, x'])U(\gamma')^{-1}P_c(U)^{-1}$ of (4.1) is a product of bond variables and their inverses, continuous in U, so $\mathcal{O}_c$, the set where finitely many such loops lie in the chart domain, is open; the chart is analytic on the chart domain (inverse of an analytic map with invertible differential), so β_c and $e^{i\beta_c}P_c$ are analytic on $\mathcal{O}_c$, and P_c is analytic on G^E. A map that is continuous on an open set and continuous on its complement is Borel. The paths γ, $[x, x']$, γ' and the straight path of c have all their bonds with both endpoints in $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}\colon$ $\gamma \subset \mathcal{B}_{c_-}$, $\gamma' \subset \mathcal{B}_{c_+}$, and the straight path from $x \in \mathcal{B}_{c_-}$ to $x + Le_\mu \in \mathcal{B}_{c_+}$ leaves $\mathcal{B}_{c_-}$ at the bond at which it enters $\mathcal{B}_{c_+}$.

(ii) Every loop is based at c_-, so under g it is conjugated by $g(c_-)$. The Hilbert ball is Ad-invariant, h being Ad-invariant, and $e^{i\,gAg^{-1}} = ge^{iA}g^{-1}$; so the chart domain is invariant under conjugation, and for W in it $\frac{1}{i}\log(gWg^{-1}) = g(\frac{1}{i}\log W)g^{-1}$, by the injectivity of $A \mapsto e^{iA}$ on the ball. Hence $U^g \in \mathcal{O}_c$ if and only if $U \in \mathcal{O}_c$, $\beta_c(U^g) = g(c_-)\beta_c(U)g(c_-)^{-1}$ and $e^{i\beta_c(U^g)} = g(c_-)e^{i\beta_c(U)}g(c_-)^{-1}$ there, while $P_c(U^g) = g(c_-)P_c(U)g(c_+)^{-1}$ everywhere. The product gives the covariance on $\mathcal{O}_c$, and off $\mathcal{O}_c$ it is the covariance of P_c. *q.e.d.*

The right side of (1.3) is read as follows. For bounded Borel h on G^{E_1} and $W \in G^{E_1}$, $\int \mathrm{d}V\, h(V) \prod_{c\in E_1} \delta_G(W(c)V(c)^{-1}) = h(W)$; so for a nonnegative or integrable Borel function ρ on G^E, $T\rho$ is the image measure of $\rho\,\mathrm{d}U$ under the Borel map $M\colon G^E \to G^{E_1}$ of Lemma 1.1, $(T\rho)(A) = \int_{M^{-1}(A)} \rho\,\mathrm{d}U$ for Borel $A \subseteq G^{E_1}$, that is $\int h\,\mathrm{d}(T\rho) = \int h(M(U))\rho(U)\,\mathrm{d}U$. This is the meaning of (1.3) and of every group δ-function of a coarse variable in this paper: it is integrated against that variable first.

Bałaban writes [9, (0.1)] as a Haar density; for integrable ρ the finite measure $T\rho$ has a Haar density exactly when it is absolutely continuous with respect to $\mathrm{d}V$, and then $(T\rho)(V)$ denotes a version of that density. Every identity below in which $\rho_1(V)$ or $N_1(V)$ appears is an identity of measures, in which $\rho_1(V)\,\mathrm{d}V$ and $N_1(V)\,\mathrm{d}V$ stand for the measures ρ_1 and N_1 of (1.5), and Bałaban's transformations of (0.1) into (1.11)–(1.15) [9] are identities for the integrals of bounded Borel functions of V against these measures under the branch hypotheses (B1)–(B4) of Section 3.1, which record the facts about Bałaban's construction that these identities use.

The first-step objects are those of T alone, without Bałaban's additional operation **R** [9, p. 243], which changes the densities on the large-field regions.

The bare expectation is $\mathbb{E}_\varepsilon[F] := \int F\rho_0\,\mathrm{d}U / \int \rho_0\,\mathrm{d}U$ for bounded Borel F on G^E. Bałaban's cutoffs, gauge insertions and changes of variables give alternative representations of this same integral: the partitions of unity [9, (1.1), (1.4)] sum to one pointwise, so the sum over histories of the branch integrals is the full integral $\int \rho_0\,\mathrm{d}U$ whatever the value of M off $\bigcup_c \mathcal{O}_c$, the branch integrals using M on $\mathcal{O}_c$ only (Section 3.1).

For a finite or complex measure μ on G^E, $T\mu \coloneqq M_*\mu$ is the same pushforward, so that T acts on the measures defined next. The outgoing measure and the measure with the insertion are

$$\rho_1 \coloneqq T[\rho_0], \qquad N_1 \coloneqq T[\rho_0 W_\chi(C;\bullet)] \,. \tag{1.5}$$

Here ρ_1 is a finite positive measure on G^{E_1} with $\rho_1(G^{E_1}) = \int \rho_0 \,\mathrm{d}U$, and N_1 is a complex measure with $|N_1(A)| \leqslant \int_{M^{-1}(A)} \rho_0 |W_\chi| \,\mathrm{d}U \leqslant \rho_1(A)$ for every Borel A, so that $|N_1| \leqslant \rho_1$ for the total variation and $N_1 \ll \rho_1$. The outgoing Wilson function is the Radon–Nikodym derivative

$$W_1(C;V) = \frac{\mathrm{d}N_1}{\mathrm{d}\rho_1}(V) \,, \tag{1.6}$$

defined for ρ_1-almost every V, with $|W_1| \leqslant 1$ ρ_1-almost everywhere; a Borel version with $|W_1| \leqslant 1$ everywhere is fixed. Where ρ_1 and N_1 have Haar densities, $W_1(C;V) = N_1(V)/\rho_1(V)$ for ρ_1-almost every V, the density-ratio form in which [9] writes such quantities.

The intermediate integration over regular fluctuations is a function of the exterior fine field and the unintegrated fluctuations; these are integrated by the outer operation [9, (1.29)], which without the insertion gives ρ_1 (the vacuum construction is [9, (1.11)–(1.30)]).

1.1 Normalisation

The physical partition function and the expectation after the transformation are

$$\mathcal{Z}_0 \coloneqq \int \rho_0(U) \,\mathrm{d}U = \rho_1(G^{E_1}) > 0 \,, \tag{1.7}$$

$$\mathbb{E}_\varepsilon[W_\chi(C)] = \frac{1}{\mathcal{Z}_0} N_1(G^{E_1}) = \frac{1}{\mathcal{Z}_0} \int W_1(C;V) \,\mathrm{d}\rho_1(V) \,. \tag{1.8}$$

Both integrals include the small-field and the large-field regions; the normalising factors of the conditional fluctuation integrals used below are part of the construction of N_1 and ρ_1.

1.2 The Main Result

The main statement of the paper concerns $\mathbb{E}_\varepsilon[W_\chi(C)]$ for a square contour: after the first step the expectation is a perimeter factor times a corner factor, times the expectation, in the coarse ensemble enriched by the large-field branch data, of $\mathbf{1}_{A_{\mathrm{cov}}} e^{E^\circ_\xi} W_\chi(C_n;U_1)$: the fine loop at the background U_1 reconstructed from the coarse field and the exterior fine field, with a finite-geometry exponent and the indicator of the covered event, plus two error terms.

The objects are defined in Sections 3–8 and the appendices, and are listed here.

Theorem 1.13 gives the same form for the product of the Wilson functions of two vertex-disjoint squares in planes parallel to that of e_1, e_2, with the product of the two coefficients, the two finite-geometry exponents and a cross exponent bounded in terms of the distance between the squares; its objects specific to the pair are defined in Section 10 and referenced from its statement.

Notation. C_n is a coarse translate of the counterclockwise square with vertices 0, nLe_1, $nL(e_1+e_2)$, nLe_2 in fine-lattice units (Section 7.4), $n \geqslant 1$; it has $n_C = 4nL$ contour bonds, $4n$ coarse segments indexed cyclically by $j \in \{0,\dots,4n-1\}$, and $P(C_n)$ is its perimeter as it enters (8.1) (Section F.1).

$W_\chi(C;U)$ is the normalised character χ of the holonomy of the fine field U around C, ε the fine lattice spacing, and σ names the cutoff norm of (4.12), the operator or the normalised Hilbert–Schmidt norm (conventions of Section 3).

T is the operator (1.3) with the averaging operation (1.4); ρ_1, N_1, W_1 and $\mathcal{Z}_0$ are (1.5), (1.6) and (1.7), and $\mathbb{E}_\varepsilon$ is the bare expectation defined after (1.3).

$\xi = (\Omega_1, R_1, V, V_0, A_{\mathrm{e}})$ is the branch data of Section 3.1 (the small-field region Ω_1, the large-fluctuation region R_1, which with Ω_1 determines the filled region Λ_1, the coarse field, the exterior fine field and the exterior free fluctuation; at the original carrier there are no labels), $U_1 = U_1(V, V_0)$ the background of Section 3.1, $\mathbb{E}_{\mathsf{h}}$ the history expectation (D.20) (weight $\mathsf{w}_\xi = e^{c_\xi} I_{\xi;1}$, normalisation $\mathcal{Z}_0$; positive, with $\mathbb{E}_{\mathsf{h}}[1] = 1$), and A_{cov} the covered event of Lemma 3.36.

$\kappa_{C_n} := e^{-\delta m^\sigma(g_0,\varepsilon)P(C_n)}(z_{\mathrm{c}}^\sigma(g_0, \pi/2))^4$ is the scalar (8.1), with the perimeter coefficient δm^σ of (5.39) and the corner factor z_{c}^σ of (6.12), in the cutoff prescription σ fixed throughout.

The setting consists of σ, the averaging and mean prescriptions of [8, (0.4) and (0.10)–(0.11)], the gauge-fixing functional of [9, (1.5)–(1.7)], the designated-bond convention (Section F.1), and the cutoff test (1.18) at the radius $r = A_1(\log g_0^{-2})^{p_0}$ (the constants A_1, p_0 of [9], with its δ_0 as corrected in Section B.8 and (H4)).

E_{def} is the signed-defect term (E.1), E_{bdry} the exterior-response and uncovered term (D.21), and $E_\xi^\circ = E_{I,C_n,T}^\circ$ the finite-geometry exponent (F.4) of Proposition F.2, a function of the history through the carrier $I = \Lambda_1^*$ alone.

$d_j := \max(0, d_T(K_j, I^c) - R_0)$ and $\iota_T := \max(0, N_{\min}/4 - R_0)$ are the distances (F.1): K_j the union of the free-input stencils of the fine traversals of segment j, I^c the complement of the carrier in the torus free carrier, d_T the periodic fine-bond metric, $N_{\min}$ the shortest fine period, and $R_0 := 4(1 + R_* + R_H + L)$ the fixed range of Section 7 (R_* the perimeter range, not the walk range n_*R of Section B.8; this R_0 is not the scale R_0 of [9]).

The scales of the statement are τ, the marked-forest parameter of (5.18), $\tau := C\lfloor g_0(1 + r)^3 + e^{-cr^2}\rfloor$ with the cutoff radius $r = A_1(\log g_0^{-2})^{p_0}$ of (4.12); $\pi_* := \min(p_0(g_0), p_0(g_1))$ with $p_0(g) := A_0(\log g^{-2})^{p_0}$ and $g_1 = g_1(g_0)$ the first-step coupling of [9, (1.27)]; $\varepsilon_1 = g_1 p_0(g_1)$; ρ of (E.11), $\rho = O(g_0^2(1 + r)^4)$; and $b := \lfloor LMR_1/16\rfloor$ with the scale $R_1 = R_1(g_0)$ of [9] (Section B.8).

The constants C, c of Proposition F.2, C_L', c_* of (E.3) and Proposition D.18, C_χ, c of (E.29), $C_{\mathrm{resp},\chi}$ of (D.42) are fixed before the coupling and depend on L, the group, the character, M_1, M_2, M, the fixed expansion and chart choices and the constants A_0, A_1, p_0, B_3, β_0 of [9] only (the last through c_θ of (D.26), hence c_*, which involve $\delta_0 = (A_1/A_0)\varepsilon_0$, B_3 and the bound [9, (2.6)] on g_1/g_0), not on g_0, n, the torus or the history; the constants C, c of different statements are distinct and are named by their statement.

$K := 256$ is the fixed constant bounding, in units of $\ell := LMR_1$, the distance of a witness from its tested bond (Lemma D.13; distinct from the periods K_μ and the stencils K_j), and $C_K := 4(4K\ell + 9)^4$ depends on g_0 through the $R_1 = R_1(g_0)$ of [9], a power of L with $R_1 \geqslant (\log g_1^{-2})^{\mathfrak{r}}$; δ_q^I of (9.7) depends on g_0 (through τ, C_K and the one-point rate q of Section D.7), not on n, the torus or the history, and is smaller than every power of g_0 (Remark 1.3(1)).

Hypotheses. (H1) G is a compact connected simple Lie group in its fixed unitary realisation $G \subset U(N)$ and χ a normalised irreducible character (the opening of this section and the conventions of Section 3).

(H2) The parameters satisfy the standing conditions (3.1), the constants are fixed in the order of Section B.8, and g_0 lies below the threshold of Section B.8: its items 1–6 and the further conditions listed there, among them the threshold g_* of Proposition D.18 (which

contains that of Proposition D.20), the thresholds of Proposition E.26 (those of Lemmas E.1, E.9 and E.10, $R_\Upsilon \geqslant 2$, $u \leqslant u_*$, $v \leqslant 1$ and the small-coupling conditions of Sections E.9.1 and E.10.5), the common small-coupling thresholds of Proposition 7.4, the thresholds of Lemmas D.13 and D.17 (required already for (ii), through g_*) and the further threshold of (9.7); Corollaries 1.4 and 1.6 use in addition the further threshold stated in Corollary 1.4.

(H3) The torus is one of those of [8] (p.251, where the period is $2L^m$ in physical units with the lattice spacing $\varepsilon = L^{-K}$, so that the exponent of [8] is $m-K$ in terms of the m here; the family of tori is the same): the fine periods are $N_\mu = 2L^m$ fine-lattice units in every direction, with the integer m so large that N_μ is a multiple of $\ell = LMR_1$ (a power of L: M by (3.1), R_1 by [9, p. 246]; the periods are then compatible and even, Section D.7), and every coarse period $K_\mu = N_\mu/L = 2L^{m-1}$ satisfies $K_\mu \geqslant 3(n+2)$ and exceeds $\max(2, 2R_{\mathrm{p}}+8)$ ((3.1)); for (iii) in addition $N_{\min} \geqslant 4R_0 + (4/c)\log(4n)$ with the c of Proposition F.2 (Corollary 9.1). These are lower bounds on m in terms of the fixed loop and the coupling. The collar conditions (7.28) concern M, M_2 and R_1 and are among the thresholds of (H2) (item 1 of Section B.8). The results of Section D.7 and of the later sections are stated for general compatible even periods and are used here for these.

(H4) Bałaban's results are used as stated and cited, in the readings recorded where [9] fixes no convention: the branch hypotheses (B1)–(B4) of Section 3.1 with Bałaban's derivation of [9, (1.11)–(1.25)] as recorded there, [8, (0.4) and (0.10)–(0.11)] for the averaging operation and the group means, Bałaban's transfer statements [9, p. 256] (all the results of [6] used for the averaging and determining sets of [9]), [8] after (0.10) and after (0.12), through which Section D.8 takes over the constraint [6, (3)] and [3, Proposition 1, (51)] for the averaging [8, (0.4)], the results of [6] for the determining sets $\mathbf{B}_1(Q_w^{[4]})$ and $\mathbf{B}_1(\Lambda_1^{\mathrm{c}})$ of [9] in place of the spacing condition [6, (1)] for their fine layers, and the restoring bound (D.49), for the local minima and for the function of (1.21) ([9, p. 269]), [9, (1.5)–(1.7)] for the gauge-fixing functional, [9, (1.27)] for g_1, p.246 for R_1 and ε_1, and [9, (2.4)–(2.9)] for the relations between the scales R_j, g_j and ε_j (Section B.8), the conditions of [9] on the parameters as taken over in item 5 of Section B.8, and the validity of Bałaban's construction [9] for the values $M_1 < M_2 < M$ fixed there following Bałaban's prescription (the conventions after (3.1)).

The constants that Bałaban does not display are Bałaban's and are independent of the coupling and of the branch:

- B_3 depends on d and L only [6, Theorem 1, p. 279];
- the $O(L^2)$ of p.247 behind the domain statement of (B1) and item 6 of Section B.8 is that of the reasoning of [5, Lemma 1, p. 79], which [9] invokes there and whose bound $|V'-1| < 4d^2\alpha_0 + \alpha_1$ has absolute constants;
- the $O(1)$ of (B3)(iii) is obtained in the general step of [9, p. 268, (3.17)–(3.19)], from $44d^2B_3$ and [3, (106)–(108), (159)–(163)], whose constants are displayed there;
- the bound on $\mathbf{H}_{1,Ax}$ of p.251, behind the condition "MR_1 large" of item 5 of Section B.8, is derived in Lemma D.21 from [6, Proposition 9, (172), and (190)] and the block-local restoration of the axial gauge, with constants fixed before g_0;
- the $O(1)$ of (B1)(iv) is displayed in [3, p. 25], $d(2d+1)$;
- the decay rate δ_{c} of the bound of (B3)(iii) is that of the exponential decay property [6, (190)], which Bałaban invokes in [9, p. 266] for the functions $\mathbf{H}_{k,\square}$ and Lemma D.21 uses for $\mathbf{H}_{1,Ax}$: in the Landau gauge it is the rate $\delta_{\mathrm{P}}/8$ of [6, (190)] in the distance $d(y,y')$ of

[2, (2.46)], whose scale weights are fixed powers of L at the first step (Section D.8.3), with δ_{P} the common decay rate defined in [4, p. 423] after Theorem 3.12, "a positive, absolute constant depending on d and L only", for every background satisfying [4, (3.35)–(3.36)], a class that contains the minimal configurations by [6, Theorem 1, (9)–(10)].

The $\widetilde{Q}$ of [9] is LQ_{U_1}, the linear part of the constraint function [8, (2.4), p. 266] (decomposed as $L\widetilde{Q}B' + \widetilde{C}(B')$ at p.267), and the σ in [9, (1.14)–(1.15)] is the normalised Haar density σ/σ_0 of [7, (18), p. 260] ($\mathrm{d}U' = \sigma(A')\,\mathrm{d}A'$, $\sigma_0 = \sigma(0)$), the transformation to which Bałaban refers at the foot of [9, p. 248] (the step list of Section 3.1).

The terms that [9] does not define are taken in the sense Bałaban gives them:

- the "domain of integration in (1.11)", p.248, is the set to which the characteristic factors of (1.11) restrict the integral, the support of every one of them (as [8, p. 265] defines its characteristic function "in such a way that the domain of integration in (2.1) is restricted to" a set of configurations), under which $U_1 \in D^{(11)}(V, V_0)$, (B1)(iii), and Remark D.19;
- in (B1), $U_1(\Omega_1 \cap \Lambda_1^c, V)$ is the minimal configuration of (2.12) and (2.15) for the determining data named at p.251, as [9] writes the operation it calls similar to (1.21)–(1.23) at the general step, p.269, with data from both scales, so that (1.21) depends on (V, V_0);
- the rows of Ω_1 and of Ω_1^c (Section 3) are those of [7, Sect. A], which [9, p. 248] follows: a row with one block off Ω_1 is a row of Ω_1, and its average involves V_0 on the outer block as a constant ([7, (12), p. 258]: $U = V_0$ on Ω_1^c, $\bar{U} = V$ on $\Omega_1^{(1)}$; the δ-functions of Ω_1^c are those of the rows $b' \subset \Omega_1^c$, p.259), which is the form of (B2) on such rows;
- the $X^{\sim}$ of [9] adds every cube meeting X, at a face, an edge, or a vertex ([8, p. 257]: $\tilde{\square}^n$ is the cube of side $(1+2n)M$ with the centre of $\square$);
- the M_1 of [9] is a lower bound on the cube side ([4, Theorem 3.1, p. 397] holds "for $M \geqslant M_1$", and M_1 "was fixed in" that paper, [5, p. 77]);
- the "compatible" partitions of [9] (p.245, where the partitions introduced there "have to be compatible", also with π_0 and with the partitions introduced later, and p.257, "compatible with all other partitions") are nested, the cubes of the coarser of two compatible partitions being unions of cubes of the finer, the sense of [5, p. 98], "compatible with the block structure of the lattice T, i.e. for every j the cube $\square_j$ is a sum of the big blocks" (Section B.2);
- "regular" in (2.12) is membership in the class [6, (6)], to which Bałaban refers in [9, p. 256] for the precise definitions, with any radius allowed by [6, Theorem 1], which gives the same minimal orbit for each (Lemma D.12);
- the $\widetilde{D}$ in [9, (1.13)–(1.14)] is the solution of the row equations [7, (17), pp. 259–260], the transformation to which Bałaban refers in [9, p. 248], solved "separately for each function $Q(A', c)$", with "a unique solution ... for A sufficiently small", analytic with a Taylor expansion beginning with second order terms, which is the small solution φ of (B4)(i) and the only chart-class family whose fine field lies in $D^{(11)}(V, V_0)$ (existence, uniqueness, analyticity, and the positivity of the Jacobian (B4)(iii) are proved in Lemma B.11, from (B1) (with (B3)(i), through (B1)(i)) and (B2) with the readings listed here, the fine periods being at least $8M$, and threshold items 5 and 6).

This paper's own readings are the following:

- the bonds crossing from Ω_1 to Ω_1^c taken as variables of the integral over Ω_1 (Section 3), the rule of [7, (8), p. 258] for plaquettes, "plaquettes with at least one corner belonging to Ω_1", applied to bonds;
- this paper's convention $\chi_1(\square') = 0$, $\chi_1^c(\square') = 1$, $\chi_0'(\square') = 0$, $\chi_0'^c(\square') = 1$ where the configuration $U_{1,\square'}(V)$ does not exist, which keeps (1.4) and (1.8) partitions of unity;
- three corrections of the printed text of [9]: v read from (1.14) where [9, p. 249] points to "(1.15)" (before (8.5)), the "$z \neq y$" in [9, (1.5)] read as $x \neq y$, as in (1.7) ((8.7)), and the first member of the chain of [9] $\delta_0 = g_0 A_1 p_0(g_0) = (A_1/A_0)\varepsilon_0$, p.247, read as $g_0(A_1/A_0)p_0(g_0)$, the value of the last member and of $\delta_k = g_k A_1/A_0\, p_0(g_k)$ at p.265, which fixes the radius $r = \delta_0/g_0$ at which κ_{C_n} is identified (Section B.8).

The domain of the integral [9, (1.11)] is otherwise Bałaban's: its averages and means are the principal logarithms of [3, p. 21, (22)–(25)]; on the whole domain of integration, the designated bonds included, the fluctuation lies in the chart ball of the conventions paragraph of Section 3 by [9, pp. 247–248] (Bałaban's statements, justified there by the reasoning of Lemma 1 of [5]), and the arguments of the averages and means lie there by [3, pp. 24–25, (44)–(46)] with (B3)(i), p.248, both derived in (B1).

Theorem 1.2 (A Wilson Loop after the First Renormalisation Step)**.** *Under (H1)–(H4):*

(i) (Form.)

$$\mathbb{E}_\varepsilon[W_\chi(C_n)] = \kappa_{C_n}\, \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E_\xi^\circ} W_\chi(C_n; U_1)] + E_{\mathrm{def}} + E_{\mathrm{bdry}}\ , \tag{1.9}$$

with $\kappa_{C_n} \neq 0$, and every term is integrable against the history weight. The coefficients δm^σ and z_{c}^σ depend on g_0, ε, L, the group, the character, and the setting only, so κ_{C_n} depends on n through $P(C_n)$ alone and not on the torus or the history.

(ii) (Bounds.)

$$\begin{gathered} |E_\xi^\circ| \leqslant C\tau^2 \Big[\sum_{j=0}^{4n-1} e^{-cd_j} + 4ne^{-c\iota_T} + (n+1)e^{-cn} \Big] \leqslant C\tau^2(9n+1) \quad \text{on } A_{\mathrm{cov}}\ , \\ |E_{\mathrm{def}}| \leqslant C_L' n_C e^{-(c_*/2)\pi_*^2} + e^{C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})} - 1\ , \\ |E_{\mathrm{bdry}}| \leqslant C_{\mathrm{resp},\chi}\, n_C(1 + g_0 r)\, e^{-c_*\pi_*^2}\ . \end{gathered} \tag{1.10}$$

(iii) (Size of the leading term.) $|\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E_\xi^\circ} W_\chi(C_n; U_1)]| \leqslant e^{C\tau^2(9n+1)}$*, and under the additional hypotheses of (H2), (H3) for (iii),*

$$\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{|E_\xi^\circ|}] \leqslant e^Y\ , \qquad Y := C\tau^2(2 + 1/c) + n_C \delta_q^I\ , \qquad \delta_q^I \leqslant \exp[-c_*\pi_*^2/(2C_K)]\ , \tag{1.11}$$

so that the leading term of (1.9) *has modulus at most $|\kappa_{C_n}| e^Y$, and $\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} |e^{E_\xi^\circ} - 1|] \leqslant e^Y - 1$.*

Proof. (i) is Corollary F.3, whose hypotheses are the conventions of Section F.1 and those of Proposition 7.4 (the square geometry and collar conditions of (H3), the thresholds of (H2)); $\kappa_{C_n} \neq 0$ is an exponential by (5.39) and (6.12), as stated after (8.18), δm^σ is the absolutely convergent sum (5.39), characterised there as a limit, and z_{c}^σ the exponential (6.12) of a convergent corner sum, from both of which n and the torus are absent.

In (ii) the first bound is Proposition F.2 with the elementary bounds ($e^{-cd_j} \leqslant 1$, $e^{-c\iota_T} \leqslant 1$, $(n+1)e^{-cn} \leqslant n+1$); the second is (E.84) of Proposition E.26, whose hypotheses are those stated at the start of Section E.11 (Section 3, Appendix B, the conventions of Section E.8.2) and the thresholds named in (H2); the third is (D.42) of Proposition D.18, under compatible even periods and (D.2) (a further condition of Section B.8).

In (iii) the first bound is the positivity of $\mathbb{E}_{\mathrm{h}}$ (D.20), $\mathbb{E}_{\mathrm{h}}[1] = 1$, $|W_\chi| \leqslant 1$ and the crude bound of (ii); (1.11) and the two consequences are Corollary 9.4, from the refined split (9.3) of Corollary 9.1 under $4ne^{-c\iota_T} \leqslant 1$ (the torus condition of (H3)), and from (9.6) of Lemma 9.3 at $\lambda = C\tau^2$, whose joint witness bound (9.4) (Lemma 9.2) rests on Section D.7; their hypotheses are those of Proposition F.2, compatible even periods, $g_0 < g_*$, and the further threshold of (9.7), all in (H2) and (H3). *q.e.d.*

Remark 1.3 (Regime of the Statement). (1) The scales: as $g_0 \to 0$, $\tau = C[g_0(1+r)^3 + e^{-cr^2}] \to 0$ (with the single radius $r = \delta_0/g_0 = A_1(\log g_0^{-2})^{p_0}$ of the notation paragraph and Section B.8, the radius of the cutoff test Θ^σ as well), while $e^{-c_*\pi_*^2}$, $e^{-(c_*/2)\pi_*^2}$ and δ_q^I are smaller than every power of g_0 ($\pi_*^2 \geqslant A_0^2 2^{-2p_0}(\log g_0^{-2})^{2p_0}$ by (9.2), with $p_0 \geqslant 5\mathfrak{r} \geqslant 10$, and for δ_q^I the last member of (1.11) with (9.9)), and $\rho\varepsilon_1 \to 0$.

The n-dependence of the bounds is explicit: pointwise on A_{cov}, E_ξ° is at most linear in n; under the torus condition $4ne^{-c\iota_T} \leqslant 1$ the growth is carried by the $4n$ boundary terms e^{-cd_j}, each at most one and exponentially small when the carrier complement is far from segment j, while the remaining terms $4ne^{-c\iota_T} + (n+1)e^{-cn}$ of (ii) are at most $2 + 1/c$ (Corollary 9.1); in history average the exponential of $|E_\xi^\circ|$ carries the perimeter-rate factor $e^{n_C\delta_q^I}$ of (iii), where δ_q^I is governed by the C_K-th root q^{1/C_K} of the one-point rate q of Section D.7, while Corollary 1.4 below separates the deterministic torus and corner part of E_ξ° and pays its carrier part at the rate $q^{1/16}$, which gives (iii) with $\delta_q^\sharp$ in place of δ_q^I; E_{bdry} and the first term of the bound (1.10) on E_{def} are perimeter rates of the same kind; and the second term of that bound is small only when $n_C\rho\varepsilon_1 \ll 1$ ($e^x - 1 \geqslant x$).

The bounds (1.10) on E_{def} and E_{bdry} are absolute, not relative to $\kappa_{C_n} = e^{-\delta m^\sigma P(C_n)}(z_{\mathrm{c}}^\sigma)^4$; whether they lie below $|\kappa_{C_n}| = e^{-\delta m^\sigma P(C_n)}|z_{\mathrm{c}}^\sigma|^4$ (δm^σ is real, by the argument following (5.39)) depends on the sizes of δm^σ and z_{c}^σ, for which Sections 5 and 6 give two-sided bounds: $|\kappa_v^\sigma| \leqslant C\tau^2$ (the sentence following (5.33), by (5.31), with (5.39)) and $|\zeta_{v,w}^\sigma| \leqslant C\tau^2$ ((6.5), Section 6) give $|\delta m^\sigma|P(C_n) = n|\sum_v \kappa_v^\sigma| \leqslant 4Cn\tau^2$ and $e^{-4C\tau^2} \leqslant |z_{\mathrm{c}}^\sigma|^4 \leqslant e^{4C\tau^2}$, hence $e^{-4C\tau^2(n+1)} \leqslant |\kappa_{C_n}| \leqslant e^{4C\tau^2(n+1)}$, with C independent of g_0, n and the torus (an upper bound is also at hand from the expectation: at any configuration of A_{cov}, $|\kappa_{C_n}| = |S_{\xi,C_n}^\circ|e^{-\Re E_\xi^\circ} \leqslant e^{C\tau^2(9n+1)}$ by (ii) and $|S_{\xi,C_n}^\circ| \leqslant 1$, the proof of Proposition E.26).

E_{bdry} and the first term of the bound on E_{def} are smaller than every power of g_0 at fixed n, and the ratios of their bounds to $|\kappa_{C_n}|$ are, respectively, $C_{\mathrm{resp},\chi}n_C(1+g_0r)e^{-c_*\pi_*^2}/|\kappa_{C_n}|$ and $C_L'n_Ce^{-(c_*/2)\pi_*^2}/|\kappa_{C_n}|$, below one exactly when $\log(C_{\mathrm{resp},\chi}n_C(1+g_0r)) + \delta m^\sigma P(C_n) + 4\log|z_{\mathrm{c}}^\sigma|^{-1} < c_*\pi_*^2$, respectively $\log(C_L'n_C) + \delta m^\sigma P(C_n) + 4\log|z_{\mathrm{c}}^\sigma|^{-1} < (c_*/2)\pi_*^2$, conditions that involve the length of the loop through n_C and $P(C_n)$; by the lower bound on $|\kappa_{C_n}|$ they hold whenever $\log(C_{\mathrm{resp},\chi}n_C(1+g_0r)) + 4C\tau^2(n+1) < c_*\pi_*^2$, respectively $\log(C_L'n_C) + 4C\tau^2(n+1) < (c_*/2)\pi_*^2$, that is for n up to a fixed multiple of π_*^2/τ^2 at small coupling, and for every fixed n once g_0 is small ($\tau \to 0$, $\pi_* \to \infty$); when $\delta m^\sigma > 0$ the exact conditions fail at fixed g_0 for all large n, which the hypotheses allow on correspondingly large tori.

The second term of the bound on E_{def}, at most $2x$ with $x := C_\chi n_C\rho(\varepsilon_1 + e^{-cb/(LM)})$ when $x \leqslant 1$ ($e^x - 1 \leqslant 2x$ for $0 \leqslant x \leqslant 1$), is bounded only by a quantity polynomially

small in g_0 at fixed n ($\rho = O(g_0^2(1+r)^4)$); its comparison with $|\kappa_{C_n}|$ at the original carrier is made by the relative form of (E.83): the far-history term there retains $|S^\circ_{\xi,C}| = |\kappa_{C_n}||e^{E^\circ_\xi}|$ on A_{cov} (F.3), so by (1.11) (the far-history term is the $\mathbb{E}_{\mathrm{h}}$-average, with the weight w_ξ and the normaliser $\mathcal{Z}_0$ of (iii), of $\mathbf{1}_{A_{\mathrm{cov}}\cap \mathsf{N}_b^{\mathrm{c}}}|S^\circ_{\xi,C}|(e^x-1) \leqslant \mathbf{1}_{A_{\mathrm{cov}}}|\kappa_{C_n}|e^{|E^\circ_\xi|}(e^x-1)$) it is at most $|\kappa_{C_n}|e^{C\tau^2(2+1/c)+n_C\delta_q^I}(e^x-1)$, a relative bound, also with $\delta_q^\sharp$ in place of δ_q^I (Corollary 1.4); only the near-history term $C'_L n_C e^{-(c_*/2)\pi_*^2}$ is absolute.

(2) The statements above do not include a bound on the history average of $e^{E^\circ_\xi}$ literally independent of n (the proved form is the perimeter rate of (iii)), a lower bound on $|\kappa_{C_n}|$ uniform in n ($|\kappa_{C_n}| = e^{-\delta m^\sigma P(C_n)}|z_{\mathrm{c}}^\sigma|^4$ is bounded below uniformly in n if and only if $\delta m^\sigma \leqslant 0$, the sign of δm^σ is not determined here, and the lower bound $e^{-4C\tau^2(n+1)}$ of (1) decays in n), a lower bound on the modulus of the leading expectation $\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}e^{E^\circ_\xi}W_\chi(C_n;U_1)]$ uniform in n (Corollary 1.6 below gives one at a fixed loop, with an error $\eta_n = a_n + e^{n_C\delta_q^\sharp} - 1$ whose leading part for $n \leqslant \varepsilon_1^{-1}$ and nontrivial χ is the area term $2\mathrm{Lip}_\chi c_{\mathrm{h}} n^2\varepsilon_1$, and when $\eta_n < 1$ "below $|\kappa_{C_n}|$" in (1) compares with the leading term up to the factor $(1-\eta_n)^{-1}e^{C\tau^2(2+1/c)}$), or the second renormalisation step; the output of the step is the pair of the complex measure $\kappa_{C_n}^{-1}N_1$ and the measure ρ_1 of Section 8 (Conjecture 11.1 of Section 11 states the form expected after $k+1$ steps for $n = L^k$).

Corollary 1.4 (Factorisation of the Leading Term)**.** *Under (H1)–(H4), including the additional hypotheses of (H2) and (H3) for (iii) of Theorem 1.2, the exponent* (F.4) *on* A_{cov} *is* $E^\circ_\xi = E^\partial_\xi + E^\circ_{T,n}$*, with the carrier part and the torus and corner part*

$$E^\partial_\xi = F_{I,C,T} - F_{C,T}\,, \qquad E^\circ_{T,n} = (F_{C,T} - F_{C,\infty}(n)) + E^\sigma_n \tag{1.12}$$

in the notation of Proposition F.2. Then $E^\circ_{T,n}$ *does not depend on the history,* $\kappa_{C_n}e^{E^\circ_{T,n}} = \mathcal{W}_T^{\circ,\sigma}(C_n)$ *is the flat reference loop* (5.5) *of* C_n *on the torus, and*

$$\begin{aligned} \kappa_{C_n}\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}e^{E^\circ_\xi}W_\chi(C_n;U_1)] &= \mathcal{W}_T^{\circ,\sigma}(C_n)(\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}W_\chi(C_n;U_1)] + R_n)\,, \\ |E^\circ_{T,n}| &\leqslant C\tau^2[4ne^{-c\iota_T} + (n+1)e^{-cn}] \leqslant C\tau^2(2+1/c)\,, \\ |R_n| \leqslant \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}|e^{E^\partial_\xi}-1|] &\leqslant e^{n_C\delta_q^\sharp}-1\,, \qquad \delta_q^\sharp := C\tau^2 N'_{\mathrm{lat}}(c)\,e^{C\tau^2L^4C_K\Phi_c}\,q^{1/16}\,, \end{aligned} \tag{1.13}$$

with C, c *of Proposition F.2,* $N'_{\mathrm{lat}}(c)$ *and* Φ_c *of Lemma 9.3 (they depend on* L *and* c *only,* (9.5)*),* C_K *of Lemma 9.2 and the one-point rate* $q \leqslant e^{-c_*\pi_*^2}$ *of Section D.7 (*$g_0 < g_*$*, as in* (9.7)*).*

For g_0 *below a further threshold, depending on* L*,* M*,* M_1*,* M_2*, the group, the character, the source constants and* c*,* $C\tau^2L^4C_K\Phi_c \leqslant 1$ *and* $\delta_q^\sharp \leqslant eC\tau^2N'_{\mathrm{lat}}(c)\,e^{-c_*\pi_*^2/16}$*.*

Moreover $\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}e^{|E^\circ_\xi|}] \leqslant \exp[C\tau^2(2+1/c)+n_C\delta_q^\sharp]$*, the bound* (1.11) *of (iii) with* $\delta_q^\sharp$ *in place of* δ_q^I*. Consequently* $\mathbb{E}_\varepsilon[W_\chi(C_n)] = \mathcal{W}_T^{\circ,\sigma}(C_n)(\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}W_\chi(C_n;U_1)] + R_n) + E_{\mathrm{def}} + E_{\mathrm{bdry}}$ *by* (1.9)*.*

Proof. The decomposition is (F.4). In the proof of Proposition 7.4 at $D = I$, from which Proposition F.2 is obtained, the three parts are bounded separately: $|E^\partial_\xi| \leqslant C\tau^2\sum_{j=0}^{4n-1}e^{-cd_j}$ by (7.44), $|F_{C,T} - F_{C,\infty}(n)| \leqslant C\tau^2(4n)e^{-c\iota_T}$ by (7.46), and $|E^\sigma_n| \leqslant C\tau^2(n+1)e^{-cn}$ by (6.11); the bound of Proposition F.2 is their sum, with the largest of the three constants and the smallest of the three rates; these are the C, c of that proposition fixed in the notation paragraph, the c of (H3) and of Corollary 9.1 is this one, and each of the three bounds holds with them.

$F_{C,T}$ is the logarithm, on the connected source branch, of $\mathcal{W}_T^{\circ,\sigma}(C_n) = N^{\sigma}_{C_n,T}/D^{\sigma}_T$ (5.5), $F_{C,\infty}(n)$ is the bulk logarithm (6.9) and E^{σ}_n the corner-extraction error; none of them involves the carrier I or any other datum of the history, so $E^{\circ}_{T,n}$ is a number fixed by g_0, n, the torus and the setting. By the identity $F_{I,C,T} = \log \kappa_{C_n} + E^{\circ}_{\xi}$ of Proposition F.2, $\log \kappa_{C_n} + E^{\circ}_{T,n} = F_{I,C,T} - E^{\partial}_{\xi} = F_{C,T}$, whence $\kappa_{C_n} e^{E^{\circ}_{T,n}} = \mathcal{W}_T^{\circ,\sigma}(C_n)$.

The second form of the bound on $E^{\circ}_{T,n}$ uses $4ne^{-c\iota_T} \leqslant 1$, the torus condition of (H3), and $(n+1)e^{-cn} \leqslant 1 + 1/c$.

Since $E^{\circ}_{T,n}$ is a constant and the integrands are bounded ($|E^{\partial}_{\xi}| \leqslant 4C\tau^2 n$ on A_{cov}, $|W_{\chi}| \leqslant 1$), $\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^{\circ}_{\xi}} W_{\chi}(C_n; U_1)] = e^{E^{\circ}_{T,n}} \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^{\partial}_{\xi}} W_{\chi}(C_n; U_1)]$, and the identity holds with $R_n = \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}(e^{E^{\partial}_{\xi}} - 1) W_{\chi}(C_n; U_1)]$. By the positivity of $\mathbb{E}_{\mathrm{h}}$ (D.20), $|W_{\chi}| \leqslant 1$, $e^{|x|} - 1 \geqslant |e^x - 1|$ and the pointwise bound of (9.6), with $\lambda = C\tau^2$ and the marks $m_b = \mathbf{1}\{b \in E(B_{\xi})\}$, $|R_n| \leqslant \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} |e^{E^{\partial}_{\xi}} - 1|] \leqslant \mathbb{E}_{\mathrm{h}}[\exp(\lambda \sum_b \varphi_c(b) m_b)] - 1$, using $\mathbb{E}_{\mathrm{h}}[1] = 1$.

Let L^p be the smallest power of L exceeding $\sigma_K = 2K\ell + 4$ (Lemma 9.2), so $L^p \leqslant L\sigma_K$, and let $s_{\mu} \coloneqq \min(L^p, N_{\mu}/2)$; N_{μ}/s_{μ} is even, as $N_{\mu} = 2L^m$ by (H3). The sites of the torus are partitioned into the boxes $Q = \prod_{\mu} [k_{\mu} s_{\mu}, (k_{\mu}+1) s_{\mu})$, $k_{\mu} \in \mathbb{Z}/(N_{\mu}/s_{\mu})$, and Q has the colour $(k_{\mu} \bmod 2)_{\mu}$, one of sixteen. Two distinct boxes of one colour differ in some direction μ by a nonzero even index difference modulo N_{μ}/s_{μ}, which forces $N_{\mu}/s_{\mu} \geqslant 4$ and $s_{\mu} = L^p$, so their sites are at periodic ℓ^{∞} distance at least $L^p > \sigma_K$. For a finite set S of bonds let $N(\mathsf{S})$ be the number of boxes containing the initial site of a bond of S; one bond of S in each such box of the most frequent colour gives a subset S' at pairwise ℓ^{∞} distance $> \sigma_K$ with $|\mathsf{S}'| \geqslant N(\mathsf{S})/16$, so the first inequality of (9.4) and $q \leqslant 1$ give $\mathbb{E}_{\mathrm{h}}[\prod_{b \in \mathsf{S}} m_b] \leqslant q^{N(\mathsf{S})/16}$.

The expansion of $\exp(\lambda \sum_b \varphi_c(b) m_b) = \prod_b (1 + y_b m_b)$, $y_b = e^{\lambda \varphi_c(b)} - 1 \geqslant 0$, into the sum over S of $\prod_{b \in \mathsf{S}} y_b m_b$ (finitely many nonnegative terms, as in the proof of Lemma 9.3) factorises over the boxes, since $N(\mathsf{S}) = \sum_Q \mathbf{1}\{\mathsf{S} \text{ has a bond with initial site in } Q\}$:

$$\mathbb{E}_{\mathrm{h}}\Big[\prod_b (1 + y_b m_b)\Big] \leqslant \sum_{\mathsf{S}} \prod_{b \in \mathsf{S}} y_b \, q^{N(\mathsf{S})/16} = \prod_Q \Big(1 + q^{1/16}\Big(\prod_{b \in Q} (1 + y_b) - 1\Big)\Big)$$
$$\leqslant \exp\Big(q^{1/16} \sum_Q (e^{\lambda \sum_{b \in Q} \varphi_c(b)} - 1)\Big) \,,$$

the products and sums over $b \in Q$ running over the bonds with initial site in Q. A box has at most $L^{4p} \leqslant L^4 \sigma_K^4$ sites, hence at most $4L^4\sigma_K^4 \leqslant L^4 C_K$ bonds, so $\lambda \sum_{b \in Q} \varphi_c(b) \leqslant \lambda L^4 C_K \Phi_c$ by (9.5); with $e^x - 1 \leqslant x e^x$ and $\sum_b \varphi_c(b) \leqslant n_C N'_{\mathrm{lat}}(c)$ (proof of Lemma 9.3) the exponent is at most $n_C \lambda N'_{\mathrm{lat}}(c) e^{\lambda L^4 C_K \Phi_c} q^{1/16} = n_C \delta_q^{\sharp}$.

The same pointwise bound gives $\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{|E^{\partial}_{\xi}|}] \leqslant \mathbb{E}_{\mathrm{h}}[\exp(\lambda \sum_b \varphi_c(b) m_b)] \leqslant e^{n_C \delta_q^{\sharp}}$, and with $|E^{\circ}_{\xi}| \leqslant |E^{\circ}_{T,n}| + |E^{\partial}_{\xi}|$ the moment bound follows.

For the further threshold: $C_K \leqslant 4(4K+9)^4 (L^2 M)^4 2^{4\mathfrak{r}} (\log g_0^{-2})^{4\mathfrak{r}}$ and $\tau \leqslant C[g_0(1+r)^3 + e^{-cr^2}]$ with $r = A_1 (\log g_0^{-2})^{p_0}$ (by (9.2), as in the proof of Corollary 9.4), so $\tau^2 C_K \to 0$ as $g_0 \to 0$, and $q^{1/16} \leqslant e^{-c_* \pi_*^2/16}$.

The last statement is (1.9) with the identity of (1.13) substituted. *q.e.d.*

Definition 1.5 (The Lipschitz Constant of the Character)**.** π_{χ} is the irreducible unitary representation of G with $\chi = (\dim \pi_{\chi})^{-1} \operatorname{Tr} \pi_{\chi}$, and $\mathbf{Lip}_{\chi}$ is a constant, depending on the group and the character only, with

$$\|\pi_{\chi}(g) - \pi_{\chi}(h)\|_{\mathrm{op}} \leqslant \mathbf{Lip}_{\chi} \|g - h\|_{\mathrm{op}} \qquad (g, h \in G) \,;$$

it exists because π_χ is smooth and $G \subset U(N)$ is a compact submanifold.

Corollary 1.6 (The Leading Term at a Fixed Loop)**.** *Under the hypotheses of Corollary 1.4, let π_χ and Lip_χ be as in Definition 1.5, C_L the contour constant with $|\Gamma_C| \leqslant C_L n_C$ (Section D.7, the paragraph before Lemma D.9), depending on L only, and q the one-point rate of Lemma D.17. Then*

$$\begin{aligned} |\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} W_\chi(C_n;U_1)] - 1| &\leqslant a_n \coloneqq 2\mathrm{Lip}_\chi c_{\mathrm{h}} n^2 \varepsilon_1 + (2(nL+1)^2 + C_L n_C) q \;, \\ \kappa_{C_n} \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^\circ_\xi} W_\chi(C_n;U_1)] &= \mathcal{W}_T^{\circ,\sigma}(C_n)(1+R_n^\sharp) \;, \qquad |R_n^\sharp| \leqslant \eta_n \coloneqq a_n + e^{n_C \delta_q^\sharp} - 1 \;, \end{aligned} \tag{1.14}$$

with $c_{\mathrm{h}} = \sqrt{N}$, $\varepsilon_1 = g_1 p_0(g_1)$ and $\delta_q^\sharp$ of (1.13)*.*

Hence, when $\eta_n < 1$, the leading term of (1.9) *is nonzero, $(1-\eta_n)|\mathcal{W}_T^{\circ,\sigma}(C_n)| \leqslant |\kappa_{C_n} \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^\circ_\xi} W_\chi(C_n;U_1)]| \leqslant (1+\eta_n)|\mathcal{W}_T^{\circ,\sigma}(C_n)|$, and $e^{-C\tau^{2(2+1/c)}}|\kappa_{C_n}| \leqslant |\mathcal{W}_T^{\circ,\sigma}(C_n)| \leqslant e^{C\tau^{2(2+1/c)}}|\kappa_{C_n}|$ with the C, c of Corollary 1.4.*

For every fixed n, $\eta_n \to 0$ as $g_0 \to 0$; at small coupling η_n is small for n up to a small multiple of $\varepsilon_1^{-1/2}$, and the bound is not uniform in n.

Proof. Let S_n be the set of the $(nL+1)^2$ fine sites of the closed square bounded by C_n in its plane, and $A_\square = \{S_n \subseteq \Lambda_1\}$, an event of the regions. If $A_\square$ fails, some $x \in S_n$ lies in $B_\xi = \Lambda_1^c$ and the positive bond $z_x = (x, x+e_1)$ lies in $E(B_\xi)$ (notation of Section D.7); so $\mathbf{1}_{A_\square^c} \leqslant \sum_{x \in S_n} \mathbf{1}\{z_x \in E(B_\xi)\}$, and $\mathbb{P}_{\mathrm{h}}(A_\square^c) \leqslant (nL+1)^2 q$ by $p_z \leqslant q$ (Lemma D.17) and the positivity of $\mathbb{E}_{\mathrm{h}}$ (D.20).

On a branch in $A_\square$ every fine plaquette p of the square has its vertices in Λ_1, a union of blocks: the block of a vertex x lies in Λ_1, a bond with both endpoints in that block is free (a designated bond crosses between two blocks) and belongs to I, and its initial site is within ℓ^∞ distance $L-1$ of x, so every bond of p is within distance $L \leqslant 3R_{\mathrm{c}}$ of I, and Lemma B.4 gives $\|U_1(\partial p) - \mathbf{1}\|_{\mathrm{op}} \leqslant 2c_{\mathrm{h}}\varepsilon_1 L^{-2}$.

Lattice Stokes: the $(nL)^2$ plaquettes of the square are ordered by rows, starting at the base corner x_0 of C_n, the image of the vertex 0 of the square of the notation paragraph under the coarse translation that gives C_n (bottom row first, left to right in each row), and let Q_k be the union of the first k of them, whose boundary, oriented counterclockwise and based at x_0, is a closed lattice path; $Q_1 = p_1$ is the plaquette at x_0, for $k \geqslant 1$ the shared path s below does not contain x_0, and $\partial Q_{(nL)^2}$ is C_n (another base point or orientation replaces the holonomy h below by a conjugate or by h^{-1}, which changes neither $\|h-\mathbf{1}\|_{\mathrm{op}}$ nor $|\chi(h)-1|$, since $\chi(h^{-1}) = \overline{\chi(h)}$ and $\chi(\mathbf{1}) = 1$); adding the next plaquette p replaces the one or two bonds s it shares with ∂Q_k (its lower and left sides that lie in Q_k) by the complementary path s' of ∂p, and if the based boundary path of Q_k is $\alpha s \beta$, then $\alpha s' \beta$ is that of Q_{k+1} and its holonomy is $vU(\partial p)^{\pm 1} v^{-1}$ times the holonomy of $\alpha s \beta$, with v the holonomy of αs and ∂p based at the endpoint of s.

For unitaries, $\|vwv^{-1} - \mathbf{1}\| = \|w - \mathbf{1}\| = \|w^{-1} - \mathbf{1}\|$ and $\|w_1 w_2 - \mathbf{1}\| \leqslant \|w_1 - \mathbf{1}\| + \|w_2 - \mathbf{1}\|$, so on $A_\square$ the holonomy h of U_1 around C_n satisfies $\|h-\mathbf{1}\|_{\mathrm{op}} \leqslant (nL)^2 \cdot 2c_{\mathrm{h}}\varepsilon_1 L^{-2} = 2c_{\mathrm{h}} n^2 \varepsilon_1$, and, χ being a class function with $|\operatorname{Tr} X| \leqslant \dim \pi_\chi \, \|X\|_{\mathrm{op}}$, $|W_\chi(C_n;U_1) - 1| = |\chi(h) - \chi(\mathbf{1})| \leqslant \|\pi_\chi(h) - \mathbf{1}\|_{\mathrm{op}} \leqslant 2\mathrm{Lip}_\chi c_{\mathrm{h}} n^2 \varepsilon_1$. Since $\mathbb{E}_{\mathrm{h}}$ is positive with $\mathbb{E}_{\mathrm{h}}[1] = 1$,

$$\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} W_\chi(C_n;U_1)] - 1 = \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}} \cap A_\square}(W_\chi - 1)] + \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}} \cap A_\square^c}(W_\chi - 1)] - \mathbb{P}_{\mathrm{h}}(A_{\mathrm{cov}}^c) \;,$$

and the three terms are at most $2\mathrm{Lip}_\chi c_{\mathrm{h}} n^2 \varepsilon_1$, $2\mathbb{P}_{\mathrm{h}}(A_\square^c) \leqslant 2(nL+1)^2 q$ ($|W_\chi| \leqslant 1$) and $C_L n_C q$ (D.25) in modulus, which is the first bound.

By Corollary 1.4 the leading term is $\mathcal{W}_T^{\circ,\sigma}(C_n)(\mathbb{E}_{\rm h}[\mathbf{1}_{A_{\rm cov}}W_\chi(C_n;U_1)]+R_n)=\mathcal{W}_T^{\circ,\sigma}(C_n)(1+R_n^\sharp)$ with $R_n^\sharp=(\mathbb{E}_{\rm h}[\mathbf{1}_{A_{\rm cov}}W_\chi(C_n;U_1)]-1)+R_n$ and $|R_n|\leqslant e^{n_C\delta_q^\sharp}-1$, which is the second; the modulus bounds follow from $1-\eta_n\leqslant|1+R_n^\sharp|\leqslant 1+\eta_n$ and $\mathcal{W}_T^{\circ,\sigma}(C_n)=\kappa_{C_n}e^{E^\circ_{T,n}}$ with $|E^\circ_{T,n}|\leqslant C\tau^2(2+1/c)$.

As $g_0\to 0$, $\varepsilon_1\to 0$ ($g_1\leqslant(1+\beta_0)g_0$ and $g_0\leqslant(1+\beta_0)g_1$, [9, (2.6)] as quoted in Section B.8; the second inequality keeps ε_1^{-1} polynomial in g_0^{-1}), $q\leqslant e^{-c_*\pi_*^2}$ (Proposition D.18) and, below the further threshold of Corollary 1.4, $\delta_q^\sharp\leqslant eC\tau^2N'_{\rm lat}(c)e^{-c_*\pi_*^2/16}$, with $\pi_*\to\infty$; so $\eta_n\to 0$ at fixed n, and the terms other than $2\mathrm{Lip}_\chi c_{\rm h}n^2\varepsilon_1$ are smaller than every power of g_0 for $n\leqslant\varepsilon_1^{-1}$. *q.e.d.*

Definition 1.7 (The Coarse Loop)**.** Let $c_0,\dots,c_{4n-1}$ be the coarse bonds of C_n in the order of traversal from the vertex 0, with $V(c)\coloneqq V(-c)^{-1}$ and $P_c(U)\coloneqq P_{-c}(U)^{-1}$ for a coarse bond c traversed in the negative direction. The holonomy and the Wilson function of the coarse field along C_n are

$$\mathrm{Hol}_V C_n\coloneqq V(c_0)V(c_1)\cdots V(c_{4n-1})\,,\qquad W_\chi(C_n;V)\coloneqq\chi(\mathrm{Hol}_V C_n)\,,$$

so that $\mathrm{Hol}_{U_1}C_n=P_{c_0}(U_1)\cdots P_{c_{4n-1}}(U_1)$ in the convention of (1.1). The transports along C_n, the insertions, and the insertion size are

$$H_0\coloneqq\mathbf{1}\,,\qquad H_j\coloneqq P_{c_0}(U_1)\cdots P_{c_{j-1}}(U_1)\,,\qquad \widetilde{R}_j\coloneqq H_jV(c_j)P_{c_j}(U_1)^{-1}H_j^{-1}\,,$$
$$s\coloneqq 6c_{\rm h}\varepsilon_1\,,\qquad t\coloneqq e^s-1\,,$$

with $c_{\rm h}$ and $\varepsilon_1=g_1p_0(g_1)$ as in Lemma B.4.

Proposition 1.8 (The Leading Term and the Coarse Loop)**.** *Let (H1)–(H4) hold, and let $s\leqslant 1/10$ and $t\leqslant r_G$ (conditions on g_0), with s, t of Definition 1.7 and r_G the group constant of Appendix B. Then on $A_{\rm cov}$, at every branch:*

(i) for every j, with c the positive coarse bond such that $c_j=\pm c$: $U_1\in\mathcal{O}_c$, $V(c)=e^{i\beta_c(U_1)}P_c(U_1)$, and

$$\|V(c_j)P_{c_j}(U_1)^{-1}-\mathbf{1}\|_{\rm op}\leqslant t\,,\qquad \|\beta_c(U_1)\|_{\rm op}\leqslant 2\arcsin(t/2)\leqslant 1.06\,s\,;\tag{1.15}$$

(ii) the coarse holonomy is

$$\mathrm{Hol}_V C_n=\widetilde{R}_0\widetilde{R}_1\cdots\widetilde{R}_{4n-1}\,\mathrm{Hol}_{U_1}C_n\,,\qquad \|\widetilde{R}_0\cdots\widetilde{R}_{4n-1}-\mathbf{1}\|_{\rm op}\leqslant e^{4ns}-1\,,\tag{1.16}$$

where $V(c_j)P_{c_j}(U_1)^{-1}$ depends on U_1 only through the bonds with both endpoints in the two blocks of c_j;

(iii) with Lip_χ of Definition 1.5,

$$|W_\chi(C_n;V)-W_\chi(C_n;U_1)|\leqslant\mathrm{Lip}_\chi(e^{6c_{\rm h}n_C\varepsilon_1/L}-1)\,.\tag{1.17}$$

Corollary 1.9 (The Leading Term with the Coarse Loop)**.** *Under the hypotheses of Proposition 1.8,*

$$\mathbb{E}_\varepsilon[W_\chi(C_n)]=\kappa_{C_n}\mathbb{E}_{\rm h}[\mathbf{1}_{A_{\rm cov}}e^{E^\circ_\xi}W_\chi(C_n;V)]+E_{\rm coarse}+E_{\rm def}+E_{\rm bdry}\,,$$
$$|E_{\rm coarse}|\leqslant|\kappa_{C_n}|\,\mathrm{Lip}_\chi(e^{6c_{\rm h}n_C\varepsilon_1/L}-1)e^{C\tau^2(9n+1)}\,,$$

and, under the additional hypotheses of (H2) and (H3) for Theorem 1.2(iii),

$$|E_{\rm coarse}|\leqslant|\kappa_{C_n}|\,\mathrm{Lip}_\chi(e^{6c_{\rm h}n_C\varepsilon_1/L}-1)\exp[C\tau^2(2+1/c)+n_C\delta_q^I]\,,$$

with $E_{\rm def}$, $E_{\rm bdry}$ and the constants of Theorem 1.2.

Proof of Proposition 1.8 and Corollary 1.9. Location. For each j let c be the positive coarse bond with $c_j = \pm c$, and $\mathcal{E}_c$ the set of bonds with both endpoints in the two blocks $\mathcal{B}_{c_-}$, $\mathcal{B}_{c_+}$ of c. Every loop in β_c, $U(\gamma)U([x,x'])U(\gamma')^{-1}P_c(U)^{-1}$ with $\gamma \in \mathcal{G}(c_-, x)$, $\gamma' \in \mathcal{G}(c_+, x')$, uses only bonds of $\mathcal{E}_c$: γ and γ' are monotone paths inside the boxes $\mathcal{B}_{c_-}$, $\mathcal{B}_{c_+}$, and $[x,x']$ and the straight path of c run from $\mathcal{B}_{c_-}$ to $\mathcal{B}_{c_+}$. The plaquettes used below lie in the box $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$, of ℓ^∞ diameter $2L-1$.

On A_{cov}, $\Lambda_b \subseteq I = \Lambda_1^*$ for every fine contour bond b of c (Section D.6); Λ_b is $\{b\}$ or the set of free bonds of $\mathcal{E}_c$. Each block of c contains $(L-1)/2$ fine bonds of the straight path of c with both endpoints in that block; they are free (a designated bond crosses between two blocks), so they lie in I, and each has an endpoint in Λ_1. Since Λ_1 is a union of blocks (the proof of Corollary 1.6), both blocks of c lie in Λ_1. By Lemma 3.19(i), every site within ℓ^∞ distance $LMR_1 - 1 > 2L$ of Λ_1 lies in $\Omega_1^{\sim -1}$, so both blocks of c lie in Ω_1, and the constraint $\bar{U}_1 = V$ of [9, (1.12)] on $\Omega_1^{(1)}$ (Section D.8.1) gives $V(c) = M(U_1, c)$.

Every plaquette of the box has a bond within distance $8L+1 < 3R_{\mathrm{c}}$ of I ($R_{\mathrm{c}} \geqslant 2R_{\mathrm{loc}}+1 = 4L+5$), so Lemma B.4 gives $\|U_1(\partial p) - \mathbf{1}\|_{\mathrm{op}} \leqslant 2c_{\mathrm{h}}\varepsilon_1 L^{-2}$ for each of them.

Area. Let ℓ be one of the loops, with $x = c_- + \sum_\nu n_\nu e_\nu$, $|n_\nu| \leqslant (L-1)/2$ ($\mathcal{B}_{c_-}$ is the block of the odd L of (3.1) centred at c_-), and let γ_+ be the translate of γ by Le_μ, $c = (c_-, c_- + Le_\mu)$; $\gamma_+ \in \mathcal{G}(c_+, x')$. Passing from the order of directions of γ' to that of γ_+ by adjacent transpositions inverts each pair of directions at most once, and exchanging two consecutive straight segments of lengths $|n_\alpha|$, $|n_\beta|$ moves the path across $|n_\alpha||n_\beta|$ plaquettes. Moving $[x,x']$ back along γ one bond at a time crosses L plaquettes for a bond of direction $\pm e_\nu$, $\nu \neq \mu$, and none for a bond of direction $\pm e_\mu$, where the two paths coincide after cancellation of backtracks. So ℓ is reduced to the empty loop by at most

$$A_L = \sum_{\alpha<\beta} |n_\alpha||n_\beta| + L\sum_{\nu\neq\mu} |n_\nu| \leqslant \tfrac{3}{2}(L-1)^2 + \tfrac{3}{2}L(L-1) \leqslant 3L^2$$

plaquette moves, by $\sum_{\alpha<\beta} a_\alpha a_\beta \leqslant \frac{3}{8}(\sum_\alpha a_\alpha)^2$ for four nonnegative numbers and $\sum_\nu |n_\nu| \leqslant 2(L-1)$ for the first term, and by $|n_\nu| \leqslant (L-1)/2$ for each of the three $\nu \neq \mu$ for the second.

One loop. A move across a plaquette p multiplies the transport of the loop at its base point by $gU_1(\partial p)^{\pm 1}g^{-1}$, with $g \in G$ a transport along the current loop. So $U_1(\ell)$ is a product of at most A_L such conjugates. For unitary W and g, $\|gWg^{-1} - \mathbf{1}\|_{\mathrm{op}} = \|W - \mathbf{1}\|_{\mathrm{op}} = \|W^{-1} - \mathbf{1}\|_{\mathrm{op}}$, and $\|\prod_m(\mathbf{1} + a_m) - \mathbf{1}\|_{\mathrm{op}} \leqslant \prod_m(1 + \|a_m\|_{\mathrm{op}}) - 1$; hence

$$\|U_1(\ell) - \mathbf{1}\|_{\mathrm{op}} \leqslant (1 + 2c_{\mathrm{h}}\varepsilon_1 L^{-2})^{3L^2} - 1 \leqslant t\ .$$

The eigenvalues $e^{i\varphi}$ of $U_1(\ell)$, $\varphi \in (-\pi, \pi]$, satisfy $2|\sin(\varphi/2)| = |e^{i\varphi} - 1| \leqslant t$, so the Hermitian matrix A with $e^{iA} = U_1(\ell)$ whose eigenvalues are these φ has $\|A\|_{\mathrm{op}} \leqslant 2\arcsin(t/2)$. The function $\phi(s) = 2\arcsin((e^s - 1)/2)$ is convex on $[0, 1/10]$, its derivative $e^s(1 - (e^s-1)^2/4)^{-1/2}$ being increasing, so $\phi(s) \leqslant 10\phi(1/10)s \leqslant 1.06s$.

Since $U_1(\ell) \in G$ and $\|U_1(\ell) - \mathbf{1}\|_{\mathrm{op}} \leqslant t \leqslant r_G$, $U_1(\ell)$ lies in the chart domain (Section 3): $U_1(\ell) = e^{iB}$ with $B \in \mathfrak{g}$ and $h(B) < \varrho_G$, so $\|B\|_{\mathrm{op}} \leqslant \sqrt{N}h(B) < \pi$. A Hermitian B with $e^{iB} = U_1(\ell)$ and spectrum in $(-\pi, \pi)$ is the function of $U_1(\ell)$ that takes the value φ on the eigenspace of $e^{i\varphi}$, $\varphi \in (-\pi, \pi)$, since $\lambda \mapsto e^{i\lambda}$ is injective on $(-\pi, \pi)$; so $B = A$. Hence $A \in \mathfrak{g}$, $h(A) < \varrho_G$, and A is the chart value $\frac{1}{i}\log U_1(\ell)$, the chart being the inverse of $A \mapsto e^{iA}$ on the chart ball. Hence $U_1 \in \mathcal{O}_c$.

One coarse bond. $\beta_c(U_1)$ is a convex combination of the matrices $\frac{1}{i}\log U_1(\ell)$ (the weights of (4.1) are nonnegative and sum to one, $|\mathcal{B}_{c_-}| = L^4$), so $\|\beta_c(U_1)\|_{\mathrm{op}} \leqslant 2\arcsin(t/2)$,

and $V(c)P_c(U_1)^{-1} = e^{i\beta_c(U_1)}$ has $\|e^{i\beta_c(U_1)} - \mathbf{1}\|_{\mathrm{op}} \leqslant 2\sin(\arcsin(t/2)) = t$. For $c_j = -c$, $V(c_j)P_{c_j}(U_1)^{-1} = P_c(U_1)^{-1}e^{-i\beta_c(U_1)}P_c(U_1)$, a conjugate of $e^{-i\beta_c(U_1)}$, with the same bound. This is (i), and the dependence stated in (ii) is Lemma 1.1(i).

The loop. By induction on m, $V(c_0)\cdots V(c_{m-1}) = \widetilde{R}_0\cdots\widetilde{R}_{m-1}H_m$, the step being

$$\widetilde{R}_0\cdots\widetilde{R}_{m-1}H_mV(c_m) = \widetilde{R}_0\cdots\widetilde{R}_{m-1}(H_mV(c_m)P_{c_m}(U_1)^{-1}H_m^{-1})H_mP_{c_m}(U_1)\,.$$

At $m = 4n$ this is the identity in (1.16). Each $\widetilde{R}_j$ is a conjugate of $V(c_j)P_{c_j}(U_1)^{-1}$, so $\|\widetilde{R}_j - \mathbf{1}\|_{\mathrm{op}} \leqslant t$ and $\|\widetilde{R}_0\cdots\widetilde{R}_{4n-1} - \mathbf{1}\|_{\mathrm{op}} \leqslant (1+t)^{4n} - 1 = e^{4ns} - 1$.

For (iii), χ is $(\dim\pi_\chi)^{-1}\operatorname{Tr}\pi_\chi$ and $|(\dim\pi_\chi)^{-1}\operatorname{Tr}X| \leqslant \|X\|_{\mathrm{op}}$, so

$$\begin{aligned}|W_\chi(C_n;V) - W_\chi(C_n;U_1)| &\leqslant \|\pi_\chi(\mathrm{Hol}_V C_n) - \pi_\chi(\mathrm{Hol}_{U_1}C_n)\|_{\mathrm{op}}\\ &\leqslant \mathrm{Lip}_\chi\|\mathrm{Hol}_V C_n - \mathrm{Hol}_{U_1}C_n\|_{\mathrm{op}}\,,\end{aligned}$$

and $\|\mathrm{Hol}_V C_n - \mathrm{Hol}_{U_1}C_n\|_{\mathrm{op}} = \|(\widetilde{R}_0\cdots\widetilde{R}_{4n-1} - \mathbf{1})\mathrm{Hol}_{U_1}C_n\|_{\mathrm{op}} \leqslant e^{4ns} - 1$, with $4ns = 24c_{\mathrm{h}}n\varepsilon_1 = 6c_{\mathrm{h}}n_C\varepsilon_1/L$.

The expectation. On A_{cov}, $W_\chi(C_n;U_1) = W_\chi(C_n;V) + (W_\chi(C_n;U_1) - W_\chi(C_n;V))$. Inserted into (1.9), it gives the stated identity with $E_{\mathrm{coarse}} = \kappa_{C_n}\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}e^{E^\circ_\xi}(W_\chi(C_n;U_1) - W_\chi(C_n;V))]$. Since $|e^{E^\circ_\xi}| \leqslant e^{|E^\circ_\xi|}$, (1.17) with $|E^\circ_\xi| \leqslant C\tau^2(9n+1)$ on A_{cov} (1.10) and $|\mathbb{E}_{\mathrm{h}}[X]| \leqslant \sup|X|$ (Section D.6) gives the first bound on E_{coarse}, and (1.17) with (1.11) the second. *q.e.d.*

Remark 1.10 (The Fine and the Coarse Loop). (1) (1.16) is exact: after the first step the fine loop at the background is the coarse holonomy with one insertion per coarse bond, $\mathrm{Hol}_{U_1}C_n = V(c_0)Q_0V(c_1)Q_1\cdots V(c_{4n-1})Q_{4n-1}$ with $Q_j := V(c_j)^{-1}P_{c_j}(U_1)$. For $c_j = \pm c$, Q_j is $P_c(U_1)^{-1}e^{-i\beta_c(U_1)}P_c(U_1)$ if $c_j = c$ and $e^{i\beta_c(U_1)}$ if $c_j = -c$, a function of U_1 on the bonds of $\mathcal{E}_{c_j}$, with $\|Q_j - \mathbf{1}\|_{\mathrm{op}} \leqslant t$. The $\widetilde{R}_j$ of (ii) are conjugates of the $V(c_j)P_{c_j}(U_1)^{-1}$ by the transports H_j along the earlier bonds, and are not local. The insertions are of first order in the plaquette deviations of U_1, of size $O(\varepsilon_1)$ per coarse bond.

(2) Per fine contour bond the exponent in (1.17) is $6c_{\mathrm{h}}\varepsilon_1/L$, of order $L^{-1}g_0(\log g_0^{-2})^{p_0}$, while that of E_{def} is $C_\chi\rho(\varepsilon_1 + e^{-cb/(LM)})$, of order g_0^3 times a power of $\log g_0^{-2}$. For a square of fixed physical side ℓ_C, n_C is of order ℓ_C/ε, and E_{coarse} is controlled by Corollary 1.9 only while $n_C\varepsilon_1/L$ is small.

(3) Both $W_\chi(C_n;U_1)$ and $W_\chi(C_n;V)$ are functions of the coarse field and the branch data: after the first step no integral over the free variable a remains in which their difference is averaged ($\mathbb{E}_{\mathrm{h}}$ integrates over the branch data, the retained fluctuation A_{e} included). Every average of the difference is taken in the coarse ensemble $\mathbb{E}_{\mathrm{h}}$, which Theorem 1.2 leaves unevaluated, so a bound on E_{coarse} that uses cancellations of the first-order part of the insertions is a statement about the coarse measure.

Definition 1.11 (The Two-Loop Data)**.** $C = C_1 \sqcup C_2$, χ_1, χ_2, $n_C := n_{C_1} + n_{C_2}$, and $W_{\chi_1,\chi_2}(C;U) := \chi_1(\mathrm{Hol}_U C_1)\chi_2(\mathrm{Hol}_U C_2)$ are as in Section 10.1: C_i is a coarse translate of the counterclockwise square of side n_iL in a plane parallel to that of e_1, e_2, χ_i is a normalised irreducible character, and $V(C_1)\cap V(C_2) = \varnothing$.

A_{cov} is the covered event of the pair, E_{def} and E_{bdry} are the terms (E.1) and (D.21) formed with the observable and the groups of the pair (Section 10), κ_{C_i} and $E^\circ_{\xi,i}$ are the coefficient and the exponent of Theorem 1.2 for (C_i,χ_i), $E^\times_\xi$ is the cross exponent of Lemma 10.7(b), $\Delta_\times$ and d are as in (10.2), and J_{C_i} is the set of the $4n_i$ coarse segments of C_i.

Definition 1.12 (The Two-Loop Constants)**.** For the data of Definition 1.11:

- C and the c of the bounds on $E^{\circ}_{\xi,i}$ and of the moment bound are the larger, respectively the smaller, of the constants of Proposition F.2 for the two characters, and d_j is the distance (F.1) of segment j of C_i at the carrier $I = \Lambda_1^*$ (not the distance d of (10.2));
- $C_\chi := \max_i C_{\chi_i}$ and the c of the bound on E_{def} are those of Lemma 10.7, formed from the constants of (E.29), and $C_\times$, $c_\times$ are the constants of that lemma;
- ρ is (E.11) with $u = C^{\mathrm{src}}_{\chi_1\chi_2} g_0(1+r)$, the enlarged source constant of Lemma 10.4, which is distinct from this C_χ;
- $C_{\mathrm{resp},\chi_1\chi_2} := C_L \max((C_{\mathrm{resp},\chi_1} + C_{\mathrm{resp},\chi_2})S_c, 1)$, with the constants C_{resp,χ_i} of Corollary D.8 for χ_i (those of (D.8) as enlarged in Lemma D.6, before the replacement made in (D.42)) and C_L, S_c of Proposition D.18;
- C'_L, c_*, C_K, τ, π_*, ε_1, and b are those of Theorem 1.2, a constant that depends on the character being taken at its larger value for χ_1, χ_2 when it is a prefactor and at its smaller value when it is a decay rate.

The constants C, c, C_χ, $C^{\mathrm{src}}_{\chi_1\chi_2}$, $C_\times$, $c_\times$, $C_{\mathrm{resp},\chi_1\chi_2}$, C'_L, and c_* are fixed before the coupling; C_K and the scales τ, π_*, ε_1, ρ, and b depend on g_0 as in Theorem 1.2; none of them depends on n_1, n_2, the positions of the squares, the torus, or the history.

Theorem 1.13 (Two Disjoint Squares after the First Renormalisation Step)**.** *Let (H1)–(H4) hold for (C_1, χ_1) and for (C_2, χ_2) in place of (C_n, χ), and let g_0 lie below the thresholds of Section 10: those of Lemmas 10.4, 10.6 and 10.7, and for Corollary 1.14 that of* (10.10)*. Then, with the data of Definition 1.11 and the constants of Definition 1.12:*

(i) (Form.)

$$\mathbb{E}_\varepsilon[W_{\chi_1,\chi_2}(C)] = \kappa_{C_1}\kappa_{C_2}\,\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^{\circ}_{\xi,1}+E^{\circ}_{\xi,2}+E^{\times}_{\xi}} W_{\chi_1,\chi_2}(C;U_1)] + E_{\mathrm{def}} + E_{\mathrm{bdry}}\ , \quad (1.18)$$

every term being integrable against the history weight; $\kappa_{C_i} \neq 0$ depends on n_i through $P(C_i)$ alone, as in Theorem 1.2(i).

(ii) (Bounds.) On A_{cov}, $|E^{\circ}_{\xi,i}| \leqslant C\tau^2[\sum_{j\in J_{C_i}} e^{-cd_j} + 4n_i e^{-c\iota_T} + (n_i+1)e^{-cn_i}] \leqslant C\tau^2(9n_i+1)$ for $i = 1, 2$, and

$$\begin{aligned} |E^{\times}_{\xi}| &\leqslant C_\times \rho \Delta_\times \leqslant C_\times \rho\, n_C e^{-c_\times d/(LM)}\ , \\ |E_{\mathrm{def}}| &\leqslant e^{C_\times\rho\Delta_\times}\Big[C'_L n_C e^{-(c_*/2)\pi_*^2} + e^{C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)}) + C_\times\rho\Delta_\times} - 1\Big]\ , \\ |E_{\mathrm{bdry}}| &\leqslant C_{\mathrm{resp},\chi_1\chi_2}\, n_C(1+g_0 r)\, e^{-c_*\pi_*^2}\ . \end{aligned} \quad (1.19)$$

Corollary 1.14 (Size of the Two-Loop Leading Term)**.** *Under the hypotheses of Theorem 1.13, with the same data and constants,*

$$|\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^{\circ}_{\xi,1}+E^{\circ}_{\xi,2}+E^{\times}_{\xi}} W_{\chi_1,\chi_2}(C;U_1)]| \leqslant e^{C\tau^2(9n_1+9n_2+2)+C_\times\rho\Delta_\times}\ ,$$

and under the condition of (H3) for Theorem 1.2(iii) for each square,

$$\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{|E^{\circ}_{\xi,1}|+|E^{\circ}_{\xi,2}|}] \leqslant \exp[2C\tau^2(2+1/c) + n_C\delta_q^{I,2}]\ , \qquad \delta_q^{I,2} \leqslant \exp[-c_*\pi_*^2/(2C_K)]\ , \quad (1.20)$$

with $\delta_q^{I,2}$ of (10.10)*, so that the leading term of* (1.18) *has modulus at most $|\kappa_{C_1}\kappa_{C_2}|e^X$ and $\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}|e^{E^{\circ}_{\xi,1}+E^{\circ}_{\xi,2}+E^{\times}_{\xi}} - 1|] \leqslant e^X - 1$, with $X := 2C\tau^2(2+1/c) + n_C\delta_q^{I,2} + C_\times\rho\Delta_\times$.*

Proof of Theorem 1.13 and Corollary 1.14. Section 10.7. *q.e.d.*

Remark 1.15 (The Cross Terms). No condition on the distance of the squares is imposed beyond $V(C_1) \cap V(C_2) = \varnothing$. The cross factor $e^{E_\xi^\times}$ and the factors $e^{C_\times \rho \Delta_\times}$ in (1.19) and in Corollary 1.14 are close to one when $C_\times \rho \Delta_\times$ is small. Since $\Delta_\times \leqslant n_C$ and $\rho = O(g_0^2(1+r)^4)$, this holds at fixed n_1, n_2 for all positions of the squares once g_0 is small; at fixed coupling, $C_\times \rho \Delta_\times \leqslant \epsilon$ whenever $d/(LM) \geqslant c_\times^{-1}(\log(C_\times \rho n_C) + \log \epsilon^{-1})$, which the hypotheses allow on tori with correspondingly large periods. As $C_\times \rho \Delta_\times \to 0$ the bound on $E_\xi^\times$ tends to zero, and those on E_{def} and E_{bdry} take the form of (1.10) with $n_C = n_{C_1} + n_{C_2}$ and the constants C_χ, c, ρ, $C_{\text{resp},\chi_1\chi_2}$ of Definition 1.12.

2 Extraction from the Effective Interaction

Bałaban first constructs the effective interaction as a function of the remaining field. He then extracts its relevant coefficients at the identity and subtracts the corresponding local terms from that field-dependent interaction. The convention is that of [8, (1.6)–(1.7)] and [9, (1.30)], in which the Gaussian normalisation is included in the effective interaction. The intermediate definition in [8, (2.13)] keeps that contribution separate.

In the small-field construction of [8], with no large-field regions, let B be a small coarse Lie-algebra field, and let $U_1(e^{iB})$ be the constrained critical configuration with coarse field e^{iB}. With $\mathbf{E}^{(1)}$ the first-step effective interaction, including the Gaussian normalisation,

$$\mathcal{E}_1(B) = \mathbf{E}^{(1)}(g_0, U_1(e^{iB})) \,. \tag{2.1}$$

In an orthonormal basis of the Lie algebra for the action's invariant inner product, the index a denotes a basis element, μ a bond direction and x its initial coarse site. The second derivative is

$$\Pi^{ab}_{1,\mu\nu}(x,y) \coloneqq \frac{\partial^2 \mathcal{E}_1(B)}{\partial B^a_\mu(x)\, \partial B^b_\nu(y)}\bigg|_{B=0} \,. \tag{2.2}$$

After the colour reduction and the translation-invariant infinite-lattice limit, let $\Pi_{1,\mu\nu}(x)$ denote the resulting scalar kernel with its second site fixed at the origin. In Bałaban's conventions the coupling counterterm is

$$\beta_1(g_0) \coloneqq \sum_{x \in \mathbb{Z}^4} x_1 x_2 \Pi_{1,12}(g_0, x) \,. \tag{2.3}$$

The definition and the symmetry argument are given in [8, (1.20)–(1.22), (5.37)–(5.43)].

In a proper regular region Λ_1, the subtraction has the form

$$\mathbf{E}^{(1)}(\Lambda_1, g_0, U_1) - \mathbf{E}^{(1)}(\Lambda_1, g_0, \mathbf{1}) - \beta_1(g_0) S_{\phi_1}(U_1) \,. \tag{2.4}$$

Here ϕ_1 is the localisation cutoff in [9, (1.27)], supported in Λ_1 and equal to one on its specified interior. If $x(p)$ is the initial site of an oriented representative of the plaquette p, then

$$S_{\phi_1}(U) = \sum_p \phi_1(x(p))(1 - \operatorname{Re} \operatorname{tr}_N U(\partial p)) \,. \tag{2.5}$$

The boundary interaction and the exterior integrations remain in [9, (1.29)–(1.30)]. Bałaban states after (1.30) that the regular interaction is the interaction of [8] restricted to Λ_1. Within the cube-localised construction of [8], the changes of reconstruction and summation domain are controlled in [8, (4.35)–(4.37)].

For the Wilson loop, the quantities to extract are the scalar density along a straight side and the scalar contribution of a corner. We denote them by δm and z_{c}. Their construction below uses connected terms of the normalised Wilson integral at the identity background.

Section 7 identifies the straight-side coefficient with the flat projection of the conditional Wilson kernel in the first RG step, and estimates the error caused by its integration boundary. Section 8 gives the outgoing insertion with the reconstructed closed flux factors and the remaining conditional source terms retained. Section 3 writes the conditional first-step integral with the insertion, at the actual background and exterior fields, as an exact sum of local closed insertions paired with signed local measures; the estimates it states as lemmas are proved in Appendix B.

3 The Sourced Effective Interaction at the First Step

This section carries out the first step of the procedure in Section 2: the conditional first-step integral with the Wilson insertion is written as an exact sum of local closed insertions paired with signed local measures, at the actual background and exterior fields. Every displayed identity is exact at fixed volume. The existence of the objects in those identities and one cluster identity are stated here as lemmas and proved in Appendix B: the two-sided bounds on the block $H_{\xi,II}$ of the regional Hessian on the integration carrier (Lemma 3.8), the analyticity of the reconstruction on the cutoff box (Lemma 3.12), the finiteness of the selected integrals (Lemma 3.24) and the absolute convergence of the cluster sums (Lemmas 3.27 and 3.28, the latter including the cluster identity (3.74), whose proof is Appendix B, Section B.6).

Each identity below holds under those lemmas, under the standing conditions (3.1), and under the branch hypotheses (B1)–(B4), which record what is taken from Bałaban's construction of the branch density. The other inputs are Bałaban's derivation of [9, (1.11)–(1.25)], as recorded after (3.7), [8, (0.4) and (0.10)–(0.11)] for the averaging operation and the axial transports, Appendix A for the flat values, and the Ad-invariance (5.4) of the bond norm. The bounds on the terms of the final identity are Appendices D and E and Section 9.

Conventions. G is the compact connected simple gauge group fixed at the start of the paper, in its fixed unitary realisation, so that group elements near the identity are $U = e^{iA}$ with A in the real vector space $\mathfrak{g}$ of Hermitian matrices $i^{-1}\times$(Lie algebra); $\langle X, Y\rangle = \mathrm{tr}_N(XY)$ is the fixed Ad-invariant real inner product on $\mathfrak{g}$ of the flat calculation (the opening paragraph of Section 4; it is the inner product for which every constant of Appendix A, in particular γ_L, is derived), $h(X) = \langle X, X\rangle^{1/2}$, $d_G = \dim \mathfrak{g}$.

Every quadratic form of this section is defined without reference to this choice; the operators representing the forms, the Hilbert norms, the Lebesgue measures $\mathrm{d}a$, the δ-functions of [9, (1.14)], the Haar density σ and the determinants all refer to this one inner product, and a constant said to depend on L and the group only is a constant for it. The identity [9, (1.15)] is an identity between the fine integral, which involves no inner product, and an expression whose measures $\mathrm{d}A$, δ-functions and densities σ, constants σ_0, $z^{(0)}$ are taken for one and the same inner product on $\mathfrak{g}$, whichever it is; it therefore holds for the fixed inner product with the function σ and the constants σ_0, $z^{(0)}$ belonging to it, and these are the values meant below.

log is the group chart: the inverse of $A \mapsto e^{iA}$ on the Ad-invariant Hilbert ball $h(A) < \varrho_G := \pi/\sqrt{N}$, whose range in G is the chart domain. On this ball $\|A\|_{\mathrm{op}} \leqslant \sqrt{N}\, h(A) < \pi$ (for h the normalised Hilbert–Schmidt norm, $h(A)^2 = N^{-1}\sum_j \lambda_j^2 \geqslant N^{-1}\|A\|_{\mathrm{op}}^2$), so the eigenvalues of A lie in $(-\pi, \pi)$: $A \mapsto e^{iA}$ is injective there (an eigenvalue λ of A is recovered from the

eigenvalue $e^{i\lambda}$ of e^{iA} on the same eigenspace), its differential is invertible (the differential of exp at iA is singular only when two eigenvalues of iA differ by a nonzero multiple of $2\pi i$), and its inverse is the restriction to the chart domain of the principal logarithm $\frac{1}{i}\log$ used in the flat calculation ("the chart domain" of $\mathcal{O}_c$, the sentence preceding (4.1) and (1.4); "the regular chart containing the identity", the sentence following (4.5)); every logarithm below is taken in this chart.

The chart is Bałaban's: [3, p. 21, (22)–(25)] defines $\log U = iA$ for a unitary $U = \sum_j e^{i\lambda_j} P_j$ (spectral projections P_j, distinct $\lambda_j \in]-\pi,\pi]$) by $A = \sum_j \lambda_j P_j$, Hermitian with $|A| \leqslant \pi$, with the bounds $|U-1| \leqslant |\log U| \leqslant \frac{\pi}{2}|U-1|$; the chart logarithm is the restriction of this principal logarithm to the chart domain, on which it is analytic, and the bound (25) holds for the normalised Hilbert–Schmidt norm h as for the operator norm, since $|e^{i\lambda}-1| = 2|\sin(\lambda/2)| \geqslant \frac{2}{\pi}|\lambda|$ for $\lambda \in]-\pi,\pi]$: $h(\frac{1}{i}\log W) \leqslant \frac{\pi}{2} h(W-\mathbf{1})$, and a unitary $W \in G$ with $\|W-\mathbf{1}\|_{\mathrm{op}} \leqslant r_G$, in particular one with $h(W-\mathbf{1}) \leqslant r_G/\sqrt{N}$, lies in the chart domain, r_G being the group constant of Appendix B.

In particular "$\mathsf{M}(U)(c)$ is defined" means throughout that every logarithm in the averaging formula [8, (0.4)] for the row c has its argument in the chart domain, that is $U \in \mathcal{O}_c$, where the averaging operation (1.4) is [8, (0.4)] (it is defined for every U; off $\mathcal{O}_c$ it is the straight transport, a value that enters only through the constraint $\bar{V}_0 = V$ on the rows of Ω_1^c, Section 3.1), and "the local group means $\mathcal{U}_U(y,x)$ are defined" that the transports along the path family from y to x lie in the small-diameter domain of the fixed group mean selected after [8, (0.9)], with the relative logarithms of (0.10) in the chart domain. The mean is that specified analytic mean, with the left-right covariance of [8, (0.6) and (0.10)], and $\mathcal{U}_U(y,x)$ is defined by (0.11); the implicit function theorem identifies the specified mean with its near-diagonal solution on a sufficiently small neighbourhood. The same convention defines the domain $D^{(11)}$ of Section 3.1, the set $\mathcal{X}$ of the measurability paragraph and the objects of Section 3.2.

The chart being Bałaban's, the domain statement that on the domain of integration in [9, (1.11)] the arguments of these logarithms and the fluctuation lie in the chart domain is the smallness of the fluctuation in [9, pp. 247–248] (by Lemma 1 of [5]) and of the contour variables ([3, pp. 24–25]), derived in (B1) for ε_0 small; its constants $O(L^2)$ and $O(1)$ are those of [5, Lemma 1, p. 79] and [3, p. 25], independent of the coupling, as recorded in (B1) and in the hypothesis (H4) of the main theorem.

Appendix B uses besides ϱ_G an operator-norm radius r_G and the comparison constants $c_{\mathrm{h}} = \sqrt{N}$, $c_{\mathrm{n}} = 1/N$ between h, the matrix norm $|\bullet|$ of the cutoff and $\|\bullet\|_{\mathrm{op}}$ ($h \leqslant |\bullet| \leqslant \|\bullet\|_{\mathrm{op}} \leqslant \sqrt{N}h$, where $h(Y) := (N^{-1}\operatorname{tr} Y^*Y)^{1/2}$ for an arbitrary matrix Y, so that $\|Y\|_{\mathrm{op}} \leqslant c_{\mathrm{h}}|Y|$ for every matrix and either cutoff norm); they are recalled where used. The right inverse h_U and the force J_U of Section 3.2 always carry their background subscript, which distinguishes them from the norm h and the carrier J. $|X|$ is the bond norm of [9] used in (4.12), (5.3) (the operator or the normalised Hilbert–Schmidt norm, fixed once; both are Ad-invariant by (5.4)). For $B \in \mathfrak{g}^S$ the Hilbert norm is $\|B\|^2 = \sum_{b\in S} h(B_b)^2$ and the phase norm is $\|B\|_\infty = \max_b h(B_b)$.

E_* is the set of free bonds (all positive fine bonds except the designated bonds $b_0(c)$, [8, p. 267]: $b_0(c)$ is the fine bond contained in the coarse bond c whose endpoints lie in the two blocks of c, the sentence following (4.2)); $\mathcal{B}_y$ is the block with centre y and $b_\pm$ the endpoints of a bond b; distances of bonds are periodic ℓ^∞ distances of initial vertices.

Bonds and rows of a set X of sites follow [9, (1.8)–(1.9)]: the bonds of X are the positive fine bonds with at least one endpoint in X, $X^{(1)}$ is the set of block centres in X, the rows of

X are the coarse bonds with at least one endpoint in $X^{(1)}$, written $X^{[1]}$, and X^* is the set of bonds of X other than the designated bonds $b_0(c)$, $c \in X^{[1]}$ (so a bond of X^* may have one endpoint outside X, and the variables of a row of X include its designated bond).

The complement is treated differently, following this paper's reading of [9, (1.11)–(1.12)], which say only that $V_0 = U|_{\Omega_1^c}$ are the integration variables localised in Ω_1^c (p.248), read with the bonds crossing from Ω_1 to Ω_1^c as variables of the integral over Ω_1 (the rule of [7, (8), p. 258] for plaquettes, "plaquettes with at least one corner belonging to Ω_1", applied to bonds; the rows are those of [7, (12), p. 258], which [9, p. 248] follows, (H4)): the bonds of Ω_1^c, written $b \in \Omega_1^c$, are the positive fine bonds with both endpoints off Ω_1, the restriction $V_0 = U|_{\Omega_1^c}$ is the restriction to these bonds, and the rows of Ω_1^c are the coarse bonds with both block centres off Ω_1; so every positive fine bond is a bond of Ω_1 or a bond of Ω_1^c and not both, and every coarse bond is a row of Ω_1 or a row of Ω_1^c and not both.

$X^\sim$ is Bałaban's enlargement [9] of X by all cubes touching X (p.245; including the cubes meeting X only at an edge or a vertex, as in [8, p. 257], where $\tilde{\square}^n$ is the cube of side $(1+2n)M$ with the centre of $\square$, (H4); used for the distance statement at the end of this sentence and in Lemma 3.19(i)) of the partition in terms of which X is defined (p.245, where the partition into cubes $\square$ of side LM_2R_0 is introduced, $\square^\sim$ is $\square$ enlarged by one layer of M_2R_0-cubes, and "usually" $X^\sim$ enlarges by a layer of the cubes of the partition defining X; here LMR_1-cubes for Ω_1, R_1' and Λ_1, $L^2M_2R_1$-cubes for $\square'$, Q_1 and R_1, p.246), $X^{\sim-1} = ((X^c)^\sim)^c$ is the shrinking and $X^{\sim k}$ the k-fold iterate; a site of $X^{\sim-1}$ is at ℓ^∞ distance at least the cube side from X^c.

The exponent of [9] in the definition of the scales, R_j the smallest power of L with $R_j \geqslant (\log g_j^{-2})^{\mathfrak{r}}$, $\mathfrak{r} \geqslant 2$ and $p_0 \geqslant 5\mathfrak{r}$ (p.245, p.246, [9, (2.5)]), is written $\mathfrak{r}$ here, because r is the cutoff radius of (4.12). The scales of [9] satisfy $R_0 \leqslant (L+1)R_1$ by [9, (2.9), p. 256], hence $R_0 \leqslant LR_1$ as both are powers of L; and M_1 is the M_1 of [9, p. 245], the size of cubes for which the theorems of the previous papers, in particular those on the variational problem, are valid (a lower bound on the cube side: [4, Theorem 3.1, p. 397] holds "for $M \geqslant M_1$", and M_1 "was fixed in" that paper, [5, p. 77]; [9] takes $M_2 = L^2M_1$ as an example, which is consistent with the conditions below).

The identity of a matrix space is written $\mathbf{1}$; the action is S (the $A(U)$ of [9]) and the scalar normaliser E of [9] is written E: it is the vacuum scalar E_ε of (1.2), a constant independent of the configuration and of the branch, which cancels in every ratio of this paper and enters only the absolute identities for $\mathcal{Z}_0$ (Section D.7).

The standing conditions on the parameters are

$$\begin{gathered} L > 11 \text{ odd ([8, p. 251])}; \quad M_1 < M_2 < M \text{ odd powers of } L,\ M \geqslant 8LM_2; \\ \text{(3.59)}; \quad \delta_M \leqslant 1,\ h_+^2\delta_M^2/\gamma \leqslant \tfrac{1}{8},\ 4C_{\mathrm{d}}\delta_M^2(LM+1)^4 \leqslant \tfrac{1}{2}; \\ \text{every fine period a multiple of } LMR_1 \text{ and at least } 8M; \\ \text{every coarse period} > \max(2, 2R_{\mathrm{p}} + 8); \\ g_0 \text{ below the threshold of Appendix B, Section B.8,} \\ \text{which includes } LMR_1 > 4R_{\mathrm{c}} + 4L^2M_2R_1 + 8 \\ \text{and } \widehat{\mu} = C_{L,R_{\mathrm{c}}}\varepsilon_1 < \min(\tfrac{1}{2}\tau_L,\ \gamma_L/(4\max(1, c_L))) \\ \text{(the hypotheses of Lemma 3.8),} \end{gathered} \tag{3.1}$$

where R_{p} and R_{c} are the constants of that appendix ((B.7) and Lemma 3.19), γ, h_+ those of (3.56), δ_M, C_{d} those of Lemma 3.24, τ_L, c_L those of Lemma 3.7 and γ_L, $C_{L,R_{\mathrm{c}}}$ those of Lemma 3.8, all depending on L and the group only (δ_M also on M), and $\varepsilon_1 = g_1 p_0(g_1)$ the

quantity of [9, p. 246], which depends on the coupling; the threshold depends on L, the group, the character χ (through the constant K_χ of Appendix E), M_1, M_2, M, on the constants A_0, A_1, the exponents p_0 and $\mathfrak{r}$ of [9], on the constants B_3, δ_{c} and the $O(1)$ of the comparison after [9, (1.19)] recorded in (B3)(iii), and on the map $g_0 \mapsto g_1$ of [9, (1.27)], which are fixed as in [9] with A_0/A_1 large enough for (B3)(i); the order of choice of every constant and the list of conditions are those of Section B.8, whose closing paragraph also lists the further conditions imposed where the statements about the square and the appendices state them (Sections D.1, D.7, and 9).

The threshold also takes over the conditions of [9] on the parameters (item 5 of Section B.8). Those on the coupling, ε_0 small enough for Theorem 1 of [6] (reference [15] of [9], whose bracketed numbers are those of the bibliography of [8]; [9, p. 248]), MR_1 large (p.251, Lemma D.21), and M_2R_1 large enough for (B3)(ii) (p.250), are conditions that g_0 lies below a value depending on L, the group, M_1, M_2, M, and the constants of [9], met for small g_0 at fixed M_1, M_2, M since $R_1 \to \infty$.

Those on M_1, M_2, M are not listed in [9]: M_1 is the cube size for which the theorems of the previous papers hold (p.245), M_2 is "not too much larger" than M_1, e.g. $M_2 = L^2M_1$, and M is "much larger than M_2", its choice "dictated by many conditions involving this constant" (p.245). They are taken over as the hypothesis that Bałaban's construction [9] is valid for the values $M_1 < M_2 < M$ used here, which follow Bałaban's prescription (Section B.8); this paper adds to them only the conditions displayed above.

The scale integer R_1 is the power of L of [9, p. 246] (Section 3.1; the same letter denotes the large-fluctuation region of [9], and in every product LMR_1, $L^2M_2R_1$ it is the scale), so the first-step cubes of sides LMR_1 and $L^2M_2R_1$, the LM-, M- and M_1-cubes of Section 3.5 and the blocks form nested partitions of the fine torus (all sides are powers of L, and the LM-cubes nest in the $L^2M_2R_1$-cubes once $LM_2R_1 \geqslant M$, threshold item 1 of Section B.8, which holds for small g_0 since $R_1 \to \infty$); the collar inequality holds once the scale satisfies $R_1 \geqslant R_{\mathrm{c}} + 3$, since $LMR_1 \geqslant 8L^2M_2R_1$.

The condition "every coarse period > 2" is the operative hypothesis of Lemma 3.6 (it makes $b_0(c)$ the only designated bond in $\mathcal{E}_c$: with a coarse period equal to two, two coarse bonds would join the same pair of blocks).

Every statement of this section is made under (3.1). When $I = \varnothing$ below, $\mathfrak{g}^I$ is a point, $\mathrm{d}a$ the unit mass, every sum reduces to its empty term and every identity holds trivially.

Some letters carry more than one meaning, each fixed by its context, as follows.

- R_* is the walk range n_*R in Section 3.5 and in the order of choice of Section B.8; from Section 5 on it is the fixed range introduced there, bounding the label-support diameters and the source-to-stencil distances (fixed independently of the coupling), and it is this R_* that enters $R_0 = 4(1 + R_* + R_H + L)$.
- R_0 is the scale of [9] where its scales and partitions are quoted (the sentence on $R_0 \leqslant (L+1)R_1$ above, p.245, and Lemma D.12); from Section 7 on it is this paper's range constant $R_0 = 4(1 + R_* + R_H + L)$, as stated there.
- R_1 is the scale integer in every product LMR_1, $L^2M_2R_1$ and wherever it is called the scale, and the large-fluctuation region of [9] where it is called a region.
- M is the cube side; the group mean is the sans-serif $\mathsf{M}(U)(c)$; $M(W)$ is in Section 8 the level-wise collection $M^j(W)$ of [9, (2.11)] at the first step, and in Section 10 and

from Section E.5.2 on the positive group majorant of a group W, with $M_v(\mathfrak{c})$ the tree majorant of a tuple.

- ℓ is the macro side LMR_1 in Section 9 and in Appendix D, where it is so declared, and a selection label under summation signs ($\sum_\ell$, Section 3.7); $\ell_i(Y) = d_i(Y) + 5$ are the shifted tree lengths of a domain (defined with (B.1)).

- m_L is the Hessian constant of Lemma 3.7; m alone is the tuple length in $\mathfrak{c} = (D_1, \dots, D_m)$ (Section 3.6) and the segment length in coarse units in (7.30); $m(D) = \varepsilon_* e^{-\lambda_* \ell_1(D)}$ is the majorant of Lemma 3.27; δm^σ is the mass coefficient (5.39).

- $\theta \geqslant 0$ is the Mayer constant of Section B.5 (Section 3.6 on); in Appendix D, from (D.26) on, θ is the witness threshold defined there, and θ_{h}, θ_{x} are the constants of Section E.7.2; ϑ and ϑ_* are different letters throughout.

- C_L is the coercivity constant (A.24) of Appendix A in Lemma 3.8 ($\gamma_L = C_L^{-1}$) and that appendix; it is the contour constant $|\Gamma_C| \leqslant C_L n_C$ in Section D.7, at the opening of Appendix E, in Section E.2 (with $C'_L \coloneqq 8C_L$), in Corollary 1.6 and in Remark 11.2; elsewhere it is a constant depending on L (in Section D.8.1 also on N) as stated where it appears.

- $K = 256$ is the fixed integer of σ_K, C_K (Lemma 9.2); in (H3) K is the exponent of [8] in $\varepsilon = L^{-K}$; the number of steps in Section 11 is $k + 1$.

- The italic $B_\xi = \Lambda_1^c$ is the filled bad region, the calligraphic $\mathcal{B}_\xi$ the exterior response functional, and the fraktur $\mathfrak{B}$ a compatible family of the expansion; B_3 is the constant of [9]; the fraktur $\mathfrak{B}_Q$ is a determining set (Section D.8.1), the calligraphic $\mathcal{B}_y$ the block with centre y, and B without the subscript ξ a generic vector of $\mathfrak{g}^S$, the ball B_δ, the bound B_C or the host constants B_h, B_{h}.

- In Section 3.1 the letter F denotes integrands, and the logarithms $F_{I,C,T}$, $F_{C,T}$, $F_{C,\infty}(n)$ and the integrals $F_\xi(t)$ are named with their subscripts; elsewhere F_γ is a set of edges of a polymer and $F(\Gamma)$ a weight in a cluster sum.

3.1 Branch Data, the Outer Operation and the Conditional Integral

One term of the history sum of [9, (1.11)–(1.30)] is fixed: the regions Ω_1, R_1 ([9] uses R_1 both for the large-fluctuation region of (1.18) and for the scale integer of p.246, the smallest power of L with $R_1 \geqslant (\log g_1^{-2})^{\mathfrak{r}}$, which grows as $g_0 \to 0$; both uses are kept, the scale appearing in the products LMR_1, $L^2M_2R_1$, MR_1, M_2R_1 and in its defining inequality $R_1 \geqslant (\log g_1^{-2})^{\mathfrak{r}}$; here $g_1 = g_1(g_0)$ is the first-step coupling of [9], the constant value on $\Lambda_1^{\sim -1}$ of the solution of (1.27), $g_0^{-2} = g_1^{-2} + \beta_1(g_0)$ with the β_1 of [9], a function of g_0 alone, and $\varepsilon_1 = g_1 p_0(g_1)$ as on p.246), the derived Λ_1, S_1 of (1.20) with Λ_1 the filled domain of the modification following (1.20), the new field V, the integration variable V_0 on Ω_1^c, and the background $U_1 = U_1(V, V_0)$ of (1.12) in its axial mean gauge, equal to V_0 on Ω_1^c.

The integration carrier is $I = \Lambda_1^*$, the retained free carrier is $J = \Omega_1^* \setminus I$, and $A_{\mathrm{e}} \in \mathfrak{g}^J$ is the retained fluctuation. The integration variable is $a \in \mathfrak{g}^I$; both are extended by zero to Ω_1^* where needed. The branch data are $\xi = (\Omega_1, R_1, V, V_0, A_{\mathrm{e}})$.

The constraint $\delta(\bar{U}V^{-1})$ of [9, (1.11)] is a product over the coarse bonds, and it is split by rows: on the rows of Ω_1 (coarse bonds with a block centre in Ω_1) the constraint $\bar{U} = V$ is the critical-point constraint of (1.12), imposed on the integral over the bonds of Ω_1 (which include the bonds crossing from Ω_1 to Ω_1^c; the rows with one block centre off Ω_1 average

also over bonds of Ω_1^c, which enter as the constant V_0); read, like every group δ-function here, by integrating against the coarse variables it constrains, it is the fibre measure of the pushforward of the fine measure under the map $\Phi(U) = (\bar{U}|_{\Omega_1^{[1]}}, U|_{\Omega_1^c})$, a submersion on the open set $\mathcal{O}^\Phi$ of fine configurations at which the averages and means on the rows of Ω_1 are defined and $D\Phi$ is onto, which contains the set $\mathcal{X}$ of the measurability paragraph below and, on every branch, $D(V, V_0)$ (shown there).

The fibre measures of a C^1 submersion f of an open set $\mathcal{Y}$ of a manifold with a smooth positive density, here G^E with Haar measure or $\mathfrak{g}^E$ with Lebesgue measure, into such a manifold of dimension k, are defined at *every* point y of $\mathsf{f}(\mathcal{Y})$, not only almost everywhere: by the submersion theorem $\mathcal{Y}$ is covered by open sets $\mathcal{Y}_j$ with C^1 diffeomorphisms ψ_j from open sets of $\mathbb{R}^k \times \mathbb{R}^{n-k}$ onto $\mathcal{Y}_j$ and $\mathsf{f}(\psi_j(y,z)) = y$; with a continuous partition of unity (θ_j) subordinate to $(\mathcal{Y}_j)$, $\mu_y^{\mathsf{f}}(g) := \sum_j \int (\theta_j g)(\psi_j(y,z)) \,|\det D\psi_j(y,z)|\, \mathrm{d}z$ for Borel $g \geqslant 0$ on $\mathcal{Y}$, the determinants taken with respect to the densities, that of the manifold on $\mathcal{Y}_j$ and, on $\mathbb{R}^k \times \mathbb{R}^{n-k}$, the product of the target density in the variable y and Lebesgue measure $\mathrm{d}z$, so that $\mathrm{d}y$ in (i) is the target density.

Then (i) $\int_{\mathcal{Y}} \varphi(\mathsf{f}(x)) g(x)\, \mathrm{d}x = \int \varphi(y)\, \mu_y^{\mathsf{f}}(g)\, \mathrm{d}y$ for Borel $\varphi \geqslant 0$ (the change of variables formula and Fubini's theorem); (ii) $y \mapsto \mu_y^{\mathsf{f}}(g)$ is continuous for every continuous g with compact support in $\mathcal{Y}$ (dominated convergence); (iii) the family does not depend on the choices made, two families with (i) and (ii) agreeing for almost every y by (i) and hence for every y by (ii); (iv) for every $y \in \mathsf{f}(\mathcal{Y})$ and every such g, $\mu_y^{\mathsf{f}}(g) = \lim_{\delta \to 0} |E_\delta|^{-1} \int_{\mathsf{f}^{-1}(E_\delta)} g\, \mathrm{d}x$ for every family of Borel sets E_δ with $B_{c\delta}(y) \subseteq E_\delta \subseteq B_{C\delta}(y)$ for constants $0 < c \leqslant C$, by (i) with $\varphi = \mathbf{1}_{E_\delta}$ and (ii) (the mean of a continuous function over E_δ tends to its value at y); so μ_y^{f} is carried by $\mathsf{f}^{-1}(y)$, and the fibre measures of the restriction of f to an open subset of $\mathcal{Y}$ are the restrictions of the μ_y^{f}.

From (iv) follows the transformation rule, at every level: if Ξ is a C^1 diffeomorphism of an open set $\mathcal{N}_0$ onto an open subset of $\mathcal{Y}$, ψ a C^1 diffeomorphism of an open neighbourhood of $\mathsf{f}(\Xi(\mathcal{N}_0))$ onto its image, and $\mathsf{f}' := \psi \circ \mathsf{f} \circ \Xi$ on $\mathcal{N}_0$ (a submersion), then $\mu_{\psi(y)}^{\mathsf{f}'}(g \circ \Xi \,|\det D\Xi|) = |\det D\psi(y)|^{-1} \mu_y^{\mathsf{f}}(g)$ for every $y \in \mathsf{f}(\Xi(\mathcal{N}_0))$ and every Borel $g \geqslant 0$ on $\Xi(\mathcal{N}_0)$: $\int_{\mathcal{N}_0} \mathbf{1}_{\mathsf{f}'(x') \in B_\delta(\psi(y))}\, g(\Xi(x'))\, |\det D\Xi(x')|\, \mathrm{d}x' = \int_{\Xi(\mathcal{N}_0)} \mathbf{1}_{\mathsf{f}(x) \in \psi^{-1}(B_\delta(\psi(y)))}\, g(x)\, \mathrm{d}x$ exactly, for every δ (the change of variables formula for Ξ); the sets $E_\delta = \psi^{-1}(B_\delta(\psi(y)))$ satisfy the condition of (iv) with $|E_\delta|/|B_\delta| \to |\det D\psi(y)|^{-1}$; dividing by $|B_\delta|$ and letting $\delta \to 0$ gives the identity for continuous g of compact support, and two Radon measures on the fibre, a C^1 submanifold, agreeing on these agree.

For Φ on $\mathcal{O}^\Phi$: for Borel $F \geqslant 0$ supported in $\mathcal{O}^\Phi$ and Borel $\varphi \geqslant 0$ (hence for bounded Borel φ when F is integrable), $\int \varphi(\Phi(U)) F(U)\, \mathrm{d}U = \int \varphi(V, V_0) \int_{\Phi^{-1}(V,V_0)} F\, \mathrm{d}\mu_{V,V_0}\, \mathrm{d}V|_{\Omega_1^{[1]}} \mathrm{d}V_0$ with $\mu_{V,V_0} := \mu^\Phi_{(V,V_0)}$, defined for every $(V, V_0) \in \Phi(\mathcal{O}^\Phi)$, in particular for every admissible pair, and since Φ preserves the coordinate $U|_{\Omega_1^c}$ its straightening maps may be taken to preserve it, so that for every (V, V_0) the measure $\mu^\Phi_{(V,V_0)}$ is the fibre measure over V of the submersion $U|_{\Omega_1} \mapsto \bar{U}|_{\Omega_1^{[1]}}$ on the slice $U|_{\Omega_1^c} = V_0$, the Jacobians agreeing because the differentials of the straightening maps are block triangular with the identity in the V_0 block.

The chart, linearisation, and elimination steps of [9] listed after (3.7) compute μ_{V,V_0} in the variables of (1.14)–(1.15), in which the constraint is linear, by the transformation rule applied at the level of the pair itself and two facts.

First, if Θ is a C^1 diffeomorphism of an open set $\mathcal{N}_\Theta$ onto its image and $\mathcal{Q} \circ \Theta = \Lambda$ on $\mathcal{N}_\Theta$ with Λ linear and onto, then $\mathcal{Q}$ is a submersion on $\Theta(\mathcal{N}_\Theta)$ and its fibre measures there are $\mu_y^{\mathcal{Q}} = \Theta_*(|\det D\Theta|\, \mu_y^\Lambda)$ for every $y \in \Lambda(\mathcal{N}_\Theta)$, the transformation rule with $\Xi = \Theta$ and $\psi = \mathbf{1}$.

Second, the fibre measure over every y, in particular over 0, of a linear surjection Λ whose restriction Λ_{d} to the designated variables is invertible is Lebesgue measure on the free variables x_{f} times $|\det \Lambda_{\mathrm{d}}|^{-1}$, since $(y, x_{\mathrm{f}}) \mapsto (x_{\mathrm{f}}, \Lambda_{\mathrm{d}}^{-1}(y - \Lambda_{\mathrm{f}} x_{\mathrm{f}}))$ is a single straightening map of Λ with Jacobian $|\det \Lambda_{\mathrm{d}}|^{-1}$ (the sifting of $\delta(\Lambda x)$ over the designated variables); here that restriction is $\bigoplus_c \alpha_c(U_1)$ up to the normalisation of the δ-functions, which gives $z^{(0)}(U_1)$.

The transformation rule is applied first on the slice $U|_{\Omega_1^c} = V_0$ with Ξ the composition of the chart $A' \mapsto e^{iA'}$ and the right translation by U_1 on the bonds of Ω_1 ($|\det D\Xi| = \prod_b \sigma(A'(b))$, the Haar density, with respect to Lebesgue and Haar measure) and ψ the row chart $W \mapsto \frac{1}{i}\log(WV^{-1})$ on a neighbourhood of V in $G^{\Omega_1^{[1]}}$ ($|\det D\psi(V)| = \sigma_0^{-|\Omega_1^{[1]}|}$ with respect to Haar and Lebesgue measure), under which the slice submersion becomes $\mathcal{Q}_{U_1}$ of (3.9) and the level V becomes the level 0: $\mu_{V,V_0} = \sigma_0^{-|\Omega_1^{[1]}|} \Xi_* (\prod_b \sigma(A'(b))\, \mu_0^{\mathcal{Q}_{U_1}})$ on Ξ of the chart domain (the rule is applied on $\mathcal{N}_0 := \Xi^{-1}(\mathsf{f}^{-1}(\operatorname{dom}\psi))$, f the slice submersion, the part of the chart domain whose row averages lie in the domain of the row chart; every point of the fibre over V in Ξ of the chart domain lies in $\Xi(\mathcal{N}_0)$, since there $\mathsf{f} = V \in \operatorname{dom}\psi$, and the fibre measure of a restriction is the restriction of the fibre measure), which is the conversion of the row δ-functions in the step list, at the level of the pair itself.

The two facts are then applied to $\mathcal{Q} = \mathcal{Q}_{U_1}$ at fixed (V, V_0), at the level 0, in the chart variable A' of (1.13), with $\Theta(A) := A - h_{U_1}\widetilde{D}(A)$ on the open set $\mathcal{N}_\Theta$ of Lemma B.11(iv), on which $\widetilde{D} = \varphi$ is analytic, solves the row equations and has $\det(\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D}) > 0$, on which Θ takes chart-valued phases (so that $\Theta(\mathcal{N}_\Theta) \subseteq \mathcal{N}_0$, the set on which the transformation rule was applied above: the phases of $\Theta(A)$ are chart-valued, and at every row c the row phase $\Theta(A)|_{\mathcal{E}_c}$ has norm below $\vartheta_* < 2\varrho_*$, where $\mathcal{N}_{U_1,c}$, and hence $\mathcal{Q}_{U_1,c}$ of (3.9), is defined and analytic (Lemma B.8(i), which applies at every row by Lemma B.11(i)), so that the row averages lie in the domain of the row chart), and which contains the argument of every $U \in D^{(11)}(V, V_0)$ (Lemma B.11(ii)) and does not depend on the region R_1 (this step precedes the insertion of (1.18), which introduces R_1): $\mathcal{Q}_{U_1} \circ \Theta = LQ_{U_1} = \Lambda$ on $\mathcal{N}_\Theta$ (the row equations and $LQ_{U_1}h_{U_1} = \mathbf{1}$), and Θ is injective on $\mathcal{N}_\Theta$, since $\Theta(A) = \Theta(A'') = q$ gives $\widetilde{D}(A) = \mathcal{N}_{U_1}(q) = \widetilde{D}(A'')$ by the row equations $d_c = \mathcal{N}_{U_1,c}(A - h_{U_1}d)$, hence $A = q + h_{U_1}\widetilde{D}(A) = A''$; so Θ is a diffeomorphism of $\mathcal{N}_\Theta$ onto the open set $\Theta(\mathcal{N}_\Theta)$ by the inverse function theorem.

On the rows of Ω_1^c (both block centres off Ω_1) the constraint $\delta(\bar{U}V^{-1})$ of [9, (1.11)] is the constraint $\bar{V}_0 = V$ on $V_0 = U|_{\Omega_1^c}$, written $\delta(\bar{V}_0 V^{-1})$ below, and the constrained integral $\int \mathrm{d}V_0|_{\Omega_1^c}\, \delta(\bar{V}_0V^{-1})\cdots$ in the form in which Bałaban writes it in [9, (1.14)–(1.15)] (in (1.11) it is part of $\int \mathrm{d}U\, \delta(\bar{U}V^{-1})$) is taken as it stands there and read, like every group δ-function of a coarse variable in this paper (the paragraph following Lemma 1.1), by integrating it against the coarse variables it constrains: for bounded Borel h on G^{E_1} and nonnegative Borel F, $\int \mathrm{d}V\, h(V) \int \mathrm{d}V_0\, \delta(\bar{V}_0V^{-1})F(V, V_0) = \int \mathrm{d}V|_{\Omega_1^{[1]}} \int \mathrm{d}V_0\, h(V)F(V, V_0)$ with $V(c) = M(V_0, c)$ of (1.4) on the rows c of Ω_1^c (a function of the bonds of Ω_1^c alone, Lemma 1.1(i)); so $\int \mathcal{O}[F](V)\,\mathrm{d}V$ below is the integral of F against the measure in V that the outer operation defines, and $\mathcal{O}[F](V)$ is a Haar density only where that measure has one.

Let $D^{(11)} := D^{(11)}(\Omega_1, V)$ be the domain of integration in [9, (1.11)]: the fine configurations U with $\chi_1(\Omega_1)\zeta(\Omega_1^c)\chi_0'(\Omega_1) \neq 0$ on which the block averages $\mathsf{M}(U)$ are defined on every row of Ω_1 and the local group means $\mathcal{U}_U(y, x)$, $x \in \mathcal{B}_y$, $y \in \Omega_1^{(1)}$, of the gauge-fixing functional $\mathbf{G}(\Omega_1^{(1)}, U)$ of (1.11) are defined (in the chart of the preamble, Bałaban's principal logarithm; by [9, pp. 247 and 248] with Lemma 1 of [5] and [3, pp. 24–25], derived in (B1), these two conditions are implied by the nonvanishing of the cutoffs and the constraint for ε_0 small, and they give $D^{(11)} \subseteq \mathcal{O}_c$ on every row of Ω_1), subject to the constraint $\bar{U} = V$ on the rows of Ω_1 and $\bar{V}_0 = V$ on the rows of Ω_1^c. Its fibre over the restriction is $D^{(11)}(V, V_0) = \{U \in D^{(11)} :$

$U|_{\Omega_1^c} = V_0\}$. A pair (V, V_0), written (Ω_1, V, V_0) where the region is to be displayed and called a pair throughout (Ω_1 is fixed on the branch), is *admissible* when $D^{(11)}(V, V_0) \neq \varnothing$; (B1) provides the background $U_1 \in D^{(11)}(V, V_0)$ for every admissible pair.

The domain of integration of (1.15) with the cutoff (1.25) is in general smaller than $D^{(11)}(V, V_0)$ (the small- and large-fluctuation tests of (1.18) in (1.25) are further conditions; on R_1 the large-fluctuation tests exclude the configurations near U_1); it is described in the variables of (1.15) as follows. For an admissible pair, a free vector $B \in \mathfrak{g}^{\Omega_1^*}$ and a family $d = (d_c)_{c \in \Omega_1^{[1]}}$ of solutions of the row equations (3.24) at the argument $A = g_0 \mathsf{C}_{U_1} B$ (that is, $d_c = \mathcal{N}_{U_1,c}(A - h_{U_1} d)$ for every row c), let $q_d(B) \coloneqq A - h_{U_1} d \in \mathfrak{g}^{\Omega_1}$, the phase (3.25) formed with this family ([9, (1.13)–(1.14)] at the background U_1; on the free bonds $q_d(B)(b) = g_0 B(b)$), and let $U_d(B)$ be the fine field equal to $e^{i q_d(B)} U_1$ on the bonds of Ω_1 and to V_0 on the bonds of Ω_1^c. The family d is *of chart class* when $h(q_d(B)(b)) < \varrho_G$ for every bond b of Ω_1, that is, when the whole phase $q_d(B)$ is chart-valued, the class of the variable A' of [9, (1.13)]. Let

$$\begin{aligned} \mathcal{A}_\xi \coloneqq \{ B \in \mathfrak{g}^{\Omega_1^*} : \text{there is a family } d \text{ of chart class at } g_0 \mathsf{C}_{U_1} B \\ \text{with } U_d(B) \in D^{(11)}(V, V_0) \text{ and the cutoff (1.25) nonzero at } U_d(B) \} , \end{aligned} \tag{3.2}$$

the cutoff (1.25) being a function of the fine field and the branch data (its tests run over the free bonds $(\square'^{\sim 2})^*$, at which the scaled variable is $g_0^{-1} \frac{1}{i} \log(U U_1^{-1})$, by (1.18)–(1.19), p.250, and (1.25), p.252, in the variables of (1.15), and through $\chi_0'(S_1)$ over the fine field; no test involves a designated value of the linearised variable, so the cutoff does not involve $\widetilde{D}$).

For $B \in \mathcal{A}_\xi$ the family d of chart class with $U_d(B) \in D^{(11)}(V, V_0)$ is unique and equals $\varphi(g_0 \mathsf{C}_{U_1} B)$ (Lemma B.11(iii)), and $\widetilde{D}_{U_1}(g_0 \mathsf{C}_{U_1} B) \coloneqq d$, $q_{U_1}(B) \coloneqq q_d(B)$ and $U(B) \coloneqq U_d(B)$ are defined on $\mathcal{A}_\xi$ by this uniqueness; the cutoff (1.25) is then a function of B, U_1 and the branch data. Since $q_{U_1}(B)$ is chart-valued, it is the chart logarithm of the fluctuation $U(B) U_1^{-1}$, so that $B \mapsto U(B)$ is injective on $\mathcal{A}_\xi$ ($U(B)$ determines $U(B) U_1^{-1}$, hence its chart logarithm $q_{U_1}(B)$, hence B on the free bonds, where $q_{U_1}(B)(b) \coloneqq g_0 B(b)$).

Conversely, every $U \in D^{(11)}(V, V_0)$ at which the cutoff (1.25) is nonzero is of the form $U(B)$ with $B \in \mathcal{A}_\xi$: by (B1) the fluctuation $U U_1^{-1}$ lies in the chart domain bond by bond on Ω_1, so $A' \coloneqq \frac{1}{i} \log(U U_1^{-1})$ of (1.13) is defined and chart-valued; the constraint $\mathsf{M}(U)(c) = V(c) = \mathsf{M}(U_1)(c)$ on the rows of Ω_1 gives $\mathcal{Q}_{U_1,c}(A') = 0$ in (3.9), that is $L Q_{U_1} A' = -\mathcal{N}_{U_1}(A')$; with $d \coloneqq \mathcal{N}_{U_1}(A')$ the vector $A' + h_{U_1} d$ lies in the kernel of Q_{U_1} ($L Q_{U_1} h_{U_1} = \mathbf{1}$) and agrees with A' on the free bonds (h_{U_1} has its outputs at the designated bonds), so it equals $g_0 \mathsf{C}_{U_1} B$ with $B \coloneqq g_0^{-1} A'|_{\Omega_1^*}$, the kernel of Q_{U_1} being the graph of C_{U_1} over the free bonds by (3.12) ($\alpha_c(U_1)$ invertible, (B2) and Lemma 3.6); hence $A' = g_0 \mathsf{C}_{U_1} B - h_{U_1} d$, $d_c = \mathcal{N}_{U_1,c}(A') = \mathcal{N}_{U_1,c}(A - h_{U_1} d)$ solves every row equation, d is of chart class because A' is chart-valued, and $U_d(B) = e^{i A'} U_1 = U$ on Ω_1 while $U|_{\Omega_1^c} = V_0$; so $B \in \mathcal{A}_\xi$ and $U(B) = U$.

Thus $\mathcal{A}_\xi$ is the domain of integration of (1.15) with the cutoff (1.25) in the free variable B of (1.15), and $B \mapsto U(B)$ is a bijection of $\mathcal{A}_\xi$ onto $D(V, V_0) \coloneqq \{U \in D^{(11)}(V, V_0) :$ the cutoff (1.25) is nonzero at $U\} = U(\mathcal{A}_\xi)$, the same domain in the fine variable, and $D(\Omega_1, R_1, V) \coloneqq \bigcup_{V_0} D(V, V_0)$. By (B4)(ii) the differential of $B \mapsto U(B)$ is injective at every point of $\mathcal{A}_\xi$, with image the tangent space of the fibre $\Phi^{-1}(V, V_0)$ at $U(B)$.

For an admissible pair and $A_{\mathrm{e}} \in \mathfrak{g}^J$, $a \in \mathfrak{g}^I$ let $U_\xi(a) \coloneqq U(\Psi_\xi(a + A_{\mathrm{e}}))$ be the reconstructed fine field (3.26) (the inverse of the changes of variables [9, (1.13), (1.14), (1.22)] at the background U_1, defined where those are: where $(a + A_{\mathrm{e}})(b) \in \mathcal{W}_b$ for every $b \in S_1^*$, Section 3.4, and every row solution is defined at $g_0 \mathsf{C}_{U_1} \Psi_\xi(a + A_{\mathrm{e}})$), and let

$$\mathcal{D}_\xi \coloneqq \{(A_{\mathrm{e}}, a) \in \mathfrak{g}^J \times \mathfrak{g}^I : \ \Psi_\xi(a + A_{\mathrm{e}}) \text{ is defined and } \Psi_\xi(a + A_{\mathrm{e}}) \in \mathcal{A}_\xi\} \tag{3.3}$$

be the domain of integration of (1.15) with the cutoff (1.25) in the variables of (1.22); by (B4)(ii) (Lemma 3.16) Ψ_ξ maps $\mathcal{D}_\xi$ bijectively onto $\mathcal{A}_\xi$, so the reconstruction $(A_{\mathrm{e}}, a) \mapsto U_\xi(a)$ is a bijection of $\mathcal{D}_\xi$ onto $D(V, V_0)$. In particular $\mathcal{D}_\xi \neq \varnothing$ exactly when $\mathcal{A}_\xi \neq \varnothing$, that is when $D(V, V_0) \neq \varnothing$.

A *branch* is a value of the data ξ such that (Ω_1, V, V_0) is admissible and $(A_{\mathrm{e}}, a) \in \mathcal{D}_\xi$ for some a. On a branch the set $\{a : (A_{\mathrm{e}}, a) \in \mathcal{D}_\xi\}$ is the open box $\{\Theta_\xi = 1\}$ of (3.42) (Lemma 3.19(iv)), and the large-field factor $\zeta_\xi(A_{\mathrm{e}})$ of (3.6) and the retained cutoff $\chi_{\xi,\mathrm{e}}(A_{\mathrm{e}})$ of (3.43) are defined and nonzero. The branch indicator $\iota_\xi(A_{\mathrm{e}})$ is one when ξ is a branch and zero otherwise; as the fibre over a branch value is the open box, which contains $a = 0$, $\iota_\xi(A_{\mathrm{e}}) = 1$ exactly when $(A_{\mathrm{e}}, 0) \in \mathcal{D}_\xi$, a Borel condition (a section of the Borel set $\mathcal{D}_\xi$, measurability paragraph). It is a factor of the integrand of (3.4), which therefore vanishes off the branches, and every object below is defined on branches only.

The sets $\mathcal{A}_\xi$, $D(V, V_0)$ and $\mathcal{D}_\xi$ depend on ξ only through (Ω_1, R_1, V, V_0): the cutoff (1.25) in the variable B of (1.15) and the map Ψ_ξ do not involve A_{e}, which is a variable of these sets.

For a branch function $\mathfrak{M}_\xi$, the outer operation is

$$\mathcal{O}[\mathfrak{M}](V) \coloneqq \sum_{\Omega_1, R_1} \chi_1(\Omega_1) \int \delta(\bar{V}_0 V^{-1}) \left[\int \iota_\xi(A_{\mathrm{e}}) \zeta_\xi(A_{\mathrm{e}}) \chi_{\xi,\mathrm{e}}(A_{\mathrm{e}}) \kappa_\xi \mathfrak{M}_\xi \, \mathrm{d}A_{\mathrm{e}}|_J \right] \mathrm{d}V_0|_{\Omega_1^c} \, , \quad (3.4)$$

with the prefactor

$$\kappa_\xi = z^{(0)}(U_1) \, g_0^{d_G |\Omega_1^*|} \sigma_0^{|\Omega_1^*|} z^{-(L^4-1)|\Omega_1^{(1)}|} \exp\left(-g_0^{-2} S(U_1) - \mathsf{E}\right) \, . \quad (3.5)$$

Here $\chi_1(\Omega_1)$ is the background test of [9, (2.16)–(2.17)] at $k = 1$, a function of V; $\delta(\bar{V}_0 V^{-1})$ is the part of the constraint $\delta(\bar{U} V^{-1})$ of (1.11) on the rows of Ω_1^c (both blocks off Ω_1), as in [9, (1.14)–(1.15)], where it stands in front of the V_0 integral.

The large-field factor $\zeta(\Omega_1^c)$ of (1.11) is defined by the characteristic functions and gauge-fixing exponentials of (1.6) and (1.8) whose cubes or blocks lie in Ω_1^c, resummed over the admissible P_0, P_1, Q_1 (p.248). It is a sum over the admissible (P_0, P_1, Q_1) compatible with Ω_1 of products of nonnegative functions of the fine field U on the bonds near their cubes or blocks and of constants; the factors of (1.4), (1.6) and (1.8) not displayed in (1.11), and the bonds involved, are the following.

- The large-field factors $\chi_0^c(\square)$ of (1.1) for the LM_2R_0-cubes $\square \subseteq P_0$, and the factors $\chi_0(\square)$, $\square \subseteq P_0^c$, that survive the removal of (B3)(i), those of the cubes $\square \not\subseteq \Omega_1^\sim$: every such cube is at ℓ^∞ distance at least LMR_1 from Ω_1 ($\Omega_1^\sim$ adds a layer of LMR_1-cubes, the partitions are nested, and the large-field regions are at distance at least $2LMR_1$ from Ω_1, p.248), and the factor tests the plaquettes of $\square^\sim$, whose bonds are within $M_2R_0 \leqslant LM_2R_1 \leqslant MR_1/8 < LMR_1$ of $\square$ (conventions: $R_0 \leqslant LR_1$ by [9, (2.9)]; $M \geqslant 8LM_2$), so it is a function of V_0.

- The factors $\chi_1^c(\square')$, $\square' \subseteq P_1$, and $\chi_1(\square')$ of (1.4) for the cubes $\square' \subseteq P_1^c$ not in Ω_1 (those in Ω_1 form $\chi_1(\Omega_1)$) are functions of the block field V (through the local minima $U_{1,\square'}(V)$ of (1.2)).

- The gauge-fixing factors χ_{Ax} of the blocks $\mathcal{B}_y$, $y \in P_1^1$, not in $\Omega_1^\sim$ (those in $\Omega_1^\sim$ are removed by (B3)(i)), the exponentials $e^{-g_0^{-2}\mathbf{G}(P_1^1 \setminus \Omega_1^{(1)}, U)}$ of (1.6)–(1.7) of the blocks of P_1^1 off Ω_1, and the constants $z^{-(L^4-1)|P_1^1 \setminus \Omega_1^{(1)}|}$ of the normaliser of (1.5)–(1.6) for these

blocks (the exponent of (1.11) counts the blocks of $\Omega_1^{(1)} \subseteq P_1^1$ only): a block off Ω_1 lies in Ω_1^c, since Ω_1 is a union of blocks, and the axial transports $\mathcal{U}_U(y,x)$ use the bonds inside the block, which are bonds of Ω_1^c, so these are functions of V_0.

- The factors $\chi_0'^c(\square')$, $\square' \subseteq Q_1$, and $\chi_0'(\square')$ of (1.8) for the $L^2M_2R_1$-cubes $\square' \subseteq \Omega_1^c$ not in Q_1 (those in Ω_1 form $\chi_0'(\Omega_1)$) involve the fine field on $(\square'^{\sim 2})^*$, the bonds within $2L^2M_2R_1+1$ of $\square'$, together with the local minima $U_{1,\square'}(V)$ of (1.2): for $\square' \subseteq Q_1$ these are bonds of Ω_1^c (Q_1 is at distance at least $2LMR_1$ from Ω_1 by (1.10)), and for the other cubes they are bonds of Ω_1^c and bonds of Ω_1 within $2L^2M_2R_1+1$ of Ω_1^c, which lie in the layer $\Omega_1 \setminus \Omega_1^{\sim -1} \subseteq S_1$ ([9, (1.20)]), since $LMR_1 \geqslant 8L^2M_2R_1$.

So $\zeta(\Omega_1^c)$ is a nonnegative function of V, V_0 and of the fine field on the bonds of Ω_1 within $2L^2M_2R_1+1$ of Ω_1^c, the last dependence coming from the factors $\chi_0'(\square')$, $\square' \subseteq \Omega_1^c \setminus Q_1$, only. The phrase of [9] "localized in Ω_1^c" refers to the cubes and blocks, and $\zeta(\Omega_1^c)$ is not a function of V, V_0 alone. In the variables of (1.22) it is

$$\zeta_\xi(A_{\mathrm{e}}) = \zeta(\Omega_1^c)(U_\xi(a)) \,, \tag{3.6}$$

with $U_\xi(a)$ the reconstructed fine field (3.26); the right side does not depend on a, because V, V_0 are branch data and the bonds of Ω_1 on which $\zeta(\Omega_1^c)$ depends are at distance at least $LMR_1 - 2L^2M_2R_1 - 2 > 2R_{\mathrm{c}} + 2L$ from Λ_1, so their links are the links of far rows, functions of A_{e} and the branch data only (Lemma 3.9, (B4)). It is therefore a factor of the A_{e} integrand, as written in (3.4).

The constants: σ_0 is the Haar constant of (1.14), z the gauge-fixing normaliser of (1.5) and E the E of [9], all independent of the branch; $z^{(0)} = z^{(0)}(U_1)$ is the factor of the δ-function integration over the designated variables (p.249); it is $\prod_{c \in \Omega_1^{[1]}} |\det_{\mathfrak{g}} \alpha_c(U_1)|^{-1}$ times a constant fixed by the normalisation of the δ-functions in [9] and by $|\Omega_1^{[1]}|$, with α_c of (3.11), so it depends on the branch through U_1 (the linearised constraint $(\widetilde{Q}A')(c) = 0$ of p.249 is the row c of $Q_{U_1}A' = 0$, whose coefficient at $A'(b_0(c))$ is $\alpha_c(U_1)$, Lemma 3.6 and [8, p. 267]). These constants enter only through the prefactor κ_ξ of [9].

The retained cutoff $\chi_{\xi,\mathrm{e}}$ is defined in (3.43): it is the product of the retained factors of [9, (1.25)], namely the good free-bond tests of (1.18) at the bonds of J, the large-fluctuation tests on R_1 and $\chi_0'(S_1)$, all composed with the change of variables (1.22), and it depends on A_{e} and the branch data only (Lemma 3.19(ii)).

The conditional integrand $\mathcal{K}_\xi(a)$, $a \in \mathfrak{g}^I$, is the integrand of [9, (1.15)] in the variables of (1.22) at the actual branch data, divided by the factors already in κ_ξ, ζ_ξ and $\chi_{\xi,\mathrm{e}}$ and by the factors $\chi_1(\Omega_1)$, $\delta(\bar{V}_0 V^{-1})$ standing outside the A_{e} integral; it is written out in (3.44), and $\mathcal{J}_\xi^{\mathrm{a}}[f] = \int f \mathcal{K}_\xi \, \mathrm{d}a$ is the conditional functional.

Measurability and integrability. The integrand of [9, (1.15)] with the cutoff (1.25), in the variables $(V, V_0, A_{\mathrm{e}}, a)$ of (1.22), is a finite composition of indicators of sets defined by inequalities between continuous functions (the tests), of the background $U_1(V, V_0)$, of the local minima $U_{1,\square'}(V)$ of (1.2) entering χ_0' and of the field $\mathbf{H}_{1,Ax}$ of (1.21) entering the bondwise maps (1.22), of the analytic objects of [9] in (B4) and of the bondwise maps (1.22).

The background is a Borel function of the admissible pairs; the proof uses (B1), (B2), and the smoothness of the constraint. Let $\mathcal{X}$ be the open set of fine configurations U on which $\mathsf{M}(U)$ is defined on every row of Ω_1, the local group means $\mathcal{U}_U(y,x)$, $x \in \mathcal{B}_y$, $y \in \Omega_1^{(1)}$, are defined (chart convention of the preamble; they are analytic there) and every $\alpha_c(U)$, $c \in \Omega_1^{[1]}$, is invertible (Section 3.2), and let $\Phi(U) \coloneqq (\bar{U}|_{\Omega_1^{[1]}}, U|_{\Omega_1^c})$ on $\mathcal{X}$; $D\Phi(U)$ is onto, since

$LQ_U h_U = \mathbf{1}$ with h_U supported on the designated bonds of Ω_1, so each fibre $\Phi^{-1}(V, V_0) \cap \mathcal{X}$ is a C^∞ submanifold and the tangent space of the fibre is the kernel of Q_U, the range of C_U. Let $\mathsf{Crit} = \{U \in \mathcal{X} : D(S + \mathbf{G}(\Omega_1^{(1)}, \bullet))(U) \circ \mathsf{C}_U = 0\}$ be the set of critical points of the constrained functional of [9, (1.12)] on the fibres; it is relatively closed in $\mathcal{X}$, being defined by the vanishing of a continuous function; the critical points of this functional, the points of the fibre at which the differential of the functional vanishes on the tangent space of the fibre, are the points of Crit (on $\mathcal{X}$ the fibres are manifolds with tangent space the range of C_U, and $U_1 \in \mathcal{X}$ by (B2) and its axial mean gauge).

The set $\{(V, U) : U \in D^{(11)}(\Omega_1, V)\}$ is Borel in (V, U): it is a finite union of sets defined by non-strict and strict inequalities between functions of (V, U) that are continuous in U and Borel in V (such functions are jointly Borel: G^E is separable metric, and a function continuous in U and Borel in V is a pointwise limit of jointly Borel functions, the standard argument for Carathéodory functions), and by the equalities $\bar{U}(c) = V(c)$ on the rows c of Ω_1 (continuous in U on the open set where the averages are defined) and $\mathsf{M}(U|_{\Omega_1^c}, c) = V(c)$ on the rows of Ω_1^c (the global average (1.4), defined for every U and Borel in U: analytic on the open set $\mathcal{O}_c$ and, off it, the straight transport of the preamble, a continuous function) (the tests involve V, U and the Borel functions $U_{1,\square'}(V)$ of (B1); each factor $\chi_0'(\square')$ is zero for V outside the Borel set $\mathcal{V}_{\mathrm{ax}}(\square')$ of Lemma 3.2 for the determining set of $\square'$, where the configuration it compares with does not exist, by the convention of (B1), so that $\{(V, U) : \chi_0'(\square') \neq 0\} = \{(V, U) : V \in \mathcal{V}_{\mathrm{ax}}(\square'),\ \sup_{b \in (\square'^{\sim 2})^*} |U(b) U_{1,\square'}(V)(b)^{-1} - \mathbf{1}| < 2\delta_0\}$ is Borel, and $U_{1,\square'}(V)$ is never evaluated off $\mathcal{V}_{\mathrm{ax}}(\square')$; likewise the factors of (1.4), functions of V alone, are Borel: by the convention of (B1), $\chi_1(\square') = \mathbf{1}\{V \in \mathcal{V}_{\mathrm{ax}}(\square')\}\, \mathbf{1}\{\sup_{p \subset \square'^{\sim}} |U_{1,\square'}(V)(\partial p) - \mathbf{1}| < \varepsilon_1 L^{-2}\}$ and $\chi_1^c(\square') = 1 - \chi_1(\square')$), intersected with the open set where the averages and the means are defined; and by (B1) for every admissible pair the fibre $D^{(11)}(V, V_0)$ contains exactly one point of Crit, namely $U_1(V, V_0)$.

So the graph $\{(V, V_0, U) : U \in D^{(11)}(\Omega_1, V) \cap \mathsf{Crit},\ \Phi(U) = (V|_{\Omega_1^{[1]}}, V_0)\}$ of $(V, V_0) \mapsto U_1(V, V_0)$ is a Borel set (the intersection of a Borel set with the graph of the continuous map Φ on the open set $\mathcal{X}$, a Borel set, relatively closed over $\mathcal{X}$), and a function whose graph is a Borel subset of a product of Polish spaces is Borel on a Borel domain: the domain is the projection of a Borel set with singleton sections, Borel by the Lusin–Novikov theorem [13, Theorem 18.10], and a function on a Borel domain whose graph is Borel is Borel, [13, Theorem 14.12]. Hence the set of admissible pairs is Borel and U_1 is Borel on it.

The set $\mathcal{A}_\xi$ of (3.2) is Borel, and q_{U_1} is Borel on it. Let P be the set of pairs $(B, d) \in \mathfrak{g}^{\Omega_1^*} \times \mathfrak{g}^{\Omega_1^{[1]}}$ such that d solves every row equation at $g_0 \mathsf{C}_{U_1} B$, is of chart class, and $U_d(B) \in D^{(11)}(V, V_0)$ with the cutoff (1.25) nonzero at $U_d(B)$. It is Borel, jointly with (V, V_0) through the Borel background U_1: the row equations are equalities between functions continuous on the open set where the relative averages are defined, the chart-class condition is open, $U_d(B)$ is continuous in (B, d, U_1), $D^{(11)}(V, V_0)$ is a section of the Borel set above, and the cutoff is a finite product of indicators of inequalities between Borel functions. By definition $\mathcal{A}_\xi$ is the projection of P to the first factor, and every section $\mathsf{P}(B)$ has at most one point, by the uniqueness of the chart-class family (Lemma B.11(iii)); so $\mathcal{A}_\xi$ is Borel by the Lusin–Novikov theorem [13, Theorem 18.10] (the projection of a Borel set with countable sections), and $B \mapsto d$ is Borel on $\mathcal{A}_\xi$ by [13, Theorem 14.12] (its graph is P); hence q_{U_1}, $U(B)$ and the row solutions $\widetilde{D}_c$ of (B4)(i) are Borel on $\mathcal{A}_\xi$, jointly with the branch data; so is $\frac{\delta}{\delta A} \widetilde{D}$ at the arguments $A = g_0 \mathsf{C}_{U_1} B$, $B \in \mathcal{A}_\xi$, by the following formula.

With $A' = q_{U_1}(B)$, the identity $\mathcal{Q}_{U_1}(A - h_{U_1} \widetilde{D}(A)) = LQ_{U_1} A$ holds on the open set $\mathcal{N}_\Theta \ni A$ of Lemma B.11(iv), on which $\widetilde{D} = \varphi$ solves the row equations and is analytic (by (3.9) and

$LQ_{U_1}h_{U_1} = \mathbf{1}$), so by the chain rule $D\mathcal{Q}_{U_1}(A') \circ (\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D}(A)) = LQ_{U_1}$. Applied to $h_{U_1}w$, $w \in \mathfrak{g}^{\Omega_1^{[1]}}$, this gives $D\mathcal{Q}_{U_1}(A')h_{U_1} \circ (\mathbf{1} - \frac{\delta}{\delta A}\widetilde{D}(A)h_{U_1}) = LQ_{U_1}h_{U_1} = \mathbf{1}$ on $\mathfrak{g}^{\Omega_1^{[1]}}$, so the square map $D\mathcal{Q}_{U_1}(A')h_{U_1}$ is invertible, and then $\frac{\delta}{\delta A}\widetilde{D}(A) = (D\mathcal{Q}_{U_1}(A')h_{U_1})^{-1}(D\mathcal{Q}_{U_1}(A') - LQ_{U_1})$, a continuous function of (U_1, A') on the open set where the averages are defined, composed with the Borel maps U_1 and q_{U_1}; hence it is Borel. The same identity shows that $\mathbf{1} - \frac{\delta}{\delta A}\widetilde{D}(A)h_{U_1}$ is invertible, hence $\det(\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D}(A)) = \det(\mathbf{1} - \frac{\delta}{\delta A}\widetilde{D}(A)h_{U_1}) \neq 0$ (Sylvester's determinant identity), which is the nonvanishing part of (B4)(iii) at every argument of the domain of integration.

Moreover $D(V, V_0) \subseteq \mathcal{O}^\Phi$, the set on which Φ is a submersion (the paragraph defining $D^{(11)}$): for $B \in \mathcal{A}_\xi$, $A = g_0\mathsf{C}_{U_1}B$ and $A' = q_{U_1}(B)$, the identity $D\mathcal{Q}_{U_1}(A') \circ (\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D}(A)) = LQ_{U_1}$ just derived has right side onto by (B2) and second factor invertible; hence $D\mathcal{Q}_{U_1}(A')$ is onto. Since $\mathcal{Q}_{U_1,c}(A') = \frac{1}{i}\log(\mathsf{M}(e^{iA'}U_1)(c)V(c)^{-1})$ with $\mathsf{M}(e^{iA'}U_1)(c) = V(c)$, where the differential of the chart is the identity, and $A' \mapsto e^{iA'}U_1$ has invertible differential (A' is chart-valued), the differential of $U \mapsto \bar{U}|_{\Omega_1^{[1]}}$ at $U(B) = e^{iA'}U_1$ is onto, that is, $D\Phi(U(B))$ is onto.

Borel as well is $\mathcal{D}_\xi$, the preimage of $\mathcal{A}_\xi$ under Ψ_ξ, which is continuous in its argument on its open domain and depends on the pair through the Borel field $\mathbf{H}_{1,Ax}$ below, hence is jointly Borel by the argument just given, and $D(V, V_0) = U(\mathcal{A}_\xi)$ is Borel as the image of a Borel set under the injective Borel map $B \mapsto U(B)$ (injective, as shown after (3.2)), by the Lusin–Souslin theorem for Borel injections [13, Corollary 15.2]; the Jacobians and the bondwise maps are continuous in their arguments; the local minima $U_{1,\square'}(V)$ of (1.2) are Borel functions of V, and the configuration $U_1(\Omega_1 \cap \Lambda_1^c, V)$ of (1.21) is a Borel function of (V, V_0) ([9] states that it depends on $V|_{\Omega_1\cap\Lambda_1^c}$; its defining conditions as recorded in (B1) involve V_0 where the rows of $(\Lambda_1^\sim)^c$ cross to Ω_1^c, and the Borel choice is made in the pair): they are the Borel choices of Lemma 3.2 fixed in (B1); hence $\mathbf{H}_{1,Ax} = L\frac{1}{i}\log(U_1U_1(\Omega_1 \cap \Lambda_1^c, V)^{-1})$ of (1.21) is Borel in (V, V_0) on the bonds of S_1^*, where alone it is used (the chart logarithm is continuous on the chart domain, which contains the argument there by Lemma D.21 and (3.40)).

So the integrand is jointly Borel, and nonnegative. By Tonelli's theorem the section integral $\mathcal{J}_\xi^{\mathrm{a}}[1] = \int \mathcal{K}_\xi \,\mathrm{d}a$ is a $[0,\infty]$-valued Borel function of the branch data, and $\int \mathcal{O}[\mathcal{J}_\xi^{\mathrm{a}}[1]](V)\,\mathrm{d}V$ is the integral of the nonnegative integrand over all variables, equal to the fine integral $\mathcal{Z}_0$ of (3.7) below, which is finite (a bounded function on a compact group); hence the outer integrals converge, $\mathcal{J}_\xi^{\mathrm{a}}[1] < \infty$ for almost every value of the branch data, and on every branch $\mathcal{J}_\xi^{\mathrm{a}}[1] < \infty$, since the integrand (3.44) is bounded on the bounded box $\{\Theta_\xi = 1\}$ (its exponent is the sum of an analytic function bounded on the closed box, Lemma 3.12, and of a constant of the branch, Lemma 3.17). The Wilson function $f_\xi(a) = W_\chi(C; U_\xi(a))$ is jointly Borel for the same reason and $|f_\xi| \leqslant 1$, so $\mathcal{J}_\xi^{\mathrm{a}}[f_\xi]$ is a Borel function of the branch data with $|\mathcal{J}_\xi^{\mathrm{a}}[f_\xi]| \leqslant \mathcal{J}_\xi^{\mathrm{a}}[1]$, and Fubini's theorem applies to it. Then

$$\mathcal{Z}_0 = \int \mathcal{O}[\mathcal{J}_\xi^{\mathrm{a}}[1]](V)\,\mathrm{d}V, \qquad \mathbb{E}_\varepsilon[W_\chi(C)] = \frac{1}{\mathcal{Z}_0}\int \mathcal{O}[\mathcal{J}_\xi^{\mathrm{a}}[f_\xi]](V)\,\mathrm{d}V\ , \tag{3.7}$$

with $f_\xi(a) = W_\chi(C; U_\xi(a))$ the Wilson function of the reconstructed fine field $U_\xi(a)$ of (3.26). The first identity is obtained by Bałaban's steps from the fine integral to [9, (1.15)] with the cutoff (1.25), performed with $\zeta(\Omega_1^c)$ kept inside the V_0 integral and the cutoff in its exact composed form (the estimate-based removals of [9] are recorded in (B3)); the second is the same derivation with the bounded function $W_\chi(C; \bullet)$ of the fine field, invariant under all fine gauge transformations, inserted under the fine integral.

The steps, in the following order, which is that of [9] except at two places (p.248 drops

χ_0 and χ_{Ax} before the resummation (1.11), and p.249–250 integrates the δ-functions over the designated variables before inserting (1.18); the operations concerned are multiplications of the integrand by functions of the fine field equal to one, regroupings of a finite sum and the evaluation of δ-functions, and the resulting integrand, a function of the fine field, is the same in either order):

- the insertion of the block-average δ-functions and of the partitions of unity (1.1), (1.4) (multiplications by functions equal to one, invariant under the fine gauge transformations of (1.5): the tests of (1.1) are plaquette tests and those of (1.4) are functions of V), with the group Dirac measure normalised against Haar measure, $\int \delta(\bar{U}V^{-1})\,\mathrm{d}V = 1$ for every U (the sentence following (1.3); [9, (0.1)] is the blocking operator whose δ-functions these are; this is why $\mathcal{Z}_0 = \int \varrho_1\,\mathrm{d}V$ in the notation of [9]);

- the Faddeev–Popov gauge fixing (1.5)–(1.6), an identity for the integral of any function invariant under the gauge transformations u of (1.5), which introduces the normaliser $z^{-(L^4-1)|P_1^1|}$ of (1.5) for the blocks of the gauge-fixed region;

- the insertion of the partition of unity (1.8) after the gauge is fixed (its tests compare U with the axial-gauge minima $U_{1,\square'}(V)$ and are not gauge invariant);

- the resummation (1.11) over P_0, P_1, Q_1 at fixed Ω_1 (p.248, a regrouping of a finite sum; R_1 is introduced afterwards, in (1.18)), after which the gauge-fixed blocks displayed are those of $\Omega_1^{(1)}$ and the displayed normaliser is $z^{-(L^4-1)|\Omega_1^{(1)}|}$, the gauge-fixing factors and normalisers of the other blocks of P_1^1 being part of $\zeta(\Omega_1^c)$;

- the removal of (B3)(i), which replaces factors by one where the retained factors are nonzero;

- the change of variables $U = U'U_1$ on the bonds of Ω_1 at fixed (V, V_0) (a translation of the Haar measure) and $A' = \frac{1}{i}\log U'$ of (1.13), the chart being a bijection of the domain onto its image with the Haar density as Jacobian, $\mathrm{d}U' = \sigma(A')\,\mathrm{d}A'$ bond by bond with $\sigma(0) = \sigma_0$ (the domain lies in the chart domain by (B1));

- the conversion of the group δ-functions on the rows of Ω_1: by (3.9) and $\mathsf{M}(U_1)(c) = V(c)$, $\mathsf{M}(e^{iA'}U_1)(c)V(c)^{-1} = e^{i\mathcal{Q}_{U_1,c}(A')}$, and $\delta_G(e^{iX}) = \sigma(X)^{-1}\delta(X) = \sigma_0^{-1}\delta(X)$ on the support, so the row δ-functions become $\sigma_0^{-|\Omega_1^{[1]}|}\prod_c \delta(\mathcal{Q}_{U_1,c}(A'))$ (the transformation rule of the paragraph preceding (3.2), applied at the level of the pair (V, V_0), as is every later change of variables of this list);

- the linearising substitution $A' = B - h_{U_1}\widetilde{D}(B)$ of (1.14) ([9] writes both variables as A; in this step list B is the unscaled variable of (1.14) and A the scaled one of p.249, whereas Section 3.1 writes B for the scaled free vector and $A = g_0\mathsf{C}_{U_1}B$ for the unscaled kernel vector, and the lemmas follow Section 3.1), the change of variables by the diffeomorphism Θ of the paragraph preceding (3.2), defined there on the set $\mathcal{N}_\Theta$ of Lemma B.11(iv), which does not depend on R_1, since this step precedes the insertion of (1.18), which introduces R_1 ((B4)(i), (iii) and Lemma B.11(iv); Θ is injective with $\mathcal{Q}_{U_1}\circ\Theta = LQ_{U_1}$), with the Jacobian $|\det(\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D})|$ (the first fact of that paragraph, at the level 0), after which $\mathcal{Q}_{U_1}(A') = LQ_{U_1}B$ (the equation of [8, p. 267], (3.24)) and the constraint is the linear constraint $\delta(\widetilde{Q}B)$ of (1.14), the $\widetilde{Q}$ of [9] being LQ_{U_1}, the linear part of the constraint function $\widetilde{Q}$ of [8, (2.4), p. 266] (the $\widetilde{Q}$ of [8] in the equation quoted at (3.24) is Q_U, so that its $L\widetilde{Q}$ is this operator; $\delta(LQ_{U_1}B) = L^{-d_G|\Omega_1^{[1]}|}\delta(Q_{U_1}B)$, and the constant $L^{-d_G|\Omega_1^{[1]}|}$ of this rescaling of the δ-functions is part of $z^{(0)}$, p.249), while the

Haar factor $\prod_{b\in\Omega_1}\sigma(A'(b))$ is written as $\sigma_0^{\#\{\text{bonds of }\Omega_1\}}$ times the normalised density $\prod_b \sigma(A'(b))/\sigma_0$, the $\sigma(A-h\widetilde{D}(A))$ in [9, (1.14)–(1.15)] (the σ there is the normalised density, as in [7, (18), p. 260], to which Bałaban refers at the foot of [9, p. 248] and where σ/σ_0 appears with the exponent $\log\sigma_0|\Omega_1^*|$ of (1.14)): the $\sigma_0^{|\Omega_1^{[1]}|}$ of this product cancels the $\sigma_0^{-|\Omega_1^{[1]}|}$ of the conversion, and $\sigma_0^{\#\{\text{bonds of }\Omega_1\}-|\Omega_1^{[1]}|}=\sigma_0^{|\Omega_1^*|}$ is the factor in [9, (1.14) and (1.15)];

- the scaling $B=g_0A$ of p.249 (in [9]: $A=g_0A'$), with Jacobian $g_0^{d_G\times(\text{number of bonds of }\Omega_1)}$ and $\delta(\widetilde{Q}g_0A)=g_0^{-d_G|\Omega_1^{[1]}|}\delta(\widetilde{Q}A)$, so that the power of g_0 is $g_0^{d_G|\Omega_1^*|}$;
- the insertion of the partition of unity (1.18) in the scaled variable (p.250) and the removal of (B3)(ii), which together produce the cutoff (1.25);
- the integration of the δ-functions $\delta(\widetilde{Q}A)$ over the designated variables $A(b_0(c))$, $c\in\Omega_1^{[1]}$, which is the elimination of (B2) and produces the factor $z^{(0)}(U_1)$, after which the remaining Lebesgue measure is $\mathrm{d}A|_{\Omega_1^*}$ and the normalised Haar density remains at every bond of Ω_1, at the designated bonds evaluated at the eliminated values (it is the Haar primitive (3.30), part of the v of [9]);
- the bondwise change of variables (1.22) on S_1^*, a change of variables by (B4)(ii) with the Jacobian $|\det D\Psi_\xi|$.

Each step is an identity for the integral of any bounded measurable function of the fine field, and the fine field is expressed through the integration variables by the reconstruction (3.26).

The outer operation is Bałaban's expansion at the stage of [9, (1.15)] with the cutoff (1.25). The conditional integration (1.26) of [9], its cluster expansion, and the renormalisation (1.27)–(1.28) are not used; the first-step coupling $g_1(g_0)$ defined by (1.27) on $\Lambda_1^{\sim-1}$ enters the scale R_1 and the quantity ε_1 of Lemma 3.8 (Section 3.1) as a parameter depending on g_0 alone.

The following facts about the branch density are taken from [9] as stated there, except where an item records a reading, a derivation or a proof (among them the Borel choices, the convention and the derivations in (B1), the Remark under (B2), the parts of (B4)(i) and (iii) proved in Lemma B.11, and the proofs in (B4)(ii)). Together with the lemmas proved in Appendix B, the standing conditions, and the further statements of [9] named in this section (the definitions (1.1)–(1.10) and (1.18)–(1.22), the local minima $U_{1,\square'}$ of (1.2) and the field $\mathbf{H}_{1,Ax}$ of (1.21), the axial mean gauge of U_1 of p.248, the scale relations (2.6) and (2.9) of p.255–256, the coupling map (1.27) with its β_1, (2.16)–(2.17) for $\chi_1(\Omega_1)$, (2.4)–(2.5) and p.246 for ε_j and R_j, (1.17) for the gauge action on phases, and, through (B1), Theorem 1 of [6], reference [15] of [9]; (26) of [6] is cited in [9, p. 248] for the expansion of the action), they are the inputs of this section:

(B1) [9, (1.11)] is taken as it stands, with its domain of integration $D^{(11)}$ (Section 3.1); the local minima $U_{1,\square'}(V)$ of (1.2) are minimal configurations of the variational problem [9, (2.12)] for the determining set of (1.2), elements of the unique minimal orbit (p.256) taken in the axial gauge (p.247); the configuration $U_1(\Omega_1\cap\Lambda_1^c,V)$ of (1.21) is determined, p.251, by the conditions of (1.12) on the domain $(\Lambda_1^{\sim})^c$ (the constraint $\bar{U}=V$, the rows crossing to Ω_1^c involving V_0 as in (B2), and the minimality of the constrained functional, whose critical point is its minimum, p.248) and on $\Lambda_1^{\sim}\setminus\Lambda_1$ by the determining set $\mathbf{B}_1(\Lambda_1^c)$ and the configuration $Q_1^{s*}V$, in the axial gauge; [9] names the determining data only at

p.251, and the configuration is the minimal configuration in the sense of (2.12) (p.256: the functional A^η minimised over the regular configurations satisfying the constraints) for the constraints named, as [9] writes the operation it calls similar to (1.21)–(1.23) at the general step, p.269, where the function is $U(\mathbf{B}(\Lambda_{k+1}^c \cap \Lambda_k) \cup \mathbf{B}_{k+1}, \bullet)$ of (2.15) with data from both scales; the phrase of [9] "depends on V restricted to $\Omega_1 \cap \Lambda_1^c$" is accordingly its statement about the dependence on V, the crossing rows involving V_0 as for U_1, so that the Borel choice below is made in the pair (V, V_0).

The integrals over V in [9] presuppose that these configurations depend measurably on the data. By Lemma 3.2, which covers conditions that are equalities between continuous functions and the minimality of a continuous functional over a set so defined, a Borel choice exists for each of the two problems, in V for (1.2) and in (V, V_0) for (1.21); one such choice is fixed for each. Where the data are regular the chosen configuration is Bałaban's: the minimal configurations for given data form one orbit, in the regularity class [6, (6)] ([9, p. 256], "the minimal orbit, i.e., the set of minima"; [6, Theorem 1, p. 279]; the class and radius of Section D.7.1, and of Section D.8.1 for the configuration of (1.21)), of the gauge transformations equal to the identity on the set $\mathfrak{B}_k$ of determining points ([6, (4), p. 278]: every fine site of the scale-zero determining set and the block centres of the scale-one set), and the axial gauge determines a unique element in each such orbit: the axial condition $U(y, x) = \mathbf{1}$ of [9] is posed on the mean contour variables of [8, (0.11)] ([9, p. 247]), which a transformation w of the orbit changes to $w(y)U(y, x)w(x)^{-1}$ by [8, (0.6)], so that $w(y) = \mathbf{1}$ and the axial condition for both configurations force $w(x) = \mathbf{1}$ ([5, p. 78] for the single-contour axial gauge); so the axial representative is unique, and the statements of [9] about these configurations that are used here (the tests (1.4) and (1.8) and (B3)(iii)) hold for the chosen configurations; (3.40) is proved for them in Lemma D.21.

The averages and means of (1.11) are the principal logarithms of [3, p. 21, (22)–(25)], Bałaban's own convention, restricted in the preamble to the chart ball. That on the domain of integration every argument of a logarithm of [8, (0.4)] on a row of Ω_1 lies in the chart domain, every family of transports entering a mean (0.10)–(0.11) at a block centre of $\Omega_1^{(1)}$ lies in the small-diameter domain of the mean, and the fluctuation $U' = UU_1^{-1}$ lies in the chart domain on every bond of Ω_1, the designated bonds $b_0(c)$ included, so that $A' = \frac{1}{i} \log U'$ of (1.13) is defined, is the smallness statement of [9, pp. 247–248], which Bałaban justifies by the reasoning of the proof of Lemma 1 of [5]. Since the hypotheses of that lemma (the exact axial condition and [5, (1.7)] on the blocks concerned) are not those of the domain of [9], the statement is derived as follows, with that proof as the model.

(i) On the domain of integration the cutoffs χ_0 and χ_{Ax} of (1.5)–(1.6) localised in $\Omega_1^\sim$ equal one ((B3)(i), p.248): $|U(\partial p) - 1| < \varepsilon_0$ for every plaquette $p \subset \Omega_1^\sim$ and $|U(y, x) - 1| < \varepsilon_0$ for every mean (0.11) at $y \in \Omega_1^{\sim(1)}$, $x \in B(y)$.

(ii) For such U [9] states $|UU_{1,\square'}^{-1} - 1| < O(L^2)\varepsilon_0$ on every bond of $\square'^{\sim 2}$ (p.247, "on $\square'^{\sim 2}$", the designated bonds not excluded, whereas the test (1.8) runs over $(\square'^{\sim 2})^*$), "by the same reasoning as in the proof of Lemma 1 [14]", that is Lemma 1 of [5] (p. 79, (1.24)–(1.25)): if two configurations V_0 and $V'V_0$ have plaquette variables within $\alpha_0 L^{-2}$ of one on Ω_1 ((1.7) there), the same variables on the axial contours $\Gamma_{y,x}$, $x \in B(y)$, and block averages within α_1 of each other on $\Omega_1^{(1)}$, then $|V' - 1| < 4d^2\alpha_0 + \alpha_1$ on every bond of Ω_1; its proof (pp. 79–80) bounds the bonds inside a block from the plaquettes and the axial conditions, the bonds joining two blocks from these, and the designated bond b_0 of each coarse bond c from the block-average condition (pp.79–80), and the bound is local (p.80); [9] applies the reasoning with $V_0 = U_{1,\square'}$ in the axial gauge, the

block averages of both configurations equal to V on the blocks of $\square'^{\sim 4}$ ((1.2)–(1.3), p.246, and the constraint of (1.11)) and the axial condition on U holding within ε_0 by (i).

(iii) U_1 lies in the domain of integration (p.248; the "domain of integration in (1.11)" of [9] is the support of every characteristic factor of (1.11), the set to which they restrict the integral, as in [8, p. 265], (H4) and Remark D.19), so (ii) applies to U_1 too, and $U' = (UU_{1,\square'}^{-1})(U_{1,\square'}U_1^{-1})$ is within $2\,O(L^2)\varepsilon_0$ of one on every bond of $\square'^{\sim 2}$, for every cube $\square' \subset \Omega_1$ (p.248: "both fields in the parenthesis ... are small").

(iv) The arguments of [8, (0.4)] on a row of Ω_1 are closed contours in two adjacent blocks of $\Omega_1^{\sim}$, and two transports of a family $\mathbf{G}(y,x)$ differ by such a contour; by the reasoning of [3, pp. 24–25] ((44)–(46): in a local axial gauge every bond of the four blocks $\Delta(p')$ is within $dL\alpha_0$ of one and every contour Γ in them within $|\Gamma| dL\alpha_0 = O(1)L^2\alpha_0$, "for $L^2\alpha_0$ sufficiently small"), with $\alpha_0 = \sqrt{N}\varepsilon_0$ from (i) (the $|\bullet|$ of [3] is the operator norm, which the norm of (i) bounds up to the factor $\sqrt{N}$), every such closed contour variable, a conjugation-invariant quantity, is within $O(1)L^2\varepsilon_0$ of one, and so is the diameter of each family of transports.

The constant $O(1)$ of (iv) is displayed by [3], $|\Gamma| \leqslant (2d+1)L$ and the factor d, up to the elementary count of the contours of [8, (0.4) and (0.10)–(0.11)] in the four blocks and the factor $\sqrt{N}$; the constant $O(L^2)$ of (ii) is not displayed in [9]; it is that of the reasoning of [5, Lemma 1, p. 79], whose bound $4d^2\alpha_0 + \alpha_1$ has absolute constants, and, like the constants of (B3)(iii), it is independent of the coupling and of the branch ((H4)), so that for ε_0 small at fixed L the quantities of (iii) and (iv) lie below ϱ_G, $r_G/\sqrt{N}$, and the small-diameter radius of the mean ("ε_0 sufficiently small", [9, p. 248]; item 5 of Section B.8). This is the domain $D^{(11)}$ of Section 3.1.

A convention of this paper: every factor of (1.4) and (1.8) defined through the configuration $U_{1,\square'}(V)$ of (1.2) is assigned a value for the V for which that configuration does not exist ($V \notin \mathcal{V}_{\text{ax}}(\square')$, Lemma 3.2), namely $\chi_1(\square') = 0$, $\chi_1^c(\square') = 1$, $\chi_0'(\square') = 0$ and $\chi_0'^c(\square') = 1$, so that (1.4) and (1.8) remain partitions of unity ([9, (1.2)–(1.4) and (1.8), pp. 246–247] define these factors through $U_{1,\square'}(V)$ without an existence clause), and $U_{1,\square'}(V)$ enters only where it exists. Under it, where $\chi_1(\Omega_1) \neq 0$ the configuration $U_{1,\square'}(V)$ exists for every cube $\square' \subset \Omega_1$, and where a factor $\chi_0'(\square')$ is nonzero it exists for that cube.

For every admissible (Ω_1, V, V_0) the background $U_1 = U_1(V, V_0)$ of (1.12) exists, is the unique critical point of the constrained functional of [9] in the domain of integration in (1.11), that is in the fibre $D^{(11)}(V, V_0)$, and is a minimum with strictly positive second differential ([9, p. 248]: "there exists exactly one critical point" of (1.12) in the domain of integration in (1.11), by Theorem 1 of [6] (reference [15] of [9], the variational problem of the averaging operation) for ε_0 small; "it is a minimum of (1.12), with a strictly positive second order differential", stated there without further citation); the fluctuation $U' = UU_1^{-1}$ lies in the chart domain for every $U \in D^{(11)}(V, V_0)$ (p.248, derived above), so that $A' = \frac{1}{i}\log U'$ of (1.13) is defined.

(B2) For every admissible pair (Ω_1, V, V_0) the constraint $\delta(\widetilde{Q}A')$ of (1.14) is solved on every row $c \in \Omega_1^{[1]}$ for the designated variable $A'(b_0(c))$ as a linear function of the free variables of the two blocks of c ([9, p. 249], stated at the critical point U_1, which exists for every admissible pair by (B1); [9] states this for the rows of Ω_1 without distinguishing the rows with one block off Ω_1; on such a row the average involves V_0 on the outer block as a constant, as in the first-step problem [7, (12), p. 258] ($U = V_0$ on Ω_1^c, $\bar{U} = V$ on $\Omega_1^{(1)}$), which [9, p. 248] follows, (H4), and $b_0(c)$, a bond of Ω_1, is the eliminated variable,

following the choice of $A'(b_0(c))$ in [9, p. 249] as the variable solved for on every row). In the notation of Section 3.2 this is the invertibility of $\alpha_c(U_1)$ for every $c \in \Omega_1^{[1]}$, so that U_1 lies in the open set $\mathcal{X}$ of the measurability paragraph for every admissible pair.

Remark. The invertibility is also proved here, at every row $c \in \Omega_1^{[1]}$, near or far: $\alpha_c(U)$ depends on $U|_{\mathcal{E}_c}$ only, Lemma 3.6 gives its invertibility for backgrounds e^{iA} with $\|A\|_\infty \leqslant \tau_L$ on $\mathcal{E}_c$ and shows that a site gauge transformation multiplies it on both sides by invertible maps, and the first assertion of Lemma B.11(i) of Appendix B, proved from (B1)(i), (iii), (iv), the first condition of threshold item 6 (with (B.6)), the fine periods being at least $8M$, the inclusion $\mathcal{E}_c \subseteq \mathfrak{P}_c$ given by (3.10) and Lemma B.5, without (B2), gives a site gauge on the two blocks of c in which $U_1 = e^{i\phi}$ on $\mathcal{E}_c$ with $\|\phi\|_\infty \leqslant \tau_L/2$. So of (B2) only its form on the rows with one block off Ω_1 (the outer block entering as the constant V_0, and $A'(b_0(c))$ the eliminated variable) is an input.

(B3) The estimate-based simplifications in Bałaban's derivation of [9, (1.25)] hold under Bałaban's parameter hierarchy, in the following form: the dropped factors are equal to one wherever the retained factors are nonzero.

(i) The removal of the factors $\chi_0(\square)$ and χ_{Ax} of the cubes and blocks localised in $\Omega_1^\sim$ after (1.10) (A_0 sufficiently larger than A_1, p.248).

(ii) The removal of the second factor of (1.19) on the good cubes (M_2R_1 large, p.250), which leaves χ_0' on the cubes of S_1 only (p.250); the condition "M_2R_1 large enough" is a threshold on the coupling (it makes $O(1)B_3e^{-\delta_c LM_2R_1}\varepsilon_1$ much smaller than δ_0, both sides depending on g_0 through ε_1, δ_0 and R_1), taken over in item 5 of Section B.8.

(iii) The comparison on which (ii) rests, p.250: for every $L^2M_2R_1$-cube $\square' \subseteq \Omega_1^{\sim -1} \setminus R_1$ (the $\square' \subset R_1^c$ of [9]; then $\square'^{\sim 4} \subseteq \Omega_1$, p.250), (by the convention of (B1), $U_{1,\square'}(V)$ exists for every cube $\square' \subset \Omega_1$ where $\chi_1(\Omega_1) \neq 0$, so the comparison is between two existing configurations), $|U_1(b)U_{1,\square'}(b)^{-1} - \mathbf{1}| < O(1)B_3e^{-\delta_c LM_2R_1}\varepsilon_1$ for the bonds b of $\square'^{\sim 2}$, with the constant B_3, the decay rate δ_c (the δ of [9]) and the $O(1)$ independent of the coupling and of the branch (Bałaban's constants, (H4): B_3 is that of [6, Theorem 1, p. 279], δ_c the rate of the exponential decay property [6, (190)], which Bałaban invokes in [9, p. 266] for the functions $\mathbf{H}_{k,\square}$, and the $O(1)$ that of Bałaban's derivation of the same comparison [9, (3.19), p. 268] from these facts and [3, (106)–(108), (159)–(163)]; only their independence from the coupling is used, in Appendix B, Lemma B.4).

Item (iii) is used in Appendix B, Lemma B.4; item (i) is used only through its conclusion, the form (1.11) of the integrand and hence the form (1.25) of the cutoff, and item (ii) through (1.25) and directly in the proof of Lemma 3.19(iv). The retained cutoff is the exact composition (3.43), not the rewriting of the retained tests in the new variables in [9, p. 252].

(B4) For every admissible pair (Ω_1, V, V_0) and every region R_1 of the history sum, the expressions in [9, (1.14)–(1.15) and (1.22)–(1.24)] are defined on the domain of integration: the linearising correction $\widetilde{D}$ of (1.14), p.249 (at the foot of [9, p. 248] Bałaban refers to the transformations analysed between (16)–(18) of [16], or between (I.1.8)–(I.1.9) of [I], where [I] is [8] in the reference list of [9, p. 285]; [8, (1.8)–(1.9), p. 261] concern the functions J_j, and the linearisation of the unit-lattice step of [8] is the one following [8, (2.10), p. 267], where Bałaban states for its correction that its defining equation has exactly one solution, analytic in its argument, with an expansion beginning with quadratic terms; here $\widetilde{D}$ is the small solution φ of (i)); the phase $A - h_{U_1}\widetilde{D}(A)$; the Haar density $\sigma(A - h_{U_1}\widetilde{D}(A))$; the logarithm $\operatorname{Tr}\log(\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D})$ of the Jacobian of (1.14); and the bondwise maps of (1.22) with their Jacobian.

Three further properties of these objects are recorded as part of (B4); they are Bałaban's statements and facts proved in Lemma B.11.

(i) The defining equation of $\widetilde{D}$ is the system (3.24) over the rows $c \in \Omega_1^{[1]}$, which decouples row by row (Lemma 3.9), as in [7, (17), pp. 259–260], the transformation to which Bałaban refers in [9, p. 248], where the equation is solved "separately for each function $Q(A', c)$"; the row component $\widetilde{D}_c(A)$ depends on $A|_{\mathcal{E}_c}$ only.

[8, p. 267] states for the equation $L\widetilde{Q}B' + \widetilde{C}(B') = L\widetilde{Q}B$, $B' = B - h\widetilde{D}(B)$, that "there exists exactly one solution of this equation, and that it is an analytic function of B", in the unit-lattice step of [8], without naming the class in which the solution is unique, and [7, p. 260] states for (17) that "there exists a unique solution $\tilde{D}$ of this equation for A sufficiently small, and it is an analytic function of A with a Taylor expansion beginning with second order terms"; these statements are used only for the identification at the end of (i).

The row equation (3.24) involves the eliminated phase only through e^{iq}, so its solutions are determined only up to the periods of the exponential; the class used here is the chart class of Section 3.1 (the whole phase in the chart ball $h(q) < \varrho_G$, the class of the chart-valued variable A' of [9, (1.13)]) with the fine field in $D^{(11)}(V, V_0)$. In this class the row solution is determined by Lemma B.11 of Appendix B, proved from (B1) (with (B3)(i), through (B1)(i)), (B2), the compatible partitions of [9] being nested ((H4)), the fine periods being at least $8M$, Lemmas B.5, B.8 and 3.6, and threshold items 5 and 6: at every row $c \in \Omega_1^{[1]}$, near or far, Lemma B.8 applies to U_1 and gives the small solution φ_c, analytic on the ball $\|A|_{\mathcal{E}_c}\| < \vartheta_*$, real on real phases, with $\varphi_c(0) = 0$ and $D\varphi_c(0) = 0$, unique among the solutions d_c with $h(d_c) \leqslant \rho_{\mathrm{d}}$ (Lemma B.11(i)); the argument A of every $U \in D^{(11)}(V, V_0)$ lies in the ball $\|A|_{\mathcal{E}_c}\| < \vartheta_*/2$ at every row (Lemma B.11(ii)); and every chart-class family d at $g_0\mathsf{C}_{U_1}B$ with $U_d(B) \in D^{(11)}(V, V_0)$ is $\varphi(g_0\mathsf{C}_{U_1}B)$, $\varphi = (\varphi_c)_c$ (Lemma B.11(iii)).

The row solution of this paper is $\widetilde{D}_c = \varphi_c$ at every row, and $\widetilde{D} = \varphi$ is analytic and solves the row equations on the open set $\mathcal{N}_\Theta$ of Lemma B.11(iv), which contains all arguments of the domain of integration. Every solution of the row system with chart-valued phase and fine field in $D^{(11)}(V, V_0)$ is φ, by Lemma B.11(iii); and the $\widetilde{D}$ in [9, (1.13)–(1.14), pp. 248–249] is φ: for A sufficiently small both are small solutions of the row equations ($\varphi(0) = 0$), so they agree by the uniqueness stated in [7], and the $\widetilde{D}$ of [9], the analytic function so described by [7] and [8], therefore agrees with φ on the connected ball $\|A|_{\mathcal{E}_c}\| < \vartheta_*$ on which φ is analytic, a ball containing the arguments of the domain of integration (Lemma B.11(ii)).

(ii) The substitutions (1.14) and (1.22) are changes of variables: the map $B \mapsto U(B)$ of Section 3.1, the composition of (1.13) and (1.14), is a bijection of $\mathcal{A}_\xi$ onto $D(V, V_0)$ (shown there from (B1), (B2) and Lemma B.11), and its differential, that of $B \mapsto g_0\mathsf{C}_{U_1}B$ followed by the substitution $A' = A - h_{U_1}\widetilde{D}(A)$ and the exponential, is injective at every point of $\mathcal{A}_\xi$ with image the tangent space of the fibre $\Phi^{-1}(V, V_0)$ at $U(B)$ (the Jacobian $\det(\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D})$ of (1.14) is nonzero there, (iii), C_{U_1} is injective, and the image of $(\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D})\mathsf{C}_{U_1}$ is the kernel of $D\mathcal{Q}_{U_1}(A')$ by the identity $D\mathcal{Q}_{U_1}(A') \circ (\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D}) = LQ_{U_1}$ of the measurability paragraph, $LQ_{U_1}\mathsf{C}_{U_1} = 0$, and a count of dimensions, $D\mathcal{Q}_{U_1}(A')$ being onto); and Ψ_ξ of (3.38), acting bond by bond on S_1^*, is a C^∞ map which maps $\mathcal{D}_\xi$ bijectively onto $\mathcal{A}_\xi$ with invertible differential at every point of $\mathcal{D}_\xi$: this is Lemma 3.16 of Section 3.4, proved from the bound on $\mathbf{H}_{1,Ax}$ of [9, p. 251], derived in Lemma D.21 from [6, Theorem 1, (172), Proposition 9, and (190)], the application of [6] to the determining set $\mathbf{B}_1(\Lambda_1^c)$ in [9, p. 256], and the restoring

bound (D.49), from the membership $U_1 \in D^{(11)}(V, V_0)$ of (B1), that is the cutoff $\chi'_0(\Omega_1)$ at U_1 ([9, p. 248]), and from the choice of the cutoff g of (1.22) made there. The Jacobian of (1.22) is $|\det D\Psi_\xi|$, the exact Jacobian of (3.38), in place of the expression [9, (1.24)].

(iii) The determinant $\det(\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D})$ is nonzero on $\mathcal{A}_\xi$, by (i) and Sylvester's identity (the measurability paragraph of Section 3.1); [9] performs (1.14) as a change of variables and writes its Jacobian as $\exp \mathrm{Tr} \log(I - h(\frac{\delta}{\delta A}\widetilde{D})(A))$ (p.249). The determinant is positive there, so that the logarithm of [9] is the real logarithm and equals $\sum_c \log \det_{\mathfrak{g}}(\mathbf{1} - h_{U_1}(c)\partial_{A(b_0(c))}\widetilde{D}_c)$ (Lemma 3.14): this is Lemma B.11(iv), (B.17), at every row; the identities of this paper use the absolute Jacobian $|\det(\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D})|$, for which the nonvanishing suffices.

For the rows within $2R_{\mathfrak{c}} + 2L$ of I and the inputs of Lemma 3.12, the expressions of (B4) coincide with the analytic objects of that lemma, by the uniqueness of the small solution stated there; elsewhere they enter only through a-independent factors (Lemma 3.17, (3.6)).

The Borel choice fixed in (B1) exists:

Definition 3.1 (Constrained Minimisation Data)**.** Let E be a finite set of fine bonds, $\mathbf{B}$ a set of blocks, and $\mathcal{U} \subseteq G^E$ a finite intersection of open and closed subsets of the compact metric space G^E, contained in the open set on which the group means $U(y, x)$ of [8, (0.11)] at the block centres y of $\mathbf{B}$ are defined (chart convention of the preamble). Let P be a Polish space of parameters, $\mathcal{W}$ a relatively open subset of $\mathcal{U} \times P$ (the set on which the averages entering the conditions are defined; $\mathcal{W} = \mathcal{U} \times P$ is allowed), $\mathfrak{c}$ and $\mathfrak{d}$ continuous maps on $\mathcal{W}$ into metric spaces, and S a continuous real function on $\mathcal{W}$ (the action, which may involve the parameter through boundary plaquettes). For $p \in P$,

$$\mathcal{W}(p) \coloneqq \{U : (U, p) \in \mathcal{W}\}\,, \qquad \mathcal{K}(p) \coloneqq \{U \in \mathcal{W}(p) : \mathfrak{c}(U, p) = 0\}\,,$$

$\mathcal{W}(p)$ being an open subset of $\mathcal{U}$; $\mathcal{V}$ is the set of p with $\mathcal{K}(p) \neq \varnothing$ on which $S(\bullet, p)$ attains its infimum $m(p)$ over $\mathcal{K}(p)$. The set of minimal configurations satisfying the further equalities $\mathfrak{d} = 0$, in the axial gauge (p.248), is

$$\begin{aligned}\mathcal{M}_{\mathrm{ax}} \coloneqq \{(p, U) : p \in \mathcal{V},\ U \in \mathcal{K}(p),\ S(U, p) = m(p),\ \mathfrak{d}(U, p) = 0,\\ U(y, x) = \mathbf{1} \text{ for every centre } y \text{ of a block of } \mathbf{B} \text{ whose bonds lie in } E\\ \text{and } x \in \mathcal{B}_y \setminus \{y\}\}\end{aligned} \tag{3.8}$$

with projection $\mathcal{V}_{\mathrm{ax}}$ to the first factor.

Lemma 3.2 (Borel Choice of Minimal Configurations)**.** *For the data of Definition 3.1, $\mathcal{V}$, m, $\mathcal{M}_{\mathrm{ax}}$, and $\mathcal{V}_{\mathrm{ax}}$ are Borel, and there is a Borel map $\mathcal{V}_{\mathrm{ax}} \ni p \mapsto U_{\mathbf{B}}(p)$ with $(p, U_{\mathbf{B}}(p)) \in \mathcal{M}_{\mathrm{ax}}$ for every $p \in \mathcal{V}_{\mathrm{ax}}$.*

Definition 3.3 (Regular Configurations)**.** For the determining sets $\mathbf{B}$ of [9], that of (1.2) and that of (1.21), $E = E(\mathbf{B})$ is the set of fine bonds of the region determined and $\mathcal{U} = \mathcal{U}_{\mathbf{B}}$ the set of regular configurations, on which the averaging operation $M_{\mathbf{B}}$ of [9, (2.11)] is defined: the configurations of the class (2) of [6, p. 278] (reference [15] of [9]), given by finitely many strict inequalities between continuous functions of U (the plaquette variables $|U(\partial p) - 1|$ and the covariant divergences $|(D_U^{\eta *}\partial U)(b)|$, polynomials in the matrix entries of the bond variables and their inverses), with the averages (the logarithmic averages of [8, (0.4)]) and the means $U(y, x)$ defined in the chart convention, an open set; the space [6, (6)] is its intersection with the constraint set (3), which is the constraint $\mathfrak{c} = 0$ of Definition 3.4; $M_{\mathbf{B}}$ is continuous on $\mathcal{U}$.

Definition 3.4 (The Two Minimisation Problems of the Step)**.** With E and $\mathcal{U} = \mathcal{U}_{\mathbf{B}}$ of Definition 3.3, for the problem (1.2) the parameter is $p = V|_{\square'^{\sim 4}}$ (p.246), $\mathcal{W} = \mathcal{U} \times P$, $\mathfrak{d} \equiv 0$, and

$$\mathfrak{c}(U, V) := M_{\mathbf{B}}(U)\,(M^{\bullet}(Q_1^{s*}V))^{-1} - \mathbf{1}\,,$$

so that $\mathcal{K}(V) = \{U \in \mathcal{U}_{\mathbf{B}} : M_{\mathbf{B}}(U) = M^{\bullet}(Q_1^{s*}V)\}$ is the constraint set of the variational problem [9, (2.12)] for the data of (1.2).

For the configuration of (1.21), $p = (V, V_0)$, E the bonds of $\Omega_1 \cap \Lambda_1^c$ (the "function $U_1(\Omega_1 \cap \Lambda_1^c, V)$" of [9, p. 251]), $\mathcal{W}$ the relatively open set of the (U, p) at which the averages of the rows crossing to Ω_1^c, which involve V_0 on the outer block, are defined, $\mathfrak{c}$ collects the constraints $\bar{U} = V$ on the rows of $(\Lambda_1^{\sim})^c$ (with V_0 on the blocks off Ω_1) and $M_{\mathbf{B}_1(\Lambda_1^c)}(U) = M^{\bullet}(Q_1^{s*}V)$ on the bonds of the determining set in $\Lambda_1^{\sim} \setminus \Lambda_1$ (its fine layer fixed to $Q_1^{s*}V$ and its coarse bonds to the averages of $Q_1^{s*}V$; the form of the constraint of (1.2); [9] names the determining data, "the set $\mathbf{B}_1(\Lambda_1^c)$ and the configuration $Q_1^{s*}V$", p.251), $\mathfrak{d} \equiv 0$, and $S(U, p)$ is the functional A^{η} of (2.12) for the determining set $\mathbf{B}_1(\Lambda_1^c)$, in which the plaquettes meeting Ω_1^c involve V_0, as recorded in (B1) from pp.251 and 269.

For (1.21), the minimality $S(U) = m(p)$ over $\mathcal{K}(p)$ is the condition of (1.12) on $(\Lambda_1^{\sim})^c$, the critical point of (1.12) being its minimum (p.248), and no critical-point equation is imposed separately.

Lemma 3.5 (The Minimal Configurations of the Step)**.** *For the problem (1.2) of Definition 3.4, every regular V lies in $\mathcal{V}_{\mathrm{ax}}$, and $U_{1,\square'}(V) = U(\mathbf{B}_1(\square'^{\sim 4}), M^{\bullet}(Q_1^{s*}V))$ is the representative of the unique minimal orbit of (2.12) chosen in [9, p. 247], "taken in the axial gauge". For the problem (1.21), the configuration exists for every admissible pair (Ω_1, V, V_0) (Section 3.1).*

Proof of Lemma 3.2. The set $\mathcal{K} := \{(p, U) : (U, p) \in \mathcal{W},\ \mathfrak{c}(U, p) = 0\}$ is Borel in $P \times G^E$, as the zero set of a continuous function on the Borel set $\mathcal{W}$ (relatively open in the Borel set $\mathcal{U} \times P$), and so is $\mathcal{K}_t = \{(p, U) \in \mathcal{K} : S(U, p) < t\}$ for every rational t. Every section $\mathcal{K}_t(p) = \{U \in \mathcal{W}(p) : \mathfrak{c}(U, p) = 0,\ S(U, p) < t\}$ is the intersection of $\mathcal{U}$, a finite intersection of open and closed subsets of the compact metric space G^E, with the open set $\mathcal{W}(p)$, a closed set (the zero set of $\mathfrak{c}(\bullet, p)$ is relatively closed in $\mathcal{W}(p)$, the trace of a closed set) and an open set (the trace of $\{S(\bullet, p) < t\}$, relatively open in $\mathcal{W}(p)$), hence K_σ: an open subset of a compact metric space is a countable union of compact sets, and the intersection of a K_σ set with a closed set, or with an open set, is K_σ. By the Arsenin–Kunugui theorem (a Borel subset of a product of Polish spaces whose sections are K_σ has a Borel projection and a Borel uniformisation, [13, Theorem 35.46(ii)]), $\operatorname{proj} \mathcal{K}_t = \{p : \inf_{\mathcal{K}(p)} S < t\}$ is Borel for every rational t, and so is $\operatorname{proj} \mathcal{K} = \{p : \mathcal{K}(p) \neq \varnothing\}$; hence $\widetilde{m}(p) = \inf_{\mathcal{K}(p)} S$ is a Borel function on $\operatorname{proj} \mathcal{K}$.

The set $\mathcal{M} := \{(p, U) \in \mathcal{K} : S(U, p) = \widetilde{m}(p)\}$ is Borel, its sections $\mathcal{K}(p) \cap \{S = \widetilde{m}(p)\}$ are K_σ, so $\mathcal{V} = \operatorname{proj} \mathcal{M}$ is Borel by the same theorem, and $m = \widetilde{m}$ on $\mathcal{V}$.

The axial gauge conditions $U(y, x) = \mathbf{1}$ are relatively closed conditions on the open set on which the means $U(y, x)$ of [8, (0.11)] are defined ([8, p. 253] defines the mean for sets of sufficiently small diameter, and it is continuous there in the chart convention), an open set containing $\mathcal{U}$ by hypothesis; and $\mathfrak{d}(U, p) = 0$ is a relatively closed condition on $\mathcal{W}$. So $\mathcal{M}_{\mathrm{ax}} = \mathcal{M} \cap \{\mathfrak{d} = 0\} \cap \{U \text{ axial}\}$ is Borel, and its sections are K_σ, being intersections of K_σ subsets of $\mathcal{U}$ with closed sets (the trace on $\mathcal{U}$ of a relatively closed subset of an open set containing $\mathcal{U}$ is the trace of a closed set); the theorem gives that $\mathcal{V}_{\mathrm{ax}}$ is Borel and that $\mathcal{M}_{\mathrm{ax}}$ has a Borel uniformisation, a Borel map $U_{\mathbf{B}}$ on $\mathcal{V}_{\mathrm{ax}}$ with graph in $\mathcal{M}_{\mathrm{ax}}$. *q.e.d.*

Proof of Lemma 3.5. For (1.2), a regular V determines the unique minimal orbit of (2.12), whose elements are the minimal configurations [9, p. 256], so $V \in \mathcal{V}$, and the configurations $U_{1,\square'}$ are taken in the axial gauge (p.247); a gauge transformation equal to $\mathbf{1}$ at the block centres and to the mean $U(y,x)$ elsewhere brings a minimal configuration to the axial gauge, leaving the block averages of [8, (0.4)] at the block centres unchanged ([8, (0.4)] transforms at the block centres, proof of Lemma 3.6) and $S(U)$ unchanged when S is gauge invariant, as the action is; the configuration so obtained is regular, satisfies the constraint and is minimal: [6] states that its class (2) is gauge invariant and its constraint set (3) invariant under the gauge transformations equal to one on its set $\mathfrak{B}_k$ of determining points ((4), p.278), which the transformation used is: it is the identity at the block centres, and on the blocks of the fine determining layer, where the constraint fixes the configuration to the straight extension $Q_1^{s*}V$, whose intrablock contours are the identity (Section D.8.2), the mean $U(y,x)$ is the identity; and that the space (6) and the action (5) are invariant under these (p.278); so $\mathcal{M}_{\mathrm{ax}}(V) \neq \varnothing$.

For (1.21), the existence for every admissible pair is the existence of the minimal orbit of [6, Theorem 1] for these determining data, Bałaban's application of [6] to the determining sets of [9, pp. 251 and 256] (Section D.8.1); the configuration of (1.21) is that of [9, p. 251]. *q.e.d.*

3.2 The Constrained Hessian at a Regular Background and Its Stencils

Let M be Bałaban's averaging operation (4.1) (written M there; here M is the cube side), and let Ω be a region, a union of blocks, and U a configuration of the fine torus for which $\mathsf{M}(U)(c)$ is defined on every row $c \in \Omega^{[1]}$ of that region and the local group means $\mathcal{U}_U(y,x)$, $x \in \mathcal{B}_y$, are defined for every centre $y \in \Omega^{(1)}$ (chart convention of the preamble; for $U = U_1$ on Ω_1 both hold, because $U_1 \in D^{(11)}$ and U_1 is in its axial mean gauge, $\mathcal{U}_{U_1}(y,x) = \mathbf{1}$); the objects below are then defined for phases q in a neighbourhood of zero depending on U, which Lemma 3.6 and Appendix B quantify for the backgrounds that occur.

The objects of this subsection are defined relative to the region Ω: the rows are the coarse bonds $c \in \Omega^{[1]}$ (at least one block in Ω), the block sums run over the centres $y \in \Omega^{(1)}$, the free carrier is Ω^*, and the links of U on bonds with both endpoints off Ω enter as fixed constants, with the phase variable equal to zero there. The torus itself ($\Omega = T$) is written without a superscript; the branch uses $\Omega = \Omega_1$ and $U = U_1$. For $c \in \Omega^{[1]}$ the row maps are

$$\mathcal{Q}_{U,c}(q) \coloneqq \frac{1}{i} \log \left(\mathsf{M}(e^{iq}U)(c)\, \mathsf{M}(U)(c)^{-1} \right), \quad Q_U \coloneqq \frac{1}{L} D\mathcal{Q}_U(0), \quad \mathcal{N}_U(q) \coloneqq \mathcal{Q}_U(q) - LQ_U q \,, \tag{3.9}$$

so that $\mathcal{N}_U$ is the remainder denoted $\widetilde{C}$ in the linearisation following [8, (2.10)]; the logarithm is the group chart of the conventions above, defined when the relative average lies in the chart domain. For a coarse bond c with designated fine bond $b_0(c)$, and a coarse centre y, the bond stencils are

$$\mathcal{E}_c \coloneqq \{r \,|\, \{r_-, r_+\} \subset \mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}\}, \qquad \mathcal{E}_y \coloneqq \{r \,|\, \{r_-, r_+\} \subset \mathcal{B}_y\} \,, \tag{3.10}$$

r ranging over the positive fine bonds.

Lemma 3.6 (Two-Block Structure)**.** *Let $L \geqslant 3$ be odd and let every coarse period exceed two. The row components $\mathcal{Q}_{U,c}$ and $\mathcal{N}_{U,c}$ depend only on the phases and background bonds in $\mathcal{E}_c$, and $b_0(c)$ is the only designated bond in $\mathcal{E}_c$. Hence*

$$Q_{0,U}(c,c') \coloneqq \delta_{cc'} \alpha_c(U), \qquad \alpha_c(U) \coloneqq Q_U(c, b_0(c)) \,, \tag{3.11}$$

where $Q_{0,U}$ is the restriction of Q_U to the designated bonds, with $\alpha_c(\mathbf{1}) = L^{-4}\mathbf{1}$. There is a phase radius $\tau_L > 0$, depending on L and the group only, such that for $U = e^{iA}$ with

$\|A\|_\infty \leqslant \tau_L$ *every* $\alpha_c(U)$ *is analytic in* A *with smallest singular value at least* $L^{-4}/2$. *Under a site gauge transformation* u, $\alpha_c(U^u) = \mathrm{Ad}_{u(c_-)}\,\alpha_c(U)\,\mathrm{Ad}^{-1}_{u(b_0(c)_-)}$.

Proof. Every path in the row of [8, (0.4)] for c lies in the two endpoint blocks, since the row averages parallel transports from the centre of $\mathcal{B}_{c_-}$ to the centre of $\mathcal{B}_{c_+}$ along paths inside the two blocks; a designated bond crosses its own coarse interface, and two adjacent blocks share exactly the interface labelled by c, so $b_0(c)$ is the only designated bond with both endpoints in $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$ (for $L \geqslant 3$ odd the other designated bonds cross other interfaces).

At the flat field, differentiating [8, (0.4)] gives (A.2) of Appendix A: the straight-path term $(Q_1 a)(c) = L^{-5}\sum_{x\in\mathcal{B}_{c_-}} a([x,x'])$ contains $b_0(c)$ in exactly L of its L^4 translated straight paths of L bonds each (the L starting points on the line through $b_0(c)$), each with coefficient L^{-5}, and the mean-path term involves the paths Φ inside the blocks, which contain no designated bond; this gives $\alpha_c(\mathbf{1}) = L^{-4}\mathbf{1}$, the designated coefficient $Q_{0,\infty} = L^{-4}I$ computed in Appendix A (paragraph "Construction on the infinite lattice", after (A.26); the computation is local and holds on every torus).

For $U = e^{iA}$, $\alpha_c(U)$ is an analytic function of $A|_{\mathcal{E}_c}$ near zero (a composition of exponentials, the averaging and the chart logarithm) with $\alpha_c(\mathbf{1}) = L^{-4}\mathbf{1}$; the smallest singular value is continuous, so for each of the finitely many row types modulo $L\mathbb{Z}^4$ (translations commute with M) there is a radius on which α_c is analytic with smallest singular value at least $L^{-4}/2$, and τ_L is the smallest of these radii; it depends on L and the group only.

The gauge covariance is $\mathsf{M}(U^u)(c) = u(c_-)\mathsf{M}(U)(c)u(c_+)^{-1}$ ([8, (0.4)] transforms at the block centres) together with $(R_u q)(b) \coloneqq \mathrm{Ad}_{u(b_-)}\,q(b)$ for the phase ([9, (1.17)]): $\mathcal{Q}_{U^u,c}(R_u q) = \mathrm{Ad}_{u(c_-)}\,\mathcal{Q}_{U,c}(q)$, hence $Q_{U^u}R_u = \mathrm{Ad}_{u(c_-)}\,Q_U$ row by row, and the $b_0(c)$ column gives the formula. *q.e.d.*

By (B2) and Lemma 3.6, $\alpha_c(U)$ is invertible on every row of the regions used below. With $LQ_U h_U = \mathbf{1}$ for the supported right inverse h_U, whose only nonzero outputs are at the designated bonds, the elimination maps are

$$\begin{aligned}(\mathsf{C}_U B)(b_0(c)) &\coloneqq -\alpha_c(U)^{-1}\sum_{b\in\mathcal{E}_c\cap E_*} Q_U(c,b)B(b),\\ (h_U v)(b_0(c)) &\coloneqq h_U(c)v(c), \qquad h_U(c) \coloneqq \frac{1}{L}\alpha_c(U)^{-1}\,,\end{aligned} \tag{3.12}$$

and $(\mathsf{C}_U B)(r) = B(r)$ on free bonds $r \in E_*$; thus $Q_U\mathsf{C}_U = 0$. The free-input and background stencils of a fine bond are

$$(\mathcal{F}_r, \mathcal{A}_r) \coloneqq \begin{cases}(\{r\}, \varnothing), & r \in E_*\,,\\ (\mathcal{E}_c \cap E_*, \mathcal{E}_c), & r = b_0(c)\,.\end{cases} \tag{3.13}$$

The constrained quadratic form is [9, (1.14)–(1.16)] on the region Ω. With J_U and Δ_U the first derivative and Hessian at $q = 0$ of $q \mapsto S(e^{iq}U)$, $q \in \mathfrak{g}^\Omega$ (the plaquettes with a bond in Ω contribute; links off Ω are the fixed constants), and the relative gauge functional

$$\mathscr{G}^\Omega_U(q) \coloneqq \sum_{y\in\Omega}\sum_{x\in\mathcal{B}_y\setminus\{y\}}\big(1 - \mathrm{Re}\,\mathrm{tr}_N(\mathcal{U}_{e^{iq}U}(y,x)\mathcal{U}_U(y,x)^{-1})\big)\,, \qquad G^{(2),\Omega}_U(a) \coloneqq D^2\mathscr{G}^\Omega_U(0)[a,a]\,, \tag{3.14}$$

where $\mathcal{U}_U(y,x)$ is the axial transport from the block centre y to x of the gauge-fixing functional $\mathbf{G}$ in [9, (1.11)] and the sum is over the block centres of Ω, the regional Hessian H^Ω_U on $\mathfrak{g}^{\Omega^*}$

is the symmetric operator of the quadratic form (B is extended by zero to the free bonds outside Ω^*, so that $a = \mathsf{C}_U B$ is defined on every row of $\Omega^{[1]}$)

$$\langle B, \mathsf{H}_U^\Omega B\rangle = \langle a, \Delta_U a\rangle + G_U^{(2),\Omega}(a) - \langle J_U, h_U D^2\mathcal{N}_U(0)[a,a]\rangle, \qquad a = \mathsf{C}_U B\ , \tag{3.15}$$

with the rows of $\mathcal{N}_U$ and h_U those of $\Omega^{[1]}$. The right side equals $D^2(S+\mathscr{G}_U^\Omega)(0)[a,a] - DS(0)[h_U D^2\mathcal{N}_U(0)[a,a]]$, which does not involve the inner product; H_U^Ω is its operator for the fixed one, and its entries $\mathsf{H}_U^\Omega(b,b') \in \mathrm{End}(\mathfrak{g})$ are those of the polarised bilinear form.

This is [9, (1.16)], $\langle A, \Delta_1 A\rangle = \langle A, \Delta A\rangle - 2\langle hC^{(2)}(A), J_1\rangle + \mathbf{G}^{(2)}(A)$, with, as in [9], $C^{(2)}(A) = \frac{1}{2}D^2\mathcal{N}_U(0)[A,A]$ (the quadratic term of the linearising transformation, [8, p. 267], $\widetilde{D}^{(2)} = \widetilde{C}^{(2)}$) and $\mathbf{G}^{(2)}(A) = D^2\mathscr{G}(0)[A,A]$; at $\Omega = \Omega_1$, $U = U_1$ the relative gauge functional is the absolute one of [9], $\mathscr{G}_{U_1}^{\Omega_1}(q) = \mathbf{G}(\Omega_1^{(1)}, e^{iq}U_1)$ with $\mathbf{G}$ of (1.7), (1.11), because U_1 is in its axial mean gauge, $\mathcal{U}_{U_1}(y,x) = \mathbf{1}$ for $x \in \mathcal{B}_y$, $y \in \Omega_1^{(1)}$ (Section 3.1; [9, p. 248]), so that $\mathbf{G}(\Omega_1^{(1)}, U_1) = 0$; and $\mathsf{H}_{U_1}^{\Omega_1}$ is the $C^*\Delta_1 C$ of [9, (1.15)], the same quadratic form written as an operator for the fixed inner product. The precision of the branch is

$$H_\xi \coloneqq \mathsf{H}_{U_1}^{\Omega_1} \quad \text{on } \mathfrak{g}^{\Omega_1^*}, \qquad H_{\xi,II} \coloneqq \iota_I^* H_\xi \iota_I, \qquad H_{\xi,IJ} \coloneqq \iota_I^* H_\xi \iota_J\ , \tag{3.16}$$

with ι_I, ι_J the extensions by zero. With $s_{p,U}(q) \coloneqq 1 - \mathrm{Re}\,\mathrm{tr}_N((e^{iq}U)(\partial p))$, $\mathscr{G}_{y,U}$ the y-summand of (3.14), and $a = \mathsf{C}_U B$, the form decomposes into local terms,

$$\begin{aligned}
\langle B, \mathsf{H}_U^\Omega B\rangle &= \sum_{p\cap\Omega\neq\varnothing} \mathfrak{a}_p(B;U) + \sum_{y\in\Omega} \mathfrak{g}_y(B;U) - \sum_{c\in\Omega^{[1]}} \mathfrak{t}_c(B;U),\\
\mathfrak{a}_p &\coloneqq D^2 s_{p,U}(0)[a,a],\\
\mathfrak{g}_y &\coloneqq D^2\mathscr{G}_{y,U}(0)[a,a],\\
\mathfrak{t}_c &\coloneqq \tfrac{1}{L}\langle J_U(b_0(c)), \alpha_c(U)^{-1} D^2\mathcal{N}_{U,c}(0)[a,a]\rangle\ .
\end{aligned} \tag{3.17}$$

Their free-input and background sets are read off from Lemma 3.6 and (3.12): $a(r)$ depends on $B|_{\mathcal{F}_r}$ and on the background on $\mathcal{A}_r$, so $\mathfrak{a}_p$ has free-input set $\bigcup_{r\subset\partial p}\mathcal{F}_r$ and background set $\bigcup_{r\subset\partial p}(\{r\}\cup\mathcal{A}_r)$; $\mathfrak{g}_y$ involves the phases on the axial paths inside $\mathcal{B}_y$, which contain no designated bond (designated bonds cross interfaces), so its free-input and background sets are $\mathcal{E}_y \cap E_*$ and $\mathcal{E}_y$; $\mathfrak{t}_c$ involves $\mathcal{N}_{U,c}$, α_c and $J_U(b_0(c))$, the latter depending on the plaquettes through $b_0(c)$, which lie in $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$ for $L \geqslant 3$, so its sets are $\mathcal{E}_c \cap E_*$ and $\mathcal{E}_c$. The entries of H_U^Ω are the sums of the polarised entries of these terms.

For a pair of positive bonds (b,b') let $\mathcal{T}_{bb'}$ be the set of terms of (3.17) (plaquettes, centres, rows of the torus) whose free-input set contains both b and b', and let the *entry stencil* $\mathcal{S}_{bb'}$ be the union of the background sets of these terms (one of finitely many types modulo $L\mathbb{Z}^4$; it is empty unless $d(b,b') \leqslant 4L$, Lemma 3.7). The entry function is

$$f_{bb'}(U) = \sum_{\mathfrak{T}\in\mathcal{T}_{bb'}} (\text{polarised entry of } \mathfrak{T} \text{ at } (b,b')) \in \mathrm{End}(\mathfrak{g}), \qquad f_{bb'}(A) \coloneqq f_{bb'}(e^{iA})\ , \tag{3.18}$$

defined for every configuration U on $\mathcal{S}_{bb'}$ for which $\mathsf{M}(U)$ is defined and $\alpha_c(U)$ is invertible on the rows of the terms in $\mathcal{T}_{bb'}$ and the local group means $\mathcal{U}_U(y,x)$ are defined for the centres y of the terms in $\mathcal{T}_{bb'}$ (the terms involve C_U on those rows and the means of those centres), whatever the configuration elsewhere.

The regional form is the compression of the torus form: for a background U on which H_U^Ω is defined (Section 3.2: $\mathsf{M}(U)$ defined and $\alpha_c(U)$ invertible on the rows of $\Omega^{[1]}$, the means $\mathcal{U}_U(y,x)$ defined for $y \in \Omega^{(1)}$) and every pair $(b,b') \in \Omega^* \times \Omega^*$,

$$\mathsf{H}_U^\Omega(b,b') = f_{bb'}(U|_{\mathcal{S}_{bb'}})\ , \tag{3.19}$$

because a term whose free-input set contains the bond b of Ω^* has its index in Ω (a plaquette containing b, or containing a bond r with $b \in \mathcal{F}_r$, meets Ω; a centre y with $b \in \mathcal{E}_y$ lies in Ω; a row c with $b \in \mathcal{E}_c \cap E_*$ has an endpoint block containing an endpoint of b, so $c \in \Omega^{[1]}$), hence every term of $\mathcal{T}_{bb'}$ is a term of the regional sum, and the regional form is defined under the weaker requirement on U stated above (Lemma B.6).

Lemma 3.7 (Gauge Law and Locality of the Hessian)**.** *Let u be a site gauge transformation of the torus, acting on the background by $U \mapsto U^u$ and on phases by $(R_u q)(b) = \mathrm{Ad}_{u(b_-)}\, q(b)$ ([9, (1.17)]), and let $R_{u,*}$ be the restriction of R_u to the free carrier Ω^*; it is orthogonal for the invariant inner product. Then, for every region Ω,*

$$\mathsf{H}^{\Omega}_{U^u} = R_{u,*}\mathsf{H}^{\Omega}_{U} R_{u,*}^{-1}\,. \tag{3.20}$$

Every entry $\mathsf{H}^{\Omega}_{U}(b,b')$ is a finite composition of local analytic operations in the background bonds of the entry stencil $\mathcal{S}_{bb'}$, which lies within the range $\rho_L = 4L$ of b, including the fibre inverses α_c^{-1}, and it vanishes for $d(b,b') > \rho_L$ (the stencils are listed in Appendix B, Lemma B.3). The entry function $f_{bb'}$ of (3.18) *is entrywise gauge covariant: for a site gauge transformation u,*

$$f_{bb'}(U^u) = \mathrm{Ad}_{u(b_-)}\, f_{bb'}(U)\, \mathrm{Ad}^{-1}_{u(b'_-)}\,. \tag{3.21}$$

The norms are the operator norm for the Hilbert norms induced by h and, for $Df_{bb'}(A)$, the norm $\sup_{\|X\|_\infty \leqslant 1} \|Df_{bb'}(A)[X]\|$. Let τ_L denote the radius of Lemma 3.6, reduced so that these operations are analytic on $\|A\|_\infty \leqslant \tau_L$ and reduced once more in (B.6) *of Appendix B; every statement is made for this final value. The entry constants are*

$$m_L \coloneqq \max_b \sum_{b'} \sup_{\|A\|_\infty \leqslant \tau_L} \|f_{bb'}(A)\|, \qquad c_L \coloneqq \max_b \sum_{b'} \sup_{\|A\|_\infty \leqslant \tau_L} \|Df_{bb'}(A)\|\,. \tag{3.22}$$

These depend on L, the group and the invariant inner product only.

Proof. The intertwinings are $\mathsf{M}(U^u)(c) = u(c_-)\mathsf{M}(U)(c)u(c_+)^{-1}$, hence $\mathcal{Q}_{U^u} \circ R_u = R_u^{(1)}\mathcal{Q}_U$ with $(R_u^{(1)}v)(c) \coloneqq \mathrm{Ad}_{u(c_-)}\, v(c)$ (proof of Lemma 3.6), and therefore $Q_{U^u}R_u = R_u^{(1)}Q_U$, $\mathcal{N}_{U^u} \circ R_u = R_u^{(1)}\mathcal{N}_U$, $h_{U^u} = R_u h_U (R_u^{(1)})^{-1}$ (from the covariance of α_c in Lemma 3.6) and $\mathsf{C}_{U^u} = R_u \mathsf{C}_U R_{u,*}^{-1}$. The three terms of (3.15) are then each preserved: $S((e^{iR_uq})U^u) = S((e^{iq}U)^u) = S(e^{iq}U)$ gives $\Delta_{U^u} = R_u\Delta_U R_u^{-1}$ and $J_{U^u} = R_u J_U$; each relative mean $\mathcal{U}_{e^{iq}U}(y,x)\mathcal{U}_U(y,x)^{-1}$ is conjugated by $u(y)$, which preserves the trace; and $\langle J_{U^u}, h_{U^u}D^2\mathcal{N}_{U^u}(0)[R_ua, R_ua]\rangle = \langle R_uJ_U, R_uh_UD^2\mathcal{N}_U(0)[a,a]\rangle$ equals the untransformed force term because R_u is orthogonal for the invariant inner product. The same three identities hold term by term in (3.17) (each term is a function of U and $a = \mathsf{C}_U B$ on its own stencil, and $\mathsf{C}_{U^u}R_{u,*}B = R_u\mathsf{C}_U B$), so the polarised entry of each term transforms as $\mathrm{Ad}_{u(b_-)}(\bullet)\,\mathrm{Ad}^{-1}_{u(b'_-)}$, which is (3.21).

Locality: the index of a term is the lower vertex $x(p)$ of its plaquette, the centre y or the initial vertex of $b_0(c)$, and the distance of a bond to the index is that of its initial vertex. By the two-block structure of Lemma 3.6 substituted into (3.17), the free-input and background bonds of a block or row term have initial vertices in the one or two blocks of the index, within ℓ^∞ distance $2L-1$ of it (the sites of two adjacent blocks are within $2L-1$ of each other); a bond r of ∂p has initial vertex within 1 of $x(p)$, and the bonds of $\mathcal{F}_r \cup \mathcal{A}_r$ are within $2L-1$ of r, so the bonds of a plaquette term are within $2L$ of its index. Hence a term whose free-input set contains b and b' has $d(b,b') \leqslant 4L = \rho_L$, and its background bonds are within $4L$ of b. The stencils are listed in Lemma B.3. *q.e.d.*

Lemma 3.8 (Regular-Background Bounds)**.** *Let $\gamma_L \coloneqq C_L^{-1}$ be the flat lower bound* (A.25) *of Appendix A, τ_L the final radius of Lemma 3.7, and let R_{c} (the collar range, Lemma 3.19)*

and C_{L,R_c} be the constants fixed in Appendix B, Section B.2; they depend on L and the group only. Whenever (3.1) *holds and $\widehat{\mu} = C_{L,R_c}\varepsilon_1$, $\varepsilon_1 = g_1 p_0(g_1)$ ([9, p. 246], with g_1 the first-step coupling), satisfies $\widehat{\mu} < \tau_L/2$ and $\max(1, c_L)\widehat{\mu} < \gamma_L/4$, on every branch (where H_ξ is defined, by (B1) and (B2), as the operator of a quadratic form at fixed volume)*

$$\frac{\gamma_L}{2}\,\mathbf{1} \leqslant H_{\xi,II} \leqslant m_L\,\mathbf{1}\,, \tag{3.23}$$

and $H_{\xi,II}(b,b') = 0$ for $d(b,b') > \rho_L$ (the entry stencil $\mathcal{S}_{bb'}$ is empty beyond that distance, Lemma B.3). The same bounds hold for every principal restriction of $H_{\xi,II}$. Moreover, for every row $b \in I$ the row sum over all free bonds, those outside I included, obeys $\sum_{b'\in\Omega_1^*} \|(H_\xi)_{bb'}\| \leqslant m_L$, with $\|\bullet\|$ the operator norm for h and m_L of (3.22).

The proof is in Appendix B, Section B.2.

3.3 The Source Phase and the Local Primitives

On the branch $U = U_1$ and $\Omega = \Omega_1$.

For the reference background $U = \mathbf{1}$ of Section 3.8 only the near rows and the germs occur (the reference inputs vanish off I), the row solutions are those of Lemma 3.12 in its reference form (Lemma B.16), (B4) is not invoked, and v, V below are defined by the same formulas. Lemma 3.14 uses in addition that U is a critical point of S under the block-average constraint and is in the axial mean gauge, which holds for U_1 by (B1) and Section 3.1 and for $U = \mathbf{1}$ because the flat configuration minimises S and $\mathscr{G}_{\mathbf{1}}^{\Omega_1}$ outright and has $\mathcal{U}_{\mathbf{1}}(y,x) = \mathbf{1}$.

For a free vector $B \in \mathfrak{g}^{\Omega_1^*}$ let $A = g_0 \mathsf{C}_U B \in \mathfrak{g}^{\Omega_1}$. The first-step correction $\widetilde{D}_U$ (for $U = U_1$, the correction of [9, (1.13)–(1.14)], as identified in (B4)(i)) is the solution, selected below, of the system

$$\widetilde{D}_{U,c}(A) = \mathcal{N}_{U,c}(A - h_U \widetilde{D}_U(A)), \qquad c \in \Omega_1^{[1]}\,, \tag{3.24}$$

the form of the equation of [8, p. 267] for Bałaban's own step ($L\widetilde{Q}B' + \widetilde{C}(B') = L\widetilde{Q}B$ with $B' = B - h\widetilde{D}(B)$, $\widetilde{C} = \mathcal{N}_U$, $\widetilde{Q} = Q_U$) written row by row.

Let $\rho_{\mathrm{d}} > 0$ be the radius $2K_*\vartheta_*^2$ of Appendix B, Lemma B.8(ii); it depends on L and the group only. A solution d_c of the row equation of c with $h(d_c) < \rho_{\mathrm{d}}$ is called *small*.

The row solution $\widetilde{D}_{U,c}(A)$ is defined as follows:

- for $U = U_1$ on the rows not within $2R_{\mathrm{c}} + 2L$ of I (the far rows; a row is within ρ of a bond set when every bond of its two-block set is, see below) it is the row solution $\widetilde{D}_c(A) = \varphi_c(A|_{\mathcal{E}_c})$ of (B4)(i), defined on the ball $\|A|_{\mathcal{E}_c}\| < \vartheta_*$ (Lemma B.11(i)); the arguments $A = g_0 \mathsf{C}_{U_1} B$, $B \in \mathcal{A}_\xi$, of the domain of integration (3.2) lie in the ball of radius $\vartheta_*/2$, $\|A|_{\mathcal{E}_c}\| < \vartheta_*/2$ (Lemma B.11(ii)), and there its value is the component at c of the chart-class family of Section 3.1 (Lemma B.11(iii));
- on the rows within $2R_{\mathrm{c}} + 2L$ of I (the near rows), for $U = U_1$ and for the reference background, it is the unique small solution of the row equation of c at the argument $A|_{\mathcal{E}_c}$ given by Lemma 3.12 on the row polydisc $\mathcal{U}_c$;
- on a fixed neighbourhood $\mathcal{N}_c$ of $A|_{\mathcal{E}_c} = 0$, for every row, it is the germ of Lemma 3.9, where $\mathcal{N}_c$ is the neighbourhood given by that lemma, reduced so that it lies in the ball $\|A|_{\mathcal{E}_c}\| < \vartheta_*$ and the germ takes values with $h < \rho_{\mathrm{d}}$ (both hold near zero, where the germ vanishes, and both conditions are open; the phase $A - h_U(c)d_c$ then has norm below $\vartheta_* + H_*\rho_{\mathrm{d}} < 2\vartheta_* \leqslant \varrho_G$ and is chart-valued).

Where two of these definitions apply they agree: each produces a solution d_c of the row equation with $h(d_c) \leqslant \rho_{\mathrm{d}}$ at an argument in the ball $\|A|_{\mathcal{E}_c}\| < \vartheta_*$ (the far-row values by Lemma B.11(i), the near-row values by Lemma 3.12, the germ by the reduction of $\mathcal{N}_c$), and Lemma B.8(ii), which applies at every row for $U = U_1$ (Lemma B.11(i)) and for the reference background $U = \mathbf{1}$ (zero phases), gives exactly one such solution.

The reconstructed phase is the function of the free vector

$$q_U(B) = A - h_U \widetilde{D}_U(A) \in \mathfrak{g}^{\Omega_1}, \qquad A = g_0 \mathsf{C}_U B \,, \tag{3.25}$$

defined on free and designated bonds of Ω_1 wherever every row solution is defined. The integration variable of the branch is $a + A_{\mathrm{e}} \in \mathfrak{g}^{\Omega_1^*}$ in the variables of [9, (1.22)]; with Ψ_ξ the bondwise map (3.38) of Section 3.4 from these variables to those of (1.15), the reconstructed fine field is

$$U_\xi(a)(b) \coloneqq e^{iq_U(\Psi_\xi(a+A_{\mathrm{e}}))(b)} U_1(b) \quad (b \in \Omega_1), \qquad U_\xi(a)(b) \coloneqq V_0(b) \quad (b \in \Omega_1^c) \,, \tag{3.26}$$

the fine field $U = U'U_1$ of [9, (1.13)] at the actual branch data, $U_\xi(a) = U(\Psi_\xi(a + A_{\mathrm{e}}))$ in the notation of Section 3.1; it is defined for $(A_{\mathrm{e}}, a) \in \mathcal{D}_\xi$ by the definition of $\mathcal{D}_\xi$ and $\mathcal{A}_\xi$, and by Lemma 3.12 and Lemma 3.9 its links on the rows within $2R_{\mathrm{c}} + 2L$ of I are defined for every a in the enlarged box defined there, the other links not depending on a.

Let $U_\xi^0 \coloneqq U_\xi(0)$. Since $I \cap S_1^* = \varnothing$ ((3.39) below) and Ψ_ξ acts bond by bond and is the identity off S_1^*,

$$\Psi_\xi(a + A_{\mathrm{e}}) = a + \Psi_\xi(A_{\mathrm{e}}) \qquad (a \in \mathfrak{g}^I,\ A_{\mathrm{e}} \in \mathfrak{g}^J) \,. \tag{3.27}$$

A row c is said to lie within ρ of a bond set S when every bond of $\mathcal{E}_c$ is within distance ρ of S; as the bonds of $\mathcal{E}_c$ have their initial vertices in two adjacent blocks, this holds as soon as one bond of $\mathcal{E}_c$ is within $\rho - (2L - 1)$ of S.

Lemma 3.9 (Rowwise Locality of the Reconstruction)**.** *The equation of row c in* (3.24) *involves $\widetilde{D}_{U,c}$ alone and depends only on $A|_{\mathcal{E}_c}$ and $U|_{\mathcal{E}_c}$; so the system decouples row by row, and a solution d_c of the row equation at the argument $A|_{\mathcal{E}_c}$ is a function of $A|_{\mathcal{E}_c}$ and $U|_{\mathcal{E}_c}$.*

The row equation has a unique analytic germ of solutions at $(A|_{\mathcal{E}_c}, d_c) = (0, 0)$, with $d_c(0) = 0$ and $Dd_c(0) = 0$; it is the germ at zero of every solution continuous at zero with value zero, in particular, for $U = U_1$, of the row solution $\widetilde{D}_c = \varphi_c$ of (B4)(i).

Consequently, wherever the row solutions are defined (Section 3.3): on a free bond r, $q_U(B)(r) = g_0 B(r)$; on $b_0(c)$, $q_U(B)(b_0(c))$ depends only on $B|_{\mathcal{E}_c \cap E_*}$ and on $U|_{\mathcal{E}_c}$; and $q_U(B)$ vanishes on all of $\mathcal{E}_c$ whenever $B = 0$ on $\mathcal{E}_c \cap E_*$ and the row solution is the germ value there. In particular, by (3.27)*, the link $U_\xi(a)(r)$ depends on a only through $a|_{\mathcal{F}_r \cap I}$ for $r \in \Omega_1$, and not at all for $r \in \Omega_1^c$.*

Proof. $\mathcal{E}_c$ contains no designated bond other than $b_0(c)$ (Lemma 3.6), so $h_U \widetilde{D}_U(A)$ restricted to $\mathcal{E}_c$ involves only $\widetilde{D}_{U,c}$, and $\mathcal{N}_{U,c}$ depends on its argument restricted to $\mathcal{E}_c$ (Lemma 3.6); by (3.12), $A|_{\mathcal{E}_c}$ depends on $B|_{\mathcal{E}_c \cap E_*}$ only.

At $(0, 0)$ the derivative in d_c of $d_c - \mathcal{N}_{U,c}(A - h_U(c)d_c)$ is the identity, since $D\mathcal{N}_U(0) = D\mathcal{Q}_U(0) - LQ_U = 0$, and $\mathcal{N}_{U,c}(0) = 0$ (the zero phase reproduces the averages of U). The implicit function theorem gives the germ, unique among continuous solutions near $(0, 0)$, with $d_c(0) = 0$ and $Dd_c(0) = D\mathcal{N}_{U,c}(0)(\ldots) = 0$.

For $U = U_1$ the row solution φ_c of (B4)(i) is analytic with $\varphi_c(0) = 0$ and solves the row equation (Lemma B.11(i)), so its germ at zero is this one. If $B = 0$ on $\mathcal{E}_c \cap E_*$ then $A = 0$ on $\mathcal{E}_c$, so the germ value is $\widetilde{D}_{U,c}(0) = 0$ and $q_U(B) = 0$ on $\mathcal{E}_c$.

The last claim follows from (3.26), (3.27) and $\mathcal{F}_{b_0(c)} = \mathcal{E}_c \cap E_*$: the links of $U_\xi(a)$ on the bonds of $\mathcal{E}_c$ involve the row solution of c at the argument $g_0 \mathsf{C}_U \Psi_\xi(a + A_{\mathfrak{e}})|_{\mathcal{E}_c}$, which involves a only through $a|_{\mathcal{E}_c \cap E_* \cap I}$. *q.e.d.*

The analyticity of the reconstruction on the cutoff box is Lemma 3.12 below, stated on the following sets.

Definition 3.10 (Polydiscs and the Enlarged Box)**.** Let r be the cutoff radius of (3.42), $R_{\mathfrak{c}}$ the collar range of Lemma 3.19, and $\vartheta_* > 0$ the phase radius of Lemma B.8 ($\vartheta_* \leqslant \tau_L/2$; it depends on L and the group only). The enlarged box is $\{a \in \mathfrak{g}^I \,|\, |a(b)| \leqslant 2r \ \forall b \in I\}$. The row polydisc of a row c and the block polydisc of a centre y are

$$\mathcal{U}_c := \{B \in \mathfrak{g}^{\Omega_1^*} \,|\, |B(b)| < 2 + 4r \ \forall b \in \mathcal{E}_c \cap E_*\}\,, \qquad \mathcal{U}_y := \{B \,|\, |B(b)| < 2 + 4r \ \forall b \in \mathcal{E}_y\}$$

(every bond of $\mathcal{E}_y$ is free). A centre y lies within ρ of a bond set S when every bond of $\mathcal{E}_y$ does.

Definition 3.11 (The Chart Functions)**.** Let c be a row and $\widetilde{D}_{U,c}$ the small solution of the row equation of c (Lemma 3.12(i)). The row determinant of c at a phase A is $\det_{\mathfrak{g}}(\mathbf{1} - h_U(c)\partial_{A(b_0(c))}\widetilde{D}_{U,c}(A))$. For $B \in \mathcal{U}_c$ and $A = g_0 \mathsf{C}_U B$, the chart functions of c are $\widetilde{D}_{U,c}$, the phase q_U on $\mathcal{E}_c$, the normalised Haar logarithms $\log(\sigma(q_U(\bullet)(b))/\sigma(0))$ for $b \in \mathcal{E}_c$, the real logarithm of the row determinant, and, when c is within $2R_{\mathfrak{c}}$ of I, the plaquette terms $s_{p,U}(q_U(\bullet))$ of the plaquettes p with a bond in $\mathcal{E}_c$ (on the intersection of the polydiscs of the rows of p).

Let y be a centre and $B \in \mathcal{U}_y$. The relative means of y are $\mathcal{U}_{e^{iq}U}(y,x)\mathcal{U}_U(y,x)^{-1}$, $x \in \mathcal{B}_y$, at $q = q_U(B) = g_0 B$ on $\mathcal{E}_y$, and the chart functions of y are these relative means and the gauge term $\mathscr{G}_{y,U}(q_U(\bullet))$.

Lemma 3.12 (Charts)**.** *On every branch, and for the reference background $U = \mathbf{1}$ of Section 3.8, let $c \in \Omega_1^{[1]}$ be a row and $y \in \Omega_1^{(1)}$ a centre within $2R_{\mathfrak{c}} + 2L$ of I. Then:*

(i) for every real phase A with $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/2$ (Hilbert norm of $\mathfrak{g}^{\mathcal{E}_c}$), the row equation of c at A has exactly one small solution, $\widetilde{D}_{U,c}(A) = \varphi_c(A|_{\mathcal{E}_c})$, where φ_c is analytic on the complex ball $\{\|A|_{\mathcal{E}_c}\| < \vartheta_*\}$, real on real phases, with $\varphi_c(0) = 0$, $D\varphi_c(0) = 0$;

(ii) under threshold item 3 of Section B.8, $g_0 \mathsf{C}_U \mathcal{U}_c$ lies in the ball $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/2$; for $B \in \mathcal{U}_c$ the row determinant of c is positive, and the chart functions of c (Definition 3.11) are analytic functions of $B|_{\mathcal{E}_c \cap E_*}$ on $\mathcal{U}_c$; the first four vanish at $B = 0$, $\widetilde{D}_{U,c}$ and $q_U(\bullet)(b_0(c)) - g_0(\mathsf{C}_U B)(b_0(c))$ together with their first derivatives;

(iii) for every $B \in \mathcal{U}_y$ the relative means of y exist and lie within r_G of $\mathbf{1}$ in operator norm, and the gauge term of y is an analytic function of $B|_{\mathcal{E}_y}$ on $\mathcal{U}_y$ vanishing at $B = 0$;

(iv) the derivatives in B up to order five of the chart functions are bounded on the polydiscs by constants depending on L and the group only;

(v) every $a + t\Psi_\xi(A_{\mathfrak{e}})$ with a in the enlarged box and $t \in [0,1]$ lies in $\mathcal{U}_c$ and $\mathcal{U}_y$, and every near primitive $\mathfrak{p}$ is an analytic function of (a,t) there, with $|\mathfrak{p}(a + t\Psi_\xi(A_{\mathfrak{e}}))| \leqslant C_{\mathfrak{p}} g_0 (1 + 2r)^3$, $C_{\mathfrak{p}}$ depending on L and the group only, and its derivatives in a up to order four bounded as in (B.18)–(B.19)*;*

(vi) where $(A_{\mathfrak{e}}, a) \in \mathcal{D}_\xi$, these objects coincide with the objects of [9, (1.14)–(1.15)].

Proof. (i)–(iii) and (vi) are proved in Section B.3, after Lemma B.12, from Lemmas B.8 and B.9; the relative means lie in the domain of the chart logarithm $\log_G$ of Appendix B. The functions of (ii) and (iii) are the compositions of analytic functions φ of the real phase $A|_{\mathcal{E}_c}$ (or $q|_{\mathcal{E}_y}$) with the linear map $B \mapsto g_0 \mathsf{C}_U B$, where φ is analytic on a complex neighbourhood of the closed phase ball $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/2$ (which contains $g_0 \mathsf{C}_U \mathcal{U}_c$; for the row solution and the phase, on the complex ball of radius ϑ_*) and has derivatives up to order five bounded on that closed ball by constants depending on L and the group only; this gives (iv).

The inclusion in (v) is Lemma 3.19(i),(iii): $\Psi_\xi(A_{\mathfrak{e}}) = A_{\mathfrak{e}}$ and $|A_{\mathfrak{e}}(b)| < r$ on the bonds of $\mathcal{E}_c$, $\mathcal{E}_y$, which are within $2R_{\mathfrak{c}} + 4L - 1 \leqslant 3R_{\mathfrak{c}}$ of I. The bounds of (v) are Lemma B.12, (B.18), the prefactor g_0^{-2} of $E_{\xi,\alpha}$ being absorbed by the third-order Taylor remainder; under threshold item 4 of Section B.8, $g_0 s_\xi^2 \leqslant 4g_0(1+r)^3 \leqslant 4\omega_*/C_0$, so these are bounds by constants depending on L and the group only. *q.e.d.*

The centred exponent and its primitives are those of [9, (1.14)–(1.16)], written as functions of the free vector $B \in \mathfrak{g}^{\Omega_1^*}$. With $s_{p,U}(q) \coloneqq 1 - \operatorname{Re}\operatorname{tr}_N((e^{iq}U)(\partial p))$ for a plaquette p meeting Ω_1 (links off Ω_1 fixed at V_0),

$$E_{\xi,p}(B) \coloneqq -\frac{1}{g_0^2}[s_{p,U}(q_U(B)) - s_{p,U}(0)] \,, \tag{3.28}$$

and $E_{\xi,y}(B) = -g_0^{-2}\mathscr{G}_{y,U}(q_U(B))$ from the y-summand of (3.14), $y \in \Omega_1$. For either label α,

$$p_{\xi,\alpha}(B) \coloneqq E_{\xi,\alpha}(B) - DE_{\xi,\alpha}(0)[B] - \frac{1}{2}D^2E_{\xi,\alpha}(0)[B,B] \,. \tag{3.29}$$

With $\sigma(q)\,\mathrm{d}q$ the pullback under $q \mapsto e^{iq}$ of the Haar measure $\mathrm{d}U$ of [9, (1.11)] (the density σ of (1.14), for the fixed inner product as explained in the conventions; its normalisation enters only the constants κ_ξ and c_ξ), for every bond b of Ω_1,

$$p^{\mathrm{H}}_{\xi,b}(B) \coloneqq \log\sigma(q_U(B)(b)) - \log\sigma(0) \,, \tag{3.30}$$

and for every row $c \in \Omega_1^{[1]}$

$$p^{\mathrm{J}}_{\xi,c}(B) \coloneqq \log\Big|\det_{\mathfrak{g}}\Big(\mathbf{1} - h_U(c)\partial_{A(b_0(c))}\widetilde{D}_{U,c}(g_0\mathsf{C}_U B)\Big)\Big| \,, \tag{3.31}$$

where $\partial_{A(b_0(c))}\widetilde{D}_{U,c}$ is the partial derivative of the row solution, regarded as a function of the $|\mathcal{E}_c|$ independent variables $A(b)$, $b \in \mathcal{E}_c$, with respect to the variable at $b_0(c)$, evaluated at $A = g_0\mathsf{C}_U B$ (this is the designated block of the $\frac{\delta}{\delta A}\widetilde{D}$ of [9], proof of Lemma 3.14).

A primitive is defined at B when the row solutions of its rows are defined at $g_0\mathsf{C}_U B$ (Section 3.3: on the domain of integration, on the polydiscs for the rows near I, and near $B|_{\text{inputs}} = 0$); for a primitive whose row solutions are not defined at $B = 0$ on the domain of integration (which happens when $R_1 \neq \varnothing$, since the large-fluctuation tests exclude $A = 0$ on R_1), the Taylor polynomial at $B = 0$ in (3.29) is formed from the germs of Lemma 3.9; the V of [9] is the sum of the terms of order higher than two of the expansion at $A = 0$ (p.249, [8, p. 267]), and Lemma 3.14 identifies the two.

The letter $\mathfrak{p}$ denotes a primitive of any of the three kinds. By Lemma 3.9 the free-input set of $E_{\xi,p}$ and $p_{\xi,p}$ is $\bigcup_{r\subset\partial p}\mathcal{F}_r$, that of $E_{\xi,y}$ and $p_{\xi,y}$ is $\mathcal{E}_y \cap E_* = \mathcal{E}_y$, that of $p^{\mathrm{H}}_{\xi,b}$ is $\mathcal{F}_b$ and that of $p^{\mathrm{J}}_{\xi,c}$ is $\mathcal{E}_c \cap E_*$; the background set of each is the set of bonds whose links enter (bonds of Ω_1 and, for plaquettes and rows at the boundary of Ω_1, bonds of Ω_1^c carrying V_0),

as in (3.17); the rows of a primitive are the rows c with $b_0(c)$ in its background set (a gauge primitive $E_{\xi,y}$ has no rows: $\mathcal{E}_y$ contains no designated bond, and $q_U(B) = g_0 B$ on it).

A primitive is *near* if its free-input set meets I and *far* otherwise; a far primitive is a function of A_{e} and the branch data alone (Lemma 3.9, on the domain of (B4)). A near primitive has a bond $b_1 \in I$ in its free-input set, and every bond of $\mathcal{E}_c$ for a row c of the primitive, and every bond of $\mathcal{E}_y$ for a gauge primitive, is within $4L + 3 \leqslant 2R_{\mathrm{c}}$ of b_1 (two bonds of a two-block set are within $2L - 1$ of each other, the bonds of a plaquette within 1); so the rows of a near primitive, and the centre of a near gauge primitive, lie within $2R_{\mathrm{c}}$ of I, and by Lemma 3.12 every near primitive is defined and analytic in (a, t) at the inputs $a + t\Psi_\xi(A_{\mathrm{e}})$, a in the enlarged box, $t \in [0, 1]$.

An *arc* γ of the contour C is a set of consecutive bonds of C (with their orientations), and γ_+ is the set of their positive representatives; the free-input and background stencils of an arc are

$$F_\gamma \coloneqq \bigcup_{r \in \gamma_+} \mathcal{F}_r, \qquad D_\gamma \coloneqq \gamma_+ \cup \bigcup_{c \,|\, b_0(c) \in \gamma_+} \mathcal{E}_c \,. \tag{3.32}$$

Definition 3.13 (The Functions v, V, and $\mathscr{E}_\xi$)**.** v and V are the functions of [9, (1.15), p. 249]: $v(A)$ is the sum of the logarithms of the Haar densities $\sigma(A - h\widetilde{D}(A))$ over the bonds of Ω_1 and of $\log|\det(\mathbf{1} - h\frac{\delta}{\delta A}\widetilde{D})|$, the logarithm of the absolute Jacobian of (1.14), which is, in the notation of [9], $\operatorname{Tr}\log(\mathbf{1} - h\frac{\delta}{\delta A}\widetilde{D})$ where the determinant is positive (on the near rows, factor by factor, by Lemma 3.12; on all of $\mathcal{A}_\xi$ by (B4)(iii), Lemma B.11(iv)), the identities below using the absolute value throughout; $V(A)$ is the remainder after the Taylor polynomial of degree two at $A = 0$ of the action and the gauge-fixing functional at the phase $A - h\widetilde{D}(A)$ (the "terms of order higher than two" of [9]; the Taylor coefficients at $A = 0$ are those of the germs of Lemma 3.9). The exponent of the second factor of [9, (1.15)] at the free vector B, with the quadratic form written through the regional Hessian (3.16), is

$$\mathscr{E}_\xi(B) \coloneqq -\frac{1}{2}\langle B, H_\xi B\rangle + v(g_0 \mathsf{C}_U B) - g_0^{-2} V(g_0 \mathsf{C}_U B) \,. \tag{3.33}$$

Lemma 3.14 (Exact Primitive Sum)**.** *For $B \in \mathcal{A}_\xi$, and more generally for every $B \in \mathfrak{g}^{\Omega_1^*}$ at which every row solution is defined with the row solutions of the near rows those of Lemma 3.12 and those of the far rows the germs at zero argument ($B = 0$ on their two-block sets), with $A = g_0 \mathsf{C}_U B$,*

$$v(A) - v(0) - g_0^{-2} V(A) = \sum_\alpha p_{\xi,\alpha}(B) + \sum_b p^{\mathrm{H}}_{\xi,b}(B) + \sum_c p^{\mathrm{J}}_{\xi,c}(B) \,, \tag{3.34}$$

the first sum over the plaquettes p meeting Ω_1 and the centres $y \in \Omega_1^{(1)}$, the second over the bonds b of Ω_1, the third over the rows $c \in \Omega_1^{[1]}$ (the same ranges in every sum $\sum_\alpha$, $\sum_b$, $\sum_c$ below). The exponent $\mathscr{E}_\xi$ of Definition 3.13 satisfies $\mathscr{E}_\xi(0) = v(0)$ with $v(0) = \sum_b \log(\sigma(0)/\sigma_0) = 0$ (sum over the bonds of Ω_1; the Haar densities in the v of [9] are divided by their value σ_0 at the identity, as fixed in the step list after (3.7)) and

$$\begin{aligned} \mathscr{E}_\xi(B) - \mathscr{E}_\xi(0) &= -\frac{1}{2}\langle B, H_\xi B\rangle + \sum_\alpha p_{\xi,\alpha}(B) + \sum_b p^{\mathrm{H}}_{\xi,b}(B) + \sum_c p^{\mathrm{J}}_{\xi,c}(B) \\ &= \sum_\alpha E_{\xi,\alpha}(B) + \sum_b p^{\mathrm{H}}_{\xi,b}(B) + \sum_c p^{\mathrm{J}}_{\xi,c}(B) \,. \end{aligned} \tag{3.35}$$

Proof. The sum $\sum_\alpha E_{\xi,\alpha}(B)$ is $-g_0^{-2}$ times the difference of the action plus gauge-fixing functional at the phase $q_U(B)$ and at $q = 0$: here the gauge-fixing functional of [9, (1.11),

(1.14)] is $\mathbf{G}(\Omega_1^{(1)}, e^{iq}U_1) = \mathscr{G}_{U_1}^{\Omega_1}(q)$ because U_1 is in the axial mean gauge (Section 3.2, after (3.15)).

Since $D\widetilde{D}_U(0) = 0$, $Dq_U(0)[B] = g_0\mathsf{C}_U B$, and the linear term of the Taylor expansion in B at zero is $\sum_\alpha DE_{\xi,\alpha}(0)[B] = -g_0^{-1}(DS(U_1)[\mathsf{C}_U B] + D\mathscr{G}_U^{\Omega_1}(0)[\mathsf{C}_U B]) = 0$: the relative gauge functional is a sum of nonnegative terms vanishing at $q = 0$, so its first variation vanishes; by [9, (1.12)] U_1 is a critical point of $S + \mathbf{G}(\Omega_1^{(1)}, \bullet)$ on the configurations with $\mathsf{M}(U) = V$ on $\Omega_1^{(1)}$ and $U = V_0$ on Ω_1^c, and $\mathbf{G}(\Omega_1^{(1)}, \bullet) \geqslant 0$ attains its minimum 0 at U_1, so $D\mathbf{G}(\Omega_1^{(1)}, U_1) = 0$ and $DS(U_1)[\delta] = 0$ for every phase δ supported on the bonds of Ω_1 with $D\mathcal{Q}_U(0)\delta = LQ_U\delta = 0$ (the tangent space of the constraint $\mathsf{M}(e^{i\delta}U_1) = V$); and $\mathsf{C}_U B$ is such a phase by $Q_U\mathsf{C}_U = 0$ (3.12).

The quadratic term is $-\frac{1}{2}\langle B, H_\xi B\rangle$ by (3.15), with $A = g_0\mathsf{C}_U B$: the second variation of the action is $\langle A, \Delta_U A\rangle$ plus the force term $-2\langle J_U, h_U\frac{1}{2}D^2\mathcal{N}_U(0)[A, A]\rangle$ from the second-order part $-h_U\widetilde{D}_U^{(2)}(A)$ of the phase, and that of the gauge functional is $G_U^{(2),\Omega_1}(A)$. Subtracting both from $\sum_\alpha E_{\xi,\alpha}$ gives $\sum_\alpha p_{\xi,\alpha}$, which is therefore $-g_0^{-2}V(A)$.

The Haar part of v is $\sum_b \log\sigma(q_U(B)(b))$ bond by bond, so v minus its Haar part at $B = 0$ gives $\sum_b p^{\mathrm{H}}_{\xi,b}$.

For the Jacobian, $\frac{\delta}{\delta A}\widetilde{D}$ has rows indexed by $\Omega_1^{[1]}$ and columns by the bonds of Ω_1, and h_U maps row c to the bond $b_0(c)$; in the ordering (free bonds, designated bonds) the matrix $\mathbf{1} - h_U\frac{\delta}{\delta A}\widetilde{D}_U$ is block triangular with identity block on the free bonds and designated block $\mathrm{diag}_c(\mathbf{1} - h_U(c)\partial_{A(b_0(c))}\widetilde{D}_{U,c})$ by Lemma 3.6 (row c depends on $A|_{\mathcal{E}_c}$, which contains the single designated bond $b_0(c)$), so its determinant is $\prod_c \det_{\mathfrak{g}}(\mathbf{1} - h_U(c)\partial_{A(b_0(c))}\widetilde{D}_{U,c})$, positive on $\mathcal{A}_\xi$ by (B4)(iii), and at the other points admitted in the statement because each near-row factor is positive by Lemma 3.12 and each far-row factor equals one at zero argument ($D\widetilde{D}_{U,c}(0) = 0$); so Tr log of the matrix, the term in the v of [9], is the real logarithm of the positive determinant, and since each factor is a nonzero real, the logarithm of the absolute value of the product is the sum of the logarithms of the absolute values of the factors, $\sum_c \log|\det_{\mathfrak{g}}(\cdots)| = \sum_c p^{\mathrm{J}}_{\xi,c}(B)$; this vanishes at $B = 0$ since $D\widetilde{D}_U(0) = 0$; hence $v(0)$ is the Haar part at $B = 0$.

The quadratic form of (1.15) in the free variable, $\frac{1}{2}\langle B, C^*\Delta_1 CB\rangle$, is $\frac{1}{2}\langle B, H_\xi B\rangle$ by the identification $\mathsf{H}_{U_1}^{\Omega_1} = C^*\Delta_1 C$ of Section 3.2 (Δ_1 of (1.16) already contains $\mathbf{G}^{(2)}$ and the force term; the g_0^{-2} of the action cancels the g_0^2 of $A = g_0\mathsf{C}_U B$).

Subtracting $\mathscr{E}_\xi(0) = v(0)$ from (3.33) and inserting (3.34) gives the first form of (3.35); the second form restores the quadratic term to $\sum_\alpha p_{\xi,\alpha}$. *q.e.d.*

Each primitive is assigned a host Y_α in the native localisation family $\mathbf{D}_0$. Its free-input and background stencil $\widetilde{\mathcal{S}}_\alpha$ is lifted to $\mathbb{Z}^4$ about its base (plaquette lower vertex, block centre, initial vertex of its bond, or initial vertex of $b_0(c)$ for $p^{\mathrm{J}}_{\xi,c}$; the stencil has ℓ^∞ diameter at most $4L < M$, below every period, so the lift about the base is well defined); with Z_α the set of M-cubes $Q_z = Mz + [-M/2, M/2]^4$ meeting the lift, the host is

$$Y_\alpha \coloneqq \pi_T\Big(\bigcup_{z\in\prod_\mu([k^-_{\alpha,\mu}-1,k^+_{\alpha,\mu}+1]\cap\mathbb{Z})} Q_z\Big), \qquad k^-_{\alpha,\mu} \coloneqq \min_{z\in Z_\alpha} z_\mu, \quad k^+_{\alpha,\mu} \coloneqq \max_{z\in Z_\alpha} z_\mu\,, \tag{3.36}$$

with π_T the reduction modulo the periods; since the stencil has diameter at most $4L < M$, it meets at most two consecutive M-cubes in each direction, so the rectangle has side at most 4 cubes, and every period being at least $8M$, π_T is injective on it. Y_α is connected and contains the stencil in its interior.

The potential is, for $Y \in \mathbf{D}_0$ and $a \in \mathfrak{g}^I$,

$$\mathbf{V}_\xi(Y,a) = \sum_{\mathfrak{p} \text{ near},\ Y_\mathfrak{p}=Y} \mathfrak{p}(a) \quad \text{if } |a(b)| < r\ \forall b \in E(Y), \qquad \mathbf{V}_\xi(Y,a) = 0 \quad \text{otherwise}, \tag{3.37}$$

the sum over the near primitives $\mathfrak{p}$ of all three kinds assigned to Y, with $E(Y) := \{b \in I \,|\, b \cap Y \neq \varnothing\}$ and the primitives at zero retained input ($t = 0$). It depends on $a|_{E(Y)}$ only, since the free-input set of a primitive assigned to Y lies in Y, hence in $E(Y)$ (Lemma 3.9), and it is bounded on the local box by Lemma 3.12 (each summand by $C_\mathrm{p} g_0 (1+2r)^3$). On the open box $\{|a(b)| < r\ \forall b \in I\}$, $\sum_Y \mathbf{V}_\xi(Y,a) = \sum_{\mathfrak{p} \text{ near}} \mathfrak{p}(a)$.

3.4 The Boundary Law, the Exterior Interpolation and the Collar

The change of variables. Let $S_1 := (\Omega_1 \setminus \Omega_1^{\sim -1}) \cup R_1$ ([9, (1.20)]) and let S_1^* be its free bonds. [9, (1.22)] replaces the variable A of (1.15) on each bond $b \in S_1^*$ by a new variable $A'(b)$ and leaves the other bonds unchanged: $A = \Psi_\xi A'$ with

$$\begin{aligned} (\Psi_\xi A')(b) &= g_0^{-1} \log\left(e^{i g_0 A'(b)} e^{-i \omega_b(A'(b)) \mathbf{H}_{1,Ax}(b)} \right) \quad (b \in S_1^*), \\ (\Psi_\xi A')(b) &= A'(b) \quad (b \in \Omega_1^* \setminus S_1^*)\,, \end{aligned} \tag{3.38}$$

where

- ω_b is the C^∞ cutoff g of [9, p. 252] ($0 \leqslant g \leqslant 1$, $g = 0$ on $|A'| \leqslant \frac{4}{3} g_0^{-1} \delta_0$, $g = 1$ where $|e^{i g_0 A'} - 1| \geqslant \frac{5}{3}\delta_0$) for b in the set $(R_1^{\sim 2} \cap S_1)^*$ named in the definition preceding (1.22) (p.251); [9] specifies g by these properties only, and one such g with $|Dg(A')| \leqslant 6 g_0/\delta_0$ for every A' (operator norm for the cutoff norm $|\bullet|$) is fixed once: the function $\ell(A') = \min(1, \max(0, (|g_0 A'| - \frac{17}{12}\delta_0)/(\frac{1}{6}\delta_0)))$ is Lipschitz with constant $6 g_0/\delta_0$, vanishes for $|g_0 A'| \leqslant \frac{17}{12}\delta_0$ and equals one for $|g_0 A'| \geqslant \frac{19}{12}\delta_0$, and its convolution with a C^∞ probability density supported in $\{|A'| \leqslant \delta_0/(12 g_0)\}$ is C^∞ with values in $[0,1]$, has the same Lipschitz constant, vanishes for $|g_0 A'| \leqslant \frac{4}{3}\delta_0$ and equals one for $|g_0 A'| \geqslant \frac{5}{3}\delta_0$, hence wherever $|e^{i g_0 A'} - \mathbf{1}| \geqslant \frac{5}{3}\delta_0$, as $|e^{iX} - \mathbf{1}| \leqslant |X|$; and $\omega_b = 1$ for the other bonds of S_1^* (p.252 describes the second set as $S_1^* \setminus (R_1^{\sim 2})^*$, which differs from the complement of the first set in S_1^* on the bonds joining $R_1^{\sim 2} \setminus S_1$ to $S_1 \setminus R_1^{\sim 2}$; the definition of p.251 is used, on the bonds of the difference the map is the second form of [9] with g replaced by one, and the cutoff is the exact composition (3.43));
- $\mathbf{H}_{1,Ax}(b) \in \mathfrak{g}$ is the field $\mathbf{H}_{1,Ax}(\Omega_1 \cap \Lambda_1^c, \bullet)$ of [9, (1.21)] at the actual branch data, as it appears in (1.22), without the prefactor L^{-1} of (1.21) (a function of the pair (V, V_0) through $U_1 = U_1(V, V_0)$ and $U_1(\Omega_1 \cap \Lambda_1^c, V)$, the latter stated by [9, p. 251] to depend on $V|_{\Omega_1 \cap \Lambda_1^c}$; Borel in (V, V_0) by the measurability paragraph of Section 3.1; with (1.21) and $\omega_b = 1$, $e^{i g_0 (\Psi_\xi A')(b)} U_1(b) = e^{i g_0 A'(b)} e^{-i(1 - L^{-1}) \mathbf{H}_{1,Ax}(b)} U_1(\Omega_1 \cap \Lambda_1^c, b)$, which is not the second factor of [9, (1.23)]: as printed, (1.22) carries $\mathbf{H}_{1,Ax}$ where (1.21) carries $L^{-1}\mathbf{H}_{1,Ax}$, and (1.24) has $+F$ against the $-F$ of (1.22); this paper uses (1.22) as printed, with the exact Jacobian $|\det D\Psi_\xi|$ and the cutoff composed in the variable of (1.15), and does not use (1.23)–(1.24));
- log is the group chart, the inverse of $A \mapsto e^{iA}$ on the chart ball (so that log is the chart logarithm of the fluctuation, written $\frac{1}{i}\log$ in the notation of [9] for the matrix logarithm, and $g_0^{-1}\log(\cdots)$ is the variable of (1.15)).

Ψ_ξ acts bond by bond, and ψ_b denotes its component at $b \in S_1^*$, a C^∞ map (g is C^∞, not analytic) on the open set $\mathcal{W}_b \subseteq \mathfrak{g}$ of those $A'(b)$ for which $g_0 A'(b)$ lies in the chart ball and

the argument of the chart logarithm lies in the chart domain (on the bonds where (1.22) is the identity, the variable A' of (1.22) is the variable A of (1.15), which on the free bonds is g_0^{-1} times the fluctuation variable of (1.13); on S_1^* the new variable A' is restricted to the values with $g_0 A'$ in the chart ball as well, which loses no preimage, every preimage under ψ_b of a point of $\mathcal{A}_\xi$ lying in $g_0^{-1}K$ with K the ball of the proof of Lemma 3.16(b),(c)).

By Lemma 3.16 below, which is the (1.22) part of (B4)(ii), Ψ_ξ maps $\mathcal{D}_\xi$ bijectively onto $\mathcal{A}_\xi$, the domain of integration of (1.15) with the cutoff (1.25) (Section 3.1), and $D\Psi_\xi$ is invertible there, so that the change of variables formula applies with the Jacobian $|\det D\Psi_\xi| = \prod_{b\in S_1^*} |\det D\psi_b|$; $\chi_{\xi,\mathrm{e}}(A_\mathrm{e})$ below is set to zero when $A_\mathrm{e}(b) \notin \mathcal{W}_b$ for some $b \in S_1^*$. Thus $\log|\det D\Psi_\xi(A')|$ depends on $A'|_{S_1^*}$ only, and it is C^∞, finite at $(A_\mathrm{e}, 0) \in \mathcal{D}_\xi$ and independent of a (a constant on the branch, by $I \cap S_1^* = \varnothing$ and (3.27)).

The integration variable of the branch is $A' = a + A_\mathrm{e}$ with $a \in \mathfrak{g}^I$ and $A_\mathrm{e} \in \mathfrak{g}^J$. A bond of I has an endpoint in $\Lambda_1 \subseteq \Omega_1^{\sim -2} \cap (R_1'^{\sim})^c$, a bond of S_1^* has an endpoint in S_1, and $\Omega_1^{\sim -2}$ is separated from $\Omega_1 \setminus \Omega_1^{\sim -1}$, as $(R_1'^{\sim})^c$ is from $R_1 \subseteq R_1'$, by a full layer of LMR_1-cubes (the enlargement adjoins every touching cube, Section 3); hence, with distances between initial vertices,

$$I \cap S_1^* = \varnothing, \qquad d(I, S_1^*) \geqslant LMR_1 - 2 \,. \tag{3.39}$$

The filling modification following [9, (1.20)] only removes cubes from Λ_1, so (3.39) holds for the final Λ_1. This gives (3.27).

Definition 3.15 (Boundary Data)**.** Let $r_G \in (0,1]$ be the group constant of Appendix B (defined there with the chart logarithm $\log_G$): every $U \in G$ with $\|U - \mathbf{1}\|_{\mathrm{op}} \leqslant r_G$ lies in the chart domain N_G, the image of the chart ball $\{h(A) < \varrho_G\}$ under $A \mapsto e^{iA}$. Let (Ω_1, V, V_0) be an admissible pair and R_1 a large-fluctuation region of the history sum (Section 3.1), with U_1 and the configuration $U_1(\Omega_1 \cap \Lambda_1^c, V)$ of (1.21) formed with the representatives chosen in Lemma 3.2. For $b \in S_1^*$ let $\square'(b) \subseteq S_1$ be an $L^2M_2R_1$-cube with $b \in (\square'(b)^{\sim 2})^*$. Then

$$\delta_G := \min(\log(3/2)/(12\sqrt{N}),\, r_G/(10\sqrt{N})) \,, \qquad Z_b := U_1(b) U_{1,\square'(b)}(b)^{-1} \,,$$
$$\mathbf{H}_{1,Ax}(b) := L \log(U_1(b)\, U_1(\Omega_1 \cap \Lambda_1^c, V)(b)^{-1}) \,,$$
$$\mathcal{T}_b := \{A \in \mathfrak{g} : h(g_0 A) < \varrho_G,\ |e^{ig_0 A} Z_b - \mathbf{1}| < 2\delta_0\} \,,$$

$\mathbf{H}_{1,Ax}(b)$ being defined wherever this chart logarithm is.

The configurations U_1 and $U_1(\Omega_1 \cap \Lambda_1^c, V)$ exist by (B1) and Lemma 3.5. Under (3.40), $\mathbf{H}_{1,Ax}$ is the analytic $\mathbf{H}_{1,Ax}$ of [9, (1.21)], since $L^{-1}\mathbf{H}_{1,Ax}(b)$ then lies in the chart ball and (1.21) determines it there as this logarithm.

The cube $\square'(b)$ exists: S_1 is a union of such cubes, $\Omega_1 \setminus \Omega_1^{\sim -1}$ being a union of LMR_1-cubes, each a union of $L^2M_2R_1$-cubes (the cube partitions of [9, pp. 245–246] are nested: each cube of a coarser partition is a union of cubes of each finer one, the L-blocks included, the side ratios $M/(LM_2)$, a power of L by the standing choice, and LM_2R_1 being integers), and R_1 a union of them, p.250; b has an endpoint in S_1, hence in such a cube, and is not designated for a row outside $S_1^{[1]}$: if $b = b_0(c)$, the block containing its endpoint in S_1 lies in S_1 by the nesting, so $c \in S_1^{[1]}$, which $b \in S_1^*$ excludes.

The bound on $\mathbf{H}_{1,Ax}$ of [9, p. 251], proved in Lemma D.21, is used at the tolerance

$$|\mathbf{H}_{1,Ax}(b)| \leqslant \frac{\delta_0}{40} \qquad (b \in S_1^*) \,. \tag{3.40}$$

Lemma 3.16 (The Boundary Change Is a Bijection)**.** *Let $\delta_0 \leqslant \delta_G$, a condition on the coupling ($\delta_0 = (A_1/A_0)\varepsilon_0 \to 0$), and let (Ω_1, V, V_0) and R_1 be as in Definition 3.15. Then for every $b \in S_1^*$:*

(a) $B(b) \in \mathcal{T}_b$ for every $B \in \mathcal{A}_\xi$;

(b) $\mathcal{T}_b \subseteq \psi_b(\mathcal{W}_b)$;

(c) ψ_b is injective on $\psi_b^{-1}(\mathcal{T}_b)$, and $D\psi_b$ is invertible at every point of $\psi_b^{-1}(\mathcal{T}_b)$.

Consequently Ψ_ξ maps $\mathcal{D}_\xi$ bijectively onto $\mathcal{A}_\xi$, and $D\Psi_\xi$ is invertible at every point of $\mathcal{D}_\xi$.

Proof. Norm facts, for the cutoff norm $|\bullet|$ (operator or normalised Hilbert–Schmidt) and Hermitian X with $\|X\|_{\mathrm{op}} \leqslant \pi$:

- $|e^{iX} - \mathbf{1}| \leqslant |X| \leqslant \frac{\pi}{2}|e^{iX} - \mathbf{1}|$ (eigenvalue by eigenvalue, $2|\sin(\theta/2)| \leqslant |\theta| \leqslant \pi|\sin(\theta/2)|$ for $|\theta| \leqslant \pi$, in both norms);
- $|UY| = |YU| = |Y|$ for unitary U, so $|UU' - \mathbf{1}| \leqslant |U - \mathbf{1}| + |U' - \mathbf{1}|$ for unitary U, U';
- a unitary $U \in G$ with $|U - \mathbf{1}| < r_G/\sqrt{N}$ lies in the chart domain, since $\|U - \mathbf{1}\|_{\mathrm{op}} \leqslant \sqrt{N}|U - \mathbf{1}|$ for either cutoff norm and by the definition of r_G (the eigenvalue argument alone does not give this for a general compact $G \subset U(N)$: the Hermitian logarithm Y of U with eigenvalues in $(-\pi, \pi]$ need not lie in the Lie algebra $\mathfrak{g}$, so r_G is a constant of the group);
- on $\{|X| \leqslant 12\delta_0\}$, writing $e^{iX} = \mathbf{1} + iX + R(X)$, one has $\|DR(X)\| \leqslant e^{\|X\|_{\mathrm{op}}} - 1 \leqslant e^{12\sqrt{N}\delta_0} - 1 \leqslant \frac{1}{2}$ (the derivative of X^k is a sum of k terms $X^j V X^{k-1-j}$, each of norm at most $\|X\|_{\mathrm{op}}^{k-1}|V|$), so $|e^{iX} - e^{iX'}| \geqslant \frac{1}{2}|X - X'|$ for $|X|, |X'| \leqslant 12\delta_0$.

Let $b \in S_1^*$, $H = \mathbf{H}_{1,Ax}(b)$, $\omega = \omega_b$, and let $x = g_0 A'$ be the group-scale variable: $\psi_b(A') = g_0^{-1} F(g_0 A')$ with $F(x) := \log(e^{ix} W(x))$, $W(x) = e^{-i\widetilde{\omega}(x)H}$, $\widetilde{\omega}(x) = \omega(x/g_0)$, on $\mathfrak{B}_b = g_0\mathcal{W}_b = \{x : h(x) < \varrho_G,\ e^{ix}W(x) \text{ in the chart domain}\}$; $|D\widetilde{\omega}| \leqslant 6/\delta_0$, and $|W(x) - \mathbf{1}| \leqslant \widetilde{\omega}(x)|H| \leqslant |H| \leqslant \delta_0/40$.

Moreover $|Z_b - \mathbf{1}| < 2\delta_0$: by (B1) the background U_1 lies in $D^{(11)}(V, V_0)$, the domain of integration in (1.11) ([9, p. 248]), on which the factor $\chi_0'(\square'(b))$ of $\chi_0'(\Omega_1)$ is nonzero, that is $|U_1(b')U_{1,\square'(b)}(b')^{-1} - \mathbf{1}| < 2\delta_0$ for every $b' \in (\square'(b)^{\sim 2})^*$ ([9, (1.8)]), and b is one of these bonds.

(a) For $B \in \mathcal{A}_\xi$, $U(B)(b) = e^{ig_0 B(b)} U_1(b)$ (b is free, so $q_{U_1}(B)(b) = g_0 B(b)$, Lemma 3.9), and $g_0 B(b)$ is the component at b of the chart logarithm $\frac{1}{i}\log U'$ of (1.13) (the variable written A' there, not the new variable A' of (1.22)), so $h(g_0 B(b)) < \varrho_G$ by the chart-class condition in the definition (3.2) of $\mathcal{A}_\xi$; the cutoff (1.25) is nonzero at $U(B)$, in particular its factor $\chi_0'(\square'(b))$, the test $|U(B)(b')U_{1,\square'(b)}(b')^{-1} - \mathbf{1}| < 2\delta_0$ for $b' \in (\square'(b)^{\sim 2})^*$ ([9, (1.8) and p. 250]; a function of the fine field, composed with the changes of variables (1.14) and $A = g_0 A'$ of p.249, Section 3.1), and at $b' = b$ this reads $|e^{ig_0 B(b)} Z_b - \mathbf{1}| < 2\delta_0$.

(b) Let $y = g_0 A$ with $A \in \mathcal{T}_b$. Then $|e^{iy} - \mathbf{1}| \leqslant |e^{iy} Z_b - \mathbf{1}| + |Z_b - \mathbf{1}| < 4\delta_0$, and $\|y\|_{\mathrm{op}} < \pi$ since $h(y) < \varrho_G$, so $|y| \leqslant \frac{\pi}{2} \cdot 4\delta_0 < 6.3\,\delta_0$.

Let $K = \{x \in \mathfrak{g} : |x| \leqslant 7\delta_0\}$, compact and convex, with $K \subseteq \mathfrak{B}_b$: $h(x) \leqslant |x| \leqslant 7\delta_0 < \varrho_G$ (as $7\delta_0 \leqslant 7r_G/(10\sqrt{N}) < \pi/\sqrt{N} = \varrho_G$, $r_G \leqslant 1$) and $|e^{ix}W(x) - \mathbf{1}| \leqslant |x| + |H| \leqslant 7\delta_0 +$

$\delta_0/40 < 8\delta_0 \leqslant r_G/\sqrt{N}$, so $e^{ix}W(x)$, an element of G, lies in the chart domain. For $x \in K$, $e^{iF(x)} = e^{ix}W(x)$ gives $|e^{iF(x)} - e^{ix}| = |W(x) - \mathbf{1}| \leqslant |H|$ and $|e^{iF(x)} - \mathbf{1}| \leqslant 7\delta_0 + |H|$, so $|F(x)| \leqslant \frac{\pi}{2}(7\delta_0 + \delta_0/40) < 11.1\,\delta_0 < 12\delta_0$ ($\|F(x)\|_{\mathrm{op}} < \pi$, $F(x)$ lying in the chart ball) and $|F(x) - x| \leqslant 2|e^{iF(x)} - e^{ix}| \leqslant 2|H| \leqslant \delta_0/20$, both $F(x)$ and x lying in $\{|X| \leqslant 12\delta_0\}$.

The continuous map $T(x) = x - F(x) + y$ maps K into itself, $|T(x)| \leqslant \delta_0/20 + 6.3\,\delta_0 < 7\delta_0$, so by Brouwer's fixed point theorem it has a fixed point $x \in K$, that is $F(x) = y$; then $A' = x/g_0 \in \mathcal{W}_b$ and $\psi_b(A') = A$.

(c) Let $x, x' \in \mathfrak{B}_b$ with $F(x) = F(x') = y$ and $y/g_0 \in \mathcal{T}_b$. Then $e^{ix} = e^{iy}W(x)^{-1}$ gives $|e^{ix} - \mathbf{1}| \leqslant |e^{iy} - \mathbf{1}| + |H| < 4\delta_0 + \frac{1}{40}\delta_0$, so $|x| \leqslant \frac{\pi}{2} \cdot \frac{161}{40}\delta_0 < 6.4\,\delta_0$, and likewise $|x'| < 6.4\,\delta_0$; both lie in K. From $e^{ix}W(x) = e^{ix'}W(x')$, $e^{ix} - e^{ix'} = e^{ix'}(W(x')W(x)^{-1} - \mathbf{1})$ with $W(x')W(x)^{-1} = e^{i(\widetilde{\omega}(x) - \widetilde{\omega}(x'))H}$, so $|e^{ix} - e^{ix'}| \leqslant |\widetilde{\omega}(x) - \widetilde{\omega}(x')|\,|H| \leqslant \frac{6}{\delta_0}|x - x'| \cdot \frac{\delta_0}{40} = \frac{3}{20}|x - x'|$, while $|e^{ix} - e^{ix'}| \geqslant \frac{1}{2}|x - x'|$; hence $x = x'$.

For the differential at such an $x \in K$: with $P(x) := e^{ix}W(x)$, $F = \log \circ P$, and $D\log$ is invertible on the chart domain, so it suffices that $DP(x)$ is injective; $DP(x)[v] = D(e^{i\bullet})(x)[v]\,W(x) + e^{ix}DW(x)[v]$ with $|e^{ix}DW(x)[v]| = |D\widetilde{\omega}(x)[v]|\,|H| \leqslant \frac{3}{20}|v|$ and $|D(e^{i\bullet})(x)[v]\,W(x)| = |iv + DR(x)[v]| \geqslant \frac{1}{2}|v|$, so $DP(x)[v] = 0$ forces $v = 0$.

The consequences: Ψ_ξ is bondwise and the identity off S_1^*. For $B \in \mathcal{A}_\xi$, (a) and (b) give A' with $A'(b) \in \mathcal{W}_b$ for $b \in S_1^*$ and $\Psi_\xi(A') = B$, and $(A_{\mathrm{e}}, a) = (A'|_J, A'|_I) \in \mathcal{D}_\xi$ by (3.3); so $\Psi_\xi(\mathcal{D}_\xi) = \mathcal{A}_\xi$, the inclusion $\subseteq$ being the definition. If $\Psi_\xi(A') = \Psi_\xi(A'') = B \in \mathcal{A}_\xi$, then bond by bond $\psi_b(A'(b)) = \psi_b(A''(b)) = B(b) \in \mathcal{T}_b$ by (a), so $A'(b) = A''(b)$ by (c); and $D\Psi_\xi = \bigoplus_{b \in S_1^*} D\psi_b \oplus \mathbf{1}$ is invertible at every point of $\mathcal{D}_\xi$ by (c).

Moreover, by (b), (c) and the inverse function theorem, Ψ_ξ restricted to the open set $\prod_{b \in S_1^*} \psi_b^{-1}(\mathcal{T}_b) \times \mathfrak{g}^{\Omega_1^* \setminus S_1^*}$ ($\mathcal{T}_b$ is open, being defined by strict inequalities between continuous functions) is a C^∞ diffeomorphism onto the open set $\prod_{b \in S_1^*} \mathcal{T}_b \times \mathfrak{g}^{\Omega_1^* \setminus S_1^*}$, which contains $\mathcal{A}_\xi$ by (a); so $\mathcal{D}_\xi$ lies in the former open set, on which $\log|\det D\Psi_\xi|$ is C^∞. *q.e.d.*

The inputs. The bound (3.40) is Lemma D.21 (Section D.8.6).

Bałaban states in [9, p. 251] the bound $|\mathbf{H}_{1,Ax}| < O(1)B_3 \exp(-\delta M R_1)\varepsilon_1$, "much smaller than δ_0, for MR_1 large enough", for $\mathbf{H}_{1,Ax}$ "on $\square'$" for the cubes $\square' \subset S_1$, from the distance between $\square'$ and the support of the argument of $\mathbf{H}_{1,Ax}$ (one layer of M_1-cubes touching $\partial\Lambda_1$) and the exponential decay property of [6, (190)], which is stated there for the Landau gauge and which [9] transfers to the axial gauge ("on the unit lattice it holds for any gauge, for which propagators are exponentially decaying, hence for the axial gauge too").

The lemma derives a bound of this form on every bond of S_1^* from [6, Proposition 9, (172), and (190)], integrated in the Landau gauge, and from the block-local restoration of the axial gauge, with constants C_{H}, c_{H} fixed before g_0; with $\varepsilon_1/\delta_0 \leqslant (1 + \beta_0)A_0/A_1$ ([9, (2.8)] and $\delta_0 = (A_1/A_0)\varepsilon_0$), the tolerance $\delta_0/40$ is a condition "MR_1 large", met for small g_0 at fixed M since $R_1 \to \infty$ (threshold item 5 of Section B.8).

The condition on MR_1 at the tolerance $\delta_0/40$ and the condition of [9] on M_2R_1 are item 5 of Section B.8; the condition $\delta_0 \leqslant \delta_G$ is among the further conditions listed after its enumeration.

The cutoff. The small-field test of [9, (1.18)] for a cube $\square'$ of side $L^2M_2R_1$ is the indicator of $\{\sup_{b \in (\square'^{\sim 2})^*} |A(b)| < r\}$, $r = \delta_0/g_0$, in the bond norm of [9] with the normalisation (4.13),

and it is the product of one indicator per bond of $(\square'^{\sim 2})^*$. Let

$$\mathcal{G}_\xi := \bigcup_{\square' \subset R_1^c} (\square'^{\sim 2})^*, \qquad R_1^c := \Omega_1^{\sim -1} \setminus R_1 \,, \tag{3.41}$$

the union over the good cubes of (1.18). Since $\Lambda_1 \subseteq R_1^c$ and both are unions of such cubes, the first two factors of [9, (1.25)] are the product of the tests over all good cubes, which by idempotence of indicators is $\prod_{b \in \mathcal{G}_\xi} \mathbf{1}_{\{|A(b)|<r\}}$. The cutoff on the integration carrier is

$$\Theta_\xi(a) := \prod_{b \in I} \mathbf{1}_{\{|a(b)|<r\}} \,, \tag{3.42}$$

and the retained cutoff is defined as the remaining factors of (1.25), composed with the change of variables:

$$\chi_{\xi,\mathrm{e}}(A_\mathrm{e}) = \prod_{b \in \mathcal{G}_\xi \cap J} \mathbf{1}_{\{|(\Psi_\xi A_\mathrm{e})(b)|<r\}} \cdot [\chi^{(0)c}(R_1)\chi'_0(S_1)](\Psi_\xi A_\mathrm{e}) \,, \tag{3.43}$$

where $\chi^{(0)c}(R_1)$ is the product of the large-fluctuation tests of (1.18) over the cubes of R_1 and $\chi'_0(S_1)$ the product of the tests of (1.8) over the cubes of S_1, both as functions of the variable A of (1.15) on Ω_1^* (through the free links $e^{ig_0 A(b)} U_1(b)$ in the case of (1.8)) and of the branch data (V_0 on Ω_1^c, U_1 and the local minima $U_{1,\square'}$).

By Lemma 3.19(ii), (3.43) is the complement of Θ_ξ in the exactly transformed cutoff.

The conditional integrand. With $\mathscr{E}_\xi$ the exponent of the second exponential of [9, (1.15)], written out in (3.33), the conditional integrand of Section 3 is

$$\begin{aligned} \mathcal{K}_\xi(a) &:= \begin{cases} \exp\Big(\mathscr{E}_\xi(\Psi_\xi(a + A_\mathrm{e})) + \log|\det D\Psi_\xi(a + A_\mathrm{e})|\Big), & \Theta_\xi(a) = 1, \\ 0, & \Theta_\xi(a) = 0, \end{cases} \\ \mathcal{J}_\xi^{\mathrm{a}}[f] &:= \int f(a)\mathcal{K}_\xi(a)\,\mathrm{d}a \,, \end{aligned} \tag{3.44}$$

the exponent being defined on the box by (B4) and Lemma 3.12: by Lemma 3.19(ii) below, the integrand of [9, (1.15)] in the variables of (1.22), with the cutoff (1.25) in its exact composed form, is $\kappa_\xi\, \chi_{\xi,\mathrm{e}}(A_\mathrm{e})\, \mathcal{K}_\xi(a)$, as used in (3.7) ($\mathscr{E}_\xi$ and κ_ξ are the exponent and prefactor of [9] as printed, with σ, σ_0 and $z^{(0)}$ for the fixed inner product as explained in the conventions).

The change-of-variables term, Bałaban's $V^{(0)}(S_1)$ ([9, (1.24) and p. 252]), is exact inside $\mathcal{K}_\xi$ and is a constant on the branch (Lemma 3.17).

The exterior interpolation. Let $H = H_{\xi,II}$. For $t \in [0,1]$ the interpolated objects are

$$\begin{aligned} P_{\xi,t}(a) &:= \sum_{\mathfrak{p} \text{ near}} \mathfrak{p}(a + tA_\mathrm{e}) \,, \qquad j_\xi(a) := \langle a, H_{\xi,IJ} A_\mathrm{e} \rangle \,, \\ K_{\xi;t}(a) &:= \mathbf{1}_{\mathcal{D}_I}(a) \exp\left(-\frac{1}{2}\langle a, Ha \rangle - t j_\xi(a) + P_{\xi,t}(a) \right) \\ &\qquad (\mathcal{D}_I := \{a \,|\, |a(b)| < r \ \forall b \in I\} \text{ the open box}, \ \mathbf{1}_{\mathcal{D}_I} = \Theta_\xi), \\ I_{\xi;t} &:= \int K_{\xi;t}(a)\,\mathrm{d}a, \qquad \mathrm{d}\nu_{\xi;t} := \frac{K_{\xi;t}}{I_{\xi;t}}\,\mathrm{d}a \,, \end{aligned} \tag{3.45}$$

the sum running over the near primitives of Section 3.3 (all three kinds). By Lemma 3.12 and Lemma 3.17 (i), each near primitive is defined and analytic in (a,t) on the enlarged box at the inputs $a + tA_\mathrm{e}$, $t \in [0,1]$; $K_{\xi;t}$ (zero off the open box) is therefore bounded and Borel

on $\mathfrak{g}^I$, positive on the open box, and $0 < I_{\xi;t} < \infty$. On the open box $P_{\xi,0}(a) = \sum_Y \mathbf{V}_\xi(Y,a)$ is the centred potential (3.37), and $K_{\xi;0}$ is the centred kernel of Section 3.5. This is the interpolation of [9, (3.27)–(3.29)] in the exterior field, restricted to the terms that depend on a.

The Wilson function and the centred coefficient are

$$f_\xi(a) \coloneqq W_\chi(C; U_\xi(a)), \qquad F_\xi(t) \coloneqq \int f_\xi \, \mathrm{d}\nu_{\xi;t} \,, \tag{3.46}$$

with $U_\xi(a)$ the reconstructed fine field (3.26) at the actual retained data, including the change of variables; it is not interpolated. By Lemmas 3.12 and 3.9, f_ξ is defined and analytic on the enlarged box: each contour link $U_\xi(a)(r)$ depends on a only through $a|_{\mathcal{F}_r \cap I}$, through a row within $2R_{\mathrm{c}}$ of I when $\mathcal{F}_r$ meets I, and is constant in a otherwise. Off the enlarged box f_ξ is extended by zero, which affects no integral against $\nu_{\xi;t}$.

Lemma 3.17 (The Collar)**.** *On every branch:*

1. $\Psi_\xi(A_{\mathrm{e}})(b) = A_{\mathrm{e}}(b)$ *for every* $b \in J$ *with* $d(b,I) \leqslant 2R_{\mathrm{c}}$*; every near primitive and every entry of* $H_{\xi,IJ}$ *involves* A_{e} *only on such bonds, so* $\mathfrak{p}(a + t\Psi_\xi(A_{\mathrm{e}})) = \mathfrak{p}(a + tA_{\mathrm{e}})$ *for near* $\mathfrak{p}$ *and* $\langle a, H_{\xi,IJ}\Psi_\xi(A_{\mathrm{e}})\rangle = j_\xi(a)$*. For every row* c *within* $2R_{\mathrm{c}}$ *of* I *the source phase on the bonds of* $\mathcal{E}_c$ *is, componentwise,*

$$\begin{aligned} q_U(\Psi_\xi(a + A_{\mathrm{e}}))(b) &= q_U(a + A_{\mathrm{e}})(b) \\ &= [g_0 \mathsf{C}_U(a + A_{\mathrm{e}}) - h_U \widetilde{D}_U(g_0 \mathsf{C}_U(a + A_{\mathrm{e}}))](b) \qquad (b \in \mathcal{E}_c) \,. \end{aligned} \tag{3.47}$$

2. *With the branch constant*

$$c_\xi = \mathscr{E}_\xi(0) - \frac{1}{2}\langle \Psi_\xi(A_{\mathrm{e}}), H_\xi \Psi_\xi(A_{\mathrm{e}})\rangle + \sum_{\mathfrak{p} \text{ far}} \mathfrak{p}(\Psi_\xi(A_{\mathrm{e}})) + \log|\det D\Psi_\xi(A_{\mathrm{e}})| \,, \tag{3.48}$$

where $\mathscr{E}_\xi(0) = v(0)$ *and the far primitives are evaluated at the retained field alone,*

$$\mathcal{K}_\xi(a) = e^{c_\xi} K_{\xi;1}(a) \quad (a \in \mathfrak{g}^I), \qquad \mathcal{J}^{\mathrm{a}}_\xi[f] = e^{c_\xi} I_{\xi;1} \int f \, \mathrm{d}\nu_{\xi;1}, \qquad \frac{\mathcal{J}^{\mathrm{a}}_\xi[f_\xi]}{\mathcal{J}^{\mathrm{a}}_\xi[1]} = F_\xi(1) \,. \tag{3.49}$$

Proof. (i) By Lemma 3.19(i), a bond within $2R_{\mathrm{c}}$ of I is not in S_1^*, where Ψ_ξ is the identity. A near primitive has all its inputs within $4L+3 \leqslant 2R_{\mathrm{c}}$ of I (Section 3.3); an entry $(H_{\xi,IJ})_{bb'}$ with $b \in I$ vanishes unless $d(b,b') \leqslant \rho_L = 4L < 2R_{\mathrm{c}}$ (Lemma 3.7), since $R_{\mathrm{c}} \geqslant 2R_{\mathrm{loc}} + 1 = 4L + 5$.

The phase on a row c within $2R_{\mathrm{c}}$ of I depends on the free vector on $\mathcal{E}_c \cap E_*$ (Lemma 3.9), and every bond of $\mathcal{E}_c$ is within $2R_{\mathrm{c}}$ of I by the definition of the row distance (Section 3.3), so Ψ_ξ is the identity on these bonds; on $\mathcal{E}_c$ only the row c contributes to $h_U \widetilde{D}_U$ (Lemma 3.9), which gives (3.47) as an identity between the $\mathcal{E}_c$-components.

(ii) Let $A^\Psi_{\mathrm{e}} \coloneqq \Psi_\xi(A_{\mathrm{e}}) \in \mathfrak{g}^J$, so that $\Psi_\xi(a + A_{\mathrm{e}}) = a + A^\Psi_{\mathrm{e}}$ by (3.27). By Lemma 3.14,

$$\mathscr{E}_\xi(a + A^\Psi_{\mathrm{e}}) = \mathscr{E}_\xi(0) - \frac{1}{2}\langle a, Ha\rangle - \langle a, H_{\xi,IJ} A^\Psi_{\mathrm{e}}\rangle - \frac{1}{2}\langle A^\Psi_{\mathrm{e}}, H_\xi A^\Psi_{\mathrm{e}}\rangle + \sum_{\mathfrak{p} \text{ near}} \mathfrak{p}(a + A^\Psi_{\mathrm{e}}) + \sum_{\mathfrak{p} \text{ far}} \mathfrak{p}(A^\Psi_{\mathrm{e}}) \,,$$

where the far primitives do not depend on a because their free-input sets miss I. By (i) the third term is $-j_\xi(a)$ and the near sum is $P_{\xi,1}(a)$. The Jacobian factor depends on $(a + A_{\mathrm{e}})|_{S_1^*} = A_{\mathrm{e}}|_{S_1^*}$ only.

The constant c_ξ is finite: $\det D\Psi_\xi(A_\mathrm{e}) \neq 0$ by (B4)(ii) (the differential of (1.22) is invertible on $\mathcal{D}_\xi$, and $D\Psi_\xi$ at $a + A_\mathrm{e}$ depends on $A_\mathrm{e}|_{S_1^*}$ only), and the far primitives are finite: their free-input sets miss I, so their values at A_e^Ψ are their values at $a_0 + A_\mathrm{e}^\Psi = \Psi_\xi(a_0 + A_\mathrm{e}) \in \mathcal{A}_\xi$ for any a_0 with $(A_\mathrm{e}, a_0) \in \mathcal{D}_\xi$ (a branch has one), where every row solution is defined and the primitives are finite by (B4).

Collecting the a-independent terms gives $\mathcal{K}_\xi = e^{c_\xi} K_{\xi;1}$, and the two remaining identities follow by integration, dividing by $\mathcal{J}_\xi^{\mathrm{a}}[1] = e^{c_\xi} I_{\xi;1} > 0$. *q.e.d.*

The constant c_ξ depends on the branch data (through A_e, U_1, V_0, and Ω_1) and enters the outer weight $\mathcal{J}_\xi^{\mathrm{a}}[1] = e^{c_\xi} I_{\xi;1}$ of the branch in (3.7); the history expectation (D.20) compares $\mathcal{O}[\mathcal{J}_\xi^{\mathrm{a}}[1]\, F_\xi(1)]$ with $\mathcal{O}[\mathcal{J}_\xi^{\mathrm{a}}[1]]$, the same nonnegative weight on both sides.

Lemma 3.18 (Boundary Response)**.** *For bounded Borel f on $\mathfrak{g}^I$, with $\mathrm{Cov}_\nu(f,l) \coloneqq \int fl\,\mathrm{d}\nu - \int f\,\mathrm{d}\nu \int l\,\mathrm{d}\nu$,*

$$\int f\,\mathrm{d}\nu_{\xi;1} = \int f\,\mathrm{d}\nu_{\xi;0} + \mathcal{B}_\xi[f], \qquad \mathcal{B}_\xi[f] = \int_0^1 \mathrm{Cov}_{\nu_{\xi;t}}(f, -j_\xi + \partial_t P_{\xi,t})\,\mathrm{d}t\,, \tag{3.50}$$

where $\partial_t P_{\xi,t}(a) = \sum_{\mathfrak{p}\ \mathrm{near}} D\mathfrak{p}(a + tA_\mathrm{e})[A_\mathrm{e}]$.

Proof. For a in the box the exponent $E_t(a) = -\frac{1}{2}\langle a, Ha\rangle - tj_\xi(a) + P_{\xi,t}(a)$ is jointly analytic in (a,t) on a neighbourhood of $\overline{\mathcal{D}_I} \times [0,1]$, since each near primitive is analytic in the free vector on its row polydiscs (Lemma 3.12) and the input $a + tA_\mathrm{e}$ is affine in (a,t) with values in those polydiscs; its t-derivative $-j_\xi(a) + \partial_t P_{\xi,t}(a)$, with $\partial_t P_{\xi,t}(a) = \sum_\mathfrak{p} D\mathfrak{p}(a + tA_\mathrm{e})[A_\mathrm{e}]$ the derivative in the direction A_e, is continuous and hence bounded on $\overline{\mathcal{D}_I} \times [0,1]$.

Hence $t \mapsto \int f K_{\xi;t}\,\mathrm{d}a$ and $t \mapsto I_{\xi;t}$ are C^1 with derivatives $\int f\,\partial_t E_t\, K_{\xi;t}\,\mathrm{d}a$ and $\int \partial_t E_t\, K_{\xi;t}\,\mathrm{d}a$ (differentiation under the integral over the bounded box, dominated by a constant times $K_{\xi;t} \leqslant e^{\sup|E_t|}$), so $F(t) = \int f\,\mathrm{d}\nu_{\xi;t}$ has $F'(t) = \mathrm{Cov}_{\nu_{\xi;t}}(f, \partial_t E_t)$, and integration over $[0,1]$ gives (3.50). *q.e.d.*

By Lemmas 3.17 and 3.18, on every branch

$$\frac{\mathcal{J}_\xi^{\mathrm{a}}[f_\xi]}{\mathcal{J}_\xi^{\mathrm{a}}[1]} = F_\xi(1) = F_\xi(0) + \mathcal{B}_\xi[f_\xi]\,, \tag{3.51}$$

the first term being the centred expectation of Section 3.7 and the second the exterior response, estimated in Appendix D.

Lemma 3.19 (Collar Coverage)**.** *Let $R_{\mathrm{loc}} \coloneqq 2L+2$, which exceeds the ℓ^∞ diameter $2L-1$ of every two-block set $\mathcal{E}_c$, hence of the sets $\mathcal{E}_y$, $\mathcal{F}_r$, $\mathcal{A}_r$ and of the free-input and background sets of the block, row, Haar and Jacobian primitives (those of a plaquette primitive have diameter at most $4L < 2R_{\mathrm{loc}}$, proof of Lemma 3.7), and let $R_\mathrm{c} \geqslant 2R_{\mathrm{loc}} + 1$ be the collar range fixed in Appendix B after* (B.7)*; it depends on L and the group only.*

Under (3.1)*, which include $M \geqslant 8LM_2$ and $LMR_1 > 4R_\mathrm{c} + 4L^2M_2R_1 + 8$, on every branch:*

1. *Every site within ℓ^∞ distance $LMR_1 - 1$ of Λ_1 lies in $\Omega_1^{\sim -1} \setminus R_1' \subseteq R_1^c$; $I \subseteq \mathcal{G}_\xi \subseteq \Omega_1^*$; and every free bond b of Ω_1 with $d(b,I) \leqslant 3R_\mathrm{c}$ lies in $\mathcal{G}_\xi \setminus S_1^*$.*

2. *The cutoff (1.25) composed with the change of variables, as a function of* $A' = a + A_{\mathrm{e}}$, *equals* $\Theta_\xi(a)\chi_{\xi,\mathrm{e}}(A_{\mathrm{e}})$ *with* $\chi_{\xi,\mathrm{e}}$ *of* (3.43), *and* $\chi_{\xi,\mathrm{e}}$ *depends on* A_{e} *and the branch data only.*

3. $|A_{\mathrm{e}}(b)| < r$ *for every* $b \in J$ *with* $d(b, I) \leqslant 3R_{\mathrm{c}}$*; that is,*

$$r_{\mathrm{e},\xi} = \max(\{|A_{\mathrm{e}}(b)| \,|\, b \in J,\ d(b, I) \leqslant 3R_{\mathrm{c}}\} \cup \{0\}) < r\ . \tag{3.52}$$

4. *For an admissible pair and a branch value of* A_{e}, $\{a \in \mathfrak{g}^I : (A_{\mathrm{e}}, a) \in \mathcal{D}_\xi\} = \{a : \Theta_\xi(a) = 1\}$, *the open box of* (3.42).

Proof. (i) Ω_1 and R_1' are unions of LMR_1-cubes ([9, (1.10), p. 251]) and $\Lambda_1 \subseteq \Omega_1^{\sim -2} \cap (R_1'^{\sim})^c$, so every site within ℓ^∞ distance $LMR_1 - 1$ of Λ_1 lies in $\Omega_1^{\sim -1} \setminus R_1' \subseteq R_1^c$.

Let b be a free bond of Ω_1 with $d(b, I) \leqslant 3R_{\mathrm{c}}$: its initial vertex is within $3R_{\mathrm{c}}+1$ of a site of Λ_1, so the cube $\square'$ of side $L^2M_2R_1$ containing it lies within $3R_{\mathrm{c}} + 1 + L^2M_2R_1 < LMR_1 - 1$ of Λ_1 ((3.1)), hence $\square' \subseteq R_1^c$ is a good cube; b has an endpoint in $\square' \subseteq \square'^{\sim 2}$ and is not designated, so $b \in (\square'^{\sim 2})^* \subseteq \mathcal{G}_\xi$. By (3.39), $d(I, S_1^*) \geqslant LMR_1 - 2 > 3R_{\mathrm{c}}$, so $b \notin S_1^*$.

Taking $d(b, I) = 0$ gives $I \subseteq \mathcal{G}_\xi$ (a bond of Λ_1 is a bond of Ω_1, and a designated bond with an endpoint in Λ_1 belongs to a row of Λ_1 because Λ_1 is a union of blocks, so $I \subseteq \Omega_1^*$).

For $\mathcal{G}_\xi \subseteq \Omega_1^*$: a good cube $\square' \subseteq \Omega_1^{\sim -1}$ has $\square'^{\sim 2} \subseteq \Omega_1$ since $2L^2M_2R_1 < LMR_1$ ($M \geqslant 8LM_2$); a bond of $(\square'^{\sim 2})^*$ has an endpoint in Ω_1, and a designated bond with an endpoint in $\square'^{\sim 2}$, a union of blocks, belongs to a row with an endpoint in $(\square'^{\sim 2})^{(1)}$ and is excluded by the definition of * in (1.8).

(ii) The first two factors of (1.25) are $\prod_{b \in \mathcal{G}_\xi} \mathbf{1}_{\{|A(b)|<r\}}$ at $A = \Psi_\xi(a + A_{\mathrm{e}})$. For $b \in I \subseteq \mathcal{G}_\xi$, $A(b) = a(b)$ by (3.27), giving $\Theta_\xi(a)$; for $b \in \mathcal{G}_\xi \setminus I = \mathcal{G}_\xi \cap J$ (by (i)), $A(b) = (\Psi_\xi A_{\mathrm{e}})(b)$, giving the first factor of (3.43).

The factor $\chi^{(0)c}(R_1)$ involves A on $(\square'^{\sim 2})^*$ for cubes $\square' \subseteq R_1$, and $\chi_0'(S_1)$ involves the free links $e^{ig_0A(b)}U_1(b)$ on bonds $b \in (\square'^{\sim 2})^* \cap \Omega_1^*$ for cubes $\square' \subseteq S_1$, together with V_0 on the bonds of $(\square'^{\sim 2})^*$ with both endpoints off Ω_1, U_1 and $U_{1,\square'}$. All these bonds have both endpoints within $2L^2M_2R_1 + 1$ of $R_1 \cup S_1$, while every site of Λ_1 is at distance at least $LMR_1 > 2L^2M_2R_1 + 1$ from $R_1 \cup S_1$; so none of them has an endpoint in Λ_1, none lies in I, and on them $A = \Psi_\xi(A_{\mathrm{e}})$. Hence the last two factors of (1.25), composed with Ψ_ξ, are functions of $\Psi_\xi(A_{\mathrm{e}})$ and the branch data.

(iii) For $b \in J$ with $d(b, I) \leqslant 3R_{\mathrm{c}}$, (i) gives $b \in \mathcal{G}_\xi \cap J$ and $b \notin S_1^*$, so $\mathbf{1}_{\{|(\Psi_\xi A_{\mathrm{e}})(b)|<r\}} = \mathbf{1}_{\{|A_{\mathrm{e}}(b)|<r\}}$ is a factor of $\chi_{\xi,\mathrm{e}}(A_{\mathrm{e}})$, which is nonzero on a branch (the maximum over the empty set is read as 0).

(iv) Let a_0 be such that $(A_{\mathrm{e}}, a_0) \in \mathcal{D}_\xi$ (a branch), so $B_0 = \Psi_\xi(a_0 + A_{\mathrm{e}}) \in \mathcal{A}_\xi$ and, by (ii), $\Theta_\xi(a_0)\chi_{\xi,\mathrm{e}}(A_{\mathrm{e}}) \neq 0$. Let a be any point of the open box and $B = \Psi_\xi(a + A_{\mathrm{e}}) = a + \Psi_\xi(A_{\mathrm{e}})$ ((3.27); $\Psi_\xi(A_{\mathrm{e}})$ is defined, since B_0 is).

The three conditions of (3.2) hold at B. The cutoff (1.25) at B is $\Theta_\xi(a)\chi_{\xi,\mathrm{e}}(A_{\mathrm{e}}) = \chi_{\xi,\mathrm{e}}(A_{\mathrm{e}}) \neq 0$, by (ii).

Every row equation has a solution at $g_0\mathsf{C}_U B$: on the near rows by Lemma 3.12, since $B = a + A_{\mathrm{e}}$ on their two-block sets by (i) and lies in the row polydiscs by (iii); on the far rows the argument $(g_0\mathsf{C}_U B)|_{\mathcal{E}_c}$ equals $(g_0\mathsf{C}_U B_0)|_{\mathcal{E}_c}$ (Lemma 3.9: it involves B only through $B|_{\mathcal{E}_c \cap E_*}$, and $\mathcal{E}_c$ misses I for a far row, every bond of which is at distance more than $2R_{\mathrm{c}} + 2L - (2L-1) >$

0 from I), where the solution exists because $B_0 \in \mathcal{A}_\xi$. So $U(B)$ is defined, and on the far rows and the bonds of Ω_1^c its links are those of $U(B_0)$.

Finally $U(B) \in D^{(11)}(V, V_0)$. The restriction to Ω_1^c is V_0. The block averages are defined on every row of Ω_1 and satisfy the constraint $\bar{U} = V$ there: $U(B) = e^{iq_U(B)}U_1$ with $q_U(B) = A - h_U\widetilde{D}_U(A)$, $A = g_0\mathsf{C}_U B$, and on every row c the row equation gives $\mathcal{Q}_{U,c}(q_U(B)) = LQ_U q_U(B) + \mathcal{N}_{U,c}(q_U(B)) = LQ_U(A - h_U\widetilde{D}_U(A)) + \widetilde{D}_{U,c}(A) = -\widetilde{D}_{U,c}(A) + \widetilde{D}_{U,c}(A) = 0$, using $Q_U\mathsf{C}_U = 0$ and $LQ_U h_U = \mathbf{1}$; so $\mathsf{M}(U(B))(c) = \mathsf{M}(U_1)(c) = V(c)$ by (3.9) and (B1). The constraint $\bar{V}_0 = V$ on the rows of Ω_1^c involves V_0 and V only and holds because the pair is admissible.

The tests of (1.11): $\chi_1(\Omega_1)$ is a function of V, $\zeta(\Omega_1^c)$ does not depend on a ((3.6)), and both are nonzero at $U(B_0) \in D^{(11)}(V, V_0)$. The factor $\chi_0'(\Omega_1)$ is the product of $\chi_0'(\square')$ over the $L^2M_2R_1$-cubes $\square' \subseteq \Omega_1$ ([9, (1.8)]); the cubes of $S_1 = (\Omega_1 \setminus \Omega_1^{\sim -1}) \cup R_1$ ([9, (1.20)]) give the factor $\chi_0'(S_1)$ of (1.25), nonzero at B by the first condition, and every other cube $\square' \subseteq \Omega_1$ lies in $\Omega_1^{\sim -1} \setminus R_1$, so it is a good cube with $\square'^{\sim 4} \subseteq \Omega_1$ (as $4L^2M_2R_1 < LMR_1$), and its factor $\chi_0'(\square')$ is the second factor of (1.19), equal to one at B by (B3)(ii), the retained factors being nonzero there. Hence $U(B) \in D^{(11)}(V, V_0)$, so $B \in \mathcal{A}_\xi$ and $(A_\mathrm{e}, a) \in \mathcal{D}_\xi$.

Conversely $(A_\mathrm{e}, a) \in \mathcal{D}_\xi$ gives $\Psi_\xi(a + A_\mathrm{e}) \in \mathcal{A}_\xi$, so the cutoff (1.25) is nonzero there, and $\Theta_\xi(a) = 1$ by (ii). *q.e.d.*

The exterior response has Bałaban's integration-by-parts form ([9, (3.28)]). Let

$$\ell_\xi := H_{\xi,IJ}A_\mathrm{e}, \qquad m_\xi := H_{\xi,II}^{-1}\ell_\xi, \qquad D_{m_\xi}f(a) := \frac{\mathrm{d}}{\mathrm{d}s}f(a + sm_\xi)\Big|_{s=0}, \tag{3.53}$$

let $\mathcal{D}_I$ be the open box of (3.45), with outward unit normal $n(a)$ and surface measure $\mathrm{d}S$ for the Euclidean structure of $\langle \bullet, \bullet \rangle$ on $\mathfrak{g}^I$ (for which $\mathrm{d}a$ is Lebesgue measure), defined almost everywhere on its Lipschitz boundary (a product of balls of the norm $|\bullet|$), and

$$\mathsf{Q}_{\xi,t}(a) = -\frac{1}{2}\langle a, H_{\xi,II}a\rangle - t\langle \ell_\xi, a\rangle + P_{\xi,t}(a), \quad \mathcal{T}_{\xi,t}[f] = \frac{1}{I_{\xi;t}}\int_{\partial\mathcal{D}_I} f(a)e^{\mathsf{Q}_{\xi,t}(a)}\langle m_\xi, n(a)\rangle\,\mathrm{d}S(a)\,, \tag{3.54}$$

so that $K_{\xi;t} = \mathbf{1}_{\mathcal{D}_I}e^{\mathsf{Q}_{\xi,t}}$ and $j_\xi(a) = \langle \ell_\xi, a\rangle$.

Lemma 3.20 (Integration by Parts with the Hard Cutoff)**.** *Under the conclusions of Lemma 3.8, $H_{\xi,II}$ is invertible and m_ξ is defined. If f is C^1 on a neighbourhood of $\overline{\mathcal{D}_I}$, then for every $t \in [0,1]$, with expectations under $\nu_{\xi;t}$,*

$$\mathrm{Cov}(f, -j_\xi + \partial_t P_{\xi,t}) = -\mathbb{E}[D_{m_\xi}f] + \mathrm{Cov}(f, \partial_t P_{\xi,t} - D_{m_\xi}P_{\xi,t}) + \mathcal{T}_{\xi,t}[f] - \mathbb{E}[f]\mathcal{T}_{\xi,t}[1]\,. \tag{3.55}$$

Proof. $e^{\mathsf{Q}_{\xi,t}}$ is C^1 on the enlarged box, a neighbourhood of $\overline{\mathcal{D}_I}$, by Lemma 3.12. The divergence theorem on the bounded Lipschitz domain $\mathcal{D}_I$ gives $\int_{\mathcal{D}_I} D_{m_\xi}(fe^{\mathsf{Q}_{\xi,t}})\,\mathrm{d}a = I_{\xi;t}\mathcal{T}_{\xi,t}[f]$, and $D_{m_\xi}\mathsf{Q}_{\xi,t} = -j_\xi - t\langle m_\xi, \ell_\xi\rangle + D_{m_\xi}P_{\xi,t}$ because $\langle m_\xi, H_{\xi,II}a\rangle = \langle \ell_\xi, a\rangle = j_\xi(a)$. Dividing by $I_{\xi;t}$, $\mathbb{E}[D_{m_\xi}f] + \mathbb{E}[f(-j_\xi - t\langle m_\xi, \ell_\xi\rangle + D_{m_\xi}P_{\xi,t})] = \mathcal{T}_{\xi,t}[f]$. Subtracting the same identity for $f = 1$ multiplied by $\mathbb{E}[f]$ cancels the constant $-t\langle m_\xi, \ell_\xi\rangle$, and the two covariances combine as stated. *q.e.d.*

Since f_ξ is analytic on the enlarged box (after (3.46)), Lemma 3.20 applies to f_ξ. The estimate of the right-hand side is Lemma D.6 in Appendix D.

3.5 A Fixed Walk Expansion of the Centred Kernel

Let $H = H_{\xi,II}$ and $\mathsf{G} = H^{-1}$. Notation of Sections 3.5–3.8:

- $E_0 = E(Z_0)$ denotes the seed bonds (the designated bonds E_0 of (4.3) do not occur here);
- $N_i(Y)$ is a cube count (the measure N_1 of (1.5) does not occur);
- σ_Δ are weakening variables (the Haar density σ does not occur);
- B is the conditioned variable on $\mathfrak{g}^{E_0}$ (the free vector B of Sections 3.2–3.4 does not occur);
- D without argument is a component;
- J_D, J_Z are sets of bonds of the integration carrier (the retained carrier J does not occur);
- δ_M is a decay constant (the Mayer constant δ of Appendix B does not occur);
- $\gamma = \gamma_L/2$ is a spectral bound (arcs γ of the contour do not occur);
- a connected final domain is any nonempty wall-connected union of LM-cubes;
- a_j, a_*, a_W are coefficients and weights, distinct from the integration variable a.

By Lemma 3.8, on every branch

$$\gamma\mathbf{1} \leqslant H \leqslant h_+\mathbf{1}, \qquad \gamma = \frac{\gamma_L}{2}, \quad h_+ = m_L\ , \tag{3.56}$$

and by Lemma 3.7, whose entries $\mathsf{H}^{\Omega_1}_{U_1}(b,c)$ at $b,c \in I$ are those of H, $H(b,c) = 0$ when $d_{\infty,T}(b_-,c_-) > R$, where $R \geqslant \max(\rho_L, 2)$ is fixed so that $H(b,c)$ is determined by the background, by the region Ω_1 and by membership in I within distance R of b_-.

Norms on $\mathfrak{g}^I$ are those of the invariant real inner product, with Hermitian extension on the complexification.

The constants γ, h_+, R depend on L and the group only (with the invariant inner product fixed in the conventions), and every choice below is made before the branch fields and the coupling, in the order γ, h_+, R; then κ_1; then n_*, R_*, C_{w}; then M_1, $\rho_{\mathrm{w}} = R_* + R + 1 + 2M_1$ and M; then g_0. None of $\gamma, h_+, R, \kappa_1, C_{\mathrm{d}}$ depends on M_1, M or the coupling (C_{d} of Lemma 3.24 is a function of γ, h_+, d_G).

An aligned auxiliary cubical partition of side M_1 in fine units is fixed, with cubes $M_1 z + [-M_1/2, M_1/2]^4$; the M-cube partition π_0 and the LM-cube partition π_1 are nested over it, M/M_1 and L are odd integers, every fine period is a multiple of LMR_1 and at least $8M$, and M_1 and M are odd ((3.1): M_1, M_2, M are powers of the odd number L), so that cube faces lie at half-integer coordinates and no lattice site lies on a face. Cubes and neighbourhoods are periodic images; all radii used below are smaller than M, so the periods $\geqslant 8M$ make every neighbourhood a lifted copy of its $\mathbb{Z}^4$ model.

Let $s_H \coloneqq 2h_+$, $\mathsf{A}_H = \mathbf{1} - H/s_H$, $q_H = 1 - \gamma/s_H < 1$, and $V_n = 4(2nR+3)^4$. A block of A_H^n vanishes beyond distance nR, has operator norm at most q_H^n and depends only on coefficients within distance $(n+1)R$ of its first bond, so its block-norm row and column sums are at most $V_n q_H^n$.

Let $\kappa_1 \geqslant 1$ be the constant fixed in Appendix B, Section B.5 ((B.27); it depends on L and the group only), the spatial Cauchy radius being e^{κ_1}; let $a_* \coloneqq e^{81\kappa_1}$, and let n_* satisfy

$$\beta = V_{n_*} q_H^{n_*}, \qquad a_*\beta \leqslant \frac{1}{4} . \tag{3.57}$$

Let $R_* \coloneqq n_* R$, $\mathsf{P}_H = \mathsf{A}_H^{n_*}$, $V_* = V_{n_*}$, and $a_j = 4^{-j}\binom{2j}{j} \in (0,1]$. The binomial series gives

$$\mathsf{G}^{1/2} = s_H^{-1/2} \sum_{r=0}^{n_*-1} \sum_{m=0}^{\infty} a_{n_* m + r} \mathsf{A}_H^r \mathsf{P}_H^m . \tag{3.58}$$

For a free bond b let D_b be the union of the M_1-cubes meeting the closed ball of radius $R_* + R + 1$ about b_-, with one further M_1-layer; it is connected, lies in the ball of radius $\rho_{\mathrm{w}} = R_* + R + 1 + 2M_1$ about b_-, and contains the bonds and coefficient stencils of every block starting at b in (3.58) with an interior margin. The cube side satisfies

$$M \geqslant 4(\rho_{\mathrm{w}} + R + R_* + 2) . \tag{3.59}$$

Then $2\rho_{\mathrm{w}} < M \leqslant LM$, so D_b meets at most $2^4 = 16 \leqslant 81$ cubes of π_1.

A walk is a tuple $\omega = (r; b_0, \dots, b_{m+1})$ with nonzero block product

$$(\mathsf{A}_H^r)_{b_0 b_1} \prod_{j=1}^{m} (\mathsf{P}_H)_{b_j b_{j+1}} , \tag{3.60}$$

and localisation domain $D(\omega) = \bigcup_{j \leqslant m} D_{b_j}$, which is connected and contains both endpoint bonds in its interior.

For an M-cube Q let $Q^{\sim}$ be the union of the 3^4 cubes of π_0 meeting Q, so that $\operatorname{int} Q^{\sim}$ is the open ℓ^∞ neighbourhood of radius M of Q. For a union Z_0 of M-cubes (the seed, selected in Section 3.6) let $\widetilde{Z}_0 \coloneqq \bigcup_{Q \subseteq Z_0} Q^{\sim}$ be its one-M-layer enlargement and Z_0' the union of the LM-cubes meeting $\operatorname{int} Q^{\sim}$ for some cube Q of Z_0. Both are unions over the cubes of Z_0, so both commute with unions of seeds, and $\operatorname{int} Q^{\sim} \subseteq \operatorname{int} Z_0'$ for every cube Q of Z_0, since every LM-cube incident to a point of the open set $\operatorname{int} Q^{\sim}$ meets it.

The cubes $\Delta \in \pi_1$ inside Z_0' carry $\sigma_\Delta = 1$, the others carry complex variables σ_Δ, and the weight of a walk is

$$w_\omega(\sigma) \coloneqq \prod_{\Delta \in \pi_1,\ \Delta \not\subseteq Z_0',\ \Delta \cap D(\omega) \neq \varnothing} \sigma_\Delta . \tag{3.61}$$

The weighted series is

$$\begin{gathered} S_{Z_0}(\sigma)_{b_0 b_{m+1}} \coloneqq s_H^{-1/2} \sum_{r=0}^{n_*-1} \sum_{m=0}^{\infty} a_{n_* m + r} \sum_{b_1, \dots, b_m} w_\omega(\sigma)\, (\mathsf{A}_H^r)_{b_0 b_1} \prod_{j=1}^{m} (\mathsf{P}_H)_{b_j b_{j+1}}, \\ \omega \coloneqq (r; b_0, \dots, b_{m+1}) , \end{gathered} \tag{3.62}$$

the inner sum running over the intermediate bonds; $S_{Z_0}(1) = \mathsf{G}^{1/2}$ by (3.58).

Lemma 3.21 (The Walk Series)**.** *On $|\sigma_\Delta| \leqslant e^{\kappa_1}$, $|w_\omega(\sigma)| \leqslant a_*^{m+1}$, and the series $S_{Z_0}(\sigma)$ converges absolutely in both Schur norms, uniformly on the polydisc, with operator norm and absolute block row and column sums at most*

$$C_{\mathrm{w}} = s_H^{-1/2} \frac{n_* V_* a_*}{1 - a_*\beta} . \tag{3.63}$$

It is holomorphic on the polydisc of radii $e^{\kappa_1+(\log 2)/81}$. The bound is independent of Z_0, the volume and the number of its components.

Under a site gauge transformation u of the background, $S_{Z_0}(\sigma)$ formed from $\iota_I^*\mathsf{H}_{U_1^u}^{\Omega_1}\iota_I$ in place of H equals $R_{u,*}S_{Z_0}(\sigma)R_{u,*}^{-1}$, with $R_{u,*}$ restricted to I and the same domains and weights.

Proof. $D(\omega)$ meets at most $81(m+1)$ cubes of π_1, which gives the weight bound. The unweakened grouped products have absolute block row and column sums at most $V_*\beta^m$ after summing the intermediate bonds, so the weighted series is dominated by $s_H^{-1/2}n_*V_*\sum_m(a_*\beta)^m a_*$. Increasing each radius by the factor $e^{(\log 2)/81}$ multiplies a_* by 2 and keeps $a_*\beta \leqslant 1/2$. The gauge law is (3.20) restricted to I (so A_H is conjugated by $R_{u,*}$) applied to each block; the weights and domains do not involve the background. *q.e.d.*

Let $E_0 \coloneqq E(Z_0) = \{b \in I \,|\, b \cap Z_0 \neq \varnothing\}$, let ι_{E_0} be extension by zero and $\Pi_{I\setminus E_0}$ the complementary orthogonal projection, and let

$$C_{E_0} \coloneqq (\iota_{E_0}^* H \iota_{E_0})^{-1}, \qquad \Gamma_{E_0}(\sigma) \coloneqq \iota_{E_0}^* H \Pi_{I\setminus E_0} S_{Z_0}(\sigma)\,. \tag{3.64}$$

Pairings $\langle \bullet, \bullet \rangle$ involving complex vectors are the complex-bilinear extensions of the real pairing throughout.

Lemma 3.22 (Gaussian Conditioning)**.** *Let μ_0 be the standard Gaussian measure on $\mathfrak{g}^I$ and $\mu_{C_{E_0}}$ the centred Gaussian measure on $\mathfrak{g}^{E_0}$ of covariance C_{E_0}. For bounded Borel F on $\mathfrak{g}^{E_0}$,*

$$\int F(a|_{E_0})\,\mathrm{d}\mu_{\mathsf{G}}(a) = \int \left[\int F(B)\, e^{-\frac{1}{2}\langle \Gamma_{E_0}(1)X, C_{E_0}\Gamma_{E_0}(1)X\rangle} e^{-\langle B, \Gamma_{E_0}(1)X\rangle}\,\mathrm{d}\mu_{C_{E_0}}(B)\right]\mathrm{d}\mu_0(X)\,. \tag{3.65}$$

Proof. With $a = \mathsf{G}^{1/2}X$, a has law μ_{G} when X has law μ_0. Given $a|_{I\setminus E_0}$, the conditional law of $a|_{E_0}$ under μ_{G} is Gaussian with covariance C_{E_0} and mean $-C_{E_0}\iota_{E_0}^* H\Pi_{I\setminus E_0}a = -C_{E_0}\Gamma_{E_0}(1)X$ (by completing the square in $\frac{1}{2}\langle a, Ha\rangle$), and its density with respect to $\mu_{C_{E_0}}$ is the displayed factor. This is [10, (2.5)–(2.6)]. *q.e.d.*

The weakened density is

$$e^{-\frac{1}{2}\langle \Gamma_{E_0}(\sigma)X, C_{E_0}\Gamma_{E_0}(\sigma)X\rangle} e^{-\langle B, \Gamma_{E_0}(\sigma)X\rangle}\,\mathrm{d}\mu_{C_{E_0}}(B)\,\mathrm{d}\mu_0(X)\,. \tag{3.66}$$

C_{E_0} is real, positive and independent of σ; its determinant is one fixed positive number on the whole polydisc.

Lemma 3.23 (Component Blocks)**.** *Let $Z \supseteq Z_0'$ be a union of LM-cubes, with $\sigma_\Delta = 0$ for $\Delta \not\subseteq Z$. For each wall-component D of Z let $J_D \coloneqq \{b \in I \,|\, b \subset \operatorname{int} D\}$ and $E_D = E_0 \cap J_D$. Then $E_0 \subseteq \bigsqcup_D J_D$, $E_D = E(Z_0 \cap D)$, initial vertices in different E_D have distance at least $2M-2 > R$, and*

$$C_{E_0} = \bigoplus_D (\iota_{E_D}^* H \iota_{E_D})^{-1}, \qquad \Gamma_{E_0}(\sigma) = \bigoplus_D \Gamma_{E_D}(\sigma|_D)\,, \tag{3.67}$$

where $\Gamma_{E_D}(\sigma|_D)$ is constructed from the seed $Z_0 \cap D$ alone, with final domain D (so $\sigma_\Delta = 0$ for every cube Δ of π_1 not in D) and the same I, H and background. Columns of $\Gamma_{E_0}(\sigma)$ outside $\bigsqcup_D J_D$ vanish, and $\Gamma_{E_0}(\sigma)$ does not depend on σ_Δ for cubes Δ in a component containing no seed cube.

Proof. A point is interior to a union of closed grid cubes exactly when all its incident cubes belong to it, and the incident family is wall connected (any two cubes containing a point are joined by a chain of cubes containing that point, consecutive ones sharing a wall); hence $\operatorname{int} Z = \bigsqcup_D \operatorname{int} D$, a disjoint union of open sets. For an M-cube Q of Z_0 the open neighbourhood $\operatorname{int} Q^{\sim}$ is connected and lies in $\operatorname{int} Z_0' \subseteq \operatorname{int} Z$, so it lies in one component interior; neighbourhoods in different components are disjoint, so their seed cubes have distance at least $2M$, and a bond meeting a seed cube lies in the same component interior. This gives $E_0 \subseteq \bigsqcup J_D$, $E_D = E(Z_0 \cap D)$ and the separation, which by finite range makes $\iota_{E_0}^* H \iota_{E_0}$ block diagonal.

Let $(\Gamma_{E_0}(\sigma))_{eb} = \sum_{b'} H_{eb'} (\Pi_{I\setminus E_0})_{b'b'} S_{Z_0}(\sigma)_{b'b}$ with $e \in E_D$ be an entry. A nonzero term has $d(e_-, b'_-) \leqslant R$; e meets a seed cube Q, so e_- is within 1 of Q and the open ball of radius $R+2$ about e_- lies in the open M-neighbourhood $\operatorname{int} Q^{\sim}$ of Q, because $R+3 \leqslant M$ by (3.59), hence in $\operatorname{int} D$; so $b' \subset \operatorname{int} D$.

A walk from b' to b with $w_\omega \neq 0$ has every cube of π_1 meeting $D(\omega)$ inside Z, so $D(\omega) \subset \operatorname{int} Z$; $D(\omega)$ is connected and contains b', so $D(\omega) \subset \operatorname{int} D$ and $b \in J_D$. Its coefficients, domains and weakening variables are independent of the seed; the cubes with $\sigma_\Delta = 1$ inside D are those of $Z_0' \cap D = (Z_0 \cap D)'$ (Z_0' is the union over the seed cubes Q of the LM-cubes meeting $\operatorname{int} Q^{\sim}$; such a cube meets $\operatorname{int} Z$ at a point whose incident LM-cubes all lie in Z and form a wall-connected family, so it lies in the component containing Q); and $\Pi_{I\setminus E_0}$ agrees with $\Pi_{I\setminus E_D}$ on J_D. This gives the block decomposition and the vanishing of the columns outside $\bigsqcup J_D$.

Finally, a walk contributing to a row $e \in E_D$ has $D(\omega) \subset \operatorname{int} D$, so its weight involves only σ_Δ with $\Delta \subseteq D$; a component without seed cube has no row at all. *q.e.d.*

Lemma 3.24 (Gaussian Payment)**.** *Let $N_M := \lfloor M/(2R_*) \rfloor$, $\delta_M = 2C_{\mathrm{w}} 4^{-N_M}$ and $A_* = h_+(\gamma^{-1/2}+1)$; by* (3.1)*, $\delta_M \leqslant 1$ and $h_+^2 \delta_M^2/\gamma \leqslant 1/8$. $\pi_i(X)$ denotes the set of π_i-cubes of a union X of such cubes. On every branch, for every seed Z_0, every union $Z \supseteq Z_0'$ of LM-cubes with $\sigma_\Delta = 0$ for $\Delta \not\subseteq Z$, and every σ in the closed polydisc $\{|\sigma_\Delta| \leqslant e^{\kappa_1}\ \forall \Delta \in \pi_1(Z) \setminus \pi_1(Z_0')\}$:*

1. *$\|\Gamma_{E_0}(\sigma) - \Gamma_{E_0}(1)\| \leqslant h_+ \delta_M$ and $\|C_{E_0}^{1/2}(\Gamma_{E_0}(\sigma) - \Gamma_{E_0}(1))\| \leqslant h_+ \gamma^{-1/2} \delta_M$, in the operator norms for the Hilbert norms on $\mathfrak{g}_{\mathbb{C}}^I$ and $\mathfrak{g}_{\mathbb{C}}^{E_0}$.*
2. *For $0 \leqslant \alpha_G \leqslant \min(\gamma/2, \gamma^2/(16A_*^2))$ and $J_Z = \bigsqcup_D J_D$, the integral of the absolute value of* (3.66) *against $e^{\alpha_G \|B\|^2/2}$ is at most $\exp(C_{\mathrm{d}}(\alpha_G + \delta_M^2)|J_Z|)$, $C_{\mathrm{d}} = C_{\mathrm{d}}(\gamma, h_+, d_G)$; with $4C_{\mathrm{d}}\delta_M^2 (LM+1)^4 \leqslant 1/2$ (*(3.1)*) and $\alpha_G \leqslant (8C_{\mathrm{d}}(LM+1)^4)^{-1}$ this is at most $e^{N_1(Z)}$, and the integral of the absolute value factors over the components D of Z (the signed density factors by Lemma 3.23).*
3. *For $0 < \gamma_2$ and the constant $c_{\mathrm{n}} > 0$ of the group with $h(X)^2 \geqslant c_{\mathrm{n}} |X|^2$ on $\mathfrak{g}$, $\prod_{b \in \mathsf{P}} \mathbf{1}_{\{|B_b| \geqslant r\}} \leqslant e^{-a_{\mathrm{bad}}|\mathsf{P}|} \exp(\frac{\gamma_2}{2} \sum_{b \in \mathsf{P}} h(B_b)^2)$ with $a_{\mathrm{bad}} := c_{\mathrm{n}} \gamma_2 r^2/2$, for every $\mathsf{P} \subseteq E_0$ (a pointwise inequality, used in the appendix with $\gamma_2 \leqslant \alpha_G$).*

The proof is in Appendix B, Section B.4.

3.6 The Selected Integrals and the Polymer Representation

Connectivity of cubes is wall connectivity ([8], before (0.24)); two domains are compatible when they share neither a cube nor a wall ([10, (2.11)]); $N_i(Y)$ is the number of π_i-cubes of Y, $d_i(Y)$ its tree length ([8, p. 257], written out in (B.1)) and $\ell_i(Y) = d_i(Y) + 5$.

The localisation family $\mathbf{D}_0$ consists of the potential hosts (3.36) and the probe hosts (3.75) below; all are connected unions of M-cubes.

A potential family indexed by $\mathbf{D}_0$ is a collection $\{\mathbf{W}(Y,\bullet)\}_{Y\in\mathbf{D}_0}$ of Borel functions $\mathfrak{g}^I\to\mathbb{C}$, each depending on a only through $a|_{E(Y)}$, $E(Y)=\{b\in I\,|\,b\cap Y\neq\varnothing\}$, and bounded on the local box $\{|a_b|<r\ \forall b\in E(Y)\}$; only the values on the local boxes enter below, because every selected host Y comes with the factors χ_b, $b\in E(Y)\subseteq E(Y_0)$. The potential $\mathbf{V}_\xi$ of (3.37) is such a family; $\mathbf{V}_\xi(Y,\bullet)=0$ on hosts carrying no near primitive.

Definition 3.25 (Selections and Selected Integrals)**.** A selection is a triple $\mathfrak{s}=(\mathcal{D},\mathsf{P},Z)$: a finite $\mathcal{D}\subseteq\mathbf{D}_0$, a finite $\mathsf{P}\subseteq I\setminus E(Y_0)$ where $Y_0:=\bigcup_{Y\in\mathcal{D}}Y$, and a final domain Z. Its seed is $Z_0=Y_0\cup\bigcup\{Q\in\pi_0\,|\,Q\cap b\neq\varnothing\text{ for some }b\in\mathsf{P}\}$, with $E_0=E(Z_0)$ and Z_0' as in Section 3.5, and a final domain is a union $Z\supseteq Z_0'$ of LM-cubes.

The selected integral of a bounded Borel F on $\mathfrak{g}^{E_0}$ is

$$\begin{aligned}\mathcal{A}^{\mathrm{f}}_{\mathfrak{s}}[F]:=&\prod_{\Delta\in\pi_1(Z)\setminus\pi_1(Z_0')}\int_0^1\mathrm{d}s_\Delta\oint_{|\sigma_\Delta|=e^{\kappa_1}}\frac{\mathrm{d}\sigma_\Delta}{2\pi i(\sigma_\Delta-s_\Delta)^2}\\&\cdot\prod_{Y\in\mathcal{D}}\int_0^1\mathrm{d}t_Y\int\mathrm{d}\mu_0(X)\int\mathrm{d}\mu_{C_{E_0}}(B)\\&\cdot e^{-\frac12\langle\Gamma_{E_0}(\sigma)X,C_{E_0}\Gamma_{E_0}(\sigma)X\rangle}e^{-\langle B,\Gamma_{E_0}(\sigma)X\rangle}(-1)^{|\mathsf{P}|}\\&\cdot\prod_{b\in E(Y_0)}\chi_b(B)\prod_{b\in\mathsf{P}}(1-\chi_b(B))\prod_{Y\in\mathcal{D}}\mathbf{W}(Y,B)e^{t_Y\mathbf{W}(Y,B)}\,F(B)\,,\end{aligned}\tag{3.68}$$

with $\chi_b(B):=\mathbf{1}_{\{|B_b|<r\}}$ in the bond norm of [9] (the indicator of (3.42)) and $\sigma_\Delta:=0$ for $\Delta\not\subseteq Z$.

The vacuum activity of a connected final domain Z is

$$k^{\mathrm{f}}_\xi(Z):=\sum_{(\mathcal{D},\mathsf{P})\,|\,Z_0\neq\varnothing,\ Z_0'\subseteq Z}\mathcal{A}^{\mathrm{f}}_{(\mathcal{D},\mathsf{P},Z)}[1]\,,\tag{3.69}$$

and $\Xi^{\mathrm{f}}_\xi:=\sum_{\mathcal{P}\text{ compatible}}\prod_{Z\in\mathcal{P}}k^{\mathrm{f}}_\xi(Z)$ (empty family included), $(\Xi^{\mathrm{f}}_\xi)^{[Y]}$ the same sum over families compatible with Y.

The selected integral is finite: the integrand is holomorphic in σ on the polydisc of Lemma 3.21 for fixed (X,B,t), its absolute value is integrable in (X,B) uniformly on the closed polydisc by Lemma 3.24(ii) together with the boundedness of F and of each $\mathbf{W}(Y,\bullet)$ on its local box (enforced by the factors χ_b, $b\in E(Y_0)$), and the contours have positive distance from $[0,1]$.

Every selected datum depends only on B, since $E(Y)\subseteq E_0$ for $Y\in\mathcal{D}$ and $\mathsf{P}\subseteq E_0$. A fine bond meets at most two M-cubes (no site lies on a face), so at most $2|\mathsf{P}|$ cubes of the seed are not cubes of Y_0.

The seed of a union of selections (disjoint host families and bad-bond sets) is the union of the seeds, and $\widetilde{Z}_0$, Z_0' are unions over the seed cubes (Section 3.5), so all three commute with unions.

Lemma 3.26 (Polymer Representation)**.** *Let* $\mathbf{W}$ *be a potential family indexed by* $\mathbf{D}_0$ *and* $\mathfrak{n}_\xi=(2\pi)^{d_G|I|/2}\det(H)^{-1/2}$*. Then*

$$\int\Theta_\xi(a)\exp\Big(-\frac12\langle a,Ha\rangle+\sum_{Y\in\mathbf{D}_0}\mathbf{W}(Y,a)\Big)\mathrm{d}a=\mathfrak{n}_\xi\,\Xi^{\mathrm{f}}_\xi[\mathbf{W}]\,,\tag{3.70}$$

with the activities of Definition 3.25 formed from $\mathbf{W}$*, and the sum* $\Xi^{\mathrm{f}}_{\xi}[\mathbf{W}]$ *is finite (finitely many selections at fixed volume). In particular, for* $\mathbf{W} = \mathbf{V}_{\xi}$*,*

$$I_{\xi;0} = \mathfrak{n}_{\xi}\, \Xi^{\mathrm{f}}_{\xi} , \qquad \Xi^{\mathrm{f}}_{\xi} = \frac{I_{\xi;0}}{\mathfrak{n}_{\xi}} > 0 . \tag{3.71}$$

Proof. The left side is $\mathfrak{n}_{\xi} \int \Theta_{\xi} e^{\sum_Y \mathbf{W}(Y,\bullet)} \mathrm{d}\mu_{\mathsf{G}}$. The expansion of each $e^{\mathbf{W}(Y)} = 1 + \int_0^1 \mathbf{W}(Y) e^{t_Y \mathbf{W}(Y)} \mathrm{d}t_Y$ gives the sum over $\mathcal{D}$. With $\Theta_{\xi} = \prod_{E(Y_0)} \chi_b \prod_{I \setminus E(Y_0)} (1 - (1 - \chi_b))$, the expansion of the second product gives the sum over P with the sign $(-1)^{|\mathsf{P}|}$. The remaining integrand depends on $a|_{E_0}$ only (each $\mathbf{W}(Y, \bullet)$, $Y \in \mathcal{D}$, through $a|_{E(Y)}$ with $E(Y) \subseteq E_0$, and it is evaluated on the local box of Y because of the factors χ_b, $b \in E(Y_0)$), so Lemma 3.22 applies and produces (3.66) at $\sigma = 1$.

For fixed (X, B, t) the density (3.66) is holomorphic in σ on the polydisc (Lemma 3.21); for each $\Delta \in \pi_1$ not in Z_0' its value at $\sigma_\Delta = 1$ is the value at $\sigma_\Delta = 0$ plus $\int_0^1 \partial_{s_\Delta} \mathrm{d}s_\Delta$. Expanding the product over these cubes, labelling each term by $Z = Z_0' \cup \{\text{differentiated cubes}\}$, and representing each derivative by its Cauchy integral along $|\sigma_\Delta| = e^{\kappa_1}$ gives pointwise identities in (X, B, t); the resulting parameter integrals may be taken outside the Gaussian integrals by Fubini, since the absolute value of the density is integrable uniformly on the polydisc (Lemma 3.24(ii)) and the other factors are bounded. This is (3.68).

Every cube of Z_0' is wall-connected inside Z_0' to a cube containing a seed cube: a cube Δ of Z_0' meets $\operatorname{int} Q^{\sim}$ for a seed cube Q at a point x; since $\operatorname{int} Q^{\sim}$ is open, x may be taken in the interior of Δ, so x lies in an M-cube $Q' \subseteq Q^{\sim} \cap \Delta$, and Q' is joined to Q by a chain of face-adjacent M-cubes of $Q^{\sim}$, whose LM-cubes start with Δ, are consecutively equal or wall adjacent, and all belong to Z_0' (each contains an M-cube of $Q^{\sim}$, whose interior lies in $\operatorname{int} Q^{\sim}$). Hence a wall-component of Z either contains a seed cube or consists of differentiated cubes only; in the second case $\Gamma_{E_0}(\sigma)$ does not depend on its σ-variables (Lemma 3.23), the Cauchy integral of a function constant in σ_Δ vanishes, and the term is zero. So only final domains all of whose components contain a seed cube contribute.

For these, Lemma 3.23 factors the density and the Gaussian measures over the components D of Z ($\mu_{C_{E_0}} = \bigotimes_D \mu_{C_{E_D}}$ and the density is a product by the block decomposition; the X-integral factors because each block of Γ_{E_0} reads X on J_D only, and the remaining Gaussian factor of μ_0 integrates to one), and the parameter integrals are indexed by cubes and hosts, each inside one component. The cutoff, potential and sign factors factor because each selected host is a connected union of M-cubes inside Z_0, hence inside one component, and each bad bond lies in the component containing its (at most two, face-adjacent) incident cubes.

Given Z with components $D_1, \dots, D_m$, the map $(\mathcal{D}, \mathsf{P}) \mapsto ((\mathcal{D}_j, \mathsf{P}_j))_j$, with $\mathcal{D}_j$ the hosts inside D_j and P_j the bad bonds inside D_j, is a bijection from the selections with final domain Z (all $Z_0' \subseteq Z$, every component seeded) onto the tuples of selections $(\mathcal{D}_j, \mathsf{P}_j, D_j)$ with nonempty seeds and final domains D_j: the inverse is the union, because the seed rule, the enlargement and the coarse cover commute with unions and $(Z_0 \cap D_j)' \subseteq D_j$ (it is wall-connected to its seed inside $Z_0' \subseteq Z$). Summing over Z gives the compatible-family sum, in which the empty selection contributes $\int \mathrm{d}\mu_{\mathsf{G}} = 1$.

Equation (3.71) is the case $\mathbf{W} = \mathbf{V}_{\xi}$: $K_{\xi;0} = \Theta_{\xi} \exp(-\frac{1}{2}\langle a, Ha \rangle + \sum_Y \mathbf{V}_{\xi}(Y, a))$ by (3.45) and (3.37) (on the open box $\sum_Y \mathbf{V}_{\xi}(Y, a) = P_{\xi,0}(a)$), and positivity holds because $K_{\xi;0} > 0$ on the open box. *q.e.d.*

Lemma 3.27 (Activity Bounds)**.** *With the constants* $\lambda_*, \varepsilon_* > 0$ *and* $\theta \geqslant 0$ *of Appendix B, Section B.5 (fixed there before the coupling together with its auxiliary constants* $\kappa, \delta, \alpha_6, \omega_*$*,*

independently of M_1 and M), and g_0 below its threshold, for $\mathbf{W} = \mathbf{V}_\xi$ on every branch and every connected final domain Z:

$$|k^{\mathrm{f}}_\xi(Z)| \leqslant \varepsilon_* e^{-\lambda_* \ell_1(Z)}, \qquad \sum_{Z' \not\sim Z} |k^{\mathrm{f}}_\xi(Z')| e^{(1+6\theta)N_1(Z')} \leqslant \frac{1}{8} N_1(Z) \,, \tag{3.72}$$

where Z' ranges over the connected final domains and $Z' \not\sim Z$ means incompatible (a domain is incompatible with itself). The same holds for the reference data of Section 3.8 (Lemma B.16).

Lemma 3.28 (Vacuum Clusters)**.** *Under Lemma 3.27, for an ordered tuple $\mathfrak{c} = (D_1, \dots, D_m)$, $m \geqslant 1$, of connected final domains with Ursell coefficient $\varphi^T(\mathfrak{c}) = \sum_{\mathcal{G}} (-1)^{|\mathcal{G}|}$, the sum over the connected spanning subgraphs $\mathcal{G}$ of the incompatibility graph on $\{1, \dots, m\}$ ($j \neq j'$ joined when $D_j \not\sim D_{j'}$; a domain is incompatible with itself, so repeated domains are joined; $\varphi^T = 1$ for $m = 1$), let*

$$\Phi^{\mathrm{v}}_\xi(\mathfrak{c}) := \frac{\varphi^T(\mathfrak{c})}{m!} \prod_{j=1}^{m} k^{\mathrm{f}}_\xi(D_j) \,. \tag{3.73}$$

For every finite set $\mathcal{A}$ of connected final domains,

$$\sum_{\mathfrak{c} \ \mathit{hits}\ \mathcal{A}} |\Phi^{\mathrm{v}}_\xi(\mathfrak{c})| e^{2\theta \sum_j N_1(D_j)} \leqslant \frac{1}{8} \sum_{D \in \mathcal{A}} N_1(D) \,,$$

where $\mathfrak{c}$ hits $\mathcal{A}$ when some D_j is incompatible with a member of $\mathcal{A}$, and

$$\frac{(\Xi^{\mathrm{f}}_\xi)^{[\cup \mathcal{A}]}}{\Xi^{\mathrm{f}}_\xi} = \exp\Big(- \sum_{\mathfrak{c} \ \mathit{hits}\ \mathcal{A}} \Phi^{\mathrm{v}}_\xi(\mathfrak{c}) \Big) \,. \tag{3.74}$$

The proof, including the identity (3.74), is in Appendix B, Section B.6, by the Kotecký–Preiss criterion with the pinned-tree bound.

3.7 Probe Kernels and the Centred Expectation

For each $b \in I$ let $\mathcal{R}_b = \{r \text{ a bond of } \Omega_1 \,|\, b \in \mathcal{F}_r\}$ be its forward influence set (the bonds whose reconstructed link depends on $a(b)$, Lemma 3.9); it lies within ℓ^∞ distance $2L+1$ of b. The set $\{b\} \cup \mathcal{R}_b$ is lifted to $\mathbb{Z}^4$ about b_- (its diameter is below every period), and

$$\begin{aligned} Y_b = \pi_T(&\text{the smallest rectangle of } M\text{-cubes containing the one-fine-layer} \\ &\text{neighbourhood of the lift, with one } M\text{-layer adjoined}) \,; \end{aligned} \tag{3.75}$$

the lift together with its one-fine-layer neighbourhood has ℓ^∞ extent at most $2(2L+1)+3 = 4L+5 < M$ in every direction ($M \geqslant 8LM_2 \geqslant 8L^2$), so it meets at most two M-cubes per direction and the rectangle has side at most 4 cubes, of length $4M < 8M$, so π_T is injective on it. Y_b is a connected member of $\mathbf{D}_0$, contains b and $\mathcal{R}_b$ in its interior, and $N_0(Y_b) \leqslant 4^4 \leqslant s_{\mathrm{p}} = 5^4$.

For a union R of LM-cubes its eligible labels are

$$\mathsf{L}_R := \{b \in I \,|\, Y_b \subseteq R\} \,, \tag{3.76}$$

and $\overline{\mathcal{D}}_{\mathsf{L}}$ is the closed box $|a_b| \leqslant r$, $b \in \mathsf{L}$, in $\mathfrak{g}^{\mathsf{L}}$. Each Y_b is a nonempty wall-connected union of M-cubes, so $Y_b \subseteq R$ places it inside exactly one wall-component of R; hence L_R is the disjoint union of the L_D over the components D of R.

For $\lambda \in \mathfrak{g}_{\mathbb{C}}^{I}$, with the complex bilinear extension of the real pairing, the probe family is

$$L_Y(\lambda, a) \coloneqq \sum_{b \,|\, Y_b = Y} \langle \lambda_b, a_b \rangle, \qquad \widetilde{\mathbf{V}}_{\xi,\lambda}(Y, a) \coloneqq \mathbf{V}_\xi(Y, a) + L_Y(\lambda, a)\,, \tag{3.77}$$

a potential family indexed by $\mathbf{D}_0$ for each λ (both terms depend on $a|_{E(Y)}$ only, since $Y_b = Y$ gives $b \in E(Y)$, and both are bounded on the local box of Y), entire in λ. Its partition function is

$$\mathscr{Z}^{\mathrm{c}}_\xi(\lambda) \coloneqq \int e^{\sum_{b\in I}\langle \lambda_b, a_b\rangle} K_{\xi;0}(a)\,\mathrm{d}a = \mathfrak{n}_\xi \sum_{\mathcal{P}\ \text{compatible}} \prod_{D\in\mathcal{P}} k^{\mathrm{f}}_{\xi,D}(\lambda)\,, \tag{3.78}$$

by Lemma 3.26 applied to $\widetilde{\mathbf{V}}_{\xi,\lambda}$, where $k^{\mathrm{f}}_{\xi,D}(\lambda)$ denotes the activity formed from that family; it is entire in λ (a finite sum of integrals of entire functions of λ with integrable majorants on compact sets of λ) and $k^{\mathrm{f}}_{\xi,D}(0) = k^{\mathrm{f}}_\xi(D)$.

Lemma 3.29 (Probe Measures)**.** *On every branch, for every connected final domain D there is a finite complex Borel measure $\nu^{\mathrm{f}}_{\xi,D}$ on $\overline{\mathcal{D}}_{\mathsf{L}_D}$ with*

$$k^{\mathrm{f}}_{\xi,D}(\lambda) = \int e^{\sum_{b\in\mathsf{L}_D}\langle \lambda_b, a_b\rangle}\,\mathrm{d}\nu^{\mathrm{f}}_{\xi,D}(a) \quad (\lambda \in \mathfrak{g}_{\mathbb{C}}^{I}), \qquad \nu^{\mathrm{f}}_{\xi,D}(\overline{\mathcal{D}}_{\mathsf{L}_D}) = k^{\mathrm{f}}_\xi(D)\,; \tag{3.79}$$

in particular $k^{\mathrm{f}}_{\xi,D}(\lambda)$ depends on λ only through $\lambda|_{\mathsf{L}_D}$.

Explicitly, splitting each probe label from its host's ordinary label, a selection becomes $\ell = (\mathcal{D}, \mathsf{J}, \mathsf{P}, Z)$ with $\mathsf{J} \subseteq \mathsf{L}_Z$ the selected probes, $Y_0 = \bigcup_{Y\in\mathcal{D}} Y \cup \bigcup_{b\in\mathsf{J}} Y_b$, and seed, E_0 and Z_0' formed from this Y_0 and P; the Mayer variable of a probe integrates exactly,

$$\int_0^1 \langle \lambda_b, B_b\rangle e^{t_b\langle \lambda_b, B_b\rangle}\,\mathrm{d}t_b = e^{\langle \lambda_b, B_b\rangle} - 1\,, \tag{3.80}$$

and with $\Lambda^{\mathrm{f}}_{\xi;\ell}$ the complex measure on $\mathfrak{g}^{E_0}$ defined by the right side of (3.68) *for $\mathbf{W} = \mathbf{V}_\xi$, with Y_0, Z_0, E_0, Z_0' the enlarged data of ℓ throughout (the factors χ_b over $E(Y_0)$, the conditional law and the weakening over this seed), with the Mayer factors $\int_0^1 \mathrm{d}t_Y\, \mathbf{W}(Y,B) e^{t_Y \mathbf{W}(Y,B)}$ for $Y \in \mathcal{D}$ only and no probe factor,*

$$\nu^{\mathrm{f}}_{\xi,D} = \sum_{\ell\,|\,Z=D,\ Z_0\neq\varnothing} \int \Big[\bigotimes_{b\in\mathsf{J}} (\delta_{B_b} - \delta_0) \otimes \bigotimes_{b\in\mathsf{L}_D\setminus\mathsf{J}} \delta_0\Big]\,\mathrm{d}\Lambda^{\mathrm{f}}_{\xi;\ell}(B)\,. \tag{3.81}$$

Proof. At a host Y carrying the ordinary potential $\mathbf{V}_\xi(Y)$ and the probe terms $F_b = \langle \lambda_b, \bullet\rangle$, b with $Y_b = Y$, $e^{\widetilde{\mathbf{V}}_{\xi,\lambda}(Y)} = e^{\mathbf{V}_\xi(Y)} \prod_b e^{F_b} = \prod_{\ell'} (1 + (e^{F_{\ell'}} - 1))$ over the labels ℓ' of Y (the ordinary label with $F_{\ell'} = \mathbf{V}_\xi(Y)$ and one label per probe); the sum over nonempty label subsets A of $\prod_{\ell'\in A}(e^{F_{\ell'}} - 1)$ equals $e^{\widetilde{\mathbf{V}}_{\xi,\lambda}(Y)} - 1$. Every term with $A \neq \varnothing$ has the same active host Y, so the split-label expansion has the same Y_0, P, seed, cover and kernel as the one-potential-per-host expansion of Lemma 3.26, and the two activities agree term by term after the Mayer variables t_b of the probes are integrated by (3.80).

The Laplace transform of $\delta_{B_b} - \delta_0$ in the variable λ_b is $e^{\langle \lambda_b, B_b\rangle} - 1$, and an unselected label contributes the factor one, the transform of δ_0; this proves (3.79) for the measure (3.81).

Each $\Lambda^{\mathrm{f}}_{\xi;\ell}$ is a finite complex measure: it is the image under $(X, B, s, \sigma, t) \mapsto B$ of a complex density whose absolute value is integrable by Lemma 3.24(ii) and the boundedness of each $\mathbf{V}_\xi(Y, \bullet)$ on its local box, the Cauchy kernels and the cutoff factors; the Dirac tensor

has total variation at most $2^{|\mathsf{J}|}$, and there are finitely many ℓ at fixed volume. The support lies in the closed box because $\chi_b(B)$, $b \in E(Y_0) \supseteq \mathsf{J}$, forces $|B_b| < r$ on the selected probes.

At $\lambda = 0$ every term with $\mathsf{J} \neq \varnothing$ has zero mass, so $\nu^{\mathrm{f}}_{\xi,D}(\overline{\mathcal{D}}_{\mathsf{L}_D}) = k^{\mathrm{f}}_{\xi,D}(0) = k^{\mathrm{f}}_{\xi}(D)$. *q.e.d.*

For a compatible family $\mathcal{P}$ let $\mathsf{e}_{\mathcal{P}}$ embed $\prod_{D\in\mathcal{P}} \mathfrak{g}^{\mathsf{L}_D}$ into $\mathfrak{g}^I$ by zero on the remaining labels (the L_D are pairwise disjoint).

Lemma 3.30 (Measure Identity)**.** *On every branch,*

$$\frac{K_{\xi;0}(a)\,\mathrm{d}a}{\mathfrak{n}_\xi} = \sum_{\mathcal{P}\ \mathit{compatible}} (\mathsf{e}_{\mathcal{P}})_* \bigotimes_{D\in\mathcal{P}} \nu^{\mathrm{f}}_{\xi,D}\,. \tag{3.82}$$

Proof. Both sides are finite complex measures with compact support (finitely many families at fixed volume), and by (3.78) and (3.79) they have the same Laplace transform on $\mathfrak{g}^I_{\mathbb{C}}$; restricted to imaginary λ this is the Fourier transform, which determines a finite measure. *q.e.d.*

Let δ_{0,L_D} be the Dirac mass at $0 \in \mathfrak{g}^{\mathsf{L}_D}$ and

$$\widehat{\nu}^{\mathrm{f}}_{\xi,D} = \nu^{\mathrm{f}}_{\xi,D} - k^{\mathrm{f}}_{\xi}(D)\delta_{0,\mathsf{L}_D}\,, \tag{3.83}$$

a measure of zero mass (the subtracted term is the sum of the selections with $\mathsf{J} = \varnothing$). For a nonempty compatible family $\mathcal{A}$ and bounded Borel f on $\mathfrak{g}^I$ let

$$\mathfrak{Q}_\xi(\mathcal{A}; f) := \int f(\mathsf{e}_{\mathcal{A}}a)\,\mathrm{d}\Big(\bigotimes_{D\in\mathcal{A}} \widehat{\nu}^{\mathrm{f}}_{\xi,D}\Big)(a), \qquad q_\xi(\mathcal{A}) := \frac{(\Xi^{\mathrm{f}}_\xi)^{[\cup\mathcal{A}]}}{\Xi^{\mathrm{f}}_\xi}\,. \tag{3.84}$$

Proposition 3.31 (Centred Expectation)**.** *On every branch, for bounded Borel f on $\mathfrak{g}^I$,*

$$\int f\,\mathrm{d}\nu_{\xi;0} = \frac{\int f(a)K_{\xi;0}(a)\,\mathrm{d}a}{I_{\xi;0}} = f(0) + \sum_{\varnothing\neq\mathcal{A}\ \mathit{compatible}} q_\xi(\mathcal{A})\mathfrak{Q}_\xi(\mathcal{A}; f)\,, \tag{3.85}$$

and, with Lemmas 3.17 and 3.18,

$$\mathcal{J}^{\mathrm{a}}_\xi[f] = e^{c_\xi} I_{\xi;1}\Big[f(0) + \sum_{\varnothing\neq\mathcal{A}\ \mathit{compatible}} q_\xi(\mathcal{A})\mathfrak{Q}_\xi(\mathcal{A}; f) + \mathcal{B}_\xi[f]\Big]\,. \tag{3.86}$$

Proof. f is integrated against (3.82), with each $\nu^{\mathrm{f}}_{\xi,D}$ expanded as $\widehat{\nu}^{\mathrm{f}}_{\xi,D} + k^{\mathrm{f}}_\xi(D)\delta_{0,\mathsf{L}_D}$. A term is indexed by $\mathcal{P}$ and the subfamily $\mathcal{A} \subseteq \mathcal{P}$ carrying the zero-mass factors; $f(\mathsf{e}_{\mathcal{P}}a)$ with $a = 0$ on the labels of $\mathcal{P}\setminus\mathcal{A}$ is $f(\mathsf{e}_{\mathcal{A}}a)$. Summing over $\mathcal{P} \supseteq \mathcal{A}$ gives $(\Xi^{\mathrm{f}}_\xi)^{[\cup\mathcal{A}]}\mathfrak{Q}_\xi(\mathcal{A}; f)$, and $\mathcal{A} = \varnothing$ gives $\Xi^{\mathrm{f}}_\xi f(0)$; division by $I_{\xi;0} = \mathfrak{n}_\xi\Xi^{\mathrm{f}}_\xi$ gives (3.85). The second display is $\mathcal{J}^{\mathrm{a}}_\xi[f] = e^{c_\xi}I_{\xi;1}\int f\,\mathrm{d}\nu_{\xi;1}$ (3.49) with (3.50). *q.e.d.*

With $f = f_\xi$ (extended by zero off the enlarged box; the measures live on the closed box, inside the enlarged box, where f_ξ is the Wilson function (3.46)), the first display is $F_\xi(0)$, and (3.86) is the exact decomposition of the conditional Wilson numerator on every branch: the loop at $a = 0$, the sum of closed insertions paired with the signed probe measures, and the exterior response.

3.8 Joint Extraction of the Scalar Reference

Let r_n be the positive representative of traversal $n \in \{1, \dots, n_C\}$ of C, the traversals numbered cyclically from a fixed base point. For a contour bond r that is not a bond of Ω_1 let $\mathcal{F}_r := \varnothing$: its link is $V_0(r)$ by (3.26), independent of a. By Lemma 3.9 and this convention, the link $U_\xi(a)(r_n)$ depends on a only through $a|_{\mathcal{F}_{r_n} \cap I}$.

For a connected final domain D the contact set is

$$\mathsf{T}_D := \{ n \,|\, \mathcal{F}_{r_n} \cap \mathsf{L}_D \neq \varnothing \} \,, \tag{3.87}$$

and D is marked when $\mathsf{T}_D \neq \varnothing$.

If a family $\mathcal{A}$ contains an unmarked domain D, then for a in the closed box $f_\xi(\mathsf{e}_{\mathcal{A}} a) = W_\chi(C; U_\xi(\mathsf{e}_{\mathcal{A}} a))$ does not depend on $a|_{\mathsf{L}_D}$ (every contour link depends on $a|_{\mathcal{F}_{r_n} \cap I}$, and $\mathcal{F}_{r_n} \cap \mathsf{L}_D = \varnothing$ for every n), and $\mathfrak{Q}_\xi(\mathcal{A}; f_\xi) = 0$ because $\widehat{\nu}^{\mathrm{f}}_{\xi, D}$ has zero mass and the measures are supported in the closed box; so only families of marked domains contribute to (3.85) for $f = f_\xi$.

By Lemma 3.28,

$$q_\xi(\mathcal{A}) = \sum_{k \geqslant 0} \frac{(-1)^k}{k!} \sum_{\mathfrak{c}_1, \dots, \mathfrak{c}_k \text{ each hits } \mathcal{A}} \prod_{j=1}^{k} \Phi^{\mathrm{v}}_\xi(\mathfrak{c}_j) \,, \tag{3.88}$$

the sum being absolutely convergent: $\sum_{\mathfrak{c} \text{ hits } \mathcal{A}} |\Phi^{\mathrm{v}}_\xi(\mathfrak{c})| \leqslant \frac{1}{8} \sum_{D \in \mathcal{A}} N_1(D)$ by Lemma 3.28, so the k-fold sums are dominated by the exponential series.

A tuple whose incompatibility graph is disconnected has $\varphi^T(\mathfrak{c}) = 0$ (it has no connected spanning subgraph), so the terms containing such a tuple vanish, and every cluster occurrence $\mathfrak{c}_j$ below has a connected incompatibility graph.

The graph of a term has as vertices the marked domains of $\mathcal{A}$ and the k cluster occurrences: a marked vertex is joined to each cluster that hits it, two clusters are joined when some constituent domains are incompatible. Every component has a marked root; its contacts are the union of the contact sets of its roots.

The hull of a nonempty set of traversals is its shortest cyclic interval when that interval has fewer than $n_C/2$ positions, and the whole contour otherwise; a shortest interval of fewer than $n_C/2$ positions is unique (two such intervals containing the set have together fewer than n_C positions, so they intersect in a cyclic interval, which contains the set and is shorter than one of them unless they coincide), and if $S \subseteq S'$ and the hull of S' is short then the hull of S is contained in it (the hull of S' is a short interval containing S, so the hull of S is short and, by the uniqueness just proved, is the shortest interval containing S; the intersection of the two short hulls is again an interval, their union having fewer than n_C positions, and it contains S, so it is the hull of S).

Let $\approx$ be the transitive closure of the relation "the hulls of the contacts intersect" on the components, and for a class K let $\mathcal{H}(K)$ be the hull of the union of the contacts of its members.

If the union of the members' hulls has fewer than $n_C/2$ positions, $\mathcal{H}(K)$ equals that union: the union is a cyclic interval (the members' hulls are chained by intersections, and two intersecting intervals with together fewer than n_C positions have an interval union), it contains the union of the contacts, so it contains the hull $\mathcal{H}(K)$ by minimality; conversely each member's contacts lie in the union of the contacts, whose hull is short, so by monotonicity

each member's hull lies in $\mathcal{H}(K)$. If the union of the members' hulls has at least $n_C/2$ positions, $\mathcal{H}(K)$ is the whole contour: a short $\mathcal{H}(K)$ would contain every member's hull by monotonicity.

The groups are the classes if every $\mathcal{H}(K)$ is short, and the single class of all components otherwise; the intervals of distinct groups are disjoint, since intersecting intervals would relate their classes.

Definition 3.32 (Groups)**.** A group W consists of its nonempty compatible marked-root set $\mathcal{A}(W)$, its ordered list of cluster occurrences $(\mathfrak{c}_1, \dots, \mathfrak{c}_{k(W)})$, $k(W) \geqslant 0$, each hitting a root of W, and its interval $\mathcal{H}(W)$, either a cyclic interval of fewer than $n_C/2$ traversals or the whole contour read from the base point; the data arise from a term of (3.88) as above; conversely, the term of (3.88) with root family $\mathcal{A}(W)$ and cluster list $(\mathfrak{c}_1, \dots, \mathfrak{c}_{k(W)})$ has W as its only group (its graph has the components of W, which form one class with the interval $\mathcal{H}(W)$).

Its weight is

$$w_\xi(W) \coloneqq \frac{(-1)^{k(W)}}{k(W)!} \prod_{j=1}^{k(W)} \Phi^{\mathrm{v}}_\xi(\mathfrak{c}_j) \,. \tag{3.89}$$

Two groups are compatible when all their constituent domains (roots and cluster members) are mutually compatible and their intervals are disjoint; a family containing a whole-contour group is a singleton. For a compatible group family $\mathfrak{B}$ and $\mathcal{S} \subseteq \mathfrak{B}$,

$$\mathcal{A}(\mathcal{S}) \coloneqq \bigcup_{W \in \mathcal{S}} \mathcal{A}(W), \qquad A_\xi(\mathcal{S}) \coloneqq \Big(\prod_{W \in \mathcal{S}} w_\xi(W) \Big) \mathfrak{Q}_\xi(\mathcal{A}(\mathcal{S}); f_\xi), \qquad A_\xi(\varnothing) \coloneqq f_\xi(0) \,. \tag{3.90}$$

Lemma 3.33 (Grouped Source)**.** *On every branch,*

$$F_\xi(0) = \sum_{\mathfrak{B} \ \mathit{compatible}} A_\xi(\mathfrak{B}) \,, \tag{3.91}$$

an absolutely convergent sum over compatible group families (including the empty family; there are finitely many root families at fixed volume but infinitely many cluster lists).

Proof. After substitution of (3.88) into (3.85) for $f = f_\xi$, each term, indexed by a family $\mathcal{A}$ of marked domains and an ordered list of k cluster occurrences, is cut into its groups; the groups are pairwise compatible (roots in different groups are compatible as members of $\mathcal{A}$, clusters in different groups are compatible with each other and with the other groups' roots because the graph has no edge between components, and the intervals are disjoint by construction).

The regrouping is reversible: a compatible group family determines $\mathcal{A} = \mathcal{A}(\mathfrak{B})$ and the multiset of cluster occurrences, and the term's graph has as components exactly the components of the groups' own graphs (no edges between groups, by compatibility), with the same hulls, so the classes and intervals are recovered; identical occurrences cannot split between groups because their domains are self-incompatible, so a group's occurrences are those of its term.

The ordered list of the term is an interleaving of the groups' lists, and the interleavings number $k!/\prod_W k(W)!$, which turns the factor $(-1)^k/k!$ into $\prod_W w_\xi(W)/\prod_W \Phi$-products.

The rearrangement is of an absolutely convergent sum: there are finitely many root families $\mathcal{A}$ at fixed volume, $|\mathfrak{Q}_\xi(\mathcal{A}; f_\xi)| \leqslant \|f_\xi\|_\infty \prod_{D \in \mathcal{A}} \|\widehat{\nu}^{\mathrm{f}}_{\xi,D}\|$ with the total variations finite, and $\sum_k \frac{1}{k!} \big(\sum_{\mathfrak{c} \text{ hits } \mathcal{A}} |\Phi^{\mathrm{v}}_\xi(\mathfrak{c})|\big)^k \leqslant \exp(\frac{1}{8} \sum_{D \in \mathcal{A}} N_1(D))$; hence the term sum converges absolutely, and so does the group-family sum, being a rearrangement of it. *q.e.d.*

A superscript $\circ$ denotes the reference data: the same Ω_1, Λ_1, I, J, cubes, hosts, constants and cutoffs, with the background $U_1^\circ = \mathbf{1}$ on every bond, $V_0^\circ = \mathbf{1}$, zero retained field $A_{\mathfrak{e}}^\circ = 0$ and $\Psi_\xi^\circ = \mathbf{1}$; thus $H_\xi^\circ = \mathsf{H}_{\mathbf{1}}^{\Omega_1}$, the reference phase is $q_{\mathbf{1}}$, the reference primitives are those of Section 3.3 at $U = \mathbf{1}$, $f_\xi^\circ(a) = W_\chi(C; e^{iq_{\mathbf{1}}(a)})$ with the links off Ω_1 equal to $\mathbf{1}$, and $f_\xi^\circ(0) = \chi(\mathbf{1}) = 1$ since $q_{\mathbf{1}}(0) = 0$. These data need not be a branch of the history sum.

Lemmas 3.6, 3.7, 3.9 and 3.14 apply to $U = \mathbf{1}$ directly (Section 3.3, opening paragraph; Lemma 3.6 at $A = 0$; the row solutions are those of Lemma 3.12 in its reference form, on the row polydiscs $\mathcal{U}_c$, and (B4) is not invoked), and the estimate lemmas used below (Lemmas 3.8, 3.12, 3.24, 3.27, 3.28) hold for the reference data by Appendix B, Lemma B.16; in particular the reference near primitives are real on the open box (Lemma 3.12: the analytic branches of the logarithms are real at real arguments, the row solution being real), so $K_{\xi;0}^\circ > 0$ there and $I_{\xi;0}^\circ > 0$.

Every lemma of Sections 3.5–3.8 stated on every branch (Lemmas 3.21, 3.22, 3.23, 3.26, 3.29, 3.30, 3.33 and the first display of Proposition 3.31) holds verbatim for the reference data: their proofs use only the field-independent geometry and the lemmas just listed. Every object of Sections 3.5–3.8 formed from the reference data carries the superscript $\circ$; in particular $K_{\xi;0}^\circ$, $I_{\xi;0}^\circ$, $k_\xi^{\mathrm{f},\circ}$, $\widehat{\nu}_{\xi,D}^{\mathrm{f},\circ}$, $w_\xi^\circ(W)$.

The index set of groups is the same for the branch and for the reference data: roots, cluster lists (the clusters are tuples of domains), contact sets and intervals are field-independent data (the sets $\mathcal{F}_{r_n}$ and L_D do not involve the background), and only the weights w_ξ, w_ξ° and the measures $\widehat{\nu}_{\xi,D}^{\mathrm{f}}$, $\widehat{\nu}_{\xi,D}^{\mathrm{f},\circ}$ differ.

For a group W let $P_W^\circ(a)$ be the ordered product of the reference links along the consecutive traversals of $\mathcal{H}(W)$, in the cyclic order of the contour starting from the first traversal of the interval (for a whole-contour group, from the base point), the link of a traversal being $e^{iq_{\mathbf{1}}(\mathsf{e}_{\mathcal{A}(W)}a)(r)}$ when its bond r is a bond of Ω_1 and $\mathbf{1}$ otherwise, as in f_ξ° (oriented representatives inverted), with the integration variable $\mathsf{e}_{\mathcal{A}(W)}a$, zero off the labels of $\mathcal{A}(W)$; then $P_W^\circ(0) = \mathbf{1}$. Let

$$a_W \coloneqq w_\xi^\circ(W) \int \chi(P_W^\circ(a))\, \mathrm{d}\Big(\bigotimes_{D \in \mathcal{A}(W)} \widehat{\nu}_{\xi,D}^{\mathrm{f},\circ} \Big)(a) \,. \tag{3.92}$$

Lemma 3.34 (Schur Identity)**.** *For $h \in G$,*

$$w_\xi^\circ(W) \int \chi(P_W^\circ(a)h)\, \mathrm{d}\Big(\bigotimes_{D \in \mathcal{A}(W)} \widehat{\nu}_{\xi,D}^{\mathrm{f},\circ} \Big)(a) = a_W \chi(h) \,. \tag{3.93}$$

Consequently

$$S_{\xi,C}^\circ = \sum_{\mathfrak{B}\ \mathit{compatible}} \prod_{W \in \mathfrak{B}} a_W = \frac{\int f_\xi^\circ(a) K_{\xi;0}^\circ(a)\, \mathrm{d}a}{I_{\xi;0}^\circ} \,. \tag{3.94}$$

Proof. For constant $u \in G$ let $(R_u a)_b \coloneqq \mathrm{Ad}_u\, a_b$ on $\mathfrak{g}^I$, an orthogonal map for the invariant inner product. At the identity background the constant gauge transformation u fixes the background, so Lemma 3.7 gives $H_\xi^\circ = R_u H_\xi^\circ R_u^{-1}$; hence all walk coefficients, $C_{E_0}^\circ$, $\Gamma_{E_0}^\circ(\sigma)$ (the domains and weights do not involve the fields) and the Gaussian density (3.66) are invariant under $(X, B) \mapsto (R_u X, R_u B)$, and μ_0, $\mu_{C_{E_0}^\circ}$ are invariant. The cutoff factors are invariant because the bond norm is Ad-invariant ((5.4)).

The reconstruction is covariant: by the intertwinings in the proof of Lemma 3.7 at $U = \mathbf{1}$, $\mathcal{N}_{\mathbf{1}} \circ R_u = R_u^{(1)} \mathcal{N}_{\mathbf{1}}$, $h_{\mathbf{1}} R_u^{(1)} = R_u h_{\mathbf{1}}$ and $\mathsf{C}_{\mathbf{1}} R_{u,*} = R_u \mathsf{C}_{\mathbf{1}}$, so the unique small solution on the row

polydiscs of Lemma 3.12 (reference version; the polydiscs are R_u-invariant because the bond norm is Ad-invariant) satisfies $\widetilde{D}_{\mathbf{1}}(R_u A) = R_u^{(1)} \widetilde{D}_{\mathbf{1}}(A)$ and $q_{\mathbf{1}}(R_u B) = R_u q_{\mathbf{1}}(B)$, that is, every reconstructed link is conjugated by u; the primitives are invariant, being traces of products of links, Haar densities of conjugated phases (σ is conjugation invariant) and determinants of conjugated operators.

The probe term transforms as $L_Y(R_u\lambda, R_u a) = L_Y(\lambda, a)$. Hence $k^{\mathrm{f},\circ}_{\xi,D}(R_u\lambda) = k^{\mathrm{f},\circ}_{\xi,D}(\lambda)$, and by Laplace uniqueness $(R_u)_* \widehat{\nu}^{\mathrm{f},\circ}_{\xi,D} = \widehat{\nu}^{\mathrm{f},\circ}_{\xi,D}$. Since the links are conjugated, $P^\circ_W(R_u a) = u P^\circ_W(a) u^{-1}$.

Let π be the irreducible unitary representation with $\chi = \dim(\pi)^{-1} \operatorname{tr} \pi$ (the normalised character, $\chi(\mathbf{1}) = 1$). The left side of (3.93), as a function of h, is $\dim(\pi)^{-1} \operatorname{tr}(\Upsilon \pi(h))$ for the matrix $\Upsilon = w^\circ_\xi(W) \int \pi(P^\circ_W(a)) \, \mathrm{d}(\bigotimes \widehat{\nu}^\circ)(a)$; by the invariance of the measure and the covariance of P°_W, $\Upsilon = \pi(u) \Upsilon \pi(u)^{-1}$ for all $u \in G$, so Υ is a multiple of the identity by Schur's lemma, and $h = \mathbf{1}$ identifies the multiple as a_W.

For (3.94), Lemma 3.33 applies to the reference data (its sum converges absolutely, as shown there). Let $\mathfrak{B} \coloneqq \{W_1, \dots, W_m\}$ be a compatible family. If $m = 1$ and W_1 is a whole-contour group, the reference Wilson function at $\mathsf{e}_{\mathcal{A}(W_1)} a$ is $\chi(P^\circ_{W_1}(a))$. Otherwise all intervals are short and pairwise disjoint; the groups are numbered in cyclic order starting from W_1, and by the cyclicity of the trace the ordered link product starts at the first traversal of $\mathcal{H}(W_1)$. The reference Wilson function at $\mathsf{e}_{\mathcal{A}(\mathfrak{B})} a$ is then $\chi(P^\circ_{W_1}(a) \cdots P^\circ_{W_m}(a))$: a traversal outside every interval is outside every contact set, so its link is $e^{i q_{\mathbf{1}}(0)} = \mathbf{1}$ (or $\mathbf{1}$ by convention when its bond is not a bond of Ω_1); a traversal in $\mathcal{H}(W_j)$ lies in no contact set of a root of another group, so its link depends only on $a|_{\mathcal{A}(W_j)}$, and the links of the consecutive traversals of $\mathcal{H}(W_j)$ multiply to $P^\circ_{W_j}(a)$ in the order of the product.

The integral against $\bigotimes_j \bigotimes_{D \in \mathcal{A}(W_j)} \widehat{\nu}^\circ_D$ of this character times $\prod_j w^\circ(W_j)$ is evaluated by Fubini one group at a time: integrating over the variables of W_1 with $h \coloneqq P^\circ_{W_2}(a) \cdots P^\circ_{W_m}(a)$ fixed gives $a_{W_1} \chi(P^\circ_{W_2}(a) \cdots P^\circ_{W_m}(a))$ by (3.93), and so on, ending with $\prod_j a_{W_j} \chi(\mathbf{1}) = \prod_j a_{W_j}$. Since $|\chi| \leqslant 1$, $|a_W| \leqslant |w^\circ_\xi(W)| \prod_{D \in \mathcal{A}(W)} \|\widehat{\nu}^{\mathrm{f},\circ}_{\xi,D}\|$, and the sum in (3.94) converges absolutely by the bound of Lemma 3.33. *q.e.d.*

Proposition 3.35 (Joint Extraction)**.** *For a compatible group family $\mathfrak{B}$ the joint defect is*

$$R_\xi(\mathfrak{B}) \coloneqq \sum_{\mathcal{S} \subseteq \mathfrak{B}} (-1)^{|\mathfrak{B} \setminus \mathcal{S}|} \Big(\prod_{W \in \mathfrak{B} \setminus \mathcal{S}} a_W \Big) A_\xi(\mathcal{S}) \,, \qquad R_\xi(\varnothing) \coloneqq f_\xi(0) \,. \tag{3.95}$$

Then, with $S^{\circ,[\mathfrak{B}]}_{\xi,C}$ the reference sum over group families compatible with every member of $\mathfrak{B}$,

$$F_\xi(0) = S^\circ_{\xi,C} f_\xi(0) + \sum_{\varnothing \neq \mathfrak{B} \ \mathit{compatible}} S^{\circ,[\mathfrak{B}]}_{\xi,C} R_\xi(\mathfrak{B}) \,, \tag{3.96}$$

and on every branch

$$\mathcal{J}^{\mathrm{a}}_\xi[f_\xi] = e^{c_\xi} I_{\xi;1} \Big[S^\circ_{\xi,C} f_\xi(0) + \sum_{\varnothing \neq \mathfrak{B} \ \mathit{compatible}} S^{\circ,[\mathfrak{B}]}_{\xi,C} R_\xi(\mathfrak{B}) + \mathcal{B}_\xi[f_\xi] \Big] \,. \tag{3.97}$$

Proof. Boolean inversion of (3.95) gives $A_\xi(\mathfrak{B}) = \sum_{\mathcal{S} \subseteq \mathfrak{B}} (\prod_{\mathfrak{B} \setminus \mathcal{S}} a_W) R_\xi(\mathcal{S})$; after substitution into (3.91) and summation over $\mathfrak{B} \supseteq \mathcal{S}$, the families $\mathfrak{B} \setminus \mathcal{S}$ range over the group families compatible with every member of $\mathcal{S}$, giving $S^{\circ,[\mathcal{S}]}_{\xi,C}$.

The interchange of the sums is justified by absolute convergence. Let $\beta_W = (|w_\xi(W)| + |w^\circ_\xi(W)|) \prod_{D \in \mathcal{A}(W)} \max(\|\widehat{\nu}^{\mathrm{f}}_{\xi,D}\|, \|\widehat{\nu}^{\mathrm{f},\circ}_{\xi,D}\|)$; then $|a_W| \leqslant \beta_W$, $|A_\xi(\mathcal{S})| \leqslant \|f_\xi\|_\infty \prod_{W \in \mathcal{S}} \beta_W$ and

$|R_\xi(\mathcal{S})| \leqslant \|f_\xi\|_\infty \prod_{W\in\mathcal{S}} 2\beta_W$. The sum $\sum_W \beta_W$ over all groups is finite: a group is determined by its root family, its cluster list and its interval, there are finitely many root families at fixed volume, and $\sum_{W\,|\,\mathcal{A}(W)=\mathcal{A}} |w_\xi(W)| \leqslant \exp(\frac{1}{8}\sum_{D\in\mathcal{A}} N_1(D))$ by Lemma 3.28, likewise for w_ξ° by Lemma B.16.

A compatible group family has at most N_{m} members, the number of marked domains (its roots are pairwise distinct marked domains), so $\sum_{\mathfrak{B}} \sum_{\mathcal{S}\subseteq\mathfrak{B}} \prod_{W\in\mathfrak{B}\setminus\mathcal{S}} |a_W|\,|R_\xi(\mathcal{S})| \leqslant \|f_\xi\|_\infty \sum_{\mathfrak{B}} \prod_{W\in\mathfrak{B}} 3\beta_W \leqslant \|f_\xi\|_\infty \sum_{n\leqslant N_{\mathrm{m}}} (3\sum_W \beta_W)^n < \infty$; in particular $S^\circ_{\xi,C}$ and every $S^{\circ,[\mathcal{S}]}_{\xi,C}$ converge absolutely.

The second identity is (3.86) with $f = f_\xi$. *q.e.d.*

At the reference data every nonempty defect vanishes: there $A^\circ_\xi(\mathcal{S}) = \prod_{W\in\mathcal{S}} a_W$ for every compatible $\mathcal{S}$ (proof of Lemma 3.34), so $R^\circ_\xi(\mathfrak{B}) = \prod_{W\in\mathfrak{B}} a_W \sum_{\mathcal{S}\subseteq\mathfrak{B}} (-1)^{|\mathfrak{B}\setminus\mathcal{S}|} = 0$ for $\mathfrak{B} \neq \varnothing$.

Lemma 3.36 (Covered Branches)**.** *A branch ξ is covered when every bond of the contour is a bond of Ω_1 and every free-input stencil F_γ of the contour* (3.32) *lies in I, that is, $\mathcal{F}_r \subseteq I$ for every contour bond r, with $\mathcal{F}_r$ of* (3.13)*. On a covered branch $U_\xi(0)(r) = U_1(r)$ for every contour bond r, hence $f_\xi(0) = W_\chi(C; U_1)$.*

Proof. At $a = 0$ the free vector is $B = \Psi_\xi(A_{\mathrm{e}})$, which vanishes on I. For a contour bond r, a bond of Ω_1 on a covered branch, $U_\xi(0)(r) = e^{iq_U(B)(r)} U_1(r)$ by (3.26), and by Lemma 3.9 $q_U(B)(r)$ depends on $B|_{\mathcal{F}_r}$ only and vanishes when $B|_{\mathcal{F}_r} = 0$ (for a free bond directly, for $r = b_0(c)$ because $\mathcal{F}_r = \mathcal{E}_c \cap E_*$). Since $\mathcal{F}_r \subseteq I$ on a covered branch, $U_\xi(0) = U_1$ on the contour. *q.e.d.*

Appendix B proves Lemmas 3.8, 3.12, 3.24, 3.27 and 3.28, and Lemma B.16 there transfers them to the reference data; Appendix D bounds $\mathcal{B}_\xi[f_\xi]$ and the uncovered branches; Appendix F identifies $S^\circ_{\xi,C}$ at the carrier I with the flat coefficient $\kappa_C e^{E^\circ_\xi}$, Appendix E bounds the defects, and Section 9 bounds the history average of $e^{|E^\circ_\xi|}$.

4 The Flat Regular-Region Calculation

This section evaluates the first step at the identity background on an all-regular periodic lattice; Section 7 treats a proper conditional integration region.

The group is in its fixed faithful unitary realisation $G \subset U(N)$, its Lie algebra $\mathfrak{g}$ is identified with Hermitian matrices, so that its group exponential is e^{iX}, $\mathrm{tr}_N = N^{-1}\,\mathrm{Tr}$, and the real inner product is $\langle X, Y\rangle = \mathrm{tr}_N(XY)$. The observable is (1.1), with the normalised character $\chi(\mathbf{1}) = 1$, and $d = \dim \mathfrak{g}$.

The first derivative of χ vanishes: the trace of its Lie algebra representation vanishes on commutators, and a semisimple Lie algebra is its own commutator algebra.

4.1 The Covariance and the Normalised Insertion

The operators of this section are in fine-lattice units. Let E be the positively oriented bonds of T, let $T^{(1)}$ be its coarse centres, and let E_1 be the positively oriented coarse bonds. The blocking factor is the odd $L > 11$ used by Bałaban ([8, p. 251]; the lemmas of this paper that need only $L \geqslant 3$ are stated so). For the covariance bounds below, $T = \prod_{\mu=1}^4 (\mathbb{Z}/N_\mu\mathbb{Z})$ with $N_\mu = LK_\mu$ and $K_\mu \geqslant 3$.

The block $\mathcal{B}_y$ consists of the fine sites at displacement n from $y \in T^{(1)}$, with $n \in \mathbb{Z}^4$ and $|n_\mu| \leqslant (L-1)/2$ in each direction. For $x \in \mathcal{B}_y$, the path family $\mathcal{G}(y,x)$ traverses each nonzero coordinate displacement in one uninterrupted run, in every ordering of those runs; the path is empty for $x = y$. These are the centred blocks and path family of [8, Section 0].

The averaging operation is (1.4); the formula below is [8, (0.4)], its value on the set $\mathcal{O}_c$ where every logarithm has its argument in the chart domain (Lemma 1.1). [9, p. 243] permits this choice in the paragraph following (0.1).

For a positive coarse bond $c = (y, y + Le_\mu)$ and $x \in \mathcal{B}_y$, let $x' \coloneqq x + Le_\mu$. The vectors e_μ are the fine-lattice unit vectors, $[x,x']$ is the straight fine path, and $P_c(U)$ is the transport along the straight fine path from c_- to c_+. On the regular logarithm domain, [8, (0.4)] is

$$\begin{aligned} \beta_c(U) &\coloneqq L^{-4} \sum_{x \in \mathcal{B}_{c_-}} \frac{1}{|\mathcal{G}(c_-,x)|\,|\mathcal{G}(c_+,x')|} \sum_{\substack{\gamma \in \mathcal{G}(c_-,x) \\ \gamma' \in \mathcal{G}(c_+,x')}} \frac{1}{i} \log\left(U(\gamma)U([x,x'])U(\gamma')^{-1}P_c(U)^{-1}\right) , \\ M(U,c) &= e^{i\beta_c(U)}P_c(U) \qquad (U \in \mathcal{O}_c) . \end{aligned} \tag{4.1}$$

Every logarithm is applied to a closed relative transport based at c_-. The logarithmic constraint map and its derivative are

$$\mathcal{Q}(q) \coloneqq \frac{1}{i}\log M(e^{iq}), \qquad Q \coloneqq \frac{1}{L}D\mathcal{Q}(0), \qquad \mathcal{N}(q) \coloneqq \mathcal{Q}(q) - LQq . \tag{4.2}$$

The maps $\mathcal{Q}$ and $\mathcal{N}$ take a neighbourhood of zero in $\mathfrak{g}^E$ to $\mathfrak{g}^{E_1}$.

The designated bond $b_0(c)$ is the fine bond on the straight coarse bond c crossing between its two blocks. Let $E_0 \coloneqq \{b_0(c) \,|\, c \in E_1\}$ and $E_* = E \setminus E_0$, let ι_0 and ι_* extend fields on these two label sets by zero to E, and let $Q_0 \coloneqq Q\iota_0$. The designated coefficient matrix Q_0 is invertible in this regular construction. The maps are

$$h \coloneqq \frac{1}{L}\iota_0 Q_0^{-1}, \qquad \mathsf{C} \coloneqq (I - \iota_0 Q_0^{-1} Q)\iota_* . \tag{4.3}$$

Thus $h\colon \mathfrak{g}^{E_1} \to \mathfrak{g}^E$, $\mathsf{C}\colon \mathfrak{g}^{E_*} \to \mathfrak{g}^E$, $LQh = I$ and $Q\mathsf{C} = 0$. The free variable is $B \in \mathfrak{g}^{E_*}$, and $\mathsf{C}B$ agrees with B on E_*. These are the designated-bond construction in [8] following (2.10) and the elimination in [9, (1.14)–(1.15)].

The gauge-fixing functional is the exponential one in [9, (1.7)], not a delta-function axial gauge. For the transports along $\mathcal{G}(y,x)$, let $\mathcal{U}_U(y,x)$ be their local group mean: it is the unique solution near their diagonal value of

$$\sum_{\gamma \in \mathcal{G}(y,x)} \log(U(\gamma)\mathcal{U}_U(y,x)^{-1}) = 0 . \tag{4.4}$$

This is [8, (0.10)–(0.11)]. The gauge-fixing functional and the action used in the constrained reference integral are

$$\begin{aligned} \mathscr{G}(U) &\coloneqq \sum_{y \in T^{(1)}} \sum_{x \in \mathcal{B}_y \setminus \{y\}} (1 - \operatorname{Re}\operatorname{tr}_N \mathcal{U}_U(y,x)) , \\ \mathscr{A}(U) &\coloneqq S(U) + \mathscr{G}(U) . \end{aligned} \tag{4.5}$$

Here ∂p is the oriented boundary of the plaquette p. All group means and logarithms are used in the regular chart containing the identity.

For $a, b \in \mathfrak{g}^E$, the flat Hessian is defined by

$$\langle a, \Delta_1 b\rangle := \frac{\partial^2}{\partial s\,\partial t}\mathscr{A}(e^{i(sa+tb)})\Big|_{s=t=0} . \tag{4.6}$$

The first derivative is zero at the identity. This is the $J_1 = 0$ specialisation of [9, (1.16)].

[9, (1.12)] states strict positivity of the second differential at the regular constrained minimum. With the exponential gauge fixing and the constrained extension above, it makes the quadratic form on the finite free carrier strictly positive. The quadratic form and its inverse on that carrier are

$$\mathsf{H} := \mathsf{C}^*\Delta_1\mathsf{C}, \qquad \mathsf{K}_B := \mathsf{H}^{-1}, \qquad \mathsf{K} := \mathsf{C}\mathsf{K}_B\mathsf{C}^* . \tag{4.7}$$

The adjoint and Lebesgue measure use the stated real inner products, summed over the respective bond labels. Thus K is the covariance of the linearised fine field $\mathsf{C}B$.

The normalised Gaussian measure on the free carrier is denoted by μ_{K_B}. In an orthonormal basis of real dimension $D = d|E_*|$, it has density

$$\mathrm{d}\mu_{\mathsf{K}_B}(B) = (2\pi)^{-D/2}(\det \mathsf{K}_B)^{-1/2}e^{-\frac{1}{2}\langle B,\mathsf{H}B\rangle}\,\mathrm{d}B . \tag{4.8}$$

Bałaban's nonlinear constraint solution gives the fine phase

$$q_{g_0}(B) = g_0\mathsf{C}B - h\widetilde{D}(g_0\mathsf{C}B), \qquad U_{g_0}(b) = e^{iq_{g_0}(B)(b)} . \tag{4.9}$$

The correction $\widetilde{D}\colon \mathfrak{g}^E \to \mathfrak{g}^{E_1}$ is defined near zero by the unique small analytic solution

$$\widetilde{D}(A) = \mathcal{N}(A - h\widetilde{D}(A)), \qquad \widetilde{D}(0) = 0 . \tag{4.10}$$

Its construction is that of [8] following (2.10). Since $\mathcal{N}(0) = D\mathcal{N}(0) = 0$, its expansion starts at degree two. Equation (4.9) therefore solves the nonlinear averaging constraint and has $q_{g_0}(B) = g_0\mathsf{C}B + O(g_0^2)$ at fixed B.

For a fixed $\sigma \in \{\mathrm{op}, \mathrm{HS}\}$, the corresponding matrix norm is used for both group-matrix and Lie-algebra cutoff tests:

$$|X|_{\mathrm{op}} = \sup_{|v|_2=1} |Xv|_2, \qquad |X|_{\mathrm{HS}} = (\mathrm{tr}_N(X^*X))^{\frac{1}{2}} . \tag{4.11}$$

Here $|v|_2$ is the Euclidean vector norm. The dependence on the fixed σ is suppressed in ϑ_{g_0} and $\mathcal{W}_{g_0}$ below.

In the all-regular branch, $U_1 = U_{1,\square'} = \mathbf{1}$. [9, (1.19)] removes the approximate relative-field tests on the good fluctuation cubes. The other retained chart tests in [9, (1.25)] have support in $S_1 = \varnothing$. The guard statement following [9, (1.10)] removes the earlier plaquette and gauge cutoffs under its prescribed parameter hierarchy. With that hierarchy retained, [9, (1.18)–(1.19), (1.25)–(1.26)] gives the following bond product when its bond norm is read as $|\bullet|_\sigma$:

$$\vartheta_{g_0}(B) = \prod_{b\in E_*} \mathbf{1}_{\{\,|B(b)|_\sigma < r(g_0)\,\}}, \qquad r(g_0) = A_1(\log g_0^{-2})^{p_0} . \tag{4.12}$$

Here $A_1 > 0$ and $p_0 \geqslant 5\mathfrak{r}$ ($\mathfrak{r} \geqslant 2$ the logarithmic exponent of the macro-cube sizes, the convention of Section 3; Section B.8) are the first-step cutoff parameters, and $\delta_0 = g_0 r(g_0)$.

The normalisation is the inductive one following [9, (3.4)], $\delta_k = g_k A_1/A_0\, p_0(g_k)$ as printed at p.265, together with [9, (2.4)], also at the first step:

$$\begin{aligned} p_0(g_k) &\coloneqq A_0(\log g_k^{-2})^{p_0} , \\ \varepsilon_k &\coloneqq g_k p_0(g_k) , \\ \delta_k &\coloneqq g_k \frac{A_1}{A_0} p_0(g_k) = \frac{A_1}{A_0}\varepsilon_k . \end{aligned} \tag{4.13}$$

For the first step this gives $\delta_0/g_0 = A_1(\log g_0^{-2})^{p_0}$.

The application to [9] assumes its compatible cube partitions and admissible first-step parameters: sufficiently large cutoff constants, sufficiently small A_1/A_0 and coupling, and $p_0 \geqslant 5\mathfrak{r}$, where $\mathfrak{r} \geqslant 2$ is the logarithmic exponent defining the macro-cube sizes (Section 3). Under these hypotheses the guard statement following [9, (1.10)] removes χ_0 and χ_{Ax} in the regular region. The reference integral below is specified separately for each of the two matrix-norm prescriptions; its identification with a conditional kernel uses that same prescription.

Let $\sigma_H(q)$ be the density of normalised Haar measure in exponential coordinates relative to the stated Lie-algebra Lebesgue measure, and let $j_H(q) \coloneqq \sigma_H(q)/\sigma_H(0)$. The Jacobian and residual exponent are the pullbacks of the measure and the action:

$$\begin{aligned} J_{g_0}(B) &\coloneqq \left|\det_{\mathfrak{g}^E}(I - hD\widetilde{D}(g_0 \mathsf{C} B))\right| \prod_{b\in E} j_H(q_{g_0}(B)(b)) , \\ \mathcal{P}_{g_0}(B) &\coloneqq \log J_{g_0}(B) - \frac{1}{g_0^2}\mathscr{A}(U_{g_0}) + \frac{1}{2}\langle B, \mathsf{H} B\rangle . \end{aligned} \tag{4.14}$$

The determinant is on the real fine carrier. The nonlinear change of variables is made before the linear constraints are eliminated, as in [9, (1.14)].

The subsequent linear-constraint factor, Haar density at zero, coupling rescaling, gauge normaliser and vacuum scalar are common to the numerator and denominator in this branch. After the cutoff reductions above, (4.14) retains the action, Haar and Jacobian factors of [9, (1.14)–(1.16)], with the same exponential gauge fixing. Its constant coefficient at $g_0 = 0$ is zero.

The regular sourced ratio is

$$\mathcal{W}_{g_0}(C) \coloneqq \frac{\displaystyle\int \vartheta_{g_0}(B) e^{\mathcal{P}_{g_0}(B)} W_\chi(C; U_{g_0})\, \mathrm{d}\mu_{\mathsf{K}_B}(B)}{\displaystyle\int \vartheta_{g_0}(B) e^{\mathcal{P}_{g_0}(B)}\, \mathrm{d}\mu_{\mathsf{K}_B}(B)} . \tag{4.15}$$

5 The Straight-Edge Coefficient from Gaussian Marginals

The normalised flat integral has a connected expansion. Its extensive contribution along a long straight side defines δm; the contributions remaining near the corners define $z_{\mathfrak{c}}$. These terms are constructed from Gaussian principal marginals and the local interaction (C.2); Section 7 compares the straight-side coefficient with the conditional kernel in a proper regular region.

For a square C_n on the compatible finite torus T, the flat reference numerator and de-

nominator are

$$N_{n,T} \coloneqq \int \vartheta_{g_0}(B) e^{\mathcal{P}_{g_0}(B)} W_\chi(C_n; U_{g_0}) \, \mathrm{d}\mu_{\mathsf{K}_B}(B) \,,$$
$$D_T \coloneqq \int \vartheta_{g_0}(B) e^{\mathcal{P}_{g_0}(B)} \, \mathrm{d}\mu_{\mathsf{K}_B}(B) \,. \tag{5.1}$$

These are the identity-background regular integrals. In the two-norm construction below the cutoff is $\vartheta^\sigma_{g_0}$, and $N_{n,T} = N^\sigma_{C_n,T}$, $D_T = D^\sigma_T$ for the chosen prescription σ.

5.1 Connected Terms of the Normalised Integral

Let E_* be the free-bond carrier and let $C_T \coloneqq \mathsf{K}_B$ on a compatible finite torus T. The flat gap and exponential decay proved in Appendix A give positive constants λ, Λ, A, μ, independent of T, such that

$$\lambda I \leqslant C_T \leqslant \Lambda I, \qquad \|C_T(b,c)\| \leqslant A e^{-\mu d_T(b,c)} \,. \tag{5.2}$$

For positive bonds $b = (x,i)$ and $c = (y,j)$, let $d_T(b,c) \coloneqq d_{\infty,T}(x,y) + \mathbf{1}_{\{i \neq j\}}$, where the first term is periodic maximum-norm site distance in fine-lattice units. The same bounds hold for every principal marginal.

One may take $\lambda = h_L^{-1}$ and $\Lambda = \gamma_L^{-1}$ from (A.25) and (A.27). The free-inverse Neumann series (A.28) applies on the compatible tori as well, with the same gap and row-sum bound.

Let $d_{\mathrm{block}}(b,c)$ be the periodic maximum-norm distance between the coarse indices of the blocks containing the initial sites of b, c. The block-distance decay converts to the fine distance using $d_T(b,c) \leqslant L d_{\mathrm{block}}(b,c) + 2L$; decreasing the decay exponent by the factor L and enlarging A suffices.

The metric on each Lie algebra factor is the fixed invariant real metric. All constants below may depend on this metric, G, L, the fixed local reconstruction and χ, but not on the square size or torus volume.

Throughout this construction, $r_0(g_0) = r(g_0) = A_1(\log g_0^{-2})^{p_0}$ is exactly the threshold in (4.12). Restoring the fixed norm index gives the two cutoff prescriptions

$$\vartheta^\sigma_{g_0}(B) = \prod_{b \in E_*} \mathbf{1}_{\{\, |B_b|_\sigma < r_0(g_0) \,\}} \,. \tag{5.3}$$

The result holds separately for each prescription, with that cutoff retained in both numerator and denominator. With the bond tests of [9] interpreted in the same norm, the regular-branch reduction above identifies its reference kernel.

For unitary u, the required invariance follows from

$$|uXu^{-1}|_{\mathrm{op}} = \sup_{|v|_2=1} |uXu^{-1}v|_2 = \sup_{|w|_2=1} |Xw|_2 = |X|_{\mathrm{op}} \,,$$
$$|uXu^{-1}|^2_{\mathrm{HS}} = \mathrm{tr}_N(uX^*Xu^{-1}) = \mathrm{tr}_N(X^*X) = |X|^2_{\mathrm{HS}} \,. \tag{5.4}$$

Thus each good- or bad-bond test is invariant for the same fixed norm and threshold; a supremum over the bonds of a cube and products of such tests preserve this invariance.

In the regular branch the starred carrier excludes the designated bonds, $A = \mathsf{C}B$ agrees with B on that carrier, and $g_0^{-1}\delta_0 = r_0$ by (4.13). Repeated cube tests reduce to the bond product (4.12) by idempotence, as in the stated regular-branch reduction.

The numerator and denominator are

$$\begin{aligned} N^{\sigma}_{C,T} &\coloneqq \int \vartheta^{\sigma}_{g_0}(B) e^{\mathcal{P}_{g_0}(B)} W_{\chi}(C; U_{g_0}) \, \mathrm{d}\mu_{C_T}(B) \, , \\ D^{\sigma}_{T} &\coloneqq \int \vartheta^{\sigma}_{g_0}(B) e^{\mathcal{P}_{g_0}(B)} \, \mathrm{d}\mu_{C_T}(B) \, , \qquad \mathcal{W}^{\circ,\sigma}_{T}(C) \coloneqq N^{\sigma}_{C,T} / D^{\sigma}_{T} \, . \end{aligned} \tag{5.5}$$

In the proof σ is fixed and its index is suppressed on the cutoffs, integrals, and cluster weights; $r = r_0(g_0)$ and $\vartheta_b = \mathbf{1}_{\{|B_b|_\sigma < r\}}$. Let $\mathcal{V}_T$ index the local terms v in (C.2), with supports Λ_v. The bounded measurable functions on the full real carrier are

$$f_v(B) \coloneqq \left(\prod_{b \in \Lambda_v} \vartheta_b(B) \right) (e^{v(B)} - 1), \qquad h_b(B) \coloneqq -(1 - \vartheta_b(B)) \, . \tag{5.6}$$

The first function is zero where its cutoff vanishes, so the local analytic function is never evaluated outside its chart. On the original cutoff the exact identity is

$$\vartheta_{g_0}(B) e^{\mathcal{P}_{g_0}(B)} = \prod_{b \in E_*} (1 + h_b(B)) \prod_{v \in \mathcal{V}_T} (1 + f_v(B)) \, . \tag{5.7}$$

For a square $C = (b_a)_{a \in \mathcal{I}_C}$, let $\mathcal{I}_C$ be its ordered finite traversal set and let $U_0 \coloneqq \mathbf{1}$. For $R \subseteq \mathcal{I}_C$, let $H_R(B)$ be the ordered transport around C using $U_{g_0}(b_a)$ on R and $U_0(b_a)$ elsewhere. The closed source differences are

$$G_S(B) \coloneqq \sum_{R \subseteq S} (-1)^{|S|-|R|} \chi\left(H_R(B) (\mathrm{Hol}_{U_0} C)^{-1} \right) , \qquad \sum_{S \subseteq \mathcal{I}_C} G_S(B) = W_{\chi}(C; U_{g_0}) \, . \tag{5.8}$$

The last equality is Boolean inversion and uses $\mathrm{Hol}_{U_0} C = \mathbf{1}$. Every summand is a closed contraction, invariant under simultaneous gauge transformation of field and reference. The identity-gauge representative is used to evaluate its flat reference integral.

A selected traversal a has its fixed local free-input support Λ_a. Multiplying G_S by $\prod_{a \in S} \prod_{b \in \Lambda_a} \vartheta_b$ leaves its contribution in the original cutoff integral unchanged. This product is zero off that protected box, so only the reconstruction rows selected by the mixed word are evaluated.

A label selection is $J = (D, P, S)$, with $D \subseteq \mathcal{V}_T$, $P \subseteq E_*$ and S a subset of contour traversals. Its active carrier and scalar local factor are

$$\begin{aligned} A_J &\coloneqq \bigcup_{v \in D} \Lambda_v \cup P \cup \bigcup_{a \in S} \Lambda_a \, , \\ F_J(B) &\coloneqq \prod_{v \in D} f_v(B) \prod_{b \in P} h_b(B) \prod_{a \in S} \prod_{b \in \Lambda_a} \vartheta_b(B) \, . \end{aligned} \tag{5.9}$$

The corresponding numerator term is exactly $\int F_J G_S \, \mathrm{d}\mu_{(C_T)_{A_J}}$. This is a marginal of the original Gaussian: its coordinates in $E_* \setminus A_J$ have been integrated out, while every interaction and cutoff is still present in the sum over J.

There are fixed integers s_*, q_* and a fixed range R_* such that every label support has at most s_* bonds and diameter at most R_*, and at most q_* possible labels meet any bond. R_* is enlarged so that every bond in Λ_a is within R_* of the source traversal a; a reversed traversal uses the free-input stencil of its positive representative. These are consequences of the bounded overlap of the primitive supports and the fact that a square has no repeated traversal.

The source cutoff is inserted before expanding the full product of cutoff factors in (5.7); an individual expanded term need not satisfy the full cutoff.

For a finite carrier A, let $C_A \coloneqq (C_T)_A$. With one scalar variable $x_{bc} = x_{cb}$ for each pair of distinct bonds of A, $C_A(x)$ keeps the diagonal blocks of C_A and multiplies its bc block by x_{bc}. For a forest F on A and parameters $w \in [0,1]^F$, let w^F_{bc} be the minimum of the parameters on the unique forest path from b to c, or zero if there is no path; its diagonal is one. Let $\partial_F \coloneqq \prod_{\{b,c\}\in F} \partial_{x_{bc}}$. The Taylor forest formula gives

$$\int F_J G_S \,\mathrm{d}\mu_{C_{A_J}} = \sum_{F \text{ forest on } A_J} \int_{[0,1]^F} \int F_J(B) G_S(B)\, \partial_F p_{C_{A_J}(x)}(B)\Big|_{x=w^F} \,\mathrm{d}B\,\mathrm{d}w\,. \tag{5.10}$$

For a covariance on $\mathfrak{g}^A$, the density used here is

$$p_M(B) \coloneqq (2\pi)^{-d|A|/2} (\det M)^{-1/2} \exp\left(-\frac{1}{2} B^{\mathsf{T}} M^{-1} B\right)\,. \tag{5.11}$$

The determinant square root is positive for positive real M and is continued analytically on the accretive covariance domain.

Let P_π be the matrix whose bc entry is one when b, c belong to the same block of a partition π of A, and zero otherwise. Let $\mathcal{K}_A$ be the convex hull of these partition matrices. If π_t is the partition into components of the forest edges with $w_e \geqslant t$, then $w^F = \int_0^1 P_{\pi_t}\,\mathrm{d}t \in \mathcal{K}_A$. Block restriction and convexity give $\lambda I \leqslant C_A(x) \leqslant \Lambda I$ throughout $\mathcal{K}_A$.

For the fixed integrand $F_J G_S$, its Gaussian integral is smooth on the open set $\{x \mid C_A(x) > (\lambda/2) I\}$: the protected local factors are bounded, and derivatives of the density are polynomials times a uniformly integrable Gaussian on compact subsets. This set contains the compact set $\mathcal{K}_A$. A smooth multiplier with compact support inside this open set and equal to one near $\mathcal{K}_A$ gives a smooth extension by zero to the full interpolation space.

The Taylor forest formula applies to this extension. At the all-one matrix and at every w^F, its value and all the required derivatives agree with those of the original Gaussian integral. The extension therefore gives exactly (5.10); no hard cutoff or Wilson function is differentiated. This is the Taylor forest construction of Abdesselam–Rivasseau, Theorems III.1 and IV.5 and Section V.A [1].

Two bonds are also joined when they belong to the same selected label support; this finite-range graph is Q_J. A connected component of $F \cup Q_J$, together with its selected labels and forest edges, is a component label γ. Let $A_\gamma, D_\gamma, P_\gamma, S_\gamma, F_\gamma$ be these restrictions. Its scalar integration functional is

$$\mathcal{I}_\gamma[H] \coloneqq \int_{[0,1]^{F_\gamma}} \int F_{J_\gamma}(B) H(B)\, \partial_{F_\gamma} p_{C_{A_\gamma}(x)}(B)\Big|_{x=w^{F_\gamma}} \,\mathrm{d}B\,\mathrm{d}w\,. \tag{5.12}$$

The covariance and every differentiated edge in (5.10) are block diagonal across distinct components of $F \cup Q_J$. Their normalised densities, including determinants, consequently factor exactly into (5.12). Every component uses the principal marginal of the same C_T; it is independent of the labels outside its carrier.

At the flat reference C_T commutes with simultaneous constant conjugation. Principal restriction, scalar covariance interpolation, inversion and differentiation preserve this law. Every local scalar factor in F_J is invariant as well. Thus each component integration is equivariant.

For $S_\gamma \neq \varnothing$, let h_γ be the shortest cyclic interval of traversal positions containing S_γ, resolving ties by the first initial position encountered from the fixed contour base in its orientation. A singleton has an interval of length one. On a line the interval runs from the first to the last selected traversal.

Components whose intervals intersect are merged into webs; the web interval is their union, possibly the whole circle. This union is an interval because its interval-intersection graph is connected. Distinct web intervals are disjoint.

A web is integrated as the single closed source $G_{\bigcup_{\gamma\in W} S_\gamma}$ against the product of its component functionals. Its scalar activity is

$$z_W \coloneqq \left(\bigotimes_{\gamma\in W} \mathcal{I}_\gamma\right)\left[G_{\bigcup_{\gamma\in W} S_\gamma}\right] \ . \tag{5.13}$$

The vacuum activity is $z_\gamma = \mathcal{I}_\gamma[1]$ when $S_\gamma = \varnothing$. Components inside a web have disjoint carriers. Distinct webs must also have disjoint contour intervals. Vacuum objects are incompatible only through intersection of their carriers with another object's carrier.

Independent constant conjugation in the component integrals and irreducibility give the integrated Schur factorisation for distinct webs. It is applied inside the closed Wilson contraction.

For the logarithm domain, each contour traversal carries a complex variable t_a, and G_S is replaced by $(\prod_{a\in S} t_a)G_S$. The source polydisc is $|t_a| \leqslant 2$. Its factor $2^{|S|}$ is included in the constants of the source bounds below.

At zero source the numerator is D_T, and at $t_a = 1$ it is $N_{C,T}$. The finite forest expansion and the unique component and web partitions consequently give

$$N_{C,T} = \Xi_{C,T}, \qquad D_T = \Xi_{0,T}, \qquad \mathcal{W}_T^\circ(C) = \frac{\Xi_{C,T}}{\Xi_{0,T}} \ . \tag{5.14}$$

Both gases are sums over unordered compatible families of the activities just defined, with the empty family contributing one.

Conversely, let a compatible family of vacuum components and webs be given. Their constituent carriers are disjoint, so their label selections and forests have unambiguous unions. Every support edge and forest edge remains within its specified component. The component decomposition of this union recovers the original components, and the interval-intersection decomposition recovers precisely the webs, since distinct web intervals are disjoint. Thus the reassembly is inverse to the decomposition. If a web interval crosses the contour base, cyclicity of the trace moves that base between two web intervals before the Schur factorisation; the resulting scalar is unchanged.

5.2 The Integrated Density Estimate

Let a forest on A be fixed, let $n = |A|$, and let d_b be its vertex degrees. For an edge $e = \{b, c\}$ with nonzero covariance block, the complex Cauchy radius is

$$r_e \coloneqq \frac{\lambda}{8 d_b d_c \|C_T(b,c)\|} \ . \tag{5.15}$$

A zero covariance block gives a zero differentiated term. Only the forest variables are perturbed about w^F. The sum of the norms of the perturbation blocks in row b is at most $\lambda/8$,

since $\sum_{c:\{b,c\}\in F}(d_b d_c)^{-1} \leqslant 1$. Thus the covariance perturbation has operator norm at most $\lambda/8$. If M is any covariance on these Cauchy circles, then

$$\operatorname{Re} M \geqslant \frac{7\lambda}{8} I, \qquad \|M\| \leqslant \Lambda + \frac{\lambda}{8}, \qquad \operatorname{Re} M^{-1} \geqslant \frac{7\lambda/8}{(\Lambda+\lambda/8)^2} I = \kappa_G I \,. \tag{5.16}$$

The inverse bound follows from $\operatorname{Re} M^{-1} = M^{-*}(\operatorname{Re} M)M^{-1}$. The determinant branch is continued from the positive real covariance. Its modulus satisfies $|\det M| \geqslant (7\lambda/8)^{dn}$. Consequently

$$|p_M(B)| \leqslant (2\pi)^{-dn/2}(7\lambda/8)^{-dn/2} e^{-\kappa_G \|B\|^2/2} \,. \tag{5.17}$$

Norm equivalence and a Gaussian exponential moment give a factor Ce^{-cr^2} for each distinct selected bad bond. Each regular vertex is bounded on its own cutoff by $Cg_0(1+r)^3$, using (C.3). The closed source with q selected traversals has bound $(C_\chi g_0 r)^q$; for $q = 1$ it has the improved bound $C_\chi g_0^2 r^2$, since $\mathrm{d}\chi_{\mathbf{1}} = 0$. These bounds are applied to the closed source inside the density integral.

Cauchy's formula and (5.17) therefore bound the integrated term for a fixed component by

$$K^{|A_\gamma|} \prod_{\{b,c\}\in F_\gamma} \left(\frac{8 d_b d_c}{\lambda} \|C_T(b,c)\| \right) \tau^{|D_\gamma|+|P_\gamma|+|S_\gamma|+\mathbf{1}_{\{|S_\gamma|=1\}}}, \qquad \tau = C\left[g_0(1+r)^3 + e^{-cr^2} \right] \,. \tag{5.18}$$

Here and below g_0 is small enough that $\tau < 1$ and the original chart condition holds. The same majorants multiply for a web.

For a component with exactly one source mark inside a larger web, the extra power comes from integrating that component first inside the closed contraction. Its matrix factor is a scalar by constant conjugation and irreducibility, and its scalar is the closed one-mark character difference. Components with two or more marks already supply two source powers.

The degree factors in (5.18) are paid by spatial decay. The free-bond lattice has uniformly bounded polynomial volume growth of degree four, including on its finite tori. For d distinct neighbours of a fixed bond, their distances have sum at least $c_1 d^{5/4} - c_2 d$. Hence, for each $\zeta > 0$, a constant K_ζ satisfies

$$d^d \leqslant K_\zeta^d \exp\left(\zeta \sum_{j=1}^{d} d_T(b, c_j) \right) \,. \tag{5.19}$$

Indeed $d \log d$ is bounded by any fixed positive multiple of $d^{5/4}$ plus a linear function of d; here $0^0 = 1$. Taking ζ smaller than $\mu/4$ absorbs $\prod_b d_b^{d_b}$ into part of the covariance decay. The remaining forest edge weights are bounded by $Ke^{-\mu d_T(b,c)/2}$, with a further positive part of the decay available for support-length weights.

On a carrier of n bonds there are at most $2^{q_* n}$ choices of elementary labels. A selection covering that carrier contains at least $\lceil n/s_* \rceil$ labels. The edges of Q_J have bounded range and bounded degree.

For fixed J and a fixed graph $G = F \cup Q_J$, the edges of F outside Q_J are determined; its edges inside Q_J have at most $2^{|Q_J|}$ choices. The support counts give $|Q_J| \leqslant q_*(s_* - 1)n/2$; this exponential multiplicity is included in K_1 below.

Each forest edge weight is bounded by the following pair weight, and this weight is at least one on every edge of Q_J. Thus the forest sum is bounded, with that multiplicity, by a

sum over connected graphs with pair weights

$$v(b,c) = \mathbf{1}_{\{d_T(b,c)\leqslant R_*\}} + Ke^{-\mu d_T(b,c)/4}, \qquad \sup_b \sum_{c\neq b} v(b,c) = V_* < \infty\,. \tag{5.20}$$

The unused graph edges cost at most $\prod_{b<c}(1+v(b,c)) \leqslant e^{V_* n/2}$. Keeping one spanning tree gives the remaining tree sum. For $n \geqslant 2$ and a fixed root bond b_0, summing its possible distinct vertices gives

$$\sum_{\substack{A\ni b_0\\ |A|=n}} \sum_{T \text{ tree on } A} \prod_{\{b,c\}\in T} v(b,c) \leqslant \frac{n^{n-2}}{(n-1)!} V_*^{n-1}\,. \tag{5.21}$$

The factorial accounts for unordered nonroot vertices; the bound follows by summing their positions along a rooted labelled tree. The right-hand side is bounded exponentially in n.

Combining these payments yields convergent series bounded by constants times

$$\sum_{n\geqslant 1} K_1^n \tau^{\lceil n/s_*\rceil} = O(\tau), \qquad \sum_{n\geqslant 1} K_1^n \tau^{\max(2,\lceil n/s_*\rceil)} = O(\tau^2)\,. \tag{5.22}$$

The first applies to vacuum components and the second to source components, rooted either at a spatial bond or at a specified source traversal. The coupling is chosen after the constants so that $K_1^{s_*}\tau$ is sufficiently small; exponential weights in the carrier size and in the sum of covariance-edge lengths remain available.

5.3 Extent Weights and the Square Geometry

The extent costs of a component, a web, and a cluster are

$$\ell(\gamma) := |A_\gamma| + \sum_{\{b,c\}\in F_\gamma} d_T(b,c), \qquad \ell(W) := \sum_{\gamma\in W} \ell(\gamma), \qquad \ell(\Gamma) := \sum_{p\in\Gamma} \ell(p)\,. \tag{5.23}$$

A vacuum object uses its component cost. A_p denotes an object's carrier and H_p its contour interval length, with $H_p = 0$ for vacuum objects.

On a straight line, connectivity of $F_\gamma \cup Q_{J_\gamma}$ gives $H_\gamma \leqslant c_h \ell(\gamma)$, since the support edges have length at most R_* and the source positions are within R_* of their supports; c_h is fixed at (5.24) below, where the square is treated, and the same constant covers the line, the interval length there being at most the spatial span of the marks plus one. (H_p, the contour interval length, is also called the contact-interval length below.)

Here the square C_n has vertices $0, nLe_1, nL(e_1+e_2), nLe_2$ in a fixed lift, with all other coordinates zero. Its side length is nL fine bonds, its physical perimeter is $4nL\varepsilon$, and its corners are coarse sites. The finite torus has fine periods LK_i, with $K_i \geqslant 3(n+2)$ and the native grid divisibilities. The limits taken below (the joint limit in n and the periods of (5.39), the torus limit at fixed n and the iterated limit of the corner section) are over such compatible integers n and periods. Between points of this square the periodic coordinate distances equal their distances in the stated lift. This fixes one coordinate plane, one position and one orientation.

An oriented square contour C as chosen at the start of the paper, with coarse corners and sides along two coordinate directions, is a coarse translate of the square with vertices $0, nLe_\mu, nL(e_\mu+e_\nu), nLe_\nu$, traversed in this cyclic order or in the reverse one (the base

point does not affect $\chi(\mathrm{Hol})$). The permutation of the coordinate axes carrying (e_μ, e_ν) to (e_1, e_2), with the periods permuted accordingly, is a symmetry of the fine lattice, of the Wilson measure $\rho_0(U)\,\mathrm{d}U$ and of every hypothesis on the periods (all are imposed on every period), so $\mathbb{E}_\varepsilon[W_\chi(C)]$ equals the expectation of a coarse translate of C_n, which is the contour admitted in Section F.1 and in (H3).

Reversing the orientation inverts the holonomy, and $\chi(U^{-1}) = \overline{\chi(U)}$ for the character of a unitary representation, so the expectation of the reversed square is the complex conjugate of that of the original. Every statement below about $\mathbb{E}_\varepsilon[W_\chi(C_n)]$ therefore holds for every such square, up to complex conjugation for the reversed orientation.

Let D be the maximum-norm diameter of a nonempty set of square contacts and let $s = nL$. If $D < s/2$, its contacts occupy one side or two adjacent sides near their common corner, and an arc of length at most $4D + 1$ contains them. If $D \geqslant s/2$, the whole contour has length $4s \leqslant 8D$.

A connected support network for the component has length at most $R_*|A_\gamma| + \sum_{F_\gamma} d_T(b, c)$, and each contact is within R_* of that network. With the stated bond-distance convention, $c_h = 32(1 + R_*)$ (the interval constant, italic subscript; distinct from the group constant $c_{\mathrm{h}} = \sqrt{N}$ of Appendix B) therefore gives

$$H_\gamma \leqslant c_h \ell(\gamma), \qquad H_W \leqslant \sum_{\gamma \in W} H_\gamma \leqslant c_h \ell(W)\,. \tag{5.24}$$

Let d_C count the shortest number of traversal steps between two positions, cyclically for a square. For $\theta > 0$, a position a in h_γ satisfies

$$\mathbf{1}_{\{a \in h_\gamma\}} \leqslant \sum_{b \in S_\gamma} e^{-\theta d_C(a,b)} e^{\theta H_\gamma}\,. \tag{5.25}$$

The sum of $e^{-\theta d_C(a,b)}$ over all traversal positions is bounded by $1 + 2/(e^\theta - 1)$, uniformly in the contour length. The already proved traversal-root load, with the extra extent weight, therefore gives an interval-root load of order τ^2.

For a source component let u_γ be the positive majorant on the right-hand side of (5.18), with the constants fixed there, and let $H_\gamma \coloneqq |h_\gamma|$ be its contact-interval length.

In the sum over webs, the indicator that their interval graph is connected is majorised by its spanning-tree sum. The rooted-tree recursion is bounded by $u_\gamma e^{aH_\gamma}$ whenever the interval-root load, with this extra weight, is at most a. This follows by iterating $T_\gamma = u_\gamma \exp(\sum_{\gamma' : h_{\gamma'} \cap h_\gamma \neq \varnothing} T_{\gamma'})$ from $T_\gamma = u_\gamma$, since the sum in the exponent is bounded by aH_γ. Spatial incompatibilities only reduce this majorant. The resulting whole-web loads are $O(\tau^2)$.

Together with the vacuum load $O(\tau)$, they satisfy the hard-core polymer convergence criterion [14], with size equal to carrier size plus a small positive multiple of contour interval length. This proves absolute convergence of the connected logarithm of (5.14) for sufficiently small fixed g_0. In particular the numerator has no zero on the source-scaling polydisc used by that expansion. Its logarithm is fixed by continuation from zero source, where numerator and denominator agree.

More explicitly, let $\beta \coloneqq \mu/(1024(1 + c_h))$, $\eta = \mu/1024$, and $a(p) = |A_p| + \beta H_p$. The density and graph payments above remain valid after multiplying the component majorants by $e^{4|A_\gamma| + 8\beta H_\gamma + 8\eta\ell(\gamma)}$: the added edge exponent is smaller than the unused covariance decay, and the added carrier exponent only changes K_1. With $a = \beta$ in the web recursion and the

remaining weights reserved for its interval-root conversion, there is a constant C_* such that

$$\begin{aligned} \sup_b \sum_{p \,|\, b \in A_p} |z_p| e^{2a(p)+2\eta\ell(p)} &\leqslant C_* \tau \ , \\ \sup_a \sum_{W \,|\, a \in h_W} |z_W| e^{2a(W)+2\eta\ell(W)} &\leqslant C_* \tau^2 \ . \end{aligned} \tag{5.26}$$

The source-only version of the first sum is also bounded by $C_*\tau^2$. The coupling g_0 is chosen after these constants so that $C_*\tau \leqslant 1/4$ and $C_*\tau^2 \leqslant \beta/4$. Carrier intersection and interval intersection then give the criterion

$$\sum_{q \not\sim p} |z_q| e^{a(q)+\eta\ell(q)} \leqslant C_* \tau |A_p| + C_* \tau^2 H_p \leqslant \frac{1}{4} a(p) \ . \tag{5.27}$$

A cluster $\Gamma = (p_1, \dots, p_j)$, with $j \geqslant 1$, is an ordered tuple of vacuum components or source webs with connected incompatibility graph; repetitions are allowed and an object is incompatible with itself. Its scalar weight is

$$\Phi_C(\Gamma) \coloneqq \frac{1}{j!} \left(\sum_{H \text{ connected on } \{1,\dots,j\}} \prod_{\{r,s\} \in H} (-\mathbf{1}_{\{p_r \not\sim p_s\}}) \right) \prod_{r=1}^{j} z_{p_r} \ . \tag{5.28}$$

The pinned version used here retains an activity factor. Let $\widetilde{z}_p \coloneqq z_p e^{\eta\ell(p)}$, and let $m_p(\Gamma)$ count the occurrences of p in the ordered cluster tuple. Then

$$\sum_{\Gamma} m_p(\Gamma) |\Phi_C(\Gamma)| e^{\eta\ell(\Gamma)} \leqslant |z_p| e^{\eta\ell(p)+a(p)} \ . \tag{5.29}$$

For a fixed incompatibility graph on labelled vertices, the connected subgraphs are partitioned by their breadth-first spanning tree rooted at vertex one, with the least-labelled parent at the previous level. A tree's interval of subgraphs allows all edges within a level and the edges to the previous level whose other endpoint is no earlier than the chosen parent. These intervals partition the connected graphs. The alternating sum on each interval is zero or has modulus one. Therefore the absolute Ursell coefficient is bounded by the sum over its spanning trees.

The positive generating sum of rooted trees with root species p and vertex weights $|\widetilde{z}_q|$ is the monotone limit of

$$T_p^{(0)} = |\widetilde{z}_p|, \qquad T_p^{(k+1)} = |\widetilde{z}_p| \exp\left(\sum_{q \not\sim p} T_q^{(k)} \right) \ . \tag{5.30}$$

Permutation symmetry of the ordered tuple gives $\sum_\Gamma m_p(\Gamma) F(\Gamma) = j \sum_{\Gamma \,|\, p_1 = p} F(\Gamma)$ at each fixed length j for its symmetric weight F. Hence the factor $1/j!$ becomes $1/(j-1)!$, including repeated species. The exponential in (5.30) assembles an unordered family of child trees, with its factorial for repeated choices; $T_p^{(k)}$ enumerates rooted trees of height at most k. Equation (5.27) implies inductively $T_p^{(k)} \leqslant |\widetilde{z}_p| e^{a(p)}$, proving (5.29).

For a countable polymer family, the species are first restricted to a finite subset. The same upper bound holds for every such restriction, and monotone convergence of the positive majorants proves the pinned bound for the countable family.

The bound applies to the source objects meeting a fixed coarse contour segment x. Their weighted activity sum is bounded by the source-only loads in (5.26). Thus

$$\sum_{\Gamma \,|\, x \in J_\Gamma} |\Phi_C(\Gamma)| e^{\eta\ell(\Gamma)} \leqslant C \tau^2 \ . \tag{5.31}$$

Here J_Γ consists of the coarse segments containing its source marks. A coarse segment contains L fine traversal positions, which changes only the constant. The same proof applies on the infinite lattice, using its finite principal covariance marginals.

A spanning tree of $F_\gamma \cup Q_{J_\gamma}$ connects each component carrier. Its metric length is at most $R_*|A_\gamma| + \sum_{F_\gamma} d_T(b,c)$. Paths of length at most R_* attach its source contacts; there are at most $q_*|A_\gamma|$ contacts. A web becomes connected after adjoining its contour intervals, whose total length is at most $c_h \ell(W)$ by (5.24). In a cluster, incompatible objects meet through carriers or contour intervals, so these networks form a connected union. The carrier bonds themselves add at most $|A_\gamma|$ per component, and conversion from maximum-norm distances to lattice path lengths costs at most four. Thus its length and diameter are bounded by $C_{\text{net}}\ell(\Gamma)$ with $C_{\text{net}} := 8(1+R_*)(1+q_*+c_h)$. This remains an upper bound with repeated objects. Equation (5.31) therefore controls spatial extent as well as contour span.

5.4 The Perimeter Coefficient

The infinite-lattice covariance is constructed from the same periodic block grid. Its finite principal marginals satisfy the same spectral and decay bounds, so the density, counting and weighted KP estimates above apply with the same constants.

For an orientation $v \in \mathscr{V}_\square = \{e_1, e_2, -e_1, -e_2\}$, the straight line through the coarse origin in that orientation is divided into half-open segments of L fine traversals, indexed by $\mathbb{Z}$. Every side of C_n is a coarse translation of the corresponding oriented segment of this line. Each square traversal belongs to exactly one side and one such segment. The lattice spacing ε is the fixed physical length of a fine bond.

Φ_v denotes the connected weight (5.28) on the straight line in orientation v.

For a finite set S of traversal positions on the line, let I be any finite oriented arc containing it. For $R \subseteq S$, let $u_a^R := U_{g_0}(b_a)$ on R and $u_a^R = U_0(b_a)$ otherwise, with $U_0 = \mathbf{1}$. The closed relative source on the line is explicitly

$$G_S^v(B) = \sum_{R \subseteq S} (-1)^{|S|-|R|} \chi\left(\left(\prod_{a \in I}^{\longrightarrow} u_a^R(B) \right) P_{U_0}(I)^{-1} \right) . \tag{5.32}$$

The arrow means order along the arc. The reference reverse transport closes the word, and extending the arc beyond its selected marks adds only identity factors in the reference gauge. This is the line version of the finite closed source definition used above.

Let $J_\Gamma \subset \mathbb{Z}$ be the set of coarse line segments containing at least one source mark of the cluster. The vacuum subtraction in $\log \Xi_C - \log \Xi_0$ retains exactly the clusters with $J_\Gamma \neq \varnothing$. Translation by one coarse segment preserves the reference construction. The absolutely convergent line density and its coefficient are

$$\boxed{\kappa_v(g_0) := \sum_{\Gamma \,|\, 0 \in J_\Gamma} \frac{\Phi_v(\Gamma)}{|J_\Gamma|}, \qquad \delta m_v(g_0, \varepsilon) := -\frac{\kappa_v(g_0)}{L\varepsilon}} . \tag{5.33}$$

The factor $|J_\Gamma|^{-1}$ allocates each translated connected contribution once. The weights contain the full local exponentials and exact hard-cutoff factors. The weighted rooted estimate (5.31) proves absolute convergence and $|\kappa_v| \leqslant C\tau^2$.

Length Allocation and Corners. Let $I_m := \{0, \dots, m-1\}$. The infinite-lattice logarithm for the finite straight relative word of length mL is

$$F_v(m) := \sum_{\Gamma \,|\, \varnothing \neq J_\Gamma \subseteq I_m} \Phi_v(\Gamma) \,. \tag{5.34}$$

It is the bulk limit of the logarithm of the same finite-word numerator divided by D_T. For fixed m, this follows by truncating (5.31), taking the limit of finitely many covariance-marginal integrals, and then removing the truncation. Translation invariance gives the exact difference

$$F_v(m) - m\kappa_v = -\sum_{x \in I_m} \sum_{\substack{\Gamma \,|\, x \in J_\Gamma \\ J_\Gamma \not\subseteq I_m}} \frac{\Phi_v(\Gamma)}{|J_\Gamma|} \,. \tag{5.35}$$

A cluster in the inner sum joins a source in segment x to a source outside I_m. With $R_{\mathrm{end}} = 2L + 4R_*$, the network bound gives $C_{\mathrm{net}}\ell(\Gamma) \geqslant (L \min(x+1, m-x) - R_{\mathrm{end}})_+$, where $t_+ := \max(t, 0)$. Applying (5.31) and summing that geometric tail yields

$$|F_v(m) - m\kappa_v| \leqslant C\tau^2 \,. \tag{5.36}$$

For the square in the infinite-lattice reference, clusters whose source marks all lie on one side have exactly the corresponding line weights: their contact intervals stay on that side, their closed words have the same ordered nonidentity factors, and every local primitive is translated by a multiple of L. The Gaussian marginal is unchanged by this translation.

All other source clusters touch at least two sides. Each such cluster is allocated over its source segments with the weight $|J_\Gamma|^{-1}$. A root segment at distance j from its nearest corner can be joined to another side only by a cluster with $C_{\mathrm{net}}\ell(\Gamma) \geqslant (Lj - R_{\mathrm{end}})_+$. Thus the sum over mixed-side clusters is bounded by $C\tau^2 \sum_{j \geqslant 0} e^{-c'j}$, with $c' := \eta L / C_{\mathrm{net}}$ (the extent weight $e^{\eta \ell(\Gamma)}$ of (5.31); the factor $e^{\eta R_{\mathrm{end}}/C_{\mathrm{net}}}$ is absorbed in C). With $\widehat{F}_\square(n)$ denoting its absolutely convergent infinite-lattice connected sum, this proves

$$\left| \widehat{F}_\square(n) - \sum_{v \in \mathscr{V}_\square} F_v(n) \right| \leqslant C\tau^2, \qquad \left| \widehat{F}_\square(n) - n \sum_{v \in \mathscr{V}_\square} \kappa_v \right| \leqslant C\tau^2 \,. \tag{5.37}$$

The bounded corner contribution is extracted in Section 6.

The Buffered Finite-Torus Limit. The perimeter coefficient also follows along the stated buffered tori. Let R be an integer extent cutoff. A cluster rooted at a specified source segment with $\ell(\Gamma) \leqslant R$ has all its sites and marks in a neighbourhood of radius $C_{\mathrm{net}}R + L + 2R_*$ in fine-lattice units around that segment. There are finitely many such cluster shapes, including multiplicities: each object has extent at least one, so a tuple has at most R objects. Away from the square corners by more than $\left\lceil \frac{C_{\mathrm{net}}R + 2L + 2R_*}{L} \right\rceil$ coarse segments, these are the same shapes as on the corresponding straight line.

Each such root is translated to segment zero. Let $\epsilon_R(T)$ be the sum, over the four orientations and these finitely many shapes, of $|\Phi_v^T(\Gamma) - \Phi_v(\Gamma)|/|J_\Gamma|$. The finite-torus weight here is the same finite Gaussian-marginal integral on the translated sites. For large enough periods the shape embeds without wraparound. Periodisation gives entrywise convergence of all its covariance blocks, uniformly under the allowed coarse translations. The accretive density bound supplies a common integrable majorant for its Cauchy derivatives, so each

integral and each finite product defining the cluster weight converges. Consequently $\epsilon_R(T) \to 0$ as the shortest period tends to infinity, for every fixed R.

The finite-square logarithm is allocated over its source segments. The clusters omitted by the extent truncation cost at most $C\tau^2 e^{-\eta R}$ per root. There are at most cR corner-near roots per side, and each has load at most $C\tau^2$. Thus

$$\left| \frac{1}{4n} \log \frac{N_{C_n,T}}{D_T} - \frac{1}{4} \sum_{v \in \mathscr{V}_\square} \kappa_v \right| \leqslant C\tau^2 \left(\frac{R}{n} + e^{-\eta R} \right) + \epsilon_R(T) \,. \tag{5.38}$$

The limit is taken first in n and the compatible buffered periods with R fixed, then in $R \to \infty$. Every logarithm in this limit is the analytic branch continued along $t_a = t$, $0 \leqslant t \leqslant 1$, from the value zero of the logarithm of the ratio at $t = 0$.

With the cutoff index restored, κ_v^σ is the absolutely convergent line sum (5.33) with the prescription (5.3). For each $\sigma \in \{\mathsf{op}, \mathsf{HS}\}$, the flat square perimeter coefficient exists and is

$$\boxed{\delta m^\sigma(g_0, \varepsilon) := -\frac{1}{4L\varepsilon} \sum_{v \in \mathscr{V}_\square} \kappa_v^\sigma(g_0) = -\lim_{n \to \infty} \frac{1}{4nL\varepsilon} \log \frac{N^\sigma_{C_n,T_n}}{D^\sigma_{T_n}}} \,. \tag{5.39}$$

Reversing a finite relative path conjugates its character, and translation by its coarse length identifies it with the oppositely oriented reference path. All other factors of the weights (the Gaussian density and its forest derivatives, f_v, h_b and θ) are real, the reindexing $a \mapsto -a-1$ carries the $-v$ line onto the reverse of the v line with the stencils, protected boxes, intervals and incompatibilities in bijection, and κ_v is an absolutely convergent sum, so $\kappa_{-v} = \overline{\kappa_v}$. Hence the square coefficient is real.

6 The Corner Coefficient in the Same First Step

The lattice spacing, character, reconstruction, and cutoff prescription σ are those of Section 5, and the coupling is below the common minimum of the straight and bent-path smallness thresholds of this section and the regional thresholds of Proposition 7.4, chosen after all chart radii and majorant constants. All limits in this section vary reference contours and enclosing tori at this one RG step. The coefficients below use $N^\sigma_{C,T}$ and D^σ_T of (5.5); σ is suppressed until the conclusion.

$C_\infty = \mathsf{H}_\infty^{-1}$ is the free-field covariance on $\ell^2(E_{*,\infty}; \mathfrak{g})$, constructed in (A.28); it is distinct from the reconstructed fine-field covariance $\mathsf{K}_\infty = \mathsf{C}_\infty C_\infty \mathsf{C}_\infty^*$.

The countable primitive index family $\mathcal{V}_\infty$ consists of labels $(0, \alpha)$, $(1, b)$ and $(2, c)$ for the functions v_α, j_b, d_c in (C.1). Here b ranges over positive fine bonds, c over positive coarse bonds, and α over positive plaquettes and the mean-transport pairs (y, x) with $y \in L\mathbb{Z}^4$ and $x \in \mathcal{B}_y \setminus \{y\}$. These are exactly the infinite-lattice translates of the finitely many local templates modulo the coarse grid.

6.1 A Local Corner Subtraction

Let $v, w \in \mathscr{V}_\square$ be perpendicular incoming and outgoing orientations. For $a \in \mathbb{Z}$, the bent reference path has fine vertices and oriented bonds

$$x_a^{v,w} = \begin{cases} av, & a < 0, \\ aw, & a \geqslant 0, \end{cases} \qquad b_a^{v,w} = (x_a^{v,w}, x_{a+1}^{v,w}) \,. \tag{6.1}$$

Its coarse segment $j \in \mathbb{Z}$ contains the fine traversal indices $jL, \dots, (j+1)L-1$. Thus the bend is at a coarse site, between segments -1 and 0.

For every finite set of selected traversals, the closed relative source is (5.32) on the corresponding finite arc of this bent path, with the same local vertices, protected cutoffs, and Gaussian principal marginals. Its components and webs are those of (5.12)–(5.13), with ordinary intervals in the ordered traversal indices, and its connected weight $\Phi_{v,w}(\Gamma)$ is (5.28). The set $J_\Gamma \subset \mathbb{Z}$ again records the coarse segments containing source marks. Vacuum-only clusters are excluded. These definitions involve finite closed contractions and finite-dimensional integrals only, even on the infinite reference path.

The density, label-counting and KP estimates of Section 5 apply to this path with enlarged fixed constants. Indeed, it has no repeated traversal, and the path distance between two of its points is at most twice their maximum-norm spatial distance. Each ray is part of one of the two straight lines, so at most $2q_*$ primitive or source labels meet any bond; the support sizes s_* and ranges R_* are unchanged. Thus the original counting proof applies with $q'_* = 2q_*$, and the interval/extent bound holds with $c'_h = 2c_h$. The resulting rooted estimate is

$$\sum_{\Gamma \,|\, j \in J_\Gamma} |\Phi_{v,w}(\Gamma)| e^{\eta' \ell(\Gamma)} \leqslant C\tau^2 \,. \tag{6.2}$$

Let $\beta' \coloneqq \mu/(2048(1+c'_h))$ and $\eta' = \eta/2$. The index-distance conversion (5.25) is unchanged, since the traversal indices still form $\mathbb{Z}$ and the sum of its geometric kernel is $1 + 2/(e^\theta - 1)$. The carrier-root and interval-root loads therefore obey (5.26) with the bent path, size $a'(p) = |A_p| + \beta' H_p$, exponent η' and a fixed enlarged constant C'_* obtained from the same label and positive-tree sums with q'_*. The coupling satisfies $C'_*\tau \leqslant 1/4$ and $C'_*\tau^2 \leqslant \beta'/4$. The incompatibility sum is then at most $a'(p)/4$, and the pinned proof (5.29)–(5.30) gives (6.2), for both cutoff choices.

The relative word transforms by conjugation at its base when field and reference are transformed together, and its character, component-integrated closed contractions and scalar cluster weights are gauge invariant.

The rooted density and its straight reference are

$$a_{v,w}(j) \coloneqq \sum_{\Gamma \,|\, j \in J_\Gamma} \frac{\Phi_{v,w}(\Gamma)}{|J_\Gamma|}, \qquad k_{v,w}(j) \coloneqq \begin{cases} \kappa_v, & j < 0, \\ \kappa_w, & j \geqslant 0. \end{cases} \,. \tag{6.3}$$

The straight densities are those of (5.33). The local corner coefficient is

$$\boxed{\zeta_{v,w} \coloneqq \sum_{j \in \mathbb{Z}} \left(a_{v,w}(j) - k_{v,w}(j)\right), \qquad z_{v,w} \coloneqq \exp(\zeta_{v,w})} \,. \tag{6.4}$$

A cluster with all source marks at negative indices has exactly the weight of the same cluster on the incoming straight line: its ordered nonidentity factors, source stencils, ordinary intervals, local vacuum factors and Gaussian marginals agree. The same holds on the positive half-line with orientation w. Vacuum carriers may cross the bend. Consequently only clusters with source marks on both sides of index zero survive the difference of the rooted sums. On the straight side of the subtraction these are the straight clusters straddling the same cut.

A surviving cluster rooted in segment j has extent at least $c|j| - C_0$, with $c \coloneqq L/(2C_{\text{net}})$ and $C_0 = R_*/C_{\text{net}}$, since its connected support/interval network reaches the other half-line,

so has length at least $L|j| - 2R_*$, while the bent-path network constant (C_{net} with q'_*, c'_h) is at most $2C_{\text{net}}$. The rooted bound, with part of its extent weight retained, therefore gives

$$|a_{v,w}(j) - k_{v,w}(j)| \leqslant C\tau^2 e^{-c|j|}, \qquad |\zeta_{v,w}| \leqslant C\tau^2, \qquad |z_{v,w} - 1| \leqslant C\tau^2 \,. \tag{6.5}$$

This proves absolute convergence and localisation at the corner.

Let $J_\Gamma^- \coloneqq J_\Gamma \cap \mathbb{Z}_{<0}$ and $J_\Gamma^+ = J_\Gamma \cap \mathbb{Z}_{\geqslant 0}$. In each of the following three sums, the cluster has both $J_\Gamma^- \neq \varnothing$ and $J_\Gamma^+ \neq \varnothing$, and its weight is that of the indicated bent or straight path. Then

$$\begin{aligned} \zeta_{v,w} = &\sum_\Gamma^{v,w} \Phi_{v,w}(\Gamma) - \sum_\Gamma^{v} \frac{|J_\Gamma^-|}{|J_\Gamma|} \Phi_v(\Gamma) \\ &- \sum_\Gamma^{w} \frac{|J_\Gamma^+|}{|J_\Gamma|} \Phi_w(\Gamma) \,. \end{aligned} \tag{6.6}$$

Indeed, summing the rooted allocations gives one for a mixed bent cluster and the displayed fractions for the two missing straight half-line contributions. Each sum is absolutely convergent by (6.2) and the distance-to-cut payment. In particular the corner factor includes the straight-edge endpoint deficits as well as interactions joining the two perpendicular rays.

6.2 Identification with the Square Constant

For a fixed positive integer n, let $\widehat{F}_\square(n)$ be the infinite-lattice connected sum over clusters with source marks on C_n, with its fixed cyclic traversal order and base, and let $F_{\square,\infty}(n)$ be the bulk-limit logarithm, whose existence is proved below.

The connected sum is allocated over the $4n$ coarse source segments by $|J_\Gamma|^{-1}$. Its absolute convergence follows from (5.31). The same bound holds uniformly on all compatible tori, which satisfy $K_i \geqslant 3(n+2)$, so the square geometry and weighted KP payment used to prove that bound have their original constants, including at the corners.

Both connected sums are truncated to $\ell(\Gamma) \leqslant R$. Once the shortest fine period exceeds a fixed multiple of $nL + R + R_*$, each such rooted cluster has a unique lift through its source root to the fixed neighbourhood of the lifted square. Its connected support/interval network has length at most a fixed multiple of R, so a network winding around a torus cannot occur in this truncation. The primitive functions and supports agree after lifting. The cyclic interval rule and its tie-breaking base depend only on the fixed ordered list of square traversals, so they agree as well, including intervals crossing the base.

There are finitely many lifted shapes, labels and multiplicities: the root belongs to a finite contour, the extent bounds all spatial displacements and the number of component and cluster occurrences, and each site has finitely many primitive labels.

Covariance entries converge by the periodisation of H_∞^{-1} (Appendix A: the periodisation of the free inverse solves the torus inverse equation, the paragraph preceding (A.31), and the image bound (A.33), displayed there for K, holds for H_∞^{-1} by the same argument from the decay of its entries established before (A.30)). For each shape, the differentiated Gaussian density majorant (5.17) is integrable and uniform in the torus. Dominated convergence therefore gives convergence of every component integral, web activity and finite product in its cluster weight. Let $\mathscr{G}_{n,R}$ be this finite set of lifted ordered clusters with at least one source mark, and let

$$\epsilon_{n,R}(T) \coloneqq \sum_{\Gamma \in \mathscr{G}_{n,R}} |\Phi_\square^T(\Gamma) - \Phi_\square^\infty(\Gamma)| \,. \tag{6.7}$$

For the domination just used, every fixed shape has $r_e^{-1} \leqslant 8d_b d_c A/\lambda$ in (5.15), uniformly in T. Thus the Cauchy factors multiplying (5.17) are uniformly bounded for that shape. The differentiated density is analytic, and hence continuous, in its covariance entries on a common accretive neighbourhood. This gives pointwise convergence as well as the stated integrable majorant, including when a covariance edge tends to zero. It follows that $\epsilon_{n,R}(T) \to 0$ at fixed n, R.

The omitted clusters, including any winding clusters, have total allocated weight at most $Cn\tau^2 e^{-\eta R}$ in each sum. Consequently

$$\left|\log \frac{N_{C_n,T}}{D_T} - \widehat{F}_{\square}(n)\right| \leqslant Cn\tau^2 e^{-\eta R} + \epsilon_{n,R}(T)\,. \tag{6.8}$$

Here the finite-torus logarithm is the connected expansion of the ratio, on its branch continued from zero source; the vacuum subtraction and original denominator are unchanged. First taking $T \to \infty$ at fixed n, R and then $R \to \infty$ proves

$$F_{\square,\infty}(n) = \widehat{F}_{\square}(n) = \lim_{T\to\infty} \log \frac{N_{C_n,T}}{D_T}\,. \tag{6.9}$$

Thus the infinite connected sum and the bulk-limit logarithm used below are the same object.

The four incoming/outgoing pairs of the counterclockwise square are

$$\mathscr{C}_{\square} \coloneqq \{(e_1, e_2), (e_2, -e_1), (-e_1, -e_2), (-e_2, e_1)\}\,. \tag{6.10}$$

On the branch of $F_{\square,\infty}(n)$ continued from zero source, the same closed-cluster allocation now gives positive constants C, c, independent of n, such that

$$\left|F_{\square,\infty}(n) - n\sum_{v\in\mathscr{V}_{\square}} \kappa_v - \sum_{(v,w)\in\mathscr{C}_{\square}} \zeta_{v,w}\right| \leqslant C\tau^2(n+1)e^{-cn}\,. \tag{6.11}$$

In the proof of this estimate every square cluster is allocated over its source segments by $|J_\Gamma|^{-1}$, and each side is divided into the segments nearest its initial corner and those nearest its terminal corner, a midpoint tie being broken consistently. A corner then receives the last $\lfloor n/2 \rfloor$ incoming segments and the first $n - \lfloor n/2 \rfloor$ outgoing segments, and it is translated to the coarse origin. At the base corner $(-e_2, e_1)$, trace cyclicity first puts the incoming factors before the outgoing factors, as in the cyclic reassembly after (5.14).

For a root in its assigned half-sides, every cluster of extent less than $c_1 n - C_1$ is identical to the corresponding bent-path cluster: all marked traversals and their stencils stay away from the other three corners, cyclic intervals become the ordinary local intervals, and the same global Gaussian principal marginals and vacuum labels occur. Any failure of this identification costs extent at least $c_1 n - C_1$ and hence at most $C\tau^2 e^{-c_2 n}$ per root by (5.31) and (6.2). There are $4n$ roots. Extending the four truncated rooted differences to all $j \in \mathbb{Z}$ costs another $C\tau^2 e^{-c_3 n}$ by (6.5). This proves (6.11), after enlarging constants for the finitely many short squares.

Here $c_1 = L/(4C_{\text{net}})$ and $C_1 = 2R_*/C_{\text{net}}$ (a cluster rooted in its assigned half-sides and reaching within R_* of another corner has network length at least $nL/2 - 2R_*$, and the bent-path network constant is at most $2C_{\text{net}}$), $c_2 = \eta' c_1$ and $c_3 = c/2$ with the c of (6.5) (the extension runs over $|j| \geqslant n/2$), the factors $e^{\eta' C_1}$ and the like being absorbed in C.

With the cutoff index restored, the square corner matching factor is

$$\boxed{z_{\mathrm{c}}^{\sigma}\left(g_0, \frac{\pi}{2}\right) \coloneqq \exp\left(\frac{1}{4}\sum_{(v,w)\in\mathscr{C}_{\square}} \zeta_{v,w}^{\sigma}(g_0)\right)}\,. \tag{6.12}$$

This definition fixes the fourth-root branch by the exponential of the convergent corner logarithm. It is the geometric mean of the four oriented corner factors; if the reference construction preserves the corresponding rotations, they agree and (6.12) is their common value.

For a real-valued character all the real Gaussian integrals, cluster weights and corner logarithms above are real, so these factors are positive.

For each fixed n, the inner limit in the following expression is the limit over compatible tori with their shortest period tending to infinity. Then

$$\boxed{\log z_{\mathrm{c}}^{\sigma}\Big(g_0,\frac{\pi}{2}\Big)=\frac{1}{4}\lim_{n\to\infty}\left(\lim_{T\to\infty}\log\frac{N^{\sigma}_{C_n,T}}{D^{\sigma}_{T}}+4nL\varepsilon\delta m^{\sigma}\right)}\,. \tag{6.13}$$

The denominator is the original vacuum integral throughout the inner limit, whose existence is (6.9). The outer limit follows from (6.11) and (5.39). The order of limits is part of this reference prescription.

Equivalently, with $\mathcal{W}^{\circ,\sigma}_{\infty}(C_n)=\lim_{T\to\infty}N^{\sigma}_{C_n,T}/D^{\sigma}_{T}$, there is an error E^{σ}_{n} such that

$$\begin{aligned}\mathcal{W}^{\circ,\sigma}_{\infty}(C_n)&=e^{-4nL\varepsilon\delta m^{\sigma}}\left(z_{\mathrm{c}}^{\sigma}\Big(g_0,\frac{\pi}{2}\Big)\right)^{4}e^{E^{\sigma}_{n}}\,,\\ |E^{\sigma}_{n}|&\leqslant C\tau^{2}(n+1)e^{-cn}\,.\end{aligned} \tag{6.14}$$

The corner coefficient uses the same flat integral and cutoff prescription as the perimeter coefficient. Its identification in a conditional regular region, with boundary and periodic errors, is Proposition 7.4.

7 Identification in a Regular Integration Region

The coefficient in (5.33) was computed with the translation-invariant flat covariance. A conditional integration in the first RG step instead uses the inverse of a restricted Hessian. The two connected expansions are compared with the action terms that cross the integration boundary retained; this identifies their straight-side coefficients quantitatively.

7.1 The Wilson Source before the Flat Evaluation

Let $\mathcal{I}_C$ be the ordered set of traversals of C. For $R\subseteq\mathcal{I}_C$, the mixed field U^R uses the integration link $U(b)=U'(b)U_1(b)$ on a traversal in R and the background link $U_1(b)$ on every other traversal. Here $U_1=U_1(\Omega_1,V)$ is the reconstruction before the boundary split (with the exterior determining field retained). Reversed traversals use inverse links. Define

$$\begin{aligned}\mathcal{G}_S(U,U_1)&=\sum_{R\subseteq S}(-1)^{|S|-|R|}\chi(\mathrm{Hol}_{U^R}C)\,,\\ \mathcal{W}(t;U,U_1)&=\sum_{S\subseteq\mathcal{I}_C}\left(\prod_{a\in S}t_a\right)\mathcal{G}_S(U,U_1)\,.\end{aligned} \tag{7.1}$$

Here $t=(t_a)_{a\in\mathcal{I}_C}$ is a family of scalar parameters. Boolean inversion gives

$$\mathcal{W}(1;U,U_1)=W_{\chi}(C;U),\qquad \mathcal{W}(0;U,U_1)=W_{\chi}(C;U_1)\,. \tag{7.2}$$

Every term is a closed contraction, invariant under simultaneous gauge transformations of U and U_1.

The Wilson insertion starts in (1.3). Its gauge invariance preserves it under the Faddeev–Popov insertion in [9, (1.5)–(1.6)]. It is then pulled back under $U = U'U_1$ and the constraint linearisation leading to (1.14). At that stage the polynomial (7.1) is introduced; its value at $t = 1$ is the pulled-back Wilson insertion. All subsequent changes of variables are applied to every fluctuating link in this polynomial, including the boundary change (1.22).

Let ξ denote the domain labels, the coarse field, the exterior fine field, and the exterior free fluctuation field in this construction. $K_\xi(t)$ is the conditional fluctuation integral in [9, (1.26)], with the pulled-back polynomial inserted and the change-of-variables density from (1.22)–(1.24) retained once. The same integral with insertion 1 is $K_{\xi,\mathrm{vac}}$. These definitions precede the separation of the effective interaction into its regular and boundary terms. The exterior integrations and their prefactors are those of (1.15), (1.25) and (1.29).

The flat projection of this conditional kernel sets its coarse and exterior determining fields to the identity and its exterior free fluctuation field to zero. The domain labels are held fixed. This evaluation is applied to K_ξ, not to the exterior indicators or history weights.

At the projection point $\mathcal{W}(0) = 1$, so the source logarithm is continued from $K_\xi(0) = K_{\xi,\mathrm{vac}}$.

7.2 Regional Integrals and Their Connected Densities

Fine-lattice units are used; $H = \mathsf{H}_\infty$ is (A.26), and $C_\infty = H^{-1}$. For a finite set $D \subset E_{*,\infty}$, let i_D extend a free field by zero, and let

$$H_D \coloneqq i_D^* H i_D, \qquad C_D \coloneqq H_D^{-1} . \tag{7.3}$$

The inverse exists by (A.27).

Let $\mathcal{V}_D$ consist of the primitive labels from (C.1) whose supports meet D. They keep their original labels, and $v_D(B) = v(i_D B)$. Every primitive missing D vanishes at this field. The cutoff and interaction are

$$\Theta_D^\sigma(B) \coloneqq \prod_{b \in D} \mathbf{1}_{\{|B_b|_\sigma < r\}}, \qquad P_D(B) \coloneqq \sum_{v \in \mathcal{V}_D} v(i_D B) . \tag{7.4}$$

The threshold is the same $r = r_0(g_0)$ as in (5.3).

Let I be an interval of m coarse segments on a coarse translate of the oriented line $v \in \mathscr{V}_\square$. Let $P_U(I)$ denote the ordered transport along its fine traversals. Its closed relative source uses the inverse reference transport, as in (5.32). The regional numerator and denominator are

$$\begin{aligned} N_{D,I}^\sigma &\coloneqq \int \Theta_D^\sigma(B) e^{P_D(B)} \chi(P_{U_{g_0}(i_D B)}(I) P_{U_0}(I)^{-1}) \, \mathrm{d}\mu_{C_D}(B) , \\ D_D^\sigma &\coloneqq \int \Theta_D^\sigma(B) e^{P_D(B)} \, \mathrm{d}\mu_{C_D}(B) . \end{aligned} \tag{7.5}$$

Here $U_0 = \mathbf{1}$ in the chosen frame; retaining its reverse transport specifies the gauge-invariant closed contraction.

The protected-label forest expansion and the web construction of Section 5 apply with every free-input support Λ replaced by $\Lambda \cap D$. A source mark with empty restricted stencil is zero. The other labels keep their original identities. The connected weights are $\Phi_{D,I}(\Gamma)$, with marked coarse-segment set J_Γ. The density assigned to segment j is

$$\kappa_{D,I}(j) \coloneqq \sum_{\Gamma \,|\, j \in J_\Gamma} \frac{\Phi_{D,I}(\Gamma)}{|J_\Gamma|} . \tag{7.6}$$

The prescription σ, fixed throughout each comparison, is suppressed in these weights and densities.

For positive bonds $b = (x, i)$ and $c = (y, k)$, the distance is $d(b, c) = \|x - y\|_\infty + \mathbf{1}_{\{i \neq k\}}$. Let R_* bound the label diameters and source-to-stencil distances, and let R_H be the range of H in this metric. With the interval constant c_h of (5.24), let

$$R_0 \coloneqq 4(1 + R_* + R_H + L), \qquad c_{\mathrm{reg}} \coloneqq 16(1 + R_* + c_h + L) \, . \tag{7.7}$$

The coarse segments of I are numbered $0, \dots, m-1$. Let K_j be the union of their fine-traversal input stencils on segment j, and let

$$d_j \coloneqq \max(0, \min(d(K_j, D^c), L \min(j, m - 1 - j)) - R_0) \, . \tag{7.8}$$

Proposition 7.1. *For either fixed cutoff prescription σ, and for sufficiently small g_0 after the constants in the comparison have been chosen, there are constants $C, c > 0$, independent of D, I and j, such that*

$$|\kappa_{D,I}(j) - \kappa_v| \leqslant C\tau^2 e^{-cd_j} \, . \tag{7.9}$$

Here τ is defined in (5.18) *and κ_v in* (5.33)*. If every K_j is at distance at least b from D^c, then*

$$\left| \log \frac{N^\sigma_{D,I}}{D^\sigma_D} - m\kappa^\sigma_v \right| \leqslant C\tau^2(1 + me^{-cb}) \, . \tag{7.10}$$

The logarithm is continued from zero source.

Proof. Principal restriction preserves $\gamma_L I \leqslant H_D \leqslant h_L I$. The finite-range Neumann series in (A.28) therefore gives exponential decay for C_D, uniformly in the shape of D. Projecting $HC_\infty = I$ onto the D rows gives

$$C_D - (C_\infty)_{D,D} = C_D H_{D,D^c} (C_\infty)_{D^c,D} \, . \tag{7.11}$$

Let $\mu > 0$ be a common inverse-decay exponent in the fine-bond metric. A term on the right joins $u \in D$ and $w \in D^c$ with $d(u, w) \leqslant R_H$. For $s = d(b, u) + d(w, c)$,

$$d(b, D^c) + d(c, D^c) + d(b, c) \leqslant 2s + 2R_H \, . \tag{7.12}$$

Half of $e^{-\mu s}$ pays this distance with exponent $a = \mu/4$; the remaining half is summable over u and its bounded number of neighbours w. Polynomial volume growth gives

$$\|C_D(b, c) - C_\infty(b, c)\| \leqslant Ce^{-a(d(b,D^c)+d(c,D^c)+d(b,c))} \, . \tag{7.13}$$

On D, let C_t interpolate between $(C_\infty)_{D,D}$ and C_D, and let $\mathcal{D}$ be their difference. Every interpolated covariance and each of its principal marginals has spectrum in $[h_L^{-1}, \gamma_L^{-1}]$, uniform exponential decay and scalar colour blocks. Thus the protected-source bounds and the Schur cancellation used in (5.18) hold throughout the interpolation.

For a forest term on carrier A with interpolated covariance $M_t = C_t \circ w^F$, and with $E_e(C_t)$ the off-diagonal direction associated with edge e, the density derivative has the product-rule decomposition

$$\begin{aligned} \partial_t D^{|F|} p_{M_t}[(E_e(C_t))_{e \in F}] = {} & D^{|F|+1} p_{M_t}[\mathcal{D}_A \circ w^F, (E_e(C_t))_{e \in F}] \\ & + \sum_{e \in F} D^{|F|} p_{M_t}[E_e(\mathcal{D}), (E_f(C_t))_{f \in F \setminus \{e\}}] \, . \end{aligned} \tag{7.14}$$

Each tuple lists the directions of the symmetric multilinear derivative of the density with respect to its covariance.

The two kinds of terms use separate Cauchy estimates. In the first, the direction $s(\mathcal{D}_A \circ w^F)$ is added to the covariance centre and every forest direction is fixed. The positive Schur mask contracts operator norm. With $\lambda = h_L^{-1}$, the radius $\lambda/(32\|\mathcal{D}_A\|)$ uses at most $\lambda/32$ of the accretivity margin; a zero direction gives zero derivative. The forest radii are reduced by a fixed factor so that the total stays below $\lambda/8$.

In the second kind of term, one forest direction is replaced by $E_e(\mathcal{D})$, which is bounded by (7.13). Its radius is chosen from the exponential entry majorant, without dividing by an entry of C_t. The normalised density, including its determinant, is bounded by (5.17).

Since $\|\mathcal{D}_A\| \leqslant C|A|e^{-ad(A,D^c)}$, the integrated differentiated term is bounded by the positive forest majorant, with enlarged constants and reduced edge decay, times

$$C|A|^2 e^{-ad(A,D^c)} . \tag{7.15}$$

The exponent of τ is unchanged. In particular the differentiated functional is still equivariant under constant conjugation, so a component with one source mark retains its quadratic character gain.

For each component γ, let u_γ be its positive fixed-forest bound with the uniform exponential entry majorants. This definition applies to vacuum and source components. For a web W, let $u_W \coloneqq \prod_{\gamma\in W} u_\gamma$; summing these numbers over webs gives the web load.

The positive majorants themselves satisfy the polymer loads. Restricting a support to D preserves its size bound s_*, overlap bound q_* and range R_*. On n bonds, label choices cost at most $2^{q_* n}$, the forest-to-graph multiplicity costs at most $2^{q_*(s_*-1)n/2}$, unused graph edges at most $e^{V_1 n/2}$, and the rooted spatial tree sum at most $n^{n-2}V_1^{n-1}/(n-1)!$, where V_1 is the finite row sum of the reduced-decay graph weights. Thus a constant $K_2 \geqslant 1$ bounds the vacuum and source root loads by

$$\sum_{n\geqslant 1} K_2^n \tau^{\lceil n/s_*\rceil}, \qquad \sum_{n\geqslant 1} K_2^n \tau^{\max(2,\lceil n/s_*\rceil)} . \tag{7.16}$$

Grouping n into intervals of s_* gives bounds $C\tau/(1-K_2^{s_*}\tau)$ and $C\tau^2/(1-K_2^{s_*}\tau)$, respectively. These sums bound the positive forest majorants before any signed activity is used. The interval-tree recursion then bounds the web loads.

Let $\mu_1 > 0$ be an exponent remaining after the degree payment, no greater than $\min(\mu, a)/8$, and let

$$\alpha_* \coloneqq \frac{\mu_1}{1024(1+c_h)}, \qquad \eta_1 \coloneqq \frac{\mu_1}{1024} . \tag{7.17}$$

The same sums allow the reserve $e^{4|A|+8\alpha_* H+8\eta_1\ell}$. Let C_{load} bound their weighted spatial and interval root loads after division by τ and τ^2, respectively. The coupling g_0 satisfies $C_{\mathrm{load}}\tau \leqslant 1/4$ and $C_{\mathrm{load}}\tau^2 \leqslant \alpha_*/4$. The incompatibility sum is then bounded by one quarter of $|A_p| + \alpha_* H_p$, so the pinned tree bound applies to the positive majorants u_p.

For a cluster Γ, let $U(\Gamma)$ be its absolute Ursell coefficient, including its factorial, multiplied by the product of these majorants. The product rule, counting repeated activity occurrences, yields

$$|\partial_t \Phi_t(\Gamma)| \leqslant C\ell(\Gamma)^2 e^{-ad(A_\Gamma,D^c)} U(\Gamma), \qquad \sum_{\Gamma \,|\, j\in J_\Gamma} U(\Gamma) e^{\eta_1 \ell(\Gamma)} \leqslant C\tau^2 . \tag{7.18}$$

The union of carriers is A_Γ. The inequality follows from $\sum_p \ell(p)^2 \leqslant \ell(\Gamma)^2$ and the positive-majorant KP bound. Half the extent weight absorbs the polynomial factor.

A component's carrier is connected by its forest and at most $|A|-1$ stencil edges, each of length at most R_*, and its contour interval is attached at one selected mark by a path of length at most R_*. The interval costs at most $c_h\ell$. Carrier or interval incompatibility joins these networks for a connected cluster. Their total length is at most $(1+2R_*+c_h)\ell(\Gamma)$; adding the root segment costs at most L. The chosen c_{reg} bounds the result.

For $d_j \leqslant 2(R_0+R_*+c_{\mathrm{reg}})$, the rooted bounds prove the comparison after enlarging C.

For larger d_j, clusters with $\ell(\Gamma) \geqslant d_j/(2c_{\mathrm{reg}})$ are bounded by the extent tail in each gas. Every shorter cluster rooted at j lies away from both the integration boundary and the ends of I. Its original stencils lie inside D, so its labels, closed sources, local functions and incompatibilities coincide in the two expansions. Only its covariance changes, and every carrier is at distance at least $d_j/2-R_0$ from D^c. The fixed loss is absorbed in the comparison constant. Integrating the differentiated bound gives (7.9). Division by $|J_\Gamma|$ cannot enlarge it.

Finally, allocation and the endpoint geometric series give

$$\log \frac{N^\sigma_{D,I}}{D^\sigma_D} = \sum_{j=0}^{m-1} \kappa_{D,I}(j), \qquad \sum_{j=0}^{m-1} e^{-cd_j} \leqslant C(1+me^{-cb}) \,, \tag{7.19}$$

which proves (7.10). *q.e.d.*

7.3 Periodic Regions and Bałaban's Kernel

Let $E_{*,T}$ be the periodic free carrier, H_T the Hessian in (4.7), and $i_{D,T}$ zero extension from $D \subseteq E_{*,T}$. The periodic primitives, written v_T collectively (the terms v_α, j_b and d_c of (C.1) on T, all three kinds), and the reconstructed links $U_{g_0,T}$ of (4.9) on T are those of Appendix C. Let

$$H_{D,T} \coloneqq i^*_{D,T} H_T i_{D,T}, \qquad C_{D,T} \coloneqq H^{-1}_{D,T}, \qquad P_{D,T}(B) \coloneqq \sum_{v_T \mid \operatorname{supp} v_T \cap D \neq \varnothing} v_T(i_{D,T}B) \,. \tag{7.20}$$

The primitives of Section 3.3 at the reference background $U=\mathbf{1}$ (written $p_{\xi,\alpha}$, $p^{\mathrm{H}}_{\xi,b}$, $p^{\mathrm{J}}_{\xi,c}$ as there, with the reference background in place of the branch data) and the local terms of (C.1) are the same functions, label by label.

Definition 7.2 (The Identity Background)**.** The identity background is $U=\mathbf{1}$ on the torus T with region $\Omega = T$ (Section 3.2), so that every row $c \in T^{[1]}$ and every centre $y \in T^{(1)}$ occurs. $|\bullet|$ is the cutoff norm (so $h \leqslant |\bullet|$, the conventions of Section 3), $\|\bullet\|$ the Hilbert norm of Lemma B.8, and $\|A\|_\infty \coloneqq \max_b h(A_b)$ the phase norm, in which a_{ph} and the chart radius a_L of Appendix C are measured. The row solutions, the phase $q_{\mathbf{1}}$, and the primitives $p_{\xi,\alpha}$, $p^{\mathrm{H}}_{\xi,b}$, $p^{\mathrm{J}}_{\xi,c}$ at the background $\mathbf{1}$ are those given by the formulas of Sections 3.2 and 3.3 with $\Omega = T$ and $U=\mathbf{1}$, the row solution of every row c being the unique small solution φ_c of Lemma B.8(ii), which applies to every row of the identity background (proof of Lemma 7.3(i)).

Lemma 7.3 (Primitives at the Identity Background)**.** *At the identity background, let $B \in \mathfrak{g}^{E_*}$ be real with $g_0 C_{\mathsf{C}} \max_b |B(b)| \leqslant \vartheta_*/2$ and $g_0 \max_b |B(b)| < a_L$, with C_{C}, ϑ_* of Lemma B.8. Then:*

1. *$\mathcal{Q}_{\mathbf{1}} = \mathcal{Q}$, $Q_{\mathbf{1}} = Q$, $\mathcal{N}_{\mathbf{1}} = \mathcal{N}$ (4.2), $\mathsf{C}_{\mathbf{1}} = \mathsf{C}$ and $h_{\mathbf{1}}(c) = L^3\mathbf{1}$ for every row (4.3); every row solution $\widetilde{D}_{\mathbf{1},c}(g_0\mathsf{C}B)$ is defined, is the unique small solution of Lemma B.8(ii), and equals the row c of $\widetilde{D}(g_0\mathsf{C}B)$ of (4.9); hence $q_{\mathbf{1}}(B) = q_{g_0}(B)$.*

2. $p_{\xi,p}(B) = v_p(B)$ *for every plaquette* p, *and* $p_{\xi,y}(B) = \sum_{x\in\mathcal{B}_y\setminus\{y\}} v_{(y,x)}(B)$ *for every centre* y.

3. $p^{\mathrm{H}}_{\xi,b}(B) = j_b(B)$ *for every bond* b.

4. $p^{\mathrm{J}}_{\xi,c}(B) = d_c(B)$ *for every row* c.

Consequently $\sum_\alpha p_{\xi,\alpha}(B) + \sum_b p^{\mathrm{H}}_{\xi,b}(B) + \sum_c p^{\mathrm{J}}_{\xi,c}(B) = \mathcal{P}_{g_0}(B)$ (C.2)*; and for every bond set* $D \subseteq E_*$ *and every real* $B \in \mathfrak{g}^D$ *with* $g_0 C_{\mathsf{C}} \max_b |B(b)| \leqslant \vartheta_*/2$ *and* $g_0 \max_b |B(b)| < a_L$*, the sum of the primitives whose free-input sets meet* D*, evaluated at* $i_{D,T}B$*, equals* $P_{D,T}(B)$.

Proof. (i) At $U = \mathbf{1}$ every transport is $\mathbf{1}$, so $\beta_c(\mathbf{1}) = 0$ and $\mathsf{M}(\mathbf{1})(c) = P_c(\mathbf{1}) = \mathbf{1}$ in (4.1); hence $\mathcal{Q}_{\mathbf{1},c}(q) = \frac{1}{i}\log \mathsf{M}(e^{iq})(c)$ is the row c of $\mathcal{Q}(q)$ in (4.2), and $Q_{\mathbf{1}} = Q$, $\mathcal{N}_{\mathbf{1}} = \mathcal{N}$ follow from the definitions (3.9).

By Lemma 3.6, $Q_0 = Q\iota_0$ is diagonal with entries $\alpha_c(\mathbf{1}) = L^{-4}\mathbf{1}$, so $h_{\mathbf{1}}(c) = L^{-1}\alpha_c(\mathbf{1})^{-1} = L^3\mathbf{1}$ is the row c of $h = L^{-1}\iota_0 Q_0^{-1}$; and since row c of Q is supported on $\mathcal{E}_c$ (Lemma 3.6), $(\mathsf{C}_{\mathbf{1}}B)(b_0(c)) = -\alpha_c(\mathbf{1})^{-1}\sum_{b\in\mathcal{E}_c\cap E_*} Q(c,b)B(b) = -(Q_0^{-1}Q\iota_* B)(c) = (\mathsf{C}B)(b_0(c))$, while both maps are the identity on free bonds; so $\mathsf{C}_{\mathbf{1}} = \mathsf{C}$.

The row equation (3.24) at $U = \mathbf{1}$ is the row c of the equation defining $\widetilde{D}$ after (4.9). Lemma B.8 applies to the identity background on every row (its gauge hypothesis holds with $u = \mathbf{1}$, $\phi = 0$), and by its part (i) the hypothesis on B gives $\|g_0\mathsf{C}B|_{\mathcal{E}_c}\| \leqslant g_0 C_{\mathsf{C}} \max_b |B(b)| \leqslant \vartheta_*/2$ for every row, so its part (ii) provides the unique small solution $\varphi_c(g_0\mathsf{C}B|_{\mathcal{E}_c})$, which is the row solution of Section 3.3 for the reference background, analytic on the real ball $\|A|_{\mathcal{E}_c}\| < \vartheta_*$ and agreeing with the germ of Lemma 3.9 at $A = 0$.

The row c of $\widetilde{D}$ of Appendix C is real analytic on $\{\|A\|_\infty < a_{\mathrm{ph}}\}$, a function of $A|_{\mathcal{S}_c}$, hence of $A|_{\mathcal{E}_c}$ ($\mathcal{S}_c$, the positive fine bonds with both endpoints in $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$, is contained in $\mathcal{E}_c$, Section 3.2), and agrees with the same germ at $A = 0$ (the paragraph "Locality of the first-step interaction" there). The set of real $A|_{\mathcal{E}_c}$ with $\|A|_{\mathcal{E}_c}\| < \vartheta_*$ and $\|A\|_\infty < a_{\mathrm{ph}}$ is convex, hence connected, open, and contains $g_0\mathsf{C}B|_{\mathcal{E}_c}$ by the hypothesis on B ($\|g_0\mathsf{C}B\|_\infty \leqslant \|\mathsf{C}\|_{\infty\to\infty} g_0 \max_b |B(b)| < a_{\mathrm{ph}}$ by the definition of a_L and $h \leqslant |\bullet|$); the identity theorem identifies the two analytic functions on it.

With $h_{\mathbf{1}} = h$ and $\mathsf{C}_{\mathbf{1}} = \mathsf{C}$, (3.25) becomes (4.9): $q_{\mathbf{1}}(B) = g_0\mathsf{C}B - h\widetilde{D}(g_0\mathsf{C}B) = q_{g_0}(B)$.

(ii) Let $\psi(A) \coloneqq A - h\widetilde{D}(A)$ as in Appendix C, so that $q_{g_0}(B) = \psi(g_0\mathsf{C}B)$, $\psi(0) = 0$ and $D\psi(0) = \mathbf{1}$ because $D\widetilde{D}(0) = 0$ (Lemma B.8(ii)). For a plaquette p, $s_{p,\mathbf{1}}(q) = 1 - \operatorname{Re}\operatorname{tr}_N(e^{iq}(\partial p)) = f_p(q)$ is the p-summand of $S(e^{iq})$, the function f_p of Appendix C, and $s_{p,\mathbf{1}}(0) = 0$ since $\operatorname{tr}_N(\mathbf{1}) = 1$; so $E_{\xi,p}(B) = -g_0^{-2} f_p(\psi(g_0\mathsf{C}B))$ by (i). Each $f_p \geqslant 0$ vanishes at $q = 0$, so $Df_p(0) = 0$. The chain rule then gives $DE_{\xi,p}(0)[B] = -g_0^{-2} Df_p(0)[D\psi(0)g_0\mathsf{C}B] = 0$ and

$$\begin{aligned} D^2E_{\xi,p}(0)[B,B] &= -g_0^{-2}(D^2f_p(0)[D\psi(0)g_0\mathsf{C}B, D\psi(0)g_0\mathsf{C}B] \\ &\qquad + Df_p(0)[D^2\psi(0)[g_0\mathsf{C}B, g_0\mathsf{C}B]]) \\ &= -D^2f_p(0)[\mathsf{C}B, \mathsf{C}B]\,, \end{aligned}$$

so by (3.29)

$$p_{\xi,p}(B) = -\frac{1}{g_0^2} f_p(\psi(g_0\mathsf{C}B)) + \frac{1}{2}D^2f_p(0)[\mathsf{C}B, \mathsf{C}B] = v_p(B)\,. \tag{7.21}$$

For a centre y: at $U = \mathbf{1}$ every path transport is $\mathbf{1}$, so the local group mean $\mathcal{U}_{\mathbf{1}}(y,x)$, the unique solution near the diagonal value of the equation after (4.3), is $\mathbf{1}$; at $q = q_{g_0}(B)$ the mean

$\mathcal{U}_{e^{iq}}(y,x)$ exists by Lemma B.9 at the identity background, since for a row c with endpoint block $\mathcal{B}_y$ one has $\mathcal{E}_y \subseteq \mathcal{E}_c$ and $\|q_{g_0}(B)|_{\mathcal{E}_c}\| \leqslant \|A\| + 2H_*K_*\|A\|^2 \leqslant \vartheta_*$ for $A = g_0\mathsf{C}B|_{\mathcal{E}_c}$, $\|A\| \leqslant \vartheta_*/2$, by (ii) of that lemma, $\|h_{\mathbf{1}}(c)\| \leqslant H_*$ and $4K_*H_*\vartheta_* < 1$; and the y-summand of (3.14) is $\mathscr{G}_{y,\mathbf{1}}(q) = \sum_{x\in\mathcal{B}_y\setminus\{y\}}(1-\operatorname{Re}\operatorname{tr}_N \mathcal{U}_{e^{iq}}(y,x)) = \sum_{x\in\mathcal{B}_y\setminus\{y\}} f_{(y,x)}(q)$, the sum of the (y,x)-summands of $\mathscr{G}(e^{iq})$ in (4.5). So $E_{\xi,y}(B) = -g_0^{-2}\mathscr{G}_{y,\mathbf{1}}(q_{g_0}(B)) = \sum_x(-g_0^{-2}f_{(y,x)}(\psi(g_0\mathsf{C}B)))$. Each $f_{(y,x)} \geqslant 0$ vanishes at $q=0$, so $Df_{(y,x)}(0) = 0$, and the computation just made for f_p applies to each $f_{(y,x)}$; since the passage from a function to its degree-two Taylor residual at $B=0$ in (3.29) is linear, $p_{\xi,y}(B) = \sum_{x\in\mathcal{B}_y\setminus\{y\}} v_{(y,x)}(B)$.

(iii) In (3.30), $\sigma(q)\,\mathrm{d}q$ is the pullback under $q \mapsto e^{iq}$ of the Haar measure of [9, (1.11)], and in (4.14) σ_H is the density of normalised Haar measure in the same coordinates relative to the same Lebesgue measure; the Haar measure of the compact group being unique up to a positive constant, $\sigma = \kappa\sigma_H$ with $\kappa > 0$, both are positive on the chart ball (Lemma B.8(iii) takes the real logarithm of $\sigma(q)/\sigma(0)$ there), and $\sigma(q)/\sigma(0) = \sigma_H(q)/\sigma_H(0) = j_H(q)$. With $q_{\mathbf{1}}(B) = q_{g_0}(B)$ from (i), $p^{\mathrm{H}}_{\xi,b}(B) = \log j_H(q_{g_0}(B)(b)) = j_b(B)$, both sides being real logarithms of positive numbers: the branch of j_b in (C.1) is the real logarithm on the convex domain of (i), on which the argument stays positive.

(iv) By (i) (the row solution being φ_c, which agrees with the row of $\widetilde{D}$ on an open set containing $g_0\mathsf{C}B|_{\mathcal{E}_c}$, so that their derivatives agree there, and $h_{\mathbf{1}}(c) = L^3\mathbf{1}$), $p^{\mathrm{J}}_{\xi,c}(B) = \log|\det_{\mathfrak{g}}(\mathbf{1} - L^3\partial_{A(b_0(c))}\widetilde{D}_c(g_0\mathsf{C}B))|$. The determinant is positive on $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/2$ (Lemma B.8(iii)), so the absolute value is redundant and the expression is $d_c(B)$, whose branch in (C.1) is the real logarithm of the positive determinant (the sentence after that display).

The first consequence is (ii)–(iv) summed over all labels, with (C.2).

For the second, every primitive and every term vanishes at zero input: $p_{\xi,\alpha}(0) = E_{\xi,\alpha}(0) = 0$ ($s_{p,\mathbf{1}}(0) = 0$, $\mathscr{G}_{y,\mathbf{1}}(0) = 0$), $p^{\mathrm{H}}_{\xi,b}(0) = 0$, $p^{\mathrm{J}}_{\xi,c}(0) = 0$ ($\widetilde{D}(0) = 0$, $D\widetilde{D}(0) = 0$), and $v_\alpha(0) = j_b(0) = d_c(0) = 0$ likewise. A primitive depends on B only through its free-input set (Lemma 3.9, the list after (3.31)), a term of (C.1) only through its support. For $B \in \mathfrak{g}^D$ as in the statement, $i_{D,T}B$ satisfies the hypothesis of the lemma (the maximum norm does not increase), a primitive whose free-input set misses D has zero input at $i_{D,T}B$ and vanishes, and a term whose support misses D vanishes there as well; so the sum of the primitives whose free-input sets meet D and $P_{D,T}(B)$ are both the full sums at $i_{D,T}B$, which agree by the first consequence. *q.e.d.*

$N^\sigma_{D,I,T}$ and $D^\sigma_{D,T}$ are (7.5) with C_D, P_D, i_D, U_{g_0} replaced by $C_{D,T}, P_{D,T}, i_{D,T}, U_{g_0,T}$; their cutoff is still the product over D. The same forest, web, and connected-weight definitions applied to these periodic integrals give weights $\Phi_{D,I,T}(\Gamma)$, and

$$\kappa_{D,I,T}(j) = \sum_{\Gamma\,|\,j\in J_\Gamma} \frac{\Phi_{D,I,T}(\Gamma)}{|J_\Gamma|}\,. \tag{7.22}$$

Let I have a lift with $mL < N_{\min}/3$, where $N_{\min}$ is the smallest fine period. For both periodic open-segment gases the positive cluster majorants obey a uniform rooted tail. Let $U_{D,I,T}(\Gamma)$ be the absolute Ursell coefficient, including its factorial, times the positive object majorants, and let $U_{I,T}$ be its value for $D = E_{*,T}$. There is $\eta_0 > 0$, independent of the region and periods, such that for every marked segment j,

$$\sum_{\Gamma\,|\,j\in J_\Gamma} U_{D,I,T}(\Gamma)e^{\eta_0\ell(\Gamma)} \leqslant C\tau^2\,. \tag{7.23}$$

The majorants dominate the absolute connected weights. The proof is the positive-majorant argument of Proposition 7.1: restricting supports preserves the label overlap bounds, and the periodic shell sum is bounded by the infinite-lattice shell sum. Under this lift hypothesis, index separation along the segment is its periodic separation. Thus (5.24) holds on the lifted segment. Its ordinary intervals have the straight-line interval tree sum. Hence the same load inequalities and extent reserve prove (7.23) for the full and restricted periodic gases, including the covariance interpolations used below.

On a compatible torus with fine periods N_μ, $N_{\min} = \min_\mu N_\mu$, and I has a lift with $mL < N_{\min}/3$. The set D may wrap around the torus. $\kappa_{I,T}(j)$ is the same allocation with $D = E_{*,T}$. In this paragraph d_j is (7.8) with d_T in place of d, including its endpoint term $L\min(j, m-1-j)$.

For bounded $d_j \leqslant 2(R_0 + R_* + c_{\mathrm{reg}})$, (7.23) and $|\kappa_v| \leqslant C\tau^2$ give (7.27) directly after enlarging C.

For larger d_j, the periodic version of (7.11) and its differentiated cluster estimate compare $\kappa_{D,I,T}(j)$ with $\kappa_{I,T}(j)$. Clusters of extent at least $d_j/(2c_{\mathrm{reg}})$ have total rooted weight at most $C\tau^2 e^{-cd_j}$. For smaller clusters the labels agree, and their carriers are at distance at least $d_j/2 - R_0$ from D^c. The covariance derivative therefore has the same bound, after absorbing the fixed factor e^{aR_0}. This proves

$$|\kappa_{D,I,T}(j) - \kappa_{I,T}(j)| \leqslant C\tau^2 e^{-cd_j} \; . \tag{7.24}$$

For the comparison of the second density with the infinite straight-line density, the root segment is lifted to the chosen copy of I. A cluster network of length $c_{\mathrm{reg}}\ell + R_0 < N_{\min}/4$ lifts uniquely and preserves carrier intersections. The distinct translates of I are separated by more than $2N_{\min}/3$ because $mL < N_{\min}/3$, so all its source marks lie on that copy. If also $c_{\mathrm{reg}}\ell + R_0 < L\min(j, m-1-j)$, the network avoids the two ends and has the same labels and ordered source factors as the infinite straight-line expansion. Clusters failing either restriction are bounded by the rooted extent tail. For the remaining clusters only the covariance marginals change. Periodisation and lifted separation $d(b,c) \leqslant N_{\min}/4$ give

$$\|C_T(b,c) - C_\infty(b,c)\| \leqslant Ce^{-\mu(N_{\min} - d(b,c))} \leqslant Ce^{-\mu N_{\min}/2} e^{-\mu d(b,c)} \; , \tag{7.25}$$

after expressing the decay exponent in fine-bond units. The free-inverse periodisation follows by applying H_T to the absolutely convergent periodised kernel of H^{-1}, as in the proof of (A.31). The covariance interpolation and the density-direction estimates in the proof of Proposition 7.4 apply with $h_A := e^{-\mu N_{\min}/2}$. They retain edge decay and the quadratic source gain; their polynomial carrier factors are absorbed by the reserved extent weight. With $\iota_T = \max(0, N_{\min}/4 - R_0)$, summing the short-cluster derivatives and the two tails gives

$$|\kappa_{I,T}(j) - \kappa_v| \leqslant C\tau^2 \left[e^{-c(L\min(j,m-1-j)-R_0)_+} + e^{-c\iota_T} \right] \; . \tag{7.26}$$

Here $x_+ = \max(x, 0)$.

Absolute convergence also gives $\log(N^\sigma_{D,I,T}/D^\sigma_{D,T}) = \sum_{j=0}^{m-1} \kappa_{D,I,T}(j)$, since summing the allocation weights $|J_\Gamma|^{-1}$ over the marked segments gives one for each source cluster. Adding the two comparisons proves

$$|\kappa_{D,I,T}(j) - \kappa_v| \leqslant C\tau^2 (e^{-cd_j} + e^{-c\iota_T}) \; . \tag{7.27}$$

The identification with K_ξ uses the averaging prescription [8, (0.4)], the bond cutoff $|\bullet|_\sigma$, and the threshold $\delta_0 = g_0 r_0(g_0)$ on both sides. The comparison applies separately to either specified σ.

Let I consist of consecutive traversals of one straight side of C. Let $E_{\Omega,*}$ be the free carrier after the designated-bond elimination in [9, (1.14)–(1.15)]. Let D_- consist of these bonds with both endpoints in Λ_1, and D_+ of those with at least one endpoint there. The comparison holds for every $D_- \subseteq D \subseteq D_+$, assigning the complementary free variables to the exterior integral. It therefore covers either endpoint convention for the integration boundary.

In the notation of [9] let M_2, M, R_1 be the cube-scale integers, so the test cubes have fine side $L^2 M_2 R_1$ and the macro cubes have fine side LMR_1; here R_1 denotes the scale integer, not the large-fluctuation region also denoted by R_1 in [9]. The partitions are aligned and nested, all periods are divisible by LMR_1, and the powers of L satisfy

$$M \geqslant 8LM_2, \qquad LMR_1 > 4R_0 + 4L^2 M_2 R_1 + 8 \,. \tag{7.28}$$

Here $R_0 = 4(1 + R_* + R_H + L)$ is the fixed range constant above, distinct from the scale integer in [9]. These choices lie within the compatible-partition and large-M prescription of [9, Section 1]. The collar inequality is among the common parameter conditions; it holds for every scale integer $R_1 \geqslant 1$ once $LM - 4L^2 M_2 > 4R_0 + 8$.

For macro regions the collar-distance statement following [9, (1.10)] supplies the metric separation, including near cube corners. For test-cube enlargements a conservative reach bound suffices: each added layer has width at most $L^2 M_2 R_1$. This bound also covers a layer of the L-fold finer cubes in the enlargement convention of [9, Section 1]. The parallelepiped modification after (1.20) removes macro cubes from Λ_1; the remaining region is still a union of macro cubes and hence of test cubes.

Using (1.20) and that collar-distance statement, a site within distance $LMR_1 - 1$ of Λ_1 lies in $\Omega_1^{\sim -1} \setminus R_1'$. Every stencil meeting $D \subseteq D_+$ therefore remains in the regular region. A bond with an endpoint in Λ_1 is tested by the test cube containing that endpoint, which is contained in Λ_1 and is good. A bad test or a test in S_1 uses only bonds within $2L^2 M_2 R_1 + 1$ of its cube. The collar and (7.28) keep those bonds outside D_+, including the fixed stencil enlargement.

At identity determining data the critical field is the identity: it satisfies the constraints, has zero action and gauge-fixing functional, and is unique in the regular chart by [9, (1.12)]. Its force is zero. The Hessian on D is consequently $H_{D,T}$, including the diagonal contributions of plaquettes crossing its boundary.

The nonlinear constraint in [8, (2.4)] becomes $M(e^{iB'})$ at identity background. Its principal logarithm is $\mathcal{Q}(B')$ from (4.2). The linearisation following (2.10) has linear part $L\widetilde{Q}B'$, hence $\widetilde{Q} = Q$ there. Its designated-row right inverse satisfies $LQh = I$. Uniqueness of the small solution for $\widetilde{D}$ gives exactly (4.9). Thus the action, Haar and Jacobian terms become $v_T(i_{D,T}B)$ with all primitive labels meeting D retained.

The split reconstruction in [9, (1.21)] also equals the identity at identity data, and $Q_1^{s*}\mathbf{1} = \mathbf{1}$ by (1.3). Its logarithmic argument is zero. Equation (1.21) then gives $e^{iL^{-1}\mathbf{H}_{1,Ax}(0)} = 1$; the small principal chart implies $\mathbf{H}_{1,Ax}(0) = 0$. Since the BCH remainder satisfies $\mathbf{F}(X, 0) = 0$, the boundary change (1.22) is the identity, its Jacobian is one and its additional exponent is zero at this projection.

For the cutoff, every free bond in D has an endpoint in a compatible good cube contained in Λ_1. The twice-enlarged starred test in [9, (1.18)] covers it. The designated-bond exclusion is the verbal convention following (1.8). A good cube outside Λ_1 can also test bonds in D; its product of bond indicators splits into its D and exterior factors. The D factors are redundant with the covered interior cutoff by idempotence; all exterior factors stay in the outer integral.

Bad tests and changed background tests are separated from D by the collar. At zero exterior free field the conditional cutoff is therefore Θ_D^σ.

The same norm is used on both sides. The threshold is $r = \delta_0/g_0$ by the normalisation convention (4.13). The exterior quadratic cross term and its completion-of-square compensator vanish at the projection. Gaussian and constraint normalisations cancel between the two conditional kernels and remain in the exterior integration.

Finally let $t_a \coloneqq 1$ on I and $t_a \coloneqq 0$ elsewhere. At the flat projection, (7.1) is the closed relative source in (7.5). With K_ξ^{flat} the projected conditional kernels,

$$\frac{K_\xi^{\text{flat}}(1_I)}{K_{\xi,\text{vac}}^{\text{flat}}} = \frac{N_{D,I,T}^\sigma}{D_{D,T}^\sigma} \, . \tag{7.29}$$

If $d_T(K_j, D^c) \geqslant b$ for every source segment j, then (7.27) gives

$$\left| -\frac{1}{mL\varepsilon} \log \frac{K_\xi^{\text{flat}}(1_I)}{K_{\xi,\text{vac}}^{\text{flat}}} - \delta m_v^\sigma \right| \leqslant \frac{C\tau^2}{L\varepsilon} \left(\frac{1}{m} + e^{-cb} + e^{-c\iota_T} \right) \, . \tag{7.30}$$

For every domain label ξ for which the buffer fits inside the torus and $mL < N_{\min}/3$, (7.30) bounds the distance between the per-length scalar and δm_v^σ by $C\tau^2(1/m + e^{-cb} + e^{-c\iota_T})/(L\varepsilon)$ on the fixed torus; along any family of such labels with m, b and ι_T tending to infinity this bound tends to zero, so the per-length scalar converges to δm_v^σ, which identifies it with (5.33). The limit is the one defining the coefficient, as in (5.39); the estimate (7.30) itself is a statement on each fixed torus.

The four oriented straight-side coefficients give the square coefficient by the average in (5.39). This comparison is made inside the conditional kernels.

7.4 The Regional Square Scalar

Let σ be either cutoff prescription. Let C_n be a coarse translate of the counterclockwise square with vertices $0, nLe_1, nL(e_1 + e_2), nLe_2$ in fine-lattice units, with $K_\mu \geqslant 3(n + 2)$ for every coarse period. Let D be the free bonds integrated in a covered conditional kernel: every free-input stencil of the contour lies in D. With the original primitive labels retained, including those crossing the boundary of D, let

$$S_{D,C_n,T}^\circ \coloneqq \frac{\displaystyle\int \Theta_D^\sigma(B) e^{P_{D,T}(B)} \chi(\mathrm{Hol}_{U_{g_0,T}(i_{D,T}B)} C_n) \, \mathrm{d}\mu_{C_{D,T}}(B)}{\displaystyle\int \Theta_D^\sigma(B) e^{P_{D,T}(B)} \, \mathrm{d}\mu_{C_{D,T}}(B)} \, , \quad C_{D,T} \coloneqq (i_{D,T}^* H_T i_{D,T})^{-1} \, . \tag{7.31}$$

Here H_T is the full flat free-field precision on the torus, and $P_{D,T}$ is (7.20); Θ_D^σ is the bond cutoff of (7.4). For a conditional history ξ, let $S_{\xi,C_n}^\circ \coloneqq S_{D,C_n,T}^\circ$, with D its integration carrier. This is a ratio of conditional reference integrals.

The square's $4n$ coarse segments are indexed cyclically by j. Let K_j be the union of the free-input stencils of its fine traversals. With complements taken in the torus free carrier and with the periodic fine-bond metric of (7.8), let

$$d_j \coloneqq \max(0, d_T(K_j, D^c) - R_0), \qquad \iota_T \coloneqq \max(0, N_{\min}/4 - R_0) \, . \tag{7.32}$$

The fixed R_0 includes the label, reconstruction and precision ranges. Distance to the empty set is infinite, with $e^{-\infty} = 0$. There is no distance to a side endpoint in d_j.

Proposition 7.4. *Under these hypotheses, the collar conditions of the conditional kernel, and the common small-coupling thresholds of the flat and regional expansions, the scalar* (7.31) *is nonzero. Its logarithm is continued from zero in the traversal-source polynomial* (7.1)*. There are $C, c > 0$, independent of D, n and the compatible periods, and an error $E^{\circ}_{D,C_n,T}$ such that*

$$S^{\circ}_{D,C_n,T} = e^{-\delta m^{\sigma} P(C_n)} (z^{\sigma}_{\mathrm{c}}(g_0, \pi/2))^4 e^{E^{\circ}_{D,C_n,T}} , \tag{7.33}$$

$$|E^{\circ}_{D,C_n,T}| \leqslant C\tau^2 \left[\sum_{j=0}^{4n-1} e^{-cd_j} + 4ne^{-c\iota_T} + (n+1)e^{-cn} \right] . \tag{7.34}$$

The coefficients are exactly (5.39) *and* (6.12)*, with the same cutoff prescription. The parameter τ is* (5.18)*; fixed enlargements of its majorant constants are absorbed in C and the coupling threshold.*

The collar hypotheses identify this integral with the conditional scalar of [9].

Proof. Let $F_{D,C,T} \coloneqq \log S^{\circ}_{D,C,T}$, with its branch specified below, $F_{C,T} = \log(N^{\sigma}_{C_n,T}/D^{\sigma}_{T})$, and $F_{C,\infty}(n) = F_{\square,\infty}(n)$. The first two quantities are compared, then the latter two, and the bulk corner extraction is applied; the errors from changing the kernel and summation domain are retained, as in [8, (4.35)–(4.37)].

At the conditional reference data the background and exterior determining fields are the identity and the retained free field is zero. The row precision entries are the flat torus entries on D, including the diagonal terms from crossing plaquettes. The collar contains every stencil touching D, so the precision is $i^{*}_{D,T} H_T i_{D,T}$. The primitives agree with the terms of (C.1) label by label, with the designated inverse L^3 at the identity and the gauge terms of a centre summed over its pairs (y, x), by Lemma 7.3, whose hypothesis on B is the pair of small-coupling conditions fixed in the next paragraph; by its second consequence the sum of the primitives whose free-input sets meet D, at $i_{D,T}B$, is $P_{D,T}(B)$, terms whose support misses D vanishing at $i_{D,T}B$. The cutoff is the same product of bond tests. Thus the numerator and denominator in (7.31) are those of the conditional reference scalar, with their normalisers retained in the ratio.

The local implicit solutions used in the two representations have the same analytic germ. To fix their common domain, let C_{C} and ϑ_* be the constants of Lemma B.8 (C_{C} bounds the fixed row map from the maximum free-bond norm to the row Hilbert norm, ϑ_* is its implicit-solution phase radius), and let a_L be the flat chart radius in Appendix C. The row map is finite dimensional with finitely many translated types, so $C_{\mathsf{C}} < \infty$; the analytic implicit-function theorem gives a uniform positive phase radius. The radius a_L is reduced if necessary to lie within those row charts and their nonvanishing Jacobian domains, and the coupling satisfies

$$g_0 C_{\mathsf{C}}(2 + 4r) < \frac{\vartheta_*}{2}, \qquad g_0(2 + 4r) < a_L . \tag{7.35}$$

The factor $2 + 4r$ specifies a larger free-variable polydisc containing the integration box $\|B\|_{\infty} \leqslant r$ with a positive margin. The convex intersection of this polydisc with the phase balls lies in the common analytic charts. The identity theorem therefore identifies the solutions throughout the integration box.

The common chart radius is fixed before the coupling; reducing it preserves all previous estimates by restriction to a smaller domain. These small-coupling requirements and all polymer-load inequalities are imposed together, which is possible because $g_0(1 + r)^3 \to 0$ and $\tau \to 0$.

The protected-label expansion applies with supports intersected with D and every original label kept. Zero extension commutes with constant conjugation, so the restricted primitives, cutoffs and closed source differences are invariant. The restricted support graph and forest partition the labels into components. Source components are merged using the shortest cyclic intervals of the flat expansion, with its fixed tie-breaking base; a web interval is their union, possibly the whole circle. Compatible components and webs have disjoint carriers and disjoint web intervals, so their union reverses this decomposition without losing or duplicating labels. If a web crosses the base, cyclicity of the trace moves the base between web intervals before Schur factorisation. This proves the regional counterpart of (5.14).

The required quantitative bounds hold uniformly for these restricted labels. Principal restriction and inversion preserve the spectral bounds, and the finite-range Neumann series gives uniform exponential covariance decay in the periodic metric. Let $\mu_1 > 0$ be a remaining edge-decay exponent after the degree payment, no larger than one eighth of the covariance and comparison exponents, and let

$$\alpha' \coloneqq \frac{\mu_1}{1024(1+c_h)}, \qquad \eta' \coloneqq \frac{\mu_1}{1024}, \qquad a'(p) \coloneqq |A_p| + \alpha' H_p \, . \tag{7.36}$$

Restriction preserves the support-size, overlap and range bounds s_*, q_*, R_*. The same labelled-tree count as in (5.22), with a fixed enlarged K_2, bounds vacuum and source loads by

$$\sum_{k \geqslant 1} K_2^k \tau^{\lceil k/s_* \rceil}, \qquad \sum_{k \geqslant 1} K_2^k \tau^{\max(2, \lceil k/s_* \rceil)} \, . \tag{7.37}$$

The bounds remain valid with the reserve $e^{4|A_\gamma| + 8\alpha' H_\gamma + 8\eta' \ell(\gamma)}$. Contacts remain within R_* of the restricted support network, so (5.24) applies to the square, including at its corners. The cyclic conversion (5.25) has the same uniform geometric sum $1 + 2/(e^{\alpha'} - 1)$. The web-tree recursion therefore gives positive majorants u_p with

$$\begin{aligned} \sup_b \sum_{p \,|\, b \in A_p} u_p e^{2a'(p) + 2\eta' \ell(p)} &\leqslant C'_* \tau \, , \\ \sup_x \sum_{W \,|\, x \in h_W} u_W e^{2a'(W) + 2\eta' \ell(W)} &\leqslant C'_* \tau^2 \, . \end{aligned} \tag{7.38}$$

The source-only spatial load is also $O(\tau^2)$. The coupling g_0, chosen last, satisfies $K_2^{s_*} \tau \leqslant 1/2$, $C'_* \tau \leqslant 1/4$ and $C'_* \tau^2 \leqslant \alpha'/4$. The incompatibility sum is then at most $a'(p)/4$. The pinned-tree proof of (5.29) yields

$$\sum_{\Gamma \,|\, j \in J_\Gamma} U(\Gamma) e^{\eta' \ell(\Gamma)} \leqslant C \tau^2 \, , \tag{7.39}$$

where U is the absolute Ursell coefficient times the positive object majorants. The factor $2^{|S|}$ for source variables $|t_a| \leqslant 2$ is included in K_2. Thus the regional source logarithm exists on that polydisc and vanishes at zero source. Its exponential is the ratio (7.31), proving nonvanishing.

For the boundary comparison, the periodic resolvent identity is

$$C_{D,T} - (C_T)_{D,D} = C_{D,T} (H_T)_{D,D^c} (C_T)_{D^c,D} \, . \tag{7.40}$$

If $u \in D$, $w \in D^c$ and $d_T(u,w) \leqslant R_H$, let $s = d_T(b,u) + d_T(w,c)$. The periodic triangle inequalities give

$$d_T(b, D^c) + d_T(c, D^c) + d_T(b,c) \leqslant 2s + 2R_H \, . \tag{7.41}$$

Half the inverse-decay exponent pays these distances; the remaining convolution is bounded by a fixed neighbour count times $\sup_b \sum_u e^{-\mu d_T(b,u)/2}$. Periodic shells have at most the infinite-lattice shell counts, so this sum is uniform even for wrapping or disconnected D. Therefore

$$\|C_{D,T}(b,c) - C_T(b,c)\| \leqslant Ce^{-a(d_T(b,D^c)+d_T(c,D^c)+d_T(b,c))} . \tag{7.42}$$

The principal covariances are interpolated as in (7.14). On a carrier A of size k, let $B_{bc} \coloneqq A_0 e^{-\mu_0 d_T(b,c)}$ be a common positive edge majorant and $h_A = e^{-ad_T(A,D^c)}$. The covariance direction satisfies $\|\mathcal{D}_A\| \leqslant Ckh_A$ and $\|\mathcal{D}(b,c)\| \leqslant Ch_A B_{bc}$. For forest degrees d_b, the radii are

$$r_{\{b,c\}} \coloneqq \frac{\lambda}{32 d_b d_c B_{bc}}, \qquad r_0 \coloneqq \frac{\lambda}{32 Ckh_A} . \tag{7.43}$$

The forest perturbation has norm at most $\lambda/32$, and the extra centre direction $\mathcal{D}_A \circ w^F$ has the same bound on its circle, since a positive Schur mask with unit diagonal is contractive. The total is below $\lambda/8$, so the normalised density has the Gaussian majorant (5.17). Replacing one edge direction by $E_e(\mathcal{D})$ instead uses radius $\lambda/(32 d_b d_c C h_A B_{bc})$ and costs Ch_A against the same forest majorant. Summing the centre and edge derivatives costs at most $C(k+|F|+1)^2 h_A$. Zero directions give zero derivatives, and no covariance entry occurs in a denominator.

Constant-conjugation invariance survives differentiation. A single marked component therefore retains its quadratic character gain, also when it crosses a corner. The product rule over repeated object occurrences and (7.39) give the differentiated cluster bound used in the regional proposition, with the same τ^2 gain.

Each source cluster is allocated to its marked coarse segments with weight $|J_\Gamma|^{-1}$. A network joins its carrier, stencil edges and contour intervals with length at most $C_{\mathrm{net}}\ell(\Gamma)+R_0$. The interval may turn a corner, but its fine-path length is still bounded by $c_h \ell$; it is anchored at the source mark in segment j, hence within R_0 of K_j.

For $d_j > 2R_* + 2R_0$, the clusters are split at $\ell = d_j/(2C_{\mathrm{net}})$. Short clusters have their original stencils inside D, identical labels and cyclic intervals in both expansions, and carriers at distance at least $d_j/2 - R_0$ from D^c, whose fixed loss is absorbed in the constant. Their covariance derivative is bounded by the preceding estimate. Long clusters are paid by (7.39). Bounded d_j are covered by the same rooted bound after enlarging C. This proves

$$|F_{D,C,T} - F_{C,T}| \leqslant C\tau^2 \sum_{j=0}^{4n-1} e^{-cd_j} . \tag{7.44}$$

No side-endpoint terms occur because both contours are the same closed square.

For the periodic comparison, the Neumann estimate for the free inverse gives $\|C_\infty(e,f)\| \leqslant \gamma_L^{-1} e^{-\mu_L \|\ell(e)-\ell(f)\|_\infty}$. Since $d_T(e,f) \leqslant L\|\ell(e)-\ell(f)\|_\infty + 2L$ for fixed lifts, it gives exponential decay in fine units with exponent $\nu = \mu_L/L$. Periodising this free inverse solves the torus inverse equation. For lifted separation $d \leqslant N_{\min}/4$, summing the image shells gives

$$\|C_T(e,f) - C_\infty(e,f)\| \leqslant Ce^{-\nu(N_{\min}-d)} \leqslant Ce^{-\nu N_{\min}/2} e^{-\nu d} . \tag{7.45}$$

The same density-direction radii apply, with $h_A \coloneqq e^{-\nu N_{\min}/2}$ and a reduced common decay exponent.

A rooted cluster with $C_{\mathrm{net}}\ell + R_0 < N_{\min}/4$ lifts uniquely. Projection is injective on its network, preserves carrier intersections, and preserves the fixed ordered square traversal list and its cyclic intervals. Distinct square images are farther than $2N_{\min}/3$ under the stated

period hypothesis, so all its source marks lift to the chosen square. Thus the short torus and bulk clusters correspond with identical combinatorics; only their covariance marginals differ. The other clusters, including winding ones, are paid by their extent tails. Summing over $4n$ roots gives

$$|F_{C,T} - F_{C,\infty}(n)| \leqslant C\tau^2(4n)e^{-c\iota_T} \ . \tag{7.46}$$

Finally, (6.11) applies. The straight sum equals $-\delta m^\sigma P(C_n)$ and the four oriented corner logarithms sum to $4\log z_{\mathrm{c}}^\sigma$, by their definitions. Let

$$E^\circ_{D,C_n,T} \coloneqq (F_{D,C,T} - F_{C,T}) + (F_{C,T} - F_{C,\infty}(n)) + E_n^\sigma \ . \tag{7.47}$$

Equations (7.44), (7.46) and (6.14) give the result. Every logarithm is the connected source branch, so the exponential identity has no branch ambiguity. The adjacent-side clusters and straight endpoint deficits are contained in the corner coefficient. *q.e.d.*

For fixed geometry, E°_{ξ,C_n} depends only on the regional labels and is therefore measurable under the original history sum. Its bound is finite at fixed cutoff.

The boundary sum in (7.34) can be of order n for a loop running near the integration boundary.

8 The Insertion Retained after One Step

The first step produces a function of the coarse field through (1.6). Its representation below retains the closed flux insertions generated by the block average, together with the other source terms. The scalar factor extracted from the reference integral is

$$\kappa_C \coloneqq e^{-\delta m^\sigma P(C)}(z_{\mathrm{c}}^\sigma(g_0, \pi/2))^4 \ . \tag{8.1}$$

Throughout this section the cutoff prescription is fixed, the contour is the counterclockwise (e_1, e_2) square or one of its coarse translates, and the period, collar, and coupling hypotheses of Proposition 7.4 hold. The physical conditional cutoff and its flat projection use the same fixed σ (the cutoff norm $|\bullet|_\sigma$ of (4.12), not the Haar density σ) and threshold $r = \delta_0/g_0$. The scale parameters satisfy (7.28) and the standing conditions (3.1). Closeness to the integration boundary remains in E°_ξ, the error of Proposition 7.4 at the carrier I, defined at the end of this subsection.

The first step is the operator (1.3) with the averaging operation (1.4), Borel and gauge covariant by Lemma 1.1, and ρ_1, N_1 are the measures (1.5). Besides Bałaban's construction [9] as recorded in (B1)–(B4), the representation below uses

- the Borel choices of minimal configurations of Lemma 3.2,
- the Borel background U_1 and row solutions of the measurability paragraph of Section 3.1,
- the bijection $\Psi_\xi \colon \mathcal{D}_\xi \to \mathcal{A}_\xi$ with invertible differential of Lemma 3.16, which is the (1.22) part of (B4)(ii) (its hypotheses: $\delta_0 \leqslant \delta_G$ is listed among the further conditions of Section B.8 and holds for small g_0, as $\delta_0 = g_0 r(g_0) \to 0$; the bound (3.40) on $\mathbf{H}_{1,Ax}$ ([9, p. 251], Lemma D.21) at the tolerance $\delta_0/40$ is the condition "MR_1 large" of item 5 there, met for small g_0 at fixed M since $R_1 \to \infty$, with $\varepsilon_1/\delta_0 \leqslant (1+\beta_0)A_0/A_1$ by [9, (2.8)] and the constants of that lemma fixed before g_0),

- the invertible differential of $B \mapsto U(B)$ (the (1.14) part of (B4)(ii)) and the nonvanishing Jacobian of (1.14) ((B4)(iii); its positivity on $\mathcal{A}_\xi$, Lemma B.11(iv), is used below only to match the Tr log of [9] with $\log|\det|$),
- Lemma 3.19(ii),(iv) and Lemma 3.17 for the a-independence of the exterior factors and the box form of the integration fibre,
- Lemmas 3.12 and 3.17 for the boundedness of the integrand on the box,
- the row factorisation of the Jacobian of (1.14) (proof of Lemma 3.14, from Lemmas 3.6 and 3.9),
- the list of operations after (3.7) with (1.7),
- the disintegration theorem for standard Borel spaces [13, (17.35), Theorem 15.6],
- for the coverage event, the stencils of Appendix C, Lemma 3.36 and Proposition 7.4.

8.1 The Conditional Integral

This section uses the unrestricted transformation of [9] (0.1) and the first-step identities (1.11)–(1.15) with the cutoff (1.25), as recorded in Section 3.1 under (B1)–(B4), with the averaging choice permitted after (0.1); of [9, (1.26)–(1.29)] only (1.27) enters, as the definition of the first-step coupling $g_1(g_0)$ in R_1 and ε_1 (Section 3.1). The outer data ξ consist of the region labels, the coarse field V, the exterior fine field V_0 and the exterior free fluctuation A_{e} in those identities.

For the integrations, the positive fine bonds are partitioned into $E_\Omega = \{b \in E \,|\, b_- \in \Omega_1 \text{ or } b_+ \in \Omega_1\}$ and $E \setminus E_\Omega$. This disjoint integration convention assigns crossing bonds to E_Ω and specifies V_0 only on bonds with both endpoints outside Ω_1. Let $E_{\Omega,*}$ be the free bonds remaining after the designated-bond elimination on Ω_1, and let $D \coloneqq I = \Lambda_1^*$ be the bonds integrated in Λ_1, the carrier D_+ of Section 7 (the comparison there holds for every $D_- \subseteq D \subseteq D_+$; this section is at $D = D_+ = I$, the carrier of Section F.1, which the definition of the integration fibre (8.4) as a section of $\mathcal{D}_\xi \subseteq \mathfrak{g}^J \times \mathfrak{g}^I$ requires).

Let $J = E_{\Omega,*} \setminus D$. The variable $a \in \mathfrak{g}^D$ is integrated and $A_{\mathrm{e}} \in \mathfrak{g}^J$ is held fixed. With i_D and i_J denoting extension by zero into $\mathfrak{g}^{E_{\Omega,*}}$, let $z_\xi(a) \coloneqq i_D a + i_J A_{\mathrm{e}}$. Lebesgue measure $\mathrm{d}a$ is the product measure from the fixed Lie-algebra inner product.

$U_1(\xi)$ is the critical configuration of [9, (1.12)], in its axial mean gauge ((B1)). Let C_ξ be the linear solution map from free fields to fields satisfying the linearised constraints in [9, (1.14)–(1.15)], and let $\mathsf{H}_\xi \coloneqq \mathsf{C}_\xi^* \Delta_{1,\xi} \mathsf{C}_\xi$, with $\Delta_{1,\xi}$ given by [9, (1.16)]. Let h_ξ be the designated-bond right inverse of the linearised constraint and $\widetilde{D}_\xi$ the analytic correction of [9, (1.14)], as identified in (B4)(i) (the corresponding construction for Bałaban's own step is the one following [8, (2.10), p. 267]): they are h_{U_1} and $\widetilde{D}_{U_1}$ of Section 3.3, with the row solutions and their domains defined there. The inverse boundary change Ψ_ξ is [9, (1.22)], acting on the free fields in $\mathfrak{g}^{E_{\Omega,*}}$ and equal to the identity off S_1^*. Let

$$\begin{aligned} q_\xi(s) &\coloneqq s - h_\xi \widetilde{D}_\xi(s) \,, \\ U_\xi(a,b) &\coloneqq \exp(i q_\xi(g_0 \mathsf{C}_\xi \Psi_\xi(z_\xi(a)))(b)) U_1(\xi, b) \,. \end{aligned} \tag{8.2}$$

The second formula applies on E_Ω; on $E \setminus E_\Omega$, $U_\xi = V_0$. The map C_ξ takes $\mathfrak{g}^{E_{\Omega,*}}$ into $\mathfrak{g}^{E_\Omega}$. The variable a on I, and A_{e} on $J \setminus S_1^*$, are the principal fluctuation variables of [9, (1.13)], the chart logarithms of $U U_1(\xi)^{-1}$ divided by g_0 on those free bonds (Ψ_ξ is the identity there;

$I \cap S_1^* = \varnothing$ by (3.39)); on $S_1^* \subseteq J$, $A_{\mathfrak{e}}$ is the variable of (1.22), the (1.13) variable there being $\psi_b(A_{\mathfrak{e}}(b))$; and the domain of the a-integration is specified in these variables below. With $\chi_0'(\Omega_1)$ denoting the product of the group tests (1.8) retained in (1.11), let

$$\begin{aligned} \mathcal{U}_\xi &\coloneqq \{U \in G^E \,|\, U|_{E\setminus E_\Omega} = V_0, M(U)|_{\Omega_1^{[1]}} = V|_{\Omega_1^{[1]}}, \mathcal{I}_\xi(U) > 0\} \;, \\ t_\xi(U) &\coloneqq \frac{1}{i}\log(UU_1(\xi)^{-1})|_{E_\Omega} \;, \\ \widehat{\mathcal{A}}_\xi &\coloneqq \{g_0^{-1} t_\xi(U)|_{E_{\Omega,*}} \,|\, U \in \mathcal{U}_\xi,\; UU_1(\xi)^{-1} \text{ in the chart domain on } E_\Omega\} \;. \end{aligned} \tag{8.3}$$

Here $\mathcal{I}_\xi(U)$ is the nonnegative factor multiplying the group Dirac factors in [9, (1.11)], including χ_1, ζ, χ_0' and the gauge-fixing and action exponential, and t_ξ is defined where $UU_1(\xi)^{-1}$ lies in the chart domain bond by bond on E_Ω, as the chart logarithm of [9, (1.13)]. Thus $\mathcal{U}_\xi \supseteq D^{(11)}(V, V_0)$, the fibre of the full integration domain of [9] (Section 3.1), the fibre defined by all factors of (1.11) and not by the starred group tests χ_0' alone, with equality at every outer datum satisfying the exterior constraint $V(c) = M(V_0, c)$ on the rows c of Ω_1^c ($\bar{V}_0 = V$, Section 3.1), the one condition of $D^{(11)}(V, V_0)$ absent from $\mathcal{U}_\xi$ (the outer measure ω defined below is carried by that set), when $\mathcal{I}_\xi(U) > 0$ is read as including that the averages and means in (1.11) are defined (the domain convention recorded in (B1)); the fluctuations UU_1^{-1} of $D^{(11)}(V, V_0)$ lie in the chart domain by (B1).

The linearising change (1.14) is the map $A \mapsto A - h_\xi \widetilde{D}_\xi(A)$ on the open set $\mathcal{N}_\Theta$ of Lemma B.11(iv), which contains $g_0 \mathsf{C}_\xi \mathcal{A}_\xi$, whose Jacobian is the determinant of (B4)(iii); restricted to the constraint surface, the range of $g_0\mathsf{C}_\xi$, it is $z \mapsto q_\xi(g_0\mathsf{C}_\xi z)$, and on $\mathcal{A}_\xi$ it is q_{U_1} of (3.25): $q_\xi(g_0\mathsf{C}_\xi B) = q_{U_1}(B)$ for $B \in \mathcal{A}_\xi$, with the row solutions of (B4)(i) and $\mathsf{C}_\xi = \mathsf{C}_{U_1}$ (Section 3.1). The Jacobian of q_ξ involves $\widetilde{D}_\xi$ on $\mathcal{N}_\Theta$, a neighbourhood of $g_0\mathsf{C}_\xi\mathcal{A}_\xi$ in $\mathfrak{g}^{E_\Omega}$ on which the row solutions are defined and analytic ((B4)(i), Lemma B.11(iv)); data that are not branches contribute zero.

The boundary change is Ψ_ξ of (3.38), [9, (1.22)] with the cutoff g on the bonds of $(R_1^{\sim 2} \cap S_1)^*$, the set defined at p.251, and its second form (g replaced by 1, the form of p.252) on the other bonds of S_1^*, a set differing from the one printed at p.252 as recorded in Section 3.4, from new free variables to old free variables. The change of variables on these branches is Lemma 3.16 (the (1.22) part of (B4)(ii)): Ψ_ξ maps $\mathcal{D}_\xi$ of (3.3) bijectively onto $\mathcal{A}_\xi$ of (3.2), and $\mathcal{A}_\xi \subseteq \widehat{\mathcal{A}}_\xi$, since for $B \in \mathcal{A}_\xi$ the field $U(B)$ lies in $D^{(11)}(V, V_0) \subseteq \mathcal{U}_\xi$ with $t_\xi(U(B)) = q_{U_1}(B)$ chart-valued and $g_0^{-1} q_{U_1}(B)|_{E_{\Omega,*}} = B$. The integration fibre over $A_{\mathfrak{e}}$ is

$$\mathcal{D}_\xi(A_{\mathfrak{e}}) \coloneqq \{a \in \mathfrak{g}^D \,|\, (A_{\mathfrak{e}}, a) \in \mathcal{D}_\xi\} \;, \tag{8.4}$$

the open box $\{\Theta_\xi = 1\}$ on a branch (Lemma 3.19(iv)). The real coordinate changes and their inverses are restricted to these domains. Their transformed indicators are retained in Θ_D^σ and the exterior factors (1.25). Values outside the admitted domains contribute zero.

Let v_ξ, V_ξ be the functions in [9, (1.15)–(1.16)]: v_ξ contains the Haar logarithms and the logarithm of the Jacobian of the linearising substitution (1.14) ([9, p. 249] says that v is the sum of the first two terms under the second exponential "in (1.15)"; as printed this points to (1.15) itself, whose second exponential begins with the quadratic form and v; the display meant is (1.14), whose second exponential begins with $\log\sigma(A - h\widetilde{D}(A)) + \operatorname{Tr}\log(I - h(\frac{\delta}{\delta A}\widetilde{D})(A))$, and v is read as the normalised Haar logarithm plus $\log|\det(\mathbf{1} - h_\xi \frac{\delta}{\delta A}\widetilde{D}_\xi)|$, Section 3.1 and the paragraph following (8.6)), and V_ξ contains the remaining Taylor residual of the action and gauge fixing. These are the full functions, before the boundary change. For z in their real domain let

$$p_\xi(z) \coloneqq \exp\left[-\frac{1}{2}\langle z, \mathsf{H}_\xi z\rangle + v_\xi(g_0\mathsf{C}_\xi z) - g_0^{-2} V_\xi(g_0\mathsf{C}_\xi z)\right] \;. \tag{8.5}$$

With $\Theta_D^\sigma(a) = \prod_{b\in D} \mathbf{1}_{\{|a(b)|_\sigma<r\}}$, in the bond norm of [9, (1.18)], written $|\bullet|_\sigma$ as in (4.12) with the fixed σ of this section, which is the norm of the integration-carrier cutoff (3.42) (so at $D = I$, $\Theta_D^\sigma = \Theta_\xi$ on $\mathfrak{g}^I$, the integration fibre (8.4) is the box $\{\Theta_D^\sigma = 1\}$ on a branch, and $\mathbf{1}_{\mathcal{D}_\xi(A_\mathrm{e})}\Theta_D^\sigma = \mathbf{1}_{\mathcal{D}_\xi(A_\mathrm{e})}$ there), the conditional density and integral are

$$\begin{aligned} \mathcal{K}_\xi(a) &\coloneqq \mathbf{1}_{\mathcal{D}_\xi(A_\mathrm{e})}(a)\Theta_D^\sigma(a)p_\xi(\Psi_\xi(z_\xi(a)))|\det D\Psi_\xi(z_\xi(a))| \,, \\ \mathcal{J}_\xi[f] &\coloneqq \int f(U_\xi(a))\mathcal{K}_\xi(a)\,\mathrm{d}a \,. \end{aligned} \tag{8.6}$$

Here $\mathcal{J}_\xi$ and $\mathcal{K}_\xi$ are the $\mathcal{J}_\xi^{\mathrm{a}}$ and $\mathcal{K}_\xi$ of the measurability paragraph of Section 3.1, written in the variable a of the integration fibre.

The determinant inside v_ξ, the Jacobian of [9, (1.14)] (not $|\det D\Psi_\xi|$, the Jacobian of the boundary change in (8.6)), is that of the map $A \mapsto A - h_\xi\widetilde{D}_\xi(A)$ on $\mathfrak{g}^{E_\Omega}$, the phases on all bonds of Ω_1; it equals the determinant of its designated block, the product over the rows of the determinants in (3.31) (proof of Lemma 3.14: h_ξ has outputs on the designated bonds only, so the free rows of $\mathbf{1} - h_\xi\frac{\delta}{\delta A}\widetilde{D}_\xi$ are identity rows, $b_0(c)$ is the only designated bond of $\mathcal{E}_c$, Lemma 3.6, and $\widetilde{D}_{\xi,c}$ depends on $A|_{\mathcal{E}_c}$ only, Lemma 3.9).

Every change of variables uses its absolute real Jacobian: v_ξ contains $\log|\det(\mathbf{1}-h_\xi\frac{\delta}{\delta A}\widetilde{D}_\xi)|$, the sum over the rows of (3.31), and the logarithms of the positive normalised Haar density, so $p_\xi > 0$ where it is defined and $\mathcal{K}_\xi \geqslant 0$; the determinant is positive on $\mathcal{A}_\xi$ ((B4)(iii), Lemma B.11(iv)) and the real logarithm written in [9] agrees with this. The definition retains $p_\xi \circ \Psi_\xi$ without a truncation of the boundary terms described after [9, (1.24)].

The exterior expression of [9] is the sum over region labels and integral over (V_0, A_e), with the constraint $\delta(\bar{V}_0 V^{-1})$, the retained region indicators, $\zeta(\Omega_1^c)$, Haar factors and the prefactors of (1.15), including $z^{(0)}$. Its exterior factors are evaluated at their transformed arguments. After splitting the good tests into their D and exterior bond factors, the factors remaining from (1.25) are independent of a: Lemma 3.19(ii), under the standing conditions (3.1) (its collar hypothesis is their condition $LMR_1 > 4R_\mathrm{c} + 4L^2M_2R_1 + 8$; the collar (7.28) is that of Proposition 7.4), separates the bad and changed tests from D, and factors of good tests on D are redundant by idempotence. Those exterior factors, including any dependence on A_e, remain in $\mathcal{O}_V$.

The maps in these integrals are jointly Borel in their parameters and integration variables, by the measurability paragraph of Section 3.1: the background U_1 is Borel on the Borel set of admissible pairs (the uniqueness statement of [9] for the critical point in the domain of integration in (1.11), (B1), and the theorems for Borel graphs with singleton sections); pairs that are not admissible have $D^{(11)}(V, V_0) = \varnothing$, hence zero source weight. The configuration of [9, (1.21)] is the configuration described at p.251, on its prescribed determining data, the notation $M^\bullet(W)$ being the level-wise collection $M^j(W)$ of [9, (2.11)], at the first step W and $M(W)$; the measurable choices of $U_{1,\square'}(V)$ and of $U_1(\Omega_1 \cap \Lambda_1^c, V)$ are the Borel choices of Lemma 3.2, fixed in (B1). The value of $\mathbf{H}_{1,Ax}$ used in (1.22) is $L\frac{1}{i}\log(U_1U_1(\Omega_1 \cap \Lambda_1^c, V)^{-1})$ by (1.21), the chart logarithm of a relative configuration lying in the chart domain on S_1^* (Lemma D.21), and is Borel in (V, V_0). The maps Q_1^{s*} in [9, (1.3)] and the averages on these regular data are Borel as well.

The matrix coefficients of C_ξ, h_ξ are obtained by inversion of the invertible designated coefficient operators $\alpha_c(U_1)$, so they are Borel in the background. The row solutions $\widetilde{D}_\xi$ of Section 3.3 are Borel on $\mathcal{A}_\xi$ jointly with the branch data, by the Lusin–Novikov argument of the measurability paragraph of Section 3.1 (the set P of pairs (B, d) with d a chart-class solution family at $g_0\mathsf{C}_{U_1}B$ for which $U_d(B) \in D^{(11)}(V, V_0)$ with the cutoff (1.25) nonzero is

Borel with sections of at most one point, Lemma B.11(iii)), and so is $\frac{\delta}{\delta A}\widetilde{D}_\xi$ (measurability paragraph).

In [9, (1.22)], the cutoff g is the fixed C^∞ function of the paragraph defining (3.38), each ψ_b is C^∞ on its open domain $\mathcal{W}_b$ (the chart logarithm and the exponential are analytic there), and the boundary data have just been shown Borel. Hence $\Psi_\xi(z)$ and its Jacobian are jointly Borel, and C^∞ in z. The sets $\mathcal{A}_\xi$ and $\mathcal{D}_\xi$, and the integration fibres (8.4), sections of the Borel set $\mathcal{D}_\xi$, are Borel (measurability paragraph).

The Hessian coefficients, v_ξ, V_ξ and the constraint factor $z^{(0)}$ are Borel: each is a finite sum or product of the Haar logarithm, the plaquette and gauge terms of the action and gauge fixing, the row determinants and the coefficients of $\alpha_c(U_1)^{-1}$, evaluated at the background, the phase and the row solutions, that is a continuous function of $(U_1, A, \widetilde{D}_\xi(A), \frac{\delta}{\delta A}\widetilde{D}_\xi(A))$ and of the Taylor coefficients at $A = 0$ of the germs of Lemma 3.9, themselves continuous in U_1 by the implicit function theorem, composed with maps that are Borel jointly in the branch data and a (the background, the row solutions and their derivative, measurability paragraph of Section 3.1). The vacuum exponential is a constant, and $\zeta(\Omega_1^c)$ is a finite product of small-field tests, indicators of sets defined by inequalities between continuous functions of (V, V_0) on the open set where the local means entering them are defined, and of gauge-fixing exponentials of continuous functions of (V, V_0) on that set, every factor being set to zero where a local mean it contains is undefined (the convention of this paper for $\zeta(\Omega_1^c)$, matching the chart convention of Section 3 for the tests); so $\zeta(\Omega_1^c)$ is Borel as well.

Together with the indicators, these facts prove joint Borel measurability of $U_\xi(a)$, $\mathcal{K}_\xi(a)$ and the outer weights.

For an empty carrier, its Lie-algebra space is a point with unit measure; empty sums equal zero, and empty products and zero-dimensional determinants equal one. If Ω_1 is empty, $U_1 = U_\xi = V_0$ on every bond.

The exterior group Dirac factors have a kernel interpretation as well. Let ω be the joint outer measure in [9, (1.15), (1.25)], retaining V, the region labels, V_0 and A_{e}: in the notation of (3.4), $\int F \,\mathrm{d}\omega = \int \mathcal{O}[F](V)\,\mathrm{d}V$ for nonnegative Borel F of the outer data, in the sense fixed in Section 3.1 (the integral of F against the measure in V that the outer operation defines; $\mathcal{O}[F](V)$ is not a density). It contains exactly the exterior factors specified above. Its group Dirac factors are read by first integrating them against the corresponding coarse bond variables, with Haar measure on the remaining coarse variables; this defines a positive joint measure.

Its weighted measure $\bar\eta = \mathcal{J}_\xi[1]\omega$ is finite and its V marginal is the measure ρ_1 of (1.5): the operations in [9, (1.1)–(1.25)], performed as listed after (3.7) with a bounded Borel test $h(V)$ retained (signed; $h(V) = h(\mathsf{M}(U))$ is invariant under the gauge transformations u of (1.5), which are the identity at the block centres (the printed "$z \neq y$" in [9, (1.5)] read as $x \neq y$, as in (1.7)), Lemma 1.1(ii), so the Faddeev–Popov step applies with it inserted; every integral below is finite: $\int \mathcal{J}_\xi[1]\,\mathrm{d}\omega = \mathcal{Z}_0 < \infty$ is the case $h = 1$ of the first identity below, both sides being the integral of one nonnegative Borel integrand over all variables by Tonelli, and $|\mathcal{J}_\xi[W_\chi]| \leqslant \mathcal{J}_\xi[1]$) and with insertion 1 or W_χ, give

$$\begin{aligned} \int h(V)\,\mathrm{d}\rho_1(V) &= \int h(V)\mathcal{J}_\xi[1]\,\mathrm{d}\omega(V,\xi)\,, \\ \int h(V)\,\mathrm{d}N_1(V) &= \int h(V)\mathcal{J}_\xi[W_\chi(C;\bullet)]\,\mathrm{d}\omega(V,\xi)\,. \end{aligned} \tag{8.7}$$

Taking $h = 1$ in the first equality gives $\bar\eta$ the total mass $\mathcal{Z}_0 < \infty$. Let Q_V be the regular

conditional probability of $\bar\eta$ on the outer data given V, defined for ρ_1-almost every V (disintegration of the finite measure $\bar\eta$ with respect to its marginal ρ_1 along the coordinate projection $\xi \mapsto V$: the measure disintegration theorem for standard Borel spaces, [13, (17.35)], stated there as an exercise with the standard proof, through a Borel isomorphism ϕ [13, Theorem 15.6] of the outer space with $[0,1]$ and the Radon–Nikodym derivatives with respect to ρ_1 of the countably many measures $B \mapsto \bar\eta(\{\xi : V(\xi) \in B,\ \phi(\xi) \leqslant t\})$, t rational, regularised to a Borel map $V \mapsto Q_V$ into the probability measures; the outer space, a finite union over the region labels of products of the compact groups G^{E_1}, $G^{E\setminus E_\Omega}$ and the Euclidean space $\mathfrak{g}^J$, is Polish, hence standard Borel, and on the branches $\bar\eta$ is carried by a bounded set, $g_0 A_{\mathrm{e}}(b)$ lying in the chart ball for every $b \in J$, so the compact case would suffice. The theorem gives: Q_V is a probability measure for every V off a Borel ρ_1-null set, $V \mapsto Q_V(E)$ is Borel for every Borel E, $\bar\eta(E) = \int Q_V(E)\,\mathrm{d}\rho_1(V)$, Q_V is concentrated on the fibre $\{\xi : V(\xi) = V\}$ for ρ_1-almost every V, and two such maps agree for ρ_1-almost every V; the concentration is what replaces $h(V(\xi))$ by $h(V)$ in the identity displayed below, and the uniqueness makes $\mathcal{O}_V$ below independent of the version off a ρ_1-null set).

For a nonnegative measurable outer function F, the exterior operator is

$$\mathcal{O}_V[F] \coloneqq \int_{\{0<\mathcal{J}_\xi[1]<\infty\}} \frac{F(\xi)}{\mathcal{J}_\xi[1]}\,\mathrm{d}Q_V(\xi)\,. \tag{8.8}$$

This defines a positive kernel for ρ_1-almost every V. It gives ω back on finite-positive-mass outer data: $\int h(V)\mathcal{O}_V[F]\,\mathrm{d}\rho_1(V) = \int h(V)F(\xi)\mathbf{1}_{\{0<\mathcal{J}_\xi[1]<\infty\}}\,\mathrm{d}\omega(V,\xi)$ for nonnegative Borel F and bounded Borel $h \geqslant 0$, since $\mathrm{d}\bar\eta = \mathcal{J}_\xi[1]\,\mathrm{d}\omega$, and for signed bounded Borel h whenever $\int F\mathbf{1}_{\{0<\mathcal{J}_\xi[1]<\infty\}}\,\mathrm{d}\omega < \infty$.

In this sense $\mathcal{O}_V$ represents the exterior expression of [9]: integrated against bounded Borel $h(V)$, it reproduces the ω-integral of every outer function supported on the branches.

On the Borel ρ_1-null set where Q_V is not defined, $\mathcal{O}_V = 0$, so that $V \mapsto \mathcal{O}_V[F]$ is Borel. It extends linearly to complex F whenever $\mathcal{O}_V[|F|] < \infty$. In particular $\mathcal{O}_V[\mathcal{J}_\xi[1]] = Q_V(\{0 < \mathcal{J}_\xi[1] < \infty\}) = 1$ for ρ_1-almost every V, by the next paragraph. The changes of variables are those of the unrestricted construction of [9], recorded after (3.7).

The outer data with $0 < \mathcal{J}_\xi[1] < \infty$ are exactly the branches in the sense of Section 3.1: $\mathcal{J}_\xi[1] = 0$ when the integration fibre $\mathcal{D}_\xi(A_{\mathrm{e}})$ is empty, that is off the branches of that section, while on a branch $\mathcal{J}_\xi[1] < \infty$ (measurability paragraph) and $\mathcal{J}_\xi[1] > 0$, since $\mathcal{K}_\xi > 0$ on the nonempty open box $\mathcal{D}_\xi(A_{\mathrm{e}})$ ($p_\xi > 0$, $|\det D\Psi_\xi| > 0$ by Lemma 3.16, and $\Theta_D^\sigma = 1$ there).

Zero-mass data are $\bar\eta$-null, hence Q_V-null for ρ_1-almost every V. Data with infinite conditional mass are ω-null, since $\int \mathcal{J}_\xi[1]\,\mathrm{d}\omega = \mathcal{Z}_0 < \infty$ (the case $h = 1$ of (8.7), with $\mathcal{Z}_0 = \rho_1(G^{E_1})$ by (1.7)), hence $\bar\eta$-null and Q_V-null for ρ_1-almost every V. So Q_V is carried by the branches for ρ_1-almost every V. All branch contractions below are zero off the branches.

Coverage uses the fixed geometric stencils Λ_b of Appendix C: $\Lambda_b = \{b\}$ for a free bond, and $\Lambda_{b_0(c)} = \mathcal{S}_c \cap E_*$ for a designated bond, where $\mathcal{S}_c$ contains the bonds with both endpoints in the two endpoint blocks of c. Let A_{cov} consist of the branches on which every fine contour bond is a bond of Ω_1 and $\Lambda_b \subseteq D$ for every such bond. The geometric condition depends only on region labels, and is precisely the coverage used in Proposition 7.4; at the carrier $D = I$ of Section F.1 it is the covered event of Lemma 3.36 and of Theorem 1.2, since $\Lambda_b = \mathcal{F}_b$ for every fine bond ($\mathcal{S}_c \cap E_* = \mathcal{E}_c \cap E_*$, both being the free positive bonds with both endpoints in $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$, (3.10) and (3.13)).

Since (7.28) supplies the collar condition at the carrier $D = I(\xi)$ of every branch, Proposition 7.4 defines E_ξ° on A_{cov}.

8.2 The Closed Flux Insertions

Let $\bar{C} \coloneqq (c_1, \dots, c_m)$ be the oriented coarse square, with $m = 4n$, and let C be the concatenation of its straight fine paths. Let ξ be a covered conditional history, with constrained background $U_1 = U_1(\xi)$. For each positive coarse bond traversed by $\bar{C}$, the first and last fine bonds of its straight path lie inside its initial and final blocks, respectively, since $L \geqslant 3$. Coverage puts these bonds in Ω_1. As Ω_1 is a union of blocks, both endpoint blocks lie there. In particular c is a bond of the coarse region $\Omega_1^{(1)}$, so [9, (1.12)] imposes $M(U_1, c) = V(c)$. Here a bond of a region has at least one endpoint in that region, as in [6, Introduction]; both endpoints belong in this case. Reversed traversals use the constraint on the underlying positive bond. All relative logarithms are in the regular chart used in (4.1).

For a positive coarse bond c, Y_c is defined by the first formula below. The averaging constraint and (4.1) then give the second:

$$Y_c(\xi) = e^{-i\beta_c(U_1)}, \qquad P_c(U_1) = Y_c(\xi)V(c)\,. \tag{8.9}$$

Thus Y_c depends on the closed relative contours inside the two endpoint blocks of c. Let $V(c^{-1}) \coloneqq V(c)^{-1}$. For a reversed traversal,

$$Y_{c^{-1}}(\xi) = V(c)^{-1}Y_c(\xi)^{-1}V(c), \qquad P_{c^{-1}}(U_1) = Y_{c^{-1}}(\xi)V(c)^{-1}\,. \tag{8.10}$$

The second identity follows by inverting $P_c = Y_cV(c)$. The factor is therefore based at the initial vertex of the reversed traversal as well.

The closed contraction is

$$\mathcal{L}_\xi(C; V) \coloneqq \chi\left(Y_{c_1}(\xi)V(c_1)\cdots Y_{c_m}(\xi)V(c_m)\right) = W_\chi(C; U_1(\xi))\,. \tag{8.11}$$

This equality follows by substituting (8.9) and (8.10) in the ordered fine holonomy.

Under a site gauge u, the closed logarithms in (4.1) and their exponentials are conjugated at c_-. Consequently

$$Y_{c_i} \longmapsto u(c_{i,-})Y_{c_i}u(c_{i,-})^{-1}, \qquad V(c_i) \longmapsto u(c_{i,-})V(c_i)u(c_{i,+})^{-1}\,. \tag{8.12}$$

The intermediate gauge factors cancel in the ordered product and its character is invariant. All subsequent insertions in this section are closed contractions.

Different histories at the same V can have different constrained backgrounds and different insertions.

8.3 The Numerator Carried to the Next Transformation

The exterior operator $\mathcal{O}_V$ of (8.8) and the conditional integral $\mathcal{J}_\xi$ defined before it give, for ρ_1-almost every V,

$$\mathcal{O}_V[\mathcal{J}_\xi[1]] = 1, \qquad W_1(C; V) = \mathcal{O}_V[\mathcal{J}_\xi[W_\chi(C; \bullet)]]\,, \tag{8.13}$$

that is, $\mathrm{d}N_1 = \mathcal{O}_V[\mathcal{J}_\xi[W_\chi(C; \bullet)]]\,\mathrm{d}\rho_1$.

The identities in (8.13) are identities of measures on G^{E_1}, tested against bounded Borel functions. Let $\mathfrak{M}$ be the image of $\rho_0(U)\,\mathrm{d}U$ under $U \mapsto (M(U), U)$, a finite positive measure on $G^{E_1} \times G^E$ (the blocking kernel (0.1) with bare weight $\rho_0\,\mathrm{d}U$, in the reading of (1.3)). Its V marginal is ρ_1. Disintegration on these standard Borel spaces supplies conditional

probabilities ν_V on fine fields, for ρ_1-almost every V. For bounded Borel functions f on G^E and h on G^{E_1},

$$\int h(V) f(U)\,\mathrm{d}\mathfrak{M}(V,U) = \int h(M(U)) f(U) \rho_0(U)\,\mathrm{d}U = \int h(V) \left(\int f(U)\,\mathrm{d}\nu_V(U) \right) \mathrm{d}\rho_1(V)\,. \tag{8.14}$$

In particular, with $f = W_\chi(C;\bullet)$ and (1.5), $\int h\,\mathrm{d}N_1 = \int h(V) \int W_\chi(C;U)\,\mathrm{d}\nu_V(U)\,\mathrm{d}\rho_1(V)$, so $W_1(C;V) = \int W_\chi(C;U)\,\mathrm{d}\nu_V(U)$ for ρ_1-almost every V, the conditional form of (1.6); the operations below act on the middle member of (8.14).

The averaging operation is (1.4), Borel and gauge covariant by Lemma 1.1; (4.1) is its formula on $\mathcal{O}_c$, which contains every configuration of $D^{(11)}$ on the rows of Ω_1 (the domain convention recorded in (B1)).

For $f = 1$ or $W_\chi(C;\bullet)$, the operations of [9, (1.1)–(1.25)] listed after (3.7) are applied with $h(V)$ retained. The partitions are pointwise decompositions of unity. The Faddeev–Popov step (1.5)–(1.6) is applied to the operator (0.1) with the averaging choice (1.4), whose covariance on all of G^E is Lemma 1.1(ii); the closed Wilson character ($|W_\chi| \leqslant 1$, χ being a normalised irreducible character) is invariant under its fine gauges. The changes of variables pull that character back to $W_\chi(C;U_\xi(a))$, and $h(V)$ stays fixed: the gauge transformations of (1.5) are the identity at the block centres, so $M(U)$, hence $h(M(U))$, is unchanged by them (Lemma 1.1(ii)), and the later changes of variables are at fixed V. Cutoff factors are removed only where [9] states they equal one, and the boundary change is retained as (8.6).

Thus the operations supply (8.7) against every bounded Borel h; since $|\mathcal{J}_\xi[W_\chi]| \leqslant \mathcal{J}_\xi[1]$ (nonnegativity of (8.6)), $\mathcal{J}_\xi[W_\chi]$ vanishes where $\mathcal{J}_\xi[1] = 0$, and $\{\mathcal{J}_\xi[1] = \infty\}$ is ω-null, so both ω-integrals are integrals over the branches, and with (8.8) they give $\int h\,\mathrm{d}\rho_1 = \int h(V) \mathcal{O}_V[\mathcal{J}_\xi[1]]\,\mathrm{d}\rho_1(V)$ and $\int h\,\mathrm{d}N_1 = \int h(V) \mathcal{O}_V[\mathcal{J}_\xi[W_\chi(C;\bullet)]]\,\mathrm{d}\rho_1(V)$. Equality of these finite measures gives the ρ_1-almost-everywhere identities (8.13), the second by the uniqueness of the Radon–Nikodym derivative (1.6).

Nonnegativity of (8.6) gives $|\mathcal{J}_\xi[W_\chi]| \leqslant \mathcal{J}_\xi[1]$, hence $|\mathcal{O}_V[\mathcal{J}_\xi[W_\chi]]| \leqslant \mathcal{O}_V[\mathcal{J}_\xi[1]] = 1$, the bound $|W_1| \leqslant 1$ of Section 1.

For the covered branches A_{cov} defined above, let

$$\mathcal{L}^{(0)}_\xi(C;V) := \begin{cases} \kappa_C e^{E^\circ_\xi} \mathcal{L}_\xi(C;V), & \xi \in A_{\mathrm{cov}}, \\ 0, & \xi \notin A_{\mathrm{cov}}. \end{cases} \tag{8.15}$$

Proposition F.2 (Appendix F) identifies its coefficient with $S^\circ_{\xi,C}$ on covered histories, with $E^\circ_\xi = E^\circ_{I,C_n,T}$ at the carrier $D = I = D_+$ (all free bonds with an endpoint in Λ_1, the convention of Section F.1). Its period and coupling hypotheses are fixed for the square; its boundary-distance terms allow arbitrary covered regional labels. The error E°_ξ depends only on those discrete labels; the background and branch indicator are Borel by the preceding construction, so this leading term is measurable.

The other source terms are retained as the following conditional integral, with the same denominator:

$$\mathcal{H}_\xi(C;V) = \frac{1}{\mathcal{J}_\xi[1]} \int \left[W_\chi(C;U_\xi(a)) - \mathcal{L}^{(0)}_\xi(C;V) \right] \mathcal{K}_\xi(a)\,\mathrm{d}a\,. \tag{8.16}$$

On an uncovered branch, $\mathcal{L}^{(0)}_\xi = 0$ and $\mathcal{H}_\xi$ is the normalised conditional Wilson integral. Its history average enters (1.9) through the defect and boundary terms E_{def}, E_{bdry} of Corollary F.3, which Theorem 1.2(ii) bounds.

For this fixed square, (7.34) and $|\chi| \leqslant 1$ give $|\mathcal{L}_\xi^{(0)}| \leqslant B_C$, where $B_C := |\kappa_C| \exp(C\tau^2(9n+1)) < \infty$ (the constant C of (7.34), not the contour). Consequently $\mathcal{J}_\xi[1]|\mathcal{H}_\xi| \leqslant (1+B_C)\mathcal{J}_\xi[1]$. The numerator terms below are therefore integrable for the complex extension of (8.8) at ρ_1-almost every V.

Substitution in (8.13) gives the following first-step identity for ρ_1-almost every V:

$$W_1(C;V) = \mathcal{O}_V\left[\mathcal{J}_\xi[1](\mathcal{L}_\xi^{(0)}(C;V) + \mathcal{H}_\xi(C;V))\right] . \tag{8.17}$$

More explicitly, let $\nu_\xi = (U_\xi)_*(\mathcal{K}_\xi(a)\,\mathrm{d}a/\mathcal{J}_\xi[1])$ be the probability measure on fine fields carried by the branch. Under a fine gauge transformation u (acting on V at the block centres, Lemma 1.1(ii)), ν_ξ is replaced by its pushforward $(U \mapsto U^u)_*\nu_\xi$, together with U_1^u and V^u. Both closed characters in (8.16) are invariant, E_ξ° depends only on the unchanged region labels, and change of variables preserves the normalised conditional integral. An axial-gauge representative and its conditional measure are transported together. The physical denominator is $\mathcal{Z}_0 = \rho_1(G^{E_1})$.

For ρ_1-almost every V, the scalar-extracted outgoing observable is

$$\widehat{W}_1(C;V) := \frac{W_1(C;V)}{\kappa_C} = \frac{\mathcal{O}_V\left[\mathcal{J}_\xi[1](\kappa_C^{-1}\mathcal{L}_\xi^{(0)}(C;V) + \kappa_C^{-1}\mathcal{H}_\xi(C;V))\right]}{\mathcal{O}_V[\mathcal{J}_\xi[1]]} . \tag{8.18}$$

The factor κ_C is nonzero: (5.39) defines its finite perimeter exponent and (6.12) writes z_{c} as the exponential of the convergent corner sum.

The pair of the complex measure $\kappa_C^{-1}N_1$ and the measure ρ_1 is the input to the next blocking transformation (T acting on measures by the pushforward $T\mu = M_*\mu$ of Section 1). Equation (8.11) displays its leading closed insertion; (8.16) specifies the other source terms carried with it. Iterating this representation requires local estimates for their transformed activities, including correlations between attachments and their boundary dependence.

9 The Exponential Moment of the Leading Error

This section bounds the history average of $e^{|E_\xi^\circ|}$, where $E_\xi^\circ := E_{I,C_n,T}^\circ$ is the exponent of Proposition F.2 at the carrier $I = \Lambda_1^*$.

The conventions are those of Section F.1: $C = C_n$ has $n_C = 4nL$ contour bonds, C_j is the set of contour bonds of the coarse segment $j \in J_C$, K_j is the union of their free-input stencils, d_j and ι_T are the distances (F.1), and d_T is the periodic fine-bond metric $d_T(b,c) = d_{\infty,T}(x_b, x_c) + \mathbf{1}\{\text{directions differ}\}$ fixed after (5.2), with x_b the initial site of b. The periods are compatible and even (Section D.7). $\mathbb{E}_{\mathrm{h}}$ and $\mathbb{P}_{\mathrm{h}}$ are the history expectation (D.20) and its probability (weight w_ξ, normalisation $\mathcal{Z}_0$), $B_\xi = \Lambda_1^c$ is the filled bad region and $E(X)$ the set of positive fine bonds with at least one endpoint in X (Section D.7), and $q = n_W \max(z_P, z_G)$ is the one-point rate of Lemma D.17, with $q \leqslant e^{-c_*\pi_*^2}$ for $0 < g_0 < g_*$ (Proposition D.18).

A free bond outside the carrier is marked: I is the set of all free bonds with an endpoint in Λ_1, so a free bond $b \notin I$ has both endpoints in B_ξ, and

$$\mathbf{1}\{b \notin I\} \leqslant \mathbf{1}\{b \in E(B_\xi)\} . \tag{9.1}$$

9.1 Scales

$r = \delta_0/g_0 = A_1(\log g_0^{-2})^{p_0}$, with $\varepsilon_0 = g_0 A_0(\log g_0^{-2})^{p_0}$ and $\delta_0 = (A_1/A_0)\varepsilon_0$ [9, pp. 246–247]. By [9, (2.6), p. 255] ($n = 1$, $m = 0$), $g_1 \leqslant (1+\beta_0)g_0$ and $g_0 \leqslant (1+\beta_0)g_1$, so for small g_0, $\frac{1}{2}\log g_0^{-2} \leqslant \log g_1^{-2} \leqslant 2\log g_0^{-2}$. R_1 is the smallest power of L with $R_1 \geqslant (\log g_1^{-2})^{\mathfrak{r}}$ [9, p. 246], so $R_1 \leqslant L(2\log g_0^{-2})^{\mathfrak{r}}$, and with $p_0(g) = A_0(\log g^{-2})^{p_0}$ (D.1),

$$\ell = LMR_1 \leqslant L^2 M(2\log g_0^{-2})^{\mathfrak{r}}\,, \qquad \pi_* = \min(p_0(g_0), p_0(g_1)) \geqslant A_0(\tfrac{1}{2}\log g_0^{-2})^{p_0}\,, \tag{9.2}$$

where $p_0 \geqslant 5\mathfrak{r} \geqslant 10$ [9, p. 246].

9.2 The Refined Split

Corollary 9.1 (The Refined Split)**.** *Under the hypotheses of Proposition F.2, with its constants C and c, and with $4ne^{-c\iota_T} \leqslant 1$, on A_{cov}*

$$|E_\xi^\circ| \leqslant C\tau^2(2 + 1/c) + C\tau^2 \sum_{j \in J_C} e^{-cd_j}\,. \tag{9.3}$$

For $n \geqslant 1$ the condition $4ne^{-c\iota_T} \leqslant 1$ holds when $N_{\min} \geqslant 4R_0 + (4/c)\log(4n)$.

Proof. Proposition F.2 bounds $|E_\xi^\circ|$ by $C\tau^2[\sum_j e^{-cd_j} + 4ne^{-c\iota_T} + (n+1)e^{-cn}]$, and $(n+1)e^{-cn} \leqslant 1 + \sup_{x \geqslant 0} xe^{-cx} = 1 + 1/(ec) \leqslant 1 + 1/c$. When $N_{\min} \geqslant 4R_0 + (4/c)\log(4n)$, $\iota_T = N_{\min}/4 - R_0 \geqslant (1/c)\log(4n)$, so $4ne^{-c\iota_T} \leqslant 1$. *q.e.d.*

Under the torus condition the only part of the bound that grows with n is the boundary sum $\sum_j e^{-cd_j} \leqslant 4n$. Each of its terms is at most 1 and exponentially small in $d_T(K_j, I^c) - R_0$, and the sum is zero when $I^c = \varnothing$.

9.3 The Joint Witness Bound

Lemma 9.2 (Joint Witness Bound)**.** *Let $K = 256$ be the constant of Lemma D.13, $\sigma_K := 2K\ell + 4$ and $C_K := 4(2\sigma_K + 1)^4 = 4(4K\ell + 9)^4$, an upper bound on the number of bonds whose initial site lies within ℓ^∞ distance σ_K of a given site. For g_0 below the thresholds of Lemmas D.13 and D.17, for every finite set S of bonds and every subset S' of S whose initial sites are at pairwise ℓ^∞ distance $> \sigma_K$,*

$$\begin{gathered}\mathbb{P}_{\mathrm{h}}(\mathsf{S} \subseteq E(B_\xi)) = \mathcal{Z}_0^{-1} \int \mathcal{O}\Big[\mathsf{w}_\xi \prod_{z \in \mathsf{S}} \mathbf{1}\{z \in E(B_\xi)\}\Big] \mathrm{d}V \leqslant q^{|\mathsf{S}'|}\,, \quad \textit{and} \\ \mathbb{P}_{\mathrm{h}}(\mathsf{S} \subseteq E(B_\xi)) \leqslant q^{|\mathsf{S}|/C_K}\,.\end{gathered} \tag{9.4}$$

Proof. The marks $\mathbf{1}\{z \in E(B_\xi)\}$ are functions of the regions and of the deterministic filling, they take the values 0 and 1, and $\prod_{z \in \mathsf{S}} \mathbf{1}\{z \in E(B_\xi)\} \leqslant \prod_{z \in \mathsf{S}'} \mathbf{1}\{z \in E(B_\xi)\}$. The reversal of Lemma D.15 is a change of variables for integrals of the density times a nonnegative function of U and the regions, so it applies with the product of the marks of S' inserted.

After the reversal, on the support of the cutoffs, each mark is dominated by $\mathbf{1}_{W_z}$ (Lemma D.13, (D.36)). Here W_z is taken as the union, over the plaquette labels p and the gauge-variable labels w all of whose sites (the four corners of p; the centre of w and the site of its variable) lie within ℓ^∞ distance $K\ell + 1$ of x_z, of the single-witness events $\{|U(\partial p) - \mathbf{1}|_\sigma \geqslant \theta\}$ and $\{|w - \mathbf{1}|_\sigma \geqslant \theta\}$. The domination holds with this union. The

proof of Lemma D.13 places the witness in the neighbourhood $N(Q)$ of a bad test cube Q, every site of a witnessing plaquette within $117\ell + 1$ of x_z, and the centre of a gauge label lies within L of the site of its variable; so every site of the witnessing label lies within $117\ell + L + 1 \leqslant K\ell + 1$ of x_z.

The union has at most $7(2K\ell + 3)^4 \leqslant n_W$ labels: at most $(2K\ell + 3)^4$ sites lie within $K\ell + 1$ of x_z, a plaquette is fixed by its base site and one of six plane orientations, and a gauge label by the noncentral site of its variable, which determines the centre (the centre of its block). So $\bigcap_{z\in\mathsf{S}'} W_z$ is contained in a union of at most $n_W^{|\mathsf{S}'|}$ intersections of single-witness events, one label per bond of S'.

For each such choice, two bonds of S' have initial sites at distance $> \sigma_K = 2(K\ell+1)+2$, so their labels have all their sites at distance > 2 from each other: the chosen gauge variables are distinct, and the chosen plaquettes have distinct base cubes C_x (the closed unit cube whose minimal corner x is the plaquette's base site). Under the product measure $\mu_0 \otimes q_g^{\otimes\mathcal{I}} \otimes \mathbf{Haar}^{\otimes T}$ of (D.36) the probability of the intersection factorises into the μ_0-probability of the plaquette events and the product of the q_g-probabilities of the gauge-variable events, each of the latter at most z_G (D.39).

For the plaquette events the chessboard estimate (D.41) applies with f_x, for each chosen base cube C_x, the indicator $\mathbf{1}\{|U(\partial p_0) - \mathbf{1}|_\sigma \geqslant \theta\}$ of the plaquette p_0 of C_0 whose transport $f_x^{[x]}$ is the indicator of the chosen plaquette of C_x, and $f_x = 1$ for the other cubes. Every plaquette of C_x is the transport of a plaquette of C_0 of the same plane orientation: a reflection in direction ρ carries the transverse coordinate $t \in \{0, 1\}$ of a face to $1 - t$, so the face of C_0 with transverse coordinates (t_ρ) is carried to the face of C_x with transverse coordinates $x_\rho + t_\rho$ or $x_\rho + 1 - t_\rho$ according to the parity of x_ρ, and the indicator does not depend on the orientation or the base point of the plaquette. Each root $(\int \prod_y f_x^{[y]}\, \mathrm{d}\mu_0)^{1/V_T}$ of a disseminated expectation is at most z_P by the computation in the proof of Lemma D.17, which applies to every plaquette of C_0: its dissemination contains $V_T/4$ distinct plaquettes of its orientation, each with $s_G \geqslant b_G\theta^2$ on the event, and (D.37) bounds $\mathcal{Z}_0$ below. Hence each choice has probability at most $\max(z_P, z_G)^{|\mathsf{S}'|}$, and the union bound over the choices gives $(n_W \max(z_P, z_G))^{|\mathsf{S}'|} = q^{|\mathsf{S}'|}$.

As in Lemma D.15, the product measure represents the integral before the history conditions; the factorisation is that of the physical measure after the reversal, and the correlation of distinct plaquettes under μ_0 is paid by the chessboard estimate.

For the second bound let S' be a maximal separated subset of S. Every bond of S has its initial site within σ_K of the initial site of a bond of S', and at most C_K bonds have their initial site within σ_K of a given site, so $|\mathsf{S}'| \geqslant |\mathsf{S}|/C_K$. When $q \leqslant 1$ this gives $q^{|\mathsf{S}'|} \leqslant q^{|\mathsf{S}|/C_K}$. When $q > 1$ the second bound holds because the left side is a probability, at most $1 \leqslant q^{|\mathsf{S}|/C_K}$. The count C_K is an upper bound on every torus, so the argument holds also on periods at most $2\sigma_K + 1$, where a separated subset has at most one bond. *q.e.d.*

9.4 The Exponential Moment of the Boundary Sum

Lemma 9.3 (Exponential Moment of the Boundary Sum)**.** *For $c > 0$ let $R_0' \coloneqq \lceil R_0 \rceil + 2L + 1$ and, for a bond b,*

$$\varphi_c(b) \coloneqq \sum_{r\in C} e^{-c(d_T(r,b)-R_0')_+} \leqslant \Phi_c \coloneqq 4(2R_0' + 1)(1 + 1/c) + 8/c^2\,, \tag{9.5}$$

and $N'_{\mathrm{lat}}(c) \coloneqq \sum_{s\geqslant 0} 4(2s+1)^4 e^{-c(s-R'_0)_+} < \infty$. *For* g_0 *below the thresholds of Lemmas D.13 and D.17, pointwise on every branch, and for every* $\lambda > 0$,

$$\begin{aligned} \sum_{j=0}^{4n-1} e^{-cd_j} &\leqslant \sum_b \varphi_c(b)\mathbf{1}\{b \in E(B_\xi)\}\ , \\ \mathcal{Z}_0^{-1} \int \mathcal{O}\Big[\mathsf{w}_\xi \mathbf{1}_{A_{\mathrm{cov}}} \exp\Big(\lambda \sum_{j=0}^{4n-1} e^{-cd_j}\Big)\Big] \mathrm{d}V &\leqslant \exp[n_C\, \lambda N'_{\mathrm{lat}}(c)\, e^{\lambda\Phi_c}\, q^{1/C_K}]\ , \end{aligned} \tag{9.6}$$

the sum running over all bonds b *of the torus. The second bound is uniform in the volume, and it holds also with* $\mathbf{1}_{A_{\mathrm{cov}}}$ *omitted.*

Proof. The bound in (9.5): the initial sites of the contour bonds lie on the four lines of the square, each meeting an ℓ^∞ ball of radius s in at most $2s+1$ sites, so at most $4(2s+1)$ contour bonds r have $d_T(r,b) \leqslant s$ ($d_T(r,b) \geqslant \|x_r - x_b\|_\infty$). Hence $\varphi_c(b) \leqslant 4(2R'_0+1) + \sum_{u\geqslant 1} 4(2(R'_0+u)+1)e^{-cu} \leqslant 4(2R'_0+1)(1+1/c) + 8/c^2$, using $\sum_{u\geqslant 1} e^{-cu} \leqslant 1/c$ and $\sum_{u\geqslant 1} ue^{-cu} \leqslant 1/c^2$.

The range of the stencils: let $\varrho_K \coloneqq \max_j \max_{k\in K_j} \min_{r\in C_j} d_T(k,r)$. The free-input stencil of a contour bond r is $\Lambda_r = \{r\}$ for a free bond, and $\Lambda_{b_0(c)} = \mathcal{S}_c \cap E_*$ for a designated bond, with $\mathcal{S}_c$ the positive fine bonds with both endpoints in the two endpoint blocks of c (Appendix C). For a free contour bond r, $d_T(r,r) = 0$. For a designated $r = b_0(c)$, the stencil bonds and r itself, a bond of the row c, have both endpoints in the two endpoint blocks of c. These are two adjacent blocks of side L, any two sites of which are within ℓ^∞ distance $2L-1$ of each other, so $d_T(k,r) \leqslant 2L-1+1 = 2L$. Hence $\varrho_K \leqslant 2L+1$ and $R_0 + \varrho_K \leqslant R'_0$.

The pointwise bound: $d_j = (d_T(K_j, I^c) - R_0)_+$ with $d_T(K_j, I^c) = \min_{b\notin I} d_T(K_j, b)$, and $x \mapsto e^{-c(x-R_0)_+}$ is nonincreasing, so $e^{-cd_j} \leqslant \sum_{b\notin I} e^{-c(d_T(K_j,b)-R_0)_+}$, both sides being zero if $I^c = \varnothing$. For $k \in K_j$ the triangle inequality gives $d_T(k,b) \geqslant \min_{r\in C_j} d_T(r,b) - \varrho_K$, so $e^{-c(d_T(K_j,b)-R_0)_+} \leqslant e^{-c(d_T(C_j,b)-R'_0)_+} \leqslant \sum_{r\in C_j} e^{-c(d_T(r,b)-R'_0)_+}$. The segments partition the contour bonds, so $\sum_j \sum_{r\in C_j} = \sum_{r\in C}$, and $\sum_j e^{-cd_j} \leqslant \sum_{b\notin I} \varphi_c(b)$. By (9.1) this is at most $\sum_b \varphi_c(b)\mathbf{1}\{b \in E(B_\xi)\}$, the added terms being nonnegative.

The moment: let $m_b \coloneqq \mathbf{1}\{b \in E(B_\xi)\}$. By the pointwise bound and $\mathbf{1}_{A_{\mathrm{cov}}} \leqslant 1$ the integrand is at most $\mathsf{w}_\xi \prod_b (1+(e^{\lambda\varphi_c(b)}-1)m_b)$, since $e^{\lambda wm} = 1+(e^{\lambda w}-1)m$ for $m \in \{0,1\}$; the product is finite. It expands into the sum over the sets S of bonds of $\prod_{b\in\mathsf{S}}(e^{\lambda\varphi_c(b)}-1)\prod_{b\in\mathsf{S}} m_b$. Every term is nonnegative, so the history average ($\mathsf{w}_\xi \geqslant 0$, $\mathcal{O}$ positive, Tonelli) is at most $\sum_{\mathsf{S}} \prod_{b\in\mathsf{S}}(e^{\lambda\varphi_c(b)}-1)\, q^{|\mathsf{S}|/C_K}$ by the second bound of (9.4). This sum equals $\prod_b(1+(e^{\lambda\varphi_c(b)}-1)\hat{q})$ with $\hat{q} \coloneqq q^{1/C_K}$, and $\prod_b(1+(e^{\lambda\varphi_c(b)}-1)\hat{q}) \leqslant \exp(\hat{q}\sum_b(e^{\lambda\varphi_c(b)}-1))$. Finally $e^x - 1 \leqslant xe^x$ for $x \geqslant 0$, $\varphi_c(b) \leqslant \Phi_c$, and $\sum_b \varphi_c(b) = \sum_{r\in C}\sum_b e^{-c(d_T(r,b)-R'_0)_+} \leqslant n_C N'_{\mathrm{lat}}(c)$, since at most $4(2s+1)^4$ bonds have initial site within ℓ^∞ distance s of x_r. The indicator $\mathbf{1}_{A_{\mathrm{cov}}}$ entered only through $\mathbf{1}_{A_{\mathrm{cov}}} \leqslant 1$. *q.e.d.*

9.5 The Moment of the Leading Error

Corollary 9.4 (The Moment of the Leading Error)**.** *Under the hypotheses of Proposition F.2 and Corollary 9.1, with the constants* C *and* c *of Proposition F.2 and* $N'_{\mathrm{lat}}(c)$, Φ_c *of Lemma 9.3, and for* $0 < g_0 < g_*$ *(Proposition D.18),*

$$\begin{aligned} \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{|E^\circ_\xi|}] &\leqslant \exp[C\tau^2(2+1/c) + n_C\delta^I_q]\ , \\ \delta^I_q \coloneqq C\tau^2 N'_{\mathrm{lat}}(c)\, e^{C\tau^2\Phi_c}\, q^{1/C_K} &\leqslant \exp[-c_*\pi_*^2/(2C_K)]\ , \end{aligned} \tag{9.7}$$

the last inequality for g_0 below a further threshold, depending on L, M, M_1, M_2, the group, the character, the source constants and c. Hence the leading term $\kappa_{C_n}\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}e^{E^\circ_\xi}W_\chi(C_n;U_1)]$ of (F.5) *has modulus at most $|\kappa_{C_n}|\exp[C\tau^2(2+1/c)+n_C\delta_q^I]$, and*

$$\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}|e^{E^\circ_\xi}-1|]\leqslant\exp[C\tau^2(2+1/c)+n_C\delta_q^I]-1\,. \tag{9.8}$$

Moreover, with $c'\coloneqq A_0^2 2^{-2p_0}/(4(4K+9)^4(L^2M)^4 2^{4\mathfrak{r}})$,

$$\pi_*^2/C_K\geqslant c'(\log g_0^{-2})^{2p_0-4\mathfrak{r}}\,,\qquad 2p_0-4\mathfrak{r}\geqslant 6\mathfrak{r}\geqslant 12\,, \tag{9.9}$$

so δ_q^I is smaller than every power of g_0.

Proof. On A_{cov}, (9.3) gives $e^{|E^\circ_\xi|}\leqslant e^{C\tau^2(2+1/c)}\exp(C\tau^2\sum_j e^{-cd_j})$, and the second bound of (9.6) at $\lambda=C\tau^2$ gives the first inequality of (9.7).

The leading term: $|W_\chi|\leqslant 1$ and $|e^{E^\circ_\xi}|\leqslant e^{|E^\circ_\xi|}$.

For (9.8), pointwise $\mathbf{1}_{A_{\mathrm{cov}}}|e^{E^\circ_\xi}-1|\leqslant\mathbf{1}_{A_{\mathrm{cov}}}(e^{|E^\circ_\xi|}-1)\leqslant e^{C\tau^2(2+1/c)}\exp(C\tau^2\sum_j e^{-cd_j})-1$, the last member being nonnegative; its history average is at most $\exp[C\tau^2(2+1/c)+n_C\delta_q^I]-1$ by (9.6) with the indicator omitted and $\mathbb{E}_{\mathrm{h}}[1]=1$ ($\mathcal{Z}_0=\int\mathcal{O}[\mathsf{w}_\xi]\,\mathrm{d}V$).

(9.9): $C_K=4(4K\ell+9)^4\leqslant 4(4K+9)^4\ell^4\leqslant 4(4K+9)^4(L^2M)^4 2^{4\mathfrak{r}}(\log g_0^{-2})^{4\mathfrak{r}}$ and $\pi_*^2\geqslant A_0^2 2^{-2p_0}(\log g_0^{-2})^{2p_0}$ by (9.2), and $2p_0-4\mathfrak{r}\geqslant 6\mathfrak{r}$ by $p_0\geqslant 5\mathfrak{r}$.

The smallness of δ_q^I: since $\tau<1$ (fixed after (5.18)) and $q\leqslant e^{-c_*\pi_*^2}$ for $g_0<g_*$, $\log\delta_q^I\leqslant\log C+\log N'_{\mathrm{lat}}(c)+C\Phi_c-c_*\pi_*^2/C_K$, where $N'_{\mathrm{lat}}(c)$ and Φ_c are constants of c, L and the fixed range R_0. By (9.9), $c_*\pi_*^2/(2C_K)\geqslant\log C+\log N'_{\mathrm{lat}}(c)+C\Phi_c$ for g_0 below a threshold depending on the listed quantities, and then $\delta_q^I\leqslant\exp[-c_*\pi_*^2/(2C_K)]\leqslant\exp[-(c_*c'/2)(\log g_0^{-2})^{2p_0-4\mathfrak{r}}]$, which is smaller than every power of g_0 since $2p_0-4\mathfrak{r}>1$. *q.e.d.*

10 Two Disjoint Squares

This section extends the extraction of Section 3.8 and Appendices E and F from one square to the product of the Wilson functions of two vertex-disjoint squares in planes parallel to that of e_1,e_2. The objects of this section are defined for the pair and compared with the single-square objects of each square.

10.1 Setting

Let C_1,C_2 be coarse translates of the counterclockwise squares with vertices $0,n_iLe_1,n_iL(e_1+e_2),n_iLe_2$, $i=1,2$, in fine-lattice units (so both lie in planes parallel to that of e_1,e_2); let $V(C_i)$ be the set of fine sites of C_i, the endpoints of its bonds; the squares are vertex-disjoint, $V(C_1)\cap V(C_2)=\varnothing$.

Let χ_1,χ_2 be normalised characters of irreducible unitary representations π_1,π_2 of G, $\chi_i=\dim(\pi_i)^{-1}\operatorname{tr}\pi_i$. The reversed orientation of C_i is the case $\bar\chi_i$, since $\chi_i(U^{-1})=\overline{\chi_i(U)}$. Let $C=C_1\sqcup C_2$ and

$$W_{\chi_1,\chi_2}(C;U)=\chi_1(\mathrm{Hol}_U C_1)\,\chi_2(\mathrm{Hol}_U C_2)\,. \tag{10.1}$$

The traversal set is $\mathcal{I}_C=\mathcal{I}_{C_1}\sqcup\mathcal{I}_{C_2}$, the traversals of each C_i numbered cyclically from the base point used for the single-square objects of C_i; $n_{C_i}=4n_iL$ and $n_C=n_{C_1}+n_{C_2}$. Each

square satisfies the square/torus geometry of Section 5.3 with its own side, $K_\mu \geqslant 3(n_i + 2)$ for every coarse period and $i = 1, 2$.

For $a \in \mathcal{I}_{C_i}$ let δ_a be the periodic ℓ^∞ fine distance from the initial site of the traversal a to $V(C_{3-i})$, and let

$$\Delta_\times := \sum_{a \in \mathcal{I}_C} e^{-c_\times \delta_a/(LM)} \leqslant n_C e^{-c_\times d/(LM)} , \qquad d := \min_{a \in \mathcal{I}_C} \delta_a , \tag{10.2}$$

with the constant $c_\times > 0$ of Lemma 10.7, fixed before the coupling.

A superscript or subscript i marks the single-square objects of the square C_i with the character χ_i: the contact sets $\mathsf{T}_D^i = \mathsf{T}_D \cap \mathcal{I}_{C_i}$, the groups of Definition 3.32 for the contour C_i alone, their reference scalars $a_W^{(i)}$ of (3.92), the reference gas S°_{ξ,C_i} of (3.94), and the single-square lemmas of Appendix E with $C = C_i$. All of them are formed on the same branch ξ with the same carrier, cubes, hosts, measures and cluster occurrences: the domains, probe measures $\widehat{\nu}^{\mathrm{f}}_{\xi,D}$, vacuum clusters and reference data of Sections 3.5–3.8 do not involve the observable, which enters only through the contact sets, f_ξ, f°_ξ and the reference scalars a_W.

The coupling is below the thresholds of the single-square results for χ_1 and for χ_2 and below those of this section.

Covered and far are taken for both squares: $\xi \in A_{\mathrm{cov}}$ means that ξ is covered (Lemma 3.36) for the contour $C = C_1 \sqcup C_2$, which, the conditions being imposed bond by bond, is the intersection of the covered events of C_1 and C_2; a history is far when $\operatorname{dist}(\Gamma_C, E(\Lambda_1^{\mathrm{c}})) > b$ with $\Gamma_C = \Gamma_{C_1} \cup \Gamma_{C_2}$, that is far for both squares.

10.2 Groups for Two Squares

For the pair, $f_\xi(a) = W_{\chi_1,\chi_2}(C; U_\xi(a))$, and the contact set T_D of (3.87) is taken over $n \in \mathcal{I}_C$. The argument preceding Definition 3.32 applies verbatim up to the formation of the components: unmarked domains contribute zero, $q_\xi(\mathcal{A})$ is expanded by (3.88), and the graph on the marked roots and the cluster occurrences of a term has components γ, each with a marked root and the contact set $\mathsf{T}_\gamma \subseteq \mathcal{I}_C$, the union of the contact sets of its roots.

For $i = 1, 2$ let $H^i_\gamma := \varnothing$ if $\mathsf{T}_\gamma \cap \mathcal{I}_{C_i} = \varnothing$; otherwise H^i_γ is the hull of $\mathsf{T}_\gamma \cap \mathcal{I}_{C_i}$ in the cyclic set $\mathcal{I}_{C_i}$, defined as in Section 3.8: the shortest cyclic interval containing it when that interval has fewer than $n_{C_i}/2$ positions, and $\mathcal{I}_{C_i}$ otherwise. Let $H_\gamma := H^1_\gamma \sqcup H^2_\gamma$.

Two components overlap when $H^i_\gamma \cap H^i_{\gamma'} \neq \varnothing$ for some i; the overlap classes are the connected components of the overlap graph on the components of the term. A set K of components is long on C_i when $|\bigcup_{\gamma \in K} H^i_\gamma| \geqslant n_{C_i}/2$.

Definition 10.1 (Groups for Two Squares)**.** A partition $\mathfrak{P}$ of the components of a term is admissible if (A1) every overlap class lies in one block, and (A2) a block that is long on C_i contains every component γ with $\mathsf{T}_\gamma \cap \mathcal{I}_{C_i} \neq \varnothing$. The groups of the term are the blocks of its finest admissible partition (Lemma 10.2(a)). A group W consists of its root family $\mathcal{A}(W)$, its ordered list of cluster occurrences, and its interval set $\mathcal{H}(W) := \mathcal{H}_1(W) \sqcup \mathcal{H}_2(W)$, where $\mathcal{H}_i(W) := \mathcal{I}_{C_i}$ if W is long on C_i, read from the base point of C_i, and $\mathcal{H}_i(W) = \bigcup_{\gamma \in W} H^i_\gamma$ otherwise, a finite union of cyclic intervals (a cyclic interval or $\mathcal{I}_{C_i}$ for a C_i-group, by Lemma 10.2(d)); as in Definition 3.32, these data arise from a term, and conversely the term with root family $\mathcal{A}(W)$ and cluster list of W has W as its only group, with the same interval set (Lemma 10.2(a)). Its weight $w_\xi(W)$ is (3.89).

W is a C_i-group if $\mathcal{H}_{3-i}(W) = \varnothing$ and a cross group if $\mathcal{H}_1(W) \neq \varnothing \neq \mathcal{H}_2(W)$. Every group has $\mathcal{H}(W) \neq \varnothing$ (each component has a marked root), so it is exactly one of a C_1-group, a C_2-group and a cross group.

Two groups are compatible when all their constituent domains are mutually compatible and $\mathcal{H}_i(W) \cap \mathcal{H}_i(W') = \varnothing$ for $i = 1, 2$. A compatible group family is a finite set of pairwise compatible groups; the singleton clause of Definition 3.32 is not imposed (for families of C_i-groups it is implied, Lemma 10.2(c)). For a compatible group family, $\mathcal{A}(\mathcal{S})$ and $A_\xi(\mathcal{S})$ are (3.90).

Lemma 10.2 (Two-Square Grouping)**.** *(a) Every term has a finest admissible partition; each of its blocks that is long on neither square is an overlap class. If no component of the term has a contact on C_{3-i}, it is the grouping of Section 3.8 for the square C_i. For a group W of a term, the term T_W with root family $\mathcal{A}(W)$ and the cluster list of W has W as its only group, with the same interval set.*

(b) The groups of a term are pairwise compatible, and Lemma 3.33 holds for the pair: $F_\xi(0) = \sum_{\mathfrak{B}\ \text{compatible}} A_\xi(\mathfrak{B})$, absolutely convergent.

(c) A C_i-group of the pair is a group of the single square C_i (Definition 3.32 with $C = C_i$) none of whose roots has a contact on C_{3-i}; conversely every such single-square group is a C_i-group of the pair, with the same interval, weight and footprint (the footprint of a group of the pair is defined in Section 10.5). Two C_i-groups are compatible for the pair if and only if they are compatible for the square C_i; a C_1-group and a C_2-group are compatible if and only if their constituent domains are mutually compatible.

(d) If W is a C_i-group that is not long on C_i, then $\mathcal{H}_i(W)$ is a cyclic interval of fewer than $n_{C_i}/2$ traversals.

Proof. (a) The one-block partition is admissible. If $\mathfrak{P}$ and $\mathfrak{Q}$ are admissible, so is their common refinement, whose blocks are the nonempty sets $P \cap Q$: (A1) holds because every overlap class lies in one block of each; for (A2), if $P \cap Q$ is long on C_i, then so are $P \supseteq P \cap Q$ and $Q \supseteq P \cap Q$ (the union of hulls is monotone), so both contain every component with a contact on C_i, and so does $P \cap Q$. There are finitely many components, so the common refinement of all admissible partitions is admissible and is the finest.

If a block P of the finest partition is long on neither square and contains two overlap classes, the partition with P replaced by its overlap classes satisfies (A1), its new blocks are long on neither square (they are subsets of P), and the other blocks are unchanged, so it is admissible and strictly finer, a contradiction.

If no component has a contact on C_{3-i}, condition (A2) on C_{3-i} is void; if no overlap class is long the overlap classes form an admissible partition, hence the finest, and these are the classes of Section 3.8 for C_i with the intervals $\mathcal{H}(K)$ there, which equal the unions of the members' hulls; if some overlap class is long, every admissible partition has a long block containing it, which by (A2) contains every component (each has a contact on C_i), so the finest partition is the single class of all components, the whole-contour group of Section 3.8.

The term of a group: let $\mathfrak{P}$ be the finest admissible partition of a term T and $W \in \mathfrak{P}$. Every cluster occurrence of W hits a root of T, which is joined to it by an edge and so lies in its component, hence in W; so T_W is a term of (3.88). Edges are defined pairwise, so the graph of T_W is the subgraph of that of T induced on W, and W being a union of components of T, the components of T_W are those of T in W, with the same contact sets, hulls and long-ness; by (A1) W is a union of overlap classes of T, which are the overlap classes of T_W.

Let $\mathfrak{P}'$ be an admissible partition of T_W and $\mathfrak{P}'' = \mathfrak{P}' \cup (\mathfrak{P} \setminus \{W\})$. (A1) holds for $\mathfrak{P}''$, and (A2) for its blocks outside W; if $Q \in \mathfrak{P}'$ is long on C_i, then $W \supseteq Q$ is long on C_i, so W contains every component of T with a contact on C_i by (A2) for $\mathfrak{P}$, and all of them lie in Q by (A2) for $\mathfrak{P}'$. So $\mathfrak{P}''$ is admissible for T, $\mathfrak{P}$ refines it, and $\mathfrak{P}' = \{W\}$. Long-ness and hulls being the same in T and T_W, the interval set is the same.

(b) Distinct groups of a term have disjoint intervals: if neither is long on C_i, their C_i-parts are unions of hulls of components in different blocks, hence of different overlap classes by (A1), and such hulls are disjoint; if one is long on C_i, it contains every component with a contact on C_i, so the other has $\mathcal{H}_i = \varnothing$. Domains in different groups are compatible as in the proof of Lemma 3.33 (no edge of the term's graph joins different components). Every component's hull satisfies $H^i_\gamma \subseteq \mathcal{H}_i(W)$ for its group W: trivially if W is long on C_i, and by definition otherwise.

Reversibility: let $\mathfrak{B} \coloneqq \{W_1, \dots, W_m\}$ be a compatible family, each W_j being the only group of its own term by (a), and let $\mathfrak{P}$ be the finest admissible partition of the union term, whose components are those of the W_j (compatibility excludes edges between groups). $\mathfrak{B}$ is admissible for the union term: components of different W_j have hulls in disjoint intervals, so they do not overlap and (A1) holds; if W_j is long on C_i, then $\mathcal{H}_i(W_j) = \mathcal{I}_{C_i}$, so every other W_k has $\mathcal{H}_i(W_k) = \varnothing$, no contact on C_i, and W_j contains every component with a contact on C_i. Hence $\mathfrak{P}$ refines $\mathfrak{B}$. The restriction of $\mathfrak{P}$ to the components of W_j is admissible for the term of W_j alone: its overlap classes are those of the union term inside W_j, and a block of $\mathfrak{P}$ inside W_j that is long on C_i contains every component of the union term with a contact on C_i, in particular those of W_j. Since $\{W_j\}$ is the finest admissible partition of that term, $\mathfrak{P}$ restricted to W_j is the single block W_j, so $\mathfrak{P} = \mathfrak{B}$, with the same intervals.

The rest of the proof of Lemma 3.33 (interleaving count, multiplicities of identical occurrences, absolute convergence) does not involve the intervals and applies verbatim.

(c) Let W be a C_i-group of the pair. Its roots are marked and $\mathcal{H}_{3-i}(W) = \varnothing$, so every root D has $\mathsf{T}_D = \mathsf{T}^i_D \neq \varnothing$, and the components, their hulls on C_i, the overlap graph and the long-ness on C_i of the term T_W of W are the same for the pair and for the square C_i. So T_W is a term of (3.88) for the square C_i (its roots are marked for C_i, and its clusters hit $\mathcal{A}(W)$ by (a)), no component of which has a contact on C_{3-i}; by (a) its finest admissible partition $\{W\}$ is its grouping of Section 3.8 for C_i, with $\mathcal{H}(W) = \mathcal{H}_i(W)$.

Conversely, if W is a group of C_i whose roots have no contact on C_{3-i}, its term (Definition 3.32) is a term of the pair, with $\mathsf{T}_D = \mathsf{T}^i_D \neq \varnothing$ for its roots, and by (a) its pair grouping is its grouping for C_i, the single group W, with $\mathcal{H}_i(W) = \mathcal{H}(W)$ and $\mathcal{H}_{3-i}(W) = \varnothing$. The weight (3.89) and the footprint of Section E.5 depend on the roots, the cluster list and the interval only. Compatibility for the pair of two C_i-groups is compatibility of domains and disjointness of the C_i-intervals, which is Definition 3.32 for C_i (its singleton clause for a whole-contour group being implied by the disjointness, since every group has a nonempty interval); for a C_1-group and a C_2-group the interval condition is void.

(d) W is not long on C_{3-i}, since $\mathcal{H}_{3-i}(W) = \varnothing$, so by (a) it is an overlap class; its members have hulls on C_i only, so they are chained by intersections of these hulls, and $\mathcal{H}_i(W)$ is a union of chained cyclic intervals with fewer than $n_{C_i}/2$ positions in total, which is a cyclic interval by the argument of Section 3.8. *q.e.d.*

10.3 Reference Scalars and the Joint Extraction

For a C_i-group W let $P^\circ_W(a)$ be the ordered product of the reference links along the consecutive traversals of $\mathcal{H}_i(W)$, in the cyclic order of C_i starting from the first traversal of the interval (from the base point of C_i if W is long on C_i), as in Section 3.8, and define a_W by (3.92) with χ_i in place of χ; $\mathcal{H}_i(W)$ is a cyclic interval or $\mathcal{I}_{C_i}$ by Lemma 10.2(d), and $a_W = a^{(i)}_W$ by Lemma 10.2(c), since the reference links and the measures $\widehat{\nu}^{\mathrm{f},\circ}_{\xi,D}$ do not involve the observable (Lemma 3.29). For a cross group $a_W \coloneqq 0$. With these numbers let, for the pair,

$$S^\circ_{\xi,C} = \sum_{\mathfrak{B}\ \text{compatible}} \prod_{W\in\mathfrak{B}} a_W \ , \tag{10.3}$$

in which a family containing a cross group contributes zero; the restricted sums $S^{\circ,[\mathfrak{B}]}_{\xi,C}$ run over the families compatible with every member of $\mathfrak{B}$, and the joint defect $R_\xi(\mathfrak{B})$ is (3.95).

Lemma 10.3 (Schur Identity and Joint Extraction for Two Squares)**.** *(a) For a C_i-group W and $h \in G$,* (3.93) *holds with χ_i in place of χ.*

(b) Proposition 3.35 holds for the pair: (3.96) *and* (3.97) *hold with $S^\circ_{\xi,C}$ of* (10.3)*, the sums converging absolutely.*

(c) At the reference data, $R^\circ_\xi(\mathfrak{B}) = 0$ for every nonempty compatible family $\mathfrak{B}$ without cross groups.

Proof. (a) The proof of Lemma 3.34 for the square C_i: it uses only the invariance of the reference measures under the constant gauge rotations R_u, the covariance $P^\circ_W(R_u a) = uP^\circ_W(a)u^{-1}$ and Schur's lemma for π_i.

(b) The proof of Proposition 3.35 is the Boolean inversion of (3.95) and an absolutely convergent rearrangement; it uses only $|a_W| \leqslant \beta_W$, which holds for $a_W = 0$, the finiteness of $\sum_W \beta_W$ (a group is determined by its root family, cluster list and interval, Lemma 3.28 and Lemma B.16) and Lemma 10.2(b) in place of Lemma 3.33.

(c) Let $\mathcal{S}$ be a compatible family of C_1- and C_2-groups. The contacts on C_i of the roots of a C_i-group W lie in $\mathcal{H}_i(W)$ (Lemma 10.2(b)), and the roots of a C_{3-i}-group have no contact on C_i. So, as in the proof of Lemma 3.34, a traversal outside every interval of $\mathcal{S}$ has the reference link $\mathbf{1}$, and a traversal in $\mathcal{H}_i(W)$ has at $\mathsf{e}_{\mathcal{A}(\mathcal{S})}a$ its link at $\mathsf{e}_{\mathcal{A}(W)}a$, since it lies in no contact set of a root of another group; the links of $\mathcal{H}_i(W)$ multiply to $P^\circ_W(a)$.

Number the C_i-groups of $\mathcal{S}$ in the cyclic order of C_i; by cyclicity of the trace the reference Wilson function at $\mathsf{e}_{\mathcal{A}(\mathcal{S})}a$ is $\prod_{i=1,2}\chi_i(\prod_{W\in\mathcal{S},\ C_i\text{-group}} P^\circ_W(a))$, the product over the C_i-groups in that order (an empty product being $\mathbf{1}$), and a family containing a group long on C_i has no other C_i-group. Integrate one group at a time by Fubini, with (a) for the group integrated, the product of the other P° of the same square as h, and the character of the other square as a constant factor: $A^\circ_\xi(\mathcal{S}) = \prod_{W\in\mathcal{S}} a_W$. Hence $R^\circ_\xi(\mathfrak{B}) = \prod_{W\in\mathfrak{B}} a_W \sum_{\mathcal{S}\subseteq\mathfrak{B}}(-1)^{|\mathfrak{B}\setminus\mathcal{S}|} = 0$ for $\mathfrak{B} \neq \varnothing$. *q.e.d.*

For a family containing a cross group, $R^\circ_\xi(\mathfrak{B})$ need not vanish (for instance $R^\circ_\xi(\{W\}) = A^\circ_\xi(\{W\})$ for a cross group W); the defects of such families are bounded directly (Lemmas 10.5 and 10.7). In place of the second equality of (3.94), Lemma 10.2(b) at the reference data (its proof uses only the field-independent geometry, as in Section 3.8) and the Fubini step of (c) give

$$\frac{\int f^\circ_\xi K^\circ_{\xi;0}\,\mathrm{d}a}{I^\circ_{\xi;0}} = S^\circ_{\xi,C} + \sum_{\mathfrak{B}\ \text{compatible, with a cross group}} A^\circ_\xi(\mathfrak{B}) \ .$$

The pair scalar is identified with the single-square scalars by Lemma 10.7(b).

10.4 The Defects of a Family with Cross Groups

For a C_i-group W the closed contraction $c_W(U_1; h)$ of (E.19), its defect d_W and $\mathfrak{d}_W(U_1)$ of (E.20) are formed with χ_i and the interval $\mathcal{H}_i(W)$; by Lemma 10.2(c) they are the single-square quantities of C_i.

For every group, cross groups included, $M(W)$ is the positive majorant of Section E.5.2: the product of the root weights $\rho\, m(D)$, the cluster weights $M_v(\mathfrak{c}_j)$ and $1/k(W)!$, with ρ of Lemma 10.4; for C_i-groups this only enlarges the single-square majorant, so every single-square bound by $M(W)$ still holds. Let $K_\chi := \max(K_{\chi_1}, K_{\chi_2})$ with the constants of Lemma E.7(a).

Lemma 10.4 (Sources with Two Closed Words)**.** *Let $f(a) := \prod_{i=1,2} \dim(\pi_i)^{-1} \operatorname{tr} Y_i(a)$, where $Y_i(a)$ is an ordered product of the matrices $\pi_i(U_\xi(a)(r))^{\pm 1}$ of a set of reconstructed contour links of C_i, each traversal used at most once, and of fixed unitary matrices of V_{π_i}. Then* (E.6)*,* (E.65)*–*(E.66)*, the undifferentiated bound* (E.36) *and the last statement of Lemma E.15, and Lemma E.3 hold for f, with C_χ enlarged once to a constant $C^{\mathrm{src}}_{\chi_1\chi_2}$ of (χ_1, χ_2), so that $u = C^{\mathrm{src}}_{\chi_1\chi_2} g_0(1 + r)$, and g_0 so small that $u \leqslant u_*$ and $v \leqslant 1$; in particular $|\mathfrak{Q}_\xi(\mathcal{A}; f)| \leqslant \prod_{D\in\mathcal{A}} \rho\, m(D)$ for every nonempty compatible root family $\mathcal{A}$, with ρ of* (E.11) *formed with this u.*

Proof. f is the normalised trace of $Y_1(a) \otimes Y_2(a)$ on $V_{\pi_1} \otimes V_{\pi_2}$, and $|f| \leqslant 1$; since $\operatorname{tr}(A \otimes B) = \operatorname{tr} A \operatorname{tr} B$, the normalised trace on $V_{\pi_1} \otimes V_{\pi_2}$ satisfies $|\dim^{-1} \operatorname{tr}(M_1 \cdots M_k)| \leqslant \prod_j \|M_j\|_{\mathrm{op}}$.

The proof of (E.65) uses only this inequality, the unitarity of the untouched factors, the row locality and the fixed derivative constants of the rows. For the row locality: $V(C_1) \cap V(C_2) = \varnothing$, so the bond sets of C_1 and C_2 are disjoint, each bond carries at most one traversal of $C_1 \cup C_2$, and the number of reconstructed links a free label affects is the geometric count of bonds r with the label in $\mathcal{F}_r$, at most K as for one square, each traversal being used at most once in the two words together; the row input count, which fixes the derivative order of (E.57), is unchanged. Then the product rule distributes the derivatives over the links of both words, with at most K^q row assignments, and every untouched factor, a reconstructed link or a fixed matrix, is unitary in $V_{\pi_1} \otimes V_{\pi_2}$. The remainder bounds (E.66) follow by the same Taylor argument.

The proofs of (E.36), of the last statement of Lemma E.15 and of Lemma E.3 use, besides these bounds, only properties of the selected measures (the reversal of a root with empty ordinary selection, the Cauchy difference in ζ_j, the majorants M_ℓ and (E.10)) and the vanishing of $\Delta_\jmath f$ when a selected label does not influence the words; none of them uses the form of f otherwise. *q.e.d.*

Lemma 10.5 (Defect Factorisation for Two Squares)**.** *On a branch covered for $C_1 \sqcup C_2$ (Section 10.1), for every compatible family $\mathfrak{B}$ with cross groups $\mathfrak{B}_\times \subseteq \mathfrak{B}$,*

$$|R_\xi(\mathfrak{B})| \leqslant \prod_{W\in\mathfrak{B}\setminus\mathfrak{B}_\times} K_\chi \mathfrak{d}_W(U_1) \prod_{W\in\mathfrak{B}_\times} M(W)\,. \tag{10.4}$$

Proof. As in the proof of Lemma E.7(b), a traversal of $\mathcal{H}(W)$, $W \in \mathfrak{B}$, has no contact with a root of another group of $\mathfrak{B}$ (the intervals are disjoint and contain the contacts), so its reconstructed link depends only on the root variables of W, and a traversal outside every interval has its background link U_1 (Lemma 3.36, Section E.7.1).

For a C_i-group $W \in \mathfrak{B}$ let $X_W = w_\xi(W) \int \pi_i(T_W P_W^{-1})\, \mathrm{d} \bigotimes_{D \in \mathcal{A}(W)} \widehat{\nu}^{\mathrm{f}}_{\xi,D}$ and $\Upsilon_W = X_W - a_W \mathbf{1}$, with T_W, P_W of Section E.5.1.

Since $a_W = 0$ for cross groups, the term of $\mathcal{S}$ in (3.95) vanishes unless $\mathcal{S} \supseteq \mathfrak{B}_\times$. Let $a_\times$ be the root variables of the cross groups, held fixed. For $i = 1, 2$ let Y_i be the ordered product, in the cyclic order of C_i starting at the first traversal of the interval of a C_i-group of $\mathfrak{B}$ if there is one and at the base point of C_i otherwise (the trace is cyclic; as in the proof of Lemma E.7(b)), of the following factors: for a C_i-group $W \in \mathfrak{B}$, the matrix $\Upsilon_W \pi_i(P_W)$ at the position of $\mathcal{H}_i(W)$; for a traversal in $\mathcal{H}_i(W)$ of a cross group W, its reconstructed link at $a_\times$; for every other traversal, its background link. Expanding each $\Upsilon_W = X_W - a_W \mathbf{1}$ and using Fubini over the product measures of the C_1- and C_2-groups, exactly as in the proof of Lemma E.7(b), the term of $\mathcal{S} \supseteq \mathfrak{B}_\times$ is reproduced, because f_ξ at $\mathsf{e}_{\mathcal{A}(\mathcal{S})} a$ is the product of the two normalised traces and each X_W enters exactly one of them; hence

$$R_\xi(\mathfrak{B}) = \Big(\prod_{W \in \mathfrak{B}_\times} w_\xi(W) \Big) \int \prod_{i=1,2} \frac{\operatorname{tr} Y_i(a_\times)}{\dim \pi_i}\, \mathrm{d} \bigotimes_{W \in \mathfrak{B}_\times} \bigotimes_{D \in \mathcal{A}(W)} \widehat{\nu}^{\mathrm{f}}_{\xi,D}(a_\times)\,.$$

The integrand is multilinear in the matrices Υ_W, $W \in \mathfrak{B} \setminus \mathfrak{B}_\times$. With $\Upsilon_W = \|\Upsilon_W\|_{\mathrm{op}} T'_W$ and $\|T'_W\|_{\mathrm{op}} \leqslant 1$, each contraction $T' = V|T'|$ (polar decomposition, V unitary) is $T' = \frac{1}{2}(Ve^{iA} + Ve^{-iA})$ with $A = \arccos |T'|$, an average of two unitary matrices. The integrand is then $\prod_W \|\Upsilon_W\|_{\mathrm{op}}$ times an average of $2^{|\mathfrak{B} \setminus \mathfrak{B}_\times|}$ functions of the form of Lemma 10.4, depending on $a_\times$ only through the links of the cross groups' traversals. If $\mathfrak{B}_\times \neq \varnothing$, that lemma applied to the compatible root family $\mathcal{A}(\mathfrak{B}_\times)$ bounds each such integral by $\prod_{D \in \mathcal{A}(\mathfrak{B}_\times)} \rho\, m(D)$ in modulus; if $\mathfrak{B}_\times = \varnothing$, there is no integral, and each of the unitary words has normalised traces of modulus at most one (for $\mathfrak{B} = \varnothing$ the bound is $|f_\xi(0)| \leqslant 1$).

Further $|w_\xi(W)| \leqslant \prod_j M_v(\mathfrak{c}_j)/k(W)!$: by (3.73) the cluster weight is $\varphi^T/m!$ times the product of the activities, $|\varphi^T|$ is at most the number of spanning trees of the incompatibility graph (proof of Lemma 3.28, Section B.6), and the activities are bounded by $m(D)$ by (3.72) of Lemma 3.27. Together with $\|\Upsilon_W\|_{\mathrm{op}} \leqslant K_{\chi_i} \mathfrak{d}_W(U_1) \leqslant K_\chi \mathfrak{d}_W(U_1)$ for a C_i-group (Lemma E.7(a) for the square C_i) this is (10.4); the bound holds with K_{χ_i} on the C_i-groups. *q.e.d.*

10.5 Pins for the Groups of Two Squares

For a group W of the pair, $n(W)$, $\mathfrak{s}(W) = n(W) + |\mathcal{H}(W)|/(LM)$, the footprint $F(W)$, $X(W)$, $\pi_1(\mathcal{H}(W))$ and the envelope $\mathfrak{E}(W)$ are defined as in Section E.5 and Lemma E.8, with $\mathcal{H}(W) = \mathcal{H}_1(W) \sqcup \mathcal{H}_2(W)$ and the traversals of both squares; for components, t_γ, n_γ, X_γ, $F(\gamma)$ and $\mathfrak{s}(\gamma) = n_\gamma + |H_\gamma|/(LM)$ are those of Section E.7.2 with $H_\gamma = H^1_\gamma \sqcup H^2_\gamma$, and $\sigma_K = \sum_{\gamma \in K} \mathfrak{s}(\gamma)$. Let $A'_E := 17(2R_{\mathrm{e}} + 1)^4$.

Lemma 10.6 (Pins for Two Squares)**.** *Under the hypotheses of Lemma E.9, with its constant C enlarged by a fixed factor:*

(a) Lemma E.8 holds for every group W of the pair with $16n(W) + |\mathcal{H}(W)|/(LM)$ and $17\mathfrak{s}(W)$ in place of the bounds $8 + |\mathcal{H}(W)|/(LM)$ and $9\mathfrak{s}(W)$ of its part (a), A'_E in place of A_F and A_E, and $\operatorname{dist}(\Delta, C)$ the distance to $V(C_1) \cup V(C_2)$; if W is a cross group, $\pi_1(\mathcal{H}(W)) \subseteq \mathfrak{E}(W)$ contains a cube containing a site of $V(C_1)$ and a cube containing a site of $V(C_2)$.

(b) Lemma E.21 holds for the components of the pair, with $|H^i_\gamma| \leqslant A_{\mathrm{h}} LM n_\gamma$ for $i = 1, 2$, the hull pin (E.44) *for $a \in \mathcal{I}_{C_i}$ taken with the contour distance d_{C_i} of C_i, and the sum over $a' \in \mathcal{I}_C$ in* (E.45)*.*

(c) *Lemma E.22(a) holds for the overlap classes of the pair. A class long on C_i satisfies $\sigma_K \geqslant n_{C_i}/(2LM)$ and $\sum_{K \text{ long on } C_i} t_K e^{34\eta\sigma_K} \leqslant C\rho n_{C_i} e^{-15\eta n_{C_i}/(LM)}$. The groups long on some square have total weighted mass $\sum_{W \text{ long}} M(W) e^{4\eta\mathfrak{s}(W)} \leqslant 2C\rho(n_{C_1} e^{-10\eta n_{C_1}/(LM)} + n_{C_2} e^{-10\eta n_{C_2}/(LM)}) \leqslant C'\rho$ with $C' \coloneqq 2CLM/(5e\eta)$, uniformly in n_1, n_2.*

(d) *The pins* (E.25) *hold for the groups of the pair, with $\mathcal{I}_C$ the traversals of both squares; for every group W of the pair, cross groups included, $\sum_{W' \not\sim W} |a_{W'}| e^{3\eta\mathfrak{s}(W')} \leqslant \frac{1}{8}\eta\mathfrak{s}(W)$, the sum over the groups of the pair; and Lemma E.10 and Lemma E.24 hold for the reference gas* (10.3)*, with $C\rho n_C$ in* (E.53)*.*

Proof. (a) Count. On each square, $\mathcal{H}_i(W)$ is empty, or $\mathcal{I}_{C_i}$, or, if W is not long on C_i, a nonempty union of member hulls with fewer than $n_{C_i}/2$ positions in total; in the second case each member hull is a cyclic interval (none is the whole of $\mathcal{I}_{C_i}$), each maximal cyclic interval of the union has fewer than $n_{C_i}/2$ positions because the union does, and there are at most $|W|$ of them, where $|W|$, the number of components of W, is at most $n(W) = \sum_\gamma n_\gamma$ since $n_\gamma \geqslant 1$. As in the proof of Lemma E.8(a), a cyclic interval of fewer than $n_{C_i}/2$ traversals meets each side in one run and lies in at most $8 + t/(LM)$ cubes for t traversals, and so does $\mathcal{I}_{C_i}$; summing over at most $2n(W)$ intervals, $|\pi_1(\mathcal{H}(W))| \leqslant 16n(W) + |\mathcal{H}(W)|/(LM)$, so $|X(W) \cup \pi_1(\mathcal{H}(W))| \leqslant 17\mathfrak{s}(W)$ and $|F(W)| \leqslant |\mathfrak{E}(W)| \leqslant A'_E\mathfrak{s}(W)$.

Connectivity. For a component γ, X_γ is connected (proof of Lemma E.8(b)); $\pi_1(H^i_\gamma)$ is connected, H^i_γ being a cyclic interval or $\mathcal{I}_{C_i}$, and it contains the cube of the initial vertex of a contact of γ on C_i, which is at index distance at most one from a cube of a root of γ; so $X_\gamma \cup \pi_1(H_\gamma)$ is connected. Two overlapping components share a traversal, hence the cube of its initial vertex. So $X(W) \cup \pi_1(\mathcal{H}(W))$ is connected if W is an overlap class, which by Lemma 10.2(a) covers every group long on neither square.

Let W be long on C_i. Each overlap class in W has a component with a contact on C_i, or W is also long on C_{3-i} and the class has a contact on C_{3-i}: otherwise splitting this class off W gives an admissible partition (the remainder is nonempty, since it contains the components of W with a contact on C_i; it keeps every component with a contact on a square on which W is long and keeps the same union of hulls there, the class is long on no square, being a subset of W without contacts on those squares and with the union of its hulls on the other square inside that of W), contradicting the finest choice. If W is long on C_i only, every class in W meets the connected set $\pi_1(\mathcal{I}_{C_i})$, so the union is connected. If W is long on both squares and no class in W has contacts on both, then splitting W into its classes with contacts on C_1 and those with contacts on C_2 is admissible (each part contains every component with a contact on the square on which it is long, and has no contact on the other square), a contradiction; so some class of W meets both $\pi_1(\mathcal{I}_{C_1})$ and $\pi_1(\mathcal{I}_{C_2})$, and the union is connected. Hence $\mathfrak{E}(W)$ is connected. The statements (c) and (d) of Lemma E.8 then follow by its proof. For a cross group both $\mathcal{H}_1(W)$ and $\mathcal{H}_2(W)$ are nonempty, and $\pi_1(\mathcal{H}(W))$ contains the cubes of their vertices.

(b) The proof of Lemma E.21. Part (a) there does not involve hulls. In (b) the contacts of γ on C_i lie within $2L$ of X_γ, and the square geometry of C_i (Section 5.3) gives $|H^i_\gamma| \leqslant A_{\mathrm{h}} LMn_\gamma$ and $d_{C_i}(a, a') \leqslant A_{\mathrm{h}} LMn_\gamma$ for $a \in H^i_\gamma$ and a contact a' of γ on C_i; the contour sum is over $\mathcal{I}_{C_i}$. In (E.45) the sum over the traversals of each square is bounded as there, and there are two squares. For part (c) of Lemma E.21: for the pair $\mathfrak{s}(\gamma) \leqslant (1 + 2A_{\mathrm{h}})n_\gamma \leqslant 2A_{\mathrm{s}} n_\gamma$, so $100\eta\mathfrak{s}(\gamma) \leqslant 200\eta A_{\mathrm{s}} n_\gamma < \varpi n_\gamma/40$ by $1000\eta A_{\mathrm{s}} < \varpi/8$, and the pins with the weight $e^{100\eta\mathfrak{s}(\gamma)}$ are dominated by those of (b) with the weight $e^{\varpi n_\gamma/4}$, with a margin $e^{\varpi n_\gamma/8}$; η is not changed.

In the footprint pin, at most CLM traversals of each square have stencils meeting a given cube, so at most $2CLM$ in all, and the footprint pin holds with the constant doubled.

(c) The proof of Lemma E.22(a) applies with $H_\gamma = H^1_\gamma \sqcup H^2_\gamma$: the components whose hulls meet H_γ have weighted mass at most $C\rho|H_\gamma|$ by the hull pins of (b) at the traversals of H_γ, and an overlap class long on neither square has the interval $\bigcup_\gamma H_\gamma$, of size at most $\sum_\gamma |H_\gamma| \leqslant LM\sigma_K$. A class long on C_i has $\sigma_K \geqslant \sum_\gamma |H^i_\gamma|/(LM) \geqslant n_{C_i}/(2LM)$; anchoring it at a traversal of $\bigcup_\gamma H^i_\gamma$ (n_{C_i} choices) and using the class pin with the weight $e^{64\eta\sigma_K}$ gives the stated sum, the exponent $30\eta\sigma_K$ being at least $15\eta n_{C_i}/(LM)$.

Long groups: if no overlap class of a term is long, the overlap classes are admissible and no group is long. Let W be long; it contains an overlap class K_0 long on some square C_{i_0}: otherwise replacing W by its overlap classes keeps (A1), the new blocks are long on no square, and the other blocks are unchanged, so the new partition is admissible, and it is strictly finer unless W is a single class, which is then not long; the first contradicts the finest choice, the second the assumption that W is long. W is long on C_{i_0}, and by the splitting argument of (a) it is the union of the overlap classes with a contact on a square on which it is long. As in Lemma E.22(c), the sum of $M(W)$ over the groups with a given set of components is at most $\prod_\gamma t_\gamma$, and $\mathfrak{s}(W) \leqslant \sum_{\gamma\in W} \mathfrak{s}(\gamma) + \sum_{i:\, W \text{ long on } C_i} n_{C_i}/(LM)$.

If W is long on C_{i_0} only, the classes other than K_0 are anchored at traversals of C_{i_0} and sum to at most $e^{C\rho n_{C_{i_0}}}$ with their weights $e^{64\eta\sigma}$, the interval weight is $e^{4\eta n_{C_{i_0}}/(LM)}$, and K_0 contributes $C\rho n_{C_{i_0}} e^{-15\eta n_{C_{i_0}}/(LM)}$; with $C\rho LM \leqslant \eta$ the product is at most $C\rho n_{C_{i_0}} e^{-10\eta n_{C_{i_0}}/(LM)}$. If W is long on both squares, let $j \neq i_0$; since $\sum_{\gamma\in W} |H^j_\gamma| \geqslant n_{C_j}/2$ and $|H^j_\gamma| \leqslant LM\mathfrak{s}(\gamma)$, $1 \leqslant e^{-15\eta n_{C_j}/(LM)} \prod_{\gamma\in W} e^{30\eta\mathfrak{s}(\gamma)}$; inserting this factor, the classes other than K_0 are anchored at traversals of C_1 or C_2 and sum to at most $e^{C\rho n_C}$ with the weights $e^{64\eta\sigma}$, the interval weight is $e^{4\eta n_C/(LM)}$, and K_0 contributes $C\rho n_{C_{i_0}} e^{-15\eta n_{C_{i_0}}/(LM)}$ with the weight $e^{34\eta\sigma_{K_0}}$; the product is at most $C\rho n_{C_{i_0}} e^{-10\eta n_C/(LM)} \leqslant C\rho n_{C_{i_0}} e^{-10\eta n_{C_{i_0}}/(LM)}$.

The groups long on C_i only thus contribute at most $C\rho n_{C_i} e^{-10\eta n_{C_i}/(LM)}$, and the groups long on both squares at most that for each $i_0 = i$; summing over the four pairs (set of long squares, i_0) gives the bound, and $xe^{-10\eta x/(LM)} \leqslant LM/(10e\eta)$ for $x = n_{C_1}, n_{C_2}$ gives $C' = 2CLM/(5e\eta)$.

(d) The pins over groups long on neither square are the class pins of (c), as in Lemma E.22: such a group is an overlap class K (Lemma 10.2(a)), the sum of $M(W)$ over the groups with class K is at most t_K, $\mathfrak{s}(W) \leqslant \sigma_K$ and $F(W) \subseteq \bigcup_{\gamma\in K} F(\gamma)$; the long groups are bounded by their total mass (c), so (E.25) holds with the constant $C + C'$.

For the second statement, a group $W' \not\sim W$ has a footprint cube in $\mathfrak{N}_1(F(W))$ or a traversal in $\mathcal{H}(W)$ by (a) and Lemma E.8(d), so the pins give at most $C\rho(3^4|F(W)| + |\mathcal{H}(W)|) \leqslant C\rho(3^4 A'_E + LM)\mathfrak{s}(W)$, which is at most $\eta\mathfrak{s}(W)/8$ under $C\rho LM \leqslant \eta/8$ with the constant C of (c) replaced by $C(1 + 2LM/(5e\eta))(1 + 3^4 A'_E/(LM))$; this factor, together with the doubling in (b), is the enlargement of the lemma, fixed with η before the coupling (and $e^{3\eta\mathfrak{s}} \leqslant e^{4\eta\mathfrak{s}}$, $|a_{W'}| \leqslant M(W')$).

The proofs of Lemmas E.10 and E.24 then apply verbatim; the species of the reference gas are the groups with $a_W \neq 0$, all of them C_1- or C_2-groups, and the exclusion ratio (E.27) holds for every compatible family $\mathfrak{B}$ of groups of the pair, cross groups included, because the criterion was proved for every W. *q.e.d.*

10.6 The Cross Terms

Covered and far histories are those of Section 10.1. A single-square group W of C_i touches C_{3-i} if some root D of W has $\mathsf{T}_D \cap \mathcal{I}_{C_{3-i}} \neq \varnothing$, with the contact set T_D of the pair. Let $c_\times := 2\eta/A'_E$.

Lemma 10.7 (The Cross Terms)**.** *Under the hypotheses of Lemma 10.6 and, for (c), those of Lemma E.11 for each square, with a constant $C_\times$ fixed before the coupling and independent of n_1, n_2 and the positions of the squares, and with $C_\chi := \max_i C_{\chi_i}$ and $c = \min_i c_i$ formed from the constants C_{χ_i}, c_i of* (E.29) *for the square C_i:*

(a) On every branch ξ, for every cross group W of the pair and every $a \in \mathcal{H}(W)$, $\mathfrak{s}(W) > \delta_a/(A'_E LM)$; and $\sum_{W \text{ cross}} M(W) e^{\mathfrak{a}_W/8} \leqslant C_\times \rho \Delta_\times$.

(b) Let $\mathscr{L}$ and $\mathscr{L}_i$ be the connected-cluster logarithms of the reference gas (10.3) *of the pair and of the reference gas of the single square C_i (Lemma E.24), and $E^\times_\xi = \mathscr{L} - \mathscr{L}_1 - \mathscr{L}_2$. Then, on every branch ξ,*

$$S^\circ_{\xi,C} = S^\circ_{\xi,C_1} S^\circ_{\xi,C_2} e^{E^\times_\xi} , \qquad |E^\times_\xi| \leqslant C_\times \rho \Delta_\times ; \tag{10.5}$$

in particular $|S^\circ_{\xi,C}| \leqslant e^{C_\times \rho \Delta_\times}$ on every branch.

(c) On covered far histories, with $\mathscr{R}_\xi$ the family sum (E.1) *of the pair,*

$$|\mathscr{R}_\xi| \leqslant |S^\circ_{\xi,C}| \Big[\exp\Big(C_\chi n_C \rho (\varepsilon_1 + e^{-cb/(LM)}) + C_\times \rho \Delta_\times \Big) - 1 \Big] . \tag{10.6}$$

Proof. Distance from an envelope. Let $\mathfrak{X}$ be a finite set of cubes, connected under adjacency, with $|\mathfrak{X}| \leqslant k$. As in the proof of Lemma E.8(c), any two sites in cubes of $\mathfrak{X}$ are at periodic fine ℓ^∞ distance less than kLM. Hence if $\mathfrak{X}$ contains the cube of the initial vertex of a traversal $a \in \mathcal{I}_{C_i}$ and a cube containing a site of $V(C_{3-i})$, then $\delta_a < kLM$.

(a) For a cross group W, $\mathfrak{E}(W)$ is connected, has at most $A'_E \mathfrak{s}(W)$ cubes and contains the cubes of the initial vertices of the traversals of $\mathcal{H}(W)$ and a cube containing a site of $V(C_1)$ and one containing a site of $V(C_2)$ (Lemma 10.6(a)); so $\delta_a < A'_E LM \mathfrak{s}(W)$ for every $a \in \mathcal{H}(W)$. Every group has a nonempty interval, so, pinning a cross group at a traversal of its interval and using $\mathfrak{a}_W/8 \leqslant \eta \mathfrak{s}(W)$, $e^{-3\eta\mathfrak{s}(W)} \leqslant e^{-3\eta\delta_a/(A'_E LM)} \leqslant e^{-c_\times \delta_a/(LM)}$ and the pin (E.25) of the pair (Lemma 10.6(d)),

$$\sum_{W \text{ cross}} M(W) e^{\mathfrak{a}_W/8} \leqslant \sum_{a \in \mathcal{I}_C} e^{-c_\times \delta_a/(LM)} \sum_{W \,|\, a \in \mathcal{H}(W)} M(W) e^{4\eta\mathfrak{s}(W)} \leqslant C\rho\Delta_\times .$$

The pins of Lemma 10.6 and (E.25), the criterion (10.7) below and Lemma E.24 use only $|a_W| \leqslant M(W)$ (Lemma E.3 at the reference data, Lemma B.16), the smallness of ρ and the field-independent geometry; coverage and farness enter only Lemmas 10.5 and E.11, so (a) and (b) hold on every branch.

(b) The three gases. The species of the pair gas with nonzero activity are its C_1- and C_2-groups; the species of the gas of C_i are the single-square groups of C_i, with the activities $a^{(i)}_W$. By Lemma 10.2(c) and the equality $a_W = a^{(i)}_W$ of Section 10.3, the C_i-groups of the pair are the single-square groups of C_i that do not touch C_{3-i}, with the same activities, and the compatibility relation of the pair restricted to them is that of the square C_i.

Each gas satisfies the Kotecký–Preiss criterion in the following weighted form: for every species W of that gas,

$$\sum_{W' \not\sim W} |a_{W'}| e^{3\eta\mathfrak{s}(W')} \leqslant \frac{1}{8}\eta\mathfrak{s}(W) \,, \tag{10.7}$$

for the pair by Lemma 10.6(d), and for the square C_i by the proof of (E.26), which bounds $e^{\mathfrak{a}_{W'}}$ by $e^{4\eta\mathfrak{s}(W')}$ and so bounds $e^{3\eta\mathfrak{s}(W')}$ in the same way. This is the criterion of Section B.6 for the weights $\widetilde{z}_W = |a_W| e^{2\eta\mathfrak{s}(W)}$ and sizes $\eta\mathfrak{s}(W)$, so the rooted-tree argument there gives, for every species W of each gas,

$$\sum_{\mathfrak{c}} m_W(\mathfrak{c}) |\Psi(\mathfrak{c})| e^{2\eta \sum_{W' \in \mathfrak{c}} \mathfrak{s}(W')} \leqslant |a_W| e^{3\eta\mathfrak{s}(W)} \,, \tag{10.8}$$

the sum over the cluster tuples $\mathfrak{c}$ of that gas, $\Psi(\mathfrak{c}) = \frac{\varphi^T(\mathfrak{c})}{m!} \prod_{W' \in \mathfrak{c}} a_{W'}$, $m_W(\mathfrak{c})$ the multiplicity of W in $\mathfrak{c}$. The logarithms $\mathscr{L}$, $\mathscr{L}_i$ are the absolutely convergent series $\sum_{\mathfrak{c}} \Psi(\mathfrak{c})$ of the three gases, $S^{\circ}_{\xi,C} = e^{\mathscr{L}}$ and $S^{\circ}_{\xi,C_i} = e^{\mathscr{L}_i}$ (Lemma E.24, for the pair by Lemma 10.6(d)), which gives the identity in (10.5).

Cancellation. A cluster tuple of the pair gas all of whose members are C_i-groups is a cluster tuple of the gas of C_i with the same Ψ: the Ursell coefficient depends only on the incompatibility graph of the tuple, which is the same for both gases, and the activities agree. Conversely a tuple of the gas of C_i none of whose members touches C_{3-i} is such a tuple of the pair gas. Hence $E^{\times}_{\xi} = \sum_{\mathfrak{c} \in \mathcal{X}_0} \Psi(\mathfrak{c}) - \sum_{i=1,2} \sum_{\mathfrak{c} \in \mathcal{X}_i} \Psi(\mathfrak{c})$, where $\mathcal{X}_0$ is the set of tuples of the pair gas containing a C_1-group and a C_2-group, and $\mathcal{X}_i$ the set of tuples of the gas of C_i containing a group that touches C_{3-i}; the rearrangement is justified by absolute convergence.

Extent of the bridging tuples. Let $\mathfrak{c} \in \mathcal{X}_0 \cup \mathcal{X}_1 \cup \mathcal{X}_2$, let $W \in \mathfrak{c}$ and $a \in \mathcal{H}(W)$, $a \in \mathcal{I}_{C_i}$. The union $\mathfrak{X}$ of the envelopes of the members of $\mathfrak{c}$ is connected: each envelope is connected (Lemma 10.6(a) for the pair, Lemma E.8(b) for a single square), the incompatibility graph of a tuple with $\varphi^T(\mathfrak{c}) \neq 0$ is connected, and two incompatible members have footprints at index distance at most one or a common traversal in their intervals (Lemma E.8(d), for the pair by Lemma 10.6(a)), the footprints and the cubes of the traversal vertices lying in the envelopes.

$\mathfrak{X}$ contains the cube of the initial vertex of a, and a cube containing a site of $V(C_{3-i})$: for $\mathfrak{c} \in \mathcal{X}_0$, the cube of a traversal vertex of the interval of a member that is a C_{3-i}-group; for $\mathfrak{c} \in \mathcal{X}_j$ (then $i = j$), the cube of the initial vertex of a contact $n \in \mathsf{T}_D \cap \mathcal{I}_{C_{3-i}}$ of a root D of a member W_0 touching C_{3-i}, which is within $2L$ of D (proof of Lemma E.8: the stencil bond of the contact has its endpoints in D), hence at index distance at most one from a cube of $X(W_0)$ and in $\mathfrak{E}(W_0)$, since $R_{\mathfrak{e}} \geqslant 1$. With $|\mathfrak{X}| \leqslant A'_E \sum_{W' \in \mathfrak{c}} \mathfrak{s}(W')$ ($A_E \leqslant A'_E$), the distance bound gives $\sum_{W' \in \mathfrak{c}} \mathfrak{s}(W') > \delta_a/(A'_E LM)$.

Bound. Every member of a tuple has a nonempty interval. Pinning each bridging tuple at a member W and a traversal $a \in \mathcal{H}(W)$, by (10.8) and the extent bound,

$$\begin{aligned} \sum_{\mathfrak{c} \in \mathcal{X}_0} |\Psi(\mathfrak{c})| &\leqslant \sum_{a \in \mathcal{I}_C} e^{-2\eta\delta_a/(A'_E LM)} \sum_{W \,|\, a \in \mathcal{H}(W)} \sum_{\mathfrak{c}} m_W(\mathfrak{c}) |\Psi(\mathfrak{c})| e^{2\eta \sum_{W' \in \mathfrak{c}} \mathfrak{s}(W')} \\ &\leqslant \sum_{a \in \mathcal{I}_C} e^{-c_\times \delta_a/(LM)} \sum_{W \,|\, a \in \mathcal{H}(W)} M(W) e^{3\eta\mathfrak{s}(W)} \,, \end{aligned}$$

the inner sums over the species of the pair gas, and the pin (E.25) of the pair bounds the last sum by $C\rho\Delta_\times$. The same computation in the gas of C_i, with the single-square pin (E.25) for C_i (with $|a^{(i)}_W| \leqslant M_i(W)$ and $\rho_i \leqslant \rho$, since (E.11) is increasing in u) and the traversals of $\mathcal{I}_{C_i}$, bounds $\sum_{\mathfrak{c} \in \mathcal{X}_i} |\Psi(\mathfrak{c})|$ by $C\rho \sum_{a \in \mathcal{I}_{C_i}} e^{-c_\times \delta_a/(LM)}$. Summing, $|E^{\times}_{\xi}| \leqslant 2C\rho\Delta_\times$. The series are those of the countable index sets; their absolute convergence is Lemma E.24.

Finally $|S^{\circ}_{\xi,C_i}| \leqslant 1$ on every branch, by the second equality of (3.94) and $K^{\circ}_{\xi;0} > 0$ (Section E.8.4: a normalised integral of a normalised character against a positive kernel), so $|S^{\circ}_{\xi,C}| \leqslant e^{|E^{\times}_{\xi}|} \leqslant e^{C_{\times}\rho\Delta_{\times}}$.

(c) By Lemma 10.3(b) the family sum of the pair is (E.1), $\mathscr{R}_{\xi} = \sum_{\mathfrak{B}\neq\varnothing} S^{\circ,[\mathfrak{B}]}_{\xi,C} R_{\xi}(\mathfrak{B})$ over the compatible families of groups of the pair. By the ratio bound (E.27) for the pair (Lemma 10.6(d), valid for families containing cross groups) and Lemma 10.5,

$$|S^{\circ,[\mathfrak{B}]}_{\xi,C} R_{\xi}(\mathfrak{B})| \leqslant |S^{\circ}_{\xi,C}| \prod_{i=1,2} \prod_{W\in\mathfrak{B},\ C_i\text{-group}} K_{\chi_i}\mathfrak{d}_W(U_1)e^{\mathfrak{a}_W/8} \prod_{W\in\mathfrak{B},\ \text{cross}} M(W)e^{\mathfrak{a}_W/8}\,,$$

with K_{χ_i} on the C_i-groups (the last sentence of the proof of Lemma 10.5). The C_i-groups of the pair are single-square groups of C_i with the same $\mathfrak{d}_W$ and $\mathfrak{a}_W$, and the pair's ρ dominates that of the square C_i, as in (b), so Lemma E.11 for C_i (on histories covered and far for both squares) bounds the sum of the first factors over the C_i-groups by $C_{\chi_i} n_{C_i}\rho(\varepsilon_1 + e^{-c_i b/(LM)})$; the sum over all non-cross groups is at most $C_{\chi} n_C \rho(\varepsilon_1 + e^{-cb/(LM)})$, and (a) bounds the sum of the second factors. Dropping the compatibility constraints, the sum over nonempty families is at most the exponential of the one-group sum minus one, as in the proof of Lemma E.12. The constant $C_{\times}$ is the largest of those above. *q.e.d.*

Corollary 10.8 (The Scalar of the Pair)**.** *Under the hypotheses of Lemma 10.7 and of Proposition F.2 for each square, on* A_{cov}*,*

$$S^{\circ}_{\xi,C} = \kappa_{C_1}\kappa_{C_2}\, e^{E^{\circ}_{\xi,1}+E^{\circ}_{\xi,2}+E^{\times}_{\xi}}\,, \tag{10.9}$$

with $E^{\circ}_{\xi,i}$ *the error* (F.4) *of the square* C_i*, bounded by* (F.3) *with* n_i*,* J_{C_i} *and the carrier distances* d_j *of the segments of* C_i*, and* $|E^{\times}_{\xi}| \leqslant C_{\times}\rho\Delta_{\times}$*.*

Proof. On A_{cov} the branch is covered for each square, so Proposition F.2 gives $S^{\circ}_{\xi,C_i} = \kappa_{C_i} e^{E^{\circ}_{\xi,i}}$, and (10.5) gives the result. *q.e.d.*

10.7 Proof of Theorem 1.13

The single-contour statements used by Theorem 1.2 outside the extraction (Lemmas D.1, D.3, D.6, Corollary D.8, Lemmas D.9, D.17, Proposition D.18, Lemma E.1, (E.40) and Lemma 9.3) involve the contour through its observable f_{ξ}, the set I_C of its free inputs (D.6), the set Γ_C of its contour bonds and free-input templates (Section D.7) and its segments, and, in Lemma E.1 and (E.40), through the bound $|S^{\circ}_{\xi,C}| \leqslant 1$ of (3.94), which for the pair is replaced by $|S^{\circ}_{\xi,C}| \leqslant e^{C_{\times}\rho\Delta_{\times}}$ of (b) of Lemma 10.9.

For the pair, f_{ξ} is (10.1) at the reconstructed field, $I_C = I_{C_1} \cup I_{C_2}$, $\Gamma_C = \Gamma_{C_1} \cup \Gamma_{C_2}$, so that $|\Gamma_C| \leqslant C_L n_{C_1} + C_L n_{C_2} = C_L n_C$, and $J_C = J_{C_1} \sqcup J_{C_2}$ is the set of the $4n_1 + 4n_2$ coarse segments of the two squares, d_j being the distance (F.1) of the square of the segment j at the common carrier I. The constants C, c are those of Definition 1.12.

Lemma 10.9 (Single-Contour Inputs for the Pair)**.** *Under the hypotheses of Theorem 1.13:*

(a) (D.8) *holds for the pair with* $C_{\mathrm{resp},\chi_1} + C_{\mathrm{resp},\chi_2}$ *in place of* $C_{\mathrm{resp},\chi}$*, and so do Lemma D.3, Lemma D.6 and Corollary D.8;* (D.42) *holds for the pair with* $C_{\mathrm{resp},\chi_1\chi_2}$*.*

(b) On A_{cov}*,* $f_{\xi}(0) = W_{\chi_1,\chi_2}(C;U_1)$ *and* $|S^{\circ}_{\xi,C}| \leqslant e^{C_{\times}\rho\Delta_{\times}}$*, and* (E.40) *holds for the pair, every term being integrable against the history weight.*

(c) On A_{cov}, $|\mathscr{R}_\xi| \leqslant 1 + e^{C_\times \rho \Delta_\times}$; *and* $\mathbb{P}_{\mathrm{h}}(\mathsf{N}_b) \leqslant 4C_L n_C (2b+3)^4 q_{\mathrm{pt}}$ *and* $|E^{\mathrm{near}}_{\mathrm{def}}| \leqslant e^{C_\times \rho \Delta_\times} C'_L n_C e^{-(c_*/2)\pi_*^2}$, *with* N_b *and* $E^{\mathrm{near}}_{\mathrm{def}}$ *of Section E.2 formed with* Γ_C *and* $\mathscr{R}_\xi$ *of the pair.*

(d) Let $\varphi_c(b) \coloneqq \sum_{r \in C_1 \sqcup C_2} e^{-c(d_T(r,b)-R'_0)_+}$, *the sum of the functions* φ_c *of* (9.5) *of the two squares. Pointwise on every branch,* $\sum_{j\in J_C} e^{-cd_j} \leqslant \sum_b \varphi_c(b) \mathbf{1}\{b \in E(B_\xi)\}$, *with* $\varphi_c(b) \leqslant 2\Phi_c$ *and* $\sum_b \varphi_c(b) \leqslant n_C N'_{\mathrm{lat}}(c)$; *and for every* $\lambda > 0$, *also with* $\mathbf{1}_{A_{\mathrm{cov}}}$ *omitted,*

$$\mathcal{Z}_0^{-1} \int \mathcal{O}\Big[\mathsf{w}_\xi \mathbf{1}_{A_{\mathrm{cov}}} \exp\Big(\lambda \sum_{j\in J_C} e^{-cd_j}\Big)\Big] \mathrm{d}V \leqslant \exp[n_C \lambda N'_{\mathrm{lat}}(c)\, e^{2\lambda\Phi_c}\, q^{1/C_K}] \,.$$

(e) For g_0 *below a further threshold depending on* L, M, M_1, M_2, *the group, the characters, the source constants and* c,

$$\delta_q^{I,2} \coloneqq C\tau^2 N'_{\mathrm{lat}}(c)\, e^{2C\tau^2\Phi_c}\, q^{1/C_K} \leqslant \exp[-c_*\pi_*^2/(2C_K)] \,. \tag{10.10}$$

Proof. (a) On the enlarged box the real reconstruction gives group-valued links (proof of Lemma D.3), so $|\chi_i(Y_i)| \leqslant 1$ for the ordered link product Y_i of C_i and $|f_\xi| \leqslant 1$. By the product rule, $\nabla_b f_\xi = (\nabla_b \chi_1(Y_1))\chi_2(Y_2) + \chi_1(Y_1)\nabla_b \chi_2(Y_2)$, and on a branch covered for the pair, which is covered for each square (Section 10.1), the proof of (D.8) for the square C_i gives $h(\nabla_b \chi_i(Y_i)) \leqslant C_{\mathrm{resp},\chi_i} g_0 \mathbf{1}_{I_{C_i}}(b)$, so $h(\nabla_b f_\xi) \leqslant |\chi_2(Y_2)|\, h(\nabla_b \chi_1(Y_1)) + |\chi_1(Y_1)|\, h(\nabla_b \chi_2(Y_2)) \leqslant (C_{\mathrm{resp},\chi_1} \mathbf{1}_{I_{C_1}}(b) + C_{\mathrm{resp},\chi_2} \mathbf{1}_{I_{C_2}}(b)) g_0 \leqslant (C_{\mathrm{resp},\chi_1} + C_{\mathrm{resp},\chi_2}) g_0 \mathbf{1}_{I_C}(b)$, which is (D.8) for the pair.

Lemma D.3 uses the observable through $\widetilde{f} = f_\xi \circ T$, $|\widetilde{f}| \leqslant 1$ and (D.8) composed with DT; Lemma D.6 uses it through Lemma D.3 and the support I_C; Corollary D.8 is that lemma averaged. The enlargement in the proof of Lemma D.6 multiplies the constant of (D.8) by a factor independent of the character, so the enlarged pair constant is the sum of the enlarged constants of χ_1 and χ_2.

The proof of Proposition D.18 combines Corollary D.8 with Lemmas D.9 and D.17, whose bounds are sums of one-point marks over the members of Γ_C; with $|\Gamma_C| \leqslant C_L n_C$ it gives (D.42) with $C_L \max((C_{\mathrm{resp},\chi_1} + C_{\mathrm{resp},\chi_2}) S_c, 1) = C_{\mathrm{resp},\chi_1\chi_2}$.

(b) Lemma 3.36 is proved bond by bond, so on A_{cov}, $U_\xi(0)(r) = U_1(r)$ for every bond r of C_1 and of C_2, and $f_\xi(0) = W_{\chi_1,\chi_2}(C; U_1)$. By Lemma 10.7(b), $|S^\circ_{\xi,C}| \leqslant |S^\circ_{\xi,C_1}||S^\circ_{\xi,C_2}| e^{|E^\times_\xi|} \leqslant e^{C_\times \rho \Delta_\times}$, where $|S^\circ_{\xi,C_i}| \leqslant 1$ because it is a normalised integral of a normalised character against the positive reference kernel (proof of Proposition E.26).

The derivation of (E.40) in Section E.7.1 multiplies (3.97), which holds for the pair by Lemma 10.3(b), by $\mathbf{1}_{A_{\mathrm{cov}}}$ and writes $\mathcal{J}^{\mathrm{a}}_\xi[f_\xi] = \mathsf{w}_\xi F_\xi(1)$ on A^c_{cov}; (3.7) holds with $W_{\chi_1,\chi_2}(C)$ in place of $W_\chi(C)$, f_ξ being jointly Borel with $|f_\xi| \leqslant 1$ (the measurability paragraph of Section 3.1). Its integrability uses $|F_\xi(s)| \leqslant 1$ ($F_\xi(s) = \int f_\xi \,\mathrm{d}\nu_{\xi;s}$ with $|f_\xi| \leqslant 1$), $|\mathcal{B}_\xi[f_\xi]| \leqslant (C_{\mathrm{resp},\chi_1} + C_{\mathrm{resp},\chi_2}) g_0 r |I_C|$ on A_{cov} by (a), and a bound on $S^\circ_{\xi,C} f_\xi(0)$, which is now $|S^\circ_{\xi,C} f_\xi(0)| \leqslant e^{C_\times \rho \Delta_\times}$ in place of 3.

(c) On A_{cov}, $\mathscr{R}_\xi = F_\xi(0) - S^\circ_{\xi,C} f_\xi(0)$ by (3.96) for the pair (Lemma 10.3(b)), so $|\mathscr{R}_\xi| \leqslant 1 + e^{C_\times \rho \Delta_\times} \leqslant 2e^{C_\times \rho \Delta_\times}$ by (b). The union bound of the proof of Lemma E.1, over the at most $C_L n_C$ members of Γ_C, gives the bound on $\mathbb{P}_{\mathrm{h}}(\mathsf{N}_b)$. Positivity of $\mathbb{E}_{\mathrm{h}}$ then gives $|E^{\mathrm{near}}_{\mathrm{def}}| \leqslant 2e^{C_\times \rho \Delta_\times} \cdot 4C_L n_C (2b+3)^4 q_{\mathrm{pt}}$, and the last step of that proof gives $8C_L(2b+3)^4 q_{\mathrm{pt}} \leqslant C'_L e^{-(c_*/2)\pi_*^2}$.

(d) The first inequality of (9.6) is proved segment by segment, the term of a segment being bounded through the contour bonds of its own square (proof of Lemma 9.3). The two

squares' inequalities at the common carrier add up to the pointwise bound. Each square's φ_c is at most Φ_c and has sum at most $n_{C_i} N'_{\mathrm{lat}}(c)$, by the counts of that proof. The rest of that proof applies with the weights $w_b = \varphi_c(b) \leqslant 2\Phi_c$: the expansion over the marks $\mathbf{1}\{b \in E(B_\xi)\}$, one per bond, and the joint bound (9.4), which holds for every finite set of bonds. They bound the average by $\exp(\hat{q} \sum_b (e^{\lambda w_b} - 1))$ with $\hat{q} := q^{1/C_K}$. Since $e^x - 1 \leqslant x e^x$ for $x \geqslant 0$, this is at most $\exp(\hat{q}\lambda e^{2\lambda\Phi_c} \sum_b w_b)$, and the indicator enters only through $\mathbf{1} \leqslant 1$.

(e) Since $\tau < 1$ and $q \leqslant e^{-c_* \pi_*^2}$ (Proposition D.18), $\log \delta_q^{I,2} \leqslant \log(C N'_{\mathrm{lat}}(c)) + 2C\Phi_c - c_* \pi_*^2 / C_K$. Moreover $\pi_*^2 / C_K \geqslant c' (\log g_0^{-2})^{2p_0 - 4\mathfrak{r}}$ with $2p_0 - 4\mathfrak{r} \geqslant 12$, by (9.9). So $c_* \pi_*^2 / (2C_K) \geqslant \log(C N'_{\mathrm{lat}}(c)) + 2C\Phi_c$ for small g_0, which is (10.10). *q.e.d.*

Proof of Theorem 1.13. (i) By Lemma 10.9(b),

$$\mathbb{E}_\varepsilon[W_{\chi_1,\chi_2}(C)] = \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} S^\circ_{\xi,C} W_{\chi_1,\chi_2}(C; U_1)] + E_{\mathrm{def}} + E_{\mathrm{bdry}} \,,$$

with all terms integrable. On A_{cov}, (10.9) (Corollary 10.8) applies. Its hypotheses are those of Lemma 10.7 and, for each square, those of Proposition F.2, which (H2) and (H3) for that square imply as in the proof of Theorem 1.2. After the substitution the first integrand is still exactly $\mathbf{1}_{A_{\mathrm{cov}}} S^\circ_{\xi,C} f_\xi(0)$. The statements on κ_{C_i} are Theorem 1.2(i) for (C_i, χ_i).

(ii) The pair's A_{cov} is contained in the covered event of each square, on which Proposition F.2 bounds $E^\circ_{\xi,i}$ with the constants C_i, c_i of χ_i; these bounds are at most the stated ones, since $C_i \leqslant C$ and $e^{-c_i x} \leqslant e^{-cx}$ for $x \geqslant 0$. The elementary bounds give $C\tau^2(9n_i + 1)$, as in Theorem 1.2. The bound on $E^\times_\xi$ is Lemma 10.7(b) with (10.2).

As in Section E.2, $E_{\mathrm{def}} = E^{\mathrm{near}}_{\mathrm{def}} + E^{\mathrm{far}}_{\mathrm{def}}$, both terms being integrable by Lemma 10.9(c). The near term is bounded in Lemma 10.9(c). The histories of $A_{\mathrm{cov}} \cap \mathsf{N}_b^c$ are covered and far for the pair. On them (10.6) and Lemma 10.9(b) give $|\mathscr{R}_\xi| \leqslant e^{C_\times \rho \Delta_\times} (e^{X + C_\times \rho \Delta_\times} - 1)$ with $X := C_\chi n_C \rho (\varepsilon_1 + e^{-cb/(LM)})$. So $|E^{\mathrm{far}}_{\mathrm{def}}| \leqslant e^{C_\times \rho \Delta_\times} (e^{X + C_\times \rho \Delta_\times} - 1)$, by the positivity of $\mathbb{E}_{\mathrm{h}}$ and $\mathbb{E}_{\mathrm{h}}[1] = 1$. The bound on E_{bdry} is (D.42) for the pair (Lemma 10.9(a)).

(iii) The first bound follows from the positivity of $\mathbb{E}_{\mathrm{h}}$, $\mathbb{E}_{\mathrm{h}}[1] = 1$, $|W_{\chi_1,\chi_2}| \leqslant 1$ and $|e^z| \leqslant e^{|z|}$ with (ii).

Under the condition of (H3) for (iii) for the square C_i, $4n_i e^{-c_i \iota_T} \leqslant 1$. So Corollary 9.1 for the square C_i gives $|E^\circ_{\xi,i}| \leqslant C_i \tau^2 (2 + 1/c_i) + C_i \tau^2 \sum_{j \in J_{C_i}} e^{-c_i d_j} \leqslant C\tau^2(2 + 1/c) + C\tau^2 \sum_{j \in J_{C_i}} e^{-c d_j}$ on A_{cov}. Hence $\mathbf{1}_{A_{\mathrm{cov}}} e^{|E^\circ_{\xi,1}| + |E^\circ_{\xi,2}|} \leqslant e^{2C\tau^2(2+1/c)} \mathbf{1}_{A_{\mathrm{cov}}} \exp(C\tau^2 \sum_{j \in J_C} e^{-c d_j})$, and Lemma 10.9(d) at $\lambda = C\tau^2$, with (e), gives (1.20).

The modulus of the leading term is at most $|\kappa_{C_1} \kappa_{C_2}| e^{C_\times \rho \Delta_\times} \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{|E^\circ_{\xi,1}| + |E^\circ_{\xi,2}|}]$, by $|W_{\chi_1,\chi_2}| \leqslant 1$ and (ii). Pointwise on A_{cov}, with $z = E^\circ_{\xi,1} + E^\circ_{\xi,2} + E^\times_\xi$, $|e^z - 1| \leqslant e^{|z|} - 1 \leqslant e^{2C\tau^2(2+1/c) + C_\times \rho \Delta_\times} \exp(C\tau^2 \sum_{j \in J_C} e^{-c d_j}) - 1$, a nonnegative function. Its $\mathbb{E}_{\mathrm{h}}$-average without the indicator is at most $\exp[2C\tau^2(2 + 1/c) + C_\times \rho \Delta_\times + n_C \delta_q^{I,2}] - 1$, by Lemma 10.9(d) and $\mathbb{E}_{\mathrm{h}}[1] = 1$. *q.e.d.*

11 Towards the Next Step

The statement of this section is a conjecture; nothing in the other sections uses it. It concerns loops whose side is a power of the block size, $n = L^k$ with an integer $k \geqslant 0$, after $k + 1$ steps (the letter K is the exponent of [8] in (H3) and the index of σ_K, C_K in Lemma 9.2).

On a torus of (H3) the lattice $L^{k+1}\mathbb{Z}^4$ and its plaquettes are defined ($K_\mu = 2L^{m-1} \geqslant 3(n+2)$ forces $m \geqslant k+2$, so L^{k+1} divides every period $N_\mu = 2L^m$), and a translation of C_n does not change $\mathbb{E}_\varepsilon[W_\chi(C_n)]$ (the Wilson measure $\rho_0(U)\,\mathrm{d}U$ on the periodic torus is invariant under lattice translations, its action being a sum over all plaquettes and the Haar measure a product measure; Section 5.3 uses the remaining lattice symmetries), so the corners of C_n are taken in $L^{k+1}\mathbb{Z}^4$; then C_n, of side L^{k+1} fine units, is a contour of the lattice of spacing $L^j\varepsilon$ for every $j \leqslant k+1$ and the boundary of a single plaquette of the lattice of spacing $L^{k+1}\varepsilon$. The scales τ and π_* are those of the notation paragraph of Section 1.2, at the bare coupling g_0.

Conjecture 11.1 (The Loop along the Scales)**.** *Let G, χ be as in (H1), and let the cutoff prescription σ and the constants be fixed as in (H2). There are $C, c > 0$, depending on L, the group, the character, σ and the constants fixed in (H2) only, thresholds $g_j^* > 0$ ($j \geqslant 1$) and, for $0 < g_0 < g_j^*$, real numbers δm_j and complex numbers ζ_j, depending on g_0, ε, j, L, the group, the character, σ and the fixed prescriptions listed in Theorem 1.2(i) only, with $z_j := e^{\zeta_j}$ and*

$$\delta m_1 = \delta m^\sigma(g_0,\varepsilon)\,, \qquad \zeta_1 = \frac{1}{4}\sum_{(v,w)\in\mathscr{C}_\square} \zeta^\sigma_{v,w}(g_0)\,, \qquad |\delta m_j|\, L^j\varepsilon \leqslant C\tau^2\,, \qquad |\zeta_j| \leqslant C\tau^2\,, \tag{11.1}$$

so that $z_1 = z_{\mathfrak{c}}^\sigma(g_0,\pi/2)$ (6.12), such that for every integer $k \geqslant 0$, with $n = L^k$, there is $g_{\mathrm{it}}(k) \leqslant \min_{j\leqslant k+1} g_j^*$ such that for $0 < g_0 < g_{\mathrm{it}}(k)$, on every torus of (H3) whose exponent m exceeds a bound depending on k and g_0,

$$\begin{gathered}\mathbb{E}_\varepsilon[W_\chi(C_n)] = \kappa_{C_n}^{(k+1)}(1+R) + R'\,, \qquad \kappa_{C_n}^{(k+1)} = \exp\Bigl(-P(C_n)\sum_{j=1}^{k+1}\delta m_j\Bigr)\prod_{j=1}^{k+1} z_j^4\,, \\ |R| \leqslant C[(k+2)\tau^2 + n_C e^{-c\pi_*^2}]\,, \qquad |R'| \leqslant C n_C e^{-c\pi_*^2}\,.\end{gathered} \tag{11.2}$$

Remark 11.2 (Content of the Conjecture and What a Proof Needs). (1) The j-th coefficients do not depend on the size of the loop: the perimeter part is a sum over the scales of perimeter coefficients, each of size τ^2 in units of its own lattice spacing, so $|P(C_n)\sum_j \delta m_j| \leqslant 4C\tau^2 n(1+L^{-1}+L^{-2}+\cdots)$, while the corner factors contribute one factor z_j^4 per scale, $k+1 = \log_L(nL)$ in all. For $k=0$ the statement is Theorem 1.2, with the relative form of the far-history bound of Remark 1.3(1), which places the second term of the bound on E_{def} in $\kappa_{C_1}R$, together with the claim that the leading expectation $\mathbb{E}_{\mathfrak{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E_\xi^\circ} W_\chi(C_1;U_1)]$ of the single-plaquette loop is $1+O(\tau^2+n_C e^{-c\pi_*^2})$; the first step gives the bounds (1.10), the upper bounds of (iii) and, by Corollaries 1.4 and 1.6 at $n=1$, $1+O(\tau^2+\varepsilon_1+(L^2+LC_L)q+n_C\delta_q^\sharp)$, in which the area term ε_1 of the bound, of order $g_0(\log g_0^{-2})^{p_0}$ by [9, (2.6)] as quoted in Section B.8, is not $O(\tau^2)$; whether the leading expectation itself is $1+O(\tau^2)$ is not decided here.

(2) For $j \geqslant 2$ the coefficients δm_j, z_j are not defined in this paper; they are to be the perimeter coefficient and the corner factor of the j-th step, computed from the density after $j-1$ steps as (5.39) and (6.12) compute them from the flat reference integrals $N_{C,T}^\sigma$, D_T^σ of (5.5), in which case the conjecture says that the coefficients of every step have the size that Sections 5 and 6 prove for the first.

At the stated precision the corner factors are not separately determined: $|z_j^4 - 1| = |e^{4\zeta_j} - 1| \leqslant e^{4C\tau^2} - 1$, so $|\prod_{j\leqslant k+1} z_j^4 - 1| \leqslant e^{4C(k+1)\tau^2} - 1 \leqslant 8C(k+1)\tau^2$ once $4C(k+1)\tau^2 \leqslant 1$, and replacing every z_j by 1 replaces $1+R$ by $\prod_j z_j^4(1+R)$, with $|\prod_j z_j^4(1+R)-1| \leqslant 8C(k+1)\tau^2(1+|R|)+|R|$, which a larger C in the bound on R absorbs below a

threshold depending on k; the z_j record where the term $(k+2)\tau^2$ of R comes from. With the threshold $g_{\text{it}}(k)$ depending on k, R' is not needed either: $|\kappa_{C_n}^{(k+1)}| \geqslant e^{-8C\tau^2(n+k+1)}$ $(L \geqslant 2)$, so once $8C\tau^2(n+k+1) \leqslant 1$ the bound on R' is at most $eCn_C e^{-c\pi_*^2}|\kappa_{C_n}^{(k+1)}|$ and R' can be moved into $\kappa_{C_n}^{(k+1)}R$ at the cost of replacing C by $(1+e)C$; it is substantive in the form with a threshold independent of k.

Since $\delta m_1 = \delta m^\sigma$ is given, what the statement determines beyond the first step and beyond $O((k+2)\tau^2)$ is $P(C_n)\sum_{j\geqslant 2}\delta m_j$, of size at most $4C\tau^2(n-1)/(L-1)$, modulo $O((k+2)\tau^2)$. For $k \leqslant 1$ this is at most $4C\tau^2$, so nothing beyond the first step is determined; in general the scales j with $L^{k+1-j} \leqslant k+2$ contribute at most $4C\tau^2(k+2)L/(L-1)$, also of order $(k+2)\tau^2$, and the content lies in the scales $j \geqslant 2$ with $L^{k+1-j} > k+2$.

(3) A proof by $k+1$ applications of the first step needs the following ((c) only for a threshold independent of k), none of which is proved here beyond the first step.

(a) A local form of the leading term: on the covered event, $W_\chi(C_n;U_1)e^{E_\xi^\partial}$ as a function of the coarse field V near the loop, a coarse Wilson loop times the exponential of a sum over the coarse bonds of the loop of terms depending on V within a fixed distance of the bond, analytic on the small-field chart, gauge covariant and bounded by $C\tau^2$, with the history labels absorbed into the large-field expansion of ρ_1; this is what turns the output of the step, the pair of the complex measure $\kappa_{C_n}^{-1}N_1$ and the measure ρ_1 (Remark 1.3(2)), into an input of the same type.

(b) General-step analogues of (B1)–(B4), which Section 2 of [9] does not state as such: it formulates the inductive assumption with the operations corresponding to $\mathbf{T}_1$ left undefined, only their general properties being formulated (p.254), their inductive definition being deferred to its next section and forthcoming papers (pp.254–255), and with modifications connected with new terms obtained by the **R**-operations (p.254), among them the admitted case $\Lambda_j = \Omega_j$ (p.255); the sets $S_j \subset \Omega_j \cap \Lambda_j^c$ and the functions $\chi(\Omega_j \cap \Lambda_j^c, S_j)$, $\zeta(\Omega_j^c)$ are the general-step forms of the first-step objects of [9, (1.29), pp. 254–255]; the geometry and the scales are [9, (2.1)–(2.9)], with the couplings of the first $k+1$ scales compared through the first inequality $g_n \leqslant (1+g_n^2\beta'(n-m))^{1/2}g_m$ of [9, (2.6), p. 255], $g_{k+1}^2\beta'(k+1)$ being small for $g_0 < g_{\text{it}}(k)$ (Section B.8 quotes (2.6) for the first step).

(c) The defect: the far-history part of (E.83) is relative to $|\kappa_{C_n}|$ but of order $x = C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})$ when $x \leqslant 1$, polynomially small per contour bond (Remark 1.3(1)); for a threshold independent of k it must be shown to be a perimeter-local correction, absorbed in δm_2 and z_2 at the next step. Below a threshold depending on k it is at most $C\tau^2$ relative to $|\kappa_{C_n}|$ ($\rho \leqslant C\tau^2$, $\varepsilon_1 \to 0$ and $e^{-cb/(LM)} \to 0$ at fixed n_C) and goes into R, as at $k=0$.

(d) The absolute terms E_{bdry} and the near-history part of E_{def} at the j-th step, of order $4L^{k+2-j}e^{-c\pi_{*,j}^2}$ for the loop of $4L^{k+2-j}$ bonds of the lattice of spacing $L^{j-1}\varepsilon$, with $\pi_{*,j} := \min(p_0(g_{j-1}), p_0(g_j))$ the scale of that step, $\pi_{*,j}^2 \geqslant \pi_*^2/2$ once $g_i^2\beta' i \leqslant 1$ for $i \in \{j-1, j\}$ and $(1-\log 2/\log g_0^{-2})^{2p_0} \geqslant 1/2$ (the first inequality of [9, (2.6)] with $m=0$, $n=i\geqslant 1$ gives $g_i^2 \leqslant (1+g_i^2\beta' i)g_0^2 \leqslant 2g_0^2$, hence $\log g_i^{-2} \geqslant \log g_0^{-2} - \log 2$ and $p_0(g_i)^2 \geqslant p_0(g_0)^2/2 \geqslant \pi_*^2/2$; for $i=0$ this is $p_0(g_0) \geqslant \pi_*$), and entering multiplied by the leading coefficients of the steps $1,\dots,j-1$, whose product must be bounded; so that their sum over the steps is at most $Cn_C e^{-c\pi_*^2}$; their relative size is discussed in Remark 1.3(1).

(e) At every step the remainder of Corollary 1.4, which at the first step is at most $e^{n_C \delta_q^\sharp} - 1$, and the torus and corner part $E^\circ_{T,n}$, at most $C\tau^2(2 + 1/c)$, one source of the term $(k + 2)\tau^2$.

(f) The endpoint: the leading expectation of the single-plaquette loop at the last step is $1+O(\tau^2)$; at the first step Corollary 1.6 gives $1+O(\tau^2+\varepsilon_1+(L^2+LC_L)q+n_C\delta_q^\sharp)$ only, and its area term ε_1, which comes from the deterministic bound on the plaquettes of the background, must be replaced by a bound of order τ^2.

(g) The terms of the **R**-operations at every step, by which the general representation modifies the first-step representation [9, (1.28)–(1.30), p. 254].

Appendix A The Flat Covariance

The bulk coefficient uses the flat fluctuation operator on the infinite lattice with the same periodic block grid and constraint extension. A uniform lower bound on compatible finite tori comes first; the operator is then constructed and its inverse compared with the finite-torus inverses.

The spatial dimension is four throughout this comparison. For a site field λ, let $(D\lambda)(x, x') \coloneqq \lambda(x') - \lambda(x)$. For an oriented plaquette p, let $(\operatorname{curl} a)(p) \coloneqq a(\partial p)$, where $a(\gamma)$ is the signed sum along an oriented path γ. All squared norms in this comparison sum the Lie-algebra inner product over positive bonds, plaquettes or sites of the indicated carrier. The mean path sum is

$$(\Phi a)(y, x) \coloneqq \frac{1}{|\mathcal{G}(y, x)|} \sum_{\gamma \in \mathcal{G}(y,x)} a(\gamma) \, . \tag{A.1}$$

[8, (0.8) and (2.5)] identify $G^{(2)}(a) = \|\Phi a\|^2$. Differentiating [8, (0.4)], with $x' = x + Le_\mu$ for $c = (y, y + Le_\mu)$, gives

$$\begin{aligned} (Q_1 a)(c) &= L^{-5} \sum_{x \in \mathcal{B}_y} a([x, x']) \, , \\ (Qa)(c) &= (Q_1 a)(c) + L^{-5} \sum_{x \in \mathcal{B}_y} ((\Phi a)(y, x) - (\Phi a)(y + Le_\mu, x')) \, . \end{aligned} \tag{A.2}$$

The operator Q_1 is the straight-path average in [2, (2.125)]; it differs from Q. Telescoping these paths gives $Q(D\lambda)(c) = (\lambda(c_+) - \lambda(c_-))/L$.

The lower corner of each block is $o_y = y - \frac{L-1}{2} \sum_{\mu=1}^4 e_\mu$. Let $\kappa_{y,x}$ traverse the coordinate displacements from o_y to x in the order $1, \dots, 4$. The union of these paths is a spanning tree. Let

$$\lambda(x) \coloneqq a(\kappa_{y,x}) - a(\kappa_{y,y}), \qquad b \coloneqq a - D\lambda, \qquad E(y, x) \coloneqq (\Phi a)(y, x) - \lambda(x) \, . \tag{A.3}$$

The block partition makes λ a periodic site field, and $\lambda(y) = 0$. Every tree bond of b vanishes. Moreover $Qb = Qa$ and $\operatorname{curl} b = \operatorname{curl} a$.

The Lie-algebra element $E(y, x)$ is a mean of the signed sums of a around the closed walks obtained by adjoining a mean path from y to x to the reverse tree path. Each walk lies inside its block and has length at most $12(L - 1)$. It is filled by commuting adjacent steps in different directions and cancelling opposite steps in each direction. Every commutation adds one unit plaquette. It remains inside the block, because exchanging two different directions preserves the intermediate coordinate ranges. Sorting the directions takes at most the square

of the walk length commutations. Thus a filling has integer plaquette coefficients with total absolute sum at most $J_L = 144(L-1)^2$. Cauchy–Schwarz and averaging give

$$\|E\|^2 \leqslant K_L \|\operatorname{curl} a\|^2, \qquad K_L = L^4 J_L^2 \,. \tag{A.4}$$

There are L^4 site tests per block and the interior plaquettes of different blocks are disjoint, so this constant is independent of their number. Also $\|\lambda\|^2 \leqslant 2\|\Phi a\|^2 + 2\|E\|^2$ and $\|D\lambda\|^2 \leqslant 16\|\lambda\|^2$.

For $Qa = 0$, equation (A.2) applied to b has $\Phi b = E$. Jensen's inequality and the eight incidences of a block with coarse bonds imply

$$\|Q_1 b\|^2 \leqslant 16 L^{-6} \|E\|^2 \,. \tag{A.5}$$

For the block-tree gauge, the following periodic bound holds with $A_L \coloneqq 384 L^5$:

$$\|b\|^2 \leqslant A_L(\|\operatorname{curl} b\|^2 + L^2 \|Q_1 b\|^2) \,. \tag{A.6}$$

For the proof, the coordinate permutation $R(x_1, x_2, x_3, x_4) = (x_4, x_3, x_2, x_1)$ reverses the block-tree order. Its pushforward on positive bonds is $(R_* b)(Rx, Rx + e_{5-\mu}) = b(x, x + e_\mu)$. It preserves the bond and curl norms, and (A.2) gives $(Q_1 R_* b)(Rc) = (Q_1 b)(c)$. Thus it suffices to prove the same bound for the reversed order $4, \dots, 1$, after translating a block corner to the origin. The inner product remains the fixed tr_N inner product throughout.

Direct Face Estimates in the Positive-Bond Norm. The fine lattice has $N_\mu = LK_\mu$ sites in direction μ, with odd $L \geqslant 3$ and $K_\mu \geqslant 3$. The preceding axis reversal permutes the periods to (N_4, N_3, N_2, N_1) and reduces the proof to the tree order $4, \dots, 1$. Its inverse restores the original order at the end. A local calculation uses the translated block $B_0 = \{0, \dots, L-1\}^4$. All norms and sums below count positive bond or plaquette labels once, with the fixed $\mathfrak{g}$ inner product.

For a block B, let $\mathcal{E}_B^+$ be its positive fine bonds with both endpoints in B, and let $\mathcal{P}_B^+$ be its positive plaquettes with all vertices in B. For a boundary face $\Sigma \subset B$, let $\mathcal{E}_\Sigma^+$ be its positive fine bonds with both endpoints in Σ. Let

$$I_B \coloneqq \sum_{r \in \mathcal{E}_B^+} \|b(r)\|_{\mathfrak{g}}^2, \qquad T_\Sigma \coloneqq \sum_{r \in \mathcal{E}_\Sigma^+} \|b(r)\|_{\mathfrak{g}}^2, \qquad F_B \coloneqq \sum_{p \in \mathcal{P}_B^+} \|(\operatorname{curl} b)(p)\|_{\mathfrak{g}}^2 \,. \tag{A.7}$$

These are scalar quadratic forms in the tree-gauged field. For the flat comparison, a linear gauge change $a \mapsto a + D\omega$ preserving its coarse average has constant values $\omega(y)$ on the connected coarse lattice. Equation (A.3) then leaves b unchanged. Constant colour conjugation sends b to $\operatorname{Ad}_g b$ and preserves every inner product below.

The Block-Face Trace Bound. Let $b_\nu(x) \coloneqq b(x, x + e_\nu)$ for an internal positive bond of B_0, so $x_\nu \leqslant L-2$. The source tree condition makes $b_\nu(x) = 0$ whenever $x_1 = \cdots = x_{\nu-1} = 0$; in particular $b_1 = 0$. For $k < \nu$ and integers $0 \leqslant t < x_k$, let

$$u^{(k)}(x, t) \coloneqq (0, \dots, 0, t, x_{k+1}, \dots, x_4) \,. \tag{A.8}$$

Here the t is the kth component. Let $F_{k\nu}(u)$ be the curl on the positive (k, ν) plaquette based at u. On the plane $u_1 = \cdots = u_{k-1} = 0$, both k-direction sides of that plaquette vanish by the tree condition. Successive telescoping therefore gives the identity used before [2, (2.123)]:

$$b_\nu(x) = \sum_{k=1}^{\nu-1} \sum_{t=0}^{x_k - 1} F_{k\nu}(u^{(k)}(x, t)) \,. \tag{A.9}$$

Every displayed plaquette lies inside B_0. The number of terms is at most $3(L-1)$.

Let $x_\mu = \alpha$, with $\alpha \in \{0, L-1\}$, be a face and $\nu \neq \mu$ a tangential component. For a fixed (k, ν) plaquette in (A.9), its multiplicity in the face sum is bounded by

$$\begin{array}{r|l} \mu < k & L^{k-2}\, L \leqslant L^2 \\ \mu = k & L^{k-1} \leqslant L^2 \\ \mu > k,\ \mu \neq \nu & L^{k-1}\, L \leqslant L^2 \end{array} . \tag{A.10}$$

In the first row, one of the $k-1$ otherwise free earlier components is fixed by the face. In the second row, x_k is fixed, so a given t occurs at most once. In the third row, the two distinct indices μ, ν exceed k, hence $k \leqslant 2$. In all rows, later components are fixed by the plaquette label. Without the face restriction the multiplicity is at most $L^{k-1}L \leqslant L^3$. For different ν, the plaquette pairs (k, ν) are disjoint. Cauchy–Schwarz in (A.9), followed by these counts, proves

$$I_B \leqslant 3(L-1)L^3 F_B \leqslant 4L^4 F_B, \qquad T_\Sigma \leqslant 3(L-1)L^2 F_B \leqslant 4L^3 F_B\,. \tag{A.11}$$

The first estimate is the positive-label version of [2, (2.123)]. The second is the face restriction of its telescoping calculation.

The Interface Poincare Estimate. Let $F = \{0, \dots, L-1\}^3$ and let $\mathcal{E}_F^+$ be its positive nearest-neighbour edges. The Neumann graph Laplacian $\mathcal{L}_F \colon \mathfrak{g}^F \to \mathfrak{g}^F$ acts componentwise. For $s \in F$, it is given by

$$(\mathcal{L}_F v)(s) \coloneqq \sum_{\substack{s' \in F \\ |s'-s|_1 = 1}} (v(s) - v(s'))\,. \tag{A.12}$$

It has quadratic form $\sum_{(s,s') \in \mathcal{E}_F^+} \|v(s') - v(s)\|^2$. For $m \in \{0, \dots, L-1\}^3$, its real scalar eigenbasis and eigenvalues are

$$v_m(s) = \prod_{i=1}^{3} \cos\left(\frac{\pi m_i (s_i + 1/2)}{L}\right), \qquad \lambda_m = 4 \sum_{i=1}^{3} \sin^2\left(\frac{\pi m_i}{2L}\right)\,. \tag{A.13}$$

The one-dimensional cosine identity verifies the eigenvalue at interior and endpoint vertices; finite cosine orthogonality gives a basis. Only $m = 0$ is constant. Since $\sin t \geqslant 2t/\pi$ for $0 \leqslant t \leqslant \pi/2$, every nonzero eigenvalue is at least $4/L^2$. Applied to each orthonormal colour component, with $\bar{v} \coloneqq L^{-3} \sum_{s \in F} v(s)$, this gives

$$\sum_{s \in F} \|v(s) - \bar{v}\|^2 \leqslant \frac{L^2}{4} \sum_{(s,s') \in \mathcal{E}_F^+} \|v(s') - v(s)\|^2\,. \tag{A.14}$$

Crossing Bonds and the Exact Average. For a positive coarse bond c in direction μ, its two endpoint blocks are $B_- = o + \{0, \dots, L-1\}^4$ and $B_+ = B_- + Le_\mu$. The transverse label set is $F_c = \{0, \dots, L-1\}^{\{\nu \neq \mu\}}$, identified with F by increasing transverse index. In that label set, e_ν denotes its transverse coordinate unit vector. For $s \in F_c$, let

$$x_c(s) \coloneqq o + (L-1)e_\mu + \sum_{\nu \neq \mu} s_\nu e_\nu, \qquad t_c(s) \coloneqq b(x_c(s), x_c(s) + e_\mu), \qquad J_c \coloneqq \sum_{s \in F_c} \|t_c(s)\|^2\,. \tag{A.15}$$

Let Σ_- and Σ_+ be the facing boundary faces in B_- and B_+. For $\nu \neq \mu$ and $s, s + e_\nu \in F_c$, the plaquette based at $x_c(s)$ has curl

$$t_c(s) - t_c(s + e_\nu) + b_\nu(x_c(s) + e_\mu) - b_\nu(x_c(s))\,. \tag{A.16}$$

For $\nu < \mu$ its sign is opposite to the positive plaquette convention; its squared norm is unchanged. Let F_c^{cr} be the sum of these squared curls, each transverse edge once. The inequality $\|u+v+w\|^2 \leqslant 3(\|u\|^2+\|v\|^2+\|w\|^2)$ gives

$$\sum_{\nu\neq\mu} \sum_{\substack{s\in F_c \\ s+e_\nu\in F_c}} \|t_c(s+e_\nu)-t_c(s)\|^2 \leqslant 3(F_c^{\mathrm{cr}} + T_{\Sigma_-} + T_{\Sigma_+}) \,. \tag{A.17}$$

Let $I_{B,\mu}$ be the part of I_B in direction μ, and let $\bar{t}_c := L^{-3}\sum_{s\in F_c} t_c(s)$. The straight-path formula (A.2) has the exact decomposition

$$\begin{aligned} (Q_1 b)(c) &= L^{-1}\bar{t}_c + q_c \,, \\ q_c &= L^{-5}\sum_{s\in F_c}\sum_{i=0}^{L-2}(i+1)b_\mu\left(o+ie_\mu+\sum_{\nu\neq\mu}s_\nu e_\nu\right) \\ &\quad + L^{-5}\sum_{s\in F_c}\sum_{j=0}^{L-2}(L-1-j)b_\mu\left(o+(L+j)e_\mu+\sum_{\nu\neq\mu}s_\nu e_\nu\right) \,. \end{aligned} \tag{A.18}$$

Indeed, an internal lower-block bond at position i belongs to $i+1$ starting paths, and an internal upper-block bond at position j belongs to $L-1-j$ paths. Each crossing bond belongs to L paths. This is the positive-label content of [2, (2.125)]. There are at most $2L^4$ internal bonds in this sum, and every coefficient is at most L^{-4}. Cauchy–Schwarz therefore yields

$$\|q_c\|^2 \leqslant 2L^{-4}(I_{B_-,\mu}+I_{B_+,\mu}), \qquad L^3\|\bar{t}_c\|^2 \leqslant 2L^5\|(Q_1 b)(c)\|^2 + 4L(I_{B_-,\mu}+I_{B_+,\mu}) \,. \tag{A.19}$$

Combining the exact mean/variance decomposition of J_c with (A.14), (A.17) and (A.19) gives

$$J_c \leqslant \frac{3L^2}{4}(F_c^{\mathrm{cr}} + T_{\Sigma_-} + T_{\Sigma_+}) + 2L^5\|(Q_1 b)(c)\|^2 + 4L(I_{B_-,\mu}+I_{B_+,\mu}) \,. \tag{A.20}$$

The Periodic Sum and Its Payments. All sums over c run over the positive coarse bonds E_1, and $\mu(c)$ is the direction of c. Let $\mathscr{I} := \sum_B I_B$, $\mathscr{F} = \|\operatorname{curl} b\|^2$ and $\mathscr{Q} = \|Q_1 b\|^2$. Every positive fine bond is either internal to exactly one block or crosses exactly one positive coarse interface, so $\|b\|^2 = \mathscr{I} + \sum_c J_c$.

Every plaquette counted by F_c^{cr} crosses only the μ-interface: its transverse edge remains inside the face. It is therefore counted for at most one c. Each block has eight incident faces, and an internal μ-bond enters the last term of (A.20) only for the incoming and outgoing μ-interfaces. Consequently,

$$\begin{aligned} \mathscr{I} &\leqslant 4L^4\mathscr{F} \,, \\ \sum_c F_c^{\mathrm{cr}} &\leqslant \mathscr{F} \,, \\ \sum_c (T_{\Sigma_-}+T_{\Sigma_+}) &\leqslant 32L^3\mathscr{F} \,, \\ \sum_{c\in E_1}(I_{B_-,\mu(c)}+I_{B_+,\mu(c)}) &= 2\mathscr{I} \,. \end{aligned} \tag{A.21}$$

Thus summing (A.20) gives the explicit payment

$$\begin{aligned}\|b\|^2 &\leqslant (1+8L)\mathscr{I} + \frac{3L^2}{4}\mathscr{F} + 24L^5\mathscr{F} + 2L^5\mathscr{Q} \\ &\leqslant \left(4L^4 + 56L^5 + \frac{3L^2}{4}\right)\mathscr{F} + 2L^5\mathscr{Q} \\ &\leqslant 384L^5(\mathscr{F} + L^2\mathscr{Q})\,.\end{aligned} \tag{A.22}$$

The last inequality uses $L \geqslant 3$; its coefficients are bounded by $61L^5$ and $2L^5$, respectively. This proves (A.6).

Combining the bound (A.6) with the constraint-error bound gives $\|b\|^2 \leqslant A_L(1 + 16L^{-4}K_L)\|\operatorname{curl} a\|^2$. Since $a = b + D\lambda$, the bound $\|a\|^2 \leqslant 2\|b\|^2 + 2\|D\lambda\|^2$ proves, for $Qa = 0$,

$$\|a\|_2^2 \leqslant C_L(\|\operatorname{curl} a\|_2^2 + G^{(2)}(a))\,. \tag{A.23}$$

An explicit admissible constant is

$$C_L \coloneqq \max(64,\, 2A_L(1 + 16L^{-4}K_L) + 64K_L)\,. \tag{A.24}$$

At the flat background, differentiating the original action and exponential gauge functional in (4.5) gives $\langle a, \Delta_1 a\rangle = \|\operatorname{curl} a\|^2 + \|\Phi a\|^2$. Substituting $a = \mathsf{C}B$ in (A.23) uses $Q\mathsf{C}B = 0$ and $\|\mathsf{C}B\| \geqslant \|B\|$, because the free bonds are unchanged. Consequently the free Hessian satisfies

$$\mathsf{H} \geqslant \gamma_L I, \qquad \gamma_L = C_L^{-1}\,. \tag{A.25}$$

Construction on the Infinite Lattice. Let $E_\infty \coloneqq \mathbb{Z}^4 \times \{1,2,3,4\}$ denote positive fine bonds, with (x,μ) representing $(x, x+e_\mu)$. The coarse centres are $L\mathbb{Z}^4$, with the centred blocks and path families specified above. Removing the designated bonds gives $E_{*,\infty}$. The square-summable field spaces use the same real Lie-algebra inner product as the finite-torus calculation. On finitely supported fields, Φ_∞ and Q_∞ are defined by (A.1)–(A.2), and C_∞ by (4.3). The designated coefficient is $Q_{0,\infty} = L^{-4}I$, so the inverse in that definition is fibrewise. Let

$$\Delta_\infty \coloneqq \operatorname{curl}^*\operatorname{curl} + \Phi_\infty^*\Phi_\infty, \qquad \mathsf{H}_\infty \coloneqq \mathsf{C}_\infty^*\Delta_\infty\mathsf{C}_\infty\,. \tag{A.26}$$

The coefficient $Q_{0,\infty}$ follows from (A.2). The within-block paths entering Φ contain no designated bond. For $b_0(c)$, precisely L of the translated straight paths in the sum for $Q_1(c)$ traverse that bond: their transverse coordinates equal those of the coarse centres, and their longitudinal initial positions range over the L sites of the block. No translated path for another coarse bond contains it. Hence its coefficient is $L^{-5}L = L^{-4}$ and all off-diagonal coefficients vanish.

The action normalisation also agrees with the finite-torus Hessian. At the identity, the first plaquette phase is $\operatorname{curl} a$ and the first mean-transport phase is Φa. The mixed second derivative of $1 - \operatorname{Re}\operatorname{tr}_N e^{i(sX+tY)}$ at zero is $\operatorname{tr}_N(XY)$, the prescribed real inner product. The terms from second derivatives of the phases multiply the zero first derivative of this class function. Summing over the plaquettes and mean transports gives $\Delta_1 = \operatorname{curl}^*\operatorname{curl} + \Phi^*\Phi$ with the same normalisation on every torus.

Each map has finite range, bounded coefficients and finitely many entry types modulo $L\mathbb{Z}^4$. Its row and column absolute sums are bounded, so it extends uniquely to a bounded operator on its square-summable carrier. In particular H_∞ is bounded and self-adjoint on all of $\ell^2(E_{*,\infty};\mathfrak{g})$.

For every finitely supported B, the support of $\mathsf{C}_\infty B$ and all action terms meeting that latter support embed without wraparound in a sufficiently large compatible torus. Their quadratic form and norm are unchanged. Equation (A.25) therefore applies to this same field. Density of finitely supported fields gives

$$\gamma_L I \leqslant \mathsf{H}_\infty \leqslant h_L I, \qquad h_L = \max_{e \in E_{*,\infty}/L\mathbb{Z}^4} \sum_{f \in E_{*,\infty}} \|\mathsf{H}_\infty(e,f)\| \,. \tag{A.27}$$

The maximum uses representatives of the finitely many free-bond types. Symmetry gives the same column bound. Thus h_L is finite and independent of the torus and coupling, with $h_L \geqslant \gamma_L > 0$.

Let $b_L \coloneqq h_L + \gamma_L$ and $\vartheta_L = 1 - \gamma_L/b_L$. Then $0 < \vartheta_L < 1$ and $\|I - \mathsf{H}_\infty/b_L\| \leqslant \vartheta_L$. The inverse is

$$\mathsf{H}_\infty^{-1} = \frac{1}{b_L} \sum_{j=0}^{\infty} (I - \mathsf{H}_\infty/b_L)^j \,. \tag{A.28}$$

The series converges in operator norm. The fine-field covariance is the bounded positive operator $\mathsf{K}_\infty = \mathsf{C}_\infty \mathsf{H}_\infty^{-1} \mathsf{C}_\infty^*$. At the flat background every colour component has the same scalar operator, with scalar bond kernel $K_\infty(b, b')$.

Decay with Finite Constants. Let $\ell(b) \in \mathbb{Z}^4$ be the coarse index of the block containing the initial point of b, and let $d(b,b') \coloneqq \|\ell(b) - \ell(b')\|_\infty$. Let R_H be the maximum of 1 and the distances $d(e,f)$ for nonzero entries of H_∞. Let R_C be the maximum of $d(b,e)$ for nonzero entries of C_∞. These maxima can be taken over the finitely many entry types modulo coarse translation. Let

$$c_L^{\mathsf{C}} \coloneqq \max_{b \in E_\infty/L\mathbb{Z}^4} \sum_{e \in E_{*,\infty}} \|\mathsf{C}_\infty(b,e)\|, \qquad \mu_L \coloneqq \frac{-\log \vartheta_L}{R_H}, \qquad A_L^{\mathrm{ker}} \coloneqq \frac{(c_L^{\mathsf{C}})^2}{\gamma_L} e^{2\mu_L R_C} \,. \tag{A.29}$$

The jth power in (A.28) has zero (e,f) entry if $jR_H < d(e,f)$, and its norm is at most ϑ_L^j. Summing the remaining terms bounds the inverse entry by $\gamma_L^{-1} e^{-\mu_L d(e,f)}$. Using the two finite rows of C_∞ and $d(e,f) \geqslant d(b,b') - 2R_C$ yields

$$|K_\infty(b,b')| \leqslant A_L^{\mathrm{ker}} e^{-\mu_L d(b,b')} \,. \tag{A.30}$$

Comparison with the Periodic Covariance. Let the torus be compatible, with fine periods $N_\alpha = LK_\alpha$ and $K_\alpha \geqslant 3$. For $z \in \mathbb{Z}^4$, let $Nz \coloneqq (N_1 z_1, \dots, N_4 z_4)$. The local torus definitions of $Q, \Phi, \mathsf{C}, \Delta_1$ are the periodisations of their infinite-lattice definitions. Periodisation preserves their finite-range products. The absolutely convergent periodisation of the free inverse therefore solves the finite-torus inverse equation: applying H_T gives the periodised Kronecker delta. Uniqueness of the inverse and multiplication by the two constraint-extension maps give

$$K_T(b,b') = \sum_{z \in \mathbb{Z}^4} K_\infty(b, b' + Nz) \,. \tag{A.31}$$

Bonds on the right have fixed lifts and unchanged orientations. Let $K_{\min} \coloneqq \min_\alpha K_\alpha$ and

$$B_L^{\mathrm{img}} = \sum_{z \in \mathbb{Z}^4 \setminus \{0\}} e^{-\mu_L(\|z\|_\infty - 1)} < \infty \,. \tag{A.32}$$

The shell of maximum norm j has $(2j+1)^4-(2j-1)^4$ points. Since $d(b,b'+Nz) \geqslant K_{\min}\|z\|_\infty - d(b,b')$, the image sum satisfies

$$|K_T(b,b') - K_\infty(b,b')| \leqslant A_L^{\ker} B_L^{\mathrm{img}} e^{-\mu_L(K_{\min}-d(b,b'))} . \tag{A.33}$$

Thus every fixed finite collection of covariance entries converges as all coarse periods tend to infinity. The limiting connected-cluster construction uses the infinite-lattice covariance.

Appendix B Background Estimates at the First Step

This appendix proves the estimates stated in Section 3: Lemmas 3.8, 3.12, 3.24, 3.27 and 3.28. It fixes every constant named there that Section 3.5 does not fix itself, and places all of them in the one order of choice stated in Section B.8: L and the group with the group constants $c_{\mathrm{h}} = \sqrt{N}$, $c_{\mathrm{n}} = 1/N$, $\varrho_G = \pi/\sqrt{N}$, c, r_G, c_G below; the constants of the lemmas of this appendix in the order of their proofs; M_1, M_2 and M with the constants chosen after them; the coupling g_0, through the thresholds collected there; and last the torus, whose periods satisfy the conditions of (3.1) (multiples of LMR_1, at least $8M$; the scale R_1 depends on g_0, so the period conditions are imposed after g_0).

Group constants: c is a constant with $\|e^{iX} - \mathbf{1}\|_{\mathrm{op}} \leqslant c\, h(X)$ for all $X \in \mathfrak{g}$ ($c = \sqrt{N}$ serves, as $\|e^{iX} - \mathbf{1}\|_{\mathrm{op}} \leqslant \|X\|_{\mathrm{op}} \leqslant \sqrt{N} h(X)$). For every $N \times N$ matrix Y let $h(Y) := (N^{-1} \operatorname{tr} Y^*Y)^{1/2}$, which extends the normalised Hilbert–Schmidt norm of $\mathfrak{g}$ to all matrices; then $\|Y\|_{\mathrm{op}} \leqslant (\operatorname{tr} Y^*Y)^{1/2} = \sqrt{N}\, h(Y)$, so $\|Y\|_{\mathrm{op}} \leqslant c_{\mathrm{h}}|Y|$ for every matrix Y and either cutoff norm $|\bullet|$ (the operator norm or h), not only for Hermitian Y.

The exponential map restricts to an analytic diffeomorphism from the ball $\{X \in \mathfrak{g} \mid h(X) < \varrho_G\}$ onto an open neighbourhood N_G of the identity in G (the conventions of Section 3), with inverse $\log_G$; since G is compact and $\|g - \mathbf{1}\|_{\mathrm{op}}$ vanishes on G only at the identity, there is $r_G \in (0,1]$ with $\{g \in G \mid \|g-\mathbf{1}\|_{\mathrm{op}} \leqslant r_G\} \subset N_G$. The constant c_G with $h(\log_G g) \leqslant c_G \|g-\mathbf{1}\|_{\mathrm{op}}$ on that set exists: $\log_G$ is analytic, so Lipschitz with some constant c' on $\{\|g - \mathbf{1}\|_{\mathrm{op}} \leqslant r'\}$ for some $r' \in (0, r_G]$, and on $r' \leqslant \|g - \mathbf{1}\|_{\mathrm{op}} \leqslant r_G$ one has $h(\log_G g) < \varrho_G \leqslant (\varrho_G/r')\|g - \mathbf{1}\|_{\mathrm{op}}$, so $c_G = \max(c', \varrho_G/r')$ serves.

The radius r_M of the local group mean is fixed in Lemma B.9. The phase norm of Lemmas 3.6 and 3.7 is $\|A\|_\infty = \max_b h(A_b)$, and operator norms of entries and their derivatives are taken with respect to the Hilbert norms induced by h. Throughout, a branch is a branch in the sense of Section 3.1 (an admissible pair with $(A_{\mathrm{e}}, a) \in \mathcal{D}_\xi$ for some a); the reference data of Section 3.8 are treated in Section B.7.

B.1 Cubes, Localisation Domains and Tree Lengths

The partitions π_0 (side M) and π_1 (side LM) of Section 3.5 have cubes $Q_z = Mz + [-M/2, M/2]^4$ and $\Delta_z = LMz + [-LM/2, LM/2]^4$, reduced modulo the periods, with M_1 and M odd. Cube faces therefore lie at half-integer fine coordinates: every lattice site lies in the interior of exactly one cube of each partition, a fine bond meets at most two cubes of each partition, and a fine bond lies in the interior of a union of cubes exactly when both its endpoints lie in cubes of that union. A cube has M^4 sites and meets at most $8M^4$ positive fine bonds.

Two cubes of the same partition are wall adjacent when they share a three-dimensional face; a localisation domain of π_i is a finite nonempty union of cubes of π_i that is connected under wall adjacency ([8, p. 257]). A localisation domain Y of π_i has $N_i(Y)$ cubes and,

following [8, p. 257], tree length

$$d_i(Y) = \inf\{\, |\tau| \,|\, \tau \subset Y \text{ a connected finite union of segments meeting every cube of } Y \,\}, \tag{B.1}$$

the Euclidean length $|\tau|$ being measured in units of the cube side. Let $\ell_i(Y) \coloneqq d_i(Y)+5$. Two domains are compatible when they share neither a cube nor a wall ([10, (2.11)]); a domain is incompatible with itself.

Lemma B.1 (Geometry of Localisation Domains)**.** *Let $Y, Y_1, \dots, Y_m$ be localisation domains of one partition and $C_{\mathrm{g}} = 162$.*

1. *$N_i(Y) \leqslant C_{\mathrm{g}} \ell_i(Y)$ and $\ell_i(Y) \leqslant N_i(Y) + 4$.*
2. *If $\bigcup_j Y_j$ is a localisation domain, then $\ell_i(\bigcup_j Y_j) \leqslant \sum_j \ell_i(Y_j)$.*
3. *If Z is a localisation domain containing $Y_1, \dots, Y_k$ ($k \geqslant 0$, pairwise disjoint) and $Z \setminus \bigcup_j Y_j$ consists of n cubes, then $\ell_i(Z) \leqslant \sum_j \ell_i(Y_j) + 5n$.*
4. *The number of localisation domains with n cubes containing a given cube is at most 64^{n-1}, and for $s > C_{\mathrm{g}} \log 64$*

$$\sum_{Y \ni Q} e^{-s\ell_i(Y)} \leqslant B(s) = \frac{e^{-s/C_{\mathrm{g}}}}{1 - 64 e^{-s/C_{\mathrm{g}}}} \tag{B.2}$$

for every cube Q, on every torus. For $s \geqslant C_{\mathrm{g}}(\log 64 + \log 2)$, $B(s) \leqslant 2e^{-s/C_{\mathrm{g}}}$.

Proof. (i) Let τ realise (B.1) up to $\varepsilon \in (0, 1/2)$. A closed traversal of τ (each segment twice) has length at most $2(d_i(Y) + \varepsilon)$, so sample points along it at arclength spacing one number at most $2d_i(Y) + 2$. Every cube of Y contains a point of τ, which is within arclength $1/2$, hence Euclidean and ℓ^∞ distance $1/2$, of a sample; the centre of that cube is then within ℓ^∞ distance one of the sample. Cube centres form a unit lattice, so at most $3^4 = 81$ cubes are assigned to each sample, and $N_i(Y) \leqslant 81(2d_i(Y) + 2) \leqslant 162\, \ell_i(Y)$. For the second inequality, the centres of wall-adjacent cubes are joined along a spanning tree of the wall-adjacency graph; each of the $N_i(Y) - 1$ edges has length one and lies in Y, so $d_i(Y) \leqslant N_i(Y) - 1$.

(ii) Let $\tau_j \subset Y_j$ be trees with $|\tau_j| \leqslant d_i(Y_j) + \varepsilon$. The graph on $\{Y_j\}$ with an edge when $Y_j \cap Y_k \neq \varnothing$ is connected: a wall-adjacent chain of cubes of the union passes from Y_j to Y_k through consecutive cubes that share a wall, hence through domains that intersect. For each edge of a spanning tree, let $x \in Y_j \cap Y_k$, let $Q_j \ni x$ be a cube of Y_j and $p_j \in \tau_j \cap Q_j$; the segment $[p_j, x] \subset Q_j$ (length at most the diameter 2 of a unit cube in $\mathbb{R}^4$) is added, and likewise for k. The result is a connected union of segments inside $\bigcup Y_j$ meeting every cube, of length at most $\sum_j (d_i(Y_j) + \varepsilon) + 4(m-1)$. Letting $\varepsilon \to 0$, $d_i(\bigcup Y_j) + 5 \leqslant \sum_j d_i(Y_j) + 4m + 1 \leqslant \sum_j (d_i(Y_j) + 5)$ for $m \geqslant 1$.

(iii) This is (ii) for the family consisting of $Y_1, \dots, Y_k$ and the n single cubes, each of which has $\ell_i = 5$.

(iv) A depth-first traversal of a spanning tree of the wall-adjacency graph of a domain with n cubes, started at the given cube, is a word of length $2(n-1)$ over the eight wall directions, and determines the domain; this gives 64^{n-1}. By (i), $e^{-s\ell_i(Y)} \leqslant e^{-sN_i(Y)/C_{\mathrm{g}}}$, so the rooted sum is at most $\sum_{n \geqslant 1} 64^{n-1} e^{-sn/C_{\mathrm{g}}} = B(s)$. The last claim is $64e^{-s/C_{\mathrm{g}}} \leqslant 1/2$. *q.e.d.*

Lemma B.2 (Scale Comparison)**.** *Let Z be a localisation domain of π_0, $\widetilde{Z}$ the union of the M-cubes meeting Z, and Z' the union of the LM-cubes meeting* $\operatorname{int} \widetilde{Z}$.

This is the enlargement rule of Section 3.5, which takes the LM-cubes meeting $\operatorname{int} Q^{\sim}$ *for some cube Q of Z: indeed $\widetilde{Z} = \bigcup_{Q \subseteq Z} Q^{\sim}$, an LM-cube meeting* $\operatorname{int} Q^{\sim}$ *meets* $\operatorname{int} \widetilde{Z}$*, and conversely an LM-cube Δ meeting* $\operatorname{int} \widetilde{Z}$ *contains a nonempty open subset of $\widetilde{Z}$, a finite union of the closed sets $Q^{\sim}$, one of which therefore contains a nonempty open subset of Δ (a finite union of closed sets with empty interiors has empty interior), so Δ meets* $\operatorname{int} Q^{\sim}$ *for that Q. The same argument shows, for a union Z_0 of M-cubes with wall components $Z_{0,j}$, that $Z_0' = \bigcup_j Z_{0,j}'$.*

Then $Z \subset \operatorname{int} \widetilde{Z}$, $\widetilde{Z} \subset Z'$, Z' is a localisation domain of π_1, every point of $\widetilde{Z}$ is within ℓ^∞ distance $2M$ of every tree realising $d_0(Z)$ up to any $\varepsilon > 0$, and

$$\ell_1(Z') \leqslant a_L^{\mathrm{sc}} \ell_0(Z) + b_L^{\mathrm{sc}}, \qquad a_L^{\mathrm{sc}} = \frac{2049}{L}, \quad b_L^{\mathrm{sc}} = 5 + \frac{1024}{L}\,. \tag{B.3}$$

Proof. Every M-cube meeting a point of Z belongs to $\widetilde{Z}$, so $Z \subset \operatorname{int} \widetilde{Z}$. An M-cube $\widetilde{Q}$ of $\widetilde{Z}$ lies in a single LM-cube, which meets $\operatorname{int} \widetilde{Q} \subset \operatorname{int} \widetilde{Z}$; hence $\widetilde{Z} \subset Z'$. If $\widetilde{Q}$ meets $Q \subset Z$ only at a lower-dimensional face, the cubes between them sharing walls also meet Q, so $\widetilde{Z}$ is wall connected, $\operatorname{int} \widetilde{Z}$ is connected, and the LM-cubes meeting a connected open set are wall connected (a path avoiding lower-dimensional faces passes between consecutive cubes through a common wall).

Let $\tau \subset Z$ meet every M-cube of Z with $|\tau| \leqslant M(d_0(Z) + \varepsilon)$ in fine units. For $x \in \widetilde{Z}$, let $\widetilde{Q} \ni x$, let $Q \subset Z$ be a cube meeting $\widetilde{Q}$, and let $y \in \widetilde{Q} \cap Q$ and $p \in \tau \cap Q$; then $|x - p|_\infty \leqslant |x - y|_\infty + |y - p|_\infty \leqslant 2M$.

For (B.3), samples along a closed traversal of τ at arclength spacing LM number at most $2(d_0(Z) + \varepsilon)/L + 1$, and every point of τ is within Euclidean distance $LM/2$ of a sample. For $\Delta \subset Z'$ let $x_\Delta \in \Delta \cap \operatorname{int} \widetilde{Z}$; by the preceding paragraph it is within ℓ^∞ distance $2M + LM/2 \leqslant 7LM/6$ of a sample ($L \geqslant 3$), so the centre of Δ is within ℓ^∞ distance $5LM/3 < 2LM$ of that sample. The centres of LM-cubes form a lattice of spacing LM, and an open ℓ^∞ ball of radius $2LM$ contains at most $4^4 = 256$ of its points. Hence $N_1(Z') \leqslant 256(2(d_0(Z) + \varepsilon)/L + 1)$.

A connected union of segments in Z' meeting every LM-cube of Z' consists of $\tau \subset Z \subset Z'$ and, for each $\Delta \subset Z'$ not met by τ, the segments $[x_\Delta, y] \subset \widetilde{Q} \subset Z'$ and $[y, p] \subset Q \subset Z'$ chosen above, each of length at most $2M$. Its length is at most $M(d_0(Z) + \varepsilon) + 4MN_1(Z')$, so

$$d_1(Z') \leqslant \frac{d_0(Z) + \varepsilon}{L} + \frac{4}{L} \cdot 256\Big(\frac{2(d_0(Z) + \varepsilon)}{L} + 1\Big) \leqslant \frac{2049}{L}(d_0(Z) + \varepsilon) + \frac{1024}{L}\,.$$

Letting $\varepsilon \to 0$, adding 5 to both sides, and using $d_0(Z) \leqslant \ell_0(Z)$ gives (B.3). *q.e.d.*

B.2 The Regular-Background Bounds

Lemma B.3 (Entry Range and Stencils)**.** *For free bonds b, b' let $z(b)$ be the coarse label of the block containing b_- and $f_{bb'}(U)$ the entry function* (3.18) *with its entry stencil $\mathcal{S}_{bb'}$ (the union of the background sets of the terms of* (3.17) *whose free-input set contains both b and b'). If $|z(b) - z(b')|_1 > 2$ then $\mathcal{T}_{bb'} = \varnothing$ and $f_{bb'} = 0$. The stencil of a nonzero entry consists of bonds e with $|z(e) - z(b)|_1 \leqslant 2$; in fine units all bonds and sites of $\mathcal{S}_{bb'}$, and both endpoints of b and b', lie within ℓ^∞ distance $3L$ of b_-, so $\rho_L = 4L$ is an admissible entry range in Lemma 3.7. Each row has at most $164L^4$ nonzero entries.*

Proof. By (3.17) and its support sets, a term containing the free input b is $\mathfrak{a}_p$ with $b \in \mathcal{F}_r$ for a boundary bond r of p, $\mathfrak{g}_y$ with $b \in \mathcal{E}_y$, or $\mathfrak{t}_c$ with $b \in \mathcal{E}_c \cap E_*$. The free inputs of a

bond r have block labels among the endpoint blocks of r (Lemma 3.6): $z(r)$ for a free bond, $\{c_-, c_+\}$ for $r = b_0(c)$. The four vertices of a fine plaquette in directions μ, ν have block labels differing only in the μ and ν coordinates, by at most one each, so these labels, and the endpoint blocks of its boundary bonds, lie in a set of ℓ^1 diameter at most two. The free inputs of $\mathfrak{g}_y$ lie in one block, those of $\mathfrak{t}_c$ in the two blocks of c. Hence two free inputs of one term have $|z(b) - z(b')|_1 \leqslant 2$, and the background bonds of that term (plaquette bonds, the sets $\mathcal{E}_c$, and for $\mathfrak{t}_c$ the plaquettes meeting $b_0(c)$, whose vertices lie in the two blocks of c) have block labels within ℓ^1 distance two of $z(b)$. A site of a block with label within ℓ^1 distance two of $z(b)$ is within ℓ^∞ distance $2L + (L-1) < 3L$ of b_-; the endpoints of b and b' are such sites or within one of them. The ℓ^1 ball of radius two in $\mathbb{Z}^4$ has 41 points and a block has at most $4L^4$ positive bonds with initial site in it. *q.e.d.*

Lemma B.4 (Plaquette Deviation on the Collar)**.** *Let B_3, δ_{c} and the $O(1)$ be the constants of the comparison following [9, (1.19)], as recorded in (B3)(iii): for every $L^2M_2R_1$-cube $\square' \subseteq \Omega_1^{\sim -1} \setminus R_1$ (the $\square' \subset R_1^c$ of [9], for which $\square'^{\sim 4} \subseteq \Omega_1$, p.250), $|U_1(e)U_{1,\square'}(e)^{-1} - 1| < O(1)B_3 e^{-\delta_{\mathrm{c}} LM_2R_1}\varepsilon_1$ on the bonds e of $\square'^{\sim 2}$ (p.250).*

[9] does not state the dependence of these constants, which are Bałaban's ((H4)): B_3 is that of [6, Theorem 1, p. 279] and the δ of [9] the rate of the exponential decay property [6, (190)], which Bałaban invokes in [9, p. 266]; Bałaban reuses the symbols B_3 and δ at every step (p.268, the regularity bound $|\partial U_{k+1} - 1| < 2B_3\varepsilon_{k+1}(L^{-1}\eta)^2$ and the comparison with $B_3 \exp(-\delta LM_2R_{k+1})44d^2B_3\varepsilon_{k+1}$) and derives the comparison (3.19) from these facts and the bounds (106)–(108), (159)–(163) of [3] (p.268). The property used is the one recorded in (B3)(iii) and (H4): B_3, δ_{c} and the $O(1)$ do not depend on the coupling g_0 or on the branch, so that at fixed L and M_2 the factor $e^{-\delta_{\mathrm{c}} LM_2R_1}$ tends to zero as $g_0 \to 0$. The δ of [9] is written δ_{c} here to distinguish it from the Mayer constant δ of Section B.5.

The coupling threshold is reduced until $4\,O(1)B_3L^2e^{-\delta_{\mathrm{c}} LM_2R_1} \leqslant 1$, which holds for small g_0 because the constants do not depend on g_0 and $R_1 \geqslant (\log g_1^{-2})^{\mathfrak{r}} \to \infty$ as $g_0 \to 0$ ([9, (2.5) and (2.6), p. 255]: $g_1 \leqslant (1+\beta_0)g_0$ and $g_0 \leqslant (1+\beta_0)g_1$, so $g_1 \to 0$ with g_0), and R_{c} satisfies Lemma 3.19. Here $\varepsilon_1 = g_1 p_0(g_1)$ is the constant of the same name in [9, p. 246, (2.4)] and g_1 the first-step coupling.

On every branch, every fine plaquette p having a bond within distance $3R_{\mathrm{c}}$ of I satisfies

$$\|U_1(\partial p) - \mathbf{1}\|_{\mathrm{op}} \leqslant 2c_{\mathrm{h}}\varepsilon_1 L^{-2}\,. \tag{B.4}$$

Proof. Let b be such a bond and $\square'$ the cube of the partition of [9, (1.18)] into cubes of side $L^2M_2R_1$ that contains b_- (a partition, so the cube is unique). By Lemma 3.19(i) (stated for $d(b, I) \leqslant 3R_{\mathrm{c}}$; the standing condition $LMR_1 > 4R_{\mathrm{c}} + 4L^2M_2R_1 + 8$ gives $3R_{\mathrm{c}} + 1 + L^2M_2R_1 < LMR_1 - 1$), $b_- \in \Omega_1^{\sim -1}$ and $\square'$ is a good cube of [9, (1.18)], $\square' \subseteq \Omega_1^{\sim -1} \setminus R_1' \subseteq \Omega_1^{\sim -1} \cap R_1^c$; since $M \geqslant 8LM_2$, the layer of LMR_1-cubes around $\Omega_1^{\sim -1}$ inside Ω_1 contains $\square'^{\sim 4}$, as [9] states after (1.19).

The outer weight (3.4) contains the background test $\chi_1(\Omega_1)$ of [9, (2.17)] at $k = 1$, a function of V, and it is nonzero on a branch: the pair is admissible, so $D^{(11)}(V, V_0) \neq \varnothing$ (Section 3.1), and $\chi_1(\Omega_1) \neq 0$ at every point of $D^{(11)}$. That test is written on the lattice T_η, $\eta = L^{-1}$, whose unit is the block side, so its cubes of side LM_2R_1 have fine side $L^2M_2R_1$ and the plaquette threshold $\varepsilon_1\eta^2$ is $\varepsilon_1 L^{-2}$ ([9, (1.4), p. 246], prints the step-one test of the $L^2M_2R_1$-cubes with the threshold $\varepsilon_1 L^{-2}$ directly); its partition is "compatible with all other partitions" ([9, p. 257]; the term is not defined there, pp.245 and 257, and in the sense of [5, p. 98], (H4), the cubes of the coarser of two compatible partitions are unions of cubes of the finer), in particular with the partition of (1.18) into cubes of the same side, and two compatible

partitions into cubes of equal side coincide. Hence on the branch $\sup_{p'\subset\square'^{\sim}}|U_{1,\square'}(\partial p')-1|<\varepsilon_1 L^{-2}$ for the local minimum $U_{1,\square'}$ of [9, (2.16)]; a gauge transformation conjugates the plaquette holonomy and $|\bullet|$ is Ad-invariant, so the bound holds for the axial-gauge representative of $U_{1,\square'}$ (p.247) to which the comparison (B3)(iii) refers. The plaquette p contains b, hence lies in $\square'^{\sim}$.

The comparison (B3)(iii), stated for the cubes $\square'\subset R_1^c$, for which $\square'^{\sim 4}\subset\Omega_1$ and U_1, $U_{1,\square'}$ are determined by the same averages on $\square'^{\sim 3}$, applies to this $\square'$ and gives $|U_1(e)U_{1,\square'}(e)^{-1}-1|<O(1)B_3e^{-\delta_{\mathrm{c}}LM_2R_1}\varepsilon_1$ on the bonds e of $\square'^{\sim 2}$, hence $\|U_1(e)-U_{1,\square'}(e)\|_{\mathrm{op}}=\|U_1(e)U_{1,\square'}(e)^{-1}-\mathbf{1}\|_{\mathrm{op}}\leqslant c_{\mathrm{h}}|U_1(e)U_{1,\square'}(e)^{-1}-\mathbf{1}|\leqslant c_{\mathrm{h}}O(1)B_3e^{-\delta_{\mathrm{c}}LM_2R_1}\varepsilon_1\leqslant c_{\mathrm{h}}\varepsilon_1L^{-2}/4$ there (unitary invariance of the operator norm, the norm comparison of the preamble, valid for every matrix, and the threshold above).

Writing the plaquette holonomy as an ordered product of four unitaries and telescoping, $\|U_1(\partial p)-U_{1,\square'}(\partial p)\|_{\mathrm{op}}\leqslant\sum_{e\subset\partial p}\|U_1(e)-U_{1,\square'}(e)\|_{\mathrm{op}}\leqslant c_{\mathrm{h}}\varepsilon_1L^{-2}$. With $\|U_{1,\square'}(\partial p)-\mathbf{1}\|_{\mathrm{op}}\leqslant c_{\mathrm{h}}|U_{1,\square'}(\partial p)-1|<c_{\mathrm{h}}\varepsilon_1L^{-2}$ this gives $\|U_1(\partial p)-\mathbf{1}\|_{\mathrm{op}}\leqslant 2c_{\mathrm{h}}\varepsilon_1L^{-2}$. *q.e.d.*

Lemma B.5 (Local Gauge on a Rectangle)**.** *Let $\mathfrak{P}$ be a product of fine integer intervals of side at most $s_{\mathfrak{P}}$ (a lifted rectangle, injective modulo the periods), and let U be a configuration whose plaquettes inside $\mathfrak{P}$ satisfy $\|U(\partial p)-\mathbf{1}\|_{\mathrm{op}}\leqslant 2c_{\mathrm{h}}\varepsilon_1$ with $6c_{\mathrm{h}}s_{\mathfrak{P}}\varepsilon_1\leqslant r_G$. There is a site gauge u on $\mathfrak{P}$ such that every bond $b\subset\mathfrak{P}$ has $U^u(b)=e^{iA_b}$ with $A_b\in\mathfrak{g}$ and*

$$h(A_b)\leqslant\widehat{\mu}_{\mathfrak{P}}=6c_Gc_{\mathrm{h}}s_{\mathfrak{P}}\varepsilon_1\,. \tag{B.5}$$

Proof. Let x_0 be the corner of $\mathfrak{P}$ with minimal coordinates. For $x\in\mathfrak{P}$ let $P(x)$ be the lattice path from x_0 to x that runs first in direction e_1, then e_2,e_3,e_4; it stays in $\mathfrak{P}$. These paths form a spanning tree; let $u(x_0):=\mathbf{1}$ and $u(x'):=u(x)U(x,x')$ along tree bonds oriented away from the root. Then $U^u(x,x')=u(x)U(x,x')u(x')^{-1}=\mathbf{1}$ on tree bonds, and for any bond $(x,x')\subset\mathfrak{P}$, $U^u(x,x')=U(P(x))U(x,x')U(P(x'))^{-1}$ is the holonomy of the closed loop $P(x)\cdot(x,x')\cdot P(x')^{-1}$ based at x_0.

Let (x,x') have direction e_μ, so $x'=x+e_\mu$. The two tree paths share their runs in directions $\nu<\mu$ and their first $x_\mu-x_{0,\mu}$ steps in direction μ; after conjugation by the holonomy of that common initial segment, which preserves $\|\bullet-\mathbf{1}\|_{\mathrm{op}}$, the loop becomes $B\cdot(x,x')\cdot B'^{-1}\cdot(y+e_\mu,y)$, where $B=(b_1,\dots,b_k)$ is the staircase from $y=(x_1,\dots,x_\mu,x_{0,\mu+1},\dots,x_{0,4})$ to x in directions $\nu>\mu$, B' its translate by e_μ, and $k\leqslant 3s_{\mathfrak{P}}$. With $\mathcal{L}_j$ the loop built from $(b_1,\dots,b_j)$, $\mathcal{L}_j=(b_1\cdots b_{j-1})\,\partial p_j\,(b_1\cdots b_{j-1})^{-1}\cdot\mathcal{L}_{j-1}$, where $\partial p_j=b_je_\mu b_j'^{-1}e_\mu^{-1}$ is the boundary of the plaquette spanned by b_j and e_μ, which lies in $\mathfrak{P}$. Hence the holonomy of the loop is an ordered product of k conjugates of plaquette holonomies, and telescoping gives $\|U^u(x,x')-\mathbf{1}\|_{\mathrm{op}}\leqslant 2kc_{\mathrm{h}}\varepsilon_1\leqslant 6c_{\mathrm{h}}s_{\mathfrak{P}}\varepsilon_1\leqslant r_G$.

The element $g=U^u(x,x')\in G$ therefore lies in N_G, and $A_b=\log_G g\in\mathfrak{g}$ is the unique X in the ball $h(X)<\varrho_G$ with $e^{iX}=g$ (the group chart of the conventions), with $h(A_b)\leqslant c_G\|g-\mathbf{1}\|_{\mathrm{op}}\leqslant 6c_Gc_{\mathrm{h}}s_{\mathfrak{P}}\varepsilon_1$. *q.e.d.*

Where it is applied with Lemma B.4 below, the plaquette hypothesis holds because $2c_{\mathrm{h}}\varepsilon_1L^{-2}\leqslant 2c_{\mathrm{h}}\varepsilon_1$. The proof uses ε_1 only through the plaquette hypothesis and the condition $6c_{\mathrm{h}}s_{\mathfrak{P}}\varepsilon_1\leqslant r_G$; so the lemma holds with any $\varepsilon>0$ in place of ε_1, with the bound $6c_Gc_{\mathrm{h}}s_{\mathfrak{P}}\varepsilon$ in (B.5). Lemma B.11 uses it with $\varepsilon=\varepsilon_0/2$.

The radius τ_L of Lemma 3.6 is reduced so that

$$\tau_L\leqslant\min\Big(1,\ 2c_Gr_G,\ \frac{r_G}{2(L-1)c},\ \frac{r_M}{4(L-1)c\,c_G}\Big)\,, \tag{B.6}$$

with c, r_G, c_G the group constants of the preamble (depending on the group only) and r_M the radius of Lemma B.9 (depending on L and the group); this is the final value of τ_L, used in every statement. Let

$$\begin{aligned} c_{1,L} &\coloneqq 320000\, m_L, \qquad R_{\mathrm{p}} \coloneqq \max\Big(2, \Big\lceil 2\sqrt{c_{1,L}/\gamma_L}\Big\rceil\Big), \\ s_{\mathfrak{P}} &\coloneqq L(2R_{\mathrm{p}}+5)+1, \qquad C_{L,R_{\mathrm{c}}} \coloneqq 6c_G c_{\mathrm{h}} s_{\mathfrak{P}}\,, \end{aligned} \tag{B.7}$$

and let the collar range satisfy $R_{\mathrm{c}} \geqslant \max(2R_{\mathrm{loc}}+1,\ s_{\mathfrak{P}}+2)$. Every coarse period exceeds $2R_{\mathrm{p}}+8$, so that the coarse ℓ^∞ ball of radius $R_{\mathrm{p}}+3$ about any coarse label lifts injectively, and ε_1 satisfies

$$\widehat{\mu} = C_{L,R_{\mathrm{c}}}\varepsilon_1 < \frac{\tau_L}{2}, \qquad \max(1,c_L)\widehat{\mu} < \frac{\gamma_L}{4}\,. \tag{B.8}$$

Since $\tau_L \leqslant 2c_G r_G$, the first condition gives $6c_{\mathrm{h}} s_{\mathfrak{P}}\varepsilon_1 = \widehat{\mu}/c_G < r_G$, the hypothesis of Lemma B.5.

Lemma B.6 (Regional Entries)**.** *Let Ω be a region and U a background on which the regional form H^{Ω}_U is defined (Section 3.2: $\mathsf{M}(U)$ defined and $\alpha_c(U)$ invertible on the rows of $\Omega^{[1]}$, the local group means $\mathcal{U}_U(y,x)$ defined for $y \in \Omega^{(1)}$). For all $(b,b') \in \Omega^* \times \Omega^*$,*

$$(\mathsf{H}^{\Omega}_U)_{bb'} = f_{bb'}(U|_{\mathcal{S}_{bb'}})\,, \tag{B.9}$$

the entry function (3.18) *evaluated on the background restricted to the entry stencil $\mathcal{S}_{bb'}$ of Lemma B.3, and $f_{bb'} = 0$ when $|z(b)-z(b')|_1 > 2$. In particular, on every branch, $(H_\xi)_{bb'} = f_{bb'}(U_1|_{\mathcal{S}_{bb'}})$ for all $b,b' \in \Omega_1^*$, and for $b,b' \in I$ the stencil $\mathcal{S}_{bb'}$ lies inside Ω_1 (within $3L < R_{\mathrm{c}}$ of I, Lemma 3.19(i)).*

Proof. This is the compression identity (3.19) of Section 3.2, whose proof is recalled: the regional form is the sum of the terms of (3.17) with index in Ω (plaquettes meeting Ω, centres in Ω, rows in $\Omega^{[1]}$), its entry at (b,b') is the sum of the polarised entries of those of these terms whose free-input set contains b and b', and every term whose free-input set contains a bond b of Ω^* has its index in Ω: a plaquette term $\mathfrak{a}_p$ with $b \in \mathcal{F}_r$, $r \subset \partial p$, has $r = b$ or $r = b_0(c)$ with $b \in \mathcal{E}_c$, and in either case p contains a bond with an endpoint in the block of the endpoint of b that lies in Ω (b has such an endpoint, being a bond of Ω; the bonds of $\mathcal{E}_c$ have both endpoints in the two blocks of c, and $b_0(c)$ has one endpoint in each), and that block lies in Ω, a union of blocks; a gauge term $\mathfrak{g}_y$ with $b \in \mathcal{E}_y$ has $y \in \Omega$; a row term $\mathfrak{t}_c$ with $b \in \mathcal{E}_c \cap E_*$ has an endpoint block of c containing an endpoint of b, so $c \in \Omega^{[1]}$.

Hence the set of terms contributing to the regional entry is exactly $\mathcal{T}_{bb'}$, and the entry is $f_{bb'}(U|_{\mathcal{S}_{bb'}})$, a finite formula in the background on the stencil in which the rowwise elimination (3.12) uses only the rows of the terms in $\mathcal{T}_{bb'}$, which are rows of $\Omega^{[1]}$, where $\alpha_c(U)$ is invertible: the rows of $\mathfrak{g}_y$ and $\mathfrak{t}_c$ are rows with a centre in Ω by the above, and a row c' of $\mathfrak{a}_p$ has $b_0(c') \subset \partial p$, whose endpoints are the two sites of the segment between the centres of c' next to the common face of its blocks; the other two vertices of p are their translates by one unit in a direction perpendicular to c', which stay in the same two blocks as $L \geqslant 3$; so all four vertices of p lie in the two blocks of c', one of which meets Ω since p does, hence lies in Ω, and $c' \in \Omega^{[1]}$.

The vanishing for $|z(b)-z(b')|_1 > 2$ is Lemma B.3. For the last claim, by Lemma B.3 every site of $\mathcal{S}_{bb'}$ is within $3L$ of $b_- \in I$, hence within $3R_{\mathrm{c}}$ of I and in $\Omega_1^{\sim -1}$ by Lemma 3.19(i). *q.e.d.*

Proof of Lemma 3.8. For $B \in \mathfrak{g}^I$ extended by zero, Lemma B.6 gives

$$\langle B, H_\xi B\rangle = \sum_{b,b' \in I} \langle B_b, f_{bb'}(U_1|_{\mathcal{S}_{bb'}})B_{b'}\rangle$$

with the local entries, whose stencils lie inside Ω_1; the vanishing $H_{\xi,II}(b,b')=0$ for $d(b,b')>\rho_L$ is Lemma B.3. The quantities m_L, c_L of (3.22) bound the row sums $\sum_{b'}$ of $\|f_{bb'}(A)\|$ and $\|Df_{bb'}(A)\|$ over $\|A\|_\infty\leqslant\tau_L$; since $f_{b'b}(A)=f_{bb'}(A)^*$ (the form is symmetric) and hence $Df_{b'b}(A)[X]=(Df_{bb'}(A)[X])^*$, the column sums obey the same bounds.

Partition of unity. Let $\mathcal{Z}_T$ be the coarse grid and $\psi(u)=\prod_{\mu=1}^4(1-|u_\mu|/R_{\mathrm{p}})_+$ on the lifted coarse ball of radius R_{p}, extended by zero and periodised (injectively, by the period condition). For $w\in\mathcal{Z}_T$ let

$$\chi_w(b)\coloneqq\frac{\psi(z(b)-w)}{S^{1/2}},\qquad S\coloneqq\sum_u\psi(u)^2=\Big(\sum_{k\in\mathbb{Z}}(1-|k|/R_{\mathrm{p}})_+^2\Big)^4\geqslant\Big(\frac{R_{\mathrm{p}}}{4}\Big)^4\,, \tag{B.10}$$

the last inequality because at least R_{p} integers k satisfy $|k|\leqslant R_{\mathrm{p}}/2$ and each contributes at least $1/4$. Then $\sum_w\chi_w(b)^2=1$ for every b. A one-step difference $|\psi(u)-\psi(u+e_\nu)|$ is at most $1/R_{\mathrm{p}}$, since only one factor changes and the others are at most one, and it vanishes unless u or $u+e_\nu$ lies in the support, a set of at most $(2R_{\mathrm{p}}+2)^4\leqslant(5R_{\mathrm{p}})^4$ points. Telescoping along a coarse path of $n=|z(b)-z(b')|_1$ steps and using $(\sum_{i=1}^n a_i)^2\leqslant n\sum a_i^2$,

$$\begin{aligned}\sum_w(\chi_w(b)-\chi_w(b'))^2&\leqslant\frac{1}{S}\,n\sum_{i=1}^n\sum_w(\text{one-step difference})^2\\&\leqslant\frac{256}{R_{\mathrm{p}}^4}\,n^2\,\frac{(5R_{\mathrm{p}})^4}{R_{\mathrm{p}}^2}=\frac{160000}{R_{\mathrm{p}}^2}\,|z(b)-z(b')|_1^2\,.\end{aligned} \tag{B.11}$$

Since $\sum_w\chi_w(b)\chi_w(b')-1=-\frac12\sum_w(\chi_w(b)-\chi_w(b'))^2$, Lemma B.3 ($n\leqslant2$ on nonzero entries) and the Schur bound $\sum_{b,b'}\|f_{bb'}\|h(B_b)h(B_{b'})\leqslant m_L\|B\|^2$, valid because row and column sums of $\|f_{bb'}\|$ are at most m_L (shown below), give

$$\Big|\sum_w\langle\chi_wB,H_\xi\chi_wB\rangle-\langle B,H_\xi B\rangle\Big|\leqslant\frac12\cdot\frac{160000\cdot4}{R_{\mathrm{p}}^2}\,m_L\|B\|^2=\frac{c_{1,L}}{R_{\mathrm{p}}^2}\|B\|^2\leqslant\frac{\gamma_L}{4}\|B\|^2\,. \tag{B.12}$$

Local comparison. Let $\chi_wB\neq0$; since B is supported on I, some $b\in I$ has $\chi_w(b)\neq0$, so the patch $\{b\,|\,|z(b)-w|_\infty\leqslant R_{\mathrm{p}}\}$ meets I. Let $\mathfrak{P}_w$ be the lifted rectangle of fine sites in blocks with $|z-w|_\infty\leqslant R_{\mathrm{p}}+2$, of side at most $L(2R_{\mathrm{p}}+5)+1=s_{\mathfrak{P}}$; it lies within fine distance $s_{\mathfrak{P}}+1\leqslant2R_{\mathrm{c}}-1$ of I, so Lemma B.4 applies to all its plaquettes and Lemma B.5 gives a gauge u_w on $\mathfrak{P}_w$ with phases $h(A_e)\leqslant\widehat{\mu}$, extended by the identity. By Lemma B.3 the stencils of all entries with both labels in the patch lie in $\mathfrak{P}_w$. With $T_w=R_{u_w,*}$ the orthogonal action of Lemma 3.7,

$$\begin{aligned}\langle\chi_wB,H_\xi\chi_wB\rangle&=\sum_{b,b'}\langle(T_w\chi_wB)_b,f_{bb'}(A)(T_w\chi_wB)_{b'}\rangle\\&=\langle T_w\chi_wB,\mathsf{H}_{\mathbf{1}}T_w\chi_wB\rangle+\sum_{b,b'}\langle(T_w\chi_wB)_b,(f_{bb'}(A)-f_{bb'}(0))(T_w\chi_wB)_{b'}\rangle\,,\end{aligned}$$

where $f_{bb'}(A)$ is the entry at the gauge-transformed background $U_1^{u_w}=e^{iA}$ on the stencil; the first equality is the entrywise gauge covariance (3.21), $f_{bb'}(U^u)=\mathrm{Ad}_{u(b_-)}f_{bb'}(U)\,\mathrm{Ad}^{-1}_{u(b'_-)}$, together with $(T_w\chi_wB)_b\coloneqq\mathrm{Ad}_{u_w(b_-)}\chi_w(b)B_b$ and the invariance of the inner product.

The map $A\mapsto f_{bb'}(A)$ is analytic on $\|A\|_\infty\leqslant\tau_L$ (Lemma 3.7) and $\|A\|_\infty\leqslant\widehat{\mu}<\tau_L/2$, so $\|f_{bb'}(A)-f_{bb'}(0)\|\leqslant\sup_{t\in[0,1]}\|Df_{bb'}(tA)\|\,\widehat{\mu}$, and the row and column sums of these bounds are at most $c_L\widehat{\mu}$ by (3.22); the Schur test gives

$$\Big|\langle\chi_wB,H_\xi\chi_wB\rangle-\langle T_w\chi_wB,\mathsf{H}_{\mathbf{1}}T_w\chi_wB\rangle\Big|\leqslant c_L\widehat{\mu}\,\|\chi_wB\|^2\,. \tag{B.13}$$

The operator $\mathsf{H}_{\mathbf{1}}$ is the flat Hessian H of (4.7): at the identity $J_{\mathbf{1}} = 0$, every relative mean $\mathcal{U}_{\mathbf{1}}(y,x)$ is the identity, so $\mathscr{G}_{\mathbf{1}}$ is the gauge functional of (4.5), and $\mathsf{C}_{\mathbf{1}}$ is (4.3). By (A.25) on the branch's torus, $\langle T_w\chi_w B, \mathsf{H}_{\mathbf{1}} T_w\chi_w B\rangle \geqslant \gamma_L \|\chi_w B\|^2$. Summing over w and using $\sum_w \|\chi_w B\|^2 = \|B\|^2$ and (B.8),

$$\langle B, H_\xi B\rangle \geqslant \Big(\gamma_L - \frac{c_{1,L}}{R_{\mathrm{p}}^2} - c_L\widehat{\mu}\Big)\|B\|^2 \geqslant \frac{\gamma_L}{2}\|B\|^2 \,. \tag{B.14}$$

Upper bound. For a row $b \in I$ all entries $f_{bb'}$, b' any free bond, have stencils in $\mathfrak{P}_{z(b)}$ (the patch of $w = z(b)$ meets I at b; a nonzero entry has its stencil within ℓ^∞ distance $3L$ of b_- by Lemma B.3, hence in blocks z with $|z - z(b)|_\infty \leqslant 3 \leqslant R_{\mathrm{p}} + 2$); in the gauge $u_{z(b)}$ of that patch their phases are at most $\widehat{\mu} \leqslant \tau_L$, and $\|f_{bb'}\|$ is unchanged by the fibre rotations of (3.21), which are orthogonal. Hence, by Lemma B.6 on $\Omega_1^* \times \Omega_1^*$ and (3.22), $\sum_{b'\in\Omega_1^*} \|(H_\xi)_{bb'}\| = \sum_{b'\in\Omega_1^*} \|f_{bb'}(U_1|_{\mathcal{S}_{bb'}})\| \leqslant m_L$, which is the row-sum statement of the lemma; in particular $\sum_{b'\in I} \|f_{bb'}(U_1)\| \leqslant m_L$, and by the symmetry noted at the start the column sums of $H_{\xi,II}$ obey the same bound. The Schur test gives $\|H_{\xi,II}\| \leqslant m_L$. Both bounds pass to principal restrictions: the lower bound by compression, the upper by the Schur test on fewer entries. This proves (3.23) and the range statement. *q.e.d.*

B.3 Uniform Charts and Primitive Bounds

Primitives are evaluated on the input $A = g_0\mathsf{C}_U B$ with the free vector $B = a + tA_{\mathrm{e}}$, $t \in [0,1]$ (the vacuum potentials of Section 3.6 have $t = 0$). For a coarse row c the relevant input is $A|_{\mathcal{E}_c}$, with the Hilbert norm on $\mathfrak{g}^{\mathcal{E}_c}$ (written $\|A|_{\mathcal{E}_c}\|$ or $\|A\|$).

Lemmas B.8, B.9, and B.12 prove Lemma 3.12 for every row c and every centre y within $2R_{\mathrm{c}} + 2L$ of I, with the primitive sizes (B.18).

Definition B.7 (Gauge Patches)**.** A background U admits a gauge patch on a set S of bonds when there is a site gauge u in which the phases ϕ, $U^u = e^{i\phi}$, satisfy $\|\phi\|_\infty \leqslant \tau_L/2$ on S. For a phase q, $q^u := R_u q$ with $(R_u q)(b) := \mathrm{Ad}_{u(b_-)}\, q(b)$, so that $\|q^u|_S\| = \|q|_S\|$. Norms of phases on S are the Hilbert norms of $\mathfrak{g}^S$ and of its complexification (so $\|d\| = h(d)$ for a single $d \in \mathfrak{g}$).

Lemma B.8 (Uniform Implicit Chart)**.** *There are constants $\varrho_*, \vartheta_* > 0$, $H_*, K_* \geqslant 1$, $C_{\mathsf{C}} \geqslant \max(1, (4L^3(L-1))^{1/2})$, and C_k^{a} for $0 \leqslant k \leqslant 5$, depending on L and the group only, with $2\vartheta_* < \varrho_*$, $2\vartheta_* \leqslant \varrho_G$, $\vartheta_* \leqslant \tau_L/2$ and $4K_*H_*\vartheta_* < 1$, such that for every background U admitting a gauge patch on $\mathcal{E}_c$:*

1. *$\|h_U(c)\| \leqslant H_*$; $\|\mathsf{C}_U B\|_{\mathcal{E}_c} \leqslant C_{\mathsf{C}} \max_{b\in\mathcal{E}_c\cap E_*} |B_b|$ and $\|\mathsf{C}_U B\|_{\mathcal{E}_c} \leqslant C_{\mathsf{C}}\|B|_{\mathcal{E}_c\cap E_*}\|$ (Hilbert norms); and $\mathcal{N}_{U,c}$ is analytic on the open complex ball $\|A\| < 2\varrho_*$ of $\mathfrak{g}_{\mathbb{C}}^{\mathcal{E}_c}$, real on real A, with $\mathcal{N}_{U,c}(0) = 0$, $D\mathcal{N}_{U,c}(0) = 0$ and $\|D^2\mathcal{N}_{U,c}\| \leqslant K_*$ on the closed ball $\|A\| \leqslant \varrho_*$.*

2. *On the open complex ball $\|A\| < \vartheta_*$ there is an analytic function $\varphi_c = \widetilde{D}_{U,c}$, real on real A, with $\varphi_c(0) = 0$, $D\varphi_c(0) = 0$, such that for every A in that ball $\varphi_c(A)$ is the unique solution d of the row equation (3.24) with $\|d\| \leqslant 2K_*\vartheta_*^2 =: \rho_{\mathrm{d}}$ (for real A, the small solution of Section 3.3); it agrees with the germ of Lemma 3.9 and satisfies*

$$\|\widetilde{D}_{U,c}(A)\| \leqslant 2K_*\|A\|^2, \qquad \|D_A\widetilde{D}_{U,c}(A)\| \leqslant C_{\mathsf{D}}\|A\| \,, \qquad C_{\mathsf{D}} := 4K_* \,. \tag{B.15}$$

3. *On the closed real ball $\|A\| \leqslant \vartheta_*/2$ the row determinant of c (Definition 3.11) is positive. On a complex neighbourhood of the closed complex ball $\|A\| \leqslant \vartheta_*/2$ the functions $\widetilde{D}_{U,c}$, the phase $q_U(A)|_{\mathcal{E}_c} = A - h_U\widetilde{D}_U(A)$, the normalised Haar logarithm*

$\log(\sigma(q_U(A)(b))/\sigma(0))$, $b \in \mathcal{E}_c$, *and the logarithm of the row determinant (both logarithms continued analytically from the real ones) are analytic, and on the closed ball their derivatives of order* $k \leqslant 5$ *are bounded in operator norm by* C_k^{a}*; for* $\widetilde{D}_{U,c}$ *this holds for* $k \leqslant 6$ *with* $C_k^{\mathsf{D}} := e^k k!\, 2K_*\vartheta_*^2(2/\vartheta_*)^k$*. The two logarithms vanish at* $A = 0$*, and the first derivative of the Haar logarithm vanishes there.*

Lemma B.9 (Uniform Local Means)**.** *With the constants of Lemma B.8, there are* $r_M, r'_M > 0$ *with* $r'_M \leqslant \varrho_G$*, depending on* L *and the group only, such that the following holds. Let* U *admit a gauge patch* u *on* $\mathcal{E}_y$ *for a block centre* y*. For every real* q *with* $\|q|_{\mathcal{E}_y}\| \leqslant \vartheta_*$*, the path transports* $(e^{iq^u}U^u)(\gamma)$*,* $\gamma \in \mathcal{G}(y,x)$*,* $x \in \mathcal{B}_y$*, are within* r_G *of* $\mathbf{1}$ *in operator norm, their logarithms* $\log_G$ *have* $h < r_M$*, and the local group mean* $\mathcal{U}_{e^{iq^u}U^u}(y,x)$ *of* (4.5) *([8, (0.10)–(0.11)]), the unique solution near the diagonal value, exists, of the form* e^{iX} *with* $h(X) < r'_M \leqslant \varrho_G$*; the relative mean* $\mathcal{U}_{e^{iq}U}(y,x)\mathcal{U}_U(y,x)^{-1}$ *lies within* r_G *of* $\mathbf{1}$ *in operator norm (the domain of* $\log_G$*), and it and the gauge term* $\mathscr{G}_{y,U}(q)$ *are restrictions to real* q *of functions analytic on the open complex ball* $\|q|_{\mathcal{E}_y}\| < 2\vartheta_*$*, with derivatives up to order five bounded by* C_k^{a} *on the closed complex ball of radius* ϑ_**, uniformly in the background, and* $\mathscr{G}_{y,U}(0) = 0$*,* $D\mathscr{G}_{y,U}(0) = 0$*.*

Proof of Lemmas B.8 and B.9. All formulas depending on ϑ_*, including ρ_{d} and C_k^{D}, use its final reduced value. There are finitely many row types modulo $L\mathbb{Z}^4$. For each type the maps $U \mapsto h_U(c)$, C_U and $(U,A) \mapsto \mathcal{N}_{U,c}(A)$ are compositions of exponentials, relative block averages, the fibre inverse $\alpha_c(U)^{-1}$ and the chart logarithm, all analytic for phases $\|\phi\|_\infty \leqslant \tau_L$ and A near zero (Lemma 3.6 gives the singular-value bound $L^{-4}/2$ on α_c), and real for real arguments.

Under the simultaneous gauge transformation of Lemma 3.7, Q_U, h_U, C_U and $\mathcal{N}_U$ intertwine with the orthogonal fibre action $R_{u,*}$ (proof of that lemma), which preserves the Hilbert norms and the maximum norm $\max_b |B_b|$ ($|\bullet|$ is Ad-invariant); so the bounds in (i)–(iii) may be proved in the gauge u and transferred to U.

The set of phase configurations with $\|\phi\|_\infty \leqslant \tau_L/2$ on $\mathcal{E}_c$ is compact, so a common complex radius $2\varrho_*$ of analyticity and bounds H_*, K_*, C_{C} exist (the two bounds on C_U are two operator norms of the same linear map, each bounded on the compact set; C_{C} is the larger, increased if necessary to $\max(1, (4L^3(L-1))^{1/2})$, which only weakens the inequalities); $\mathcal{N}_{U,c}(0) = 0$ because the zero fluctuation reproduces the background averages, and $D\mathcal{N}_{U,c}(0) = 0$ as in the proof of Lemma 3.9.

(ii) Let A be in the complex ball with $\|A\| \leqslant \vartheta \leqslant \vartheta_*$ and let $\Phi(d) := \mathcal{N}_{U,c}(A - h_U d)$ on the closed complex ball $\|d\| \leqslant 2K_*\vartheta^2$. Then $\|A - h_U d\| \leqslant \vartheta + 2H_*K_*\vartheta^2 \leqslant 2\vartheta \leqslant \varrho_*$; Taylor's formula with $\mathcal{N}(0) = 0$, $D\mathcal{N}(0) = 0$ gives $\|\Phi(d)\| \leqslant \frac{1}{2}K_*(2\vartheta)^2 = 2K_*\vartheta^2$, and $\|D\Phi(d)\| \leqslant \|D\mathcal{N}(A - h_U d)\|H_* \leqslant K_* \cdot 2\vartheta H_* < \frac{1}{2}$. So Φ is a contraction of the closed ball into itself, with a unique fixed point; taking $\vartheta = \vartheta_*$ gives the uniqueness in the ball $\|d\| \leqslant \rho_{\mathrm{d}}$ for every A with $\|A\| < \vartheta_*$, and taking $\vartheta = \|A\|$ gives the first bound of (B.15); the fixed point $\varphi_c(A)$ is analytic in A on the open ball as the uniform limit of the analytic iterates $\Phi^n(0)$ (each iterate is analytic in A, and the convergence is geometric, uniformly in A), and it is real for real A because the iterates are.

Differentiating the fixed-point equation, $D_A\widetilde{D} = (\mathbf{1} + D\mathcal{N}(x)h_U)^{-1}D\mathcal{N}(x)$ with $x = A - h_U\widetilde{D}$, $\|D\mathcal{N}(x)\| \leqslant K_*\|x\| \leqslant 2K_*\|A\|$ and $\|(\mathbf{1} + D\mathcal{N}(x)h_U)^{-1}\| \leqslant 2$; this is the second bound, and $D\varphi_c(0) = 0$. The germ of Lemma 3.9, constructed there from Lemma 3.6, $\mathcal{N}_{U,c}(0) = 0$, $D\mathcal{N}_{U,c}(0) = 0$ and the analyticity of $\mathcal{N}_{U,c}$ near zero, is the solution near zero, and uniqueness in the ball identifies the two.

(iii) Cauchy's estimate on the ball of radius ϑ_*, where $\|\widetilde{D}_{U,c}\| \leqslant 2K_*\vartheta_*^2$, bounds the k-th derivative at a point of the closed ball of radius $\vartheta_*/2$ in a fixed direction v, $\|v\| \leqslant 1$, by $k!\,2K_*\vartheta_*^2(\vartheta_*/2)^{-k}$ for every k, and the polarisation identity for symmetric k-linear maps bounds the operator norm of $D^k\widetilde{D}_{U,c}$ by e^k times the diagonal bound; this gives C_k^{D} for $0 \leqslant k \leqslant 6$. The constants C_k^{a} bounding the other functions are chosen at the end of the proof. The phase $q_U(A)|_{\mathcal{E}_c} = A - h_U\widetilde{D}_U(A)$ is a composition of $\widetilde{D}_{U,c}$ with the fixed linear map $h_U(c)$ of norm at most H_*, so it is analytic on the ball of radius ϑ_* and its derivatives are bounded by the chain rule.

The Haar density: $\sigma(q)\,\mathrm{d}q$ is the pullback of Haar measure under $q \mapsto e^{iq}$ on the chart ball $h(q) < \varrho_G$, so $\sigma(q) = \sigma(0)|\det D(q \mapsto e^{iq})(q)|$, the determinant of the differential taken with respect to left-invariant frames; the determinant is analytic in q and nonvanishing on the chart ball (the chart has invertible differential, conventions of Section 3), so it has constant sign on the connected ball, positive since it equals one at $q = 0$; thus σ is analytic and positive there, and even, $\sigma(-q) = \sigma(q)$, because Haar measure is invariant under $g \mapsto g^{-1}$ and $e^{-iq} = (e^{iq})^{-1}$. Hence $\log(\sigma/\sigma(0))$, the real logarithm, extends analytically to a complex neighbourhood of the real chart ball, vanishes at 0 with vanishing first derivative (evenness), and its composition with $q_U(A)(b)$ (a phase with $h(q_U(A)(b)) \leqslant \|A\| + H_*2K_*\|A\|^2 \leqslant \vartheta_*$ on the ball) is analytic on a complex neighbourhood of the closed ball $\|A\| \leqslant \vartheta_*/2$ after reducing ϑ_* so that the compositions stay in the domain of analyticity, with derivatives bounded on the closed ball by the chain rule, and its first derivative at $A = 0$ is $D\log\sigma(0) \circ Dq_U(0) = 0$.

The determinant argument $\mathbf{1} - h_U(c)\partial_{A(b_0(c))}\widetilde{D}_{U,c}(A)$ equals $\mathbf{1}$ at $A = 0$ since $D\widetilde{D}(0) = 0$, and is within $H_*C_{\mathsf{D}}\vartheta_*/2 < 1/2$ of $\mathbf{1}$ in operator norm on the ball; for real A it is a real endomorphism of $\mathfrak{g}$ within $1/2$ of $\mathbf{1}$, whose eigenvalues have positive real part, so its determinant is positive, and on a complex neighbourhood of the closed ball $\|A\| \leqslant \vartheta_*/2$ the determinant stays in the disc of radius $1/2$ about 1 (after reducing ϑ_* once more if needed, by continuity and compactness), where log is analytic; so the real logarithm of the determinant extends analytically to that neighbourhood and vanishes at $A = 0$, where the determinant is one. Its derivatives of order $k \leqslant 5$ are bounded on the closed ball by the chain rule, since the argument contains $\partial_{A(b_0(c))}\widetilde{D}_{U,c}$ and so involves the derivatives of $\widetilde{D}_{U,c}$ up to order six, bounded above.

The bounds are uniform in the background by the compactness of the set of phase configurations. All reductions of ϑ_* are by constants depending on L and the group only, over the finitely many row types.

Lemma B.9. The fibre action is orthogonal on real phases and unitary on their complexifications, so it preserves the phase balls and operator norms of multilinear derivatives. In the gauge u the links of $\mathcal{B}_y$ are $e^{i\phi}$ with $\|\phi\|_\infty \leqslant \tau_L/2$, and $h(q_b) \leqslant \|q|_{\mathcal{E}_y}\| \leqslant \vartheta_* \leqslant \tau_L/2$ for every $b \in \mathcal{E}_y$. A path of $\mathcal{G}(y,x)$ has at most $2(L-1)$ bonds (four coordinate runs of length at most $(L-1)/2$), and for unitaries $\|U_1 \cdots U_n - \mathbf{1}\|_{\mathrm{op}} \leqslant \sum_j \|U_j - \mathbf{1}\|_{\mathrm{op}}$, so every path transport $g_\gamma = (e^{iq^u}U^u)(\gamma)$ lies within $2(L-1)c(\tau_L/2 + \vartheta_*) \leqslant 2(L-1)c\,\tau_L \leqslant r_G$ of $\mathbf{1}$ in operator norm, by (B.6); hence $A_\gamma = \log_G g_\gamma$ is defined with $h(A_\gamma) \leqslant c_G \cdot 2(L-1)c\,\tau_L \leqslant r_M/2$, again by (B.6).

The local group mean of a family $(g_\gamma)_{\gamma\in\mathcal{G}(y,x)}$ is the solution $\mathcal{U}$ near the diagonal value of $\sum_\gamma \log_G(g_\gamma\mathcal{U}^{-1}) = 0$; with $g_\gamma = e^{iA_\gamma}$ and $\mathcal{U} = e^{iX}$, let $F(X,(A_\gamma)) := \sum_\gamma \log_G(e^{iA_\gamma}e^{-iX})$, an analytic map on a neighbourhood of $(0,0)$ in $\mathfrak{g}\times\mathfrak{g}^{\mathcal{G}(y,x)}$ with $F(0,0) = 0$ and $D_XF(0,0) = -n\,\mathbf{1}$, $n = |\mathcal{G}(y,x)|$ (as $D\log_G(\mathbf{1}) = \mathbf{1}$). The product neighbourhood in the complexified Lie-algebra variables is chosen so that every $e^{iA_\gamma}e^{-iX}$ lies in the analytic logarithm domain there. The analytic implicit function theorem on that neighbourhood gives radii $r_M, r'_M > 0$,

with $r'_M \leqslant \varrho_G$ (any smaller solution radius is admissible together with a correspondingly smaller r_M), and an analytic map $X = X((A_\gamma))$ on $\max_\gamma h(A_\gamma) < r_M$, with $h(X) < r'_M$, such that $X((A_\gamma))$ is the unique solution of $F = 0$ with $h(X) < r'_M$; there are finitely many values of n (one per $x \in \mathcal{B}_y$ modulo translations), so r_M, r'_M and the derivative bounds of X are constants of the group and L.

This solution is the fixed mean used in Stage 1. [8, (0.8)] places that specified analytic mean near the identity for input families near the identity, and (0.10) says it solves this equation. The radius r_M is reduced, before the final choice of τ_L, so that all families $\{e^{iA_\gamma}\}$ with $\max_\gamma h(A_\gamma) < r_M$ lie in the small-diameter domain of the specified mean, that this mean has logarithm of norm below r'_M, and that all relative logarithms use the common chart. The operator-norm diameter $\max_{\gamma,\gamma'} \|g_\gamma - g_{\gamma'}\|_{\mathrm{op}}$ is unchanged by the two-sided multiplication $g_\gamma \mapsto u(y) g_\gamma u(x)^{-1}$. Thus the original-gauge families lie in the same small-diameter domain. Their relative products with the mean transform by conjugation at y, which preserves the Ad-invariant logarithm chart, so the original-gauge relative logarithms are defined as well. Local uniqueness from the implicit function theorem then identifies it with e^{iX}. Its left-right covariance is [8, (0.6)]. Thus for any admissible gauge u its value in the original gauge is $u(y)^{-1}\mathcal{U}_{U^u}(y,x)u(x)$; the value is independent of the choice of admissible gauge because both expressions are values of the same specified mean.

Since $h(A_\gamma) \leqslant r_M/2 < r_M$ for the transport families above, the mean $\mathcal{U} = e^{iX((A_\gamma))}$ exists, lies in the chart domain, and is the unique solution near the diagonal value in the sense of (4.5).

Let $A_\gamma(q^u) := \log_G((e^{iq^u}U^u)(\gamma))$ and $X(q^u) = X((A_\gamma(q^u))_\gamma)$, and let $M_x(q) := \mathcal{U}_{e^{iqU}}(y,x)\mathcal{U}_U(y,x)^{-1}$ and $M^u_x(q^u) = \mathcal{U}_{e^{iq^uU^u}}(y,x)\mathcal{U}_{U^u}(y,x)^{-1}$. Gauge covariance gives $M^u_x(q^u) = u(y)M_x(q)u(y)^{-1} = e^{iX(q^u)}e^{-iX(0)}$, with inverse $M^u_x(q^u)^{-1} = e^{iX(0)}e^{-iX(q^u)}$; both are compositions of the analytic map X with $\log_G$, the exponentials of the phases and the fixed links, so they extend analytically to a complex ball $\|q|\mathcal{E}_y\| < 2\vartheta_*$ (after reducing ϑ_* by a constant of the group and L). For real q, $M_x(q)$ is unitary and $\|M_x(q) - \mathbf{1}\|_{\mathrm{op}} = \|e^{iX(q^u)} - e^{iX(0)}\|_{\mathrm{op}} \leqslant C\|q|\mathcal{E}_y\|$ with a constant C of L and the group (the derivatives of X on the ball $\max_\gamma h(A_\gamma) \leqslant r_M/2$, of $\log_G$ on the r_G-ball and of the products of exponentials are bounded), so after reducing ϑ_* once more $M_x(q)$ lies within r_G of $\mathbf{1}$ for $\|q|\mathcal{E}_y\| \leqslant \vartheta_*$.

The gauge term is $\mathscr{G}_{y,U}(q) = \sum_x (1 - \operatorname{Re}\operatorname{tr}_N M_x(q))$; for real q, $\operatorname{Re}\operatorname{tr}_N M_x(q) = \frac{1}{2}(\operatorname{tr}_N M_x(q) + \operatorname{tr}_N M_x(q)^{-1})$ because $M_x(q)$ is unitary, and the right side is the analytic continuation to the complex ball. The derivatives up to order five of M_x and $\mathscr{G}_{y,U}$ are bounded on the closed ball of radius ϑ_* by constants C^{a}_k uniform in the background, because they are analytic jointly in q and in the phases ϕ of the links of $\mathcal{B}_y$ in the gauge u, and the set $\|\phi\|_\infty \leqslant \tau_L/2$ is compact. $\mathscr{G}_{y,U}(0) = 0$ because $M_x(0) = \mathbf{1}$, and $D\mathscr{G}_{y,U}(0) = 0$ because each summand is nonnegative on real q and vanishes at $q = 0$.

The gauge u multiplies every transport from y to x on the left by $u(y)$ and on the right by $u(x)^{-1}$, and the mean equation is covariant under this ($\mathcal{U} \mapsto u(y)\mathcal{U}u(x)^{-1}$, so $M_x \mapsto u(y)M_x u(y)^{-1}$, which preserves $\|M_x - \mathbf{1}\|_{\mathrm{op}}$ and $\operatorname{tr}_N M_x$), so the relative-mean and gauge-functional bounds hold for U itself.

For each $0 \leqslant k \leqslant 5$, C^{a}_k is the maximum of one, C^{D}_k, and the uniform derivative bounds for the phase, the two logarithms, the relative means, and the gauge functional obtained in (iii) and in the proof of Lemma B.9; these are finite and uniform by the compact phase sets, the finitely many templates, and the stated analytic neighbourhoods. *q.e.d.*

Let c be a row within $2R_{\mathrm{c}}+2L$ of I (a near row); by Section 3.3 every row of a near primitive (a primitive whose free-input set meets I) is within $2R_{\mathrm{c}}$ of I, since its bonds are within $4L+3 \leqslant 2R_{\mathrm{c}}$ of a free-input bond in I, and every row of a plaquette with a bond in $\mathcal{E}_c$ for such a row c is within $2R_{\mathrm{c}}+2L$ of I (its bonds are within $2L-1+1$ of a bond of $\mathcal{E}_c$).

The two-block set $\mathcal{E}_c$ lies inside a lifted rectangle of side $2L+1 \leqslant s_{\mathfrak{P}}$ all of whose bonds are within $2R_{\mathrm{c}}+2L+2L+1 \leqslant 3R_{\mathrm{c}}$ of I (as $R_{\mathrm{c}} \geqslant 2R_{\mathrm{loc}}+1 = 4L+5$); Lemmas B.4 and B.5 give a gauge on it with phases at most $\widehat{\mu} < \tau_L/2$ (the bound of Lemma B.5 is monotone in the side), and Lemma 3.19(iii) (range $3R_{\mathrm{c}}$) bounds the retained inputs $A_{\mathrm{e}}(b)$, $b \in \mathcal{E}_c \cap J$, by $r_{\mathrm{e},\xi}$.

Likewise a block $\mathcal{B}_y$ with centre y within $2R_{\mathrm{c}}+2L$ of I is a lifted rectangle of side $L \leqslant s_{\mathfrak{P}}$ whose bonds are within $3R_{\mathrm{c}}$ of I, so Lemmas B.4 and B.5 apply to it directly. For real $B \in \mathcal{U}_y$, the block input satisfies $\|g_0 B|_{\mathcal{E}_y}\| \leqslant 2C_{\mathsf{C}} g_0(1+2r) \leqslant \vartheta_*/4$ by threshold item 3 and $C_{\mathsf{C}} \geqslant |\mathcal{E}_y|^{1/2}$. Hence the conclusions of Lemma B.9 apply on the entire block polydisc $\mathcal{U}_y$. So Lemmas B.8 and B.9 apply to every row and every centre within $2R_{\mathrm{c}}+2L$ of I, in particular to every row of every near primitive and of every plaquette with a bond in the two-block set of such a row.

Let $s_\xi := 1+r+r_{\mathrm{e},\xi} \leqslant 1+2r$. The next lemma shows that Lemma B.8(i)–(iii) apply at every row of $\Omega_1^{[1]}$, near or far, and that the arguments of the whole fibre $D^{(11)}(V,V_0)$ lie in its balls.

Definition B.10 (Fibre Data)**.** Let $n_{\mathcal{E}}$ be the largest number of bonds of a two-block set $\mathcal{E}_c$ of (3.10) (a constant of L), $O(L^2)$ the constant of [9, p. 247] in (B1)(ii), and (Ω_1, V, V_0) an admissible pair, with the background $U_1 = U_1(V,V_0)$ of (B1) on the bonds of Ω_1 and V_0 on the other bonds. Phases $A \in \mathfrak{g}^{\Omega_1}$ are extended by zero, and

$$\varphi(A) := \left(\varphi_c(A|_{\mathcal{E}_c})\right)_{c \in \Omega_1^{[1]}}, \qquad \Theta(A) := A - h_{U_1}\varphi(A)$$

wherever every component is defined. For $U \in D^{(11)}(V,V_0)$,

$$A' := \tfrac{1}{i}\log(UU_1^{-1}) \text{ on the bonds of } \Omega_1, \qquad d := \mathcal{N}_{U_1}(A'),$$
$$A := A' + h_{U_1} d, \qquad B := g_0^{-1} A'|_{\Omega_1^*},$$

and

$$\mathcal{N}_\Theta := \{A \in \mathfrak{g}^{\Omega_1} : \|A|_{\mathcal{E}_c}\| < \vartheta_*/2 \text{ for every } c \in \Omega_1^{[1]}\}.$$

The logarithm defining A' exists by (B1), and $A = g_0 \mathsf{C}_{U_1} B$ by the decomposition step of Section 3.1 after (3.2), which uses (B1), the constraint, $LQ_{U_1} h_{U_1} = \mathbf{1}$, and the invertibility of $\alpha_c(U_1)$ in the elimination maps (3.12), given by Lemma B.11(i) and Lemma 3.6. Threshold item 6 of Section B.8 is

$$3c_G c_{\mathrm{h}} s_{\mathfrak{P}} \varepsilon_0 \leqslant \tfrac{1}{2}\tau_L, \qquad n_{\mathcal{E}}^{1/2} c_G c_{\mathrm{h}}\, 2\, O(L^2)\, \varepsilon_0 \leqslant \tfrac{1}{4}\vartheta_*. \tag{B.16}$$

The first condition gives $3c_{\mathrm{h}} s_{\mathfrak{P}} \varepsilon_0 \leqslant \tau_L/(2c_G) \leqslant r_G$, since $\tau_L \leqslant 2c_G r_G$ by (B.6).

Lemma B.11 (Fibre Smallness)**.** *Let g_0 satisfy threshold items 5 and 6 of Section B.8, and let the data be those of Definition B.10.*

1. *For every row $c \in \Omega_1^{[1]}$, U_1 admits a gauge patch on $\mathcal{E}_c$, with a site gauge on $\mathfrak{P}_c := \mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$, the union of the two blocks of c. Hence Lemma B.8(i)–(iii) hold at c for $U = U_1$: the function $\varphi_c = \widetilde{D}_{U_1,c}$ of Lemma B.8(ii) is analytic on the ball $\|A|_{\mathcal{E}_c}\| < \vartheta_*$, real on real phases, with $\varphi_c(0) = 0$, $D\varphi_c(0) = 0$, and for every A in that ball $\varphi_c(A|_{\mathcal{E}_c})$ is the unique solution d_c of the row equation of c with $h(d_c) \leqslant \rho_{\mathrm{d}}$.*

2. *For $U \in D^{(11)}(V, V_0)$ and every row $c \in \Omega_1^{[1]}$*

$$\|A'|_{\mathcal{E}_c}\| \leqslant \tfrac{1}{4}\vartheta_*, \qquad h(d_c) \leqslant \tfrac{1}{32}K_*\vartheta_*^2 < \rho_{\mathrm{d}}, \qquad \|A|_{\mathcal{E}_c}\| < \tfrac{1}{2}\vartheta_*, \qquad d_c = \varphi_c(A|_{\mathcal{E}_c})\,.$$

3. *For every $B \in \mathfrak{g}^{\Omega_1^*}$, every family d of chart class at $A = g_0 \mathsf{C}_{U_1} B$ (Section 3.1) with $U_d(B) \in D^{(11)}(V, V_0)$ is $d = \varphi(A)$.*

4. *$\mathcal{N}_\Theta$ is open, and by (ii) it contains the argument A of every $U \in D^{(11)}(V, V_0)$. On $\mathcal{N}_\Theta$ the family φ is analytic, real on real phases, and solves the row equations* (3.24)*; the phase $\Theta(A)$ has $\|\Theta(A)|_{\mathcal{E}_c}\| < \vartheta_*$ at every row and is chart-valued, $h(\Theta(A)(b)) < \vartheta_* < \varrho_G$ on every bond b of Ω_1; for every row, $\|h_{U_1}(c)\partial_{A(b_0(c))}\varphi_c(A|_{\mathcal{E}_c})\| \leqslant 4H_*K_*\|A|_{\mathcal{E}_c}\| < 1$; and*

$$\det\Big(\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\varphi(A)\Big) = \prod_{c\in\Omega_1^{[1]}} \det{}_{\mathfrak{g}}(\mathbf{1} - h_{U_1}(c)\partial_{A(b_0(c))}\varphi_c(A|_{\mathcal{E}_c})) > 0\,. \tag{B.17}$$

Proof. (i) The set $\mathfrak{P}_c$ is a lifted rectangle of side at most $2L+1 \leqslant s_{\mathfrak{P}}$, injective modulo the periods (every fine period is at least $8M > 2L+1$), and it contains every bond of $\mathcal{E}_c$, both endpoints of such a bond lying in $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$ by (3.10). The two blocks of c lie in $\Omega_1^\sim$ ((B1)(iv)) and $U_1 \in D^{(11)}(V, V_0)$ ((B1)(iii)), so (B1)(i) gives $|U_1(\partial p) - 1| < \varepsilon_0$, hence $\|U_1(\partial p) - \mathbf{1}\|_{\mathrm{op}} \leqslant c_{\mathrm{h}}\varepsilon_0 = 2c_{\mathrm{h}}(\varepsilon_0/2)$ by the norm comparison of the preamble, for every plaquette $p \subset \mathfrak{P}_c$. By the consequence $3c_{\mathrm{h}}s_{\mathfrak{P}}\varepsilon_0 \leqslant r_G$ of (B.16) noted after it, $6c_{\mathrm{h}}s_{\mathfrak{P}}(\varepsilon_0/2) \leqslant r_G$, so Lemma B.5 with $\varepsilon_0/2$ in place of ε_1 (the remark after its proof) gives a site gauge on $\mathfrak{P}_c$ with phases $h(A_b) \leqslant 3c_G c_{\mathrm{h}} s_{\mathfrak{P}}\varepsilon_0 \leqslant \tau_L/2$, by the first condition. This is the hypothesis of Lemma B.8 at c.

The phases of $\mathfrak{g}^{\Omega_1}$, extended by zero, form the subspace of $\mathfrak{g}^{\mathcal{E}_c}$ of the phases vanishing on the bonds of $\mathcal{E}_c$ off Ω_1; for A in it and $d \in \mathfrak{g}$ the phase $A - h_{U_1}(c)d$ lies in it, since $b_0(c)$ is a bond of Ω_1, and Lemma B.8(i)–(iii) hold on it by restriction.

(ii) By (B1)(ii)–(iii), $|U'(b) - 1| < 2\,O(L^2)\varepsilon_0$ for $U' = UU_1^{-1}$ on every bond b of Ω_1: every such bond, the designated bonds included, is a bond of $\square'^{\sim 2}$ for the $L^2M_2R_1$-cube $\square' \subset \Omega_1$ that contains an endpoint of b in Ω_1 (Ω_1 is a union of LMR_1-cubes, p.248, each a union of $L^2M_2R_1$-cubes: the partitions of [9, p. 245] are compatible, that is nested, (H4), and all sides are powers of L, the paragraph following (3.1)). By item 5, $2\,O(L^2)\varepsilon_0 < r_G/\sqrt{N}$, so $\|U'(b) - \mathbf{1}\|_{\mathrm{op}} \leqslant c_{\mathrm{h}}|U'(b) - 1| < r_G$ and $h(A'(b)) \leqslant c_G\|U'(b) - \mathbf{1}\|_{\mathrm{op}} \leqslant c_G c_{\mathrm{h}}\, 2\,O(L^2)\varepsilon_0$ (preamble); $\mathcal{E}_c$ has at most $n_{\mathcal{E}}$ bonds, and A' vanishes on those off Ω_1, so the second condition of (B.16) gives $\|A'|_{\mathcal{E}_c}\| \leqslant \vartheta_*/4 < \varrho_*$.

The row component $d_c = \mathcal{N}_{U_1,c}(A')$ depends on $A'|_{\mathcal{E}_c}$ only (Lemma 3.6), and Taylor's formula with Lemma B.8(i), which applies by (i), gives $h(d_c) \leqslant \frac{1}{2}K_*(\vartheta_*/4)^2 = \frac{1}{32}K_*\vartheta_*^2 < 2K_*\vartheta_*^2 = \rho_{\mathrm{d}}$.

The only designated bond in $\mathcal{E}_c$ is $b_0(c)$ (Lemma 3.6) and h_{U_1} has its outputs at the designated bonds, so $A|_{\mathcal{E}_c} = A'|_{\mathcal{E}_c} + h_{U_1}(c)d_c$, and $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/4 + H_*K_*\vartheta_*^2/32 < \vartheta_*/2$ by $4K_*H_*\vartheta_* < 1$. Finally $A'|_{\mathcal{E}_c} = A|_{\mathcal{E}_c} - h_{U_1}(c)d_c$, so $d_c = \mathcal{N}_{U_1,c}(A|_{\mathcal{E}_c} - h_{U_1}(c)d_c)$ solves the row equation of c at A, with $h(d_c) \leqslant \rho_{\mathrm{d}}$ and $\|A|_{\mathcal{E}_c}\| < \vartheta_*$; by the uniqueness in (i), $d_c = \varphi_c(A|_{\mathcal{E}_c})$.

(iii) Since d is of chart class, $q_d(B) = A - h_{U_1}d$ is the chart logarithm of $U_d(B)U_1^{-1}$ on the bonds of Ω_1, so it is the phase A' of (ii) for $U = U_d(B)$. The row equations give $d_c = \mathcal{N}_{U_1,c}(A - h_{U_1}d) = \mathcal{N}_{U_1,c}(A')$, that is $d = \mathcal{N}_{U_1}(A')$, and $A = A' + h_{U_1}d$. So A', d and A are the quantities of (ii) for $U = U_d(B)$, and (ii) gives $d_c = \varphi_c(A|_{\mathcal{E}_c})$ for every row.

(iv) $\mathcal{N}_\Theta$ is a finite intersection of preimages of open balls under restriction maps, hence open, and it contains the arguments by (ii). By (i) each component $\varphi_c(A|_{\mathcal{E}_c})$ is analytic

and real on $\mathcal{N}_\Theta$ and solves the row equation of c at A; the row equations decouple ($(A - h_{U_1}\varphi(A))|_{\mathcal{E}_c} = A|_{\mathcal{E}_c} - h_{U_1}(c)\varphi_c(A|_{\mathcal{E}_c})$, Lemma 3.6), so φ solves the system (3.24).

By (B.15) and Lemma B.8(i), $\|\Theta(A)|_{\mathcal{E}_c}\| \leqslant \|A|_{\mathcal{E}_c}\| + 2H_*K_*\|A|_{\mathcal{E}_c}\|^2 < \vartheta_*/2 + H_*K_*\vartheta_*^2/2 < \vartheta_*$.

Every bond b of Ω_1 lies in $\mathcal{E}_c$ for a row $c \in \Omega_1^{[1]}$: b has an endpoint in Ω_1, whose block $\mathcal{B}_y$ lies in Ω_1 (Ω_1 is a union of blocks); if both endpoints of b lie in $\mathcal{B}_y$, any coarse bond c with $c_- = y$ serves, and otherwise the coarse bond joining y and the centre of the block of the other endpoint does; in both cases a block centre of c lies in Ω_1, so $c \in \Omega_1^{[1]}$, and $b \in \mathcal{E}_c$ by (3.10). Hence $h(\Theta(A)(b)) \leqslant \|\Theta(A)|_{\mathcal{E}_c}\| < \vartheta_* \leqslant \varrho_G/2$.

By (B.15) and Lemma B.8(i), $\|h_{U_1}(c)\partial_{A(b_0(c))}\varphi_c\| \leqslant H_*C_{\mathsf{D}}\|A|_{\mathcal{E}_c}\| = 4H_*K_*\|A|_{\mathcal{E}_c}\| < 2H_*K_*\vartheta_* < 1$.

The matrix $\frac{\delta}{\delta A}\varphi$ has rows indexed by $\Omega_1^{[1]}$ and columns by the bonds of Ω_1, h_{U_1} maps row c to the bond $b_0(c)$, and $\partial_{A(b_0(c'))}\varphi_c = 0$ for $c' \neq c$, since φ_c depends on $A|_{\mathcal{E}_c}$, which contains no designated bond other than $b_0(c)$ (Lemma 3.6). So, in the ordering (free bonds, designated bonds), $\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\varphi$ is block triangular, with identity block on the free bonds and designated block $\mathrm{diag}_c(\mathbf{1} - h_{U_1}(c)\partial_{A(b_0(c))}\varphi_c)$; this gives the product in (B.17). Each factor is positive by Lemma B.8(iii), which applies by (i), since $\|A|_{\mathcal{E}_c}\| < \vartheta_*/2$. *q.e.d.*

Lemma B.12 (Primitive Bounds)**.** *Let g_0 satisfy threshold item 3 of Section B.8, $4C_{\mathsf{C}}g_0(1+2r) \leqslant \vartheta_*/2$ (hence $4C_{\mathsf{C}}g_0 s_\xi \leqslant \vartheta_*/2$, and $g_0 s_\xi \leqslant \vartheta_*/(8C_{\mathsf{C}}) \leqslant 1$, since $C_{\mathsf{C}} \geqslant 1$ and $\vartheta_* \leqslant \tau_L/2 \leqslant 1/2$ by (B.6)). Then on the enlarged box $\{|a_b| \leqslant 2r\ \forall b \in I\}$ and for $t \in [0,1]$, every primitive p_α with free-input set S_α meeting I is, at the input $a + tA_{\mathrm{e}}$, the restriction to real arguments of a function analytic in (a,t) on a complex neighbourhood of the enlarged box times $[0,1]$, and with a constant C_{p} depending on L and the group only,*

$$|p_{\xi,\alpha}(a + tA_{\mathrm{e}})| \leqslant C_{\mathrm{p}}g_0 s_\xi^3, \qquad |p^{\mathrm{H}}_{\xi,b}(a + tA_{\mathrm{e}})| \leqslant C_{\mathrm{p}}g_0^2 s_\xi^2, \qquad |p^{\mathrm{J}}_{\xi,c}(a + tA_{\mathrm{e}})| \leqslant C_{\mathrm{p}}g_0 s_\xi\ , \tag{B.18}$$

and for each of the three kinds, in the Hilbert norms on $\mathfrak{g}^{S_\alpha}$,

$$\|D_a^2 p_\alpha\| \leqslant C_{\mathrm{p}}g_0 s_\xi, \qquad \|D_a p_\alpha\| \leqslant C_{\mathrm{p}}g_0 s_\xi^2, \qquad |p_\alpha(a + tA_{\mathrm{e}}) - p_\alpha(tA_{\mathrm{e}})| \leqslant C_{\mathrm{p}}g_0 s_\xi^3\ , \tag{B.19}$$

and $\|D_a^3 p_\alpha\| \leqslant C_{\mathrm{p}}g_0$, $\|D_a^4 p_\alpha\| \leqslant C_{\mathrm{p}}g_0^2$; the same four derivative bounds hold for the derivatives $D_B^k p_\alpha$, $k = 1,\dots,4$, in the full input $B = a + tA_{\mathrm{e}}$ on $\mathfrak{g}^{S_\alpha}$, retained inputs included (the input of the primitive is linear in B with the same norm bound).

Proof. The input of a primitive. On the enlarged box, $|a_b + tA_{\mathrm{e}}(b)| \leqslant 2r + r_{\mathrm{e},\xi} \leqslant 2s_\xi$ for every bond b within $3R_{\mathrm{c}}$ of I (Lemma 3.19(iii) bounds the retained inputs by $r_{\mathrm{e},\xi}$), in particular for the free bonds of the two-block sets of the near rows and for the free bonds of the plaquettes and blocks of the near primitives. The rows of a near primitive are within $2R_{\mathrm{c}}$ of I (applicability paragraph).

The *input* $\mathfrak{u}$ of a primitive is the vector $g_0\mathsf{C}_U B$ restricted to the following bond set S: for a plaquette primitive $E_{\xi,p}$, $p_{\xi,p}$ the union of the sets $\mathcal{E}_c$ over the rows c of p (at most four) together with the free bonds of ∂p not in those sets; for the Haar primitive $p^{\mathrm{H}}_{\xi,b}$ the set $\mathcal{E}_c$ when $b = b_0(c)$ and $\{b\}$ when b is free; for the Jacobian primitive $p^{\mathrm{J}}_{\xi,c}$ the set $\mathcal{E}_c$; for a gauge primitive $E_{\xi,y}$, $p_{\xi,y}$ the set $\mathcal{E}_y$ (on which C_U is the identity, all its bonds being free).

The primitive is a function of the phase $q_U(B)$ on its bonds, and that phase is a function of the input: on $\mathcal{E}_c$ it is $\mathfrak{u}|_{\mathcal{E}_c} - h_U\widetilde{D}_{U,c}(\mathfrak{u}|_{\mathcal{E}_c})$ (Lemma B.8(ii)), and on a free bond outside the row sets it is $\mathfrak{u}_b = g_0 B(b)$. With $B = a + tA_{\mathrm{e}}$, by Lemma B.8(i) each row component

satisfies $\|\mathfrak{u}|_{\mathcal{E}_c}\| \leqslant C_{\mathsf{C}} g_0 \cdot 2s_\xi \leqslant \vartheta_*/4$ and each free-bond component outside the row sets $h(\mathfrak{u}_b) \leqslant |\mathfrak{u}_b| \leqslant 2g_0 s_\xi \leqslant \vartheta_*/4$ (threshold item 3 and $C_{\mathsf{C}} \geqslant 1$), so for the plaquette, Haar and Jacobian primitives (at most four rows and four further free bonds)

$$\|\mathfrak{u}\| \leqslant 16 C_{\mathsf{C}} g_0 s_\xi, \qquad \|D_a \mathfrak{u}\| \leqslant 5 C_{\mathsf{C}} g_0, \qquad \|\partial_t \mathfrak{u}\| \leqslant 16 C_{\mathsf{C}} g_0 s_\xi \,,$$

the second in the operator norm from the Hilbert norm of $\mathfrak{g}^{S_\alpha}$ (the Hilbert bound of Lemma B.8(i) on each of at most four rows, plus the identity on the free bonds); for the gauge primitive, whose input $g_0 B|_{\mathcal{E}_y}$ has $|\mathcal{E}_y| = 4L^3(L-1)$ free bonds, $\|\mathfrak{u}\| \leqslant g_0 |\mathcal{E}_y|^{1/2} \cdot 2s_\xi \leqslant 2C_{\mathsf{C}} g_0 s_\xi \leqslant \vartheta_*/4$ (as $C_{\mathsf{C}} \geqslant (4L^3(L-1))^{1/2}$ and $h \leqslant |\bullet|$), $\|D_a \mathfrak{u}\| \leqslant g_0$ and $\|\partial_t \mathfrak{u}\| \leqslant 2C_{\mathsf{C}} g_0 s_\xi$, so the three displayed bounds hold for all four kinds.

The input is linear in B, hence affine in (a,t) and linear in a at fixed t; $D_a\mathfrak{u}$ is the restriction of $D_B \mathfrak{u} = g_0 \mathsf{C}_U|_S$ to the a-directions, and the same count gives $\|D_B \mathfrak{u}\| \leqslant 5 C_{\mathsf{C}} g_0$ (respectively $\leqslant g_0$ for the gauge primitive), so every bound below on a derivative in a holds verbatim for the derivative of the same order in B on $\mathfrak{g}^{S_\alpha}$, retained inputs included.

Each primitive is $\varphi(\mathfrak{u})$ for a function φ of the input analytic on a complex neighbourhood of the closed set $\mathcal{P} = \{\mathfrak{u} \in \mathfrak{g}^S_{\mathbb{C}} : \|\mathfrak{u}|_{\mathcal{E}_c}\| \leqslant \vartheta_*/2$ for each row c of the primitive, $h(\mathfrak{u}_b) \leqslant \vartheta_*/2$ for each free bond b of S outside the row sets$\}$ (for the gauge primitive, the closed ball $\|\mathfrak{u}\| \leqslant \vartheta_*$ of $\mathfrak{g}_{\mathbb{C}}^{\mathcal{E}_y}$), with derivatives up to order five bounded on $\mathcal{P}$ by constants of L and the group: Lemma B.8(iii) for the row solution, the phase and the two logarithms, each a function of one row component (the Haar logarithm at a free bond outside the row sets is the function of (iii) at a free bond of a two-block set, where $q_U(A)(b) = A(b)$, a function of the single component $\mathfrak{u}_b$); for the plaquette term, with $W(q) \coloneqq (e^{iq}U)(\partial p)$, $s_{p,U}(q) = 1 - \operatorname{Re}\operatorname{tr}_N W(q)$ equals $1 - \frac{1}{2}(\operatorname{tr}_N W(q) + \operatorname{tr}_N W(q)^{-1})$ for real q, $W(q)$ being unitary, and $W(q)$ and $W(q)^{-1}$ are ordered products of the fixed links and of $e^{\pm iq(b)}$, entire functions of the phases $q(b)$, which are analytic functions of one row component (with $h(q_U(A)(b)) \leqslant \|A\| + H_* 2K_* \|A\|^2 \leqslant \vartheta_*$ on the ball) or are free-bond components of $\mathfrak{u}$; this defines the analytic function $s_{p,U}(q_U(\bullet))$, with derivatives bounded on $\mathcal{P}$ by the chain rule; for the gauge term, Lemma B.9.

The segment from 0 to $\mathfrak{u}$ lies in $\mathcal{P}$, each component of $\mathfrak{u}$ being within $\vartheta_*/4$ of zero. Hence every primitive at the input $a + tA_{\mathsf{e}}$ is the restriction to real (a,t) of a function analytic in (a,t) on a complex neighbourhood of the enlarged box times $[0,1]$ (the composition of φ with the affine map), and the bounds below are uniform there.

Energy residual. $E_{\xi,\alpha}(a + tA_{\mathsf{e}}) = g_0^{-2}\varphi(\mathfrak{u})$ with $\varphi(\mathfrak{u}) \coloneqq -[s_{\alpha,U}(q_U(\mathfrak{u})) - s_{\alpha,U}(0)]$ for a plaquette and $\varphi(\mathfrak{u}) = -\mathscr{G}_{y,U}(\mathfrak{u})$ for a centre. Since $\mathfrak{u}$ is linear in B, the Taylor polynomial of degree two in B at $B = 0$ subtracted in (3.29) is that of φ at 0 composed with the linear map, so $p_{\xi,\alpha} = g_0^{-2}[\varphi(\mathfrak{u}) - D\varphi(0)\mathfrak{u} - \frac{1}{2}D^2\varphi(0)[\mathfrak{u},\mathfrak{u}]]$, the second-order Taylor residual at the input, bounded by $g_0^{-2}\frac{1}{6}\|D^3\varphi\|\|\mathfrak{u}\|^3 \leqslant C g_0 s_\xi^3$ (the segment from 0 to $\mathfrak{u}$ lies in $\mathcal{P}$).

Its first derivative in a is $g_0^{-2}(D\varphi(\mathfrak{u}) - D\varphi(0) - D^2\varphi(0)\mathfrak{u})D_a\mathfrak{u}$, bounded by $g_0^{-2}\frac{1}{2}\|D^3\varphi\|\|\mathfrak{u}\|^2 \cdot 5C_{\mathsf{C}} g_0 \leqslant C g_0 s_\xi^2$; its second derivative is $g_0^{-2}(D^2\varphi(\mathfrak{u}) - D^2\varphi(0))[D_a\mathfrak{u}, D_a\mathfrak{u}]$, bounded by $g_0^{-2}\|D^3\varphi\|\|\mathfrak{u}\|(5C_{\mathsf{C}} g_0)^2 \leqslant C g_0 s_\xi$; its third and fourth derivatives are $g_0^{-2}D^3\varphi(\mathfrak{u})[D_a\mathfrak{u}]^3$ and $g_0^{-2}D^4\varphi(\mathfrak{u})[D_a\mathfrak{u}]^4$, bounded by Cg_0 and Cg_0^2.

The Taylor polynomial subtracted is that of the germ (Section 3.3), which is the Taylor polynomial of φ at 0 by Lemma B.8(ii).

Haar logarithm. $\psi(\mathfrak{u}) = \log\sigma(q_U(\mathfrak{u})(b)) - \log\sigma(0)$, the primitive (3.30), has $\psi(0) = 0$ and $D\psi(0) = 0$ (Lemma B.8(iii): σ is even and $Dq_U(0)$ is linear). Hence $|\psi(\mathfrak{u})| \leqslant \frac{1}{2}\|D^2\psi\|\|\mathfrak{u}\|^2 \leqslant C g_0^2 s_\xi^2$, $\|D_a\psi\| \leqslant \|D^2\psi\|\|\mathfrak{u}\| \cdot 5C_{\mathsf{C}} g_0 \leqslant C g_0^2 s_\xi \leqslant C g_0 s_\xi^2$, $\|D_a^2\psi\| \leqslant \|D^2\psi\|(5C_{\mathsf{C}} g_0)^2 \leqslant C g_0 s_\xi$, and the third and fourth derivatives are bounded by $Cg_0^3 \leqslant Cg_0$ and $Cg_0^4 \leqslant Cg_0^2$.

Constraint logarithm. $\lambda(\mathfrak{u}) = \log|\det_{\mathfrak{g}}(\mathbf{1} - h_U(c)\partial_{A(b_0(c))}\widetilde{D}_{U,c}(\mathfrak{u}))| = \log\det_{\mathfrak{g}}(\cdots)$ (the determinant is positive, Lemma B.8(iii)) has $\lambda(0) = 0$ and derivatives up to order five bounded on $\mathcal{P}$. Hence $|\lambda(\mathfrak{u})| \leqslant \|D\lambda\|\|\mathfrak{u}\| \leqslant Cg_0 s_\xi$, $\|D_a\lambda\| \leqslant \|D\lambda\| \cdot 5C_{\mathsf{C}}g_0 \leqslant Cg_0 \leqslant Cg_0 s_\xi^2$, $\|D_a^2\lambda\| \leqslant \|D^2\lambda\|(5C_{\mathsf{C}}g_0)^2 \leqslant Cg_0^2 \leqslant Cg_0 s_\xi$ (as $g_0 \leqslant 1$ and $s_\xi \geqslant 1$), and the third and fourth derivatives are bounded by $Cg_0^3 \leqslant Cg_0$ and $Cg_0^4 \leqslant Cg_0^2$.

Let C'_{p} be the maximum over the finitely many templates of the constants C above and let $C_{\mathrm{p}} := 2C'_{\mathrm{p}}$. Then (B.18), the first two bounds of (B.19), and the third- and fourth-order bounds hold, and the last bound of (B.19) is the difference of two values each bounded by $C'_{\mathrm{p}}g_0 s_\xi^3$ (for all three kinds, since $s_\xi \geqslant 1$ and $g_0 \leqslant 1$), hence by $2C'_{\mathrm{p}}g_0 s_\xi^3 = C_{\mathrm{p}}g_0 s_\xi^3$.

q.e.d.

Proof of Lemma 3.12(i)–(iii) and (vi). Let c be a row within $2R_{\mathrm{c}}+2L$ of I; by the applicability paragraph Lemma B.8 applies to it, and (ii)–(iii) give, for every real A with $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/2$, the unique small solution $\varphi_c(A|_{\mathcal{E}_c})$ with φ_c analytic on the complex ball of radius ϑ_* and real on real phases, $\varphi_c(0) = 0$, $D\varphi_c(0) = 0$, the positivity of the determinant, and the analyticity of the row solution, the phase and the two logarithms on a complex neighbourhood of the closed ball of radius $\vartheta_*/2$, with derivatives up to order five bounded on that ball.

For $B \in \mathcal{U}_c$, $\max_{b\in\mathcal{E}_c\cap E_*}|B(b)| < 2+4r = 2(1+2r)$ and $\|g_0\mathsf{C}_U B\|_{\mathcal{E}_c} \leqslant 2C_{\mathsf{C}}g_0(1+2r) \leqslant \vartheta_*/4$ by threshold item 3, so $g_0\mathsf{C}_U\mathcal{U}_c$ lies in the phase ball; $B \mapsto g_0\mathsf{C}_U B$ is linear with norm at most $g_0C_{\mathsf{C}} \leqslant C_{\mathsf{C}}$ for the Hilbert norms (Lemma B.8(i)), so the derivatives in B are bounded by the same constants times C_{C}^k.

For c within $2R_{\mathrm{c}}$ of I, the plaquette term of a plaquette p with a bond in $\mathcal{E}_c$ is analytic in B on the intersection of the polydiscs of the rows of p, as above, and these rows are within $2R_{\mathrm{c}} + 2L$ of I (applicability paragraph), so the chart applies to each of them.

For a centre y within $2R_{\mathrm{c}} + 2L$ of I, Lemma B.5 applied to the block $\mathcal{B}_y$ (applicability paragraph) gives a gauge with phases at most $\widehat{\mu} < \tau_L/2$ on $\mathcal{E}_y$; every bond of $\mathcal{E}_y$ is free, so $q_U(B) = g_0B$ there and $\|g_0B|_{\mathcal{E}_y}\| \leqslant g_0|\mathcal{E}_y|^{1/2}\max_b|B(b)| \leqslant g_0(4L^3(L-1))^{1/2}\cdot 2(1+2r) \leqslant 2C_{\mathsf{C}}g_0(1+2r) \leqslant \vartheta_*/4$ under threshold item 3 (using $h \leqslant |\bullet|$, $|\mathcal{E}_y| = 4L^3(L-1)$ and $C_{\mathsf{C}} \geqslant (4L^3(L-1))^{1/2}$); Lemma B.9 then gives the existence of the relative means within r_G of $\mathbf{1}$ and the analyticity and derivative bounds of $\mathscr{G}_{y,U}(g_0B|_{\mathcal{E}_y})$ on $\mathcal{U}_y$. The values at $q = 0$ of the plaquette and gauge terms are subtracted in the primitives.

Finally, where $(A_{\mathrm{e}}, a) \in \mathcal{D}_\xi$ and c is within $2R_{\mathrm{c}} + 2L$ of I, the phase $A = g_0\mathsf{C}_U\Psi_\xi(a + A_{\mathrm{e}})$ has $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/4$, because a is in the open box and $\Psi_\xi(a + A_{\mathrm{e}}) = a + A_{\mathrm{e}}$ with $|A_{\mathrm{e}}(b)| < r$ on the two-block set (Lemma 3.19(i),(iii),(iv)), so $\Psi_\xi(a + A_{\mathrm{e}}) \in \mathcal{U}_c$ and the bound on $g_0\mathsf{C}_U\mathcal{U}_c$ applies; $\varphi_c(A|_{\mathcal{E}_c})$ is the row solution $\widetilde{D}_c(A)$ of (B4)(i), which is that of [9] by (B4)(i) and Lemma B.11(iii), and the phase, the Haar density, the Jacobian and the plaquette and gauge terms of [9, (1.14)–(1.15)] are the same functions of it. *q.e.d.*

Lemma B.13 (Host Multiplicity and Potential Size)**.** *Let $\mathfrak{T}$ be the set of primitive templates (kind, label and free-input shape, including the shapes truncated by Ω_1) modulo $L\mathbb{Z}^4$, each anchored at one of its input bonds, and let D_{h} be one plus the maximum ℓ^∞ diameter of a lifted host* (3.36) *and B_{h} the maximum number of M-cubes in one, both over $\mathfrak{T}$. With*

$$N_{\mathrm{h}} = |\mathfrak{T}|(2\lceil D_{\mathrm{h}}\rceil + 3)^4, \qquad C_0 = N_{\mathrm{h}}C_{\mathrm{p}}e^{\kappa(B_{\mathrm{h}}+4)}, \qquad \omega(g_0) = C_0g_0(1+r)^3\,, \tag{B.20}$$

at most N_{h} primitives are assigned to any host Y, $d_0(Y) \leqslant B_{\mathrm{h}} + 4$, and on the enlarged box, for every host,

$$\sup|\mathbf{V}_\xi(Y, a)| \leqslant \omega(g_0)e^{-\kappa d_0(Y)}\,. \tag{B.21}$$

Proof. Let v be a vertex of Y. Every primitive assigned to Y has its stencil, hence its anchor, in the lifted host, within ℓ^∞ distance D_h of v; there are at most $(2\lceil D_\mathrm{h}\rceil+1)^4$ such anchor positions, and a template with an anchor position determines the primitive. Projection to the torus only identifies positions. By Lemma B.1(i), $d_0(Y)\leqslant N_0(Y)-1\leqslant B_\mathrm{h}$. At $t=0$, s_ξ may be replaced by $1+r$ in Lemma B.12, since no retained input enters; each of the three kinds is then bounded by $C_\mathrm{p}g_0(1+r)^3$, and $|\mathbf{V}_\xi(Y,a)|\leqslant N_\mathrm{h}C_\mathrm{p}g_0(1+r)^3\leqslant N_\mathrm{h}C_\mathrm{p}g_0(1+r)^3e^{\kappa(B_\mathrm{h}+4)}e^{-\kappa d_0(Y)}$. *q.e.d.*

The definition gives $C_0\geqslant C_\mathrm{p}$, since $N_\mathrm{h}\geqslant 1$ and the exponential factor is at least one. The constant C_0 depends on L, the group, M and κ (through B_h, D_h), not on the volume, the branch or g_0.

The hypothesis of Lemma 3.20 holds for $f=f_\xi$ on the enlarged box, which is a neighbourhood of the closed cutoff box: the exponent $\mathsf{Q}_{\xi,t}$ is a quadratic polynomial plus $P_{\xi,t}$, a finite sum of near primitives, each analytic there by Lemma B.12; and $f_\xi(a)=W_\chi(C;U_\xi(a))$ is a character of a product of the reconstructed links (3.26) over the contour bonds, each of which depends on a through the rows of a free-input set meeting I (Lemma 3.17(i) puts $\Psi_\xi(a+A_\mathrm{e})=a+A_\mathrm{e}$ on those rows, so the phase is (3.47)), where q_U is analytic in a by Lemma B.8(ii)–(iii); contour bonds whose rows do not meet I do not depend on a. Both factors are therefore analytic on the enlarged box, uniformly in $t\in[0,1]$.

B.4 Proof of the Gaussian Payment

Proof of Lemma 3.24. Tail of the walk series. Throughout, E_0, C_{E_0}, $\Gamma_{E_0}(\sigma)$ are as in (3.64) and the sentence preceding it (their block decomposition is (3.67)), and $S_{Z_0}(1)=\mathsf{G}^{1/2}$. Let N be the set of bonds $b'\in I$ with $d_{\infty,T}(b'_-,Z_0)\leqslant R+1$. Since $\iota_{E_0}^*H$ has nonzero entries $H(b,b')$ only for $d(b_-,b'_-)\leqslant R$ and $b\in E_0$ meets Z_0, $\Gamma_{E_0}(\sigma)-\Gamma_{E_0}(1)=\iota_{E_0}^*H\Pi_{I\setminus E_0}\Pi_\mathsf{N}(S_{Z_0}(\sigma)-\mathsf{G}^{1/2})$.

A walk $\omega=(r;b_0,\dots,b_{m+1})$ with $b_0\in\mathsf{N}$ has $|w_\omega(\sigma)-1|\neq 0$ only if $D(\omega)$ meets a cube $\Delta\not\subseteq Z_0'$. The open M-neighbourhood of Z_0 is $\bigcup_{Q\subseteq Z_0}\operatorname{int}Q^\sim$, and by definition every LM-cube meeting it lies in Z_0' (Section 3.5). Hence $D(\omega)$ contains a point at distance at least M from Z_0, and at least $M-R-1$ from $b_{0,-}$.

Each block of A_H^r ($r<n_*$) or P_H vanishes beyond distance R_*, and D_b lies within ρ_w of b_-, so $D(\omega)$ lies within $(m+1)R_*+\rho_\mathrm{w}$ of $b_{0,-}$. By (3.59), $M/4\geqslant\rho_\mathrm{w}+R+R_*+2$, so $(m+1)R_*\geqslant M-R-1-\rho_\mathrm{w}\geqslant\frac34M+R_*+1$ and $m>M/(2R_*)\geqslant N_M$. The absolute block row sum of the terms with $m\geqslant N_M$ is, by the bounds in the proof of Lemma 3.21 and $|w_\omega-1|\leqslant 2a_*^{m+1}$,

$$s_H^{-1/2}\sum_{r<n_*}\sum_{m\geqslant N_M}a_{n_*m+r}\,2a_*^{m+1}V_*\beta^m\leqslant 2s_H^{-1/2}\frac{n_*V_*a_*}{1-a_*\beta}4^{-N_M}=\delta_M\,, \tag{B.22}$$

using $a_*\beta\leqslant 1/4$. The absolute block column sums of these terms obey the same bound, because every walk in the tail has $m\geqslant N_M$ whatever its first bond. Schur's test gives $\|\Pi_\mathsf{N}(S_{Z_0}(\sigma)-\mathsf{G}^{1/2})\|\leqslant\delta_M$. With $\|H\|\leqslant h_+$ this is the first bound of (i); with $\|C_{E_0}\|\leqslant\gamma^{-1}$, since $\iota_{E_0}^*H\iota_{E_0}\geqslant\gamma$, the second follows.

Reserve identity. Let $\Gamma:=\Gamma_{E_0}(\sigma)=\Gamma_1+i\Gamma_2$ with real Γ_1,Γ_2; $\Gamma_{E_0}(1)$ is real, so $\Gamma_2=\operatorname{Im}(\Gamma_{E_0}(\sigma)-\Gamma_{E_0}(1))$. The quadratic form in (3.66) is bilinear, so for real X, $\operatorname{Re}\langle\Gamma X,C_{E_0}\Gamma X\rangle=\langle\Gamma_1X,C_{E_0}\Gamma_1X\rangle-\langle\Gamma_2X,C_{E_0}\Gamma_2X\rangle$ and $|e^{-\langle B,\Gamma X\rangle}|=e^{-\langle B,\Gamma_1X\rangle}$ for real B.

For $0\leqslant\alpha_G<\gamma$ let $C_{E_0,\alpha_G}:=(\iota_{E_0}^*H\iota_{E_0}-\alpha_G)^{-1}$. The Gaussian integral over B with density $(2\pi)^{-d_G|E_0|/2}\det(C_{E_0})^{-1/2}e^{-\frac12\langle B,(C_{E_0}^{-1}-\alpha_G)B\rangle-\langle B,\Gamma_1X\rangle}$ equals $\det_{E_0}(\mathbf{1}-$

$\alpha_G C_{E_0})^{-1/2} e^{\frac{1}{2}\langle \Gamma_1 X, C_{E_0,\alpha_G}\Gamma_1 X\rangle}$. Collecting the X-exponent,

$$\begin{aligned}\int \Big| e^{-\frac{1}{2}\langle \Gamma X, C_{E_0}\Gamma X\rangle} e^{-\langle B,\Gamma X\rangle}\Big| e^{\frac{\alpha_G}{2}\|B\|^2} \mathrm{d}\mu_{C_{E_0}}(B) \\ = \det_{E_0}(\mathbf{1}-\alpha_G C_{E_0})^{-1/2} e^{\frac{1}{2}\langle X, T_{\alpha_G} X\rangle}, \\ T_{\alpha_G} = \Gamma_1^{\mathsf{T}}(C_{E_0,\alpha_G}-C_{E_0})\Gamma_1 + \Gamma_2^{\mathsf{T}} C_{E_0}\Gamma_2\ ,\end{aligned} \tag{B.23}$$

a real symmetric positive semidefinite matrix on $\mathfrak{g}^I$ ($C_{E_0,\alpha_G} \geqslant C_{E_0}$ for $\alpha_G \geqslant 0$). By Lemma 3.23 the columns of Γ outside J_Z vanish, so T_{α_G} is supported on J_Z, and $\int e^{\frac{1}{2}\langle X, T_{\alpha_G} X\rangle} \mathrm{d}\mu_0(X) = \det_{J_Z}(\mathbf{1}-T_{\alpha_G})^{-1/2}$ when $\|T_{\alpha_G}\| < 1$. Integrating (B.23) against μ_0 therefore gives the integral of (ii) in closed form, $\det_{E_0}(\mathbf{1}-\alpha_G C_{E_0})^{-1/2} \det_{J_Z}(\mathbf{1}-T_{\alpha_G})^{-1/2}$, once $\|T_{\alpha_G}\| < 1$ is shown.

Norm of T_{α_G}. The resolvent identity gives $C_{E_0,\alpha_G} - C_{E_0} = \alpha_G C_{E_0,\alpha_G} C_{E_0}$, of norm at most $\alpha_G/(\gamma(\gamma-\alpha_G))$. By (i) and $\delta_M \leqslant 1$, $\|\Gamma_1\| \leqslant \|\Gamma_{E_0}(1)\| + h_+\delta_M \leqslant h_+\gamma^{-1/2} + h_+ = A_*$, since $\|\Gamma_{E_0}(1)\| \leqslant \|H\|\|\mathsf{G}^{1/2}\| \leqslant h_+\gamma^{-1/2}$ ($\mathsf{G} = H^{-1}$ and $H \geqslant \gamma$); and $\|\Gamma_2\| \leqslant h_+\delta_M$. Hence, for $\alpha_G \leqslant \min(\gamma/2, \gamma^2/(16A_*^2))$,

$$\|T_{\alpha_G}\| \leqslant \frac{\alpha_G A_*^2}{\gamma(\gamma-\alpha_G)} + \frac{h_+^2\delta_M^2}{\gamma} \leqslant \frac{2\alpha_G A_*^2}{\gamma^2} + \frac{1}{8} \leqslant \frac{1}{4}\ . \tag{B.24}$$

Determinant bounds. The eigenvalues of $\alpha_G C_{E_0}$ lie in $[0,1/2]$ and $-\frac{1}{2}\log(1-t) \leqslant t$ for $t \leqslant 1/2$, so $\log\det_{E_0}(\mathbf{1}-\alpha_G C_{E_0})^{-1/2} \leqslant \alpha_G \operatorname{tr} C_{E_0} \leqslant (\alpha_G/\gamma) d_G |E_0|$. The eigenvalues of T_{α_G} lie in $[0,1/4]$ and $-\frac{1}{2}\log(1-t) \leqslant \frac{2}{3}t$ there, so $\log\det_{J_Z}(\mathbf{1}-T_{\alpha_G})^{-1/2} \leqslant \frac{2}{3}\operatorname{tr} T_{\alpha_G} \leqslant \frac{2}{3} d_G |J_Z| \|T_{\alpha_G}\|$. With $E_0 \subseteq J_Z$ (Lemma 3.23),

$$\begin{aligned}\log[\det_{E_0}(\mathbf{1}-\alpha_G C_{E_0})^{-1/2} \det_{J_Z}(\mathbf{1}-T_{\alpha_G})^{-1/2}] &\leqslant d_G|J_Z|\Big(\frac{\alpha_G}{\gamma} + \frac{4\alpha_G A_*^2}{3\gamma^2} + \frac{2h_+^2\delta_M^2}{3\gamma}\Big) \\ &\leqslant C_{\mathrm{d}}(\alpha_G + \delta_M^2)|J_Z|, \\ C_{\mathrm{d}} = d_G \max\Big(\frac{1}{\gamma} + \frac{4A_*^2}{3\gamma^2}, \frac{2h_+^2}{3\gamma}\Big)\ .\end{aligned} \tag{B.25}$$

A bond of J_Z has its initial site in a cube of Z, so $|J_Z| \leqslant 4(LM+1)^4 N_1(Z)$. Since n_*, R_*, C_{w} are fixed before M and $\delta_M = 2C_{\mathrm{w}} 4^{-\lfloor M/(2R_*)\rfloor}$ decays exponentially in M, there is M with $4C_{\mathrm{d}}\delta_M^2 (LM+1)^4 \leqslant 1/2$; then $\alpha_G \leqslant (8C_{\mathrm{d}}(LM+1)^4)^{-1}$ gives the bound $e^{N_1(Z)}$.

Both C_{E_0} and Γ_{E_0} are block diagonal over the components D of Z by (3.67), hence so are T_{α_G} and both determinants, which proves the factorisation.

Bad-bond reserve. On $\{|B_b| \geqslant r\}$, $h(B_b)^2 \geqslant c_{\mathrm{n}}|B_b|^2 \geqslant c_{\mathrm{n}} r^2$, so $\mathbf{1}_{\{|B_b|\geqslant r\}} \leqslant e^{\frac{\gamma_2}{2}(h(B_b)^2 - c_{\mathrm{n}} r^2)}$; the product over P is (iii). *q.e.d.*

B.5 The Mayer Constants and the Activity Bounds

The following constants depend on L and the group only and are fixed in this order: $C_{\mathrm{g}} = 162$ and $\delta = 1/40$; a decay rate $\kappa \geqslant 3$ with

$$\delta\kappa \geqslant 2C_{\mathrm{g}}\log 128, \qquad \lambda' = (1-5\delta)\frac{L\kappa}{2049}, \qquad \lambda_* = \lambda' - 2C_{\mathrm{g}} \geqslant 2C_{\mathrm{g}}\log 128\ ; \tag{B.26}$$

then

$$\begin{aligned}\alpha_6 = \min\Big(\frac{1}{8}, \frac{\delta\kappa}{2C_{\mathrm{g}} B(\delta\kappa/2)}\Big), \qquad \kappa_1 - 1 \geqslant 5\lambda' + 1, \\ \theta = \frac{\lambda_*}{16 C_{\mathrm{g}}}, \qquad C_{\mathrm{s}} = \frac{2049 L^3 C_{\mathrm{g}} B(\delta\kappa) e^{L\kappa/2}}{\delta\kappa}\ ,\end{aligned} \tag{B.27}$$

and a number $\omega_* > 0$ with

$$\eta_* = 2e^{1/2+5\kappa}\omega_* \leqslant \frac{\alpha_6}{4}, \qquad \varepsilon_* = \frac{C_{\mathrm{s}}\eta_*}{\alpha_6} \leqslant \min\Big(\frac{1}{4}, \frac{1}{72B(\lambda_*/2)}\Big)\,; \tag{B.28}$$

these conditions involve only κ, α_6, C_{s} and λ_*, so ω_*, η_* and $\varepsilon_* = C_{\mathrm{s}}t_0$ are fixed before M_1, M and the Gaussian reserve α_G, as Lemma 3.27 states.

Let $t_0 \coloneqq \eta_*/\alpha_6 \leqslant 1/4$ and, once α_G of Lemma 3.24 is fixed after M, $\gamma_2 = \alpha_G/2$, so $a_{\mathrm{bad}} = c_{\mathrm{n}}\alpha_G r^2/4$. The coupling is then taken so small that $\omega(g_0) \leqslant \omega_*$, the chart condition of Lemma B.12 holds, and

$$a_{\mathrm{bad}} \geqslant \max\Big(160(\kappa + \log(8M^4+8) + 1),\ 64\log\frac{1}{\eta_*}\Big)\,. \tag{B.29}$$

These are compatible because $r = A_1(\log g_0^{-2})^{p_0}$ (the radius of the literal cutoff (4.12)) tends to infinity while $g_0(1+r)^3 \to 0$.

The bounds $B(\delta\kappa/2) \leqslant 2e^{-\delta\kappa/(2C_{\mathrm{g}})}$, $B(\delta\kappa) \leqslant 2e^{-\delta\kappa/C_{\mathrm{g}}}$ and $B(\lambda_*/2) \leqslant 2e^{-\lambda_*/(2C_{\mathrm{g}})}$ hold by Lemma B.1(iv) and (B.26).

Lemma B.14 (Cover Sum)**.** *For every nonempty union Y_0 of M-cubes,*

$$\sum_{\mathcal{D}\subseteq\mathbf{D}_0\,|\,\bigcup_{Y\in\mathcal{D}}Y=Y_0}\ \prod_{Y\in\mathcal{D}} \alpha_6 e^{-\delta\kappa\ell_0(Y)} \leqslant 1\,. \tag{B.30}$$

Proof. The members of $\mathbf{D}_0$ are localisation domains of π_0 and $\mathcal{D}$ is a set of distinct domains. For a cover, $\sum_{Y\in\mathcal{D}}\ell_0(Y) \geqslant \sum_Y N_0(Y)/C_{\mathrm{g}} \geqslant N_0(Y_0)/C_{\mathrm{g}}$ by Lemma B.1(i), so $\prod_Y e^{-\delta\kappa\ell_0(Y)/2} \leqslant e^{-\delta\kappa N_0(Y_0)/(2C_{\mathrm{g}})}$. Dropping the cover constraint, the remaining sum over sets of domains contained in Y_0 is at most $\prod_{Y\subseteq Y_0}(1+\alpha_6 e^{-\delta\kappa\ell_0(Y)/2}) \leqslant \exp(\alpha_6\sum_{Q\subseteq Y_0}\sum_{Y\ni Q} e^{-\delta\kappa\ell_0(Y)/2}) \leqslant \exp(\alpha_6 B(\delta\kappa/2)N_0(Y_0))$ by (B.2). The product of the two bounds is at most one by the choice of α_6. *q.e.d.*

Lemma B.15 (Selected Majorant)**.** *On every branch, with $\mathbf{W} = \mathbf{V}_\xi$, $\omega(g_0) \leqslant \omega_*$ and the chart condition, every selected integral of Definition 3.25 with $F = 1$ satisfies*

$$|\mathcal{A}^{\mathrm{f}}_{(\mathcal{D},\mathsf{P},Z)}[1]| \leqslant e^{-(\kappa_1-1)N_1(Z\setminus Z_0')+N_1(Z)-a_{\mathrm{bad}}|\mathsf{P}|}\prod_{Y\in\mathcal{D}}\omega_* e^{1/2}e^{-\kappa d_0(Y)}\,. \tag{B.31}$$

Proof. The weakening in (3.68) is, for each cube Δ of $Z\setminus Z_0'$, the operation

$$G \mapsto \int_0^1 \mathrm{d}s \oint_{|\sigma_\Delta|=e^{\kappa_1}} \frac{G\,\mathrm{d}\sigma_\Delta}{2\pi i(\sigma_\Delta - s)^2}$$

applied to the inner integral (the t_Y-, X- and B-integrals) as a function of σ_Δ on the circle; it uses the measurability of the inner integral in σ on the distinguished boundary (the integrand is continuous in σ, since $\Gamma_{E_0}(\sigma)$ is the walk series of Lemma 3.21, and Fubini applies to the iterated integral because the integral of the modulus of the inner integrand is bounded uniformly on the distinguished boundary, by the bounds below) and the bound $\sup_{s\in[0,1],|\sigma|=e^{\kappa_1}}|\sigma-s|^{-2}\cdot e^{\kappa_1} \leqslant e^{\kappa_1}(e^{\kappa_1}-1)^{-2} \leqslant e^{-(\kappa_1-1)}$, the last step because $e^{\kappa_1} \geqslant 3$.

Applying it for each cube of $Z\setminus Z_0'$ bounds (3.68) by $e^{-(\kappa_1-1)N_1(Z\setminus Z_0')}$ times the supremum over the distinguished boundary $\{|\sigma_\Delta| = e^{\kappa_1}\ \forall\Delta \in \pi_1(Z)\setminus\pi_1(Z_0')\}$ (with $\sigma_\Delta = 0$ off Z, $\sigma_\Delta = 1$ on Z_0') of the integral of the inner integrand's modulus, which lies in the closed polydisc of Lemma 3.24.

There, the Mayer integrals $\int_0^1 \mathrm{d}t_Y$ have mass one; $\chi_b \leqslant 1$; by Lemma 3.24(iii) with $\gamma_2 \leqslant \alpha_G$, $\prod_{b\in\mathsf{P}}(1-\chi_b) \leqslant e^{-a_{\mathrm{bad}}|\mathsf{P}|}e^{\frac{\alpha_G}{2}\|B\|^2}$; on the box $|\mathbf{V}_\xi(Y,B)e^{t_Y\mathbf{V}_\xi(Y,B)}| \leqslant \omega(g_0)e^{-\kappa d_0(Y)}e^{\omega(g_0)} \leqslant \omega_* e^{1/2}e^{-\kappa d_0(Y)}$ by Lemma B.13 and $\omega_* \leqslant 1/2$ (which follows from $\eta_* \leqslant 1/32$); and the integral of the absolute density against $e^{\alpha_G\|B\|^2/2}$ is at most $e^{N_1(Z)}$ by Lemma 3.24(ii). *q.e.d.*

Proof of Lemma 3.27. Let Z be a connected final domain. Each selection $(\mathcal{D},\mathsf{P})$ in (3.69) determines $Y_0 = \bigcup\mathcal{D}$, the seed Z_0 with wall components $Z_{0,1},\dots,Z_{0,k'}$, the components Y_i of Y_0 (each inside one $Z_{0,j}$), $n_j = N_0(Z_{0,j}\setminus Y_0)$ and $n = \sum_j n_j$. By Lemma B.15 and $\omega_* e^{1/2}e^{-\kappa d_0(Y)} \leqslant \eta_* e^{-\kappa\ell_0(Y)}$, the sum over selections is organised as $\sum_{Y'\subseteq Z}\sum_{Z_0\,|\,Z_0'=Y'}\sum_{Y_0\subseteq Z_0}\sum_{\mathcal{D}\,|\,\bigcup\mathcal{D}=Y_0}\sum_{\mathsf{P}}$ over unions of cubes Y' of π_1 and Y_0 of π_0, and each inner sum is bounded in turn.

Potential labels. The label weight factors as $\eta_* e^{-\kappa\ell_0(Y)} = t_0 e^{-(1-\delta)\kappa\ell_0(Y)}\cdot\alpha_6 e^{-\delta\kappa\ell_0(Y)}$. Each $Y\in\mathcal{D}$ lies in one component Y_i, and $\sum_{Y\subseteq Y_i}\ell_0(Y) \geqslant \ell_0(Y_i)$ by Lemma B.1(ii), so with $t_0 \leqslant 1$ and at least one label per component, $\prod_{Y\in\mathcal{D}} t_0 e^{-(1-\delta)\kappa\ell_0(Y)} \leqslant \prod_i t_0 e^{-(1-\delta)\kappa\ell_0(Y_i)}$. Lemma B.14 sums the second factors to at most one.

Bad-bond labels. Given Y_0 and Z_0, every $b\in\mathsf{P}$ has $b\cap Y_0 = \varnothing$ and both endpoints in cubes of Z_0, hence in $Z_0\setminus Y_0$; there are at most $4M^4 n$ such bonds. Every cube of $Z_0\setminus Y_0$ meets a bond of P and a bond meets at most two cubes, so $|\mathsf{P}| \geqslant n/2$. Hence

$$\sum_{\mathsf{P}} e^{-a_{\mathrm{bad}}|\mathsf{P}|/2} \leqslant e^{-a_{\mathrm{bad}}n/8}\sum_{\mathsf{P}} e^{-a_{\mathrm{bad}}|\mathsf{P}|/4} \leqslant \exp\Big[-\frac{a_{\mathrm{bad}}}{8}n + 4M^4 e^{-a_{\mathrm{bad}}/4}n\Big] \leqslant e^{-a_{\mathrm{bad}}n/16}\,, \tag{B.32}$$

since $4M^4e^{-a_{\mathrm{bad}}/4} \leqslant a_{\mathrm{bad}}/16$: by (B.29), $a_{\mathrm{bad}} \geqslant 160(\kappa+1) > 1$ and $64M^4e^{-a_{\mathrm{bad}}/4} \leqslant 1$ (as $a_{\mathrm{bad}}/4 \geqslant 40\log(8M^4+8) \geqslant \log(64M^4)$).

The unused half $e^{-a_{\mathrm{bad}}|\mathsf{P}|/2} \leqslant e^{-a_{\mathrm{bad}}n/4}$ leaves in total $e^{-5a_{\mathrm{bad}}n/16} = \prod_j e^{-5a_{\mathrm{bad}}n_j/16}$, of which $e^{-a_{\mathrm{bad}}n_j/32}$, $e^{-a_{\mathrm{bad}}n_j/64}$ and $e^{-a_{\mathrm{bad}}n_j/64}$ are used below.

Component reserve. For a component $Z_{0,j}$ containing $k_j \geqslant 1$ components Y_i of Y_0, Lemma B.1(iii) gives $\ell_0(Z_{0,j}) \leqslant \sum_i \ell_0(Y_i) + 5n_j$, so with $a_{\mathrm{bad}}/32 \geqslant 5\kappa$,

$$t_0^{k_j} e^{-(1-\delta)\kappa\sum_i \ell_0(Y_i)} e^{-a_{\mathrm{bad}}n_j/32} \leqslant t_0 e^{-(1-\delta)\kappa\ell_0(Z_{0,j})}\,. \tag{B.33}$$

If $k_j = 0$ then $n_j \geqslant 1$, $\ell_0(Z_{0,j}) \leqslant 5n_j$, and $e^{-a_{\mathrm{bad}}n_j/32}e^{-a_{\mathrm{bad}}n_j/64} \leqslant e^{-5\kappa n_j}\eta_* \leqslant t_0 e^{-(1-\delta)\kappa\ell_0(Z_{0,j})}$, using $e^{-a_{\mathrm{bad}}/64} \leqslant \eta_* \leqslant t_0$.

Choice of Y_0. With the reserve $e^{-a_{\mathrm{bad}}n/64}$, $\sum_{Y_0\subseteq Z_0} e^{-a_{\mathrm{bad}}N_0(Z_0\setminus Y_0)/64} = (1+e^{-a_{\mathrm{bad}}/64})^{N_0(Z_0)} \leqslant \exp(e^{-a_{\mathrm{bad}}/64}C_{\mathrm{g}}\sum_j \ell_0(Z_{0,j})) \leqslant \prod_j e^{\delta\kappa\ell_0(Z_{0,j})}$, since $C_{\mathrm{g}}e^{-a_{\mathrm{bad}}/64} \leqslant 162e^{-5(\kappa+1)/2} \leqslant \kappa/40$ for $\kappa\geqslant 3$. Altogether, writing $\mathfrak{M}(Z_0,Z)$ for the sum of (B.31) over the selections with seed Z_0,

$$\mathfrak{M}(Z_0,Z) \leqslant e^{-(\kappa_1-1)N_1(Z\setminus Z_0')+N_1(Z)}\prod_j t_0 e^{-(1-2\delta)\kappa\ell_0(Z_{0,j})}\,. \tag{B.34}$$

Scale change. By Lemma B.2, the enlargement Z_0' of Section 3.5 (the LM-cubes meeting $\operatorname{int}Q^\sim$ for some cube Q of Z_0) equals $\bigcup_j Z_{0,j}'$, where $Z_{0,j}'$ is the enlargement of the component $Z_{0,j}$ in the sense of that lemma, so (B.3) applies to each $Z_{0,j}$. Each $Z_{0,j}'$ is connected, so every component Y_i' of $Y' = Z_0'$ is the union of the $Z_{0,j}'$ it contains.

With $e^{-(1-2\delta)\kappa\ell_0(Z_{0,j})} = e^{-\delta\kappa\ell_0(Z_{0,j})}e^{-(1-3\delta)\kappa\ell_0(Z_{0,j})}$ and (B.3) in the form $\ell_0(Z_{0,j}) \geqslant \frac{L}{2049}\ell_1(Z'_{0,j}) - \frac{5L+1024}{2049}$, where $(5L+1024)/2049 \leqslant L/2$ for $L \geqslant 3$,

$$\prod_{j \,|\, Z'_{0,j} \subseteq Y'_i} e^{-(1-3\delta)\kappa\ell_0(Z_{0,j})} \leqslant e^{L\kappa/2 \cdot k'_i} \prod_j e^{-(1-3\delta)\frac{L\kappa}{2049}\ell_1(Z'_{0,j})} \leqslant e^{L\kappa k'_i/2} e^{-(1-3\delta)\frac{L\kappa}{2049}\ell_1(Y'_i)} ,$$

by Lemma B.1(ii), where k'_i counts the fine components in Y'_i.

The remaining factors $t_0 e^{L\kappa/2} e^{-\delta\kappa\ell_0(Z_{0,j})}$ pay for the sum over the set of fine components inside Y'_i: each contains an M-cube of Y'_i, of which there are $L^4 N_1(Y'_i)$, so by (B.2) the sum over single components is $x_i \leqslant L^4 N_1(Y'_i) t_0 e^{L\kappa/2} B(\delta\kappa) \leqslant \varepsilon_* \frac{\delta L\kappa}{2049}\ell_1(Y'_i)$, using $N_1 \leqslant C_{\mathrm{g}}\ell_1$ and the definitions of C_{s} and $\varepsilon_* = C_{\mathrm{s}} t_0$, and the sum over nonempty sets is at most $e^{x_i} - 1 \leqslant x_i e^{x_i}$. Let $\Lambda_i := \frac{L\kappa}{2049}\ell_1(Y'_i)$ and $y = \delta\Lambda_i$, so $x_i \leqslant \varepsilon_* y \leqslant y$. Then $x_i e^{x_i} e^{-(1-3\delta)\Lambda_i} \leqslant \varepsilon_* y e^y e^{-(1-3\delta)\Lambda_i} = \varepsilon_* y e^{-(1-4\delta)\Lambda_i} = \varepsilon_* (y e^{-y}) e^{-(1-5\delta)\Lambda_i} \leqslant \varepsilon_* e^{-\lambda'\ell_1(Y'_i)}$, using $y e^{-y} \leqslant 1$, so

$$\sum_{Z_0 \,|\, Z'_0 = Y'} \mathfrak{M}(Z_0, Z) \leqslant e^{-(\kappa_1 - 1)N_1(Z \setminus Y') + N_1(Z)} \prod_{Y'_i \in \mathrm{cc}(Y')} \varepsilon_* e^{-\lambda'\ell_1(Y'_i)} . \tag{B.35}$$

Choice of Y'. For nonempty $Y' \subseteq Z$ with $n' := N_1(Z \setminus Y')$ cubes outside, Lemma B.1(iii) gives $\ell_1(Z) \leqslant \sum_i \ell_1(Y'_i) + 5n'$, so $\prod_i e^{-\lambda'\ell_1(Y'_i)} \leqslant e^{-\lambda'\ell_1(Z)} e^{5\lambda' n'}$, and $\varepsilon_*^{\#\,\mathrm{cc}(Y')} \leqslant \varepsilon_*$. With $\kappa_1 - 1 \geqslant 5\lambda' + 1$,

$$\begin{aligned} |k^{\mathrm{f}}_\xi(Z)| &\leqslant \varepsilon_* e^{N_1(Z)} e^{-\lambda'\ell_1(Z)} \sum_{\varnothing \neq Y' \subseteq Z} e^{-N_1(Z \setminus Y')} \\ &\leqslant \varepsilon_* e^{(1+e^{-1})N_1(Z)} e^{-\lambda'\ell_1(Z)} \leqslant \varepsilon_* e^{-(\lambda' - 2C_{\mathrm{g}})\ell_1(Z)} , \end{aligned}$$

which is the first inequality of (3.72).

Criterion. Let Z be any union of LM-cubes, connected or not. A domain Z' incompatible with Z contains one of the at most $9N_1(Z)$ cubes of Z or wall adjacent to Z. Since $(1 + 6\theta)N_1(Z') \leqslant (1+6\theta)C_{\mathrm{g}}\ell_1(Z') \leqslant \frac{\lambda_*}{2}\ell_1(Z')$ by $C_{\mathrm{g}} \leqslant \lambda_*/8$ and $6\theta C_{\mathrm{g}} = 3\lambda_*/8$, (B.2) gives

$$\sum_{Z' \not\sim Z} |k^{\mathrm{f}}_\xi(Z')| e^{(1+6\theta)N_1(Z')} \leqslant 9N_1(Z)\varepsilon_* B(\lambda_*/2) \leqslant \frac{1}{8} N_1(Z) . \tag{B.36}$$

The reference data are treated in Section B.7. *q.e.d.*

B.6 Proof of the Vacuum-Cluster Bounds

Proof of Lemma 3.28. Species are the connected final domains D, with weights $\widetilde{z}_D = |k^{\mathrm{f}}_\xi(D)| e^{2\theta N_1(D)}$ and sizes $a(D) = N_1(D)/8$; two species are incompatible when they share a cube or a wall, and each with itself. By (B.36) and $2\theta + \frac{1}{8} \leqslant 1 + 6\theta$,

$$\sum_{D' \not\sim D} \widetilde{z}_{D'} e^{a(D')} \leqslant \sum_{D' \not\sim D} |k^{\mathrm{f}}_\xi(D')| e^{(1+6\theta)N_1(D')} \leqslant a(D) .$$

Tuples of species are written $\mathfrak{c} = (D_1, \dots, D_m)$ in this proof (γ is the spectral bound $\gamma_L/2$). The argument of Section 5 following (5.29) applies verbatim to these species, weights and sizes: the absolute Ursell coefficient $|\varphi^T(\mathfrak{c})|$ is bounded by the number of spanning trees of the incompatibility graph of $\mathfrak{c}$, the rooted-tree generating sum with root D is the monotone limit of the recursion (5.30) with $\widetilde{z}$ in place of $|\widetilde{z}_p|$, the displayed criterion gives $T_D^{(k)} \leqslant \widetilde{z}_D e^{a(D)}$

inductively, and the factor $1/m!$ in (3.73) together with the multiplicity $m_D(\mathfrak{c})$ of D in the tuple reproduces the rooted sum. Hence, for every connected final domain D,

$$\sum_{\mathfrak{c}} m_D(\mathfrak{c})|\Phi^{\mathrm{v}}_\xi(\mathfrak{c})|e^{2\theta\sum_j N_1(D_j)} \leqslant |k^{\mathrm{f}}_\xi(D)|e^{2\theta N_1(D)+N_1(D)/8}\,. \tag{B.37}$$

Here $\mathcal{A}$ is any finite set of unions of LM-cubes, connected or not; the criterion (B.36) was proved for every such union. A tuple hitting $\mathcal{A}$ contains a member D' incompatible with some $D\in\mathcal{A}$, so

$$\begin{aligned}\sum_{\mathfrak{c}\text{ hits }\mathcal{A}}|\Phi^{\mathrm{v}}_\xi(\mathfrak{c})|e^{2\theta\sum_j N_1(D_j)} &\leqslant \sum_{D\in\mathcal{A}}\sum_{D'\not\sim D}\sum_{\mathfrak{c}} m_{D'}(\mathfrak{c})|\Phi^{\mathrm{v}}_\xi(\mathfrak{c})|e^{2\theta\sum_j N_1(D_j)}\\ &\leqslant \sum_{D\in\mathcal{A}}\sum_{D'\not\sim D}|k^{\mathrm{f}}_\xi(D')|e^{(2\theta+\frac18)N_1(D')} \leqslant \frac18\sum_{D\in\mathcal{A}}N_1(D)\,.\end{aligned}$$

For (3.74), at fixed volume there are finitely many species. Let $\mathcal{S}_+$ be the set of species with $k^{\mathrm{f}}_\xi(D)\neq 0$; species with $k^{\mathrm{f}}_\xi(D)=0$ contribute to neither side (a tuple containing one has $\Phi^{\mathrm{v}}_\xi=0$, and the gas does not see them), so they are discarded.

For complex activities $z=(z_D)_{D\in\mathcal{S}_+}$, the hard-core gas $\Xi(z)=\sum_{\mathcal{P}\text{ compatible}}\prod_{D\in\mathcal{P}}z_D$ is a polynomial with $\Xi(0)=1$: the polymer partition function (5.3) of [11, Chapter 5] for the polymers $\mathcal{S}_+$ with the hard-core interaction $\zeta(D,D')=0$ on incompatible and $\zeta(D,D')=1$ on compatible pairs (which satisfies (5.1)–(5.2) there).

Let $S(z)=\sum_{\mathfrak{c}}\frac{\varphi^T(\mathfrak{c})}{m!}\prod_j z_{D_j}$, the sum over all ordered tuples $\mathfrak{c}=(D_1,\dots,D_m)$, $m\geqslant 1$, of members of $\mathcal{S}_+$; this is the cluster series of that gas, since the Ursell function of [11, (5.6)] is $\frac{1}{m!}\sum_G\prod_{\{i,j\}\in G}(\zeta(D_i,D_j)-1)=\frac{\varphi^T(\mathfrak{c})}{m!}$, the sum over the connected graphs G on $\{1,\dots,m\}$, a factor $\zeta-1$ being -1 on an incompatible pair and 0 otherwise.

[11, Proposition 5.3] is the identity $\Xi(z)=\exp(S(z))$ between formal power series in the variables z_D (its proof, (5.4)–(5.8) there, rearranges finitely many terms for each monomial).

By (B.37) with z_D in place of $k^{\mathrm{f}}_\xi(D)$ (the bound is monotone in the weights and only uses $|z_D|\leqslant|k^{\mathrm{f}}_\xi(D)|$ through the criterion; equivalently [11, Theorem 5.4], whose hypothesis (5.10) is the first display of this proof and whose conclusion (5.9) is this absolute convergence), S converges absolutely at $|z_D|=|k^{\mathrm{f}}_\xi(D)|$, hence absolutely and uniformly on the closed polydisc $\overline{P}=\{|z_D|\leqslant|k^{\mathrm{f}}_\xi(D)|\ \forall D\in\mathcal{S}_+\}$ (Weierstrass M-test), so S is continuous on $\overline{P}$, analytic on the open polydisc P, and its Taylor series at 0 is the formal series S grouped by monomials.

The analytic functions Ξ and $\exp\circ S$ on the connected open set $P\ni 0$ have the same Taylor series at 0, by the formal identity, so they coincide on P (identity theorem) and, by continuity, on $\overline{P}$; in particular at $z_D=k^{\mathrm{f}}_\xi(D)$, where $\Xi(k^{\mathrm{f}}_\xi)=\Xi^{\mathrm{f}}_\xi$ and $S(k^{\mathrm{f}}_\xi)=\sum_{\mathfrak{c}}\Phi^{\mathrm{v}}_\xi(\mathfrak{c})$ by (3.73).

The restricted gas $(\Xi^{\mathrm{f}}_\xi)^{[\cup\mathcal{A}]}$ is the same gas with $z_D=0$ for the D incompatible with $\bigcup\mathcal{A}$, which are exactly the D incompatible with some member of $\mathcal{A}$; its expansion is the sum over the tuples none of whose members hits $\mathcal{A}$. The quotient is therefore $\exp(-\sum_{\mathfrak{c}\text{ hits }\mathcal{A}}\Phi^{\mathrm{v}}_\xi(\mathfrak{c}))$. *q.e.d.*

B.7 The Reference Data

The reference data of Section 3.8 consist of the identity background $U^\circ_1=\mathbf{1}$ on every bond (the constrained minimiser for identity averages, with zero action and zero gauge functional), $V^\circ_0=\mathbf{1}$, zero retained input $A^\circ_{\mathfrak{e}}=0$ with $\Psi^\circ_\xi=\mathbf{1}$, and the same I, Ω_1, partitions, hosts, cutoff

radius r and constants as the underlying branch. They are not a branch, so the conclusions of the lemmas are checked directly.

The reference Hessian is the regional Hessian (3.16) at the identity background on the branch's Ω_1, $H^\circ_\xi = \mathsf{H}^{\Omega_1}_{\mathbf{1}}$, and H° denotes its restriction to I.

Since I, Ω_1 and R_{c} are those of the branch, the geometric facts used in Lemma B.6 (for $b \in I$ every site of a stencil $\mathcal{S}_{bb'}$ lies within $3L$ of b_-, hence within $3R_{\mathrm{c}}$ of I and in $\Omega_1^{\sim -1}$, by Lemmas B.3 and 3.19(i)) hold for the reference data as well; Lemma B.6 is stated for every background on which the regional form is defined, so it applies to the identity background.

Lemma B.16 (Reference Data)**.** *The reference precision H° satisfies* (3.23)*, as do its principal restrictions, and has the range and gauge law of Lemma 3.7; Lemma 3.12 holds for the reference data (at $U = \mathbf{1}$, $A_{\mathrm{e}} = 0$); Lemma 3.24 holds for H° with the same $\gamma, h_+, R, M, \alpha_G$; and* (3.72) *and Lemma 3.28 hold for the reference activities $k^{\mathrm{f},\circ}_\xi$ with the same constants. The reference row solutions, phases and primitives are real, and the reference normalisations $K^\circ_{\xi;0}$, $I^\circ_{\xi;0}$ of Section 3.8 are positive.*

Proof. By Lemma B.6 with the identity background, the entries of H° with a label in I are $f_{bb'}(0)$, the entries of the flat Hessian H of (4.7) on the branch's torus. The compression of $\mathsf{H} \geqslant \gamma_L I$ (A.25) to I, and further to any subset of I, gives the lower bound $\gamma_L \geqslant \gamma_L/2$ for H° and for every principal restriction, and the Schur bound with row sums $\sum_{b'} \|f_{bb'}(0)\| \leqslant m_L$ gives the upper bound for H° and every principal restriction (a principal restriction has smaller row sums); the range is ρ_L by Lemma B.3.

The gauge law of Lemma 3.7 is the entrywise covariance (3.21), proved there for every background, hence for $\mathbf{1}$: $f_{bb'}(\mathbf{1}^u) = \mathrm{Ad}_{u(b_-)} f_{bb'}(\mathbf{1}) \mathrm{Ad}^{-1}_{u(b'_-)}$.

Lemma 3.24 and Lemmas 3.21, 3.22, 3.23 use only (3.56), the range and the partitions, so they hold for H°.

Lemma B.8 applies to the identity background (its hypothesis, a site gauge with phases at most $\tau_L/2$ on the two-block sets and blocks concerned, is satisfied by $u = \mathbf{1}$, $\phi = 0$; the geometric facts of the applicability paragraph depend only on I, Ω_1, R_{c}), so Lemma 3.12 holds at $U = \mathbf{1}$, and Lemma B.12 holds with $r^\circ_{\mathrm{e}} = 0$ and $s^\circ = 1 + r \leqslant s_\xi$; the hosts and the host multiplicity are those of Lemma B.13, so (B.21) holds for $\mathbf{V}^\circ_\xi$. The proofs of Lemmas B.15, 3.27 and 3.28 use nothing else about the branch.

Reality: the row solutions are the limits of the contraction iterates of Lemma B.8(ii), which are real for real phases, so for real inputs the phases, the Haar logarithms, the determinant logarithms (positive determinants) and the plaquette and gauge terms are real, and so are the primitives and $\mathbf{V}^\circ_\xi$.

Positivity: $K^\circ_{\xi;0} = \mathbf{1}_{\mathcal{D}_I} \exp(-\frac{1}{2}\langle a, H^\circ a\rangle + P^\circ_{\xi,0}(a))$, the function (3.45) for the reference data at $t = 0$, is a real exponential on the open box, the reference primitives being real, hence positive there; and $I^\circ_{\xi;0} = \int K^\circ_{\xi;0}\, \mathrm{d}a$ is positive and finite ($H^\circ \geqslant \gamma$ and the exponent is bounded on the box by Lemma 3.12). *q.e.d.*

B.8 The Thresholds

Before the coupling, in this order:

- $L > 11$ odd ([8, p. 251], the range in which Bałaban's results are stated);

- the group constants $c_{\mathrm{h}} = \sqrt{N}$, $c_{\mathrm{n}} = 1/N$, $\varrho_G = \pi/\sqrt{N}$, c, r_G, c_G, and r_M, r'_M (Lemma B.9; these two depend on L and the group);
- τ_L by (B.6), with m_L, c_L (Lemma 3.7) and γ_L (Appendix A);
- $R_{\mathrm{loc}} = 2L + 2$, R_{p}, $s_{\mathfrak{P}}$, $R_{\mathrm{c}} \geqslant \max(2R_{\mathrm{loc}} + 1, s_{\mathfrak{P}} + 2, 4L + 5)$ and $C_{L,R_{\mathrm{c}}}$;
- the two-block size $n_{\mathcal{E}}$ of Lemma B.11 (a constant of L);
- the ranges R_* and R_H of Section 5 (the perimeter range bounding the primitive-label diameters and the source-to-stencil distances, distinct from the walk range $R_* = n_* R$ below, and the range of the flat Hessian, whose entries couple the bonds of a common plaquette or block; both are constants of L), hence $R_0 = 4(1 + R_* + R_H + L)$;
- $\vartheta_* \leqslant \tau_L/2$, $\rho_{\mathrm{d}} = 2K_* \vartheta_*^2$, C_{C}, C_{D}, C_k^{a} $(0 \leqslant k \leqslant 5)$ and C_k^{D} $(0 \leqslant k \leqslant 6)$ of Lemma B.8 and C_{p} of Lemma B.12;
- $\gamma = \gamma_L/2$, $h_+ = m_L$, R;
- C_{g}, δ, κ, λ', λ_*, α_6, κ_1, θ, C_{s} by (B.26)–(B.27), then ω_*, η_*, ε_*, t_0 by (B.28), with $\omega_* \leqslant 1/2$ (used in Appendix E) (these involve only κ, α_6, C_{s}, λ_*, so they are fixed before M_1 and M, as Lemma 3.27 states);
- n_*, R_*, C_{w} (Section 3.5), A_*, C_{d} (Lemma 3.24);
- M_1 as the M_1 of [9, p. 245], M_2 and M odd powers of L with $M_1 < M_2 < M$, M/M_1 integral, $M \geqslant 8LM_2$, the requirement of [9, p. 245] that M be large, $\rho_{\mathrm{w}} = R_* + R + 1 + 2M_1$, (3.59), $\delta_M \leqslant 1$, $h_+^2 \delta_M^2/\gamma \leqslant 1/8$ and $4C_{\mathrm{d}}\delta_M^2(LM+1)^4 \leqslant 1/2$;
- α_G with $0 < \alpha_G \leqslant \min(\gamma/2, \gamma^2/(16A_*^2), (8C_{\mathrm{d}}(LM+1)^4)^{-1})$, $\gamma_2 = \alpha_G/2$;
- then N_{h} and C_0 of (B.20), whose hosts depend on the chosen M;
- then the constants of the signed-defect estimates (Appendix E, where they are introduced), in this order: the character constant K_χ, the dummy weight u_* of the host comparison, the support reserve η of the positive group summation, its pin constant C_{pin}, and the constant C_M of Lemma E.14, all fixed before the coupling, each depending on L, M, κ, ω_*, the group, the character and the constants earlier in this order only.

Then the constant β_0 of [9, (2.6)] (p.255; the coupling is then taken small enough for (2.6) to hold with it), the mean radius ρ_{av} of Section D.8.4, the constants $C_{L,G}$, $c_{L,G}$ of Proposition D.20, and C_{H}, c_{H} of Lemma D.21, which depend only on L, the group, M_1, M_2, and the constants of [9] and [6].

The coupling g_0 is then taken below a threshold, and the torus is chosen last: every fine period a multiple of LMR_1 and at least $8M$ ((3.1); the partitions of Section 3.5 need a multiple of LM, the hosts of Section 3.3 need $8M$, and the partitions of [9] into LMR_1-cubes need the multiple of LMR_1; $R_1 = R_1(g_0)$, so the period condition is imposed after the coupling) and every coarse period exceeding $\max(2, 2R_{\mathrm{p}} + 8)$.

The statements about a square C_n of side nL (Proposition 7.4, Appendices E and F, Section 9) assume in addition the square/torus geometry of Section 5.3: every coarse period $K_\mu \geqslant 3(n+2)$, so that the square and its fixed neighbourhood have unique lifts and the loop is contractible; and Appendix D (Section D.7) and Section 9 assume compatible even fine periods. These are lower bounds on the periods in terms of the fixed loop and the coupling; for a given loop and coupling they hold on every sufficiently large compatible torus.

In the threshold conditions, $\varepsilon_1 = g_1 p_0(g_1)$ with $g_1 = g_1(g_0)$ by [9, (1.27)]; R_1 is that of [9], the smallest power of L with $R_1 \geqslant (\log g_1^{-2})^{\mathfrak{r}}$ (p.246, with Bałaban's exponent written $\mathfrak{r}$; $\mathfrak{r} \geqslant 2$ at p.245, $p_0 \geqslant 5\mathfrak{r}$ at p.246, and [9, (2.5)] for the general step); and the cutoff radius is $r = A_1(\log g_0^{-2})^{p_0}$, the radius of the literal cutoff (4.12); it is δ_0/g_0 for the $\delta_k = g_k A_1/A_0\, p_0(g_k)$ of [9] as printed at p.265, which identifies Θ_ξ with the test (1.18) (the first member of the equality chain printed for δ_0 at p.247 is corrected in (H4)). The threshold conditions are:

1. $LMR_1 > 4R_{\mathrm{c}} + 4L^2 M_2 R_1 + 8$ (Lemma 3.19), $LM_2R_1 \geqslant M$ (so that the LM-cubes nest in the $L^2M_2R_1$-cubes, (3.1)) and $4\,O(1)B_3L^2 e^{-\delta_{\mathrm{c}} LM_2R_1} \leqslant 1$ (Lemma B.4); both depend on the coupling through R_1;

2. (B.8), the hypotheses of Lemma 3.8, which (3.1) lists as well;

3. $4C_{\mathsf{C}} g_0(1+2r) \leqslant \vartheta_*/2$ (Lemma B.12);

4. the activity budgets of Section B.5: $\omega(g_0) = C_0 g_0 (1+r)^3 \leqslant \omega_*$ (B.20) and (B.29) with $a_{\mathrm{bad}} \coloneqq c_{\mathrm{n}} \alpha_G r^2/4$;

5. Bałaban's own conditions in [9], taken over as they stand:

   - ε_0 small (Theorem 1 of [6], reference [15] of [9], behind (B1); and, the constants $O(L^2)$ of p.247 and $O(1)$ of [3, p. 25] being independent of the coupling ((B1), (H4)), so small that $2\,O(L^2)\varepsilon_0$ and $O(1)L^2\varepsilon_0$ lie below ϱ_G, $r_G/\sqrt{N}$, and the small-diameter radius of the group mean, the domain statement of (B1));
   - MR_1 large (p.251, the bound on $\mathbf{H}_{1,Ax}$ at the tolerance $\delta_0/40$ of (3.40), behind the change of variables (1.22), Lemma 3.16: by Lemma D.21 it is the condition $\frac{\pi}{2} L C_{\mathrm{H}} (1+\beta_0)(A_0/A_1) e^{-c_{\mathrm{H}} LMR_1} \leqslant \frac{1}{40}$, with C_{H} and c_{H} fixed before g_0, together with the conditions of that lemma's threshold);
   - A_0/A_1 and M_2R_1 large (the two qualitative conditions of (B3), pp.248 and 250; the second requires $O(1)B_3 e^{-\delta_{\mathrm{c}} LM_2R_1}\varepsilon_1$ to be much smaller than δ_0, and $\varepsilon_1/\delta_0 \leqslant (1+\beta_0)A_0/A_1$ stays bounded by [9, (2.8)] and $\delta_0 = (A_1/A_0)\varepsilon_0$, so $R_1 \to \infty$ at fixed L, M_2 suffices, the constants being independent of the coupling by (B3)(iii)).

   The conditions on MR_1 and M_2R_1 depend on the coupling through R_1 and are met for small g_0 at fixed M, M_2, since $R_1 \to \infty$; the condition on A_0/A_1 is met by the choice before the coupling; Bałaban's conditions on M_1, M_2, M themselves are not listed in [9] (p.245: M_1 the cube size of the previous papers, M_2 not too much larger, M much larger than M_2, "dictated by many conditions involving this constant") and are taken over as the hypothesis that the construction of [9] is valid for the values chosen above, which follow its prescription.

6. the fibre conditions (B.16) of Lemma B.11: $3c_G c_{\mathrm{h}} s_{\mathfrak{P}} \varepsilon_0 \leqslant \tau_L/2$ (which gives $3c_{\mathrm{h}} s_{\mathfrak{P}} \varepsilon_0 \leqslant r_G$, by $\tau_L \leqslant 2c_G r_G$) and $n_{\mathcal{E}}^{1/2} c_G c_{\mathrm{h}}\, 2\,O(L^2)\varepsilon_0 \leqslant \vartheta_*/4$, with $n_{\mathcal{E}}$ the largest number of bonds of a two-block set and $O(L^2)$ the constant of p.247 of item 5.

All hold for small g_0; for item 4 because $a_{\mathrm{bad}} \to \infty$ and $g_0(1+r)^3 \to 0$, and for item 6 because $\varepsilon_0 = g_0 A_0 (\log g_0^{-2})^{p_0} \to 0$ while its other constants are fixed before the coupling, $O(L^2)$ being independent of the coupling and of the branch by [5, Lemma 1, p. 79] ((B1), (H4)).

The first-step coupling is defined by [9, (1.27)], $g_0^{-2} = g_1^{-2} + \beta_1(g_0)$, and [9, (2.6), p. 255] (with $n = 1$, $m = 0$) states $g_1 \leqslant (1+\beta_0) g_0$ and $g_0 \leqslant (1+\beta_0) g_1$, where $\beta_0 > 0$ is a constant that "can be chosen arbitrarily small, if g is sufficiently small" (p.255); [9] derives (2.6)

from the renormalisation group equations (0.20) of [I] and the properties of the β-functions (p.255). Hence $g_1 \to 0$ with g_0, so $R_1 \to \infty$; and [9, (2.8), p. 256] (with $n = 1$, $m = 0$) gives $\varepsilon_1 \leqslant (1 + \beta_0)\varepsilon_0$, so $\varepsilon_1 = g_1 A_0 (\log g_1^{-2})^{p_0} \to 0$ with $\varepsilon_0 = g_0 A_0 (\log g_0^{-2})^{p_0}$; while $r \to \infty$ and $g_0(1+r)^3 \to 0$. (The p_0 of [9] denotes both the function $p_0(g) = A_0(\log g^{-2})^{p_0}$ and the integer exponent in it.)

The threshold depends on L, the group (through c_{h}, c_{n}, ϱ_G, c, r_G, c_G, r_M, d_G), the choices τ_L, ϑ_*, R_{c}, κ, α_G, ω_* made above, M_1, M_2, M, and, through ε_1, R_1 and r, on the constants A_0, A_1, p_0, $\mathfrak{r}$ of [9], the map $g_0 \mapsto g_1$ of [9, (1.27)], the constants B_3, δ_{c} and the $O(1)$ of the comparison after (1.19) (p.268; Bałaban's constants of (B3)(iii) and (H4)), the threshold ε_0 of Theorem 1 of [6] (reference [15] of [9]) behind (B1), the constant $O(L^2)$ of p.247 (items 5 and 6), the constants C_{H} and c_{H} of Lemma D.21 behind (B4)(ii), and the tolerances "much smaller than δ_0" of p.250 behind (B3)(ii).

Further conditions are imposed where the appendices and Section 9 state them; each is a condition that g_0 lie below a threshold depending on L, the group, the character, M_1, M_2, M, the constants chosen above and those of [9], holding for all small g_0, or a lower bound on the periods in terms of the loop and the coupling, holding on every sufficiently large compatible torus. They are:

- $\delta_0 \leqslant \delta_G$ (Lemma 3.16);
- (D.2) ($r \geqslant 1$ and $C_P g_0(1 + 2r) \leqslant \gamma/2$, Section D.1);
- the threshold g_* of Proposition D.18, which contains the threshold of Proposition D.20, the coupling conditions of [9, (2.6)] and the smallness giving $q \leqslant e^{-c_* \pi_*^2}$ (Section D.7);
- the thresholds of Lemma E.1 (its last inequality), the smallness conditions of Lemma E.9 on the root weight and the smallness of Lemma E.10 (Appendix E, Proposition E.26);
- the further threshold of (9.7) (Corollary 9.4);
- the thresholds of Lemmas D.12, D.13 and D.17 (Section D.7; every condition there is an upper bound on the witness threshold θ of (D.26) or a lower bound on R_1);
- the collar conditions (7.28), which are item 1 with $R_0 = 4(1 + R_* + R_H + L)$ in place of R_{c}, the small-coupling conditions $g_0 C_{\mathsf{C}}(2 + 4r) < \vartheta_*/2$ and $g_0(2 + 4r) < a_L$ with the polymer-load inequalities of the proof of Proposition 7.4, and the torus condition $4ne^{-c\iota_T} \leqslant 1$ of Corollary 9.1;
- the coupling conditions of the flat and regional expansions where they are imposed ($\tau < 1$, the original chart condition, the smallness of $K_1^{s_*}\tau$ and the straight- and bent-path thresholds of Sections 5 and 6, and the threshold of Proposition 7.1);
- $R_\Upsilon \geqslant 2$, $u \leqslant u_*$, $v \leqslant 1$ and the small-coupling conditions of Sections E.9.1 and E.10.5 (Proposition E.26);
- the further threshold of Corollary 1.4 ($C\tau^2 L^4 C_K \Phi_c \leqslant 1$ and the bound on $\delta_q^\sharp$),

each stated at the place cited.

Appendix C The Local Interaction at the Identity

Locality of the First-Step Interaction. The trace in the bare action is $\mathrm{tr}_N = N^{-1}\,\mathrm{Tr}$ in its fixed unitary embedding, so $\mathrm{tr}_N(\mathbf{1}) = 1$. Let a_{ph} be a radius for the input A of $\widetilde{D}$, measured in the maximum fine-bond Hilbert–Schmidt norm, on which the local implicit solutions, principal logarithms and composed action and Haar primitives are analytic, with nonzero Jacobian determinants and Haar densities.

The radius a_L is a free-input radius with $a_L \leqslant a_{\mathrm{ph}}/\max(1, \|\mathsf{C}\|_{\infty\to\infty})$, where the operator norm maps the maximum free-bond Hilbert–Schmidt norm to the maximum fine-bond Hilbert–Schmidt norm. This norm is finite and depends on L only, by the fixed row stencils. The Lie-algebra norm is the normalised Hilbert–Schmidt norm, and $|A|_{\mathrm{HS}} \leqslant |A|_{\mathrm{op}}$. Thus $c_{\mathrm{cmp}} = 1$ is a common norm-comparison constant for both cutoff prescriptions, and $g_0 r(g_0) < a_L$ suffices.

The componentwise implicit solutions assemble to a small solution in this maximum norm. Uniqueness of the analytic germ in [8] identifies it with $\widetilde{D}$ on the smaller common chart.

The following calculation concerns the identity-background, all-regular integral below.

Let $\mathcal{S}_c$ contain the positive fine bonds with both endpoints in $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$, and let $\Lambda_c \coloneqq \mathcal{S}_c \cap E_*$. At the present torus sizes, the only designated bond in $\mathcal{S}_c$ is $b_0(c)$. Indeed, an internal path in either block crosses no block interface, and the translated length-L path in [8, (0.4)] crosses only the interface between these two blocks. The reference coarse path crosses the same interface. Thus $\mathcal{Q}_c$ and $\mathcal{N}_c$ depend only on the fine bonds in $\mathcal{S}_c$. In the defining equation for $\widetilde{D}_c(A)$, the only unknown correction entering those bonds is $\widetilde{D}_c(A)$ itself. The analytic implicit-function theorem therefore gives a function of $A|_{\mathcal{S}_c}$ alone, with its radius and constants independent of the number of blocks. Here $h_c = L^3 I$ at the identity, since $Q_0 = L^{-4} I$.

For a free fine bond b, let $\Lambda_b \coloneqq \{b\}$; for a designated bond $b = b_0(c)$, let $\Lambda_b \coloneqq \Lambda_c$. Both $(\mathsf{C}B)(b)$ and $q_{g_0}(B)(b)$ depend only on $B|_{\Lambda_b}$.

The index α runs over the plaquettes and the pairs (y, x) in (4.5). Let $f_\alpha(q)$ be the corresponding summand of $\mathscr{A}(e^{iq})$, a real analytic scalar function near zero in $\mathfrak{g}^E$, extended holomorphically to complex arguments. Let Λ_α be the union of the sets Λ_b over the bonds in that summand's plaquette or path family. Each Λ_α lies in at most four adjacent blocks. Each free bond belongs to a bounded number of the sets $\Lambda_\alpha, \Lambda_b, \Lambda_c$, with a bound depending on L, not on the torus.

Let $\psi(A) \coloneqq A - h\widetilde{D}(A)$. The local scalar functions are the real analytic branches at zero of

$$\begin{aligned} v_\alpha(B) &= -\frac{1}{g_0^2}\left(f_\alpha(\psi(g_0\mathsf{C}B)) - \frac{g_0^2}{2} D^2 f_\alpha(0)[\mathsf{C}B, \mathsf{C}B]\right) , \\ j_b(B) &= \log j_H(q_{g_0}(B)(b)) , \\ d_c(B) &= \log \det_{\mathfrak{g}} \left(I - L^3 \partial_{A(b_0(c))} \widetilde{D}_c(g_0 \mathsf{C}B)\right) . \end{aligned} \tag{C.1}$$

Their supports are respectively $\Lambda_\alpha, \Lambda_b, \Lambda_c$. The full fine Jacobian is block triangular: its nonidentity rows are designated rows, and its designated-to-designated block is diagonal in c. Its determinant is consequently the product of the determinants in d_c. Each real determinant is positive in the chart connected to zero, so this branch agrees with the absolute determinant

in (4.14). Summing the action Hessians gives H, and hence

$$\mathcal{P}_{g_0}(B) = \sum_{\alpha} v_\alpha(B) + \sum_{b\in E} j_b(B) + \sum_{c\in E_1} d_c(B)\,. \tag{C.2}$$

There is no linear term in either j_b or d_c. Haar density in exponential variables has zero derivative at zero. For d_c, the quadratic part of $\widetilde{D}_c$ is the quadratic part of $\mathcal{N}_c$. Expanding [8, (0.4)] to degree two gives a linear combination of Lie brackets of the bond variables. Differentiating with respect to the designated variable gives a linear combination of adjoint maps. Their traces vanish for a compact Lie algebra. Multiplication by the scalar L^3 does not change this conclusion. Each f_α and its first derivative vanish at zero, and $\psi(A) = A+O(A^2)$, so the subtracted action term begins at degree three.

In the positive Lie-algebra scalar-product norm, there are constants $a_L, C_L^{\mathrm{loc}} > 0$, depending on L, G and the fixed averaging prescription, such that, on the complex polydisc $g_0 \max_b \|B(b)\| < a_L$, each term v of (C.2), with its indicated support Λ, satisfies

$$|v(B)| \leqslant C_L^{\mathrm{loc}} \left(g_0 \max_{b\in\Lambda} \|B(b)\| + g_0^2 \right) \sum_{b\in\Lambda} \|B(b)\|^2\,. \tag{C.3}$$

On a fixed smaller complex chart for the finitely many local templates, their convergent Taylor series and the vanishing orders just proved give a cubic remainder divided by g_0^2 for v_α and quadratic remainders for j_b, d_c. Equivalence of the finite-dimensional local maximum and Euclidean norms gives the displayed inequality. The bounded overlap then yields a torus-independent quadratic majorant on the cutoff. If c_{cmp} is the fixed norm-equivalence constant from the bond norm of [9] to this norm (the constant of the preceding paragraph), then

$$\sum_{\alpha} |v_\alpha(B)| + \sum_{b} |j_b(B)| + \sum_{c} |d_c(B)| \leqslant C_L^{\mathrm{ov}} (c_{\mathrm{cmp}} g_0 r(g_0) + g_0^2) \|B\|^2\,. \tag{C.4}$$

The constant C_L^{ov} includes the bounded overlap and is independent of the torus.

The local scalars are evaluated in the identity-background frame and are invariant under simultaneous constant colour conjugation. This follows from equivariance of the logarithm, averages, and implicit correction of [8], cyclicity of the action traces, and conjugation invariance of Haar density and determinants.

Their extension to the gauge orbit of this background is unambiguous: if $U_0^u(b) = u(b_-)u(b_+)^{-1}$ and $A^u(b) = u(b_-)A(b)u(b_-)^{-1}$, the transported scalar at (U_0^u, A^u) is its identity-frame value at A. Two choices of u representing the same background differ by a constant right factor on the connected torus, and give the same scalar by the stated constant-conjugation invariance.

Appendix D The Exterior Response and Its History Average

This appendix bounds the exterior response $\mathcal{B}_\xi[f_\xi]$ of (3.51) on every branch (Lemma D.6, Section D.5), and pays its history average, together with the uncovered branches of Lemma 3.36, by one-point witness probabilities on compatible even fine periods (Proposition D.18, Section D.7). Section D.8 proves the deterministic comparison of the two minima used by the witness argument and, by the same argument, the bound on the boundary field of [9, (1.21)] used in Section 3.4 (Lemma D.21). The inputs are the Hessian, Schur, chart and primitive bounds of Appendix B and the collar statements of Section 3.4; Remark D.19 records the scope of these bounds.

D.1 Objects and the Conditional Law

Let ξ be the data of a branch (Section 3.1): the integration carrier I, the retained free carrier J, the exterior field $A_{\mathrm{e}} \in \mathfrak{g}^J$, extended by zero on every other free bond, the reconstructed field $U_\xi(a)$ of (3.26), the cutoff norm $|\bullet|_\sigma$ (the bond norm $|\bullet|$ of (4.12), operator or normalised Hilbert–Schmidt, fixed once). The inner product $\langle \bullet, \bullet \rangle$, the norm h and the product Hilbert norms are those of Section 3; $h \leqslant |\bullet|_\sigma$ for both permitted norms. The cutoff radius is

$$r := \delta_0/g_0 = (A_1/A_0)\, p_0(g_0) = A_1(\log g_0^{-2})^{p_0} \,, \tag{D.1}$$

(the first-step cutoff normalisation (4.13), in which p_0 denotes both the function $p_0(g) = A_0(\log g^{-2})^{p_0}$ and its exponent, as in [9]), so $r \to \infty$ and $g_0 r = \delta_0 \to 0$ as $g_0 \to 0$.

Every derivative, overlap, range and Hessian constant below is independent of g_0 once L, the group, the character and the source constants are fixed; the two conditions

$$r \geqslant 1 \qquad \text{and} \qquad C_P g_0(1+2r) \leqslant \gamma/2 \,, \tag{D.2}$$

with C_P the constant of Lemma D.1 as enlarged in Lemma D.3 (fixed there independently of g_0; (D.2) is imposed with the enlarged constant) and $\gamma = \gamma_L/2$, therefore hold for all g_0 below a threshold (the exponent p_0 is positive, $p_0 \geqslant 5\mathfrak{r}$ by Section B.8, so $r \to \infty$ while $g_0 r \to 0$), and they are assumed throughout this appendix, together with the standing conditions (3.1) and the coupling thresholds of Section B.8 under which Lemmas 3.8, 3.19 and B.12 hold (in particular threshold item 3 there, $4C_{\mathsf{C}} g_0(1+2r) \leqslant \vartheta_*/2$).

If $I = \varnothing$ then $F_\xi(1) = F_\xi(0)$ ((3.46)) and the response (3.51) is zero; below $I \neq \varnothing$.

Let

$$H := H_{\xi,II}, \qquad \ell_\xi := H_{\xi,IJ} A_{\mathrm{e}}, \qquad j_\xi(a) := \langle \ell_\xi, a \rangle, \qquad s_\xi := 1 + r + r_{\mathrm{e},\xi} \leqslant 1 + 2r \,, \tag{D.3}$$

with $r_{\mathrm{e},\xi} < r$ the retained radius (3.52), the maximum of $|A_{\mathrm{e}}(b)|_\sigma$ over the retained bonds within $3R_{\mathrm{c}}$ of I (zero if there is none). Every retained input entering $H_{\xi,IJ}$ or a near primitive lies in this collar (Lemma 3.17(i)).

The collar bound holds on every branch, independently of contour coverage: by Lemma 3.19(i) every retained bond within $3R_{\mathrm{c}}$ of I lies in $\mathcal{G}_\xi \setminus S_1^*$ ($\mathcal{G}_\xi$ the union of the good test carriers, (3.41)), its factor in $\chi_{\xi,\mathrm{e}}$ is $\mathbf{1}\{|\Psi_\xi(A_{\mathrm{e}})(b)| < r\}$ ((3.43)) and Ψ_ξ, the identity off S_1^* ((3.38)), is the identity there, so a nonzero retained weight forces $|A_{\mathrm{e}}(b)| < r$; this is Lemma 3.19(iii).

On the open box $\mathcal{D}_I = \{a \in \mathfrak{g}^I \,|\, |a_b|_\sigma < r \ \forall b \in I\}$ of (3.45) the interpolating law of that display, which at $t = 1$ is the conditional law of the branch (Lemma 3.17(ii)), is, exactly, for $t \in [0,1]$,

$$\mathrm{d}\nu_{\xi;t}(a) = I_{\xi;t}^{-1} e^{\mathsf{Q}_{\xi,t}(a)}\, \mathrm{d}a \,, \qquad \mathsf{Q}_{\xi,t}(a) = -\tfrac{1}{2}\langle a, Ha \rangle - t\langle \ell_\xi, a \rangle + P_{\xi,t}(a) \,, \tag{D.4}$$

zero off $\mathcal{D}_I$ (the indicator in (3.45)), with $\mathsf{Q}_{\xi,t}$ as in (3.54); $I_{\xi;t}$ is the normaliser of (3.45), and $\mathrm{d}a$ is the product Lebesgue measure of the fixed inner product.

Throughout $f_\xi(a) = W_\chi(C; U_\xi(a))$ ((3.46)) with the exterior data fixed at their branch values, defined where $U_\xi(a)$ is (Lemma 3.12), which includes $\mathcal{D}_I$, and extended by zero elsewhere (immaterial below, since $\nu_{\xi;t}$ vanishes off $\mathcal{D}_I$); it is bounded by one, and

$$L_{\xi,t} = -j_\xi + \partial_t P_{\xi,t} \,, \qquad \mathcal{B}_\xi[f] = \int_0^1 \mathrm{Cov}_{\nu_{\xi;t}}(f, L_{\xi,t})\, \mathrm{d}t \tag{D.5}$$

(Lemma 3.18); $L_{\xi,t} = \partial_t \mathsf{Q}_{\xi,t}$ is called the score below.

Derivatives and operator norms refer to the product Hilbert structure, complexified for complex scalar gradients; the fixed matrix-norm equivalence factors of Appendix B are absorbed into the constants before g_0 is chosen.

The inputs are Lemmas 3.8 and B.12 under their stated hypotheses (in force by the preceding paragraph): $\gamma \mathbf{1} \leqslant H \leqslant m_L \mathbf{1}$ with $\gamma = \gamma_L/2$; finite interaction range; and bounded local first and second derivatives of the near primitives on the enlarged box $|a_b|_\sigma \leqslant 2r$ at the inputs $a + tA_\mathrm{e}$, $t \in [0,1]$. They are used at the background U_1 of the branch.

D.2 Local Derivative Bounds

Let S_α be the full free-input stencil, retained inputs included, of a near primitive $\mathfrak{p}_\alpha$ (the p_α of Lemma B.12), let N_in bound its size and N_ov the number of primitive stencils meeting any free bond; both are fixed by L and the group.

The distance $d(b,b')$ of two bonds is the periodic ℓ^∞ distance of their initial sites, plus one for differing directions (the sum of two pseudometrics; a metric, because a positive bond is determined by its initial site and its direction); $d(b,S) = \inf_{b' \in S} d(b,b')$ for a bond set S, with $d(b,\varnothing) = +\infty$, and this d is the only bond distance used in this appendix.

Let $R = 8L + 8$, a range dominating in this metric every near-primitive stencil diameter (at most $4L+1$, Lemma 3.19, $R_\mathrm{loc} = 2L+2$) and the Hessian range (at most $\rho_L + 1 = 4L+1$ in this metric, $\rho_L = 4L$ being the ℓ^∞ range of the initial sites of the nonzero entries of H_ξ: Lemmas B.3 and 3.7). Let

$$\mathsf{K}_\xi \coloneqq \{b \in I \,:\, d(b,J) \leqslant R\}\,, \qquad I_C \coloneqq \{b \in I \,:\, b \in \mathcal{F}_r \text{ for some traversal } r \text{ of } C\}\,, \tag{D.6}$$

with $\mathcal{F}_r$ the free-input stencil (3.13) of the contour bond r (a traversal bond, as in Section 3.3; the letter r denotes a traversal bond only inside (D.6) and in the stencils $\mathcal{F}_r$, and the cutoff radius r of (D.1) everywhere else); the source depends on the integration field only through I_C (Lemma 3.9).

Lemma D.1 (Local Bounds)**.** *There are constants C_P, C_S (depending on L and the group only) and $C_{\mathrm{resp},\chi}$ (depending on L, the group, the character and the maximal multiplicity with which C traverses a link only, the latter 1 for the square C_n) such that on the enlarged box $\{|a_b|_\sigma \leqslant 2r \ \forall b \in I\}$, uniformly in $t \in [0,1]$ and in the branch (the fourth bound on every covered branch, Lemma 3.36),*

$$\|D_a^2 P_{\xi,t}\| \leqslant C_P g_0 s_\xi\,, \qquad h(\nabla_b \partial_t P_{\xi,t}) \leqslant C_S g_0 s_\xi\, r_{\mathrm{e},\xi} \mathbf{1}_{\mathsf{K}_\xi}(b)\,, \qquad h((\ell_\xi)_b) \leqslant m_L r_{\mathrm{e},\xi} \mathbf{1}_{\mathsf{K}_\xi}(b)\,, \tag{D.7}$$

$$\sup_a h(\nabla_b f_\xi(a)) \leqslant C_{\mathrm{resp},\chi} g_0 \mathbf{1}_{I_C}(b)\,. \tag{D.8}$$

Under (D.2) *the function $-\mathsf{Q}_{\xi,t}$ is strongly convex on $\mathcal{D}_I$ with Hessian at least $\rho = \gamma/2$.*

Proof. The first bound of (D.7) is the second-derivative bound of Lemma B.12 on the enlarged box with the finite overlap N_ov.

For the second, $\partial_t P_{\xi,t}(a) = \sum_\alpha D\mathfrak{p}_\alpha(a + tA_\mathrm{e})[A_\mathrm{e}]$ (by (3.45), $P_{\xi,t}$ has no other t-dependence), so each summand of $\nabla_b \partial_t P_{\xi,t}$ is $D^2\mathfrak{p}_\alpha$, the full second derivative on $\mathfrak{g}^{S_\alpha}$ (retained inputs included) bounded by Lemma B.12, applied to a direction at $b \in I$ and to $A_\mathrm{e}|_{S_\alpha}$; it vanishes unless S_α contains a retained input, which lies in the collar (stencil

diameter at most $4L+1<3R_{\mathrm{c}}$), so $h(A_{\mathrm{e}}(b'))\leqslant|A_{\mathrm{e}}(b')|_\sigma\leqslant r_{\mathrm{e},\xi}$ for each such input ($h\leqslant|\bullet|_\sigma$ for both permitted norms, Section D.1) and $h(A_{\mathrm{e}}|_{S_\alpha})\leqslant\sqrt{N_{\mathrm{in}}}\,r_{\mathrm{e},\xi}$, and summing the at most N_{ov} stencils meeting b gives the bound; the support statement follows from the range R of the stencils.

The third bound is the row-sum bound $\sum_{b'\in\Omega_1^*}\|(H_\xi)_{bb'}\|\leqslant m_L$ for the rows $b\in I$, the row-sum statement of Lemma 3.8, applied to A_{e} on the collar: $(H_{\xi,IJ})_{bb'}=0$ unless the initial sites are within ℓ^∞ distance ρ_L (Lemma B.3), so a nonzero entry has $d(b',I)\leqslant\rho_L+1=4L+1<3R_{\mathrm{c}}$ ($R_{\mathrm{c}}\geqslant 2R_{\mathrm{loc}}+1=4L+5$) and $h(A_{\mathrm{e}}(b'))\leqslant|A_{\mathrm{e}}(b')|_\sigma\leqslant r_{\mathrm{e},\xi}$, as for the second bound, and $\rho_L+1\leqslant R$ gives the support statement.

For (D.8), on a covered branch every contour bond r lies in Ω_1 with $\mathcal{F}_r\subseteq I$ (Lemma 3.36), so each contour link is reconstructed by a row within $R_{\mathrm{loc}}=2L+2\leqslant 2R_{\mathrm{c}}+2L$ of I, a row of the charts of Lemma 3.12. For the derivative of the normalised character of the ordered link product: by the rowwise reconstruction (Lemma 3.9) and the uniform charts (Lemma B.8) on the enlarged box with the exterior fixed, whose inputs lie in the phase ball of the charts by Lemma 3.12, each affected link has a phase derivative bounded by Cg_0, a fixed input affects a bounded number of links, each link is traversed by C at most as often as its maximal link multiplicity (once for the square C_n, a simple closed contour; the multiplicity enters $C_{\mathrm{resp},\chi}$ as a factor), the derivative of the character is bounded in the invariant tangent norm on the compact group, and the other links remain unitary inside the closed contraction.

The convexity statement is $\gamma\,\mathbf{1}\leqslant H$ (Lemma 3.8) together with $\|D^2P_{\xi,t}\|\leqslant C_Pg_0s_\xi\leqslant C_Pg_0(1+2r)\leqslant\gamma/2$, using $s_\xi\leqslant 1+2r$ (Lemma 3.19(iii)) and (D.2). *q.e.d.*

D.3 An Extension within the Proved Chart

Let $B_\sigma:=\{x\in\mathfrak{g}:|x|_\sigma\leqslant 1\}$, let Π_r be the Hilbert metric projection onto the closed convex set $\frac{3}{2}rB_\sigma$, let ϕ be a fixed nonnegative smooth h-radial mollifier on $\mathfrak{g}$ supported in the unit h-ball with integral one (in this appendix ϕ denotes this mollifier), let $\phi_s(y):=s^{-d_G}\phi(y/s)$ for $s>0$ with $d_G=\dim\mathfrak{g}$, let $\epsilon:=1/(4\sqrt{N})$, and let

$$T_r(x):=\int_{\mathfrak{g}}\Pi_r(x-y)\,\phi_{\epsilon r}(y)\,\mathrm{d}y\qquad(x\in\mathfrak{g})\,. \tag{D.9}$$

Lemma D.2 (The Extension Map)**.** *T_r is smooth and* Ad*-equivariant, and*

$$T_r(x)=x\ \ \textit{if } |x|_\sigma\leqslant r,\qquad |T_r(x)|_\sigma\leqslant\tfrac{3}{2}r,\qquad \|DT_r\|\leqslant 1,\qquad \|D^2T_r\|\leqslant C_\sigma/r\,, \tag{D.10}$$

with $C_\sigma:=4\sqrt{N}\,\|\nabla\phi\|_{L^1}$, which depends on N and the fixed mollifier only.

Proof. Both norms are Ad-invariant, so B_σ is Ad-invariant and Ad is h-orthogonal; hence metric projection and the h-radial mollifier commute with the adjoint action, which gives the equivariance.

If $h(y)\leqslant\epsilon r$ then $|y|_\sigma\leqslant\sqrt{N}\,h(y)\leqslant r/4$ (the conventions of Section 3: $|\bullet|_\sigma\leqslant\sqrt{N}\,h$), so for $|x|_\sigma\leqslant r$ one has $|x-y|_\sigma\leqslant\frac{5}{4}r<\frac{3}{2}r$ and $\Pi_r(x-y)=x-y$ (indeed for all $|x|_\sigma<\frac{5}{4}r$); the convolution then equals x because the mollifier has integral one and, being h-radial, zero first moment.

The range statement is the convexity of $\frac{3}{2}rB_\sigma$: $T_r(x)$ is a probability average of its points.

Metric projection is 1-Lipschitz for h and convolution preserves this, giving $\|DT_r\|\leqslant 1$ in the h-operator norm; moving one derivative onto the mollifier of scale ϵr bounds $\|D^2T_r\|$ by $\|D\Pi_r\|_\infty\|\nabla\phi_{\epsilon r}\|_{L^1}\leqslant\|\nabla\phi\|_{L^1}/(\epsilon r)=C_\sigma/r$. *q.e.d.*

Let T be the product map on $\mathfrak{g}^I$ that applies T_r to each bond of I, and let

$$\widetilde{P}_t(a) \coloneqq \sum_{\alpha \text{ near}} [\mathfrak{p}_\alpha(Ta + tA_{\mathrm{e}}) - \mathfrak{p}_\alpha(tA_{\mathrm{e}})], \qquad \widetilde{f}(a) \coloneqq f_\xi(Ta), \qquad \widetilde{L}_t(a) \coloneqq -\langle \ell_\xi, a\rangle + \partial_t \widetilde{P}_t(a)\,. \tag{D.11}$$

Lemma D.3 (The Extended Objects)**.** *$\widetilde{P}_t$, $\widetilde{f}$ and $\widetilde{L}_t$ are defined and smooth on all of $\mathfrak{g}^I$, $|\widetilde{f}| \leqslant 1$, and on $\mathcal{D}_I$: $\widetilde{f} = f_\xi$, $\widetilde{P}_t = P_{\xi,t} - P_{\xi,t}(0)$ (the scalar $P_{\xi,t}(0)$ is removed only in the normalised conditional law; the unnormalised integral keeps it) and $\widetilde{L}_t - L_{\xi,t} = -\partial_t P_{\xi,t}(0)$, a constant, so that $\mathrm{Cov}_{\nu_{\xi;t}}(\widetilde{f}, \widetilde{L}_t) = \mathrm{Cov}_{\nu_{\xi;t}}(f_\xi, L_{\xi,t})$. After enlarging C_P (before g_0 is chosen; the enlarged constant depends on L, the group, C_σ, C_{p}, N_{ov} and N_{in}, all fixed before the coupling, and* (D.2) *is imposed with it), uniformly in $t \in [0,1]$ and $a \in \mathfrak{g}^I$: $\|D^2\widetilde{P}_t\| \leqslant C_P g_0 s_\xi$, the block Schur row and column sums of $D^2\widetilde{P}_t$ are at most $C_P g_0 s_\xi$ with range R,*

$$h(\nabla_b \partial_t \widetilde{P}_t) \leqslant C_S g_0 s_\xi\, r_{\mathrm{e},\xi} \mathbf{1}_{\mathsf{K}_\xi}(b)\,, \qquad h(\nabla_b \widetilde{f}) \leqslant C_{\mathrm{resp},\chi} g_0 \mathbf{1}_{I_C}(b)\,. \tag{D.12}$$

Proof. The image Ta has cutoff norm at most $\frac{3}{2}r < 2r$ on every bond of I, so it lies in the enlarged box of Lemma B.12 (threshold item 3 of Section B.8, in force by Section D.1), where every near primitive and every reconstruction row within $2R_{\mathrm{c}} + 2L$ of I is analytic, the exterior fields being fixed; the real reconstruction gives group-valued links, whence $|\widetilde{f}| \leqslant 1$.

On $\mathcal{D}_I$, T is the identity (Lemma D.2), which gives the three identities; a constant shift of one operand leaves the covariance unchanged.

The Hessian of $\widetilde{P}_t$ is $D^2\mathfrak{p}_\alpha(DT\bullet, DT\bullet) + D\mathfrak{p}_\alpha\, D^2T$ summed over the near primitives: the first term is $O(g_0 s_\xi)$ in operator norm by Lemma D.1 and $\|DT\| \leqslant 1$, and per stencil by the second-derivative bound of Lemma B.12, the second is $O(g_0 s_\xi^2/r) = O(g_0 s_\xi)$ by the first-derivative bound of Lemma B.12, $\|D^2T\| \leqslant C_\sigma/r$, $r \geqslant 1$ and $s_\xi \leqslant 1 + 2r \leqslant 3r$. Each input belongs to at most N_{ov} stencils and each stencil has at most N_{in} blocks, so summing at most $N_{\mathrm{ov}} N_{\mathrm{in}}$ blocks with this chain-rule bound gives the operator and Schur bounds; the supports and the range are those of the original stencils.

The mixed derivative $\nabla_b \partial_t \widetilde{P}_t$ contains no second derivative of T, since T does not depend on t and is bondwise: $\nabla_b \partial_t \widetilde{P}_t(a) = DT_r(a_b)^*(\nabla_b \partial_t P_{\xi,t})(Ta)$, so (D.7) of Lemma D.1 at Ta and $\|DT\| \leqslant 1$ give the first bound of (D.12); the second is (D.8) composed with DT. *q.e.d.*

D.4 Covariance Decay at the Hard Boundary

For a positive integer n, the barrier and the barrier laws are

$$\beta_n(a) \coloneqq n \sum_{b \in I} [\mathrm{dist}_h(\bullet, rB_\sigma)^2 * \phi_{n^{-2}}](a_b)\,, \tag{D.13}$$

$$V_{n,t}(a) \coloneqq \tfrac{1}{2}\langle a, Ha\rangle + t\langle \ell_\xi, a\rangle - \widetilde{P}_t(a) + \beta_n(a)\,, \qquad \mathrm{d}\nu_{n,t} \coloneqq Z_{n,t}^{-1} e^{-V_{n,t}(a)}\,\mathrm{d}a \quad \text{on } \mathfrak{g}^I\,.$$

The squared distance to a convex set is convex and convolution with a nonnegative mollifier preserves convexity, so the barrier Hessian is positive semidefinite; it is diagonal in the bond index because β_n is a sum of functions of the single bond variables.

For fixed n the potential is smooth with bounded Hessian (that of the barrier is bounded by $2n$: the gradient $2(x - \Pi_r(x))$ of the squared distance is 2-Lipschitz, and mollification is an average of translates) and Gaussian confinement, its Hessian is at least ρI with $\rho = \gamma/2$ as in Section D.2 ($\gamma \mathbf{1} \leqslant H$ by Lemma 3.8, and $\|D^2\widetilde{P}_t\| \leqslant C_P g_0 s_\xi \leqslant C_P g_0(1 + 2r) \leqslant \gamma/2$ by Lemma D.3 and (D.2)–(D.3)), and its off-diagonal block row and column sums are at most

$$M_* = m_L + \gamma/2\,, \tag{D.14}$$

with range at most R in the bond distance d of Section D.2, independently of n, t, the volume and the branch (the barrier is diagonal; the row sums of H are at most m_L by the row-sum statement of Lemma 3.8, its column sums equal its row sums since $H_{bb'} = H_{b'b}^{\mathsf{T}}$ have equal norms, and its range is $\rho_L = 4L \leqslant R$ by Lemma B.3; the Schur sums of $D^2\widetilde{P}_t$ are at most $C_P g_0 s_\xi \leqslant \gamma/2$ with range R by Lemma D.3).

Lemma D.4 (Helffer–Sjöstrand Identity and Decay)**.** *Let $\mathscr{A}_{n,t} \coloneqq \mathscr{L}_{n,t} \otimes I + D^2 V_{n,t}$ with $\mathscr{L}_{n,t} \coloneqq -\Delta + \nabla V_{n,t} \cdot \nabla$, on the Hilbert direct sum over $b \in I$ of $L^2(\nu_{n,t}; \mathfrak{g})$. Then $\mathscr{A}_{n,t} \geqslant \rho I$ ($\mathscr{L}_{n,t} = \nabla^*\nabla \geqslant 0$ on $L^2(\nu_{n,t})$ and $D^2 V_{n,t} \geqslant \rho I$), and for real u, v in the weighted space $H^1(\nu_{n,t})$*

$$\mathrm{Cov}_{\nu_{n,t}}(u, v) = \langle \nabla u, \mathscr{A}_{n,t}^{-1} \nabla v \rangle_{L^2(\nu_{n,t})} \,. \tag{D.15}$$

With $\mu = R^{-1} \log(1 + \gamma/(4M_*))$ and Π_b the projection onto the b-component,

$$\|\Pi_b \mathscr{A}_{n,t}^{-1} \Pi_c\| \leqslant \frac{4}{\gamma} e^{-\mu d(b,c)} \,, \qquad |\mathrm{Cov}_{\nu_{n,t}}(u, v)| \leqslant \frac{4}{\gamma} \sum_{b,c \in I} e^{-\mu d(b,c)} \|\nabla_b u\|_{L^2(\nu_{n,t})} \|\nabla_c v\|_{L^2(\nu_{n,t})} \,. \tag{D.16}$$

For complex u and real v, applying (D.16) *to the real and imaginary parts of u gives the same bound with the gradient norm of u read as the sum of the gradient norms of its two parts; each is at most the gradient norm of u, so the bound holds with an extra factor two.*

Proof. The identity (D.15) is [12, Proposition 4.5] (Theorem 1.3 there at $k = 0$) for the real-valued C^2 potential $V_{n,t} + \log Z_{n,t}$ (normalised; the additive constant leaves the derivatives unchanged; $V_{n,t}$ is real because its quadratic, linear and barrier terms are real and $\widetilde{P}_t$ is real, the primitives being real on real inputs with a positive determinant, Lemma B.8(iii) and Lemma B.12). Its conditions are a real C^2 normalised potential and the closed range of the operator A_0 there; the latter follows from $A_1 \geqslant \rho I$, which is $\mathscr{A}_{n,t} \geqslant \rho I$ here, by [12, Theorem 1.3], as explained there after (4.39). The theorem gives the projection onto the orthogonal complement of the constants as $d_0^*(A_1|_{\overline{\mathrm{ran}}\, d_0})^{-1} d_0$, and at fixed n the bounded Hessian identifies A_1 with the displayed operator $\mathscr{A}_{n,t}$: A_1 is the Friedrichs extension of [12, Lemma 4.2] with the action (4.22) there, $\mathscr{L}_{n,t} \otimes I + D^2 V_{n,t}$ on gradient fields, and since $D^2 V_{n,t}$ is a bounded multiplication operator at fixed n its form domain is that of $\mathscr{L}_{n,t} \otimes I$ and A_1 is the operator sum on the domain of $\mathscr{L}_{n,t} \otimes I$, a domain the bounded field-independent weights W_c below preserve.

The operands are the source $\widetilde{f}$, bounded with bounded gradient (Lemma D.3), and the score $\widetilde{L}_t = -\langle \ell_\xi, \bullet \rangle + \partial_t \widetilde{P}_t$, which grows at most linearly and has bounded gradient by (D.12); both lie in $H^1(\nu_{n,t})$.

For a fixed $c \in I$ let W_c multiply the component b by $e^{\mu d(b,c)}$; it is constant in field space, hence commutes with the scalar generator $\mathscr{L}_{n,t} \otimes I$ and with every diagonal Hessian block, the barrier included, and the off-diagonal Schur bounds give $\|W_c \mathscr{A}_{n,t} W_c^{-1} - \mathscr{A}_{n,t}\| \leqslant M_*(e^{\mu R} - 1) \leqslant \rho/2$ by the choice of μ. W_c and its inverse are bounded at finite volume, so a resolvent Neumann series gives $\|W_c \mathscr{A}_{n,t}^{-1} W_c^{-1}\| \leqslant (\rho - \rho/2)^{-1} = 4/\gamma$; compressing by Π_b on the left and Π_c on the right, where W_c acts as $e^{\mu d(b,c)}$ and as $e^{\mu d(c,c)} = 1$, gives the first bound of (D.16); the second follows from (D.15) and the Cauchy–Schwarz inequality. *q.e.d.*

Lemma D.5 (The Barrier Limit)**.** *For u, v bounded or of linear growth with bounded gradients, $\mathrm{Cov}_{\nu_{n,t}}(u, v) \to \mathrm{Cov}_{\nu_{\xi;t}}(u, v)$ as $n \to \infty$. Consequently the second bound of* (D.16) *holds for $\nu_{\xi;t}$ on the hard box $\mathcal{D}_I$, with the gradient norms taken as global suprema, $\|\nabla_b u\|_\infty \|\nabla_c v\|_\infty$ in place of the L^2 norms.*

Proof. At finite volume $\beta_n \to 0$ on the closed cutoff body $\{|a_b|_\sigma \leqslant r\ \forall b\}$ and $\beta_n \to \infty$ outside it, the boundary of the body has Lebesgue measure zero, and $\beta_n \geqslant 0$ together with the Gaussian confinement supplies an integrable majorant for u, v and uv (bounded or of linear growth); dominated convergence gives the convergence of the unnormalised integrals, and of the normaliser to $\int_{\mathcal{D}_I} e^{-V_t^\circ}\,\mathrm{d}a > 0$ (the box has nonempty interior), where $V_t^\circ := V_{n,t} - \beta_n$ is the potential without the barrier, independent of n, hence of the covariances, to those of the law with density $e^{-V_t^\circ}$ on $\mathcal{D}_I$, which is $\nu_{\xi;t}$ by Lemma D.3 ($\widetilde{P}_t = P_{\xi,t} - P_{\xi,t}(0)$ on $\mathcal{D}_I$, and (D.4)).

On the right of (D.16) the $L^2(\nu_{n,t})$ norms depend on n, but they are bounded by the global suprema $\|\nabla_b u\|_\infty$, $\|\nabla_c v\|_\infty$, which do not, and μ and $4/\gamma$ are independent of n, so the bound passes to the limit. *q.e.d.*

D.5 The Response

Let

$$D_\xi := \sum_{b \in I_C} e^{-\mu d(b,J)/2}\,, \tag{D.17}$$

with μ of Lemma D.4 and the bond distance d of Section D.2; $D_\xi = 0$ when $J = \varnothing$ (by $d(b,\varnothing) = +\infty$; then also $A_\mathrm{e} = 0$ and the score vanishes).

Lemma D.6 (The Exterior Response)**.** *Under* (D.2)*, on every covered branch (Lemma 3.36) and for every $t \in [0,1]$,*

$$|\operatorname{Cov}_{\nu_{\xi;t}}(f_\xi, L_{\xi,t})| \leqslant C_{\mathrm{resp},\chi}\, g_0 r \sum_{b \in I_C} e^{-\mu d(b,J)/2}\,, \qquad |\mathcal{B}_\xi[f_\xi]| \leqslant C_{\mathrm{resp},\chi}\, g_0 r\, D_\xi\,, \tag{D.18}$$

after enlarging the constant $C_{\mathrm{resp},\chi}$ of Lemma D.1 (the enlarged constant keeps the name); it depends on L, γ, m_L, M_*, R, N_{in}, N_{ov}, C_P, C_S, C_σ, the character, the group and the fixed source parameters, and is independent of g_0, the branch, the volume, t and the barrier index.

Proof. By Lemma D.3, $\operatorname{Cov}_{\nu_{\xi;t}}(f_\xi, L_{\xi,t}) = \operatorname{Cov}_{\nu_{\xi;t}}(\widetilde{f}, \widetilde{L}_t)$, and Lemma D.5 bounds it by $(8/\gamma)\sum_{b,c\in I} e^{-\mu d(b,c)} \|\nabla_b \widetilde{f}\|_\infty \|\nabla_c \widetilde{L}_t\|_\infty$: the bound (D.16) is applied to the real and imaginary parts of the complex $\widetilde{f}$ and the real $\widetilde{L}_t$ (Lemma D.4), the sum of the two gradient norms being at most $2\|\nabla_b \widetilde{f}\|_\infty$.

By (D.12) and (D.7), $\|\nabla_c \widetilde{L}_t\|_\infty \leqslant (m_L + C_S g_0 s_\xi) r_{\mathrm{e},\xi} \mathbf{1}_{\mathsf{K}_\xi}(c) \leqslant (m_L + C_S\gamma/(2C_P))\, r\, \mathbf{1}_{\mathsf{K}_\xi}(c)$ (as $r_{\mathrm{e},\xi} < r$ and $C_S g_0 s_\xi \leqslant C_S g_0(1+2r) \leqslant C_S\gamma/(2C_P)$ by (D.2); the coefficient is a constant fixed before g_0) and $\|\nabla_b \widetilde{f}\|_\infty \leqslant C_{\mathrm{resp},\chi} g_0 \mathbf{1}_{I_C}(b)$ with the constant of Lemma D.1.

On K_ξ, $d(c,J) \leqslant R$, so $d(b,c) \geqslant d(b,J) - R$ and

$$e^{-\mu d(b,c)} = e^{-\mu d(b,c)/2} e^{-\mu d(b,c)/2} \leqslant e^{\mu R/2} e^{-\mu d(b,J)/2}\, e^{-\mu d(b,c)/2}\,;$$

the sum of $e^{-\mu d(b,c)/2}$ over all periodic bonds c is bounded by the infinite-lattice shell sum $S_{\mu/2} = \sup_b \sum_c e^{-\mu d(b,c)/2}$ (the nearest-lift map from torus bonds to lattice bonds is injective and the terms are nonnegative), uniformly in the periods.

Collecting the constants, $(8/\gamma)(m_L + C_S\gamma/(2C_P)) e^{\mu R/2} S_{\mu/2}$ times the constant of Lemma D.1, which depends on μ through $e^{\mu R/2}$ and $S_{\mu/2}$ and hence on M_*, gives the first bound of (D.18) with the enlarged $C_{\mathrm{resp},\chi}$; integrating over $t \in [0,1]$ in (D.5) gives the second. *q.e.d.*

Remark D.7 (The Surface Terms Are Retained). In the notation (3.53)–(3.54), Lemma 3.20 (its hypothesis holds: f_ξ is C^1 on the enlarged box $\{|a_b|_\sigma \leqslant 2r\}$, a neighbourhood of $\overline{\mathcal{D}_I}$, by the charts of Lemma 3.12 and Lemma B.8, as in the proof of Lemma D.1) writes the integrand of the response as

$$-\mathbb{E}_{\nu_{\xi;t}}[D_{m_\xi} f_\xi] + \mathrm{Cov}_{\nu_{\xi;t}}(f_\xi, \partial_t P_{\xi,t} - D_{m_\xi} P_{\xi,t}) + \mathcal{T}_{\xi,t}[f_\xi] - \mathbb{E}_{\nu_{\xi;t}}[f_\xi]\,\mathcal{T}_{\xi,t}[1]\,, \tag{D.19}$$

with the hard-cutoff flux $\mathcal{T}_{\xi,t}[u] = I_{\xi;t}^{-1} \int_{\partial\mathcal{D}_I} u\, e^{\mathsf{Q}_{\xi,t}} \langle m_\xi, n\rangle \,\mathrm{d}S$. Lemma D.6 bounds this whole expression.

The extension T_r of Lemma D.2 is the identity on the neighbourhood $\{|a_b|_\sigma < \frac{5}{4}r\}$ of $\overline{\mathcal{D}_I}$ and the barrier limit of Lemma D.5 returns the original measure, so the boundary traces and the normalised covariance in (D.19) are those of Section 3.4.

D.6 The Physical Substitution

Let $\mathsf{w}_\xi := e^{c_\xi} I_{\xi;1} = \mathcal{J}^{\mathrm{a}}_\xi[1]$ be the branch weight (3.49) and $\mathcal{O}$ the outer operation of (3.4), so that $\mathcal{Z}_0 = \int \mathcal{O}[\mathsf{w}_\xi](V)\,\mathrm{d}V$ ((3.7)). The history expectation and probability are

$$\mathbb{E}_{\mathrm{h}}[X] := \mathcal{Z}_0^{-1} \int \mathcal{O}[\mathsf{w}_\xi X](V)\,\mathrm{d}V\,, \qquad \mathbb{P}_{\mathrm{h}}(A) := \mathbb{E}_{\mathrm{h}}[\mathbf{1}_A]\,, \tag{D.20}$$

for nonnegative or bounded Borel functions X of the branch data; $\mathcal{O}$ has a nonnegative kernel (its Haar and constraint factors are measures and its cutoffs and exponentials are nonnegative, (3.4)) and $0 < \mathcal{Z}_0 < \infty$ (the measurability paragraph of Section 3.1), so $\mathbb{E}_{\mathrm{h}}$ is positive, $|\mathbb{E}_{\mathrm{h}}[X]| \leqslant \sup|X|$ and $\mathbb{E}_{\mathrm{h}}[1] = 1$.

Let A_{cov} be the set of covered branches of Lemma 3.36: every contour bond is a bond of Ω_1 and $\mathcal{F}_r \subseteq I$ for every contour bond r. The event A_{cov} is determined by Ω_1, R_1 and the fixed contour templates ($I = \Lambda_1^*$, the free bonds with an endpoint in Λ_1, Section 3.1, with Λ_1 derived from Ω_1 and R_1 by [9, (1.20)] and its filling; the stencils $\mathcal{F}_r$ are fixed); it and D_ξ are measurable functions of the outer branch data, constant in the integration over a, and $0 \leqslant D_\xi \leqslant |I_C|$, a bound by the number of free inputs of the contour.

The combined exterior-response and uncovered contribution to $\mathbb{E}_\varepsilon[W_\chi(C)]$ of (3.7) is

$$E_{\mathrm{bdry}} := \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} \mathcal{B}_\xi[f_\xi] + \mathbf{1}_{A_{\mathrm{cov}}^c} F_\xi(1)]\,, \tag{D.21}$$

the exterior response on the covered branches and the whole conditional expectation on the uncovered ones ((3.51), (3.97)); the integrand is a bounded Borel function of the branch data ($F_\xi(t) = \int f_\xi K_{\xi;t}\,\mathrm{d}a / I_{\xi;t}$ is Borel in the branch data for each t, by the composition argument of Section 3.1 applied to the interpolated kernel of (3.46) and Fubini's theorem, and $|\mathcal{B}_\xi[f_\xi]| \leqslant C_{\mathrm{resp},\chi} g_0 r D_\xi$ is bounded), so (D.20) applies to it. The quantities $\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} D_\xi]$ and $\mathbb{P}_{\mathrm{h}}(A_{\mathrm{cov}}^c)$ are bounded in Section D.7 (Proposition D.18).

Corollary D.8. *Under* (D.2)*,*

$$\Big|\mathcal{Z}_0^{-1} \int \mathcal{O}[\mathsf{w}_\xi \mathbf{1}_{A_{\mathrm{cov}}} \mathcal{B}_\xi[f_\xi]](V)\,\mathrm{d}V\Big| \leqslant C_{\mathrm{resp},\chi}\, g_0 r\, \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} D_\xi]\,, \tag{D.22}$$

$$|E_{\mathrm{bdry}}| \leqslant C_{\mathrm{resp},\chi}\, g_0 r\, \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} D_\xi] + \mathbb{P}_{\mathrm{h}}(A_{\mathrm{cov}}^c)\,.$$

Proof. $\mathsf{w}_\xi \geqslant 0$ and $\mathcal{O}$ is positive, so Lemma D.6 gives the first bound pointwise on the branches; the triangle inequality, (D.22), positivity and $|F_\xi(1)| \leqslant 1$ give the second. *q.e.d.*

D.7 The History Average by One-Point Witnesses

This section assumes compatible even fine periods: every fine period is a multiple of $\ell = LMR_1$ and at least $8M$ ((3.1)) and even. This is a restriction relative to the compatible periods of Section 3; it includes the torus with periods $2L^m$ when the partitions fit.

Let $\ell \coloneqq LMR_1$ be the macro side (R_1 the scale integer), $B_\xi = \Lambda_1^c$ the filled bad region, $E(X)$ the set of positive fine bonds with at least one endpoint in a set X of sites, and n_C the number of contour traversals.

Let Γ_C be the deterministic set of the contour bonds together with the free-input templates of their reconstruction rows, projected periodically from the torus templates before any history is selected (so $\mathcal{F}_r \subseteq \Gamma_C$ for every contour bond r, with $\mathcal{F}_r$ the free-input set of (3.13), the set appearing in the covered event of Lemma 3.36); there is a constant C_L, depending on L only (the templates are fixed lattice stencils), with $|\Gamma_C| \leqslant C_L n_C$, and $I_C \subseteq \Gamma_C$ on every branch (the designated variables of the integration carrier are eliminated, so their free-input stencils, not the designated bonds, are the members of Γ_C).

For a positive bond z the marked probability is

$$p_z \coloneqq \mathcal{Z}_0^{-1} \int \mathcal{O}[\mathsf{w}_\xi \mathbf{1}\{z \in E(B_\xi)\}](V)\,\mathrm{d}V = \mathbb{P}_{\mathrm{h}}(z \in E(B_\xi))\,; \tag{D.23}$$

the mark depends only on the regions and on the deterministic filling.

Lemma D.9 (Reduction to One-Point Marks)**.** *Let $c = \mu/2$ and $S_c = \sup_b \sum_z e^{-cd(b,z)}$, the periodic bond shell sum, bounded by its infinite-lattice counterpart uniformly in the periods. Pointwise on every branch*

$$D_\xi \leqslant \sum_{b \in \Gamma_C} \sum_z e^{-cd(b,z)} \mathbf{1}\{z \in E(B_\xi)\}\,, \qquad \mathbf{1}_{A_{\mathrm{cov}}^c} \leqslant \sum_{b \in \Gamma_C} \mathbf{1}\{b \in E(B_\xi)\}\,. \tag{D.24}$$

Hence, if $p_z \leqslant q$ for every positive bond z,

$$\mathbb{E}_{\mathrm{h}}[D_\xi] \leqslant C_L S_c n_C q\,, \qquad \mathbb{P}_{\mathrm{h}}(A_{\mathrm{cov}}^c) \leqslant C_L n_C q\,. \tag{D.25}$$

Proof. $J \subseteq E(B_\xi)$: a free bond not in I has no endpoint in Λ_1. So each $b \in I_C \subseteq \Gamma_C$ contributes $e^{-cd(b,J)} \leqslant \sum_{z \in J} e^{-cd(b,z)} \leqslant \sum_z e^{-cd(b,z)} \mathbf{1}\{z \in E(B_\xi)\}$.

If coverage fails, either a contour bond is not a bond of Ω_1, and then it meets B_ξ since $\Lambda_1 \subseteq \Omega_1$, or a free input of the contour is not in I (some $b \in \mathcal{F}_r \setminus I$ for a contour bond r, and $\mathcal{F}_r \subseteq \Gamma_C$), and then it is retained in J or lies outside Ω_1 and meets B_ξ; in both cases some $b \in \Gamma_C$ lies in $E(B_\xi)$. Taking $\mathbb{E}_{\mathrm{h}}$ of (D.24) (Tonelli; all integrands nonnegative) and using $p_z \leqslant q$ and $|\Gamma_C| \leqslant C_L n_C$ gives (D.25). *q.e.d.*

D.7.1 Cutoff Failures Give Local Witnesses

The witness argument operates on the original unresummed cutoffs [9, (1.1), (1.4), (1.8), (1.18)], before the boundary substitution; it is proved in Lemmas D.10, D.12 and D.13 below. Its conclusion: a bad eligible test forces a fine plaquette with $|U(\partial p) - \mathbf{1}|_\sigma \geqslant \theta$ in its neighbourhood, or an active mean-transport variable with $|w - \mathbf{1}|_\sigma \geqslant \theta$ there (the P_0 and P_1 alternatives need no gauge-variable witness), with the threshold

$$\theta = \min\Big(\varepsilon_0,\ \frac{\varepsilon_1}{4B_3 N^{3/2} L^2},\ \frac{\delta_0}{2C_Q}\Big)\,, \qquad C_Q = (9K_L + 2)(1 + 2B_3 N^{3/2} L^2)\,, \tag{D.26}$$

where K_L is the plaquette-filling constant of the loops of [8, (0.4), (0.11)] and of the bond loops of one block or of two adjacent blocks, defined in the paragraph "Constants of the block comparison" below.

The norm is the cutoff norm $|\bullet|_\sigma$, and the mean and comparison constants are chosen for it. The seed is used on the original earlier-cutoff and averaging-constraint support. Its common local width w is LM_2R_1 or $L^2M_2R_1$, with one convention throughout, and its macro width is ℓ; these are the conventions of the branch hypotheses of Section 3.1.

For a cube Q and $k \geqslant 1$, $Q_w^{[k]}$ denotes Q enlarged by k layers of width w (the $\square'^{\sim k}$ of [9] when $w = L^2M_2R_1$), so that $\mathrm{side}(Q_w^{[k]}) = \mathrm{side}(Q) + 2kw$ and $Q_w^{[2]}$ is at ℓ^∞ distance $2w$ from the complement of $Q_w^{[4]}$.

An eligible cube is a cube $\square'$ of the partition of [9] into $L^2M_2R_1$-cubes carrying the test (1.18), which Bałaban performs for $\square' \subset \Omega_1^{\sim -1}$ (p.250), so every eligible cube satisfies the hypothesis $Q \subseteq \Omega_1^{\sim -1}$ of Proposition D.20.

Constants of the block comparison. For a block centre y and a site x of its block let $\mathbf{G}(y, x)$ be Bałaban's finite family of shortest paths from y to x ([8, p. 252], the definition preceding (0.4), used in (0.11)): for each permutation π of the axes the path moves first along $e_{\pi(1)}$ by the full coordinate difference, then along $e_{\pi(2)}$, and so on; each coordinate is monotone along the path, so the path stays in the block $B(y)$. Let $\Gamma_x \in \mathbf{G}(y, x)$ be a fixed representative (Γ_y the empty path, with transport $\mathbf{1}$).

Let K_L bound the number of plaquettes needed to fill each closed loop of the following finite family, each filling chosen inside the union of the block or blocks containing the loop (a lattice box, in which every closed lattice loop bounds a plaquette filling):

- the difference of two paths of one $\mathbf{G}(y, x)$;
- the loop $\Gamma_x \cup b \cup (-\Gamma_{x'})$ of a block-internal bond $b = (x, x')$;
- the loop $\Gamma_x \cup b \cup (-\Gamma_{x'}) \cup (-c)$ of an interface bond $b = (x, x')$ of the coarse bond c, $x \in B(c_-)$, $x' \in B(c_+)$, $\Gamma_x \in \mathbf{G}(c_-, x)$, $\Gamma_{x'} \in \mathbf{G}(c_+, x')$;
- the loops $\Gamma \cup [x, x'] \cup (-\Gamma') \cup (-c)$ of [8, (0.4)], for a site $x \in B(c_-)$, with $[x, x']$ the parallel transport of the coarse bond c to x ([8, p. 252], the sentence preceding the definition of the averages: the straight contour of L fine bonds in the direction of c from x to $x' = x + (c_+ - c_-) \in B(c_+)$) and paths $\Gamma \in \mathbf{G}(c_-, x)$, $\Gamma' \in \mathbf{G}(c_+, x')$.

Each loop lies in one block or in the union $B(c_-) \cup B(c_+)$ of two adjacent blocks (the contour $[x, x']$ runs inside this union, a box of sides $2L, L, L, L$), so K_L depends on L only.

The transports of a unitary field F along two paths with the same endpoints differ in $|\bullet|_\sigma$ by at most the sum of $|F(\partial p) - \mathbf{1}|_\sigma$ over the plaquettes filling their loop (unitary invariance of the norm and telescoping), hence by at most $K_L\eta$ when $|F(\partial p) - \mathbf{1}|_\sigma \leqslant \eta$ for every plaquette p with its four vertices in $B(c_-) \cup B(c_+)$ ("the plaquettes of the two blocks"; this includes the plaquettes straddling the interface).

For the group mean M of [8, (0.5)–(0.10)] let $\mathcal{N}_L$ be the finite set of cardinalities $|\mathbf{G}(y, x)|$ and, for $n \in \mathcal{N}_L$, $\mathfrak{m}_n(A_1, \dots, A_n) = (1/i) \log M(e^{iA_1}, \dots, e^{iA_n})$; it is analytic near zero with $\mathfrak{m}_n(0) = 0$ and derivative $A \mapsto n^{-1} \sum_j A_j$ at zero ([8, (0.8)]). Let $r_{\mathrm{M}} > 0$ be a radius such that every $\mathfrak{m}_n$, $n \in \mathcal{N}_L$, is analytic on the closed maximum-norm ball of n-tuples of radius r_{M} (the minimum over the finite set $\mathcal{N}_L$ of such radii), and C_{M} half the maximum over

$n \in \mathcal{N}_L$ of the second-derivative norms on these balls; Taylor's theorem gives $|\mathfrak{m}_n(A)|_\sigma \leqslant \max_j |A_j|_\sigma + C_{\mathrm{M}}(\max_j |A_j|_\sigma)^2$ for $\max_j |A_j|_\sigma \leqslant r_{\mathrm{M}}$.

On the principal unitary chart $|(1/i)\log W|_\sigma \leqslant |W-\mathbf{1}|_\sigma + C_{\log}|W-\mathbf{1}|_\sigma^2$ with a fixed $C_{\log}$: eigenvalue by eigenvalue, $2\arcsin(t/2) \leqslant t + Ct^2$ for $0 \leqslant t \leqslant 2$, $t_j = |e^{i\varphi_j} - 1|$, which gives the operator-norm case directly and the normalised Hilbert–Schmidt case with a constant depending on N ($\max_j t_j \leqslant \sqrt{N}|W-\mathbf{1}|_\sigma$).

Let $C'_{\mathrm{M}} := C_{\log} + 4C_{\mathrm{M}}$. If $|W_j W_k^{-1} - \mathbf{1}|_\sigma \leqslant \rho$ for all j, with $\rho \leqslant 1/\max(1, C_{\log})$ and $2\rho \leqslant r_{\mathrm{M}}$, then $|M(W_1, \dots, W_n) - W_k|_\sigma \leqslant \rho + C'_{\mathrm{M}}\rho^2$: after translation by W_k^{-1} ([8, (0.6)]) the inputs $A_j = (1/i)\log(W_j W_k^{-1})$ have norm at most $\rho + C_{\log}\rho^2 \leqslant 2\rho$; the phase $H = \mathfrak{m}_n(A)$ is Hermitian ([8, (0.9)], the W_j being unitary), so $|e^{iH} - \mathbf{1}|_\sigma \leqslant |H|_\sigma \leqslant \rho + C_{\log}\rho^2 + 4C_{\mathrm{M}}\rho^2$.

Let $\eta_0 > 0$ satisfy $C'_{\mathrm{M}} K_L \eta_0 \leqslant 1$, $2K_L\eta_0 \leqslant r_{\mathrm{M}}$, $K_L\eta_0 \leqslant 1$, $C_{\log}K_L\eta_0 \leqslant 1$ (so $\rho = K_L\eta_0 \leqslant 1/\max(1, C_{\log})$) and $2K_L\eta_0 \leqslant r_G/\sqrt{N}$, so that every $W \in G$ with $|W-\mathbf{1}|_\sigma \leqslant 2K_L\eta_0$ lies in the chart domain: $\|W-\mathbf{1}\|_{\mathrm{op}} \leqslant \sqrt{N}|W-\mathbf{1}|_\sigma \leqslant r_G$, and r_G is the radius of Appendix B with this property (every logarithm above is taken at such a W: the loops have $|W-\mathbf{1}|_\sigma \leqslant K_L\eta_0$, the translated means $|W_j W_k^{-1} - \mathbf{1}|_\sigma \leqslant \rho \leqslant K_L\eta_0$); for $\eta \leqslant \eta_0$ and $\rho = K_L\eta$ this gives $\rho + C'_{\mathrm{M}}\rho^2 \leqslant 2K_L\eta$. The constants K_L, r_{M}, C_{M}, $C_{\log}$, η_0 depend on L, the group and the norm only.

Lemma D.10 (Block Comparison)**.** *Let F be a unitary field on the bonds of two blocks $B(c_-)$, $B(c_+)$ joined by the coarse bond c (or of one block), with $|F(\partial p) - \mathbf{1}|_\sigma \leqslant \eta \leqslant \eta_0$ for every plaquette with its four vertices in $B(c_-)\cup B(c_+)$ and $|F(y,x)-\mathbf{1}|_\sigma \leqslant \eta$ for its averaged contour variables $F(y,x) = M(\{F(\Gamma)\}_{\Gamma\in\mathbf{G}(y,x)})$ ([8, (0.11)]), $y \in \{c_-, c_+\}$. Then, with $C_{\mathrm{G}} := 9K_L + 2$ and $M(F,c)$ the block average of [8, (0.4)],*

$$\begin{aligned} |F(b) - \mathbf{1}|_\sigma &\leqslant C_{\mathrm{G}}\eta \quad \text{for a block-internal bond } b, \\ |F(b) - M(F,c)|_\sigma &\leqslant C_{\mathrm{G}}\eta \quad \text{for an interface bond } b \in B(c)\,. \end{aligned} \tag{D.27}$$

Proof. Two paths of $\mathbf{G}(y,x)$ have transports within $K_L\eta$ of each other, so the group-mean estimate with $\rho = K_L\eta$ puts each $F(\Gamma)$, $\Gamma \in \mathbf{G}(y,x)$, within $2K_L\eta$ of $F(y,x)$, hence within $(2K_L+1)\eta$ of the identity.

For an internal bond $b = (x,x')$ the loop $\Gamma_x \cup b \cup (-\Gamma_{x'})$ gives $|F(\Gamma_x)F(b)F(\Gamma_{x'})^{-1} - \mathbf{1}|_\sigma \leqslant K_L\eta$, and replacing $F(\Gamma_x)$, $F(\Gamma_{x'})$ by the identity costs $2(2K_L+1)\eta$, so $|F(b)-\mathbf{1}|_\sigma \leqslant (5K_L+2)\eta$.

For an interface bond $b = (x,x')$ the loop $\Gamma_x \cup b \cup (-\Gamma_{x'}) \cup (-c)$ gives in the same way $|F(b) - F(c)|_\sigma \leqslant (5K_L+2)\eta$, $F(c)$ the transport along the straight centre-to-centre path.

By [8, (0.4)], $M(F,c) = e^{iH}F(c)$ with H the average, with weights summing to one (the weight L^{-4} of each site $x \in B(c_-)$ and the path weights of (0.4)), of the logarithms $(1/i)\log F(\Gamma \cup [x,x'] \cup (-\Gamma') \cup (-c))$ over the sites $x \in B(c_-)$ and the path pairs $\Gamma \in \mathbf{G}(c_-, x)$, $\Gamma' \in \mathbf{G}(c_+, x')$, $[x,x']$ the transported coarse bond; each of these loops belongs to the family defining K_L, so each logarithm has norm at most $K_L\eta + C_{\log}(K_L\eta)^2 \leqslant 2K_L\eta$ and is Hermitian, so $|M(F,c) - F(c)|_\sigma \leqslant |H|_\sigma \leqslant 2K_L\eta \leqslant 4K_L\eta$. Together $|F(b) - M(F,c)|_\sigma \leqslant (9K_L + 2)\eta$. *q.e.d.*

Definition D.11 (Seed Constants)**.** $k_N := \sqrt{N}$ and $\alpha := 2k_N^2 L^2 \theta$, with θ of (D.26). For a cube Q, $N(Q)$ is the set of fine sites within distance 6ℓ of Q. The block stencil of a bond b is the block or the pair of adjacent blocks containing its endpoints, with the plaquettes having their four vertices in their union and the averaged contour variables of their centres. The

seed conditions on g_0 are

$$\begin{gathered} LM_2R_1 > 4LM_1 + 4L + 8\,, \qquad \theta \leqslant \eta_0\,, \qquad 2B_3k_N^3L^2\theta \leqslant \eta_0\,, \\ k_N\theta \leqslant c_2'\,, \qquad C_0L^2k_N\theta \leqslant 1\,, \qquad \alpha \leqslant a_1\,, \end{gathered} \tag{D.28}$$

with η_0, c_2', C_0 the constants of [3, Proposition 1, (51)] and a_1 the constant of [6, Theorem 1].

$|\bullet|_\sigma$ and the norms of [3] and [6] are within factors k_N of each other: each of these is the operator norm or the Hilbert–Schmidt norm in one of its two normalisations, and $h \leqslant |\bullet|_\sigma \leqslant \|\bullet\|_{\mathrm{op}} \leqslant \sqrt{N}\,h$ with the unnormalised Hilbert–Schmidt norm equal to $\sqrt{N}\,h$ (the conventions of Section 3). Under (D.28), $B_3\alpha \leqslant B_3a_1 \leqslant a_0$. Every seed condition is an upper bound on θ or a lower bound on R_1, and all hold for small g_0 since $\theta \leqslant \varepsilon_0 \to 0$ and $R_1 \to \infty$ ([9, (2.4)–(2.8)], the definition of R_1).

Lemma D.12 (Witnesses for Bad Tests)**.** *Let g_0 satisfy the seed conditions* (D.28)*. On the support of the averaging constraint $\delta(\bar{U}V^{-1})$ of [9, (1.6), (1.11)], for the gauge-fixed field $\widetilde{U}$ of that integrand:*

(a) a P_0 test is good whenever every plaquette of its defining enlargement has $|U(\partial p)-\mathbf{1}|_\sigma < \theta$;

(b) if every plaquette $p \subseteq N(Q)$ has $|U(\partial p) - \mathbf{1}|_\sigma < \theta$, where Q is any cube of the partition of [9] into $L^2M_2R_1$-cubes (in particular any cube carrying the test (1.4)), the P_1 test of Q is good, and $|U_{1,Q}(\partial p) - \mathbf{1}|_\sigma \leqslant 2B_3k_N^3\theta \leqslant \varepsilon_1L^{-2}/2$ on $Q_w^{[1]}$ and on $Q_w^{[2]}$, its incident bonds and their block stencils;

(c) if moreover every active averaged contour variable $w_{yx} = \widetilde{U}(y,x)$ with $x \in N(Q)$ has $|w_{yx} - \mathbf{1}|_\sigma < \theta$, then, for a Q_1- or R_1-eligible cube Q,

$$\sup_{b \in (Q_w^{[2]})^*} |\widetilde{U}(b)U_{1,Q}(b)^{-1} - \mathbf{1}|_\sigma \leqslant C_Q\theta \leqslant \frac{\delta_0}{2} < 2\delta_0\,, \tag{D.29}$$

so the Q_1 test of Q is good.

The R_1 test is treated in the proof of Lemma D.13.

Proof. (a) is $\theta \leqslant \varepsilon_0$.

Collar payments: $w \leqslant a = L^2M_2R_1$ and $M \geqslant 8LM_2$ give $4w + 2L + 4 \leqslant 4a + 2L + 4 < \ell$, and $w \geqslant LM_2R_1 > 4LM_1 + 4L + 8$ gives $2w - 2LM_1 > 2L + 4$. The $\sim$ operations on P_0', P_1' and $B(P_1^1)^c$ are by LMR_1-cubes ([9, pp. 246 and 248]), so the first inequality places $Q_w^{[4]}$ and the blocks meeting it inside $B(P_1^1)$ for a Q_1-eligible cube $Q \subseteq B(P_1^1)^{\sim -1}$ ([9, (1.8)]) and, by [9, (1.10)] ($\Omega_1 \subseteq B(P_1^1)^{\sim -3}$), for an R_1-eligible cube $Q \subseteq \Omega_1^{\sim -1}$ as well; the second places $Q_w^{[2]}$, its incident bonds and their block stencils at distance more than $2L+4$ inside $Q_w^{[4]}$ eroded by $2LM_1$. All these sets lie in $N(Q)$, since they are within $4w + 2L + 4 \leqslant 4a + 2L + 4 < \ell < 6\ell$ of Q.

(b) The block average is gauge covariant, $M(U^v, c) = v(c_-)M(U,c)v(c_+)^{-1}$ for a gauge transformation v: in [8, (0.4)] every loop is based at c_-, so its holonomy is conjugated by $v(c_-)$, the chart logarithm is conjugation covariant, and $U(c)$ is a transport from c_- to c_+. So the coarse plaquette deviations $|\bar{U}(\partial p') - \mathbf{1}|_\sigma$ of $\bar{U} = M(U, \bullet)$ coincide for U and its gauge transform $\widetilde{U}$, as do the fine plaquette deviations, and the plaquette hypothesis on $N(Q)$ may be read for either field.

[3, Proposition 1, (51)], in the averaging norm (at most $k_N|\bullet|_\sigma$) with $\alpha_0 = k_N\theta \leqslant c_2'$ gives $|\bar U(\partial p') - \mathbf{1}| \leqslant L^2 k_N\theta + C_0(L^2k_N\theta)^2 \leqslant 2L^2k_N\theta$ for every coarse plaquette p' whose blocks lie in $N(Q)$; a further factor k_N converts this to the norm of [6], so on the support of $\delta(\bar U V^{-1})$ the coarse data V on $Q_w^{[4]}$ satisfy the data bound of [6] with $\alpha = 2k_N^2L^2\theta$.

The determining data of $U_{1,Q}$ are those of [9, (1.2)]: the minimal determining set $\mathbf{B}_1(Q_w^{[4]})$ with the data $M^\bullet(Q_1^{s*}V)$, no exterior data. The extension (1.3) has identity block-internal links and interface links $V(c)$; its fine plaquettes are the identity (two internal and two interface links across one coarse bond) or coarse plaquette holonomies (four interface links at a corner of four blocks), and its block averages are V (the loops of [8, (0.4)] are the identity on it). So the ordinary and mixed determining plaquettes of [6, (7)] obey the bound $\alpha \leqslant a_1$, and [6, Theorem 1] gives a minimal critical orbit in the class (8) with bound $B_3\alpha$.

[9, p. 256, (2.12)] defines $U_{1,Q}$ as an element of the unique minimal orbit of the local action over the regular configurations U with $M_{\mathbf{B}}(U) = V$, and refers for "regular" and for the variational theory to [6]; "regular" is membership in the class [6, (6)], the space [6, (2)] with a radius ε satisfying $B_3\alpha \leqslant \varepsilon \leqslant a_0$, the setting of [6, Theorem 1] (such ε exist, as $B_3\alpha \leqslant B_3a_1 \leqslant a_0$). Theorem 1 states that the minimal orbit of the class (8) is the unique critical orbit of the class (6), for every such ε, so the choice of the radius is immaterial; a minimal configuration is critical, so the minimal orbit over the regular configurations is this orbit, $U_{1,Q}$ satisfies the bounds (8) with the parameter of [6] (written ε_1 there; not the $\varepsilon_1 = g_1p_0(g_1)$ of [9]) equal to α, and the axial gauge condition ([9, p. 247]) fixes the representative.

The construction of $\mathbf{B}_1$ in [9, pp. 256–257] (after (2.12); $Q_w^{[4]}$ is a union of w-cubes, $w \in \{LM_2R_1, L^2M_2R_1\}$ a multiple of LM_1, hence a union of LM_1-cubes and of M_1-cubes, $M_1 < M_2$ being powers of L) gives that the coarse determining interior contains $(Q_w^{[4]})^{\sim -2}$ with the erosion in M_1-cubes of $T_{L^{-1}}$, of side LM_1 in fine units, so the loss is at most $2LM_1$, and by the collar payment $Q_w^{[2]}$, its incident bonds and their block stencils lie inside it. There [6, (8)] bounds the fine plaquettes of the minimum by $B_3\alpha L^{-2}$ in the norm of [6], that is by $k_NB_3\alpha L^{-2} = 2B_3k_N^3\theta \leqslant \varepsilon_1L^{-2}/2$ in $|\bullet|_\sigma$ by (D.26); on $Q_w^{[1]}$ this is the good P_1 test of [9, (1.4)].

(c) The blocks meeting $(Q_w^{[2]})^*$ are active by the collar payments. Lemma D.10 applies to $\widetilde U$ with $\eta = \theta$ (plaquettes and averaged contour variables below θ) and to $U_{1,Q}$ with $\eta = 2B_3k_N^3L^2\theta \leqslant \eta_0$ (its plaquette deviations are at most $2B_3k_N^3\theta$ by (b), and its averaged contour variables are the identity in the axial gauge). On the support of the averaging constraint $M(\widetilde U, c) = V(c)$, and $M(U_{1,Q}, c) = V(c)$ on the coarse bonds of the determining interior (the constraint of the local problem, data (1.3)). Comparing both fields with the identity for an internal bond, and with $V(c)$ for an interface bond, $|\widetilde U(b)U_{1,Q}(b)^{-1} - \mathbf{1}|_\sigma = |\widetilde U(b) - U_{1,Q}(b)|_\sigma \leqslant C_{\mathrm G}\theta(1 + 2B_3k_N^3L^2) = C_Q\theta \leqslant \delta_0/2$ by (D.26), the good Q_1 test of [9, (1.8)]. *q.e.d.*

Lemma D.13 (A Bad Test Forces a Witness)**.** *For g_0 below a threshold depending on L, the group, M_1, M_2, M and the source constants, on the original cutoff support: every bad P_0, P_1, Q_1 or R_1 test forces a plaquette or gauge-variable witness at threshold θ within distance $K\ell$ of every bond $z \in E(B_\xi)$ whose membership it causes, with the fixed $K = 256$; and the number of plaquette and gauge-variable labels in such a neighbourhood is at most*

$$n_W \coloneqq 7(2K\ell + 7)^4 \,. \tag{D.30}$$

Proof. The P_0, P_1 and Q_1 alternatives are Lemma D.12(a)–(c). For the R_1 test, Lemma D.12(c) is combined with the comparison of the two minima on the twice enlarged

carrier of every eligible cube Q, uniformly in its label, proved in Proposition D.20 below:

$$\sup_{b\in(Q_w^{[2]})^*} |U_1(b)U_{1,Q}(b)^{-1} - \mathbf{1}|_\sigma \leqslant C_{L,G}\varepsilon_1 e^{-c_{L,G}LM_2R_1} < \delta_0/10\,. \tag{D.31}$$

Its only input from the characteristic factors of [9, (1.11)] is their nonvanishing at U_1 itself, the membership $U_1 \in D^{(11)}(V, V_0)$ of (B1) ([9, p. 248], in the sense recorded there and in (H4)).

On the native principal chart a bad R_1 label of the cube Q forces $|g_0A(b)|_\sigma \geqslant \delta_0$ at a bond $b \in (Q_w^{[2]})^*$, the carrier of the test in [9, (1.18)]; the eigenvalue inequality $|e^{iX} - \mathbf{1}|_\sigma \geqslant (2/\pi)|X|_\sigma$, valid when every eigenvalue of X lies in $[-\pi, \pi]$ (as on the chart) in both permitted norms, and (D.31) give, for the gauge-fixed field $\widetilde{U} = e^{ig_0A}U_1$ at the free bond b ([9, (1.13)]: $U' = UU_1^{-1} = e^{iA'}$ with $A' = g_0A$ the fluctuation field; the variable tested in (1.18) is the free variable $A(b)$ of (1.15), and the fluctuation at a free bond is exactly $e^{ig_0A(b)}$, because the substitution (1.14), $A' = B - h_{U_1}\widetilde{D}(B)$ with $B = g_0\mathsf{C}_{U_1}A$, changes the phase only at the designated bonds: h_{U_1} is supported on them and $\mathsf{C}_{U_1}A$ agrees with A on the free bonds, (4.3) at the background, Section 3.2),

$$|\widetilde{U}(b)U_{1,Q}(b)^{-1} - \mathbf{1}|_\sigma \geqslant (2/\pi - 1/10)\delta_0 > \delta_0/2\,. \tag{D.32}$$

If every witness in $N(Q)$ were below θ, (D.29) (Lemma D.12(c) for the R_1-eligible cube Q; the earlier cutoffs and Ω_1 are fixed term by term in the sum (1.11), which resums P_0, P_1, Q_1 at fixed Ω_1, and $\Omega_1 \subseteq B(P_1^1)^{\sim -3}$ holds for each term; the comparison (D.31) holds whatever the R_1 label of Q) would give $|\widetilde{U}(b)U_{1,Q}(b)^{-1} - \mathbf{1}|_\sigma \leqslant C_Q\theta \leqslant \delta_0/2$, a contradiction; so the label is excluded.

[9, (1.1), (1.4), (1.8), (1.18)] are sharp characteristic partitions at their printed thresholds, so these inequalities make the corresponding bad factor vanish; (1.18) is introduced after the logarithmic reconstruction (1.13)–(1.15).

For the distances: before filling, [9, (1.10), (1.20)] give the bad region as $Q_1'^{\sim 4} \cup (B(P_1^1)^c)^{\sim 5} \cup R_1'^{\sim 1}$, with $B(P_1^1)^c \subseteq P_0'^{\sim 2} \cup P_1'^{\sim 1}$ (the definition of P_1^1 following [9, (1.4)], as printed, gives $B(P_1^1)^c = P_0'^{\sim 2}$; the reading consistent with the active-region construction of [9, (3.2)–(3.3)] gives $P_0'^{\sim 2} \cup P_1'^{\sim}$; both are contained in $P_0'^{\sim 2} \cup P_1'^{\sim 1}$), so a point of this region is within 10ℓ of a bad test.

The modification after (1.20) fills only components contained in cubes of side less than 100ℓ, so every added point is within 100ℓ of an original unfilled bad point; this also holds if small merged components are filled again, since by induction every current component contains an original bad point and each fill is performed inside a cube of side less than 100ℓ containing that component (every point of the rectangular hull of such a component is within 100ℓ of a chosen original bad point of it, and overlapping hulls or later merges preserve the presence of an original point), so there is no cumulative distance loss. [9, (2.9)] gives, for its scale R_0 (the range of its expansions, distinct from the range constant R_0 of (7.28)), $R_0 \leqslant (L+1)R_1$, hence $R_0 \leqslant LR_1$ as both are powers of L (the conventions of Section 3).

Including each test's witness neighbourhood and incident bonds, every $z \in E(B_\xi)$ forces a witness within $K\ell$ of z with $K = 256$: a point of the filled bad region is within 100ℓ of an original bad point, which is within 10ℓ of a bad test cube Q of side at most $a \leqslant \ell/8$, whose witness lies in $N(Q)$, within 6ℓ of Q; an incident bond adds one; in all $100\ell + 10\ell + a + 6\ell + 1 \leqslant 117\ell < 256\ell$. The count (D.30) bounds the plaquette and gauge-variable labels in that neighbourhood, also on a small periodic torus. *q.e.d.*

D.7.2 The Marked Probability Retains the Physical Normaliser

Definition D.14 (The Comparison Measures)**.** $A_{\mathrm{e}}^{\Psi} \coloneqq \Psi_\xi(A_{\mathrm{e}})$, and $\mathscr{E}_\xi(B)$ is the pre-boundary-change exponent of [9, (1.15)], its outer prefactor excluded ((3.33)). With $s_G(w) \coloneqq 1 - \operatorname{Re}\operatorname{tr}_N w$, the bare probability and the gauge-variable law are

$$\mu_0(\mathrm{d}U) \coloneqq \mathcal{Z}_0^{-1} e^{-\mathsf{E}} e^{-S(U)/g_0^2}\,\mathrm{d}U\ , \qquad q_g(\mathrm{d}w) \coloneqq z_g^{-1}\mathbf{1}\{|w-\mathbf{1}|_\sigma < \varepsilon_0\} e^{-s_G(w)/g_0^2}\,\mathrm{d}w\ , \tag{D.33}$$

with z_g the integral of the displayed density, the gauge normaliser of [9]. For a positive bond z, W_z is the event that some plaquette or some auxiliary gauge variable in the deterministic $K\ell$-neighbourhood of z is a witness at threshold θ; $\mathcal{I}$ is the set of pairs (y,x) of a block centre y and a noncentral site x of its block, and T the set of centres and unused sites.

Lemma D.15 (The Marked Density and Its Reversal)**.** *With c_ξ of* (3.48)*, pointwise on $\mathfrak{g}^I$,*

$$e^{c_\xi} K_{\xi;1}(a) = \mathbf{1}_{\mathcal{D}_I}(a)\,|\det D\Psi_\xi(A_{\mathrm{e}})|\,\exp \mathscr{E}_\xi(\Psi_\xi(a+A_{\mathrm{e}}))\ , \tag{D.34}$$

and the composed retained cutoff $\chi_{\xi,\mathrm{e}}$ and ζ_ξ multiply both sides; integrating over a,

$$\mathsf{w}_\xi = e^{c_\xi} I_{\xi;1} = \int e^{c_\xi} K_{\xi;1}(a)\,\mathrm{d}a\ , \tag{D.35}$$

the branch constant c_ξ and every far primitive present ((3.49)*). Together with Lemma D.13, on compatible even fine periods, for every positive bond z, with the measures and the event of Definition D.14,*

$$p_z \leqslant (\mu_0 \otimes q_g^{\otimes\mathcal{I}} \otimes \mathrm{Haar}^{\otimes T})(W_z)\ . \tag{D.36}$$

Proof. Ψ_ξ is the identity in every near stencil and its support S_1^* is disjoint from I, so $\Psi_\xi(a+A_{\mathrm{e}}) = a + A_{\mathrm{e}}^{\Psi}$ ((3.27)). Expanding the quadratic form of $\mathscr{E}_\xi$ into its II, IJ, JJ blocks, leaving every far primitive at A_{e}^{Ψ} and retaining the boundary determinant gives (D.34), which is the computation of Lemma 3.17(ii); the retained cutoff and ζ_ξ are factors of the integrand outside the a-integral. Together with the outer prefactor κ_ξ ((3.5)) the right side recombines into the transformed density of [9, (1.15)], boundary Jacobian included.

The changes of variables of Section 3.1 are reversed with the mark $\mathbf{1}\{z \in E(B_\xi)\}$ inserted; the reversal is a change of variables for the integral of the density times any nonnegative measurable function of U and of the regions, performed term by term before the characteristic partitions are summed. The reversal is performed in each term of the history sum separately, where the regions, the filling and the labels are fixed, and each change of variables is an identity between integrals of a density times a nonnegative measurable integrand (the term's partitions and cutoffs multiply the integrand and are carried along), so it holds for the mark, and equally for any nonnegative measurable function of U, the regions and the labels in its place; the partitions are summed only after the domination by a label-free event below.

The map Ψ_ξ, the free-variable constraint elimination, the scaling, the nonlinear constraint linearisation of (1.14), the logarithmic chart of (1.13) with its Haar density, and the translation about U_1 are undone in this order; the retained cutoff stays composed with Ψ_ξ and ζ remains in that composition, boundary-fluctuation dependence included. The marked integral is then [9, (1.6)] with its four original partitions and the site-gauge integration, no determinant or normalisation discarded; the unmarked chain is $\mathcal{Z}_0 = e^{-\mathsf{E}} \int e^{-S(U)/g_0^2}\,\mathrm{d}U$.

For each active block centre y and noncentral x let w_{yx} be its mean transport after the gauge change; [8, (0.6)] and [9, (1.5)] give the Haar substitution $v(x) = w_{yx}^{-1} h_y U(y,x)$, $v(y) = h_y$, with Jacobian one, and the law of w is (D.33). Independent q_g variables of integral

one are added for the inactive indices, and independent Haar variables for the centres and unused sites. The earlier P_0, P_1 choices depend on U and its coarse gauge orbit, not on these auxiliary variables.

By Lemma D.13 the mark is dominated by $\mathbf{1}_{W_z}$, and W_z does not depend on the later Q_1, R_1 labels; only after this enlargement are the characteristic partitions summed to one. This gives (D.36).

The plaquette witness norms and the conjugation-invariant gauge-variable norms are gauge-invariant scalars under the simultaneous gauge transformations. *q.e.d.*

D.7.3 One-Point Tails and Summation

Let $d_G \coloneqq \dim G$ and $b_G \coloneqq 1/(2N)$. The action is $S(U) = \sum_p s_G(U(\partial p))$ over the unoriented plaquettes, (1.2), with the normalised trace $\mathrm{tr}_N = N^{-1}\,\mathrm{Tr}$, and $s_G(w) \geqslant b_G|w-\mathbf{1}|_\sigma^2$. By (1.7), $\mathcal{Z}_0 = \int \rho_0\,\mathrm{d}U = e^{-\mathsf{E}}\int e^{-S/g_0^2}\,\mathrm{d}U$ with $\mathsf{E} = E_\varepsilon$ the vacuum scalar of (1.2); $\mathrm{d}U$, $\mathrm{d}w$ are normalised Haar measures, and μ_0 and q_g are probability measures by their normalisations $\mathcal{Z}_0$ and z_g (D.33).

There are constants c_H, $g_H > 0$ depending on the group only such that the Haar measure of the operator ball of radius g about $\mathbf{1}$ is at least $c_H g^{d_G}$ for $0 < g \leqslant g_H$: in the chart, $\mathrm{d}w = \sigma(A)\,\mathrm{d}A$ with the continuous Haar density σ, $\sigma(0) = \sigma_0 > 0$ (the steps listed after (3.7)); let $g_H < \varrho_G$ satisfy $\sigma \geqslant \sigma_0/2$ on $\{\|A\|_{\mathrm{op}} \leqslant g_H\}$, a subset of the chart ball as $h \leqslant \|\bullet\|_{\mathrm{op}}$; for $g \leqslant g_H$ the map $A \mapsto e^{iA}$ sends $\{\|A\|_{\mathrm{op}} < g\}$ injectively into the operator ball of radius g ($\|e^{iA}-\mathbf{1}\|_{\mathrm{op}} \leqslant \|A\|_{\mathrm{op}}$), so that ball has Haar measure at least $(\sigma_0/2)\,\mathrm{Leb}\{\|A\|_{\mathrm{op}} < 1\}\,g^{d_G} =: c_H g^{d_G}$.

The operator ball of radius g_0 lies in $\{|w-\mathbf{1}|_\sigma < \varepsilon_0\}$ for small g_0, since the two norms are comparable with fixed constants and $\varepsilon_0/g_0 = A_0(\log g_0^{-2})^{p_0} \to \infty$, $A_0 > 0$ ((4.13), [9, p. 246]: $\varepsilon_0 = g_0 A_0(\log g_0^{-2})^{p_0}$).

Definition D.16 (Single-Witness Rates)**.** With c_H as above and the number n_W of witness labels of Lemma D.13, (D.30), for $\theta' \in (0, \varepsilon_0]$,

$$z_P(\theta') \coloneqq \exp\Big[-\frac{b_G\theta'^2}{4g_0^2} + 4d_G\log g_0^{-1} + 4\log c_H^{-1} + 48\Big]\,, \qquad z_G(\theta') \coloneqq e^{1/2}(c_H g_0^{d_G})^{-1}e^{-b_G\theta'^2/g_0^2}\,,$$

and, at the threshold $\theta \leqslant \varepsilon_0$ of (D.26), $z_P \coloneqq z_P(\theta)$, $z_G \coloneqq z_G(\theta)$, and the one-point rate $q \coloneqq n_W \max(z_P, z_G)$.

Lemma D.17 (Single-Witness Tails)**.** *Let $g_0 \leqslant g_H$ be small enough that the operator ball of radius g_0 lies in $\{|w-\mathbf{1}|_\sigma < \varepsilon_0\}$. With V_T the number of fine sites,*

$$\mathcal{Z}_0 \geqslant e^{-\mathsf{E}}(c_H g_0^{d_G})^{4V_T}e^{-48V_T}\,. \tag{D.37}$$

On compatible even fine periods, for every threshold $\theta' \in (0, \varepsilon_0]$, every plaquette p and every auxiliary gauge variable w,

$$\mu_0(|U(\partial p)-\mathbf{1}|_\sigma \geqslant \theta') \leqslant z_P(\theta')\,, \tag{D.38}$$

$$q_g(|w-\mathbf{1}|_\sigma \geqslant \theta') \leqslant z_G(\theta')\,, \tag{D.39}$$

and therefore, for g_0 below the threshold of Lemma D.13 (which (D.36) *inherits through Lemma D.15), $p_z \leqslant q$ for every positive bond z, with q of Definition D.16.*

Proof. On the event that every fine link lies in the operator ball of radius g_0 every plaquette has $\|U(\partial p)-\mathbf{1}\|_{\mathrm{op}} \leqslant 4g_0$ (telescoping over its four links) and contributes $s_G(U(\partial p)) = \frac{1}{2N}\sum_j |e^{i\varphi_j}-1|^2 \leqslant \frac{1}{2}\|U(\partial p)-\mathbf{1}\|_{\mathrm{op}}^2 \leqslant 8g_0^2$ to S ($e^{i\varphi_j}$ the eigenvalues of $U(\partial p)$), and there are at most $6V_T$ plaquettes and $4V_T$ links, which gives (D.37).

Let $\theta' \in (0, \varepsilon_0]$. The bound (D.38) follows from reflection positivity and the chessboard estimate of μ_0 on the even fine periods N_μ.

For a coordinate reflection ϑ through the planes $x_\mu = 0$, $x_\mu = N_\mu/2$ let $(\vartheta U)(x,x') \coloneqq U(\vartheta x, \vartheta x')$ (inverse values on reversed bonds; $s_G(W)$ and $|W-\mathbf{1}|_\sigma$ are invariant under $W \mapsto W^{-1}$ and under conjugation, so the plaquette terms of S and the plaquette indicators below do not depend on the orientation or the base point of a plaquette) and $\Theta F = F \circ \vartheta$. Every plaquette lies in one of the two closed half-tori; let U_∂ be the links in the two planes, U_+ the remaining links of one half, and S_+ the sum of $s_G(U(\partial p))$ over the plaquettes of that closed half, the plaquettes lying in the planes (functions of U_∂) counted with weight $1/2$, so that $S = S_+ + \Theta S_+$. For a bounded F of the links (U_+, U_∂), since Θ fixes U_∂ and carries the Haar measure of U_+ to that of the other half,

$$\int \overline{F}\,\Theta F \,\mathrm{d}\mu_0 = \frac{e^{-\mathsf{E}}}{\mathcal{Z}_0} \int \Big| \int F e^{-S_+/g_0^2}\,\mathrm{d}U_+ \Big|^2 \,\mathrm{d}U_\partial \geqslant 0\,. \tag{D.40}$$

Let $C_x \coloneqq x + [0,1]^4$, x a fine site, be the closed unit cubes and, for a bounded function f of the links of C_0, let $f^{[x]}$ be its transport to C_x by successive reflections (the four coordinate reflections commute and the periods are even, so $f^{[x]}$ depends neither on the order of the directions nor on the winding of the path of reflections; and the reflection through any site plane $x_\mu = m$ carries the transport to the slab at position k in direction μ to the transport to position $2m-1-k$, since every composition of integer-plane reflections carrying position 0 to position k has orientation $(-1)^k$ in that direction and so coincides with the defining transport). For nonnegative bounded functions f_x of the links of C_0, indexed by the fine sites x,

$$\int \prod_x f_x^{[x]}\,\mathrm{d}\mu_0 \leqslant \prod_x \Big(\int \prod_y f_x^{[y]}\,\mathrm{d}\mu_0 \Big)^{1/V_T}\,, \tag{D.41}$$

the chessboard estimate. Its proof is the argument of [11, Theorem 10.11, p. 445], with (D.40) as the reflection positivity and the link variables with site reflections in place of the spin system there.

In one direction, with $N_\mu = 2h$: by translation invariance of μ_0, (D.40) holds for the reflection through any pair of antipodal site planes $x_\mu = k$, $x_\mu = k+h$; let $\Phi(f_0, \dots, f_{2h-1})$ be the μ_0-expectation of the reflected product of functions of the $2h$ closed unit slabs (slab j carrying $f_j^{[j]}$, the transport of f_j to slab j); adjacent slabs share only plane links; the two planes bounding h consecutive slabs are fixed by the reflection through them and the plane links interior to these slabs belong to U_+, so the product over any h consecutive slabs is a function of (U_+, U_∂) for the reflection through the planes bounding it, and (D.40) with the Cauchy–Schwarz inequality for the sesquilinear form $(F, F') \mapsto \int \overline{F}\,\Theta F' \mathrm{d}\mu_0$ bounds Φ^2 by the product of the two values of Φ obtained by reflecting either half of the tuple onto the other.

Each function is increased by a positive constant and divided by the $2h$-th root of its fully disseminated expectation (the normalised functions $\widetilde{f}_j$ have $\Phi(\widetilde{f}_j, \dots, \widetilde{f}_j) = 1$); among the finitely many tuples with entries from $\{\widetilde{f}_0, \dots, \widetilde{f}_{2h-1}\}$, let a maximiser of Φ have a longest constant run (runs counted cyclically), of length r; both tuples obtained by reflecting either half onto the other are maximisers as well (the maximum is positive, every function being bounded below by its added constant, and the Cauchy–Schwarz bound, the square of the

maximum being at most the product of the two reflected values, forces both to equal the maximum). If $r \leqslant h$, reflecting the half that ends with the run onto the other half gives a maximiser with a constant run of length $2r > r$, contradicting the choice of r; so $r > h$, a half lying inside the run is reflected, and the resulting maximiser is constant: the maximum is $\Phi(\widetilde{f}_j, \dots, \widetilde{f}_j) = 1$ for some j, which gives the one-dimensional inequality for the normalised functions; letting the added constants decrease to zero (dominated convergence) gives it for the f_j.

Applying it successively in the four directions (the reflections commute, and the slab functions produced after disseminating in the earlier directions are again nonnegative and bounded) gives the exponent $1/V_T$.

Every plaquette is a translate of a plaquette of C_0 and μ_0 is translation invariant, so it suffices to apply (D.41) with $f_0 = \mathbf{1}\{|U(\partial p) - \mathbf{1}|_\sigma \geqslant \theta'\}$ for a plaquette p of C_0 and $f_x = 1$ for $x \neq 0$: the dissemination $\{f_0^{[y]}\}_y$ contains $V_T/4$ distinct plaquettes of the plane orientation $(\mu\nu)$ of p (the two transverse coordinates of a copy range over one parity class; repeated copies of an indicator do not change the product), each with $s_G \geqslant b_G\theta'^2$ on the event, and $S \geqslant 0$ termwise, so the disseminated expectation is at most $e^{-\mathsf{E}}e^{-b_G\theta'^2 V_T/(4g_0^2)}/\mathcal{Z}_0$; dividing by (D.37) and taking the $1/V_T$-th root gives $z_P(\theta')$, a normalised probability estimate.

For (D.39), the restriction of one gauge variable to its operator ball of radius g_0, where its action is at most $g_0^2/2$ and its cutoff factor is one (the ball lies in $\{|w - \mathbf{1}|_\sigma < \varepsilon_0\}$ by the hypothesis on g_0), bounds z_g from below by $c_H g_0^{d_G} e^{-1/2}$; on the event $|w - \mathbf{1}|_\sigma \geqslant \theta'$ the action is at least $b_G\theta'^2$, which gives $z_G(\theta')$.

The union bound over the at most n_W witness labels of Lemma D.13 and (D.36) at the threshold θ give $p_z \leqslant q$. *q.e.d.*

Proposition D.18 (The Exterior-Response and Uncovered Term)**.** *Let*

$$\pi_* = \min(p_0(g_0), p_0(g_1)) \,.$$

There are c_θ, c_*, $g_* > 0$, fixed after L, the group, the character, M_1, M_2, M and the source constants, such that $\theta \geqslant c_\theta g_0 \pi_*$ and $q \leqslant e^{-c_*\pi_*^2}$ for $0 < g_0 < g_*$, and on compatible even fine periods, for $0 < g_0 < g_*$ and under (D.2)*,*

$$|E_{\text{bdry}}| \leqslant C_{\text{resp},\chi}\, n_C(1 + g_0 r)\, e^{-c_*\pi_*^2} \,, \tag{D.42}$$

after replacing $C_{\text{resp},\chi}$ by $C_L \max(C_{\text{resp},\chi} S_c, 1)$; g_* is at most the thresholds of Lemma D.13, the g_H and the smallness condition of Lemma D.17, the threshold of Proposition D.20 and the coupling below which the comparison of [9, (2.6)] used in the proof holds. The constants are uniform in the compatible even periods, the contour location and the first-step histories.

Proof. [9, (2.6)–(2.9)] and $\delta_0 = (A_1/A_0)\varepsilon_0$ give $\theta \geqslant c_\theta g_0 \pi_*$ for small coupling: each of the three terms of (D.26) is at least a fixed constant times $g_0 p_0(g_0)$ or $g_1 p_0(g_1)$, and g_1/g_0 is bounded above and below by [9, (2.6)].

Since $\ell = LMR_1$ grows as a fixed power of $\log g_1^{-2}$ ((9.2)), $\log n_W = O(\log\log g_0^{-2})$; with the exponent of [9], $p_0 \geqslant 5\mathfrak{r} \geqslant 10$, $\pi_*^2 \geqslant A_0^2(\frac{1}{2}\log g_0^{-2})^{2p_0}$ dominates $4d_G \log g_0^{-1}$ and $\log n_W$ for small g_0, so $q = n_W \max(z_P, z_G) \leqslant e^{-c_*\pi_*^2}$ with $c_* \coloneqq b_G c_\theta^2/8$, for $g_0 < g_*$.

Then Corollary D.8, Lemma D.9 and Lemma D.17 give $|E_{\text{bdry}}| \leqslant C_{\text{resp},\chi} g_0 r\, C_L S_c n_C q + C_L n_C q \leqslant C_L \max(C_{\text{resp},\chi} S_c, 1)\, n_C(1 + g_0 r)\, q$, and $q \leqslant e^{-c_*\pi_*^2}$ gives (D.42). *q.e.d.*

Remark D.19 (Scope). Lemma D.6 is proved from the Hessian, Schur, enlarged-chart and primitive derivative bounds of Appendix B and the collar cutoff of Section 3.4, with both hard-cutoff surface terms inside the bounded covariance; it is applied on A_{cov}.

Proposition D.18 holds on compatible even-period tori and uses the single-witness tails of Lemma D.17 and the reconstruction and measure identities of Section 3; the filling proximity of Lemma D.13 holds for a simultaneous or an iterated filling of small components.

The R_1-label step of Lemma D.13 rests on Proposition D.20, which uses local data of [6], with the local hypotheses of the existence and uniqueness theorem [6, Theorem 1] verified where this paper's inputs allow and otherwise taken over as the application of [6] to the determining sets of [9, p. 256], uses the derivative-decay results of [6] and the identification [9, (3.17)–(3.19)] as stated, and accepts Bałaban's gauge-conversion results at their stated regular-chart hypotheses through the extension statements of [8]; of the characteristic factors of [9, (1.11)] it uses their nonvanishing at the minimum U_1 itself, the membership $U_1 \in D^{(11)}(V, V_0)$ of (B1) ([9, p. 248], in the sense recorded in (B1)(iii) and (H4)).

This appendix controls the exterior-response and uncovered term only.

D.8 The Minimum Comparison from Local Data

This section proves the deterministic comparison (D.31) of the two minima from local data and, in Section D.8.6, the bound on the boundary field of [9, (1.21)] used in Section 3.4 and (B4) (Lemma D.21).

Of the branch hypothesis (B1) of Section 3.1 it uses the membership $U_1 \in D^{(11)}(V, V_0)$ of the minimum in the domain of integration of [9, (1.11), p. 248] (by [6, Theorem 1]) with the smallness statement (i) derived from it in (B1), the averaging constraint and the source axial mean gauge, which place U_1 in the regularity class of [6] near Ω_1 and so identify it with the regular critical orbit constructed below.

The sources are [6], [9], [8], and the averaging paper [3]; Bałaban's existence and uniqueness result ([6, Theorem 1]) is used at local hypotheses verified below where this paper's inputs allow, and otherwise as the application of [6] to the determining sets of [9, p. 256] (where Bałaban states that all the results of [6] are used for the averaging); his derivative-decay results ([6, Proposition 9, (172), and (190)]), the gauge transformation of [6, Proposition 2] and the identification [9, (3.17)–(3.19)] are used as stated, and his gauge-conversion results are accepted at their stated regular-chart hypotheses through the extension statements of [8].

Proposition D.20 (Comparison of the Global and Local Minima)**.** *There are constants $C_{L,G}$, $c_{L,G} > 0$, depending on L, G, M_1, M_2 and the source parameters and fixed before g_0, independent of the fluctuation, its R_1 label, the exterior history and the torus, such that, on compatible even fine periods (every fine period is then at least 2ℓ, which the lifts below need), for every admissible (Ω_1, V, V_0) and every eligible cube $Q \subseteq \Omega_1^{\sim -1}$, on its twice enlarged free-bond carrier,*

$$\sup_{b \in (Q_w^{[2]})^*} |U_1(b) U_{1,Q}(b)^{-1} - \mathbf{1}|_\sigma \leqslant C_{L,G} \varepsilon_1 e^{-c_{L,G} L M_2 R_1} < \delta_0/10 \,, \tag{D.43}$$

for g_0 below a threshold depending on L, G, M_1, M_2 and the source constants (the conditions $\alpha \leqslant a_1$, $B_3 \alpha \leqslant a_0$, the smallness of α at [6, (172) and (187)], $CL^2 \varepsilon_0 \leqslant 1$, $\ell > \max(2L + 4, RM_1)$, $LM_2R_1 \geqslant 2LM_1 + C_L$ and the chart reductions of the proof; this threshold is part of g_* of Proposition D.18), the last inequality by Section D.8.5.

The fine periods are multiples of ℓ (compatibility, Section 3) and, by the hypothesis of the proposition, even (the restriction of Section D.7), and $\ell = LMR_1$ is odd (L is odd and M, R_1 are powers of L, (3.1)), so every compatible fine period is at least 2ℓ; since $w \leqslant L^2 M_2 R_1$ and $M \geqslant 8LM_2$ ((3.1)), $\mathrm{side}(Q_w^{[4]}) \leqslant 9L^2 M_2 R_1 \leqslant 9\ell/8 < 2\ell$, so this determining box and its fixed-range stencils have lifted copies before the next periodic image for small coupling.

If Ω_1 is the full torus there is no exterior collar to freeze and the same argument uses the periodic problem of [6].

The quantity in (D.43) concerns a simultaneously gauge-covariant pair of links: their relative product transforms by conjugation at the initial endpoint, so its norm is gauge invariant. The proof occupies the rest of this section.

D.8.1 Source Inputs and the Local-Domain Verification

U_1 is the critical minimum of [9, (1.12)], the unique critical point in the domain of integration in (1.11), supplied by [6, Theorem 1] with the prescribed V, V_0 and the source axial mean gauge ([9, p. 248], recorded in (B1) as $U_1 \in D^{(11)}(V, V_0)$). This membership gives, by (B1)(i), $|U_1(\partial p) - \mathbf{1}| < \varepsilon_0$ for every plaquette $p \subset \Omega_1^{\sim}$ and the axial condition $|U_1(y, x) - \mathbf{1}| < \varepsilon_0$ at the block centres of $\Omega_1^{\sim(1)}$, so that U_1 lies in the regularity class [6, (6)] with radius $C_L \varepsilon_0$ on the outer domain Ω_0 of [6] below, by the same conjugation and counting argument as for $\widehat{U}$ in the local-domain verification ([6] counts as plaquettes of a region those with a corner in it, so Ω_0 is taken to be the enlarged domain Ω^+ shrunk by $\ell/2$, a union of M_1-blocks every plaquette with a corner in which lies in $\Omega_1^{\sim}$; the remaining separation $\ell/2$ from Ω_1 keeps the spacing condition [6, (1)]); the localisation below verifies the hypotheses of [6, Theorem 1] for the localised problem near Ω_1. The plaquette variables of U_1 on $\Omega_1^{\sim}$ are thus within $\varepsilon_0 \leqslant (1 + \beta_0)^2 \varepsilon_1$ of the identity ([9, (2.8)]).

The local minimum $U_{1,Q}$ is [9, (1.2)] with determining set $\mathfrak{B}_Q \coloneqq \mathbf{B}_1(Q_w^{[4]})$ and reference data $D_Q^0 = \dot{M}(Q_1^{s*} V)|_{\mathfrak{B}_Q}$; it has the corresponding mixed-scale regularity of [6], with the scale-zero boundary bound described below, and the same coarse averages V on the interior determining region (the interior coarse labels of $Q_w^{[4]}$ lie in Ω_1, where the constraint $\mathsf{M}(U_1) = V$ holds: $Q \subseteq \Omega_1^{\sim -1}$ is at ℓ^∞ distance at least ℓ from Ω_1^c, and $4w \leqslant 4L^2 M_2 R_1 \leqslant \ell/2$ by $M \geqslant 8LM_2$, (3.1); [9, p. 250] states this containment).

These are the hypotheses of the derivation [9, (3.17)–(3.19)], whose only field is the minimum; at $k = 0$ the fine spacing is $\eta = 1$ and $V^{(0)} = U_1$, $V_Q^{(0)} = U_{1,Q}$.

The local-domain verification. The argument runs before the P_0, P_1, Q_1 resummation, at any supported history contributing to the fixed Ω_1, V, V_0, with fine field $\widehat{U}$. Let $\Omega^+ \coloneqq \Omega_1^{\sim 1}$ be the enlarged fine domain (the macro enlargement of [9, (1.10)]), with $\widehat{U} = V_0$ fixed on the collar $\Omega^+ \setminus \Omega_1$. Outside Ω^+ the original field is retained and its action terms are omitted from the variational problem: once $\ell > 2L + 4$, every plaquette or block-gauge term containing an unfrozen variable lies in Ω^+, so the omitted terms are constants, and the constrained first variation is exactly that of [9, (1.12)] (the gauge functional G of (1.12) has zero first variation at U_1, each of its nonnegative summands vanishing there since $U_1(y, x) = \mathbf{1}$ in the source axial mean gauge, [9, p. 248], so the first variations of $A + G$ and of A agree on the constraint tangent space at U_1).

[9, (1.10)] and the following sentence separate Ω_1 from the earlier bad regions by at least two macro layers, so every plaquette and averaging stencil in Ω^+ lies in the P_0-good region, where at this unresummed stage $|\widehat{U}(\partial p) - \mathbf{1}|_\sigma < \varepsilon_0$.

If the canonical outer operation has already omitted these factors, they equal one on the support of the retained original-field factors ((B3)(i), the form in which the removal statement of [9, p. 248] before (1.11) is recorded), so restoring them changes no integral; since the history summands are nonnegative, a point of nonzero resummed density has at least one supported unresummed history. This restoration concerns $\widehat{U}$ and controls the prescribed collar V_0.

Conjugation and the bounded number of incident plaquettes give $|D^{*}_{\widehat{U}}\partial\widehat{U}| \leqslant C_{\mathrm{inc}}\varepsilon_0$ on the collar and the variable region (C_{inc} the incidence count), and multiplying by the largest of the fixed scale weights L^2, L^3 puts $\widehat{U}$ in the class [6, (6)] with radius $C_L\varepsilon_0$. Its fixed data are exactly V_0 on the fine collar and V on the interior coarse labels, since the original averaging constraint is retained.

The carriers are the positive-bond sets $E(X)$. The fixed fine collar carrier is $E(\Omega^{+})\setminus E(\Omega_1)$, so no crossing bond that is a variable of [9, (1.12)] is fixed. The hierarchy for [6] has outer fine domain Ω^{+}, inner block domain Ω_1 and determining set this bond-set difference at scale zero together with Ω_1 at scale one, with the mixed-bond convention of [6, (7)]; its spacing condition (1) holds because the separating macro layer has width ℓ, exceeding the fixed RM_1 of [6] for small coupling.

For a mixed determining plaquette p', every coarse bond of p' outside the scale-one determining set is replaced by the block average of the scale-zero data, as prescribed in [6, (7)]: the original constraint makes the resulting loop an ordinary fine plaquette of $\widehat{U}$ or a coarse plaquette of its block average, and [3, Proposition 1, (51)], bounds the latter, in the operator norm of [3], by $L^2\alpha_0 + C(L^2\alpha_0)^2 \leqslant 2L^2\alpha_0$ with $\alpha_0 = \sqrt{N}\varepsilon_0$ for g_0 small ($CL^2\alpha_0 \leqslant 1$; the conversion $\|\bullet\|_{\mathrm{op}} \leqslant \sqrt{N}h$ of the conventions covers the case of the normalised Hilbert–Schmidt cutoff norm), which is also the explicit bound of [9] before (1.2).

The constraint [6, (3)] and [3, Proposition 1] are stated for Bałaban's earlier averaging; their use with the averaging [8, (0.4)] of [9, (1.12)] is Bałaban's own transfer in [9, p. 256] (all the results of [6] used for its averaging) and the extension statement of [8] after (0.12), accepted as such. So the entire determining set of the localised problem has data regularity in the sense of [6] at most $C_L\varepsilon_0$ (C_L including $\sqrt{N}$).

[6, Theorem 1] therefore constructs the regular critical orbit for these data, at radius $B_3C_L\varepsilon_0$, unique in the regularity class (6) of radius $\varepsilon_0^{\mathrm{V}} \coloneqq \max(C_L\varepsilon_0, B_3C_L\varepsilon_0) \leqslant a_0$ for small g_0, a class containing U_1, with both the plaquette and the divergence bounds (2).

The minimum U_1 of [9, (1.12)] lies on this orbit: it is critical for the localised problem, since the localisation changes the action by constants only, for every admissible variation, and it lies in the regularity class of [6] on Ω^{+} by its membership $U_1 \in D^{(11)}(V, V_0)$ (the first paragraph of this subsection); the uniqueness in the class identifies the two.

Restricted to $Q_w^{[4]}$ the same bound gives membership in the local uniqueness class, with fixed scale and norm conversions (that the results of [6] hold for the determining set $\mathbf{B}_1(Q_w^{[4]})$ of [9, (1.2)], whose fine layer has width between LM_1 and $2LM_1$, p.256, in place of the spacing condition [6, (1)] for that layer, is Bałaban's application of [6] to the determining sets of [9], p.256, taken over as such); since $\varepsilon_0 \leqslant (1+\beta_0)^2\varepsilon_1$ by [9, (2.8), p. 256] (with $n = 1$, $m = 0$), the radius is $C_{L,G}\varepsilon_1$, below the a_0 of [6] for small coupling.

The fine boundary layer of $U_{1,Q}$ has the scale-zero bound of [6]: the plaquettes of its straight-extension data are trivial or equal to coarse plaquette holonomies of V, bounded by $2L^2\varepsilon_0$, so its mixed-scale data are regular with the same parameter $C_{L,G}\varepsilon_1$.

D.8.2 The Boundary Data and Their Size

Let $D_Q \coloneqq \dot{M}(U_1)|_{\mathfrak{B}_Q}$ and, on the determining set, let

$$B_Q(e) \coloneqq \tfrac{1}{i}\log(D_Q(e)D_Q^0(e)^{-1}) \,. \tag{D.44}$$

Both data are defined by the source averaging operation on this regular configuration; the determining interior carries scale-one coarse bonds and its boundary layer the finer determining variables, as defined after [9, (2.12)]; the mixed determining plaquettes and the local averages are those of [9, (1.2), (2.12)–(2.16)].

For the straight extension of [9, (1.3)] every intrablock contour is the identity, every centre-to-centre path crossing a specified interface has transport $V(c)$, and every relative closed holonomy of [8, (0.4)] is the identity, so the logarithmic correction vanishes and $\mathsf{M}(Q_1^{s*}V)(c) = V(c)$; on the interior both data equal V, hence $B_Q = 0$ there, and $\operatorname{supp} B_Q$ lies in the boundary layer of width $2LM_1$ of $Q_w^{[4]}$, the support statement preceding [9, (3.18)].

The principal logarithm is legitimate because the reasoning after [9, (3.17)] bounds the argument of the logarithm by $C_{L,G}\varepsilon_1$ (the printed coefficient $44d^2B_3$ of [9] belongs to its own regularity coefficient; the enlarged local coefficient is kept inside $C_{L,G}$). Thus

$$\|B_Q\|_\infty \leqslant C_{L,G}\varepsilon_1 \,. \tag{D.45}$$

For [6, (172)] with its absolute constant C_1, its small parameter is $\alpha = K_\alpha\varepsilon_0$, with K_α dominating all determining-data, uniqueness-class and B_Q bounds (for the last, $\varepsilon_1 \leqslant (1+\beta_0)\varepsilon_0$ by [9, (2.8)]); then $|e^{itB_Q} - \mathbf{1}| \leqslant |tB_Q| \leqslant \alpha$ and $C_1 = 1$ suffices.

[6, Proposition 9 and (190)] are applied with this α, not with the identically named ε_1 of [9]; for small g_0, $\alpha \leqslant a_1$, $B_3\alpha \leqslant a_0$, and α lies below the thresholds of [6] "for ε_1 sufficiently small" at (172) and (187) (pp.305 and 308); and $\varepsilon_0 \leqslant (1+\beta_0)^2\varepsilon_1$ ([9, (2.8)]) converts the resulting $C\alpha$ bounds into $C'\varepsilon_1$.

D.8.3 Integrating the Local Derivative Estimate

Let $\mathscr{H}_Q(B)$ be the relative Landau-gauge logarithm of the local minimal-configuration map of [6, Proposition 9] on $\mathfrak{B}_Q$ with reference $U_{1,Q}$, normalised by $\mathscr{H}_Q(0) = 0$; this is the map of [9, (3.17)], with its L^{-1} scale factor at $k = 0$.

[9, (3.17)] identifies the map at B_Q with the restriction of U_1 up to the source gauge restoration: both lie on the unique regular critical orbit with determining data D_Q, the former satisfying the constraint up to the gauge transformation of [6, Proposition 2].

For the criticality, a local admissible variation fixes the determining fine layer and the averages on the interior coarse rows; extended by zero outside $Q_w^{[4]}$ it preserves every global average and exterior value in [9, (1.12)], because the fixed layer is wider than the plaquette and averaging range. The first variation of the global action vanishes and its local part is the local action variation; the gauge functional of the source axial mean gauge, in which the variational problem [9, (1.12)] is posed (as recalled in the local-domain verification of Section D.8.1), has zero first variation at an axial mean configuration, where each of its nonnegative summands attains zero. So the restriction of U_1 is critical for the local action, it lies in the uniqueness class of Section D.8.1 by its regularity there (from $U_1 \in D^{(11)}(V, V_0)$), and the uniqueness identifies its orbit with the local analytic solution.

[6, (190), Proposition 9], gives the functional derivative with the mixed-scale weights; at one step the scale labels are 0 and 1, so the powers of L in those weights are fixed constants, and a determining bond e of the outer layer and an interior bond b satisfy the bound $C_{L,G}e^{-c_{L,G}d(b,e)}$, the unit-density measure at the finer level and the scale-one determining measure contributing fixed L-dependent factors.

Here $d(b,e)$ is the ℓ^∞ distance between the initial vertices of b and e on the fine lattice. The distance in [6, (190)] is that of [2, (2.46)] ([6, p. 288]), which charges one unit for each fine bond of a contour in the scale-zero region and one unit for each coarse bond, of fine length L, in the scale-one region, and is evaluated at the block points of the two endpoints; so it is at least $L^{-1}(d(b,e)-L)$, and the rate $\delta_{\mathrm{P}}/8$ of (190), with δ_{P} the decay rate of [4, p. 423] ((H4)), gives $c_{L,G}=\delta_{\mathrm{P}}/(8L)$, the factor $e^{\delta_{\mathrm{P}}/8}$ entering $C_{L,G}$. Integrating along tB_Q, $t\in[0,1]$, and summing the labels,

$$|\mathscr{H}_Q(B_Q;b)|_\sigma \leqslant C_{L,G}\varepsilon_1 \sum_{e\in\operatorname{supp} B_Q} e^{-c_{L,G}d(b,e)} \leqslant C'_{L,G}\varepsilon_1 e^{-c'_{L,G}d(b,\operatorname{supp} B_Q)}\,, \tag{D.46}$$

where for the second inequality each exponential is split into two halves, $e^{-c_{L,G}d(b,e)} \leqslant e^{-c_{L,G}d(b,\operatorname{supp} B_Q)/2}e^{-c_{L,G}d(b,e)/2}$, and the remaining sum over e is bounded by the infinite-lattice sum $\sum_e e^{-c_{L,G}d(b,e)/2}$ over the bonds of $\mathbb{Z}^4$ (four per site) through injective nearest lifts, a constant depending on $c_{L,G}$ only; so $c'_{L,G}=c_{L,G}/2$ and no volume-dependent constant enters. For a bond $b\in(Q_w^{[2]})^*$ (its initial vertex is within one unit of $Q_w^{[2]}$, a difference absorbed in C_L),

$$d(b,\operatorname{supp} B_Q) \geqslant 2w - 2LM_1 - C_L \geqslant LM_2R_1 \tag{D.47}$$

once $w\geqslant LM_2R_1$ (the collar-width convention of Section D.7) and $LM_2R_1\geqslant 2LM_1+C_L$, which holds for small g_0 since $R_1\to\infty$ (Section B.8); the first inequality is the distance from $Q_w^{[2]}$ to the complement of $Q_w^{[4]}$ less the width $2LM_1$ of the boundary layer carrying $\operatorname{supp} B_Q$ and the bond offsets. This holds for either candidate common width w.

D.8.4 Restoring the Source Gauge

The transformation of [6, Proposition 2] is not centre-fixing: its condition is $(\overline{R_0u})^j=\mathbf{1}$ on Λ_j, in particular $u=\mathbf{1}$ on the fine determining layer ([6, p. 281]), and at one step its interior condition $\overline{R_0u}(y)=\mathbf{1}$ does not imply $u(y)=\mathbf{1}$.

The restoring transformation u_Q is that of [9, (3.17)], defined by this source normalisation: the configuration $e^{iL^{-1}\mathscr{H}_Q(B_Q)}U_{1,Q}$ of Proposition 9 is in the relative gauge of Proposition 2 and satisfies the constraint with data D_Q up to that gauge, so it lies on the orbit of the restriction of U_1, and u_Q is the transformation carrying it to U_1, whose values at the block centres need not be the identity and are retained. At the first step, with the $\eta=1$ of [9] and its convention (1.17), on the required bonds

$$U_1(b) = u_Q(b_-)^{-1}e^{iL^{-1}\mathscr{H}_Q(B_Q;b)}U_{1,Q}(b)\,u_Q(b_+)\,. \tag{D.48}$$

The local inputs of the transferred source estimates are the uniqueness and regularity class of Section D.8.1, the small supported B_Q of Section D.8.2 and the response bound (D.46) on both endpoint blocks and the crossing bond.

Let $\rho_{\mathrm{av}}>0$ be an analytic mean neighbourhood radius for the finite source contour families, with derivative bounds on its closed half-radius neighbourhood; the coupling g_0 is chosen after it so that the reference path-family diameter and its perturbation are both less than $\rho_{\mathrm{av}}/4$ (the regular-chart reduction).

The block-local gauge inequality (D.49) below is the transferred source result; the following path description gives its one-step support and its constants. In [3] the equations determining u_Q are (77)–(87), with its expression in the relative fields given in (106)–(108): at one step the product in (106) has a single $j = 0$ factor, a reference-unitary conjugation of the inverse block average followed by a local relative path transport, and the centre factor (87) is retained. Unitary conjugations preserve the norm, each path has length at most $(2d+1)L$, so the telescoping product identity bounds the relative deviation by C_L times the largest relative-link deviation in the block, and the analytic averaging bound of (161)–(163) bounds the block-average deviation by the same quantity after reducing the chart. The transferred result reads

$$\sup_{x\in\mathcal{B}_y} |u_Q(x) - \mathbf{1}|_\sigma \leqslant C_{L,G} \sup_{b\in E(\mathcal{B}_y)} |\mathscr{H}_Q(B_Q; b)|_\sigma \,, \tag{D.49}$$

the factor L^{-1} of [9, (3.17)] absorbed into the constant.

The restoring map is relative to the reference pair: under a simultaneous gauge change h of field and reference its representative is $u_Q^h(x) = h(x)u_Q(x)h(x)^{-1}$ (by conjugation of the relative action, [6, (16)–(17)], and of its reference-dependent gauge condition), so $|u_Q(x)-\mathbf{1}|_\sigma$ is a conjugation-invariant scalar of the relative construction; the centre contribution $u_Q(y)$ is retained.

[3] uses its older averaging definition; the application here rests on Bałaban's extension statements, [8] after (0.10) (the considerations and results of [8], and those of the earlier papers, hold for every group mean with the properties (0.5)–(0.9)) and [8] after (0.12) (the average (0.12) has properties similar to those of (0.4), all results of [3] hold for it, with proofs mostly unchanged and otherwise with minor modifications, and both definitions are equally good for the purposes of [8]; the coverage of (0.4) rests on this last remark). On these statements (D.49) is the transfer of the results of [3] to the averaging and mean gauge of [8]; [9, (3.17)–(3.19)] uses the transferred results in the same way for the later scheme, as does [8] for its gauge conversions ((3.26)–(3.27) with [3, (77)], and (3.39) with [3, (97)], retaining the centre-block gauge factors; reference [12] of [8] is [3]).

For the bond b of (D.43), (D.46) applies on both endpoint blocks, on b itself and on the paths entering the gauge factors, which enlarges the input set by at most $2L + 2$, already included in the C_L of (D.47). The relative exponential and both endpoint factors of (D.48) then give $|U_1(b)U_{1,Q}(b)^{-1} - \mathbf{1}|_\sigma \leqslant C_{L,G}\varepsilon_1 e^{-c_{L,G}LM_2R_1}$; both endpoint gauge factors of [9, (3.18)] are retained.

D.8.5 Conclusion of the Proof

[9, (2.6), (2.8)] bound ε_1/δ_0 by fixed source constants and R_1 grows as a power of $\log g_1^{-2}$, so the bound is $< \delta_0/10$ for small g_0, which completes the proof of Proposition D.20.

D.8.6 The Boundary Field of Bałaban's Equation (1.21)

The same argument bounds the field $\mathbf{H}_{1,Ax}$ of [9, (1.21)] on the bonds of S_1^*, where it enters the change of variables (1.22) through Lemma 3.16 and (3.40).

Lemma D.21 (The Boundary Field on S_1)**.** *There are constants C_{H}, $c_{\mathrm{H}} > 0$, depending on L, G, M_1, M_2, and the source constants and fixed before g_0, independent of the fluctuation, its R_1 label, the exterior history, and the torus, such that, for g_0 below a threshold depending on the same quantities (the conditions of Sections D.8.1, D.8.2, and D.8.4 and those named*

in the proof), for every admissible (Ω_1, V, V_0)*, every large-fluctuation region* R_1 *of the history sum, and every* $b \in S_1^*$*, with* $U_1' := U_1(\Omega_1 \cap \Lambda_1^c, V)$ *the representative fixed in (B1),*

$$|U_1(b)U_1'(b)^{-1} - \mathbf{1}|_\sigma \leqslant C_{\mathrm{H}}\varepsilon_1 e^{-c_{\mathrm{H}}LMR_1} \,. \tag{D.50}$$

Consequently $U_1(b)U_1'(b)^{-1}$ *lies in the chart domain,* $\mathbf{H}_{1,Ax}(b) = L\log(U_1(b)U_1'(b)^{-1})$ *satisfies* $|\mathbf{H}_{1,Ax}(b)|_\sigma \leqslant \frac{\pi}{2}LC_{\mathrm{H}}\varepsilon_1 e^{-c_{\mathrm{H}}LMR_1}$*, and* (3.40) *holds once* $\frac{\pi}{2}LC_{\mathrm{H}}(1+\beta_0)(A_0/A_1)e^{-c_{\mathrm{H}}LMR_1} \leqslant \frac{1}{40}$ *(item 5 of Section B.8).*

Proof. This is the proof of Proposition D.20, Sections D.8.1–D.8.4, for the pair (U_1, U_1') in place of $(U_1, U_{1,Q})$, without the cube localisation: the reference problem is that of [9], so no lift and no condition on the periods enters. The norm $|\bullet|_\sigma$ is the cutoff norm of (3.40).

The problem and the argument. U_1' is the minimal configuration for the determining data named at p.251 ((B1)): the conditions of (1.12) on the rows of $(\Lambda_1^\sim)^c$, and on $\Lambda_1^\sim \setminus \Lambda_1$ the determining set $\mathbf{B}_1(\Lambda_1^c)$ with the data $D^0 = M^\bullet(Q_1^{s*}V)$. The configuration, and with it the first equality of (1.21), lives on $\Lambda_1^c = (\Lambda_1^\sim)^c \cup (\Lambda_1^\sim \setminus \Lambda_1)$, the union of the two domains on which p.251 determines it, and Λ_1 carries no variable of the problem; every plaquette with a corner in Λ_1 has its bonds in Λ_1 or in the fixed layer, which is at least LM_1 wide, so these terms are constants of the problem and are omitted, as are those off Ω^+.

The hierarchy for [6] is that of Section D.8.1 with Λ_1 removed from its domains: V_0 is fixed on the collar $\Omega^+ \setminus \Omega_1$, $\Omega^+ = \Omega_1^{\sim 1}$, and the action terms off Ω^+ are omitted as constants; the scale-zero determining set is the collar together with the fine layer of $\mathbf{B}_1(\Lambda_1^c)$, and the scale-one determining set is the set of coarse bonds of $\Omega_1 \cap \Lambda_1^c$ off that layer. The data are regular as there: the collar lies in the P_0-good region, the plaquettes of $Q_1^{s*}V$ on the layer are trivial or coarse plaquette holonomies of V, bounded by $2L^2\varepsilon_0$, and the mixed determining plaquettes are bounded by [3, Proposition 1, (51)].

That the results of [6] hold for this determining set, whose fine layer has width between LM_1 and $2LM_1$ (p.256), in place of the spacing condition [6, (1)] for the layer, is Bałaban's application of [6] to the determining sets of [9] (p.256, and p.251 for this problem), taken over as such, as for $\mathbf{B}_1(Q_w^{[4]})$ in Section D.8.1.

By the first equality of (1.21), $U_1 = U_1(\Omega_1 \cap \Lambda_1^c, M^\bullet(U_1))$, U_1 lies on the minimal orbit of the same problem with the data $D = M^\bullet(U_1)$ (the criticality and uniqueness argument of Section D.8.3 gives the same identification). [9, p. 251] applies [6, Proposition 9] to this problem ("The dependence is analytic, and the function $\mathbf{H}_{1,Ax}$ has the exponential decay property. It was formulated in (190) Sect. G of [15] for the Landau gauge"); let $\mathscr{H}$ be its relative Landau-gauge function with reference U_1', normalised by $\mathscr{H}(0) = 0$, and let B be the chart logarithm of $D(D^0)^{-1}$ (the $\frac{1}{i}\log$ of [9]), the argument of $\mathbf{H}_{1,Ax}$ in (1.21).

Support and size of B*.* On the rows of $(\Lambda_1^\sim)^c$ the two data coincide, the conditions there being those of (1.12) for both. On the coarse bonds of $\mathbf{B}_1(\Lambda_1^c)$ in $\Lambda_1^\sim \setminus \Lambda_1$, the only ones entering (p.251), both data equal V: these bonds lie in Ω_1, where $\bar{U}_1 = V$, and $M(Q_1^{s*}V) = V$ for the straight extension (Section D.8.2). So B vanishes off the fine layer of $\mathbf{B}_1(\Lambda_1^c)$, on which D and D^0 are the bond variables of U_1 and of $Q_1^{s*}V$ ([9, (2.11)] at the first step).

That layer lies within ℓ^∞ distance $2LM_1$ of Λ_1 ([9, p. 256] for the domain Λ_1^c: the coarse part of $\mathbf{B}_1(\Omega)$ contains $\Omega^{\sim -2}$, the enlargement taken for M_1-cubes, of side LM_1 in fine units); its width is between one and two M_1-cubes (p.256), and [9, p. 251] describes it as "one layer of M_1-cubes touching $\partial\Lambda_1$ and contained in Λ_1^c". Since Ω_1 and R_1' are unions of LMR_1-cubes and $\Lambda_1 \subseteq \Omega_1^{\sim -2} \cap (R_1'^\sim)^c$, every site within ℓ^∞ distance $LMR_1 - 1$ of Λ_1 lies in $\Omega_1^{\sim -1} \setminus R_1'$; for small g_0 ($2LM_1 + 1 < LMR_1 - 1$) the layer therefore lies in $\Omega_1^{\sim -1}$, where U_1 is regular

and in the source axial mean gauge ((B1)(i)), and the two-block comparison of Section D.8.2 gives $\|B\|_\infty \leqslant C_{L,G}\varepsilon_1$. [6, (172)] is met as there, with $\alpha = K_\alpha\varepsilon_0$, K_α dominating also this bound, and $C_1 = 1$.

Distance. Every site of $S_1 = (\Omega_1 \setminus \Omega_1^{\sim -1}) \cup R_1$ is thus at ℓ^∞ distance at least LMR_1 from Λ_1 ($R_1 \subseteq R_1'$), and the filling modification after (1.20) only removes cubes from Λ_1. For $b \in S_1^*$ (it has an endpoint in S_1) and b' equal to b or a bond of one of its two endpoint blocks, the initial vertex of b' is within $2L+1$ of a site of S_1, and the initial vertex of a bond of the layer is within $2LM_1 + 1$ of Λ_1; so, with d the ℓ^∞ distance between initial vertices of Section D.8.3,

$$d(b', \operatorname{supp} B) \geqslant LMR_1 - 2LM_1 - 2L - 2 \geqslant \tfrac{1}{2}LMR_1 \tag{D.51}$$

once $LMR_1 \geqslant 4LM_1 + 4L + 4$, which holds for small g_0 since $R_1 \to \infty$.

The Landau-gauge function. [6, (190)], integrated along tB, $t \in [0,1]$, and summed over $\operatorname{supp} B$ as in (D.46), gives $|\mathscr{H}(B;b')|_\sigma \leqslant C'_{L,G}\varepsilon_1 e^{-c'_{L,G}d(b',\operatorname{supp} B)}$ for every bond b' of the problem, with the constants of (D.46); the rate $c'_{L,G} = \delta_{\mathrm{P}}/(16L)$ is fixed before g_0, coming from the rate $\delta_{\mathrm{P}}/8$ of [6, (190)] in the distance [2, (2.46)], converted to the fine ℓ^∞ distance in Section D.8.3, with δ_{P} the decay rate defined in [4, p. 423] after Theorem 3.12, a constant of d and L.

[6] states (190), and the representation (189) behind it, without the calculation (p.308, "We do not perform these calculations here"); (190) is Bałaban's statement, taken over as stated.

Restoring the axial gauge. As in Section D.8.4, the configuration $e^{iL^{-1}\mathscr{H}(B)}U_1'$ of [6, Proposition 9] lies on the orbit of U_1, and the restoring transformation u carrying it to U_1, with its centre values retained, gives (D.48) with $(U_1', \mathscr{H})$ in place of $(U_{1,Q}, \mathscr{H}_Q)$ and satisfies the block-local bound (D.49), $\sup_{x\in\mathcal{B}_y}|u(x)-\mathbf{1}|_\sigma \leqslant C_{L,G}\sup_{b'\in E(\mathcal{B}_y)}|\mathscr{H}(B;b')|_\sigma$: at one step the transformation on a block is fixed by the relative transports along the contours inside that block and by its centre factor ([3, (77)–(87) and (106)–(108)], as recorded there), which is the locality of the axial gauge on the unit lattice; (D.49) is the accepted transfer of Section D.8.4, applied to the function of (1.21) as [9] applies (3.17)–(3.18) to the corresponding function at the general step, "with similar bounds" (p.269).

The target is U_1 itself: the minimal configurations with the data D form one orbit, in the class of Section D.8.1, of the gauge transformations equal to the identity on the set $\mathfrak{B}_k$ of [6] ([9, p. 256]; [6, (4) and Theorem 1, pp. 278–279]), in which the axial gauge determines a unique element ([8, (0.6) and (0.11)], as recorded in (B1)), and U_1 is in the axial gauge ([9, p. 248]), so it is that element.

At an endpoint off Ω_1, a point of the fixed exterior field V_0 (scale-zero data, as in the hierarchy of Section D.8.1), $u = \mathbf{1}$ ([6, p. 281] for the transformation of [6, Proposition 2], and [6, (4)] for the orbit element, the collar being part of $\mathfrak{B}_k$). For $b \in S_1^*$, unitary invariance and conjugation give

$$\begin{aligned}|U_1(b)U_1'(b)^{-1} - \mathbf{1}|_\sigma &\leqslant |u(b_-) - \mathbf{1}|_\sigma + L^{-1}|\mathscr{H}(B;b)|_\sigma + |u(b_+) - \mathbf{1}|_\sigma \\ &\leqslant (2C_{L,G}+1)C'_{L,G}\varepsilon_1 e^{-c'_{L,G}LMR_1/2}\end{aligned}$$

by (D.51), which is (D.50) with $C_{\mathrm{H}} \coloneqq (2C_{L,G}+1)C'_{L,G}$ and $c_{\mathrm{H}} = c'_{L,G}/2$.

The consequences. For small g_0 the right side of (D.50) is below $r_G/\sqrt{N}$, so $U_1(b)U_1'(b)^{-1}$ lies in the chart domain (the norm facts at the start of the proof of Lemma 3.16), and its chart logarithm $X = L^{-1}\mathbf{H}_{1,Ax}(b)$ satisfies $|X|_\sigma \leqslant \frac{\pi}{2}|e^{iX} - \mathbf{1}|_\sigma$, which is the bound on $\mathbf{H}_{1,Ax}(b)$.

With $\varepsilon_1/\delta_0 \leqslant (1+\beta_0)A_0/A_1$ ([9, (2.8)] and $\delta_0 = (A_1/A_0)\varepsilon_0$), it is at most $\delta_0/40$ under the stated condition, which holds for small g_0 at fixed L, M, and M_1 since $R_1 \to \infty$. *q.e.d.*

Appendix E The Signed Integrated Wilson Defects

This appendix bounds the defect term of the branch identity (3.97) at the carrier I of Section 3, under the original history law: the joint defects $R_\xi(\mathfrak{B})$ of (3.95) are summed with their restricted reference sums before absolute values are taken, each closed group is compared with its flat value by one background derivative along a local coefficient path, and the comparison is paid by the Gaussian reserve, the positive host comparison and the positive group summation of Sections E.4, E.6 and E.7. The result is Proposition E.26: on compatible even fine periods and for the square geometry of Section 5.3,

$$|E_{\mathrm{def}}| \leqslant C'_L n_C e^{-(c_*/2)\pi_*^2} + e^{C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})} - 1 \,,$$

with $\rho = O(g_0^2(1+r)^4)$, $\varepsilon_1 = g_1 p_0(g_1)$, $b = \lfloor LMR_1/16 \rfloor$, $\pi_* = \min(p_0(g_0), p_0(g_1))$, $C'_L = 8C_L$ with the contour constant C_L of Section D.7 and the tail exponent c_* of Proposition D.18, and C_χ, c fixed before the coupling.

The regional scalar $S^{\circ}_{\xi,C}$ remains inside the expectation; its identification with the perimeter/corner coefficient is Appendix F.

E.1 Operand and Retained Normalisation

The groups W and the compatible families $\mathfrak{B}$ are those of Definition 3.32; the closed pairings $A_\xi(\mathcal{S})$, the scalar references a_W, the restricted reference sums $S^{\circ,[\mathfrak{B}]}_{\xi,C}$ and the joint defects $R_\xi(\mathfrak{B})$ are those of (3.90), (3.92) and Proposition 3.35. Every source term, the Wilson contraction f_ξ and each group's closed contraction of Definition 3.32, is a gauge-invariant closed contraction (Section E.10.2). The signed family sum and the defect term are

$$\mathscr{R}_\xi \coloneqq \sum_{\substack{\mathfrak{B}\neq\varnothing \\ \mathfrak{B}\ \text{compatible}}} S^{\circ,[\mathfrak{B}]}_{\xi,C} R_\xi(\mathfrak{B}), \qquad E_{\mathrm{def}} \coloneqq \frac{1}{\mathcal{Z}_0}\int \mathcal{O}[\mathsf{w}_\xi \mathbf{1}_{A_{\mathrm{cov}}} \mathscr{R}_\xi](V)\,\mathrm{d}V \,. \tag{E.1}$$

The weight is the branch weight $\mathsf{w}_\xi = e^{c_\xi} I_{\xi;1}$ of (3.49), the outer operation $\mathcal{O}$ of (3.4) contains the original cutoffs, ζ_ξ with its boundary-fluctuation dependence, the gauge and Haar factors, the Jacobians, the vacuum prefactor and the deferred integrations, $\mathcal{Z}_0 = \int \mathcal{O}[\mathsf{w}_\xi]\,\mathrm{d}V$, and $\mathbb{E}_{\mathrm{h}}$, $\mathbb{P}_{\mathrm{h}}$ are the history expectation and probability (D.20) for this positive weight; A_{cov} is the covered event of Lemma 3.36 (every contour bond is a bond of Ω_1 and every free input of the contour lies in I).

The weight is positive: $K_{\xi;1}$ is the exponential of a real function times indicators (Section 3.4) and c_ξ is real, the logarithm of the positive a-independent factor collected in the proof of (3.49).

At fixed finite volume the signed sum (E.1) converges absolutely: the family sum (3.91) converges absolutely, and the proof of Proposition 3.35 obtains (E.1) from it by Boolean inversion over finitely many root families, each with an absolutely convergent cluster-list sum.

E.2 Histories near the Bad Region

Let $B_\xi \coloneqq \Lambda_1^c$ be the filled bad region and $E(B_\xi)$ the set of positive fine bonds with at least one endpoint in B_ξ. Let Γ_C be the deterministic set of contour bonds and free-input templates

of Section D.7, with $|\Gamma_C| \leqslant C_L n_C$, n_C the number of contour traversals; distances are those of the periodic fine-bond metric d_T of (5.2), $d_T(b,c) = d_{\infty,T}(x,y) + \mathbf{1}\{i \neq j\}$ for $b = (x,i)$, $c = (y,j)$. For an integer $b \geqslant 0$ define

$$\mathsf{N}_b := \{\xi \mid \operatorname{dist}(\Gamma_C, E(B_\xi)) \leqslant b\}\,, \tag{E.2}$$

with distance to the empty set infinite. This is a measurable history event (Λ_1 is a function of the history labels).

The near and far parts are $E^{\text{near}}_{\text{def}} = \mathbb{E}_{\text{h}}[\mathbf{1}_{A_{\text{cov}}}\mathbf{1}_{\mathsf{N}_b}\mathscr{R}_\xi]$ and $E^{\text{far}}_{\text{def}} = \mathbb{E}_{\text{h}}[\mathbf{1}_{A_{\text{cov}}}\mathbf{1}_{\mathsf{N}_b^c}\mathscr{R}_\xi]$, and $E_{\text{def}} = E^{\text{near}}_{\text{def}} + E^{\text{far}}_{\text{def}}$ (both integrable, by the bound $|\mathscr{R}_\xi| \leqslant 2$ of Lemma E.1).

Lemma E.1 (Near Histories)**.** *Let $b \geqslant 0$ be an integer and let q_{pt} be the one-point bound of Lemma D.17 ($\mathbb{P}_{\text{h}}(z \in E(B_\xi)) = p_z \leqslant q_{\text{pt}}$ for every bond z, on compatible even fine periods and for $g_0 \leqslant g_H$), with $q_{\text{pt}} \leqslant e^{-c_*\pi_*^2}$ for $0 < g_0 < g_*$ by Proposition D.18. Then $|\mathscr{R}_\xi| \leqslant 2$ pointwise on every branch, and*

$$\mathbb{P}_{\text{h}}(\mathsf{N}_b) \leqslant 4C_L n_C (2b+3)^4 q_{\text{pt}}, \qquad |E^{\text{near}}_{\text{def}}| \leqslant 8C_L n_C (2b+3)^4 q_{\text{pt}} \leqslant C'_L n_C e^{-(c_*/2)\pi_*^2}\,, \tag{E.3}$$

the last inequality for $b = \lfloor LMR_1/16 \rfloor$, $C'_L := 8C_L$ and g_0 below a threshold depending on L, M_1, M_2, M, the group, the character and the source constants.

Proof. By (3.96), $\mathscr{R}_\xi = F_\xi(0) - S^\circ_{\xi,C} f_\xi(0)$. Both $F_\xi(0) = \int f_\xi \,\mathrm{d}\nu_{\xi;0}$ ((3.46)) and $S^\circ_{\xi,C}$ ((3.94)) are normalised integrals of a character of modulus at most one against the real positive kernels $K_{\xi;0}$ (the exponential of a real function times indicators, Section 3.4) and $K^\circ_{\xi;0}$ (Lemma B.16), and $|f_\xi(0)| \leqslant 1$; hence $|\mathscr{R}_\xi| \leqslant 2$.

The members of Γ_C are bonds (contour bonds and the bonds of the free-input templates), and there are at most $4(2b+3)^4$ positive fine-bond labels within d_T-distance b of a fixed fine bond (an overcount on short periods), so a union bound over Γ_C and the one-point estimate $p_z \leqslant q_{\text{pt}}$ of Lemma D.17 (p_z of (D.23)) give the first bound; positivity of w_ξ and $\mathcal{O}$ and $|\mathscr{R}_\xi| \leqslant 2$ give the second.

For the last: $b = \lfloor LMR_1/16 \rfloor$ exceeds every fixed source stencil for small g_0, and $\log(2b+3) = O(\log\log g_0^{-2})$ since $\ell = LMR_1 \leqslant L^2M(2\log g_0^{-2})^{\mathfrak{r}}$ ((9.2)), while $\pi_* = \min(p_0(g_0), p_0(g_1)) \geqslant A_0(\frac{1}{2}\log g_0^{-2})^{p_0}$ with $p_0 > 0$, by $p_0(g) = A_0(\log g^{-2})^{p_0}$ ((D.1)) and $\log g_1^{-2} \geqslant \frac{1}{2}\log g_0^{-2}$ (as at (9.2)); so $4\log(2b+3) \leqslant (c_*/2)\pi_*^2$, that is $8C_L(2b+3)^4 e^{-c_*\pi_*^2} \leqslant C'_L e^{-(c_*/2)\pi_*^2}$ with $C'_L := 8C_L$, for small g_0 at fixed source constants.

The constants are uniform in the compatible even-period tori and the histories. *q.e.d.*

E.3 The Interior Operand

The remaining operand is $E^{\text{far}}_{\text{def}}$. The Wilson contraction of a group W depends on the background through its random-field support and through the transports along its contour interval $\mathcal{H}(W)$. The compared word of a group is the product of these transports, in their order, closed back to the base of the interval (Remark E.6, Lemma E.23).

Section E.4 bounds the background derivative of the closed source-containing selected integrals and of their joint vacuum ratio; Sections E.5, E.7 and E.8 sum the groups; Sections E.6 and E.9 supply the derivative bounds.

E.4 The Joint Signed Selected-Integral Estimate

Let $\mathcal{A} \coloneqq \{D_1, \dots, D_m\}$ be a compatible root family. For each D_j the selected measures are $\Lambda^{\mathrm{f}}_{\xi;\ell_j}$ of (3.81) with $\ell_j \coloneqq (\mathcal{D}_j, \mathsf{J}_j, \mathsf{P}_j, Z_j)$, $Z_j = D_j$ and $\mathsf{J}_j \neq \varnothing$, and $q_j = |\mathsf{J}_j|$, $q = \sum_j q_j$.

The centred measure $\widehat{\nu}^{\mathrm{f}}_{\xi,D} = \nu^{\mathrm{f}}_{\xi,D} - k^{\mathrm{f}}_{\xi}(D)\delta_{0,\mathsf{L}_D}$ of (3.83) is the sum of the nonempty-probe selections of (3.81): the empty-probe selections give the mass times δ_0 and disappear under centring (Lemma 3.29).

The finite-difference operator on a bounded function f of the free variables is

$$\Delta_{\mathsf{J}} f(B) \coloneqq \sum_{K \subseteq \mathsf{J}} (-1)^{|\mathsf{J}\setminus K|} f(\mathbf{1}_K B)\,, \tag{E.4}$$

every unselected free label being zero; for several roots $\Delta_{\mathbf{J}} f$ is the composition of these operators on their disjoint label sets.

The integrand is a sum of closed Wilson contractions; the source may carry a fixed complementary closed holonomy $h \in G$, transforming by conjugation at the base, and all bounds are uniform in h. With (3.84),

$$\mathfrak{Q}_{\xi}(\mathcal{A}; f) = \sum_{(\ell_j)_{j\leqslant m} \,|\, Z_j = D_j,\ \mathsf{J}_j \neq \varnothing} \int \Delta_{\mathbf{J}} f(B)\, \mathrm{d} \bigotimes_{j=1}^{m} \Lambda^{\mathrm{f}}_{\xi;\ell_j}(B_j)\,, \tag{E.5}$$

the integral of each Dirac tensor $\bigotimes_{b\in\mathsf{J}}(\delta_{B_b} - \delta_0)$ against f being the finite difference (E.4).

E.4.1 The Powers Supplied by the Signed Source

Let $u = C_\chi g_0(1+r)$, the fixed row/character constant C_χ being increased once (Section E.9.4). On the real protected box the row reconstruction depends on $g_0 B$ and has analytic derivative bounds on a larger fixed phase chart (Lemma 3.12, Lemma B.8).

Each selected free label affects a bounded number of reconstructed contour links, each link has a bounded input set, and for the square the traversal multiplicity is bounded; applying the product rule inside the closed character therefore gives C_χ^q derivative assignments.

Derivative orders on any one row are bounded by twice its fixed input count plus one, including the remainders below (Section E.9.4); the higher row derivatives are obtained from the bounded analytic phase map on its larger fixed ball by Cauchy estimates (Lemma B.8(i)–(ii): $\widetilde{D}_{U,c}$ is analytic on the ball of radius ϑ_* with $\|\widetilde{D}_{U,c}(A)\| \leqslant 2K_*\|A\|^2$ and $\|h_U(c)\| \leqslant H_*$, so Cauchy's estimates on the ball of radius $\vartheta_*/2$ bound every order), and the maximum order used is fixed by L. All untouched transports are unitary matrices inside the same closed contraction.

For the closed character f of the source (Lemma E.3 below) the mixed integral remainder formula (E.64) gives

$$\sup |\Delta_{\mathbf{J}} f| \leqslant u^q\,, \tag{E.6}$$

the supremum over the real protected box, which contains the support of every selected measure in the selected variables (the measure (3.81) carries the cutoffs χ_b over $E(Y_0)$ in the variables it integrates; its bad cutoffs $1-\chi_b$ act on the labels of P, which are integrated out, not on the selected probes J).

For a root with odd q_j take the homogeneous term of degree q_j in its selected variables B_{D_j}, the other root variables fixed; it is odd under $B_{D_j} \to -B_{D_j}$, and the difference from

it carries an additional factor u ((E.66); the choice of the extra increment is bounded by $q_j \leqslant 2^{q_j}$, absorbed in C_χ). The same holds simultaneously for any set of odd roots, the product-rule allocation count remaining exponential in q. Every term is a closed contraction.

Each selected Gaussian kernel (3.66) with its cutoffs is invariant under $(B_{D_j}, X_{D_j}) \to (-B_{D_j}, -X_{D_j})$ once its ordinary Mayer factors are removed: the determinant, the quadratic cross term, the compensator, the real cutoffs χ_b, $1-\chi_b$ and the spatial Cauchy integrations are even. The tensor product in (E.5) gives separate integration variables and a separate reversal for each root. Thus the leading odd part integrates to zero in the signed selected integral, also when several roots are present.

Let $\omega(g_0) \coloneqq C_0^\partial g_0(1+r)^3$ with the host derivative constant $C_0^\partial \geqslant C_0$ of Section E.6.1, let ω_* be the fixed host budget of (B.28), and let $R_\Upsilon \coloneqq \omega_*/(2\omega(g_0))$, with g_0 so small that $R_\Upsilon \geqslant 2$.

In root j the selected ordinary potentials $\mathbf{V}_\xi(Y)$, $Y \in \mathcal{D}_j$, are multiplied by an independent complex parameter ζ_j, $|\zeta_j| \leqslant R_\Upsilon$, the Gaussian kernel, cutoffs and probes being unchanged. If $\mathcal{D}_j \neq \varnothing$ the selected integral vanishes at $\zeta_j = 0$ (each Mayer factor $\int_0^1 \zeta \mathbf{V} e^{t\zeta\mathbf{V}} \mathrm{d}t = e^{\zeta\mathbf{V}} - 1$ does), so the Cauchy difference between $\zeta_j = 0$ and $\zeta_j = 1$ of the analytic function of ζ_j costs at most $2/(R_\Upsilon - 1)$ times its supremum on the disc; if $\mathcal{D}_j = \varnothing$ the density is the even Gaussian/cutoff kernel and the reversal applies. Independent parameters on disjoint roots fit the same local potential budget simultaneously (Section E.6.1). Therefore an odd q_j costs

$$v = u + \frac{2}{R_\Upsilon - 1} \,. \tag{E.7}$$

Both u and v tend to zero as $g_0 \to 0$, since $g_0(1+r)^3 \to 0$ by (D.1). (ω as a background direction, with $\|\omega\|_\infty$ and $\mathscr{D}_\omega$, and $\omega(g_0)$, ω_* as host bounds are distinguished by their arguments.)

E.4.2 Numerical Majorants Paid with the Same Gaussian Reserve

For a selected root ℓ define

$$M_\ell \coloneqq \sup_{|\zeta| \leqslant R_\Upsilon} \|\Lambda^{\mathrm{f}}_{\xi;\ell,\zeta}\|_{\mathrm{TV}} \,, \tag{E.8}$$

where $\Lambda^{\mathrm{f}}_{\xi;\ell,\zeta}$ is the measure (3.81) with each selected ordinary potential replaced by ζ times that potential; the cutoffs, Gaussian density and normalisers, compensator, integration carriers, weakening contours and seed data are unchanged.

Lemma E.2 (Dummy Comparison)**.** *There are at most $P_M = 4(4M+3)^4$ probe labels with a given host (a label b lies in the interior of its host Y_b, a rectangle of at most four M-cubes per side, (3.75)), and each probe host has $N_0(Y_b) \leqslant 4^4 \leqslant s_{\mathrm{p}} = 5^4$ cubes ((3.75)). Let $R_\Upsilon \geqslant 2$ and let g_0 be below the thresholds of Lemmas 3.27 and B.15, and let*

$$u_* \coloneqq \min\Big(\frac{1}{4},\ \frac{\omega_* e^{-\kappa s_{\mathrm{p}}}}{8P_M}\Big) \,, \tag{E.9}$$

a constant fixed after M and before g_0. Then for every connected final domain D, with $m(D) = \varepsilon_* e^{-\lambda_* \ell_1(D)}$ the majorant of Lemma 3.27,

$$\sum_{\ell \,|\, Z(\ell) = D,\ \mathsf{J}(\ell) \neq \varnothing} u_*^{q(\ell)} M_\ell \leqslant m(D) \,. \tag{E.10}$$

Proof. For every probe label b the numerical comparison inserts the dummy potential $p_b^+ = u_*$ on its protected host box and zero outside; its selected Mayer factor $e^{u_*} - 1$ is at least u_*. The summed dummy potential at a host is at most $P_M u_* \leqslant \omega_* e^{-\kappa s_{\mathrm{p}}}/8 \leqslant \omega_* e^{-\kappa d_0(Y)}/8$, the input B_Y of (E.32) (for a probe host $d_0(Y) \leqslant N_0(Y) - 1 < s_{\mathrm{p}}$: its cubes form a rectangular array, and a path through the centres of wall-adjacent cubes visiting each once meets every cube and has length $N_0(Y) - 1$ in (B.1)), while the complex ordinary potentials use at most half the budget $\omega_* e^{-\kappa d_0(Y)}$ on $|\zeta| \leqslant R_\Upsilon$ (Lemma B.13 and $R_\Upsilon \omega(g_0) = \omega_*/2$).

The labels with a common host are split as in the proof of Lemma 3.29; the seeds, conditioning carrier and Gaussian kernel depend on the active hosts only. The split selections with the same active-host set are summed first: Lemma E.14 proves that their many-to-one positive sum is dominated hostwise by the undifferentiated majorant (B.31). That majorant pays its determinant and compensator with the Gaussian reserve α_G (Lemma 3.24(ii)), and the cover and seed sums (B.34), (B.35) with the final Y'-sum in the proof of Lemma 3.27 (Appendix B) sum the seeds and final domains to $m(D)$. *q.e.d.*

Lemma E.3 (Joint Selected-Integral Bound)**.** *Let f be a normalised character of an ordered closed reconstructed word, optionally closed with its inverse background interval and a fixed complementary h (Section E.6.2). For g_0 small enough that $u \leqslant u_*$ and $v \leqslant 1$, put*

$$\rho \coloneqq \Big(\frac{u}{u_*}\Big)^2 + \frac{u}{u_*}\, v \,. \tag{E.11}$$

Then for every compatible root family $\mathcal{A}$ and every choice of selections (ℓ_j) with $\mathsf{J}_j \neq \varnothing$,

$$\Big|\int \Delta_{\mathsf{J}} f(B)\, \mathrm{d} \bigotimes_{j=1}^{m} \Lambda^{\mathrm{f}}_{\xi;\ell_j}(B_j)\Big| \leqslant \prod_{j=1}^{m} u^{q_j} v^{q_j \bmod 2} M_{\ell_j}, \qquad |\mathfrak{Q}_\xi(\mathcal{A}; f)| \leqslant \prod_{D \in \mathcal{A}} \rho\, m(D) \,. \tag{E.12}$$

Proof. The first bound combines the reversal, the Cauchy differences and (E.6)–(E.7) (Lemma E.15): the roots with odd q_j and empty ordinary selection lose their leading odd part to the even kernel and keep the remainder u^{q_j+1}, the roots with odd q_j and nonempty ordinary selection pay $u^{q_j} \cdot 2/(R_\Upsilon - 1)$ by the Cauchy difference in ζ_j (the supremum on the disc being $u^{q_j} M_{\ell_j}$ by (E.8)), the roots with even q_j pay $u^{q_j} M_{\ell_j}$.

For every nonempty probe set $(u/u_*)^{q_j} v^{q_j \bmod 2} \leqslant \rho$ ($q_j = 1$ gives $(u/u_*)v$, $q_j \geqslant 2$ gives $(u/u_*)^2$). Summing over the selections of every root with (E.5) and (E.10) gives the second bound. *q.e.d.*

By (E.11), $\rho = O(g_0^2(1+r)^4)$ with constants fixed before the coupling.

E.4.3 Background Derivative and the Joint Vacuum Ratio

For the flat subtraction, ω is a real background direction along one of the globally small analytic extensions of Section E.8.1, with the geometry, carriers, cutoff norm, means, Gaussian reserve and walk recipe unchanged. Let L_f be the number of occurrences of contour background links in the closed word whose local coefficient arguments vary; when a group's relative interval is compared this includes the interval and its inverse closing transport. Complementary closed holonomies, outer history variables and physical weights are held fixed, and $\mathscr{D}_\omega$ denotes this derivative.

Lemma E.4 (One Background Derivative)**.** *On the extensions of Section E.8.1, with ρ, $m(D)$ of Lemma E.3 and a constant C fixed before the coupling,*

$$|\mathscr{D}_\omega \mathfrak{Q}_\xi(\mathcal{A}; f)| \leqslant C\|\omega\|_\infty \Big(L_f + \sum_{D\in\mathcal{A}} N_1(D)\Big) \prod_{D\in\mathcal{A}} \rho\, m(D)\,. \tag{E.13}$$

Proof. The derivative of the selected Gaussian factors (density, normaliser, compensator) inserts a scalar polynomial of degree at most two in (B, X), including the determinant trace; its moment is paid by the joint coercive Gaussian bound of Section E.9.3 at cost $CN_1(D)\|\omega\|_\infty$. Differentiated walk coefficients sum in the convergent differentiated series of Section E.9.2.

The derivative of a host Mayer t-integrand is $\zeta(\mathscr{D}_\omega \mathbf{V})e^{t\zeta\mathbf{V}}(1 + t\zeta\mathbf{V})$, whose t-integral $\zeta(\mathscr{D}_\omega \mathbf{V})e^{\zeta\mathbf{V}}$ vanishes at $\zeta = 0$; its size is bounded by the differentiated host comparison of Lemma E.14.

A Gaussian derivative is even under each component reversal and the leading source derivative has the parity of the undifferentiated term ((E.67)), so the signed powers of Section E.4.1 survive one derivative, with the additional factor $C\|\omega\|_\infty(L_f + \sum_D N_1(D))$: the source derivative contributes L_f plus the number of selected labels, the latter at most $4(LM + 1)^4 \sum_D N_1(D)$ (Section E.6.1), and the host, Gaussian and walk derivatives contribute one factor $N_1(D)$ each by Lemma E.14. The factor $N_1(D)$ is paid by $m(D)$ in Section E.6. *q.e.d.*

The vacuum ratio is the joint ratio of the whole root family: by (3.84) and (3.74),

$$q_\xi(\mathcal{A}) = \exp\Big[-\sum_{\mathfrak{c} \text{ hits } \mathcal{A}} \Phi^{\mathrm{v}}_\xi(\mathfrak{c})\Big]\,, \tag{E.14}$$

a cluster hitting several roots being counted once, and its derivative is the derivative of this exponential.

Lemma E.5 (The Joint Ratio and Its Derivative)**.** *With the constant C_M of Lemma E.14,*

$$|q_\xi(\mathcal{A})| \leqslant e^{\sum_{D\in\mathcal{A}} N_1(D)/8}, \qquad |\mathscr{D}_\omega q_\xi(\mathcal{A})| \leqslant \frac{C_M}{2}\|\omega\|_\infty \sum_{D\in\mathcal{A}} N_1(D)\, e^{\sum_{D\in\mathcal{A}} N_1(D)/8}\,, \tag{E.15}$$

and consequently

$$|\mathscr{D}_\omega[q_\xi(\mathcal{A})\mathfrak{Q}_\xi(\mathcal{A}; f)]| \leqslant C\|\omega\|_\infty \Big(L_f + \sum_{D\in\mathcal{A}} N_1(D)\Big) \prod_{D\in\mathcal{A}} [\rho\, m(D) e^{N_1(D)/8}]\,. \tag{E.16}$$

Proof. The first bound is Lemma 3.28. The second is Lemma E.17. The third follows from the product rule with Lemmas E.3 and E.4 and the two bounds of (E.15), with the constant C of (E.13) increased by $C_M/2$. *q.e.d.*

E.5 The Closed Group Subtraction

Remark E.6 (The Closed Group Contraction). The closed contraction of a group is formed with the interval transport relative to its background interval, closed back to its base, and an arbitrary complementary closed holonomy h; its flat value is $a_W\chi(h)$ by Lemma 3.34. Products of such closed contractions are bounded by the norm equivalence of Lemma E.7, and the multi-group factorisation is (3.90).

Geometry. The square/torus geometry of Section 5.3 is in force: C is a coarse square of side $s = nL$ fine bonds on fine periods LK_i with $K_i \geqslant 3(n+2)$, and $n_C = 4s$. For a group W, $\mathcal{H}(W)$ is its interval (Definition 3.32: a cyclic interval of fewer than $n_C/2$ contour traversals, or the whole contour read from the base point) and $|\mathcal{H}(W)|$ the number of traversals in it. Contour hull length is paid by connected spatial extent as in (5.24): contacts of diameter $< s/2$ occupy one side or two adjacent sides and fit in an arc of length at most $4\,\mathrm{diam}+1$; otherwise $n_C \leqslant 8\,\mathrm{diam}$. Periodic distances between square contacts equal their lifted distances.

For a group W let $n(W)$ be the sum of N_1 over its roots and over every constituent domain of every occurrence in its cluster list, with multiplicities; the size and the footprint of W are

$$\mathfrak{s}(W) \coloneqq n(W) + \frac{|\mathcal{H}(W)|}{LM}\,, \tag{E.17}$$

$$F(W) = \Big\{ \Delta \in \pi_1 \,\Big|\, \Delta \text{ meets a constituent domain of } W, \text{ its coefficient stencil halo, or } \mathcal{F}_r \cup \{r\} \text{ for a traversal } r \in \mathcal{H}(W) \Big\}\,, \tag{E.18}$$

where π_1 is the partition into LM-cubes, a cube meets a set of sites or bonds when it contains a site or a bond endpoint of the set, and $\mathcal{F}_r$ is the free-input stencil of r. The ranges are made explicit in Lemma E.8, which gives $|F(W)| \leqslant C\mathfrak{s}(W)$.

For a component of the roots/cluster-occurrence graph the union of its domains is wall connected, so its diameter is at most CLM times its cube count, and its contact hull is bounded by the square estimate above; overlapping hulls have a connected interval union, which pays the hull of a group by the sum of its component counts. If all components are merged because some hull is long (Section 3.8), the sum of the component sizes in that class is at least $cn_C/(LM)$. Hence $|\mathcal{H}(W)|/(LM) \leqslant Cn(W)$. Coefficients whose local rectangle fails to lift are kept in the large-domain class of Section E.7.4.

E.5.1 The Closed Relative Interval Functional

In the reconstructed links along $\mathcal{H}(W)$ only the root variables of W are nonzero. $T_W(a;U_1)$ is the ordered product of these links (the interval transport) and $P_W(U_1)$ the same product at zero free variable, both with their source orientations and with the same two endpoint vertices. The relative product $T_W(a;U_1)P_W(U_1)^{-1}$ is based and closed at the initial vertex. The scalar closed contraction of W is

$$c_W(U_1;h) \coloneqq w_\xi(W) \int \chi(T_W(a;U_1)P_W(U_1)^{-1}h)\,\mathrm{d}\Big(\bigotimes_{D\in\mathcal{A}(W)} \widehat{\nu}^{\mathrm{f}}_{\xi,D}\Big)(a)\,, \tag{E.19}$$

$h \in G$ a complementary closed holonomy at that base, transforming by simultaneous conjugation; the product $T_W P_W^{-1}$ and h transform at one base, so the displayed scalar is gauge invariant. The group defect and its size are

$$d_W(U_1;h) \coloneqq c_W(U_1;h) - a_W\chi(h), \qquad \mathfrak{d}_W(U_1) \coloneqq \sup_{h\in G} |d_W(U_1;h)|\,. \tag{E.20}$$

At the reference data, where the same formula is taken with the reference weights and measures, $c_W(\mathbf{1};h) = a_W\chi(h)$ by Lemma 3.34 (there the interval transport at the root variables is the $P^\circ_W(a)$ of that lemma and the background transport is the identity). The weights and measures in c_W are those of the joint extraction, including every cluster occurrence of the group.

Lemma E.7 (Factorisation and Product Inequality)**.** *Let π be the irreducible unitary representation with $\chi = \dim(\pi)^{-1}\operatorname{tr}\pi$.*

(a) There is a constant K_χ of the character such that

$$\|\Upsilon\|_{\mathrm{op}} \leqslant K_\chi \sup_{h\in G} |\dim(\pi)^{-1}\operatorname{tr}(\Upsilon\pi(h))|$$

for every $\Upsilon \in \operatorname{End}(V_\pi)$.

(b) For a compatible family $\mathfrak{B} := \{W_1,\dots,W_m\}$ on a covered branch, $R_\xi(\mathfrak{B})$ is the normalised trace of the ordered product of the matrices $\Upsilon_{W_j}\pi(G_j)$, where $\Upsilon_W = w_\xi(W)\int \pi(T_W P_W^{-1})\,\mathrm{d}\bigotimes \widehat{\nu}_\xi^{\mathrm{f}} - a_W\mathbf{1}$ and G_j is the background transport of the gap after $\mathcal{H}(W_j)$ composed with P_{W_j}.

(c) Consequently

$$|R_\xi(\mathfrak{B})| \leqslant \prod_{W\in\mathfrak{B}} K_\chi \mathfrak{d}_W(U_1)\,. \tag{E.21}$$

Proof. (a) $\Upsilon \mapsto (h \mapsto \operatorname{tr}(\Upsilon\pi(h)))$ is linear and injective on $\operatorname{End}(V_\pi)$, since $\pi(G)$ spans $\operatorname{End}(V_\pi)$ for an irreducible representation (Burnside); so the supremum over h is a norm on the finite-dimensional space $\operatorname{End}(V_\pi)$, equivalent to the operator norm, and K_χ is the equivalence constant.

(b) In a compatible family a traversal of $\mathcal{H}(W)$ has no contact with a root of a different group (that contact hull would meet $\mathcal{H}(W)$), so its reconstructed link uses the root variables of W only (Lemma 3.9); a traversal outside every interval has its background link. The groups are numbered in cyclic order, and the ordered link product starts at the first traversal of $\mathcal{H}(W_1)$ (cyclicity of the trace). Expanding the ordered product $\prod_j (X_j - a_{W_j}\mathbf{1})\pi(G_j)$ with $X_j = w_\xi(W_j)\int \pi(T_{W_j}P_{W_j}^{-1})\mathrm{d}\bigotimes\widehat{\nu}^{\mathrm{f}}$ over subsets $\mathcal{S}$ of slots, a slot of the first kind contributes the links of $\mathcal{H}(W_j)$ at the root variables followed by the gap transport, a slot of the second kind contributes $-a_{W_j}$ times the links of $\mathcal{H}(W_j)$ at zero free variable followed by the gap; by multilinearity of the trace and Fubini over the product measure (as in the proof of Lemma 3.34) the normalised trace of the term of $\mathcal{S}$ is $(\prod_{W\notin\mathcal{S}}(-a_W))\,(\prod_{W\in\mathcal{S}} w_\xi(W))\,\mathfrak{Q}_\xi(\mathcal{A}(\mathcal{S}); f_\xi) = (\prod_{W\notin\mathcal{S}}(-a_W))A_\xi(\mathcal{S})$ by (3.90), since on a covered branch the links outside the intervals are the background links at U_1 (Section E.7.1; Lemma 3.36 gives $f_\xi(0) = W_\chi(C;U_1)$). Summing over $\mathcal{S}$ reproduces (3.95).

(c) $|\operatorname{tr}(Y_1\cdots Y_{2m})| \leqslant \dim(\pi)\prod\|Y_i\|_{\mathrm{op}}$, the gap transports are unitary, and $\|\Upsilon_W\|_{\mathrm{op}} \leqslant K_\chi\mathfrak{d}_W(U_1)$ by (a), since $\dim(\pi)^{-1}\operatorname{tr}(\Upsilon_W\pi(h)) = d_W(U_1;h)$. *q.e.d.*

E.5.2 Positive Graph Sums for the Reference Factors

A cluster occurrence $\mathfrak{c}$, defined by the ordered tuple $(D_1,\dots,D_p)$, has cube cost $n(\mathfrak{c}) = \sum_{D\in\mathfrak{c}} N_1(D)$ and positive weight

$$M_v(\mathfrak{c}) = \frac{1}{p!}\sum_\tau \prod_{i=1}^{p} m(D_i)\,, \tag{E.22}$$

the sum over the spanning trees τ of the incompatibility graph of the tuple: the tree majorant of its Ursell coefficient (the positive tree sum of the proof of Lemma 3.28, Section B.6). Source roots have the weights $\rho\, m(D)$ of Lemma E.3.

For a group W with root family $\mathcal{A}(W)$ and cluster list $(\mathfrak{c}_1, \dots, \mathfrak{c}_{k(W)})$,

$$M(W) = \frac{1}{k(W)!} \prod_{D \in \mathcal{A}(W)} \rho\, m(D) \prod_{j=1}^{k(W)} M_v(\mathfrak{c}_j)\ ; \tag{E.23}$$

it bounds $|a_W|$, the pairings $|w_\xi(W)\mathfrak{Q}_\xi(\mathcal{A}(W); f)|$ for the words of Lemma E.3, and the undifferentiated closed source functional $|c_W(U_1; h)|$, the physical and reference data having the same numerical majorant (Lemma E.3 applies at the reference data too, whose hypotheses Lemma B.16 verifies).

The cubes are those of the partition π_1 (side LM, Section B.1): $\Delta_z = LMz + [-LM/2, LM/2]^4$ reduced modulo the fine periods, which are multiples of LM by (3.1); every site lies in exactly one cube, and the index distance of two cubes is the periodic maximum-norm distance of their indices z.

For a set S of sites, $\pi_1(S)$ is the set of cubes containing a site of S, and $\pi_1(\mathcal{H}(W))$ abbreviates π_1 of the vertex set of the traversals of $\mathcal{H}(W)$. For a set X of cubes, $\mathfrak{N}_R(X)$ is the set of cubes at index distance at most R from a cube of X, so $|\mathfrak{N}_R(X)| \leqslant (2R+1)^4 |X|$.

R_{h} is the maximal ℓ^∞ distance from the base of a host (3.36), (3.75) to a point of the host, a constant fixed by the host constructions of Section E.6.1. The coefficient stencil halo of a domain D is the union of the hosts whose base lies within R_{h} of D. The envelope of a group W is defined with

$$\begin{aligned} R_{\mathrm{e}} &= \Big\lceil \frac{2\max(R_{\mathrm{h}}, L)}{LM} \Big\rceil, \qquad A_F = A_E = 9(2R_{\mathrm{e}}+1)^4, \\ X(W) &= \bigcup_D \pi_1(D), \qquad \mathfrak{E}(W) = \mathfrak{N}_{R_{\mathrm{e}}}\big(X(W) \cup \pi_1(\mathcal{H}(W))\big)\ , \end{aligned} \tag{E.24}$$

the union over the constituent domains D of W (roots and cluster constituents); $X(W)$ is a set, so $|X(W)| \leqslant n(W)$, which counts with multiplicity.

Lemma E.8 (Footprints and Envelopes)**.** *For every group W:*

(a) $|\pi_1(\mathcal{H}(W))| \leqslant 8 + |\mathcal{H}(W)|/(LM)$, $|X(W) \cup \pi_1(\mathcal{H}(W))| \leqslant 9\mathfrak{s}(W)$, $F(W) \subseteq \mathfrak{E}(W)$ *and* $|F(W)| \leqslant |\mathfrak{E}(W)| \leqslant A_E \mathfrak{s}(W)$*;*

(b) $\mathfrak{E}(W)$ *is connected under maximum-norm adjacency of cubes (two distinct cubes are adjacent when their index distance is one);*

(c) $\mathcal{H}(W) \neq \varnothing$*, and if* $\Delta \in F(W)$ *then* $\operatorname{dist}(\Delta, C) \leqslant A_E LM\, \mathfrak{s}(W)$*, that is* $\mathfrak{s}(W) \geqslant \operatorname{dist}(\Delta, C)/(A_E LM)$*, where* $\operatorname{dist}(\Delta, C)$ *is the ℓ^∞ distance in fine units from the nearest site of* Δ *to the vertex set of* C*;*

(d) two incompatible groups have footprints in contact (sharing a cube, or adjacent across a face, edge or corner, that is at index distance at most one) or overlapping intervals, so an incompatible W' has a footprint cube in $\mathfrak{N}_1(F(W))$, a set of at most $3^4 A_F \mathfrak{s}(W)$ cubes, or a traversal in $\mathcal{H}(W)$.

Proof. Ranges, in fine ℓ^∞ units: the free-input stencil $\mathcal{F}_r$ of a bond r (3.13) is $\{r\}$ or $\mathcal{E}_c \cap E_*$, bonds with both endpoints in two adjacent blocks of side L (3.10), or empty (for a contour bond that is not a bond of Ω_1, the convention of Section 3.8), so every endpoint of a bond of $\mathcal{F}_r \cup \{r\}$ is within $2L$ of the initial vertex of r. The active hosts of a selection with final

domain D have their bases in the R_{h}-neighbourhood of D, and each host lies within R_{h} of its base, so the halo of D lies within $2R_{\mathrm{h}}$ of D. A root contact $n \in \mathsf{T}_D$ (3.87) has a bond $b \in \mathcal{F}_{r_n} \cap \mathsf{L}_D$, whose endpoints lie in D ($Y_b \subseteq D$ by (3.76), and Y_b contains b by its definition (3.75)), so the initial vertex of r_n is within $2L$ of D.

Index conversion: if sites $x \in \Delta_z$ and $y \in \Delta_{z'}$ are at fine distance at most ρ, then, for lifts of x, y realising that distance and the indices z, z' of their lifted cubes, $|LM(z-z')|_\infty \leqslant \rho + 2 \cdot \frac{LM-1}{2} < \rho + LM$, so the index distance of Δ_z and $\Delta_{z'}$ is at most $\lceil \rho/(LM) \rceil$; in particular the cube of a site within $2L$ of a cube Δ is at index distance at most $\lceil 2L/(LM) \rceil = 1$ from Δ ($M \geqslant 8$), and a cube meeting the halo of D is at index distance at most $\lceil 2R_{\mathrm{h}}/(LM) \rceil \leqslant R_{\mathrm{e}}$ from a cube of D.

Every root D of W is marked, that is $\mathsf{T}_D \neq \varnothing$ ((3.87) and the construction preceding Definition 3.32), and $\mathsf{T}_D \subseteq \mathcal{H}(W)$, because $\mathcal{H}(W)$ is the hull of the union of the contacts of the group's components, which include the contact sets of its roots. In particular $\mathcal{H}(W) \neq \varnothing$.

(a) $\mathcal{H}(W)$ is a cyclic interval of fewer than $n_C/2$ traversals of the square C, or the whole contour (Definition 3.32), so the vertices of its traversals on each side form one run of consecutive collinear sites (the whole side, for the whole contour), and C does not wrap around the torus ($LK_i \geqslant 3L(n+2) > nL$). A run of t traversals along a side has $t+1$ consecutive collinear vertices, which lie in at most $\lfloor t/(LM) \rfloor + 2$ cubes, since the cubes cut the side into runs of LM consecutive sites; summing over the four sides, $|\pi_1(\mathcal{H}(W))| \leqslant 8 + |\mathcal{H}(W)|/(LM)$. Hence $|X(W) \cup \pi_1(\mathcal{H}(W))| \leqslant n(W) + 8 + |\mathcal{H}(W)|/(LM) = \mathfrak{s}(W) + 8 \leqslant 9\mathfrak{s}(W)$ by (E.17) and $\mathfrak{s}(W) \geqslant n(W) \geqslant 1$ (a group has a nonempty root family, Definition 3.32, and each root is a connected final domain, a nonempty union of cubes, Section 3.5). A cube of $F(W)$ (E.18) is a cube of a constituent domain, in $X(W)$; or meets the halo of a constituent domain, at index distance at most R_{e} from $X(W)$; or contains an endpoint of a bond of the stencil of a traversal of $\mathcal{H}(W)$, within $2L$ of the initial vertex of that traversal, hence at index distance at most $1 \leqslant R_{\mathrm{e}}$ from $\pi_1(\mathcal{H}(W))$. So $F(W) \subseteq \mathfrak{E}(W)$ and $|\mathfrak{E}(W)| \leqslant (2R_{\mathrm{e}}+1)^4 \cdot 9\mathfrak{s}(W) = A_E \mathfrak{s}(W)$.

(b) The set $X(W) \cup \pi_1(\mathcal{H}(W))$ is connected under adjacency: the two vertices of a traversal are at fine distance one and consecutive traversals of $\mathcal{H}(W)$ share a vertex, so the vertices lie in the same or in adjacent cubes, and $\pi_1(\mathcal{H}(W))$ is connected; each constituent domain is wall connected (Section 3.5), so its cubes are connected; within a cluster occurrence the constituent domains are linked by incompatibilities (a tuple with a disconnected incompatibility graph has $\varphi^T = 0$ and does not occur in (3.88)), a cluster occurrence of W hits a root of W (Definition 3.32: some constituent is incompatible with a root), and incompatible domains share a cube or a wall (Section 3.6), that is contain cubes at index distance at most one; finally every root D has a contact $n \in \mathsf{T}_D \subseteq \mathcal{H}(W)$ (shown above), the initial vertex of r_n being within $2L$ of D, so a cube of $\pi_1(\mathcal{H}(W))$ is at index distance at most one from a cube of D. Thus every constituent domain is joined to $\pi_1(\mathcal{H}(W))$ through adjacent cubes, and $\mathfrak{E}(W)$, the union of the connected boxes $\mathfrak{N}_{R_{\mathrm{e}}}(\{\Delta\})$ over the cubes Δ of a connected set, each containing its centre Δ, is connected.

(c) $\mathcal{H}(W) \neq \varnothing$ was shown above. A connected set of k cubes contains a path of at most k cubes between any two of them, so the cubes of any two of its sites are at index distance at most $k-1$ and the sites are at fine distance less than kLM; with $k \leqslant A_E \mathfrak{s}(W)$, the ℓ^∞ diameter of $\mathfrak{E}(W)$ is at most $A_E LM\, \mathfrak{s}(W)$. $\mathfrak{E}(W)$ contains $\Delta \in F(W)$ and the nonempty set $\pi_1(\mathcal{H}(W))$, whose cubes contain vertices of C.

(d) Two groups are compatible when all their constituent domains are mutually compatible and their intervals are disjoint, and a family containing a whole-contour group is a singleton (Definition 3.32). So if W and W' are incompatible, either one of them is a whole-contour

group, and then the intervals overlap, as $\mathcal{H}(W)$ and $\mathcal{H}(W')$ are nonempty by (c); or the intervals overlap, so W' has a traversal in $\mathcal{H}(W)$; or some constituent domains D of W and D' of W' share a cube or a wall, so a cube of $D' \subseteq F(W')$ lies at index distance at most one from a cube of $D \subseteq F(W)$, that is in $\mathfrak{N}_1(F(W))$, and $|\mathfrak{N}_1(F(W))| \leqslant 3^4|F(W)| \leqslant 3^4 A_F \mathfrak{s}(W)$. *q.e.d.*

Lemma E.9 (Positive Group Sums)**.** *Let $\eta > 0$ be chosen after the fixed geometry and forest decay constants, with $1000\eta A_{\mathrm{s}} < \varpi/8$ as in Section E.7.2. Under the smallness conditions $C\rho LM \leqslant \eta/8$ and the root-incidence bound of Section E.7.2, which hold for g_0 small,*

$$\sup_{n \in \mathcal{I}_C} \sum_{W \,|\, n \in \mathcal{H}(W)} M(W) e^{4\eta \mathfrak{s}(W)} + \sup_{\Delta \in \pi_1} \sum_{W \,|\, \Delta \in F(W)} M(W) e^{4\eta \mathfrak{s}(W)} \leqslant C\rho \,, \tag{E.25}$$

with a constant C fixed before the coupling (the constants of Lemma E.21(b) and of Section E.7.3), where $\mathcal{I}_C$ is the set of contour traversals.

Proof. This is Lemma E.22(d). *q.e.d.*

Lemma E.10 (The Reference Group Gas)**.** *Let $\mathfrak{a}_W := \eta \mathfrak{s}(W)$. Under the hypotheses of Lemma E.9, for every group W,*

$$\sum_{W' \not\sim W} |a_{W'}| e^{\mathfrak{a}_{W'}} \leqslant C\rho(|F(W)| + |\mathcal{H}(W)|) \leqslant \frac{1}{8}\mathfrak{a}_W \tag{E.26}$$

for g_0 small. Hence the reference group gas (3.94) *has a convergent connected-cluster logarithm, $S^{\circ}_{\xi,C} \neq 0$, and for every compatible family $\mathfrak{B}$*

$$\Big|\frac{S^{\circ,[\mathfrak{B}]}_{\xi,C}}{S^{\circ}_{\xi,C}}\Big| \leqslant \exp\Big(\frac{1}{8}\sum_{W \in \mathfrak{B}} \mathfrak{a}_W\Big) \,. \tag{E.27}$$

Proof. Incompatibility of W and W' implies contact of their footprints or overlap of their intervals, so the two pin sums (E.25) (with $|a_{W'}| \leqslant M(W')$ and $e^{\mathfrak{a}_{W'}} \leqslant e^{4\eta\mathfrak{s}(W')}$) give the first inequality of (E.26), and $|F(W)| + |\mathcal{H}(W)| \leqslant C_1 LM\mathfrak{s}(W)$ with $C\rho LM \leqslant \eta/8$, the constant C there taken to include C_1, gives the second. This is the Kotecký–Preiss criterion for the species W with activities a_W and costs $\mathfrak{a}_W$; the pinned-tree argument of Section B.6 gives the convergent logarithm and the exclusion identity, whose countable-species form is Lemma E.24. The restricted ratio is the single joint exclusion ratio of that gas, and the pinned cluster estimate counts each cluster hitting $\mathfrak{B}$ once, giving (E.27). *q.e.d.*

E.5.3 Subtraction and Its Curvature Payment

For a group W the comparison box Q_W is a fine rectangle containing every constituent domain's coefficient halo and every background transport along $\mathcal{H}(W)$, including the inverse interval closing the source; its lifted side is at most $CLMn(W)$ (Section E.7.4). For a fine rectangle Q and a unitary field U,

$$\delta_Q(U) = \max_{p \subset Q} |U(\partial p) - \mathbf{1}|_\sigma \,, \tag{E.28}$$

the maximum over the plaquettes of Q.

A group is small interior when $n(W) < b/(4C_{\mathrm{box}}LM)$ (E.48); on far histories its box lies in Λ_1 and misses S_1, and $\mathfrak{d}_W(U_1)$ is bounded through the coefficient path of Lemma E.23 and the tree gauge of Lemma E.25. Every other group is bounded by its two undifferentiated endpoint coefficients, and the reserved exponential pays this tail (Section E.11.2).

Lemma E.11 (The Group Pin Sum)**.** *On covered far histories, with* $\varepsilon_1 = g_1 p_0(g_1)$*,* $b = \lfloor LMR_1/16 \rfloor$ *and constants* C_χ*,* c *fixed before the coupling,*

$$\sum_W K_\chi \mathfrak{d}_W(U_1) e^{\mathfrak{a}_W/8} \leqslant C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)}) \,. \tag{E.29}$$

The proof is in Section E.11.2.

E.5.4 The Signed Integrated Bound

Lemma E.12 (The Family Sum)**.** *On covered far histories,*

$$|\mathscr{R}_\xi| \leqslant |S^\circ_{\xi,C}| \Big[\exp\Big(C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)}) \Big) - 1 \Big] \,. \tag{E.30}$$

Proof. In the sum (E.1), by (E.21) and (E.27), $|S^{\circ,[\mathfrak{B}]}_{\xi,C} R_\xi(\mathfrak{B})| \leqslant |S^\circ_{\xi,C}| \prod_{W \in \mathfrak{B}} K_\chi \mathfrak{d}_W(U_1) e^{\mathfrak{a}_W/8}$, and the sum over nonempty compatible families is bounded by the exponential series of the one-group sum (E.29) minus its zeroth term. The countable-group limit is Lemma E.24. *q.e.d.*

Together with Lemma E.1 this gives the defect bound (E.83).

E.6 The Joint Kernel Payment

This section proves the quantitative inputs of Section E.4.

E.6.1 Host Budget and Fixed Parameter Order

C_{p} bounds one primitive (Lemma B.12) and $C_0 = N_{\mathrm{h}} C_{\mathrm{p}} e^{\kappa(B_{\mathrm{h}}+4)}$ bounds their sum at a host (Lemma B.13). Let C^∂_{p} be the maximum of the finite local coefficient-derivative bounds of Section E.9.1: the energy primitive is g_0 times the third-order Taylor remainder in the protected variables, and its background derivative has the same factor $g_0(1+r)^3$ because its local analytic coefficients and their background derivatives are bounded on the larger fixed row chart; the Haar primitive and its background derivative start at order $g_0^2(1+r)^2$; the constraint-Jacobian primitive and its derivative are bounded by $Cg_0(1+r)$. Let

$$C^\partial_0 \coloneqq N_{\mathrm{h}} \max(C_{\mathrm{p}}, C^\partial_{\mathrm{p}}) e^{\kappa(B_{\mathrm{h}}+4)}, \qquad \omega(g_0) \coloneqq C^\partial_0 g_0 (1+r)^3, \qquad R_\Upsilon \coloneqq \frac{\omega_*}{2\omega(g_0)} \,, \tag{E.31}$$

C^∂_0 being fixed after M and before g_0. Summing the same finite host assignments proves $\sup |\mathscr{D}_\omega \mathbf{V}_\xi(Y)| \leqslant \omega(g_0) e^{-\kappa d_0(Y)} \|\omega\|_\infty$ on the background neighbourhood of Section E.9.1 (phases within $\tau_{\mathfrak{b}}$ of the branch background, Lemma E.16), where the primitive bounds hold with the same constants, so that $\omega(g_0) \geqslant C_0 g_0 (1+r)^3$ also bounds $\sup |\mathbf{V}_\xi(Y)| e^{\kappa d_0(Y)}$ there ((B.21)); in Sections E.4 and E.5 this $\omega(g_0)$ replaces the $\omega(g_0)$ of (B.20), which it dominates. The coupling threshold is $\omega(g_0) \leqslant \omega_*/4$, that is $R_\Upsilon \geqslant 2$ (Lemma E.2). Let N_{type} be the number of tagged ordinary primitive and probe types based in one fine translation cell of the fixed L-periodic template and R_{h} the maximal base-to-host offset, both fixed by the host constructions (3.36), (3.75). Distinct active hosts of a selection with final domain D have a labelled base in the fixed R_{h}-neighbourhood of D, so their number is at most $H_M N_1(D)$ with $H_M \coloneqq N_{\mathrm{type}}(LM + 2R_{\mathrm{h}} + 3)^4$, and the selected probe count is at most the number of fine bonds in D, $q \leqslant 4(LM+1)^4 \sum_D N_1(D)$.

Definition E.13 (Host Weights)**.** For a host Y let n_Y be the number of probe labels with host Y, and let

$$A_Y \coloneqq \frac{\omega_*}{2} e^{-\kappa d_0(Y)}, \qquad B_Y \coloneqq n_Y u_*, \qquad v_{\text{ordinary}} \coloneqq e^{A_Y} - 1, \qquad v_{\text{probe}} \coloneqq u_* \,. \tag{E.32}$$

Lemma E.14 (Positive Host Comparison)**.** *For every host Y, with the weights of Definition E.13, $n_Y \leqslant P_M$ and $B_Y \leqslant \frac{\omega_*}{8} e^{-\kappa d_0(Y)}$. Moreover:*

(a) the sum over the nonempty label subsets at Y satisfies

$$\begin{aligned} \sum_{\varnothing \neq S \subseteq \{\text{ordinary,probes}\}} \prod_{\ell' \in S} v_{\ell'} &\leqslant e^{A_Y}(1+u_*)^{n_Y} - 1 \leqslant e^{A_Y + B_Y} - 1 \\ &\leqslant (A_Y + B_Y) e^{A_Y + B_Y} \leqslant \omega_* e^{1/2} e^{-\kappa d_0(Y)} \,. \end{aligned} \tag{E.33}$$

(b) For a differentiated host, the summed weight of the label subsets at Y is bounded by

$$R_\Upsilon \sup |\mathscr{D}_\omega \mathbf{V}_\xi(Y)| \, e^{A_Y}(1+u_*)^{n_Y} \leqslant A_Y e^{A_Y + B_Y} \|\omega\|_\infty \leqslant \omega_* e^{1/2} e^{-\kappa d_0(Y)} \|\omega\|_\infty \,. \tag{E.34}$$

(c) There is a constant C_M, fixed after M and before g_0, such that for every connected final domain D

$$\sum_{\ell \,|\, Z(\ell) = D} u_*^{q(\ell)} \sup_{|\zeta| \leqslant R_\Upsilon} \|\mathscr{D}_\omega \Lambda^{\mathrm{f}}_{\xi;\ell,\zeta}\|_{\mathrm{TV}} \leqslant C_M N_1(D) \|\omega\|_\infty \, \varepsilon_* e^{-\lambda_* \ell_1(D)} \,, \tag{E.35}$$

the derivative including every Gaussian normaliser and the compensator; and $|\mathscr{D}_\omega k^{\mathrm{f}}_\xi(D)| \leqslant C_M N_1(D) \|\omega\|_\infty m(D)$ for the vacuum activities.

Proof. The bound $n_Y \leqslant P_M$ holds because a probe label b lies in its host, whose fine side is at most $4M$, and there are four positive directions per site. The bound on B_Y follows from $n_Y \leqslant P_M$, $u_* \leqslant \omega_* e^{-\kappa s_{\mathrm{p}}}/(8P_M)$ ((E.9)), and $d_0(Y) \leqslant N_0(Y) - 1 < s_{\mathrm{p}}$ for a probe host ((B.1) and Lemma E.2).

(a) After taking absolute values inside the real Mayer t-integral the ordinary factor is at most $e^{A_Y} - 1$ on $|\zeta| \leqslant R_\Upsilon$ (as $R_\Upsilon \omega(g_0) = \omega_*/2$). The last inequality of (E.33) uses $\omega_* \leqslant 1/2$ (the standing choice of Section B.8) and $A_Y + B_Y \leqslant \frac{5}{8}\omega_* e^{-\kappa d_0(Y)}$.

All split selections with the same active-host set have the same Y_0, seed, conditioning carrier and Gaussian kernel and, at fixed bad-bond choice P and final domain Z, the same final-domain weakening (proof of Lemma 3.29), so their many-to-one sum is dominated at each host by (E.33) before the cover and seed sums of Appendix B are applied.

(b) is the same comparison with the differentiated host bound of Section E.6.1.

(c) There are at most $H_M N_1(D)$ active hosts; the Gaussian and walk derivatives have the same factor after their polynomial moment is paid (Sections E.9.2, E.9.3): the uniform Gaussian coercivity gives polynomial moments at most $C_M N_1(D)$ times the same exponential volume majorant $e^{N_1(D)}$. The differentiated sum with the positive host comparison (b) gives (E.35), and the same sum over the selections with $\mathsf{J} = \varnothing$ gives the vacuum-activity bound. *q.e.d.*

E.6.2 The Multi-root Cancellation

The input f is a single normalised character of the ordered closed reconstructed word, optionally closed with its inverse background interval and a fixed complementary h; for a linear combination the bound is multiplied by the sum of the absolute coefficients. On covered branches its contour input labels lie in I, so the retained A_{e} and the boundary map do not enter the source rows (Lemma 3.36); the centred kernel $K_{\xi;0}$ likewise uses zero retained input, and the reference group construction uses the same row and mean conventions.

Lemma E.15 (Commuting Projections)**.** *Let O_0 be the set of roots with odd q_j and empty ordinary selection, O_1 the set of roots with odd q_j and nonempty ordinary selection. Then*

$$\Big|\int \Delta_{\mathfrak{J}} f \,\mathrm{d} \bigotimes_j \Lambda^{\mathrm{f}}_{\xi;\ell_j}\Big| \leqslant u^q \prod_{j\in O_0\cup O_1} \Big(u + \frac{2}{R\Upsilon - 1}\Big) \prod_j M_{\ell_j} \,, \tag{E.36}$$

with the exponential-in-q constants absorbed in C_χ; after one background derivative the same holds with the factor M_{ℓ_j} of the differentiated root replaced by $\sup_{|\zeta|\leqslant R\Upsilon} \|\mathscr{D}_\omega \Lambda^{\mathrm{f}}_{\xi;\ell_j,\zeta}\|_{\mathrm{TV}}$, which (E.35) *sums, and with the additional factor of Lemma E.4. Moreover $\Delta_{\mathfrak{J}} f = 0$ if some selected label does not influence the closed source word, so every surviving root has a contact with Γ_C.*

Proof. For $j \in O_0$, P_j is the projection onto the homogeneous part of degree q_j of $\Delta_{\mathfrak{J}} f$ in the selected variables of root j, the other root variables fixed; these projections commute. In the expansion of $\prod_{j\in O_0}(P_j + (1 - P_j))$ a term containing P_j is odd under the reversal of exactly root j and integrates to zero against that root's even source-free density; only $\prod_{j\in O_0}(1 - P_j)$ remains, with one extra u per such root by the mixed Taylor remainder (E.66) and the fixed row-overlap count.

For $j \in O_1$ the independent ζ_j-integral vanishes at $\zeta_j = 0$, so the multivariable Cauchy difference supplies $\prod_{j\in O_1} 2/(R\Upsilon - 1)$ times the supremum $\prod M_{\ell_j}$ of (E.8). Together this is (E.36), and for even q_j only $u^{q_j} M_{\ell_j}$ is paid.

After one background derivative the Gaussian derivative is even, the leading source derivative has the same parity in the selected variables (a background derivative hits a row coefficient or a background occurrence of the closed word, not the selected root variables, Section E.9.4; the resulting bound is (E.67)) and a differentiated Mayer factor still vanishes at $\zeta_j = 0$, so the argument repeats.

The last statement is the pairwise cancellation of K with $K \cup \{b\}$ in the K-sum (E.4) for a label b of a root that the word does not see. *q.e.d.*

E.6.3 The Common Hessian Bounds on the Extension

Lemma E.16 (Hessian Bounds along the Coefficient Path)**.** *Let $\tau_{\mathfrak{b}} > 0$ be fixed after τ_L, c_L, γ_L and before g_0, with $\tau_{\mathfrak{b}} \leqslant \tau_L/4$, $c_L\tau_{\mathfrak{b}} \leqslant \gamma_L/4$ and the chart conditions of Section E.9.1 (the matrix-logarithm neighbourhood and the entry-chart norm, each a smallness condition on $\tau_{\mathfrak{b}}$ in terms of constants fixed before g_0). For every $t \in [0,1]$ and every globally small extension $U = e^{itB}U^\circ$ with $\|B\|_\infty \leqslant \tau_{\mathfrak{b}}$ of the flat background on a comparison box, every compression of the regional Hessian to the same I satisfies*

$$\frac{3}{4}\gamma_L \mathbf{1} \leqslant H_{e^{itB}} \leqslant m_L \mathbf{1} \,. \tag{E.37}$$

Proof. On physical branches Lemma 3.8 gives $\gamma_L/2 \leqslant H_{\xi,II} \leqslant m_L$ with the force–constraint and soft-gauge terms.

The local entry formula $f_{bb'}(A)$ of Lemma B.6 has row sums at most m_L on the phase box τ_L by the definition of m_L in (3.22), column sums equal to its row sums since $f_{bb'} = f_{b'b}^{\mathsf{T}}$ have equal norms, entries vanishing beyond the range of that lemma, and Lipschitz bound c_L in the phase; so the extension has operator difference from $H^\circ = \mathsf{H}_{\mathbf{1}}^{\Omega_1}$ at most $c_L\tau_{\mathfrak{b}} \leqslant \gamma_L/4$ by the Schur test, uniformly in t. The flat lower bound γ_L ((A.25)) passes to every compression to I by the compression argument in the proof of Lemma B.16, and with the difference bound this gives (E.37).

The recipe $\gamma = \gamma_L/2$, $h_+ = m_L$ of Section 3.5 therefore applies at the physical endpoint and throughout the extension. *q.e.d.*

E.6.4 Differentiating the Joint Vacuum Ratio

$\Phi_\xi^{\mathrm{v}}(\mathfrak{c})$ is the Ursell expression $\varphi^T(\mathfrak{c})\prod_j k_\xi^{\mathrm{f}}(D_j)/m!$ of (3.73) with the fixed-kernel vacuum activities.

Lemma E.17 (Derivative of the Cluster Sum)**.** *With the majorant $M_v(\mathfrak{c})$ of* (E.22) *and $n(\mathfrak{c}) = \sum_{D\in\mathfrak{c}} N_1(D)$,*

$$|\mathscr{D}_\omega \Phi_\xi^{\mathrm{v}}(\mathfrak{c})| \leqslant C_M \|\omega\|_\infty \, n(\mathfrak{c}) M_v(\mathfrak{c}) \,, \tag{E.38}$$

and for every finite set $\mathcal{A}$ of connected final domains

$$\sum_{\mathfrak{c}\ \mathit{hits}\ \mathcal{A}} M_v(\mathfrak{c}) e^{n(\mathfrak{c})/2} \leqslant \frac{1}{8}\sum_{D\in\mathcal{A}} N_1(D), \qquad \sum_{\mathfrak{c}\ \mathit{hits}\ \mathcal{A}} |\mathscr{D}_\omega \Phi_\xi^{\mathrm{v}}(\mathfrak{c})| \leqslant \frac{C_M}{2}\|\omega\|_\infty \sum_{D\in\mathcal{A}} N_1(D)\,. \tag{E.39}$$

Proof. (E.38) is the product rule over the constituents with the activity derivative bound of Lemma E.14(c), summed over the differentiated vertex.

The numerical criterion of Lemma 3.27, (B.36), is stated for $|k_\xi^{\mathrm{f}}|$; its proof uses only $|k_\xi^{\mathrm{f}}(D)| \leqslant m(D) = \varepsilon_* e^{-\lambda_* \ell_1(D)}$, so it gives $\sum_{D'\not\asymp D} m(D') e^{(1+6\theta)N_1(D')} \leqslant N_1(D)/8$ as well. The rooted-tree proof of Lemma 3.28 (Appendix B.6), applied to the activity $m(D)e^{N_1(D)/2}$ with node cost $N_1(D)/8$, the combined exponent $5/8$ being below $1+6\theta$, gives the first bound of (E.39); the family version follows by summing the single-root bound over $D \in \mathcal{A}$ (clusters hitting several roots being overcounted). Since $n \leqslant 4e^{n/4} \leqslant 4e^{n/2}$, inserting (E.38) gives the second bound.

Inserting the second bound into the derivative of the exponential (E.14) proves (E.15).

The same $m(D)$ controls the undifferentiated activity along the whole extension ($|k_\xi^{\mathrm{f}}(D)| \leqslant m(D)$ there by Lemma E.16 and the host bounds of Section E.9) and its one-derivative polynomial majorant, the polynomial being paid by the displayed exponential moment. *q.e.d.*

E.7 Quantitative Group Estimates

E.7.1 Carrier, Source and Physical Weights

All group comparisons and flat-subtraction identities are made on A_{cov}. There the source rows have their free inputs in I, so Lemma 3.36 gives $U_\xi(0)(r) = U_1(r)$ for every contour bond r, and the free vector $B = a + \Psi_\xi(A_{\mathfrak{e}})$ equals a on I ($\Psi_\xi(A_{\mathfrak{e}})$ vanishes on I, proof of that lemma), so neither $A_{\mathfrak{e}}$ nor Ψ_ξ enters the fluctuating source rows; Lemma 3.17 and

(3.49) give $K_{\xi;1}$, and $K_{\xi;0}$ is the $s = 0$ member of the interpolation (E.71), with $H = H_{\xi,II}$, $P_{\xi,0}(a) = \sum_{\mathfrak{p}\ \text{near}} \mathfrak{p}(a)$ and the chosen r-box, every primitive argument having zero retained input. The boundary Jacobian, the far primitives and the retained quadratic term are in c_ξ, and every fixed-kernel operation, probe measure and vacuum activity is built from the centred objects. Hence the local background data of a coefficient are U_1, the carrier geometry and the fixed coupling/cutoff/mean prescription; V_0 enters these local coefficients only through U_1.

The local-data comparison of Section E.10 applies on a box containing all source and coefficient stencils and lying in Λ_1 outside S_1. The fixed-kernel support statements (Section E.9.2) place every surviving walk coefficient, seed inverse and primitive stencil there, and the regional I and Ω_1 are kept on both sides of the interpolation. At the flat endpoint these local formulas are the reference formulas for a_W, since their arguments contain no retained field, boundary map or additional V_0. The force–constraint and relative soft-gauge terms are those of the regional Hessian (3.16) and are included throughout.

Under a simultaneous site gauge each Hessian entry, restricted inverse and fixed walk coefficient is conjugated by the free-field adjoint action (Lemma 3.7, Lemma 3.21), the protected cutoffs are invariant, and the row solution is covariant; orthogonal changes of the integration variables leave the closed selected integrals unchanged, including their spatial Cauchy integrals, and their Laplace transforms give the covariance of each probe measure; the vacuum activities and Ursell products are scalar invariants; finally $T_W P_W^{-1}$ and h are simultaneously conjugated at the interval base. This is the gauge law of the group closed contraction factor by factor (Section E.10.2).

$\mathcal{O}$ is the positive outer operation (3.4): the Haar and constraint factors are measures and the displayed cutoffs and exponentials are nonnegative, and $0 < \mathcal{Z}_0 < \infty$ (finite by the measurability paragraph of Section 3.1; $\mathcal{Z}_0 = \int \mathcal{O}[\mathcal{J}_\xi^{\mathrm{a}}[1]]\,\mathrm{d}V = \int \mathcal{O}[\mathsf{w}_\xi]\,\mathrm{d}V$ is the normaliser of (3.7), the integral of a positive function, since $\nu_{\xi;1}$ is a probability measure and so $\mathcal{J}_\xi^{\mathrm{a}}[1] = \mathsf{w}_\xi$ by (3.49)), so $\mathbb{E}_{\mathrm{h}}$ is a positive normalised functional.

In (3.97), multiplied by $\mathbf{1}_{A_{\mathrm{cov}}}$ and with $\mathcal{J}_\xi^{\mathrm{a}}[f_\xi] = \mathsf{w}_\xi F_\xi(1)$ on A_{cov}^c, every term is integrable against the finite positive measure $\mathcal{O}[\mathsf{w}_\xi \bullet]\,\mathrm{d}V$, because $|S^\circ_{\xi,C} f_\xi(0)| = |F_\xi(0) - \mathscr{R}_\xi| \leqslant 3$ by (3.96), $|F_\xi(0)| \leqslant 1$ and Lemma E.1, $|F_\xi(1)| \leqslant 1$, and $|\mathcal{B}_\xi[f_\xi]| \leqslant C_{\mathrm{resp},\chi} g_0 r |I_C|$ on A_{cov} (Section D.6). With E_{bdry} of (D.21), the full identity is

$$\mathbb{E}_\varepsilon[W_\chi(C)] = \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} S^\circ_{\xi,C} f_\xi(0)] + E_{\mathrm{def}} + E_{\mathrm{bdry}}\ , \tag{E.40}$$

by (3.97) and (3.7).

The far uncovered event is empty: if coverage fails, either a contour bond r is not a bond of $\Omega_1 \supseteq \Lambda_1$, so that r has an endpoint in $\Lambda_1^c = B_\xi$, or a bond of a stencil $\mathcal{F}_r \not\subseteq I$ is a free bond with no endpoint in Λ_1 ($I = \Lambda_1^*$, Section 3.1); in both cases an element of Γ_C (a contour bond or a bond of a free-input stencil) lies in $E(B_\xi)$, so $\mathrm{dist}(\Gamma_C, E(B_\xi)) = 0$ and $A_{\mathrm{cov}}^c \subseteq \mathsf{N}_0 \subseteq \mathsf{N}_b$ ((E.2); the events N_b increase with b).

The exterior response is E_{bdry} of (D.21), bounded in Appendix D.

E.7.2 Two Quantitative Pins

Definition E.18 (Components, Classes, and Their Weights)**.** A component γ consists of a nonempty unordered root family $\mathcal{A}(\gamma)$ of marked domains and an ordered list $(\mathfrak{c}_1, \dots, \mathfrak{c}_k)$ of cluster occurrences with connected incidence graph as in Section 3.8. Its cube count n_γ counts

every constituent cube with its occurrence multiplicity; T_γ is the union of the contact sets of its roots, H_γ the hull of T_γ, X_γ the union of its domains (roots and cluster constituents), and $F(\gamma)$ its footprint, defined by (E.18) with the domains of X_γ and the traversals of H_γ. The weight of a component is

$$t_\gamma \coloneqq \frac{\rho^{|\mathcal{A}(\gamma)|}}{k!} \prod_{D\in\mathcal{A}(\gamma)} m(D) \prod_{j=1}^{k} M_v(\mathfrak{c}_j)\,. \tag{E.41}$$

A class K is an unordered family of components with connected hull-overlap graph, with weight $t_K \coloneqq \prod_{\gamma\in K} t_\gamma$. A class is long when the union of the hulls of its members has at least $n_C/2$ positions.

Every component has at least one marked root, so $\mathsf{T}_\gamma \neq \varnothing$ (only families of marked domains contribute, Section 3.8), and its hull is a cyclic interval of fewer than $n_C/2$ traversals or the whole contour (Section 3.8). For a long class all components merge into the whole-contour group (Section 3.8).

Summing unordered distinct roots is overcounted by ordered root tuples divided by their factorial; for the positive recursion the sum over classes is overcounted by ordered tuples divided by the tuple factorial, dropping the disjointness restrictions. The exponent ϖ satisfies

$$0 < \varpi \leqslant \min\Big(\frac{1}{16}, \frac{\lambda_*}{64 C_{\mathrm{g}}}\Big)\,, \tag{E.42}$$

with C_{g} of Lemma B.1.

Lemma E.19 (Hull and Spread Constants)**.** *There are constants A_{h}, A_{x}, fixed before the coupling, such that for every component γ, $|H_\gamma| \leqslant A_{\mathrm{h}} L M n_\gamma$, and the distance from a source contact of γ to any cube of X_γ, halos included, is at most $A_{\mathrm{x}} L M n_\gamma$.*

Proof. Both bounds follow from the square geometry and the wall connectivity of a component's domains (the geometry paragraph of Section E.5). *q.e.d.*

Definition E.20 (Pin Rates)**.** With A_{h}, A_{x} of Lemma E.19, let

$$\theta_{\mathrm{h}} \coloneqq \frac{\varpi}{4A_{\mathrm{h}} LM}, \qquad \theta_{\mathrm{x}} \coloneqq \frac{\varpi}{4A_{\mathrm{x}} LM}, \qquad A_{\mathrm{s}} \coloneqq 1 + A_{\mathrm{h}}, \qquad \mathfrak{s}(\gamma) \coloneqq n_\gamma + \frac{|H_\gamma|}{LM}\,,$$

and fix $\eta > 0$ with $1000\eta A_{\mathrm{s}} < \varpi/8$. Further, d_C is the contour distance and $d_T(a', \Delta)$ the periodic fine distance (Section E.2) from the bond of the traversal a' to the cube Δ.

Lemma E.21 (Component Pins)**.** *There are constants C_{pin} and C, fixed before the coupling, such that, for ρ below a threshold depending on them and with the rates of Definition E.20:*

(a) for every root D_0,

$$\sum_{\gamma\,|\,D_0\in\mathcal{A}(\gamma)} t_\gamma e^{\varpi n_\gamma} \leqslant \rho\, m(D_0) e^{(\varpi+1/4)N_1(D_0)}, \qquad \sup_{a\in\mathcal{I}_C} \sum_{\gamma\,|\,a\in\mathsf{T}_\gamma} t_\gamma e^{\varpi n_\gamma} \leqslant C_{\mathrm{pin}}\rho\,; \tag{E.43}$$

(b) for every traversal a and every cube Δ,

$$\sum_{\gamma\,|\,a\in H_\gamma} t_\gamma e^{\varpi n_\gamma/4} \leqslant \sum_{a'\in\mathcal{I}_C} e^{-\theta_{\mathrm{h}} d_C(a,a')} \sum_{\gamma\,|\,a'\in\mathsf{T}_\gamma} t_\gamma e^{\varpi n_\gamma/2} \leqslant C_{\mathrm{pin}}\rho\Big(1 + \frac{2}{e^{\theta_{\mathrm{h}}} - 1}\Big)\,, \tag{E.44}$$

$$\sum_{\gamma\,|\,\Delta\subset X_\gamma} t_\gamma e^{\varpi n_\gamma/4} \leqslant C_{\mathrm{pin}}\rho \sum_{a'\in\mathcal{I}_C} e^{-\theta_{\mathrm{x}} d_T(a',\Delta)} \leqslant C\rho\,, \tag{E.45}$$

and the footprint pin $\sum_{\gamma\,|\,\Delta\in F(\gamma)} t_\gamma e^{\varpi n_\gamma/4} \leqslant C\rho$ *holds, with* C *enlarged by the stencil factor* CLM*;*

(c) the three pins of (b) hold with the weight $e^{100\eta\mathfrak{s}(\gamma)}$*, leaving a fixed exponential margin.*

Proof. (a) The sum is a two-species positive tree recursion, the recursion of the proof of Lemma 3.28 (Appendix B.6) with two kinds of nodes: the roots D, with weight $\rho\, m(D)e^{\varpi N_1(D)}$ and cost $N_1(D)/4$, and the cluster occurrences $\mathfrak{c}$, with weight $M_v(\mathfrak{c})e^{\varpi n(\mathfrak{c})}$ and cost $n(\mathfrak{c})/4$; a connected incidence graph is bounded by the sum of its spanning trees, the factorials are the exponential-generating factorials of the unordered roots and the ordered occurrence list, and the iterates $T_v^{(k+1)} = t_v \exp(\sum_{v'\sim v} T_{v'}^{(k)})$ satisfy $T_v^{(k)} \leqslant t_v e^{a_v}$ once $\sum_{v'\sim v} t_{v'} e^{a_{v'}} \leqslant a_v$ for every node v (weights t_v, costs a_v).

The criterion holds. The occurrences hitting a node have weighted mass at most one half of its cost, by (E.39) applied to the node's constituents, since $e^{\varpi n}e^{n/4} \leqslant e^{n/2}$ ($\varpi+1/4 < 1/2$). A root hitting a node is incompatible with one of its constituents, so it contains one of the at most $9N_1$ cubes meeting a constituent cube or sharing a wall with it (N_1 the node's cube count); for a fixed cube Q,

$$\sum_{D\ni Q} m(D)e^{(\varpi+1/4)N_1(D)} \leqslant \varepsilon_* \sum_{D\ni Q} e^{-(\lambda_*-(\varpi+1/4)C_{\mathrm{g}})\ell_1(D)} \leqslant \varepsilon_* B(\lambda_*/2) =: C_{\mathrm{a}}\,,$$

by $N_1(D) \leqslant C_{\mathrm{g}}\ell_1(D)$ (Lemma B.1(1); the final domains are localisation domains) and the animal bound (B.2), applicable because $\lambda_* - (\varpi+1/4)C_{\mathrm{g}} \geqslant \lambda_* - \frac{5}{16}C_{\mathrm{g}} \geqslant \lambda_*/2 > C_{\mathrm{g}}\log 64$ by $\varpi \leqslant 1/16$ and $\lambda_* \geqslant 2C_{\mathrm{g}}\log 128$ ((B.26)). So the roots hitting a node have weighted mass at most $9\rho C_{\mathrm{a}}N_1 \leqslant N_1/32$, one eighth of the cost, once $9\rho C_{\mathrm{a}} \leqslant 1/32$, a condition on ρ alone, that is on g_0. The rooted sum at D_0 is therefore at most $\rho\, m(D_0)e^{\varpi N_1(D_0)}e^{N_1(D_0)/4}$, the first bound of (E.43). A root D_0 with source contact a has a bond of L_{D_0} in the stencil $\mathcal{F}_{r_a}$ ((3.87)), so it contains one of the at most C_{st} cubes meeting that stencil (C_{st} a constant of L and M); summing the first bound over these roots by the display gives the second bound with $C_{\mathrm{pin}} := C_{\mathrm{st}}C_{\mathrm{a}}$.

(b) By Lemma E.19, if a traversal a lies in H_γ and a' is a contact of γ, then $d_C(a,a') \leqslant A_{\mathrm{h}}LMn_\gamma$, so $e^{\varpi n_\gamma/4} \leqslant e^{\varpi n_\gamma/2}e^{-\theta_{\mathrm{h}}d_C(a,a')}$, and the contour sum $\sum_{a'} e^{-\theta_{\mathrm{h}}d_C(a,a')} \leqslant 1+2/(e^{\theta_{\mathrm{h}}}-1)$ is uniform in n_C. For a cube $\Delta \subset X_\gamma$ the analogous inequality holds with θ_{x} and the torus distance; for each of the four straight sides the sum over a' is at most $1 + 2/(e^{\theta_{\mathrm{x}}} - 1)$ up to the fixed cube/halo offset, bounded by the full one-dimensional periodic lattice sum independently of n and the periods. A cube $\Delta \in F(\gamma)$ either meets a domain of X_γ or its stencil halo (a fixed offset, absorbed as above) or meets the stencil of a traversal of H_γ, and at most CLM traversals have stencils meeting a given cube; so the footprint pin is at most (E.45) plus CLM times the hull pin (E.44), which is the stated bound with the enlarged constant.

(c) $100\eta\mathfrak{s}(\gamma) \leqslant 100\eta A_{\mathrm{s}}n_\gamma < \varpi n_\gamma/8$, so the enhanced weights are dominated by the $e^{\varpi n_\gamma/4}$ of (b) with a margin $e^{\varpi n_\gamma/8}$. *q.e.d.*

E.7.3 Short Classes and the Exceptional Whole-Contour Merge

Lemma E.22 (Class Pins)**.** *Let* C *be the constant of the pins (b)–(c) of Lemma E.21 and assume* $C\rho LM \leqslant \eta/8$*. Let* $\sigma_K := \sum_{\gamma\in K}\mathfrak{s}(\gamma)$ *for a class* K*. Then:*

(a) for every overlap-connected class K, before the promotion of a long class to the whole contour, the hull and footprint pin sums $\sup_{a\in\mathcal{I}_C}\sum_{K\,|\,a\in\bigcup_{\gamma\in K}H_\gamma} t_K e^{64\eta\sigma_K}$ *and* $\sup_{\Delta}\sum_{K\,|\,\Delta\in\bigcup_{\gamma\in K}F(\gamma)} t_K e^{64\eta\sigma_K}$ *are at most* $C\rho$*;*

(b) a long class satisfies

$$\sigma_K \geqslant \frac{1}{LM}\sum_{\gamma\in K}|H_\gamma| \geqslant \frac{n_C}{2LM}, \qquad \sum_{K\ \text{long}} t_K e^{4\eta\sigma_K} \leqslant C\rho n_C e^{-30\eta n_C/(LM)}\,; \tag{E.46}$$

(c) the whole-contour groups created by the exceptional merge have total weighted mass $\sum_{W\ \text{whole}} M(W)e^{4\eta\mathfrak{s}(W)}$ *at most*

$$C\rho n_C e^{-30\eta n_C/(LM)} e^{4\eta n_C/(LM)} e^{C\rho n_C} \leqslant C\rho n_C e^{-25\eta n_C/(LM)} \leqslant C'\rho\,, \tag{E.47}$$

uniformly in n_C;

(d) the two weighted group pins (E.25) *hold.*

Proof. (a) The positive overlap-tree recursion runs on the decorated components with the weights enhanced by $e^{64\eta\mathfrak{s}(\gamma)}$. The pin of Lemma E.21(c) controls the overlap sum of a hull: the components whose hulls meet H_γ have weighted mass at most $C\rho|H_\gamma| \leqslant \eta\mathfrak{s}(\gamma)/8$ by the hypothesis $C\rho LM \leqslant \eta/8$, so the rooted recursion over components (node weight $t_\gamma e^{64\eta\mathfrak{s}(\gamma)}$, node cost $\eta\mathfrak{s}(\gamma)/8$; the pins of Lemma E.21(c) carry the weight $e^{100\eta\mathfrak{s}(\gamma)} \geqslant e^{64\eta\mathfrak{s}(\gamma)}e^{\eta\mathfrak{s}(\gamma)/8}$, which covers the node cost) gives the hull and footprint pins for every class before promotion: the sum over the classes containing a component pinned by the hull pin at a, or by the footprint pin at Δ, is the rooted sum with that component as root. No property of the union interval enters, so the class pins hold for long classes as well; for a short class the interval $\mathcal{H}(K)$ is the union of its hull intervals (Section 3.8), so its size is bounded by the additive class cost σ_K.

(b) The first inequality is the definition of $\mathfrak{s}$; the second is the definition of a long class (the union of its hulls has at least $n_C/2$ positions, Section 3.8). A class is anchored at a traversal of its union interval; there are at most n_C anchors, and the class pin (a) with the weight $e^{64\eta\sigma_K}$ gives the second bound of (E.46), the remaining exponent $60\eta\sigma_K$ being at least $30\eta n_C/(LM)$.

(c) Once a long class occurs, every other component is merged into the whole-contour group; the other components with their 4η weights sum to at most $e^{C\rho n_C}$ (each has a contact on the contour and is pinned there by Lemma E.21(c), so the sum over the families of further components is at most $\prod_{a\in\mathcal{I}_C}\exp(C\rho)$), and the whole-contour interval weight costs $e^{4\eta n_C/(LM)}$. The middle inequality of (E.47) uses $C\rho LM \leqslant \eta$, implied by the hypothesis, and the final maximum is finite uniformly in n_C because LM and η are fixed.

If a spatial pin lies in an additional component, that component is rooted by its footprint pin at a further cost $C\rho$, the anchored long-class factor and remaining exponential being paid by (E.47); if it lies in the long class, the footprint pin replaces the hull pin; pins on the whole contour are covered by the total whole-group bound. Polynomial moments use a smaller part of the same enhanced-weight margin.

(d) A group W with a short interval determines a class K (Section 3.8), and the groups with class K are the interleavings of the cluster lists of its components: by the interleaving

count of Lemma 3.33 there are at most $k(W)!/\prod_{\gamma\in K} k_\gamma!$ of them, $k(W)=\sum_{\gamma\in K} k_\gamma$, and by (E.23) each has

$$M(W)=\prod_{\gamma\in K}(\rho^{|\mathcal{A}(\gamma)|}\prod_{D\in\mathcal{A}(\gamma)} m(D)\prod_j M_v(\mathfrak{c}_j))/k(W)!\,,$$

so the sum of $M(W)$ over the groups with class K is at most t_K. Every such W has $\mathfrak{s}(W)\leqslant\sigma_K$, since $n(W)=\sum_{\gamma\in K} n_\gamma$ and $\mathcal{H}(W)$, the union of the overlapping hulls H_γ, has at most $\sum_{\gamma\in K}|H_\gamma|$ traversals, and $F(W)\subset\bigcup_{\gamma\in K}F(\gamma)$. Hence the sums of (E.25) over short groups are bounded by the pin sums of (a), and those over whole-contour groups by (c). *q.e.d.*

E.7.4 Interior Boxes, Endpoints and the Differentiated Group Weight

Let C_{box} be a fixed geometric constant for the group footprint and the row/contour halos such that a liftable enclosing comparison rectangle has diameter at most $C_{\mathrm{box}}LMn(W)$, increased so that a group with

$$n(W)<\frac{b}{4C_{\mathrm{box}}LM} \tag{E.48}$$

has its entire rectangle and stencils within $b/2$ of a source contact and an injective lift (Section E.10.4). On a far history this box lies in Λ_1, hence in Ω_1, and outside $S_1=(\Omega_1\setminus\Omega_1^{\sim-1})\cup R_1$; its plaquettes satisfy $\delta_{Q_W}(U_1)\leqslant C\varepsilon_1$ by Lemma B.4 and the distance check of Section E.10.5, and, for a group passing the phase test of Lemma E.23 (the groups failing it are treated in Section E.11.2), the Hessian bounds (E.37) of Lemma E.16 hold on the globally small extension used for its comparison.

All other groups keep their undifferentiated difference, $|c_W|+|a_W\chi|\leqslant 2M(W)$ (Section E.5.2); their cube count $n(W)$ is at least $b/(4C_{\mathrm{box}}LM)$, and if a rectangle fails to lift the same lower bound holds, since $N_{\min}=\min_i N_i\geqslant 2LMR_1>32b$ (Section E.8.2) and a wrapping connected network has length at least a fixed fraction of $N_{\min}$.

For a good box, by the coefficient locality and covariance of Sections E.7.1 and E.10, with the tree gauge rooted at the initial interval base and h fixed, $c_W(U_1;h)$ is compared with $c_W(\mathbf{1};h)=a_W\chi(h)$ by the small-phase extension and the fundamental theorem of calculus (Lemma E.23, for a group passing the phase test). By the product rule,

$$\mathscr{D}_\omega w_\xi(W)=\frac{(-1)^{k(W)}}{k(W)!}\sum_{j=1}^{k(W)}\mathscr{D}_\omega\Phi^{\mathrm{v}}_\xi(\mathfrak{c}_j)\prod_{l\neq j}\Phi^{\mathrm{v}}_\xi(\mathfrak{c}_l)\,, \tag{E.49}$$

bounded by Lemma E.17 by $C_M\|\omega\|_\infty\sum_j n(\mathfrak{c}_j)$ times the positive product majorant. With the joint source derivative of Lemma E.4, whose factor $L_f+\sum_D N_1(D)$ is at most $C(|\mathcal{H}(W)|+n(W))$ because the number L_f of varying background occurrences in the relative word is at most $2|\mathcal{H}(W)|$ times the fixed reconstruction-row overlap constant (Section E.8.1) and $\sum_D N_1(D)\leqslant n(W)$, this gives $|\mathscr{D}_\omega c_W|\leqslant C\|\omega\|_\infty(n(W)+|\mathcal{H}(W)|)M(W)$ uniformly in h ((E.78)); the curvature phase bound contributes $C\,\mathrm{diam}(Q_W)^2\delta_{Q_W}$, and with $|\mathcal{H}(W)|\leqslant CLMn(W)$ all these losses are a fixed constant times ε_1 times a fixed polynomial in $\mathfrak{s}(W)$, absorbed by the enhanced pin ($x^ke^{-\eta x}\leqslant k!\eta^{-k}$).

For a large group the factor $e^{-\eta b/(8C_{\mathrm{box}}LM)}$ is extracted from its reserved weight $e^{\eta\mathfrak{s}(W)/2}$. Combining the two classes and summing over a contour pin gives (E.29) (Section E.11.2).

E.8 The Coefficient Extension and the Countable-Gas Limit

E.8.1 The Small-Phase Extension

Sections E.4 and E.6 give the joint source and one-derivative bounds at the physical centred background and on every globally small real extension described here, for each closed character $\chi(T_W P_W^{-1} h)$ uniformly in h.

Let W be a good interior group of Section E.7.4, with enclosing fine rectangle Q_W, which contains all coefficient, reconstruction, averaging and interval/closing stencils. By the support statements of Section E.9.2, for the fixed selection indices and the fixed spatial Cauchy coefficient defining the group only the background links on Q_W are used: the restricted seed inverse uses entries on E_0, every walk coefficient has its domains in the final component, and the potential and source rows have their finite stencils. The regional I, Ω_1 and every auxiliary parameter are held fixed.

g_W is the rooted tree gauge of U_1 on Q_W, with $g_W(\mathsf{base}) = \mathbf{1}$ (Lemma E.25). If $C_{\mathrm{tg}} \operatorname{diam}(Q_W)^2 \delta_{Q_W} \leqslant \tau_{\mathrm{b}}/2$ (the phase test; since $\operatorname{diam}(Q_W)$ is at least the largest side R of Q_W, it implies the hypothesis $C_{\mathrm{tg}} R^2 \delta_{Q_W} \leqslant \tau_{\mathrm{b}}/2$ of Lemma E.25), the principal phases $B_W(b) = \frac{1}{i} \log U_1^{g_W}(b)$ on the rectangle, extended by $B_W = 0$ on every other bond, define the coefficient argument and its direction:

$$U_{W,t}(b) = e^{itB_W(b)}, \qquad \omega_W(b) = B_W(b), \qquad \|\omega_W\|_\infty = \max_b h(B_W(b)), \qquad 0 \leqslant t \leqslant 1 \, ; \tag{E.50}$$

the norm is the invariant norm h, the comparison with the cutoff norm being absorbed in C. L_f is the number of occurrences of background links in the relative word $T_W P_W^{-1}$ on which this argument varies, and $L_f \leqslant 2|\mathcal{H}(W)|$ times the fixed reconstruction-row overlap constant. The complementary h is held fixed at the root base.

At each t the operator $H_t = \iota_I^* \mathsf{H}_{U_{W,t}}^{\Omega_1} \iota_I$, the row reconstruction $q_{U_{W,t}}$ and the near primitives at $U_{W,t}$ are formed with zero retained input and the same protected r-box. For a generic noncritical $U_{W,t}$ the energy primitive is defined by subtracting its constant, linear and quadratic Taylor terms as in (3.29), the Hessian keeping the force–constraint term; the Haar and constraint-Jacobian primitives use their analytic row formulas. These definitions specify an analytic coefficient extension.

For $t \in (0,1)$ the extension is an auxiliary analytic kernel without the fixed-V constraint. At the physical endpoint each compensated primitive agrees with its Taylor-subtracted formula by locality, criticality entering only through the primitive-sum identity (3.34); at the flat endpoint the formulas agree with the reference formulas.

Lemma E.16 gives $\gamma_L/2 \leqslant H_t \leqslant m_L$; the designated inverses, reconstruction and means use the fixed smaller row charts since $\|B_W\|_\infty \leqslant \tau_{\mathrm{b}}/2$ and $\tau_{\mathrm{b}} \leqslant \tau_L/4$; the host norms and their derivatives are bounded by $C_0^\partial g_0 (1+r)^3$ on these charts; so the Gaussian payment, the positive forest budget and the source estimates hold uniformly in t. The force term is one of the local entries off the minimum.

Lemma E.23 (The Coefficient Path)**.** *For a small interior group (*(E.48)*, a good box of Section E.7.4) passing the phase test of Section E.8.1, and for each $h \in G$ in the closed character $\chi(T_W P_W^{-1} h)$,*

$$d_W(U_1; h) = \int_0^1 \mathscr{D}_{\omega_W} c_W(U_{W,t}; h) \, \mathrm{d}t \, , \tag{E.51}$$

and $|\mathscr{D}_{\omega_W} c_W(U_{W,t}; h)| \leqslant C \|\omega_W\|_\infty (L_f + n(W)) M(W)$ uniformly in t and h.

Proof. At $t = 1$ the group coefficient agrees with the original one by coefficient locality on Q_W and simultaneous gauge covariance (Section E.10.3): the values outside Q_W can be replaced in the argument because no term of the spatial Cauchy coefficient consults them, and the root gauge $g_W(\mathsf{base}) = \mathbf{1}$ leaves h unchanged. At $t = 0$ the coefficient is exactly the regional reference one, with the same I and the same group index W, so $c_W(U_{W,0}; h) = a_W \chi(h)$.

The fundamental theorem of calculus applies: by the dominated-convergence majorants of Section E.10.3 and the termwise differentiation statement of Lemma E.24, $t \mapsto c_W(U_{W,t}; h)$ is continuously differentiable on $[0, 1]$ with derivative $\mathscr{D}_{\omega_W} c_W(U_{W,t}; h)$. The derivative of c_W differentiates $\prod_j \Phi^{\mathrm{v}}(\mathfrak{c}_j)$, giving the occurrence sum (E.49) with majorant $C\|\omega\|_\infty(\sum_j n(\mathfrak{c}_j))M(W)$, and the root integral, giving $C\|\omega\|_\infty(L_f + \sum_D N_1(D))M(W)$ by Lemmas E.4 and E.14; the sum is the displayed bound. *q.e.d.*

E.8.2 Parameter and Period Conventions

All results inherit the standing conditions (3.1) and the compatible even fine periods of Section D.7: every fine period N_i is an even integer multiple of $\ell = LMR_1$, and ℓ is odd (L odd, M and R_1 powers of L). If $N_i = q_i \ell$ then q_i is even and positive, so

$$N_{\min} = \min_i N_i \geqslant 2LMR_1 > 32\lfloor LMR_1/16 \rfloor = 32b\,, \tag{E.52}$$

the strict middle inequality because ℓ is odd. The square-fit condition $K_i \geqslant 3(n+2)$ and the source-row and coarse-grid conditions of Section 3 are also in force. Thus the good comparison box and every fixed stencil have a unique lift under the small-group criterion (E.48), and the periodic failure lies in the large-group class.

Γ_C consists of the individual positive contour bonds and all their fixed input templates, C_L including the number and size of the templates, and the radius-b ball of Lemma E.1 is around each individual bond, so the count $4(2b+3)^4$ has no n_C-dependent factor.

The input of Appendix D is the one-point probability $p_z \leqslant q \leqslant e^{-c_* \pi_*^2}$ of Lemma D.17 and Proposition D.18 (this q is the q_{pt} of Lemma E.1) with $p_0(g) = A_0(\log g^{-2})^{p_0}$, $p_0 \geqslant 5\mathfrak{r} \geqslant 10$.

E.8.3 Countably Many Group Species and Differentiation

Lemma E.24 (Countable Species)**.** *At a fixed finite torus let $\mathcal{W}_N := \{W \mid n(W) \leqslant N\}$, a finite set, with partition sum S_N and connected-cluster logarithm $\mathscr{L}_N$, $S_N = e^{\mathscr{L}_N}$. Under the hypotheses of Lemma E.10 ($C\rho LM \leqslant \eta/8$ and g_0 small),*

$$\sum_W |a_W| e^{\mathfrak{a}_W/8} \leqslant C\rho n_C < \infty, \qquad \mathscr{L}_N \to \mathscr{L}, \qquad S_N \to S^{\circ}_{\xi,C} = e^{\mathscr{L}} \neq 0\,, \tag{E.53}$$

the restricted identity $S^{\circ,[\mathfrak{B}]}_{\xi,C} = e^{\mathscr{L}^{[\mathfrak{B}]}}$ ($\mathscr{L}^{[\mathfrak{B}]}$ the connected-cluster logarithm of the species compatible with $\mathfrak{B}$) and the ratio bound (E.27) *hold for the countable index set, and the background derivatives of Lemmas E.4, E.14 and E.17 may be taken termwise and integrated in t before the countable limit.*

Proof. $\mathcal{W}_N$ is finite: the domain set is finite, and $n(W)$ bounds the number of roots, occurrences and members of every occurrence. Its gas satisfies the same Kotecký–Preiss inequality, deleting species only decreasing the positive sums. Every group has a nonempty contour interval, $|a_W| \leqslant M(W)$ (Section E.5.2) and $e^{\mathfrak{a}_W/8} = e^{\eta \mathfrak{s}(W)/8} \leqslant e^{4\eta \mathfrak{s}(W)}$, so the pin (E.25) summed

over the n_C traversals gives the first bound of (E.53); the rooted-tree estimate bounds the total absolute connected-cluster series by this finite sum up to a fixed constant; each finite cluster tuple is present for all large N, so absolute domination gives $\mathscr{L}_N \to \mathscr{L}$; independently the full gas sum over finite compatible families is absolutely bounded by $\exp(\sum_W |a_W|)$, and dominated convergence gives $S_N \to S^\circ_{\xi,C}$.

The same argument after excluding every species incompatible with a fixed compatible family $\mathfrak{B}$ gives the restricted identity; the logarithmic difference is the connected-cluster sum of tuples hitting $\mathfrak{B}$, counted once, with absolute sum at most $\sum_{W\in\mathfrak{B}} \mathfrak{a}_W/8$ by the pinned estimate.

For the derivatives, Lemmas E.14 and E.17 give summable positive majorants independent of t on the extension; on each finite truncation the product and integral derivatives are justified by the uniform Gaussian coercivity (Section E.9.3) and the fixed protected box, and the walk derivative series converges on its unchanged contours (Section E.9.2); the cluster majorant $n(\mathfrak{c})M_v(\mathfrak{c})$ is paid by its exponential moment and the group majorant has only the polynomial in $\mathfrak{s}(W)$, paid by the reserved weight. Uniform absolute convergence permits termwise differentiation and t-integration before the countable limit. *q.e.d.*

E.8.4 Integrability of the Far-History Sum

The near-history contribution is bounded after forming $\mathscr{R}_\xi$ (Lemma E.1). On covered far histories the proof of Lemma E.12 bounds $\sum_{\mathfrak{B}\neq\varnothing} |S^{\circ,[\mathfrak{B}]}_{\xi,C} R_\xi(\mathfrak{B})|$ by $|S^\circ_{\xi,C}|$ times the exponential of the one-group sum (E.30) minus one; since $|S^\circ_{\xi,C}| \leqslant 1$ ($S^\circ_{\xi,C}$ is the normalised integral (3.94) of f°_ξ, a normalised character of modulus at most one, against the positive reference kernel $K^\circ_{\xi;0}$) and the exponential is uniform in the branch fields and retained variables, it is integrable against the normalised positive history law at each fixed loop and cutoff, which justifies the far-history sum/integral exchanges.

E.9 Analytic Inputs for the Coefficient Extension

E.9.1 Joint Background/Phase Analyticity and Primitive Derivatives

For each of the finitely many row/primitive stencil types, in its translated fine-lattice frame, the background on the stencil is $U_Z(b) = e^{iZ_b}$; background directions use the maximum h-norm on this finite label set. The radii $\tau_{\mathfrak{b}}$ and $\theta_0 > 0$ are such that $\|Z\| < 2\tau_{\mathfrak{b}}$ and $\|A\| < 2\theta_0$ keep every finite path product and relative product entering the bond average and the mean of [8, (0.4), (0.11)] ((4.1)) inside the matrix-logarithm neighbourhood $\|W - \mathbf{1}\|_{\mathrm{op}} < 1/2$; this is possible uniformly over the finite types, every word having bounded length and $e^{iA}e^{iZ}$ being analytic in both variables.

The radii $\tau_{\mathfrak{b}}$, θ_0 used below are those of the final joint domain, after the reductions of this section and of Section E.10.5; in particular $2\tau_{\mathfrak{b}} \leqslant \tau_L/2$ (Lemma E.16) and $2\theta_0 \leqslant \vartheta_*/2$ (Section E.10.5), so that Lemma B.8(ii)–(iii) and Lemma B.9 apply at each real Z with $\|Z\| < 2\tau_{\mathfrak{b}}$ and each A with $\|A\| < 2\theta_0$.

The mean of [8, (0.11)] solves $\sum_j \log(W_j M^{-1}) = 0$. At $W_j = \mathbf{1}$, $M = \mathbf{1}$ its derivative in the Lie-algebra parameter of M is $-n\,\mathbf{1}$, so the parametric analytic implicit function theorem gives a jointly analytic local mean in the background and fluctuating phases; it agrees with the mean of Lemma B.9, both taking the value $\mathbf{1}$ at zero and solving that equation in the same smaller neighbourhood. The joint radii are shrunk so that the mean stays in that

neighbourhood.

The bond constraint on a row c is the jointly analytic function

$$\mathcal{Q}_Z(A) = \frac{1}{i}\log(\mathsf{M}(e^{iA}U_Z, c)\mathsf{M}(U_Z, c)^{-1}), \quad LQ_Z = D_A\mathcal{Q}_Z(0), \quad N_Z(A) = \mathcal{Q}_Z(A) - LQ_Z A\ , \tag{E.54}$$

M the bond average of [8, (0.4)], with $N_Z(0) = D_A N_Z(0) = 0$ for every Z.

The coefficient of the designated phase in LQ_Z is uniformly invertible on a smaller background polydisc, by the identity-background designated inverse (Lemma 3.6) and continuity, so the designated inverse h_Z with $LQ_Z h_Z = \mathbf{1}$ and $\mathsf{C}_Z = \iota_{\text{free}} - h_Z LQ_Z \iota_{\text{free}}$ are analytic in Z with uniformly bounded first background derivatives; these are the maps of Section 3.2. The row solution is defined by

$$\widetilde{D}_Z(A) = N_Z(A - h_Z\widetilde{D}_Z(A))\ . \tag{E.55}$$

On smaller joint balls the derivative of the right side in $\widetilde{D}$ has norm at most $1/2$ uniformly in Z, since $D_A N_Z(A) = O(\|A\|)$; uniform analytic contraction (the parametric implicit function theorem) gives joint analyticity and $\|\widetilde{D}_Z(A)\| \leqslant C\|A\|^2$ (Lemma B.8(ii) at each real Z). Cauchy estimates on the larger joint balls, followed by the order-two zero in A, give for $l = 0, 1$ and each fixed k needed below

$$\|D_Z^l D_A^k \widetilde{D}_Z(A)\| \leqslant C_{l,k}\|A\|^{\max(2-k,0)}\ . \tag{E.56}$$

All constants are maxima over finitely many types, independent of the volume, the history and g_0; the dimensions of the types depend on L and G only.

The protected free input satisfies $g_0\|\mathsf{C}_Z a\| < \theta_0/2$ after reducing the coupling, and the phase $q_Z(a) = g_0\mathsf{C}_Z a - h_Z\widetilde{D}_Z(g_0\mathsf{C}_Z a)$ satisfies

$$\|D_Z q_Z(a)\| \leqslant Cg_0(1+r), \qquad \|D_Z^l D_a^k q_Z(a)\| \leqslant C_{l,k}g_0^k \tag{E.57}$$

for $l = 0, 1$ and $1 \leqslant k \leqslant 2K+1$, K the fixed row input count of Section E.9.4.

Analytic traces of X and X^{-1} extend $\operatorname{Re}\operatorname{tr} X$ from unitary fields, where it equals $\operatorname{tr}(X + X^{-1})/2$; the energy and relative gauge expressions are therefore jointly analytic finite word functions. For the local energy function $E_Z(s)$ in the dimensionless free phase s (the action and gauge-fixing functional of (4.5) at the background U_Z), the compensated primitive has the Taylor-remainder form

$$p_Z^{\mathrm{E}}(a) = -\frac{g_0}{2}\int_0^1 (1-t)^2 D_s^3 E_Z(tg_0 a)[a, a, a]\,\mathrm{d}t\ . \tag{E.58}$$

Its first background derivative has the same factor g_0, since $D_Z D_s^3 E_Z$ is uniformly bounded on the smaller joint domain; hence $|p^{\mathrm{E}}| + \|D_Z p^{\mathrm{E}}\| \leqslant Cg_0(1+r)^3$, the relative means and the soft-gauge term being included in E_Z.

The Haar function $\log(\sigma(q)/\sigma(0))$ is analytic in q with an order-two zero, so composed with q_Z it and its Z-derivative are at most $Cg_0^2(1+r)^2$. The constraint-Jacobian logarithm is $\log\det(\mathbf{1} - h_Z D_A\widetilde{D}_Z)$ on a matrix ball of radius $< 1/2$; its value and Z-derivative are at most $Cg_0(1+r)$, as $D_A\widetilde{D}_Z = O(g_0(1+r))$ and $D_Z D_A\widetilde{D}_Z$ has the same order, and the real determinant is positive on that ball, so the analytic logarithm restricts to the real logarithm of Lemma B.8(iii).

Summing the finite primitive types assigned to one host gives the host derivative bound with $C_0^\partial = N_{\mathrm{h}}\max(C_{\mathrm{p}}, C_{\mathrm{p}}^\partial)e^{\kappa(B_{\mathrm{h}}+4)}$ of Section E.6.1; this constant and the finite source derivative constants are fixed after M and $\tau_{\mathfrak{b}}$ and before g_0, and no Mayer, walk or reserve constant depends on them.

E.9.2 Differentiated Fixed Walks and Exact Locality

The series (3.62) has the parameters $s_H = 2h_+$, $\mathsf{A}_H = \mathbf{1} - H/s_H$, $\mathsf{P}_H = \mathsf{A}_H^{n_*}$, $q_H = 1 - \gamma/s_H$, $\beta = V_* q_H^{n_*}$ ($V_* = V_{n_*}$), $a_* = e^{81\kappa_1}$ and n_*, fixed from the common input bounds for all backgrounds. Of the background these constructions use only the input bounds (3.56), $\gamma \leqslant H \leqslant h_+$, and the R-locality of the entries; both hold along $U_{W,t}$, the two-sided bound by Lemma E.16 and the locality by the entry statement of Lemma 3.7, whose radius R does not depend on the background.

The index set contains every geometrically allowed tuple of bond indices, including those whose coefficients vanish at a particular background, so it does not change under differentiation. The weights $w_\omega(\sigma)$ and domains $D(\omega)$ depend only on the labels and the fixed geometry.

Let c_H bound the Schur row and column norms of $\mathscr{D}_\omega H/\|\omega\|_\infty$, supplied by the finite analytic entries of Section E.9.1. Then $\mathscr{D}_\omega \mathsf{A}_H = -\mathscr{D}_\omega H/s_H$ and $\mathscr{D}_\omega \mathsf{A}_H^j = \sum_{l=0}^{j-1} \mathsf{A}_H^l (\mathscr{D}_\omega \mathsf{A}_H) \mathsf{A}_H^{j-1-l}$, a block sum of which is bounded by $V_j j q_H^{j-1} c_H \|\omega\|_\infty / s_H$. Let $d_* \coloneqq V_* n_* q_H^{n_*-1} c_H / s_H$ and let d_{ini} be the maximum of the corresponding bounds for $j < n_*$. Differentiating the grouped walk products gives the absolute Schur bound

$$\|\mathscr{D}_\omega S_{Z_0}(\sigma)\|_{\mathrm{Schur}} \leqslant C_{\mathrm{w}}^{\partial} \|\omega\|_\infty, \qquad C_{\mathrm{w}}^{\partial} = s_H^{-1/2} n_* \Big(\frac{a_* d_{\mathrm{ini}}}{1 - a_*\beta} + \frac{a_*^2 V_* d_*}{(1 - a_*\beta)^2} \Big) , \tag{E.59}$$

the denominators positive because $a_*\beta \leqslant 1/4$ ((3.57)); the same bound holds on the spatial Cauchy circles.

The derivative of C_{E_0} is $-C_{E_0}(\mathscr{D}_\omega H_{E_0}) C_{E_0}$, with $H_{E_0} \coloneqq \iota_{E_0}^* H \iota_{E_0}$, and that of Γ_{E_0} is the product rule on $\iota_{E_0}^* H \Pi_{I \setminus E_0} S_{Z_0}$ ((3.64)); their operator norms are fixed constants times $\|\omega\|_\infty$.

For a selected final domain Z the spatial variables outside Z are zero and the protected seed enlargement Z_0' lies in Z; a walk with a coefficient domain meeting a cube outside Z has a zero spatial factor, which remains zero after background differentiation. Every surviving walk, including every hidden fine step in its grouped powers, has its coefficient stencil inside its declared domains by the construction of D_b (Section 3.5).

The restricted inverse $C_{E_0} = H_{E_0}^{-1}$ consults only entries on the seed E_0, included in the comparison rectangle, and the normalised Gaussian determinant is $\det C_{E_0}$. All potential and source stencils and every constituent vacuum domain of c_W lie in Q_W. Hence no surviving selected coefficient consults the background outside Q_W; this is the locality used at $t = 1$ in Lemma E.23.

The common full Gaussian prefactor $\mathfrak{n}_\xi$ (Section 3.5) cancels in the conditional ratio defining $F_\xi(0)$ ((3.85)).

E.9.3 The Complex-Gaussian Moment Payment

On a selected component, E is its seed carrier E_D, J_Z its auxiliary carrier J_D (the block of $\bigsqcup_D J_D$ belonging to this component, Lemma 3.23), $H_E = \iota_E^* H \iota_E$, $C_E = H_E^{-1}$, and $\Gamma = \Gamma_1 + i\Gamma_2$ is the block of $\Gamma_{E_0}(\sigma)$, the pairings being complex bilinear. Before cutoffs and Mayer factors the joint density with respect to real Lebesgue measure $\mathrm{d}B\,\mathrm{d}X$ is the two centred Gaussian normalisations times

$$G_\sigma(B, X) = \exp\Big[-\frac{1}{2}\langle B, H_E B\rangle - \langle B, \Gamma X\rangle - \frac{1}{2}\langle \Gamma X, C_E \Gamma X\rangle - \frac{1}{2}\|X\|^2 \Big] . \tag{E.60}$$

The columns of Γ outside J_Z, and those of its derivative, vanish identically by Lemma 3.23, so the auxiliary standard Gaussians outside J_Z integrate to one.

With $\alpha = \alpha_G$ and $C_\alpha = (H_E - \alpha\mathbf{1})^{-1}$, the absolute density times $e^{\alpha\|B\|^2/2}$ is a positive real Gaussian with the original determinant prefactor; in the variables $y = B + C_\alpha\Gamma_1 X$ its exponent is

$$-\frac{1}{2}\langle y, (H_E - \alpha)y\rangle - \frac{1}{2}\langle X, (\mathbf{1} - T_\alpha)X\rangle\ , \qquad T_\alpha \coloneqq \Gamma_1^{\mathsf{T}}(C_\alpha - C_E)\Gamma_1 + \Gamma_2^{\mathsf{T}} C_E \Gamma_2\ , \tag{E.61}$$

the reserve matrix of Appendix B.4. With $H_E - \alpha \geqslant \gamma/2$ (from $H \geqslant \gamma$, (3.56), and $\alpha_G \leqslant \gamma/2$, Lemma 3.24(ii)), $\|T_\alpha\| \leqslant 1/4$ ((B.24)) and $\|\Gamma_1\| \leqslant A_*$ (the norm paragraph of Appendix B.4) and

$$c_{\mathrm{G}} = \frac{\min(\gamma/2, 3/4)}{2(1 + 2A_*/\gamma)^2} > 0\ , \tag{E.62}$$

the real quadratic form is at least $c_{\mathrm{G}}(\|B\|^2 + \|X\|^2)$. Its total unnormalised mass is the determinant product bounded by $e^{N_1(Z)}$ in Lemma 3.24(ii), and its normalised covariance is at most $c_{\mathrm{G}}^{-1}\mathbf{1}$. With $d = d_G(|E| + |J_Z|)$,

$$\int (1 + \|B\|^2 + \|X\|^2) e^{\alpha_G\|B\|^2/2} |G_\sigma(B, X)|\, \mathrm{d}B\, \mathrm{d}X \leqslant (1 + d/c_{\mathrm{G}}) e^{N_1(Z)}\ , \tag{E.63}$$

and $d \leqslant 8d_G(LM + 1)^4 N_1(Z)$, so the loss is linear in the cube count $N_1(Z)$ with a fixed M-dependent constant, paid by the margin $3/4$ in $\mathbf{1} - T_\alpha$.

The logarithmic derivative of the Gaussian normalisation contributes $\mathrm{Tr}(C_E \mathscr{D}_\omega H_E)/2$; the exponent derivative contributes $-\langle B, (\mathscr{D}_\omega H_E)B\rangle/2$, $-\langle B, (\mathscr{D}_\omega\Gamma)X\rangle$ and $-\mathscr{D}_\omega\langle \Gamma X, C_E\Gamma X\rangle/2$ with $\mathscr{D}_\omega C_E = -C_E(\mathscr{D}_\omega H_E)C_E$; each is bounded by $C\|\omega\|_\infty(d + \|B\|^2 + \|X\|^2)$ by Section E.9.2. Hence (E.63) proves the differentiated Gaussian payment $C_M N_1(Z)\|\omega\|_\infty e^{N_1(Z)}$ on the same Cauchy circles and carriers, all normalisers and the negative compensator $-\frac{1}{2}\langle \Gamma X, C_E\Gamma X\rangle$ differentiated. The protected cutoff domain does not depend on the background, so the derivative has no surface term.

E.9.4 Source Remainder Bounds

Let $\mathsf{J} \coloneqq (b_1, \dots, b_q)$ be distinct selected labels and let a' denote all other selected root variables. The mixed fundamental theorem of calculus gives the exact identity

$$\Delta_{\mathsf{J}} f(B, a') = \int_{[0,1]^q} D_{b_1} \cdots D_{b_q} f\Big(a' + \sum_{j=1}^{q} s_j B_{b_j}\Big)[B_{b_1}, \dots, B_{b_q}]\, \mathrm{d}\mathbf{s}\ , \tag{E.64}$$

the sum in the argument meaning the canonical embedding into the free carrier. Let K be a fixed bound for the number of source rows affected by one free label and for the number of inputs of one row.

The source is $f_W(a; h) = \chi(T_W(a)P_W^{-1}h)$ or the original closed Wilson character, a normalised character, so $|f| \leqslant 1$; untouched factors are unitary in its representation. The derivative product rule has at most K^q row assignments, and the derivatives within a row have order at most $2K + 1$, whose constants (E.57) are fixed. Consequently, with a fixed C_χ enlarged once,

$$\|D_{b_1} \cdots D_{b_q} f\| \leqslant (C_\chi g_0)^q, \qquad \sup|\Delta_{\mathsf{J}} f| \leqslant (C_\chi g_0 r)^q\ . \tag{E.65}$$

For root j let P_j be its homogeneous degree-q_j projection after taking Δ_{J_j}. Subtracting the integrand of (E.64) at zero root input and integrating its first derivative along the segment

$t(\mathbf{s}B)$ gives q_j terms, each with one additional derivative and one additional selected B factor. Thus, for any set O of roots,

$$\sup\Big|\prod_{j\in O}(1-P_j)\Delta_{\mathbf{J}}f\Big| \leqslant u^{q+|O|}, \qquad u = C_\chi g_0(1+r)\,, \tag{E.66}$$

the choices of the extra increment, at most $\prod_{j\in O} q_j \leqslant 2^q$, being absorbed in C_χ. The row-order bound $2K+1$ covers these remainders simultaneously because the roots have disjoint selected labels and a row has at most K inputs.

A single background derivative hits either a row coefficient or one of the L_f background occurrences of the closed word; (E.57) and the product rule give

$$\sup\Big|\mathscr{D}_\omega \prod_{j\in O}(1-P_j)\Delta_{\mathbf{J}}f\Big| \leqslant C_\chi\|\omega\|_\infty(L_f+q)\,u^{q+|O|}\,. \tag{E.67}$$

E.9.5 Tree Gauge on a Comparison Rectangle

Lemma E.25 (Tree Gauge)**.** *Let Q be an injectively lifted fine rectangle of largest side $R \geqslant 1$ with a root $o \in Q$, let U be a unitary field, and let $\delta_Q(U)$ be as in* (E.28)*. For $x \in Q$ let P_x run from o to x in coordinate order, each coordinate run oriented by its displacement; these paths form a rooted tree. Let $g(o) := \mathbf{1}$ and $g(x) := U(P_x)$, so that $U^g(b) = U(P_x)U(b)U(P_{x'})^{-1}$ for $b = (x, x')$ is the based closed transport at o of the fundamental loop of b. Then*

$$U^g(b) = \prod_{j=1}^{m_b} G_j U(\partial p_j)^{\epsilon_j} G_j^{-1}, \qquad m_b \leqslant 3R, \qquad \epsilon_j \in \{1,-1\}\,, \tag{E.68}$$

with every plaquette p_j in Q, and

$$|U^g(b)-\mathbf{1}|_\sigma \leqslant 3R\,\delta_Q(U)\,. \tag{E.69}$$

There is a constant C_{tg}, depending only on the fixed group chart and τ_{b}, such that $C_{\mathrm{tg}}R^2\delta_Q(U) \leqslant \tau_{\mathrm{b}}/2$ implies that every $U^g(b)$, $b \subset Q$, lies in the group chart, with principal phase B_b satisfying

$$\max_{b\subset Q} h(B_b) \leqslant C_{\mathrm{tg}}R^2\delta_Q(U) \leqslant \tau_{\mathrm{b}}/2\,. \tag{E.70}$$

Proof. For a bond of direction μ, its last μ-step is moved past the later coordinate runs of P_x. Each interchange with a perpendicular fine edge replaces the ordered word by one plaquette transport conjugated by the preceding word; all plaquettes stay in Q, and the number of interchanges is at most $\sum_{\nu>\mu}|x_\nu - o_\nu| \leqslant 3R$. Negative coordinate runs reverse a plaquette orientation without altering this count. Induction on the interchanges gives the exact nonabelian word (E.68), the order being that of the step interchanges and G_j the preceding transport to the plaquette, so every factor is a conjugated closed plaquette.

Unitary invariance of $|\bullet|_\sigma$ and telescoping give (E.69), with a constant independent of R. Norm conversion to the operator norm and the fixed local group logarithm give $h(\log U^g(b)) \leqslant 3C_G\sqrt{N}\,R\,\delta_Q$ whenever that distance is inside the group chart; since $R \geqslant 1$, requiring $C_{\mathrm{tg}} \geqslant 3C_G\sqrt{N}$ and $C_{\mathrm{tg}} \geqslant 6\sqrt{N}\tau_{\mathrm{b}}/r_G$ makes $C_{\mathrm{tg}}R^2\delta_Q \leqslant \tau_{\mathrm{b}}/2$ sufficient for the chart condition and for (E.70). The constants do not grow with b or the scale R_1.

The root $g(o) = \mathbf{1}$ leaves the complementary h fixed in every closed contraction. *q.e.d.*

E.10 Identification of the Centred Coefficient and Its Local Extension

This section concerns c_W of Section E.5.1, with W fixed on a covered branch, under the hypotheses of Section 3 (its Section 3.8 included) and Appendix B, with their fixed charts, carriers, cutoffs and parameters.

E.10.1 The Exterior Dependence

The exterior-response parameter is s, and the background-path parameter is t. Section 3 defines

$$K_{\xi;s}(a) = \Theta_\xi(a) \exp\Big(-\frac{1}{2}\langle a, H_{\xi,II} a\rangle - s\langle a, H_{\xi,IJ} A_{\mathrm{e}}\rangle + \sum_{\mathfrak{p} \text{ near}} \mathfrak{p}(a + sA_{\mathrm{e}})\Big) , \tag{E.71}$$

with U_1 held fixed in the s-interpolation. The collar identity and the boundary-response identity give, with $F_\xi(s)$ defined with the same Wilson function f_ξ,

$$\mathcal{J}^{\mathrm{a}}_\xi[1] = e^{c_\xi} I_{\xi;1}, \qquad F_\xi(1) = F_\xi(0) + \mathcal{B}_\xi[f_\xi] . \tag{E.72}$$

On a covered branch every contour row has its entire free-input set in I. Since Ψ_ξ is the identity on I and A_{e} vanishes there, row locality implies, for every a in the integration box, that every contour link in $f_\xi(a)$ is its reconstruction from a alone at background U_1: this is the argument of the proof of Lemma 3.36 with a retained instead of set to zero, using Lemma 3.9. Thus the source and $K_{\xi;0}$ have zero retained free input; U_1 itself still depends on the exterior determining data.

Every selected measure in c_W is constructed from $K_{\xi;0}$: its potentials are the near primitives at a, its Hessian is $H_{\xi,II}$ and its cutoff is Θ_ξ. The factors involving $\Psi_\xi(A_{\mathrm{e}})$, the far primitives and the boundary-map Jacobian are in c_ξ, and the interaction between a and A_{e} is accounted for by $\mathcal{B}_\xi$. Consequently the physical comparison is

$$\frac{1}{\mathcal{Z}_0}\int \mathcal{O}[e^{c_\xi} I_{\xi;1} F_\xi(0)]\mathrm{d}V + \frac{1}{\mathcal{Z}_0}\int \mathcal{O}[e^{c_\xi} I_{\xi;1} \mathcal{B}_\xi[f_\xi]]\mathrm{d}V , \tag{E.73}$$

with $I_{\xi;1}$; the term with $\mathcal{B}_\xi$ plays the role of the boundary separation in [9, (1.26)–(1.30)].

E.10.2 Covariance of the Selected Coefficient

For a site gauge g let $R_g a(b) \coloneqq \mathrm{Ad}_{g(b_-)}\, a(b)$ on free bonds; R_g is orthogonal for h, preserves each free bond subspace and commutes with geometric restrictions. The intertwinings in the proof of Lemma 3.7, $Q_{U^g} R_g = R_g^{(1)} Q_U$, $\mathcal{N}_{U^g} \circ R_g = R_g^{(1)} \mathcal{N}_U$, $h_{U^g} = R_g h_U (R_g^{(1)})^{-1}$ and $\mathsf{C}_{U^g} = R_g \mathsf{C}_U R_{g,*}^{-1}$ with $(R_g^{(1)} v)(c) \coloneqq \mathrm{Ad}_{g(c_-)}\, v(c)$ (the row covariance of Section 3.2, used in the same way in the proof of Lemma 3.34), give $q_{U^g}(R_g a) = R_g q_U(a)$, branch included: $R_g q_U(a)$ solves the transformed row equation and lies in the same invariant small ball, and the uniqueness of the small row solution there (Lemma 3.9; the uniqueness ball of Lemma B.8(ii)) identifies it with the selected solution.

The joint fixed point of Section E.9.1 is identified with this solution by choosing its phase radius so that its $O(\|A\|^2)$ bound lies inside the uniqueness ball of Lemma B.8 (Section E.10.5); the inverse and mean branches are those fixed in the row construction.

The action of each plaquette is invariant and each relative soft-gauge mean is conjugated at its base, so the constant, linear and quadratic Taylor coefficients of the energy primitives

transform with their arguments, the force–constraint term included; subtracting them preserves the covariance of the compensated primitives even at noncritical backgrounds. The Haar density is conjugation invariant. The constraint-Jacobian endomorphism transforms by conjugation between the same free and designated spaces, so its determinant and its selected logarithm are unchanged, the logarithm branch being fixed by continuation from zero within the nonvanishing chart. Hence each primitive and each host sum $\mathbf{V}_{\xi}(Y)$ is invariant under the simultaneous transformation of background and free variables.

The Hessian, each restricted seed inverse and every fixed walk transform by R_g on their indices, by the entry identities of Lemma 3.7 and the geometric walk formula; all geometric choices, Cauchy variables and Mayer parameters are unchanged. The change of variables $B \mapsto R_g B$, $X \mapsto R_g X$ in the selected integral preserves Lebesgue and standard Gaussian measures, the two allowed cutoff norms ((5.4)), the conditional Gaussian determinant and the cross and compensator terms, so each selected measure is pushed forward by R_g, and in the probe formula $\delta_B - \delta_0$ is pushed to $\delta_{R_g B} - \delta_0$. Summing selections,

$$\widehat{\nu}^{\mathrm{f}}_{U^g,D} = (R_g)_* \widehat{\nu}^{\mathrm{f}}_{U,D} \, . \tag{E.74}$$

The vacuum activities, their Ursell products and $w_{\xi}(W)$ are scalar invariants, the incompatibility graph and every numerical combinatorial factor being field independent. Both $T_W P_W^{-1}$ and the complementary h transform at the initial interval vertex o, so

$$c_W(U^g; g(o) h g(o)^{-1}) = c_W(U; h) \, . \tag{E.75}$$

The rooted comparison gauge has $g(o) = \mathbf{1}$.

E.10.3 Domain and Endpoint

Section E.7.4 defines the small interior groups by (E.48). The support of such a group includes every root domain, each vacuum-cluster domain occurrence, their coefficient halos and the whole contour interval $\mathcal{H}(W)$ with its closing transport. Under (E.48) the enclosing rectangle has diameter at most $C_{\mathrm{box}} LMn(W)$, lies within $b/2$ of a source contact and is injectively liftable under the period convention of Section E.8.2; on the far-history event the box and its halos lie in the regular region. All other groups, including winding or boundary-reaching groups, are in the large-domain tail of Section E.7.4, as in the small-domain/large-domain separation of [8, (3.5)–(3.9)] and the boundary separation of [9].

A small interior group is taken in its rooted tree gauge; if $C_{\mathrm{tg}} \operatorname{diam}(Q_W)^2 \delta_{Q_W}(U_1) \leqslant \tau_{\mathfrak{b}}/2$, then B_W and $U_{W,t}$ are defined by (E.50). With $\tau_{\mathfrak{b}} \leqslant \tau_L/4$ and $c_L \tau_{\mathfrak{b}} \leqslant \gamma_L/4$ (Lemma E.16) the entry Lipschitz estimate and the flat compression gap supply the two-sided Hessian bounds throughout the path.

For each selected coefficient Section E.9.2 fixes all walk parameters and restricts every surviving walk and coefficient stencil to its final domain. Thus, on the supports included in Q_W, the $t = 1$ recipe equals the gauge-transformed physical centred recipe, its values on bonds outside Q_W cannot enter the coefficient, and (E.75) returns the coefficient to the original gauge. At $t = 0$ the recipe is the regional reference recipe with exactly the same I, Ω_1, W, cutoffs and constants. This yields the endpoint identities

$$c_W(U_{W,1}; h) = c_W(U_1; h), \qquad c_W(U_{W,0}; h) = a_W \chi(h) \, . \tag{E.76}$$

For differentiation under the selected integrals, the norm of X in Section E.9.3 is the norm of X restricted to J_Z; all other auxiliary columns integrate out with mass one and

have zero derivative. The common strictly positive c_{G} of (E.62) supplies a background-independent Gaussian majorant for that finite-dimensional integral and its first derivative; the selected sums use the summable majorants of Lemma E.14 and Section E.6.4, and the vacuum occurrence sums those of Lemma E.17 and Section E.7.4. These majorants give dominated convergence for the derivative of the group coefficient, that of $w_{\xi}(W)$ included.

E.10.4 The Site-Gauge Extension and Lifting

The endpoint statement concerns covered far histories and the small interior groups of Section E.7.4. The rooted site gauge is extended from Q_W to the whole torus by the identity on sites outside Q_W. Every background bond consulted by the coefficient has both endpoints inside Q_W: in defining Q_W the enclosing rectangle is enlarged to contain both endpoints of each Hessian stencil, row set, block-mean path, plaquette and interval or closing-path bond, an enlargement that is part of the fixed halo included in C_{box}. Boundary bonds of the site-gauge extension are not consulted by the coefficient.

Along $U_{W,t}$ only the finite product $w_{\xi}(W)$, $(-1)^{k(W)}/k(W)!$ times the product of the specified vacuum cluster occurrences, varies; the global exclusion ratio $q_{\xi}(\mathcal{A})$ is expanded at the original background before the group construction. Lemma E.17 supplies the sum of the occurrence derivative majorants, and (E.49) applies the finite product rule.

The regional scalar a_W is defined in Section 3.8 from $w_{\xi}^{\circ}(W)$ and the same root measures at the identity background ((3.92)); its closed Schur identity (3.93) is the $t = 0$ identity.

For the lifting, each constituent connected domain is represented by a tree of its constituent coarse cubes and adjoining faces, and the union is rooted at a source contact. Each root or cluster-domain tree has at most its number of cubes minus one edges; incompatibility joins have bounded range, and the finite reconstruction halos add a fixed range at their contacts. The contour-hull extent bound bounds the additional contour edges joining the components by $CLMn(W)$; the network contains the whole interval $\mathcal{H}(W)$ and its closing path. The resulting connected bond network has a spanning traversal of length at most $CLMn(W)$ with C fixed.

If this is less than $N_{\min}/4$, the traversal is lifted from its root: every lifted vertex lies within that distance of the root, and two lifts of the same vertex could differ only by a nonzero period vector, of length at least $N_{\min}$, while the two paths giving them have total length less than $N_{\min}/2$, a contradiction. Every point of the enclosing coordinate rectangle and of its stipulated halos is within $\operatorname{diam}(Q_W) \leqslant C_{\mathrm{box}} LMn(W)$ of the root, and two lifts of such a point would differ by a nonzero period vector, of length at least $N_{\min}$, while being joined through the root by two segments of total length at most $2C_{\mathrm{box}} LMn(W) < b < N_{\min}$ ((E.48), (E.52)); so the rectangle and its halos project injectively. C_{box} includes these constants, so by (E.48) and (E.52) every small group lifts.

In applying the entry estimates, τ_{b} is taken small enough for both the h-maximum and the stated entry-chart norm, using $h \leqslant |\bullet|_{\sigma} \leqslant \sqrt{N} h$, before g_0 is chosen. Geometrically allowed zero walks re-index the same series and do not change its value or support domains.

E.10.5 Source and Radius Specifications

The imported mean covariance is [8, (0.6)], $\mathsf{M}(\{uU_j v\}) = u\mathsf{M}(\{U_j\})v$ for the specified mean on its small-diameter domain, which [8, (0.10)] asserts for the selected mean; it is used for

unitary site gauges with no smallness restriction on the gauge values, the relative distances and the real chart domains being preserved. The cutoff invariance is (5.4).

The curvature input of Section E.7.4 is Lemma B.4. Every vertex x of a plaquette of Q_W lies in Λ_1, since $Q_W \subset \Lambda_1$, and the L-block containing x contains a neighbouring vertex ($L \geqslant 3$). That internal fine bond is free, since designated bonds join distinct blocks, and belongs to I because it has the endpoint $x \in \Lambda_1$; so the plaquette has a bond within two fine-bond metric units of I, within the $3R_{\mathrm{c}}$ the lemma requires. This yields the bound $\delta_{Q_W}(U_1) \leqslant C\varepsilon_1$, with the fixed conversion between the lemma's operator norm and the norm $|\bullet|_\sigma$ of Lemma E.25.

For the branch identification of Section E.10.2 let $C_{\mathrm{D,joint}}$ be the constant in $\|\widetilde{D}_Z(A)\| \leqslant C_{\mathrm{D,joint}}\|A\|^2$ on the joint chart of Section E.9.1. After that chart is fixed, θ_0 is reduced so that $2\theta_0 \leqslant \vartheta_*/2$ and $4C_{\mathrm{D,joint}}\theta_0^2 < \rho_{\mathrm{d}}$; the norms of Z and A are those of Lemma B.8, in which ϑ_*, ρ_{d} and τ_L are measured, and the background radius satisfies $2\tau_{\mathrm{b}} \leqslant \tau_L/2$ (Lemma E.16), so every real background with $\|Z\| < 2\tau_{\mathrm{b}}$ lies in the chart of that lemma. At real backgrounds the joint solution for $\|A\| < 2\theta_0$ then lies in the uniqueness ball of Lemma B.8 and is its selected solution. The coupling threshold is fixed after these radii so that the phase inputs $g_0\mathsf{C}_U B$ of the enlarged free box remain in $\|A\| < \theta_0$.

E.11 Quantitative Assembly

This section proves the first-step estimate for E_{def} of (E.1). It uses the hypotheses of Section 3 (its Section 3.8 included) and Appendix B, the compatible even-period and square-fit conventions of Section E.8.2, and the same irreducible normalised character. Constants may depend on the fixed group, the character, L, M and the fixed expansion and chart choices, but not on the coupling, the square length, the compatible torus size or the history; the coupling is chosen last.

E.11.1 The Joint Derivative Supplied to the Group Coefficient

The quantitative derivative is obtained from Lemmas E.14 and E.15 and Section E.9. For each selected root the source is $\Delta_\jmath f$, a closed character difference; (E.66) bounds its mixed Taylor remainder by u^q and supplies an extra u for each odd leading term removed by reversal.

The root's Gaussian derivative is even in its joint (B, X) variables, including its determinant and compensator derivatives, and a derivative of a selected ordinary potential carries a factor ζ and vanishes at $\zeta = 0$; so the independent-root parity/Cauchy argument of Lemma E.15 applies after one background derivative.

Lemma E.14(b),(c) ((E.34), (E.35)) bounds a differentiated potential; (E.59) sums the derivatives of all geometrically indexed walks on the same spatial contours; (E.63) pays each Gaussian derivative by $C_M N_1(D)$ times the same $e^{N_1(D)}$ determinant majorant, with X restricted to J_Z, all other auxiliary variables integrating to one. Host and Gaussian derivatives therefore insert one factor $N_1(D)$. The source derivative inserts the number L_f of varying contour/closing links plus the number of selected labels, which Section E.6.1 bounds by $C_M \sum_D N_1(D)$. The resulting joint estimate is

$$|\mathscr{D}_\omega \mathfrak{Q}_\xi(\mathcal{A}; f)| \leqslant C_\chi \|\omega\|_\infty \Big(L_f + \sum_{D\in\mathcal{A}} N_1(D)\Big) \prod_{D\in\mathcal{A}} \rho\, m(D)\,, \tag{E.77}$$

with ρ of (E.11) and $m(D) = \varepsilon_* e^{-\lambda_* \ell_1(D)}$ of Lemma E.2; the pairing $\mathfrak{Q}_\xi(\mathcal{A}; f)$ and its source

are evaluated at the background $U_{W,t}$ of the fixed coefficient path of Section E.10 in place of U_1 (the subscript ξ is kept for brevity).

For a vacuum occurrence $\mathfrak{c}$ the selected-integral derivative gives (E.38); differentiating the finite product $w_\xi(W)$ by (E.49), its occurrences $\Phi^{\mathsf{v}}(\mathfrak{c}_j)$ likewise evaluated at $U_{W,t}$, and adding the root-integral derivative yields, uniformly for $h \in G$,

$$|c_W(U;h)| \leqslant M(W), \qquad |\mathscr{D}_\omega c_W(U;h)| \leqslant C_\chi \|\omega\|_\infty (n(W) + |\mathcal{H}(W)|) M(W) \,. \tag{E.78}$$

The undifferentiated estimate holds at the original physical background and at the regional flat reference. The factor from the relative interval's inverse word, at most a fixed multiple of $|\mathcal{H}(W)|$, is included in C_χ.

E.11.2 Payment of the Local and Tail Losses

Proof of Lemma E.11. Let $s = \mathfrak{s}(W) \geqslant 1$, so that $\mathfrak{a}_W = \eta s$. The fixed hull and box bounds give $n(W) + |\mathcal{H}(W)| \leqslant CLMs$ and $\operatorname{diam}(Q_W) \leqslant C_{\mathrm{box}} LMn(W) \leqslant C_{\mathrm{box}} LMs$ (Section E.7.4). The endpoint identities (E.76), the path integral (E.51) with the derivative bound (E.78) along it, and the plaquette estimate $\delta_{Q_W}(U_1) \leqslant C\varepsilon_1$ of Section E.10.5 with the phase bound (E.70) therefore give, for a small interior group passing the phase test,

$$\mathfrak{d}_W(U_1) \leqslant C_\chi \varepsilon_1 s^3 M(W) \,. \tag{E.79}$$

If the phase test of such a group fails, then $C_{\mathrm{tg}} \operatorname{diam}(Q_W)^2 \delta_{Q_W} > \tau_{\mathfrak{b}}/2$; with $\delta_{Q_W} \leqslant C\varepsilon_1$ and $\operatorname{diam}(Q_W) \leqslant CLMs$ this gives $1 \leqslant C\varepsilon_1 s^2$ with constants fixed before the coupling, so the two undifferentiated endpoint bounds (E.78) give (E.79) again, as $s \geqslant 1$. For a group outside the small interior class

$$\mathfrak{d}_W(U_1) \leqslant 2M(W), \qquad s \geqslant T_b \coloneqq \frac{b}{4C_{\mathrm{box}} LM} \,. \tag{E.80}$$

For the local polynomial payment let

$$C_{\eta,3} \coloneqq \sup_{s \geqslant 1} s^3 e^{-(31\eta/8)s} < \infty \,, \tag{E.81}$$

so that $s^3 e^{\eta s/8} \leqslant C_{\eta,3} e^{4\eta s}$; for the tail, $s \geqslant T_b$ gives $e^{\eta s/8} \leqslant e^{-2\eta T_b} e^{4\eta s}$. Hence the two weighted pin estimates of Lemmas E.21 and E.22 imply

$$\begin{gathered} \sum_W K_\chi \mathfrak{d}_W(U_1) e^{\mathfrak{a}_W/8} \leqslant C_\chi (\varepsilon_1 + e^{-2\eta T_b}) \sum_W M(W) e^{4\eta \mathfrak{s}(W)} \leqslant C_\chi n_C \rho (\varepsilon_1 + e^{-cb/(LM)}), \\ c = \frac{\eta}{2C_{\mathrm{box}}} \,. \end{gathered} \tag{E.82}$$

The last summation uses that every W has a nonempty contour interval, so summing the contour-pin estimate (E.25) over the n_C traversals counts every group at least once; the whole-contour groups are included by Lemma E.22(d). This is (E.29). *q.e.d.*

E.11.3 The Signed Family Sum and the Physical Expectation

Proposition E.26 (The Signed Defect Bound)**.** *Under the hypotheses stated at the start of this section, for g_0 below the thresholds of Lemmas E.1, E.9 and E.10, with $R_\Upsilon \geqslant 2$*

(Section E.6.1, Lemma E.2), $u \leqslant u_*$ and $v \leqslant 1$ (Lemma E.3) and the small-coupling conditions of Sections E.9.1 and E.10.5, with ρ of (E.11)*, $\rho = O(g_0^2(1+r)^4)$, $\varepsilon_1 = g_1 p_0(g_1)$, $b = \lfloor LMR_1/16 \rfloor$, $\pi_* = \min(p_0(g_0), p_0(g_1))$ and the constants C'_L, c_* of* (E.3) *and C_χ, c of* (E.29)*,*

$$|E_{\mathrm{def}}| \leqslant C'_L n_C e^{-(c_*/2)\pi_*^2} + \frac{1}{\mathcal{Z}_0} \int \mathcal{O}\Big[\mathsf{w}_\xi \mathbf{1}_{A_{\mathrm{cov}} \cap \mathsf{N}_b^c} |S^\circ_{\xi,C}| \Big(e^{C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})} - 1\Big)\Big](V)\, \mathrm{d}V \;, \tag{E.83}$$

and consequently

$$|E_{\mathrm{def}}| \leqslant C'_L n_C e^{-(c_*/2)\pi_*^2} + e^{C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})} - 1 \;. \tag{E.84}$$

Proof. Lemma E.10 supplies the single joint ratio (E.27). Combined with the closed-contraction product bound (E.21), formed after each d_W, the nonempty compatible-family sum is bounded by the exponential series of the summable one-group weight (E.29), which on covered far histories is (E.30) (Lemma E.12); the countable-group limit is the dominated finite-species limit of Lemma E.24, with the same positive majorant and the nonzero reference partition function supplied by its convergent logarithm.

Combining the near-history estimate (E.3) with the positivity of the original history measure gives (E.83).

Since $|S^\circ_{\xi,C}| \leqslant 1$ (Section E.8.4), the physical normalisation yields (E.84). *q.e.d.*

When $C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)}) \leqslant 1$, the exponential term of (E.84) is at most $2C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})$.

E.11.4 Fixed Reserve Conventions

The exponential vacuum estimate used in Lemma E.21 holds with $n(\mathfrak{c})/2$ as well as with $n(\mathfrak{c})/4$: the rooted-tree proof of Lemma 3.27 applied to the numerical activity $m(D)e^{N_1(D)/2}$ with node cost $N_1(D)/8$ has combined exponent $5/8$, below the $1+6\theta$ of the activity payment (3.72), which gives the first bound of (E.39).

In the footprint-contact sums of Lemma E.9 ((E.25)) and Lemma E.10, footprint contact includes adjacency of two LM-cubes (a shared face, edge or corner). A cube has at most $3^4 - 1$ such neighbours, so the spatial pin is summed over them as well, which inserts a fixed factor in the constant C multiplying ρ; the smallness condition $C\rho LM \leqslant \eta/8$ of Lemma E.9 is imposed with this factor.

Appendix F The Regional Scalar of the Covered Branch

This appendix identifies the scalar $S^\circ_{\xi,C}$ of the branch identity (3.97), defined in Section 3.8 as the reference-gas sum (3.94), with the regional square scalar (7.31) at the carrier I, and thereby with the perimeter/corner coefficient κ_C of (8.1) up to the error of Proposition 7.4. The quantitative comparison of the regional square scalar with the bulk square is the proof of Proposition 7.4, in particular (7.44) and (7.46).

F.1 Conventions

The cutoff prescription σ, the bare coupling g_0, the irreducible normalised character χ, the averaging and mean prescription of [8] and the designated-bond convention are those of the rest of the paper, with the standing conditions (3.1) and the collar conditions (7.28). The fine periods are $N_\mu = LK_\mu$, and $C = C_n$ is a coarse translate of the counterclockwise square with vertices 0, nLe_1, $nL(e_1+e_2)$, nLe_2, with n coarse segments of length L on each side and $K_\mu \geqslant 3(n+2)$ for every coarse period; its physical perimeter is $P(C_n) = 4nL\varepsilon$.

Let A_{cov} be the set of covered branches of Lemma 3.36 and let $\xi \in A_{\mathrm{cov}}$. The carrier is $D = I = \Lambda_1^*$ (Section 3), the set of free bonds with an endpoint in Λ_1, which is the carrier D_+ of Section 7 and contains its D_-; D^c is its complement in the torus free carrier $E_{*,T}$; distances are those of the periodic fine-bond metric d_T of (5.2).

For a coarse segment j of the cyclic index set $J_C := \{0,\dots,4n-1\}$ let K_j be the union of the fixed free-input stencils of its fine traversals, and with the fixed range $R_0 = 4(1+R_*+R_H+L)$ of Section 7 let

$$d_j := \max(0, d_T(K_j, I^c) - R_0), \qquad \iota_T := \max(0, N_{\min}/4 - R_0), \qquad N_{\min} := \min_\mu N_\mu\ , \tag{F.1}$$

the distance to the empty set being infinite with $e^{-\infty} = 0$. The constants below depend on the character, the group, L and the forest parameters only, not on n, the compatible periods or the shape of I; g_0 is chosen after them, below the common flat and regional thresholds. τ is the marked-forest parameter of (5.18); enlarging a fixed majorant constant multiplies τ by a fixed factor, absorbed in the constants and the coupling threshold.

F.2 Identification with the Regional Square Integral

Lemma F.1 (The Reference-Gas Scalar at the Carrier I)**.** *On A_{cov}, under the conventions of Section F.1 and the thresholds displayed in the proof of Proposition 7.4,*

$$S^\circ_{\xi,C_n} = S^\circ_{I,C_n,T} = \frac{\displaystyle\int \Theta_I^\sigma(B) e^{P_{I,T}(B)} \chi(\mathrm{Hol}_{U_{g_0,T}(i_{I,T}B)}\, C_n)\, \mathrm{d}\mu_{C_{I,T}}(B)}{\displaystyle\int \Theta_I^\sigma(B) e^{P_{I,T}(B)}\, \mathrm{d}\mu_{C_{I,T}}(B)}\ , \tag{F.2}$$

the right side being (7.31) *at $D = I$, with every original primitive label retained, including the labels crossing the boundary of I, and with the exterior link value the identity.*

Proof. By (3.94), $S^\circ_{\xi,C_n} = \int f^\circ_\xi K^\circ_{\xi;0}\, \mathrm{d}a / I^\circ_{\xi;0}$, the normalised expectation of the full reference Wilson loop under the centred kernel at the identity background, whose precision is $H^\circ_\xi = \iota_I^* \mathsf{H}_{\mathbf{1}}^{\Omega_1} \iota_I$, whose potentials are the near primitives evaluated at zero retained input and whose cutoff is the bond cutoff on I. Both sides of (F.2) are ratios of two integrals over $\mathfrak{g}^I$, so it suffices to identify the factors of the integrands.

Precision: by Lemma B.6 at the identity background the entries of H°_ξ are $f_{bb'}(0)$, the entries of the flat Hessian (4.7) on the branch's torus (as in the proof of Lemma B.16), and the collar puts every precision and primitive stencil touching I inside the regular region (the identification paragraph of Section 7 preceding (7.29)), so $H^\circ_\xi = i_{I,T}^* H_T i_{I,T}$, the diagonal contributions of the plaquettes crossing the boundary of I included; the Gaussian normaliser of $\mu_{C_{I,T}}$ is common to numerator and denominator and cancels within the ratio. This is the full-square form of (7.29), which uses only that the source is a closed contraction with the given input stencils.

Reconstruction: the row solutions $\widetilde{D}_{U,c}$ of Section 3.3 (Lemma 3.12) at the identity background and those of Section 7 agree as analytic germs at zero by their defining equation and uniqueness; the Hilbert-ball domain of Lemma 3.12 (reference version) and the maximum-norm domain of the flat expansion have a convex common phase domain containing the closed cutoff box at the stated thresholds, and analytic continuation on that domain identifies the two solutions throughout the box.

Primitives: Section 3.3 groups the gauge primitives by their centre y while (7.20) indexes them by the pairs (y, x); the sums coincide, the degree-two Taylor residual being linear in the local term, and every primitive whose free support misses I vanishes at $i_{I,T}B$ while every primitive meeting I is retained on both sides.

Cutoff: at zero exterior free field the conditional cutoff is Θ_I^σ at the threshold $r = \delta_0/g_0$ in the norm $|\bullet|_\sigma$, as in Section 7.

Wilson factor: on a covered branch every contour traversal has its free-input set in I, so the proof of Lemma 3.36 applies verbatim with a retained instead of set to zero and gives that each contour link of $f_\xi^\circ(a)$ is its reconstruction from a alone, that is, $U_{g_0,T}(i_{I,T}B)$ at $B = a$; the boundary map is the identity by the reference-data convention $\Psi^\circ = \mathbf{1}$ of Section 3.8.

Numerator and denominator agree factor by factor, so the ratios agree. *q.e.d.*

F.3 The Scalar of the Branch Identity

Proposition F.2 (The Regional Scalar of the Covered Branch)**.** *Under the conventions of Section F.1 and the hypotheses of Proposition 7.4, on* A_{cov}*,*

$$S^\circ_{\xi,C_n} = e^{-\delta m^\sigma(g_0,\varepsilon)P(C_n)}\big(z_{\mathrm{c}}^\sigma(g_0,\pi/2)\big)^4 e^{E^\circ_\xi} = \kappa_{C_n} e^{E^\circ_\xi}\,, \tag{F.3}$$
$$|E^\circ_\xi| \leqslant C\tau^2\Big[\sum_{j\in J_C} e^{-cd_j} + 4ne^{-c\iota_T} + (n+1)e^{-cn}\Big]\,,$$

with $\kappa_{C_n} \coloneqq e^{-\delta m^\sigma(g_0,\varepsilon)P(C_n)}(z_{\mathrm{c}}^\sigma(g_0,\pi/2))^4$ *the coefficient of* (8.1)*, built from the coefficients of* (5.39) *and* (6.12)*, and*

$$E^\circ_\xi = E^\circ_{I,C_n,T} = (F_{I,C,T} - F_{C,T}) + (F_{C,T} - F_{C,\infty}(n)) + E_n^\sigma\,, \tag{F.4}$$

the error of Proposition 7.4 at $D = I$*: as in its proof,* $F_{I,C,T} = \log S^\circ_{I,C_n,T}$ *on the connected source branch,* $F_{C,T} = \log(N^\sigma_{C_n,T}/D^\sigma_T)$*,* $F_{C,\infty}(n) = F_{\square,\infty}(n)$ *is the bulk logarithm* (6.9)*, and* $E_n^\sigma \coloneqq F_{\square,\infty}(n) - n\sum_{v\in\mathscr{V}_\square}\kappa_v - \sum_{(v,w)\in\mathscr{E}_\square}\zeta_{v,w}$ *is the corner-extraction error,* $|E_n^\sigma| \leqslant C\tau^2(n+1)e^{-cn}$ *by* (6.11)*, with* $n\sum_v\kappa_v = -\delta m^\sigma P(C_n)$ *((5.39)) and* $\sum_{(v,w)}\zeta_{v,w} = 4\log z_{\mathrm{c}}^\sigma$ *((6.12)), so that* $F_{I,C,T} = \log\kappa_{C_n} + E^\circ_\xi$*.*

A segment with $d_j = 0$ *contributes a boundary term of size* $C\tau^2$*.*

Proof. Lemma F.1 and Proposition 7.4 at $D = I$, which is covered on A_{cov} and is the carrier D_+ itself (Section F.1), so $D_- \subseteq D \subseteq D_+$ holds with the collar; the three terms of (F.4) are (7.44), (7.46) and the corner extraction (6.11), the first two retaining the changes of kernel and of summation range as errors before the translation-invariant coefficients are identified, as in [8, (4.35)–(4.37)]; the exterior operation and its boundary terms are those of Section 8. *q.e.d.*

Corollary F.3 (The Physical Identity with the Scalar Substituted)**.** *Under the hypotheses of Proposition F.2, with* E_{def} *of* (E.1) *and* E_{bdry} *of* (D.21)*,*

$$\mathbb{E}_\varepsilon[W_\chi(C_n)] = \kappa_{C_n}\,\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^\circ_\xi} W_\chi(C_n;U_1)] + E_{\mathrm{def}} + E_{\mathrm{bdry}}\,, \tag{F.5}$$

where $\mathbb{E}_{\mathrm{h}}$ *uses the original outer operation, the weight* $\mathsf{w}_\xi = e^{c_\xi} I_{\xi;1}$ *and the physical* $\mathcal{Z}_0$*. Every term is integrable against the normalised history weight.*

Proof. By (E.40), $\mathbb{E}_\varepsilon[W_\chi(C_n)] = \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} S^\circ_{\xi,C_n} f_\xi(0)] + E_{\mathrm{def}} + E_{\mathrm{bdry}}$, with $f_\xi(0) = W_\chi(C_n; U_1)$ on A_{cov} (Lemma 3.36), into which (F.3) is substituted. E°_ξ is a function of the finite geometry at fixed torus and reference parameters, hence of the history labels through I alone, and is independent of the field values at fixed labels, so the first term is measurable; its bound has at most $4n$ boundary terms, $|E^\circ_\xi| \leqslant C\tau^2(9n+1)$, so the factor $e^{E^\circ_\xi}$ is bounded and the term integrable.

For the other terms, $|S^\circ_{\xi,C_n}| \leqslant 1$ because it is the normalised integral of a character of modulus at most one against the positive reference kernel $K^\circ_{\xi;0}$ (Lemma B.16); $|F_\xi(s)| \leqslant 1$ since $F_\xi(s) = \int f_\xi \, \mathrm{d}\nu_{\xi;s}$ ((3.46)) with $\nu_{\xi;s}$ a probability measure, and $|f_\xi(0)| \leqslant 1$; hence $|\mathcal{B}_\xi[f_\xi]| = |F_\xi(1) - F_\xi(0)| \leqslant 2$ ((3.51)) and the summed signed defect $|F_\xi(0) - S^\circ_{\xi,C_n} f_\xi(0)| \leqslant 2$, and both are integrable against the probability history weight. *q.e.d.*

Remark F.4 (Scope). The identity (F.5) concerns the reconstructed fine loop at the background U_1 after the first step; the bare coarse loop and the later scales are not covered.

References

[1] Abdelmalek Abdesselam and Vincent Rivasseau, *Trees, Forests and Jungles: A Botanical Garden for Cluster Expansions*, Constructive Physics, Lecture Notes in Physics, vol. 446, Springer, Berlin, 1995, arXiv:hep-th/9409094.

[2] Tadeusz Bałaban, *Propagators and Renormalization Transformations for Lattice Gauge Theories. II*, Comm. Math. Phys. **96** (1984), 223–250.

[3] ———, *Averaging Operations for Lattice Gauge Theories*, Comm. Math. Phys. **98** (1985), 17–51.

[4] ———, *Propagators for Lattice Gauge Theories in a Background Field*, Comm. Math. Phys. **99** (1985), 389–434.

[5] ———, *Spaces of Regular Gauge Field Configurations on a Lattice and Gauge Fixing Conditions*, Comm. Math. Phys. **99** (1985), 75–102.

[6] ———, *The Variational Problem and Background Fields in Renormalization Group Method for Lattice Gauge Theories*, Comm. Math. Phys. **102** (1985), 277–309.

[7] ———, *Ultraviolet Stability of Three-Dimensional Lattice Pure Gauge Field Theories*, Comm. Math. Phys. **102** (1985), 255–275.

[8] ———, *Renormalization Group Approach to Lattice Gauge Field Theories. I. Generation of Effective Actions in a Small Field Approximation and a Coupling Constant Renormalization in Four Dimensions*, Comm. Math. Phys. **109** (1987), 249–301.

[9] ———, *Convergent Renormalization Expansions for Lattice Gauge Theories*, Comm. Math. Phys. **119** (1988), 243–285.

[10] ———, *Renormalization Group Approach to Lattice Gauge Field Theories. II. Cluster Expansions*, Comm. Math. Phys. **116** (1988), 1–22.

[11] Sacha Friedli and Yvan Velenik, *Statistical Mechanics of Lattice Systems: A Concrete Mathematical Introduction*, Cambridge University Press, Cambridge, 2017.

[12] Jon Johnsen, *On the Spectral Properties of Witten-Laplacians, Their Range Projections and Brascamp–Lieb's Inequality*, Integral Equations Operator Theory **36** (2000), 288–324, arXiv:1703.10011v2; the numbering of Proposition 4.5, Theorem 1.3, and (4.39) used in Appendix D is that of v2.

[13] Alexander S. Kechris, *Classical Descriptive Set Theory*, Graduate Texts in Mathematics, vol. 156, Springer, New York, 1995.

[14] Roman Kotecký and David Preiss, *Cluster Expansion for Abstract Polymer Models*, Comm. Math. Phys. **103** (1986), 491–498.